

T3 matched EFT Lagrangian terms: EFT_1_after_F

EFT order: 5; loop order: 1. Rows are model configurations and columns are field configurations. This stage integrates out F. Remaining active T3 fields: S1, S2.

Field combinations 1 of 10

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	S_1	$S_1 \ D$	S_2
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	$-\frac{2\hbar\overline{y_{1,i_1,r_1}y_{1,i_2,r_2}}S_1^2y_{1,i_1,r_2}y_{1,i_2,r_1}\left(S_1^\dagger\right)^2}{\epsilon}$ $+C_{S_1^\dagger S_1}S_1S_1^\dagger$ $-C_{S_1^\dagger S_1}\hbar\overline{y_{1,i_1,r_1}S_1y_{1,i_1,r_1}S_1^\dagger}$ $2C_{S_1^\dagger S_1}\hbar\overline{y_{1,i_1,r_1}}S_1y_{1,i_1,r_1}S_1^\dagger$ $-\frac{\epsilon}{2}S_1^2\left(S_1^\dagger\right)^2\lambda_{S_1}^{(1)}$ $-\frac{2\hbar\overline{y_{1,i_1,r_1}}S_1^2y_{1,i_1,r_1}\left(S_1^\dagger\right)^2\lambda_{S_1}^{(1)}}{\epsilon}$ $-\hbar\overline{y_{1,i_1,r_1}}S_1^2y_{1,i_1,r_1}\left(S_1^\dagger\right)^2\lambda_{S_1}^{(1)}$ $-\frac{2\hbar\overline{y_{1,i_1,r_1}}S_1^2y_{1,i_1,r_1}\left(S_1^\dagger\right)^2\lambda_{S_1}^{(1)}}{\epsilon}$	$+D_{\mu_1}\left(S_1\right)^\dagger D_{\mu_1}\left(S_1\right)$	$+\frac{2\hbar\overline{y_{2,i_1,r_1}y_{2,i_2,r_2}}S_{2,j_2}^2y_{2,i_1,r_2}y_{2,i_2,r_1}\left(S_{2,j_1}^\dagger\right)^2}{3\epsilon}$ $+C_{S_2^\dagger S_2}S_{2,j_1}S_{2,j_1}^\dagger$ $-\frac{C_{S_2^\dagger S_2}\hbar\overline{y_{2,i_1,r_1}}S_{2,j_1}y_{2,i_1,r_1}S_{2,j_1}^\dagger}{\sqrt{3}}$ $-\frac{2C_{S_2^\dagger S_2}\hbar\overline{y_{2,i_1,r_1}}S_{2,j_1}y_{2,i_1,r_1}S_{2,j_1}^\dagger}{\sqrt{3}\epsilon}$ $-\frac{4\hbar\overline{y_{2,i_1,r_1}y_{2,i_2,r_2}}S_{2,j_1}S_{2,j_2}y_{2,i_1,r_2}y_{2,i_2,r_1}S_{2,j_1}^\dagger S_{2,j_2}^\dagger}{3\epsilon}$ $+\frac{1}{2}S_{2,j_3}S_{2,j_4}S_{2,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{S_{2,j_3,j_1,j_4,j_2}}{}^1\lambda_{S_2}^{(1)}$ $-\frac{2\hbar\overline{y_{2,i_1,r_1}}S_{2,j_3}S_{2,j_4}y_{2,i_1,r_1}S_{2,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{S_{2,j_3,j_1,j_4,j_2}}{}^1\lambda_{S_2}^{(1)}}{\sqrt{3}\epsilon}$ $-\frac{\hbar\overline{y_{2,i_1,r_1}}S_{2,j_3}S_{2,j_4}y_{2,i_1,r_1}S_{2,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{S_{2,j_3,j_1,j_4,j_2}}{}^1\lambda_{S_2}^{(1)}}{\sqrt{3}}$ $+\frac{1}{2}S_{2,j_3}S_{2,j_4}S_{2,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{S_{2,j_3,j_1,j_4,j_2}}{}^2\lambda_{S_2}^{(2)}$ $\frac{2\hbar\overline{y_{2,i_1,r_1}}S_{2,j_3}S_{2,j_4}y_{2,i_1,r_1}S_{2,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{S_{2,j_3,j_1,j_4,j_2}}{}^2\lambda_{S_2}^{(2)}}{\sqrt{3}\epsilon}$ $-\frac{\hbar\overline{y_{2,i_1,r_1}}S_{2,j_3}S_{2,j_4}y_{2,i_1,r_1}S_{2,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{S_{2,j_3,j_1,j_4,j_2}}{}^2\lambda_{S_2}^{(2)}}{\sqrt{3}}$
cont.										
cont.										
cont.										
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$-\frac{2\hbar\overline{y_{1,i_1,r_1}y_{1,i_2,r_2}}S_1^2y_{1,i_1,r_2}y_{1,i_2,r_1}\left(S_1^\dagger\right)^2}{\epsilon}$ $+C_{S_1^\dagger S_1}S_1S_1^\dagger$ $-C_{S_1^\dagger S_1}\hbar\overline{y_{1,i_1,r_1}}S_1y_{1,i_1,r_1}S_1^\dagger$ $2C_{S_1^\dagger S_1}\hbar\overline{y_{1,i_1,r_1}}S_1y_{1,i_1,r_1}S_1^\dagger$ $-\frac{\epsilon}{2}S_1^2\left(S_1^\dagger\right)^2\lambda_{S_1}^{(1)}$ $-\frac{2\hbar\overline{y_{1,i_1,r_1}}S_1^2y_{1,i_1,r_1}\left(S_1^\dagger\right)^2\lambda_{S_1}^{(1)}}{\epsilon}$ $-\hbar\overline{y_{1,i_1,r_1}}S_1^2y_{1,i_1,r_1}\left(S_1^\dagger\right)^2\lambda_{S_1}^{(1)}$ $-\frac{2\hbar\overline{y_{1,i_1,r_1}}S_1^2y_{1,i_1,r_1}\left(S_1^\dagger\right)^2\lambda_{S_1}^{(1)}}{\epsilon}$	$+D_{\mu_1}\left(S_1\right)^\dagger D_{\mu_1}\left(S_1\right)$	$+\frac{2\hbar\overline{y_{2,i_1,r_1}y_{2,i_2,r_2}}S_{2,j_2}^2y_{2,i_1,r_2}y_{2,i_2,r_1}\left(S_{2,j_1}^\dagger\right)^2}{3\epsilon}$ $+C_{S_2^\dagger S_2}S_{2,j_1}S_{2,j_1}^\dagger$ $-\frac{C_{S_2^\dagger S_2}\hbar\overline{y_{2,i_1,r_1}}S_{2,j_1}y_{2,i_1,r_1}S_{2,j_1}^\dagger}{\sqrt{3}}$ $-\frac{2C_{S_2^\dagger S_2}\hbar\overline{y_{2,i_1,r_1}}S_{2,j_1}y_{2,i_1,r_1}S_{2,j_1}^\dagger}{\sqrt{3}\epsilon}$ $-\frac{4\hbar\overline{y_{2,i_1,r_1}y_{2,i_2,r_2}}S_{2,j_1}S_{2,j_2}y_{2,i_1,r_2}y_{2,i_2,r_1}S_{2,j_1}^\dagger S_{2,j_2}^\dagger}{3\epsilon}$ $+\frac{1}{2}S_{2,j_3}S_{2,j_4}S_{2,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{S_{2,j_3,j_1,j_4,j_2}}{}^1\lambda_{S_2}^{(1)}$ $-\frac{2\hbar\overline{y_{2,i_1,r_1}}S_{2,j_3}S_{2,j_4}y_{2,i_1,r_1}S_{2,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{S_{2,j_3,j_1,j_4,j_2}}{}^1\lambda_{S_2}^{(1)}}{\sqrt{3}\epsilon}$ $-\frac{\hbar\overline{y_{2,i_1,r_1}}S_{2,j_3}S_{2,j_4}y_{2,i_1,r_1}S_{2,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{S_{2,j_3,j_1,j_4,j_2}}{}^1\lambda_{S_2}^{(1)}}{\sqrt{3}}$ $+\frac{1}{2}S_{2,j_3}S_{2,j_4}S_{2,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{S_{2,j_3,j_1,j_4,j_2}}{}^2\lambda_{S_2}^{(2)}$ $\frac{2\hbar\overline{y_{2,i_1,r_1}}S_{2,j_3}S_{2,j_4}y_{2,i_1,r_1}S_{2,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{S_{2,j_3,j_1,j_4,j_2}}{}^2\lambda_{S_2}^{(2)}}{\sqrt{3}\epsilon}$ $-\frac{\hbar\overline{y_{2,i_1,r_1}}S_{2,j_3}S_{2,j_4}y_{2,i_1,r_1}S_{2,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{S_{2,j_3,j_1,j_4,j_2}}{}^2\lambda_{S_2}^{(2)}}{\sqrt{3}}$
cont.										
cont.										
cont.										
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$-\frac{2\hbar\overline{y_{1,i_1,r_1}y_{1,i_2,r_2}}S_1^2y_{1,i_1,r_2}y_{1,i_2,r_1}\left(S_1^\dagger\right)^2}{\epsilon}$ $+C_{S_1^\dagger S_1}S_1S_1^\dagger$ $-C_{S_1^\dagger S_1}\hbar\overline{y_{1,i_1,r_1}}S_1y_{1,i_1,r_1}S_1^\dagger$ $-\frac{2\hbar\overline{y_{1,i_1,r_1}}S_1^2y_{1,i_1,r_1}\left(S_1^\dagger\right)^2\lambda_{S_1}^{(1)}}{\epsilon}$	$+D_{\mu_1}\left(S_1\right)^\dagger D_{\mu_1}\left(S_1\right)$	$+\frac{2\hbar\overline{y_{2,i_1,r_1}y_{2,i_2,r_2}}S_{2,j_2}^2y_{2,i_1,r_2}y_{2,i$

[illegible]

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	S_1	D	S_2
cont.										$-\frac{2h\overline{y_{2,i_1,r_1}}S_{2,j_3}S_{2,j_4}y_{2,i_1,r_1}S_{2,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{S_2,j_3,j_1,j_4,j_2}}{\sqrt{3}\epsilon} {}^2\lambda_{S_2}^{(2)}$
										$-\frac{h\overline{y_{2,i_1,r_1}}S_{2,j_3}S_{2,j_4}y_{2,i_1,r_1}S_{2,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{S_2,j_3,j_1,j_4,j_2}}{\sqrt{3}} {}^2\lambda_{S_2}^{(2)}$
								$+C_{S_1^\dagger S_1}S_{1,j_1}S_{1,j_1}^\dagger$		$+C_{S_2^\dagger S_2}S_{2,j_1}S_{2,j_1}^\dagger$
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	$-\frac{1}{2}C_{S_1^\dagger S_1}h\overline{y_{1,i_1,r_1}}S_{1,j_1}y_{1,i_1,r_1}S_{1,j_1}^\dagger$	$+D_{\mu_1}(S_{1,j_1})^\dagger D_{\mu_1}(S_{1,j_1})$	$-\frac{1}{2}C_{S_2^\dagger S_2}h\overline{y_{2,i_1,r_1}}S_{2,j_1}y_{2,i_1,r_1}S_{2,j_1}^\dagger$
								$-\frac{C_{S_1^\dagger S_1}h\overline{y_{1,i_1,r_1}}S_{1,j_1}y_{1,i_1,r_1}S_{1,j_1}^\dagger}{\epsilon}$		$-\frac{C_{S_2^\dagger S_2}h\overline{y_{2,i_1,r_1}}S_{2,j_1}y_{2,i_1,r_1}S_{2,j_1}^\dagger}{\epsilon}$
								$-\frac{h\overline{y_{1,i_1,r_1}}y_{1,i_2,r_2}S_{1,j_1}S_{1,j_2}y_{1,i_1,r_2}y_{1,i_2,r_1}S_{1,j_1}^\dagger S_{1,j_2}^\dagger}{\epsilon}$		$-\frac{h\overline{y_{2,i_1,r_1}}y_{2,i_2,r_2}S_{2,j_1}S_{2,j_2}y_{2,i_1,r_2}y_{2,i_2,r_1}S_{2,j_1}^\dagger S_{2,j_2}^\dagger}{\epsilon}$
cont.								$+\frac{1}{2}S_{1,j_3}S_{1,j_4}S_{1,j_1}^\dagger S_{1,j_2}^\dagger \mathcal{I}_{S_1,j_1,j_3,j_2,j_4} {}^1\lambda_{S_1}^{(1)}$		$+\frac{1}{2}S_{2,j_3}S_{2,j_4}S_{2,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{S_2,j_1,j_3,j_2,j_4} {}^1\lambda_{S_2}^{(1)}$
								$\frac{h\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{1,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{1,j_2}^\dagger \mathcal{I}_{S_1,j_1,j_3,j_2,j_4} {}^1\lambda_{S_1}^{(1)}}{\epsilon}$		$\frac{h\overline{y_{2,i_1,r_1}}S_{2,j_3}S_{2,j_4}y_{2,i_1,r_1}S_{2,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{S_2,j_1,j_3,j_2,j_4} {}^1\lambda_{S_2}^{(1)}}{\epsilon}$
cont.								$-\frac{1}{2}h\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{1,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{1,j_2}^\dagger \mathcal{I}_{S_1,j_1,j_3,j_2,j_4} {}^1\lambda_{S_1}^{(1)}$		$-\frac{1}{2}h\overline{y_{2,i_1,r_1}}S_{2,j_3}S_{2,j_4}y_{2,i_1,r_1}S_{2,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{S_2,j_1,j_3,j_2,j_4} {}^1\lambda_{S_2}^{(1)}$
								$+C_{S_1^\dagger S_1}S_{1,j_1}S_{1,j_1}^\dagger$		$+C_{S_2^\dagger S_2}S_{2,j_1}S_{2,j_1}^\dagger$
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-\frac{1}{2}C_{S_1^\dagger S_1}h\overline{y_{1,i_1,r_1}}S_{1,j_1}y_{1,i_1,r_1}S_{1,j_1}^\dagger$	$+D_{\mu_1}(S_{1,j_1})^\dagger D_{\mu_1}(S_{1,j_1})$	$-\frac{1}{2}C_{S_2^\dagger S_2}h\overline{y_{2,i_1,r_1}}S_{2,j_1}y_{2,i_1,r_1}S_{2,j_1}^\dagger$
								$-\frac{C_{S_1^\dagger S_1}h\overline{y_{1,i_1,r_1}}S_{1,j_1}y_{1,i_1,r_1}S_{1,j_1}^\dagger}{\epsilon}$		$-\frac{C_{S_2^\dagger S_2}h\overline{y_{2,i_1,r_1}}S_{2,j_1}y_{2,i_1,r_1}S_{2,j_1}^\dagger}{\epsilon}$
								$\frac{h\overline{y_{1,i_1,r_1}}y_{1,i_2,r_2}S_{1,j_1}S_{1,j_2}y_{1,i_1,r_2}y_{1,i_2,r_1}S_{1,j_1}^\dagger S_{1,j_2}^\dagger}{\epsilon}$		$\frac{h\overline{y_{2,i_1,r_1}}y_{2,i_2,r_2}S_{2,j_1}S_{2,j_2}y_{2,i_1,r_2}y_{2,i_2,r_1}S_{2,j_1}^\dagger S_{2,j_2}^\dagger}{\epsilon}$
cont.								$+h\overline{y_{1,i_1,r_1}}y_{1,i_2,r_1}S_{1,j_1}S_{1,j_2}y_{1,i_1,r_2}y_{1,i_2,r_2}S_{1,j_1}^\dagger S_{1,j_2}^\dagger$		$+h\overline{y_{2,i_1,r_1}}y_{2,i_2,r_1}S_{2,j_1}S_{2,j_2}y_{2,i_1,r_2}y_{2,i_2,r_2}S_{2,j_1}^\dagger S_{2,j_2}^\dagger$
								$+\frac{1}{2}S_{1,j_3}S_{1,j_4}S_{1,j_1}^\dagger S_{1,j_2}^\dagger \mathcal{I}_{S_1,j_1,j_3,j_2,j_4} {}^1\lambda_{S_1}^{(1)}$		$+\frac{1}{2}S_{2,j_3}S_{2,j_4}S_{2,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{S_2,j_1,j_3,j_2,j_4} {}^1\lambda_{S_2}^{(1)}$
								$\frac{h\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{1,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{1,j_2}^\dagger \mathcal{I}_{S_1,j_1,j_3,j_2,j_4} {}^1\lambda_{S_1}^{(1)}}{\epsilon}$		$\frac{h\overline{y_{2,i_1,r_1}}S_{2,j_3}S_{2,j_4}y_{2,i_1,r_1}S_{2,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{S_2,j_1,j_3,j_2,j_4} {}^1\lambda_{S_2}^{(1)}}{\epsilon}$
cont.								$-\frac{1}{2}h\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{1,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{1,j_2}^\dagger \mathcal{I}_{S_1,j_1,j_3,j_2,j_4} {}^1\lambda_{S_1}^{(1)}$		$-\frac{1}{2}h\overline{y_{2,i_1,r_1}}S_{2,j_3}S_{2,j_4}y_{2,i_1,r_1}S_{2,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{S_2,j_1,j_3,j_2,j_4} {}^1\lambda_{S_2}^{(1)}$
								$+C_{S_1^\dagger S_1}S_{1,j_1}S_{1,j_1}^\dagger$		$+C_{S_2^\dagger S_2}S_{2,j_1}S_{2,j_1}^\dagger$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$-\frac{1}{2}C_{S_1^\dagger S_1}h\overline{y_{1,i_1,r_1}}S_{1,j_1}y_{1,i_1,r_1}S_{1,j_1}^\dagger$	$+D_{\mu_1}(S_{1,j_1})^\dagger D_{\mu_1}(S_{1,j_1})$	$-\frac{1}{2}C_{S_2^\dagger S_2}h\overline{y_{2,i_1,r_1}}S_{2,j_1}y_{2,i_1,r_1}S_{2,j_1}^\dagger$
								$-\frac{C_{S_1^\dagger S_1}h\overline{y_{1,i_1,r_1}}S_{1,j_1}y_{1,i_1,r_1}S_{1,j_1}^\dagger}{\epsilon}$		$-\frac{C_{S_2^\dagger S_2}h\overline{y_{2,i_1,r_1}}S_{2,j_1}y_{2,i_1,r_1}S_{2,j_1}^\dagger}{\epsilon}$
								$\frac{h\overline{y_{1,i_1,r_1}}y_{1,i_2,r_2}S_{1,j_1}S_{1,j_2}y_{1,i_1,r_2}y_{1,i_2,r_1}S_{1,j_1}^\dagger S_{1,j_2}^\dagger}{\epsilon}$		$\frac{h\overline{y_{2,i_1,r_1}}y_{2,i_2,r_2}S_{2,j_1}S_{2,j_2}y_{2,i_1,r_2}y_{2,i_2,r_1}S_{2,j_1}^\dagger S_{2,j_2}^\dagger}{\epsilon}$
cont.								$+\frac{1}{2}S_{1,j_3}S_{1,j_4}S_{1,j_1}^\dagger S_{1,j_2}^\dagger \mathcal{I}_{S_1,j_1,j_3,j_2,j_4} {}^1\lambda_{S_1}^{(1)}$		$+\frac{1}{2}S_{2,j_3}S_{2,j_4}S_{2,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{S_2,j_1,j_3,j_2,j_4} {}^1\lambda_{S_2}^{(1)}$
								$\frac{h\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{1,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{1,j_2}^\dagger \mathcal{I}_{S_1,j_1,j_3,j_2,j_4} {}^1\lambda_{S_1}^{(1)}}{\epsilon}$		$\frac{h\overline{y_{2,i_1,r_1}}S_{2,j_3}S_{2,j_4}y_{2,i_1,r_1}S_{2,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{S_2,j_1,j_3,j_2,j_4} {}^1\lambda_{S_2}^{(1)}}{\epsilon}$
cont.								$-\frac{1}{2}h\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{1,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{1,j_2}^\dagger \mathcal{I}_{S_1,j_1,j_3,j_2,j_4} {}^1\lambda_{S_1}^{(1)}$		$-\frac{1}{2}h\overline{y_{2,i_1,r_1}}S_{2,j_3}S_{2,j_4}y_{2,i_1,r_1}S_{2,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{S_2,j_1,j_3,j_2,j_4} {}^1\lambda_{S_2}^{(1)}$
								$+C_{S_1^\dagger S_1}S_{1,j_1}S_{1,j_1}^\dagger$		$+C_{S_2^\dagger S_2}S_{2,j_1}S_{2,j_1}^\dagger$
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	$-\frac{1}{2}C_{S_1^\dagger S_1}h\overline{y_{1,i_1,r_1}}S_{1,j_1}y_{1,i_1,r_1}S_{1,j_1}^\dagger$	$+D_{\mu_1}(S_{1,j_1})^\dagger D_{\mu_1}(S_{1,j_1})$	$-\frac{1}{2}C_{S_2^\dagger S_2}h\overline{y_{2,i_1,r_1}}S_{2,j_1}y_{2,i_1,r_1}S_{2,j_1}^\dagger$
								$-\frac{C_{S_1^\dagger S_1}h\overline{y_{1,i_1,r_1}}S_{1,j_1}y_{1,i_1,r_1}S_{1,j_1}^\dagger}{\epsilon}$		$-\frac{C_{S_2^\dagger S_2}h\overline{y_{2,i_1,r_1}}S_{2,j_1}y_{2,i_1,r_1}S_{2,j_1}^\dagger}{\epsilon}$
								$\frac{h\overline{y_{1,i_1,r_1}}y_{1,i_2,r_2}S_{1,j_1}S_{1,j_2}y_{1,i_1,r_2}y_{1,i_2,r_1}S_{1,j_1}^\dagger S_{1,j_2}^\dagger}{\epsilon}$		$\frac{h\overline{y_{2,i_1,r_1}}y_{2,i_2,r_2}S_{2,j_1}S_{2,j_2}y_{2,i_1,r_2}y_{2,i_2,r_1}S_{2,j_1}^\dagger S_{2,j_2}^\dagger}{\epsilon}$
cont.								$+\frac{1}{2}S_{1,j_3}S_{1,j_4}S_{1,j_1}^\dagger S_{1,j_2}^\dagger \mathcal{I}_{S_1,j_1,j_3,j_2,j_4} {}^1\lambda_{S_1}^{(1)}$		$+\frac{1}{2}S_{2,j_3}S_{2,j_4}S_{2,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{S_2,j_1,j_3,j_2,j_4} {}^1\lambda_{S_2}^{(1)}$
								$\frac{h\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{1,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{1,j_2}^\dagger \mathcal{I}_{S_1,j_1,j_3,j_2,j_4} {}^1\lambda_{S_1}^{(1)}}{\epsilon}$		$\frac{h\overline{y_{2,i_1,r_1}}S_{2,j_3}S_{2,j_4}y_{2,i_1,r_1}S_{2,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{S_2,j_1,j_3,j_2,j_4} {}^1\lambda_{S_2}^{(1)}}{\epsilon}$
cont.								$-\frac{1}{2}h\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{1,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{1,j_2}^\dagger \mathcal{I}_{S_1,j_1,j_3,j_2,j_4} {}^1\lambda_{S_1}^{(1)}$		$-\frac{1}{2}h\overline{y_{2,i_1,r_1}}S_{2,j_3}S_{2,j_4}y_{2,i_1,r_1}S_{2,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{S_2,j_1,j_3,j_2,j_4} {}^1\lambda_{S_2}^{(1)}$

[illegible]

[illegible]

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	S_1	S_1	D	S_2
cont.											
T3-D	0	3	0	1	1	2	$\frac{1}{2}$				
cont.											
cont.											
cont.											
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1				
cont.											
cont.											
cont.											
T3-D	2	3	1	1	2	2	$\frac{3}{2}$				

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	S_1	$S_1 D$	S_2
										$-\frac{2C_{S_1^\dagger S_1} \overline{h\bar{y}_{1,i_1,r_1}} S_{1,j_1} y_{1,i_1,r_1} S_{1,j_1}^\dagger}{\sqrt{3}\epsilon}$ $-\frac{4h\bar{y}_{1,i_1,r_1} \overline{y_{1,i_2,r_2}} S_{1,j_1} S_{1,j_2} y_{1,i_1,r_2} y_{1,i_2,r_1} S_{1,j_1}^\dagger S_{1,j_2}^\dagger}{3\epsilon}$ $+\frac{1}{2} S_{1,j_3} S_{1,j_4} S_{1,j_1}^\dagger S_{1,j_2}^\dagger \mathcal{I}_{S_1,j_3,j_1,j_4,j_2} {}^1\lambda_{S_1}^{(1)}$ $-\frac{2h\bar{y}_{1,i_1,r_1} S_{1,j_3} S_{1,j_4} y_{1,i_1,r_1} S_{1,j_1}^\dagger S_{1,j_2}^\dagger \mathcal{I}_{S_1,j_3,j_1,j_4,j_2} {}^1\lambda_{S_1}^{(1)}}{\sqrt{3}\epsilon}$ $-\frac{h\bar{y}_{1,i_1,r_1} S_{1,j_3} S_{1,j_4} y_{1,i_1,r_1} S_{1,j_1}^\dagger S_{1,j_2}^\dagger \mathcal{I}_{S_1,j_3,j_1,j_4,j_2} {}^1\lambda_{S_1}^{(1)}}{\sqrt{3}}$ $+\frac{1}{2} S_{1,j_3} S_{1,j_4} S_{1,j_1}^\dagger S_{1,j_2}^\dagger \mathcal{I}_{S_1,j_3,j_1,j_4,j_2} {}^2\lambda_{S_1}^{(2)}$ $-\frac{2h\bar{y}_{1,i_1,r_1} S_{1,j_3} S_{1,j_4} y_{1,i_1,r_1} S_{1,j_1}^\dagger S_{1,j_2}^\dagger \mathcal{I}_{S_1,j_3,j_1,j_4,j_2} {}^2\lambda_{S_1}^{(2)}}{\sqrt{3}\epsilon}$ $-\frac{h\bar{y}_{1,i_1,r_1} S_{1,j_3} S_{1,j_4} y_{1,i_1,r_1} S_{1,j_1}^\dagger S_{1,j_2}^\dagger \mathcal{I}_{S_1,j_3,j_1,j_4,j_2} {}^2\lambda_{S_1}^{(2)}}{\sqrt{3}}$
cont.										$-\frac{2C_{S_2^\dagger S_2} \overline{h\bar{y}_{2,i_1,r_1}} S_{2,j_2} y_{2,i_1,r_1} S_{2,j_2}^\dagger}{\epsilon}$ $+\frac{1}{2} S_2^2 \left(S_2^\dagger\right)^2 \lambda_{S_2}^{(1)}$ $\frac{2h\bar{y}_{2,i_1,r_1} S_{2,j_2}^2 y_{2,i_1,r_1} \left(S_2^\dagger\right)^2 \lambda_{S_2}^{(1)}}{\epsilon}$
cont.										$-\frac{h\bar{y}_{2,i_1,r_1} S_{2,j_2}^2 y_{2,i_1,r_1} \left(S_2^\dagger\right)^2 \lambda_{S_2}^{(1)}}{\epsilon}$
cont.										$-\frac{h\bar{y}_{2,i_1,r_1} S_{2,j_2}^2 y_{2,i_1,r_1} \left(S_2^\dagger\right)^2 \lambda_{S_2}^{(1)}}{\epsilon}$
										$+\frac{2h\bar{y}_{1,i_1,r_1} y_{1,i_2,r_2} S_{1,j_2}^2 y_{1,i_1,r_2} y_{1,i_2,r_1} \left(S_{1,j_1}^\dagger\right)^2}{3\epsilon}$ $+C_{S_1^\dagger S_1} S_{1,j_1} S_{1,j_1}^\dagger$ $-\frac{C_{S_1^\dagger S_1} \overline{h\bar{y}_{1,i_1,r_1}} S_{1,j_1} y_{1,i_1,r_1} S_{1,j_1}^\dagger}{\sqrt{3}}$ $-\frac{2C_{S_1^\dagger S_1} \overline{h\bar{y}_{1,i_1,r_1}} S_{1,j_1} y_{1,i_1,r_1} S_{1,j_1}^\dagger}{\sqrt{3}\epsilon}$ $-\frac{4h\bar{y}_{1,i_1,r_1} \overline{y_{1,i_2,r_2}} S_{1,j_1} S_{1,j_2} y_{1,i_1,r_2} y_{1,i_2,r_1} S_{1,j_1}^\dagger S_{1,j_2}^\dagger}{3\epsilon}$ $+\frac{1}{2} S_{1,j_3} S_{1,j_4} S_{1,j_1}^\dagger S_{1,j_2}^\dagger \mathcal{I}_{S_1,j_3,j_1,j_4,j_2} {}^1\lambda_{S_1}^{(1)}$ $-\frac{2h\bar{y}_{1,i_1,r_1} S_{1,j_3} S_{1,j_4} y_{1,i_1,r_1} S_{1,j_1}^\dagger S_{1,j_2}^\dagger \mathcal{I}_{S_1,j_3,j_1,j_4,j_2} {}^1\lambda_{S_1}^{(1)}}{\sqrt{3}\epsilon}$ $-\frac{h\bar{y}_{1,i_1,r_1} S_{1,j_3} S_{1,j_4} y_{1,i_1,r_1} S_{1,j_1}^\dagger S_{1,j_2}^\dagger \mathcal{I}_{S_1,j_3,j_1,j_4,j_2} {}^1\lambda_{S_1}^{(1)}}{\sqrt{3}}$ $+\frac{1}{2} S_{1,j_3} S_{1,j_4} S_{1,j_1}^\dagger S_{1,j_2}^\dagger \mathcal{I}_{S_1,j_3,j_1,j_4,j_2} {}^2\lambda_{S_1}^{(2)}$ $-\frac{2h\bar{y}_{1,i_1,r_1} S_{1,j_3} S_{1,j_4} y_{1,i_1,r_1} S_{1,j_1}^\dagger S_{1,j_2}^\dagger \mathcal{I}_{S_1,j_3,j_1,j_4,j_2} {}^2\lambda_{S_1}^{(2)}}{\sqrt{3}\epsilon}$ $-\frac{h\bar{y}_{1,i_1,r_1} S_{1,j_3} S_{1,j_4} y_{1,i_1,r_1} S_{1,j_1}^\dagger S_{1,j_2}^\dagger \mathcal{I}_{S_1,j_3,j_1,j_4,j_2} {}^2\lambda_{S_1}^{(2)}}{\sqrt{3}}$
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$		$+D_{\mu_1} \left(S_{1,j_1}\right)^\dagger D_{\mu_1} \left(S_{1,j_1}\right)$	$+\frac{2h\bar{y}_{2,i_1,r_1} y_{2,i_2,r_2} S_{2,j_2}^2 y_{2,i_1,r_2} y_{2,i_2,r_1} \left(S_{2,j_1}^\dagger\right)^2}{3\epsilon}$ $+C_{S_2^\dagger S_2} S_{2,j_1} S_{2,j_1}^\dagger$ $-\frac{C_{S_2^\dagger S_2} \overline{h\bar{y}_{2,i_1,r_1}} S_{2,j_1} y_{2,i_1,r_1} S_{2,j_1}^\dagger}{\sqrt{3}}$ $-\frac{2C_{S_2^\dagger S_2} \overline{h\bar{y}_{2,i_1,r_1}} S_{2,j_1} y_{2,i_1,r_1} S_{2,j_1}^\dagger}{\sqrt{3}\epsilon}$ $-\frac{4h\bar{y}_{2,i_1,r_1} \overline{y_{2,i_2,r_2}} S_{2,j_1} S_{2,j_2} y_{2,i_1,r_2} y_{2,i_2,r_1} S_{2,j_1}^\dagger S_{2,j_2}^\dagger}{3\epsilon}$ $+\frac{1}{2} S_{2,j_3} S_{2,j_4} S_{2,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{S_2,j_3,j_1,j_4,j_2} {}^1\lambda_{S_2}^{(1)}$ $-\frac{2h\bar{y}_{2,i_1,r_1} S_{2,j_3} S_{2,j_4} y_{2,i_1,r_1} S_{2,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{S_2,j_3,j_1,j_4,j_2} {}^1\lambda_{S_2}^{(1)}}{\sqrt{3}\epsilon}$ $-\frac{h\bar{y}_{2,i_1,r_1} S_{2,j_3} S_{2,j_4} y_{2,i_1,r_1} S_{2,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{S_2,j_3,j_1,j_4,j_2} {}^1\lambda_{S_2}^{(1)}}{\sqrt{3}}$ $+\frac{1}{2} S_{2,j_3} S_{2,j_4} S_{2,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{S_2,j_3,j_1,j_4,j_2} {}^2\lambda_{S_2}^{(2)}$ $-\frac{2h\bar{y}_{2,i_1,r_1} S_{2,j_3} S_{2,j_4} y_{2,i_1,r_1} S_{2,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{S_2,j_3,j_1,j_4,j_2} {}^2\lambda_{S_2}^{(2)}}{\sqrt{3}\epsilon}$ $-\frac{h\bar{y}_{2,i_1,r_1} S_{2,j_3} S_{2,j_4} y_{2,i_1,r_1} S_{2,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{S_2,j_3,j_1,j_4,j_2} {}^2\lambda_{S_2}^{(2)}}{\sqrt{3}}$
cont.										$-\frac{4h\bar{y}_{2,i_1,r_1} \overline{y_{2,i_2,r_2}} S_{2,j_1} S_{2,j_2} y_{2,i_1,r_2} y_{2,i_2,r_1} S_{2,j_1}^\dagger S_{2,j_2}^\dagger}{3\epsilon}$ $+\frac{1}{2} S_{2,j_3} S_{2,j_4} S_{2,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{S_2,j_3,j_1,j_4,j_2} {}^1\lambda_{S_2}^{(1)}$ $-\frac{2h\bar{y}_{2,i_1,r_1} S_{2,j_3} S_{2,j_4} y_{2,i_1,r_1} S_{2,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{S_2,j_3,j_1,j_4,j_2} {}^1\lambda_{S_2}^{(1)}}{\sqrt{3}\epsilon}$ $-\frac{h\bar{y}_{2,i_1,r_1} S_{2,j_3} S_{2,j_4} y_{2,i_1,r_1} S_{2,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{S_2,j_3$

[illegible]

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	S_1	$S_1 D$	S_2
cont.									$-\frac{2h\overline{y}_{1,i_1,r_1} S_{1,j_3} S_{1,j_4} y_{1,i_1,r_1} S_{1,j_1}^\dagger S_{1,j_2}^\dagger \mathcal{I}_{S_1,j_3,j_1,j_4,j_2} 2\lambda_{S_1}^{(2)}}{\sqrt{3}\epsilon}$	$-\frac{2h\overline{y}_{2,i_1,r_1} S_{2,j_3} S_{2,j_4} y_{2,i_1,r_1} S_{2,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{S_2,j_3,j_1,j_4,j_2} 2\lambda_{S_2}^{(2)}}{\sqrt{3}\epsilon}$
									$-\frac{h\overline{y}_{1,i_1,r_1} S_{1,j_3} S_{1,j_4} y_{1,i_1,r_1} S_{1,j_1}^\dagger S_{1,j_2}^\dagger \mathcal{I}_{S_1,j_3,j_1,j_4,j_2} 2\lambda_{S_1}^{(2)}}{\sqrt{3}}$	$-\frac{h\overline{y}_{2,i_1,r_1} S_{2,j_3} S_{2,j_4} y_{2,i_1,r_1} S_{2,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{S_2,j_3,j_1,j_4,j_2} 2\lambda_{S_2}^{(2)}}{\sqrt{3}}$

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Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	S_2	D	$B_{\mu\nu}$	$W_{\mu\nu}$
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	$+D_{\mu_1}(S_{2,j_1})^\dagger D_{\mu_1}(S_{2,j_1})$		$-\frac{\hbar B_{\mu_1\mu_2}^2}{2\epsilon}$	$-\frac{\hbar(W_{\mu_1\mu_2}^{A_1})^2}{2\epsilon}$
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$+D_{\mu_1}(S_{2,j_1})^\dagger D_{\mu_1}(S_{2,j_1})$			$-\frac{\hbar(W_{\mu_1\mu_2}^{A_1})^2}{2\epsilon}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$+D_{\mu_1}(S_{2,j_1})^\dagger D_{\mu_1}(S_{2,j_1})$		$-\frac{\hbar B_{\mu_1\mu_2}^2}{2\epsilon}$	$-\frac{\hbar(W_{\mu_1\mu_2}^{A_1})^2}{2\epsilon}$
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$+D_{\mu_1}(S_{2,j_1})^\dagger D_{\mu_1}(S_{2,j_1})$		$-\frac{2\hbar B_{\mu_1\mu_2}^2}{\epsilon}$	$-\frac{\hbar(W_{\mu_1\mu_2}^{A_1})^2}{2\epsilon}$
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	$+D_{\mu_1}(S_{2,j_1})^\dagger D_{\mu_1}(S_{2,j_1})$		$-\frac{9\hbar B_{\mu_1\mu_2}^2}{2\epsilon}$	$-\frac{\hbar(W_{\mu_1\mu_2}^{A_1})^2}{2\epsilon}$
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	$+D_{\mu_1}(S_{2,j_1})^\dagger D_{\mu_1}(S_{2,j_1})$		$-\frac{\hbar B_{\mu_1\mu_2}^2}{4\epsilon}$	
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+D_{\mu_1}(S_{2,j_1})^\dagger D_{\mu_1}(S_{2,j_1})$			
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$+D_{\mu_1}(S_{2,j_1})^\dagger D_{\mu_1}(S_{2,j_1})$		$-\frac{\hbar B_{\mu_1\mu_2}^2}{4\epsilon}$	
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	$+D_{\mu_1}(S_{2,j_1})^\dagger D_{\mu_1}(S_{2,j_1})$		$-\frac{\hbar B_{\mu_1\mu_2}^2}{\epsilon}$	
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	$+D_{\mu_1}(S_{2,j_1})^\dagger D_{\mu_1}(S_{2,j_1})$		$-\frac{9\hbar B_{\mu_1\mu_2}^2}{4\epsilon}$	
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	$+D_{\mu_1}(S_{2,j_1})^\dagger D_{\mu_1}(S_{2,j_1})$		$-\frac{3\hbar B_{\mu_1\mu_2}^2}{4\epsilon}$	$-\frac{2\hbar(W_{\mu_1\mu_2}^{A_1})^2}{\epsilon}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+D_{\mu_1}(S_{2,j_1})^\dagger D_{\mu_1}(S_{2,j_1})$			$-\frac{\hbar(W_{\mu_1\mu_2}^{A_1})^2}{\epsilon}$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	$+D_{\mu_1}(S_{2,j_1})^\dagger D_{\mu_1}(S_{2,j_1})$		$-\frac{3\hbar B_{\mu_1\mu_2}^2}{4\epsilon}$	$-\frac{2\hbar(W_{\mu_1\mu_2}^{A_1})^2}{\epsilon}$
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	$+D_{\mu_1}(S_{2,j_1})^\dagger D_{\mu_1}(S_{2,j_1})$		$-\frac{3\hbar B_{\mu_1\mu_2}^2}{\epsilon}$	$-\frac{2\hbar(W_{\mu_1\mu_2}^{A_1})^2}{\epsilon}$
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	$+D_{\mu_1}(S_{2,j_1})^\dagger D_{\mu_1}(S_{2,j_1})$		$-\frac{27\hbar B_{\mu_1\mu_2}^2}{4\epsilon}$	$-\frac{2\hbar(W_{\mu_1\mu_2}^{A_1})^2}{\epsilon}$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$+D_{\mu_1}(S_2)^\dagger D_{\mu_1}(S_2)$		$-\frac{\hbar B_{\mu_1\mu_2}^2}{2\epsilon}$	$-\frac{\hbar(W_{\mu_1\mu_2}^{A_1})^2}{2\epsilon}$
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	$+D_{\mu_1}(S_2)^\dagger D_{\mu_1}(S_2)$			$-\frac{\hbar(W_{\mu_1\mu_2}^{A_1})^2}{2\epsilon}$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$+D_{\mu_1}(S_2)^\dagger D_{\mu_1}(S_2)$		$-\frac{\hbar B_{\mu_1\mu_2}^2}{2\epsilon}$	$-\frac{\hbar(W_{\mu_1\mu_2}^{A_1})^2}{2\epsilon}$
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	$+D_{\mu_1}(S_2)^\dagger D_{\mu_1}(S_2)$		$-\frac{2\hbar B_{\mu_1\mu_2}^2}{\epsilon}$	$-\frac{\hbar(W_{\mu_1\mu_2}^{A_1})^2}{2\epsilon}$
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	$+D_{\mu_1}(S_2)^\dagger D_{\mu_1}(S_2)$		$-\frac{9\hbar B_{\mu_1\mu_2}^2}{2\epsilon}$	$-\frac{\hbar(W_{\mu_1\mu_2}^{A_1})^2}{2\epsilon}$
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	$+D_{\mu_1}(S_{2,j_1})^\dagger D_{\mu_1}(S_{2,j_1})$		$-\frac{\hbar B_{\mu_1\mu_2}^2}{2\epsilon}$	$-\frac{\hbar(W_{\mu_1\mu_2}^{A_1})^2}{2\epsilon}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	S_2	D	$B_{\mu\nu}$	$W_{\mu\nu}$
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$+D_{\mu_1}$	$(S_{2,j_1})^\dagger D_{\mu_1}(S_{2,j_1})$		$-\frac{\hbar \left(W_{\mu_1\mu_2}^{A_1}\right)^2}{2\epsilon}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+D_{\mu_1}$	$(S_{2,j_1})^\dagger D_{\mu_1}(S_{2,j_1})$	$-\frac{\hbar B_{\mu_1\mu_2}^2}{2\epsilon}$	$-\frac{\hbar \left(W_{\mu_1\mu_2}^{A_1}\right)^2}{2\epsilon}$
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$+D_{\mu_1}$	$(S_{2,j_1})^\dagger D_{\mu_1}(S_{2,j_1})$	$-\frac{2\hbar B_{\mu_1\mu_2}^2}{\epsilon}$	$-\frac{\hbar \left(W_{\mu_1\mu_2}^{A_1}\right)^2}{2\epsilon}$
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	$+D_{\mu_1}$	$(S_{2,j_1})^\dagger D_{\mu_1}(S_{2,j_1})$	$-\frac{9\hbar B_{\mu_1\mu_2}^2}{2\epsilon}$	$-\frac{\hbar \left(W_{\mu_1\mu_2}^{A_1}\right)^2}{2\epsilon}$

Field combinations 3 of 10

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$H \ell$	$S_1 H$	$S_1 S_2$
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	$+\frac{2\hbar y_{1,i_1,r_1} y_{2,i_2,r_1} \overline{L_{a_4,i_2}}^T C P_L L_{a_2,i_1} H_{a_1} H_{a_3} \lambda_{T3} \epsilon_{a_1,a_2} \epsilon_{a_3,a_4}}{\sqrt{3} \epsilon M_F}$	$-\frac{2\hbar y_{1,i_1,r_1} \overline{Y_{e,i_2,i_3}} H_{a_1} S_1 y_{1,i_2,r_1} Y_{e,i_1,i_3} H_{a_1} \dagger S_1 \dagger}{\epsilon}$	$-\frac{4\hbar y_{1,i_1,r_1} y_{2,i_2,r_2} S_1 S_2 j_1 y_{1,i_2,r_1} y_{2,i_1,r_2} S_1 \dagger S_{2,j_1} \dagger}{\sqrt{3} \epsilon}$
								$+\frac{2\hbar y_{1,i_1,r_1} y_{2,i_2,r_1} \overline{L_{a_4,i_2}}^T C P_L L_{a_2,i_1} H_{a_1} H_{a_3} \lambda_{T3} \epsilon_{a_1,a_2} \epsilon_{a_3,a_4}}{\sqrt{3} M_F}$	$-2\hbar y_{1,i_1,r_1} \overline{Y_{e,i_2,i_3}} H_{a_1} S_1 y_{1,i_2,r_1} Y_{e,i_1,i_3} H_{a_1} \dagger S_1 \dagger$	$+\frac{4\hbar y_{1,i_1,r_1} y_{2,i_2,r_1} S_1 S_2 j_1 y_{1,i_2,r_1} y_{2,i_1,r_2} S_1 \dagger S_{2,j_1} \dagger}{\sqrt{3}}$
								$+\frac{2\hbar \lambda_{T3} \overline{L_{a_3,i_1}} C P_R \overline{L_{a_4,i_2}}^T y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_3} \epsilon_{a_2,a_4} H_{a_1} \dagger H_{a_2} \dagger}{\sqrt{3} \epsilon M_F}$	$+H_{a_2} S_1 H_{a_1} \dagger S_1 \dagger \mathcal{I}_{H1,a_1,a_2} {}^1\lambda_{H1}^{(1)}$	$+\frac{4\hbar y_{1,i_1,r_1} y_{2,i_2,r_1} S_1 S_2 j_1 y_{1,i_1,r_2} y_{2,i_2,r_2} S_1 \dagger S_{2,j_1} \dagger}{\sqrt{3}}$
cont.								$+\frac{2\hbar \lambda_{T3} \overline{L_{a_3,i_1}} C P_R \overline{L_{a_4,i_2}}^T y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_3} \epsilon_{a_2,a_4} H_{a_1} \dagger H_{a_2} \dagger}{\sqrt{3} M_F}$	$-\frac{2\hbar y_{1,i_1,r_1} H_{a_2} S_1 y_{1,i_1,r_1} H_{a_1} \dagger S_1 \dagger \mathcal{I}_{H1,a_1,a_2} {}^1\lambda_{H1}^{(1)}}{\epsilon}$	$+\frac{4\hbar y_{1,i_1,r_1} y_{2,i_2,r_1} S_1 S_2 j_2 y_{1,i_1,r_1} S_1 \dagger S_{2,j_1} \dagger \mathcal{I}_{12,j_2,j_1} {}^1\lambda_{12}^{(1)}}{\epsilon}$
cont.									$-\hbar y_{1,i_1,r_1} H_{a_2} S_1 y_{1,i_1,r_1} H_{a_1} \dagger S_1 \dagger \mathcal{I}_{H1,a_1,a_2} {}^1\lambda_{H1}^{(1)}$	$-\frac{2\hbar y_{1,i_1,r_1} S_1 S_2 j_2 y_{1,i_1,r_1} S_1 \dagger S_{2,j_1} \dagger \mathcal{I}_{12,j_2,j_1} {}^1\lambda_{12}^{(1)}}{\epsilon}$
										$-\frac{\hbar y_{1,i_1,r_1} S_1 S_2 j_2 y_{1,i_1,r_1} S_1 \dagger S_{2,j_1} \dagger \mathcal{I}_{12,j_2,j_1} {}^1\lambda_{12}^{(1)}}{\sqrt{3} \epsilon}$
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$+\frac{2\hbar y_{1,i_1,r_1} y_{2,i_2,r_1} \overline{L_{a_4,i_2}}^T C P_L L_{a_2,i_1} H_{a_1} H_{a_3} \lambda_{T3} \epsilon_{a_1,a_2} \epsilon_{a_3,a_4}}{\sqrt{3} \epsilon M_F}$	$-\frac{2\hbar y_{1,i_1,r_1} \overline{Y_{e,i_2,i_3}} H_{a_1} S_1 y_{1,i_2,r_1} Y_{e,i_1,i_3} H_{a_1} \dagger S_1 \dagger}{\epsilon}$	$-\frac{4\hbar y_{1,i_1,r_1} y_{2,i_2,r_2} S_1 S_2 j_1 y_{1,i_2,r_1} y_{2,i_1,r_2} S_1 \dagger S_{2,j_1} \dagger}{\sqrt{3} \epsilon}$
								$+\frac{2\hbar y_{1,i_1,r_1} y_{2,i_2,r_1} \overline{L_{a_4,i_2}}^T C P_L L_{a_2,i_1} H_{a_1} H_{a_3} \lambda_{T3} \epsilon_{a_1,a_2} \epsilon_{a_3,a_4}}{\sqrt{3} M_F}$	$-2\hbar y_{1,i_1,r_1} \overline{Y_{e,i_2,i_3}} H_{a_1} S_1 y_{1,i_2,r_1} Y_{e,i_1,i_3} H_{a_1} \dagger S_1 \dagger$	$+\frac{4\hbar y_{1,i_1,r_1} y_{2,i_2,r_1} S_1 S_2 j_1 y_{1,i_2,r_1} y_{2,i_1,r_2} S_1 \dagger S_{2,j_1} \dagger}{\sqrt{3}}$
								$+\frac{2\hbar \lambda_{T3} \overline{L_{a_3,i_1}} C P_R \overline{L_{a_4,i_2}}^T y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_3} \epsilon_{a_2,a_4} H_{a_1} \dagger H_{a_2} \dagger}{\sqrt{3} \epsilon M_F}$	$+H_{a_2} S_1 H_{a_1} \dagger S_1 \dagger \mathcal{I}_{H1,a_1,a_2} {}^1\lambda_{H1}^{(1)}$	$+\frac{4\hbar y_{1,i_1,r_1} y_{2,i_2,r_1} S_1 S_2 j_1 y_{1,i_1,r_2} y_{2,i_2,r_2} S_1 \dagger S_{2,j_1} \dagger}{\sqrt{3}}$
cont.								$+\frac{2\hbar \lambda_{T3} \overline{L_{a_3,i_1}} C P_R \overline{L_{a_4,i_2}}^T y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_3} \epsilon_{a_2,a_4} H_{a_1} \dagger H_{a_2} \dagger}{\sqrt{3} M_F}$	$-\frac{2\hbar y_{1,i_1,r_1} H_{a_2} S_1 y_{1,i_1,r_1} H_{a_1} \dagger S_1 \dagger \mathcal{I}_{H1,a_1,a_2} {}^1\lambda_{H1}^{(1)}}{\epsilon}$	$+\frac{4\hbar y_{1,i_1,r_1} y_{2,i_2,r_1} S_1 S_2 j_2 y_{1,i_1,r_1} S_1 \dagger S_{2,j_1} \dagger \mathcal{I}_{12,j_2,j_1} {}^1\lambda_{12}^{(1)}}{\epsilon}$
cont.									$-\hbar y_{1,i_1,r_1} H_{a_2} S_1 y_{1,i_1,r_1} H_{a_1} \dagger S_1 \dagger \mathcal{I}_{H1,a_1,a_2} {}^1\lambda_{H1}^{(1)}$	$-\frac{2\hbar y_{1,i_1,r_1} S_1 S_2 j_2 y_{1,i_1,r_1} S_1 \dagger S_{2,j_1} \dagger \mathcal{I}_{12,j_2,j_1} {}^1\lambda_{12}^{(1)}}{\epsilon}$
										$-\frac{\hbar y_{1,i_1,r_1} S_1 S_2 j_2 y_{1,i_1,r_1} S_1 \dagger S_{2,j_1} \dagger \mathcal{I}_{12,j_2,j_1} {}^1\lambda_{12}^{(1)}}{\sqrt{3} \epsilon}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$+\frac{2\hbar y_{1,i_1,r_1} y_{2,i_2,r_1} \overline{L_{a_4,i_2}}^T C P_L L_{a_2,i_1} H_{a_1} H_{a_3} \lambda_{T3} \epsilon_{a_1,a_2} \epsilon_{a_3,a_4}}{\sqrt{3} \epsilon M_F}$	$-\frac{2\hbar y_{1,i_1,r_1} \overline{Y_{e,i_2,i_3}} H_{a_1} S_1 y_{1,i_2,r_1} Y_{e,i_1,i_3} H_{a_1} \dagger S_1 \dagger}{\epsilon}$	$-\frac{4\hbar y_{1,i_1,r_1} y_{2,i_2,r_2} S_1 S_2 j_1 y_{1,i_2,r_1} y_{2,i_1,r_2} S_1 \dagger S_{2,j_1} \dagger}{\sqrt{3} \epsilon}$
								$+\frac{2\hbar y_{1,i_1,r_1} y_{2,i_2,r_1} \overline{L_{a_4,i_2}}^T C P_L L_{a_2,i_1} H_{a_1} H_{a_3} \lambda_{T3} \epsilon_{a_1,a_2} \epsilon_{a_3,a_4}}{\sqrt{3} M_F}$	$-2\hbar y_{1,i_1,r_1} \overline{Y_{e,i_2,i_3}} H_{a_1} S_1 y_{1,i_2,r_1} Y_{e,i_1,i_3} H_{a_1} \dagger S_1 \dagger$	$+\frac{4\hbar y_{1,i_1,r_1} y_{2,i_2,r_1} S_1 S_2 j_1 y_{1,i_2,r_1} y_{2,i_1,r_2} S_1 \dagger S_{2,j_1} \dagger}{\sqrt{3}}$
								$+\frac{2\hbar \lambda_{T3} \overline{L_{a_3,i_1}} C P_R \overline{L_{a_4,i_2}}^T y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_3} \epsilon_{a_2,a_4} H_{a_1} \dagger H_{a_2} \dagger}{\sqrt{3} \epsilon M_F}$	$+H_{a_2} S_1 H_{a_1} \dagger S_1 \dagger \mathcal{I}_{H1,a_1,a_2} {}^1\lambda_{H1}^{(1)}$	$+\frac{4\hbar y_{1,i_1,r_1} y_{2,i_2,r_1} S_1 S_2 j_1 y_{1,i_1,r_2} y_{2,i_2,r_2} S_1 \dagger S_{2,j_1} \dagger}{\sqrt{3}}$
cont.								$+\frac{2\hbar \lambda_{T3} \overline{L_{a_3,i_1}} C P_R \overline{L_{a_4,i_2}}^T y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_3} \epsilon_{a_2,a_4} H_{a_1} \dagger H_{a_2} \dagger}{\sqrt{3} M_F}$	$-\frac{2\hbar y_{1,i_1,r_1} H_{a_2} S_1 y_{1,i_1,r_1} H_{a_1} \dagger S_1 \dagger \mathcal{I}_{H1,a_1,a_2} {}^1\lambda_{H1}^{(1)}}{\epsilon}$	$+\frac{4\hbar y_{1,i_1,r_1} y_{2,i_2,r_1} S_1 S_2 j_2 y_{1,i_1,r_1} S_1 \dagger S_{2,j_1} \dagger \mathcal{I}_{12,j_2,j_1} {}^1\lambda_{12}^{(1)}}{\epsilon}$
cont.									$-\hbar y_{1,i_1,r_1} H_{a_2} S_1 y_{1,i_1,r_1} H_{a_1} \dagger S_1 \dagger \mathcal{I}_{H1,a_1,a_2} {}^1\lambda_{H1}^{(1)}$	$-\frac{2\hbar y_{1,i_1,r_1} S_1 S_2 j_2 y_{1,i_1,r_1} S_1 \dagger S_{2,j_1} \dagger \mathcal{I}_{12,j_2,j_1} {}^1\lambda_{12}^{(1)}}{\epsilon}$
										$-\frac{\hbar y_{1,i_1,r_1} S_1 S_2 j_2 y_{1,i_1,r_1} S_1 \dagger S_{2,j_1} \dagger \mathcal{I}_{12,j_2,j_1} {}^1\lambda_{12}^{(1)}}{\sqrt{3} \epsilon}$
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$+\frac{2\hbar y_{1,i_1,r_1} y_{2,i_2,r_1} \overline{L_{a_4,i_2}}^T C P_L L_{a_2,i_1} H_{a_1} H_{a_3} \lambda_{T3} \epsilon_{a_1,a_2} \epsilon_{a_3,a_4}}{\sqrt{3} \epsilon M_F}$	$-\frac{2\hbar y_{1,i_1,r_1} \overline{Y_{e,i_2,i_3}} H_{a_1} S_1 y_{1,i_2,r_1} Y_{e,i_1,i_3} H_{a_1} \dagger S_1 \dagger}{\epsilon}$	$-\frac{4\hbar y_{1,i_1,r_1} y_{2,i_2,r_2} S_1 S_2 j_1 y_{1,i_2,r_1} y_{2,i_1,r_2} S_1 \dagger S_{2,j_1} \dagger}{\sqrt{3} \epsilon}$
								$+\frac{2\hbar y_{1,i_1,r_1} y_{2,i_2,r_1} \overline{L_{a_4,i_2}}^T C P_L L_{a_2,i_1} H_{a_1} H_{a_3} \lambda_{T3} \epsilon_{a_1,a_2} \epsilon_{a_3,a_4}}{\sqrt{3} M_F}$	$-2\hbar y_{1,i_1,r_1} \overline{Y_{e,i_2,i_3}} H_{a_1} S_1 y_{1,i_2,r_1} Y_{e,i_1,i_3} H_{a_1} \dagger S_1 \dagger$	$+\frac{4\hbar y_{1,i_1,r_1} y_{2,i_2,r_1} S_1 S_2 j_1 y_{1,i_2,r_1} y_{2,i_1,r_2} S_1 \dagger S_{2,j_1} \dagger}{\sqrt{3}}$
								$+\frac{2\hbar \lambda_{T3} \overline{L_{a_3,i_1}} C P_R \overline{L_{a_4,i_2}}^T y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_3} \epsilon_{a_2,a_4} H_{a_1} \dagger H_{a_2} \dagger}{\sqrt{3} \epsilon M_F}$	$+H_{a_2} S_1 H_{a_1} \dagger S_1 \dagger \mathcal{I}_{H1,a_1,a_2} {}^1\lambda_{H1}^{(1)}$	$+\frac{4\hbar y_{1,i_1,r_1} y_{2,i_2,r_1} S_1 S_2 j_1 y_{1,i_1,r_2} y_{2,i_2,r_2} S_1 \dagger S_{2,j_1} \dagger}{\sqrt{3}}$
cont.								$+\frac{2\hbar \lambda_{T3} \overline{L_{a_3,i_1}} C P_R \overline{L_{a_4,i_2}}^T y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_3} \epsilon_{a_2,a_4} H_{a_1} \dagger H_{a_2} \dagger}{\sqrt{3} M_F}$	$-\frac{2\hbar y_{1,i_1,r_1} H_{a_2} S_1 y_{1,i_1,r_1} H_{a_1} \dagger S_1 \dagger \mathcal{I}_{H1,a_1,a_2} {}^1\lambda_{H1}^{(1)}}{\epsilon}$	$+\frac{4\hbar y_{1,i_1,r_1} y_{2,i_2,r_1} S_1 S_2 j_2 y_{1,i_1,r_1} S_1 \dagger S_{2,j_1} \dagger \mathcal{I}_{12,j_2,j_1} {}^1\lambda_{12}^{(1)}}{\epsilon}$
cont.									$-\hbar y_{1,i_1,r_1} H_{a_2} S_1 y_{1,i_1,r_1} H_{a_1} \dagger S_1 \dagger \mathcal{I}_{H1,a_1,a_2} {}^1\lambda_{H1}^{(1)}$	$-\frac{2\hbar y_{1,i_1,r_1} S_1 S_2 j_2 y_{1,i_1,r_1} S_1 \dagger S_{2,j_1} \dagger \mathcal{I}_{12,j_2,j_1} {}^1\lambda_{12}^{(1)}}{\epsilon}$
										$-\frac{\hbar y_{1,i_1,r_1} S_1 S_2 j_2 y_{1,i_1,r_1} S_1 \dagger S_{2,j_1} \dagger \mathcal{I}_{12,j_2,j_1} {}^1\lambda_{12}^{(1)}}{\sqrt{3} \epsilon}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$H \ell$	S_1	H	S_2
cont.											$-\frac{2h\bar{y}_{2,i_1,r_1} S_1 S_{2,j_2} y_{2,i_1,r_1} S_1^\dagger S_{2,j_1}^\dagger \mathcal{I}_{12,j_2,j_1}{}^1 \lambda_{12}^{(1)}}{\sqrt{3}\epsilon}$
											$-\frac{h\bar{y}_{2,i_1,r_1} S_1 S_{2,j_2} y_{2,i_1,r_1} S_1^\dagger S_{2,j_1}^\dagger \mathcal{I}_{12,j_2,j_1}{}^1 \lambda_{12}^{(1)}}{\sqrt{3}}$
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	$+\frac{2h\bar{y}_{1,i_1,r_1} \bar{y}_{2,i_2,r_1} L_{a_4,i_2}{}^T C P_L L_{a_2,i_1} H_{a_1} H_{a_3} \lambda_{T3} \epsilon_{a_1,a_2} \epsilon_{a_3,a_4}}{\sqrt{3}\epsilon M_F}$ $+\frac{2h\bar{y}_{1,i_1,r_1} \bar{y}_{2,i_2,r_1} L_{a_4,i_2}{}^T C P_L L_{a_2,i_1} H_{a_1} H_{a_3} \lambda_{T3} \epsilon_{a_1,a_2} \epsilon_{a_3,a_4}}{\sqrt{3}M_F}$ $+\frac{2h\lambda_{T3} L_{a_3,i_1} C P_R \overline{L_{a_4,i_2}}{}^T y_{1,i_1,r_1} \bar{y}_{2,i_2,r_1} \epsilon_{a_1,a_3} \epsilon_{a_2,a_4} H_{a_1}^\dagger H_{a_2}^\dagger}{\sqrt{3}\epsilon M_F}$	$-\frac{2h\bar{y}_{1,i_1,r_1} \overline{Y_{e,i_2,i_3}} H_{a_1} S_1 y_{1,i_2,r_1} Y_{e,i_1,i_3} H_{a_1}^\dagger S_1^\dagger}{\epsilon}$ $-2h\bar{y}_{1,i_1,r_1} \overline{Y_{e,i_2,i_3}} H_{a_1} S_1 y_{1,i_2,r_1} Y_{e,i_1,i_3} H_{a_1}^\dagger S_1^\dagger$ $+H_{a_2} S_1 H_{a_1}^\dagger S_1^\dagger \mathcal{I}_{H1,a_1,a_2}{}^1 \lambda_{H1}^{(1)}$	$-\frac{4h\bar{y}_{1,i_1,r_1} \bar{y}_{2,i_2,r_2} S_1 S_{2,j_1} y_{1,i_2,r_1} y_{2,i_1,r_2} S_1^\dagger S_{2,j_1}^\dagger}{\sqrt{3}\epsilon}$ $+\frac{4h\bar{y}_{1,i_1,r_1} \bar{y}_{2,i_2,r_1} S_1 S_{2,j_1} y_{1,i_2,r_1} y_{2,i_1,r_2} S_1^\dagger S_{2,j_1}^\dagger}{\sqrt{3}}$ $+\frac{4h\bar{y}_{1,i_1,r_1} \bar{y}_{2,i_2,r_1} S_1 S_{2,j_1} y_{1,i_1,r_2} y_{2,i_2,r_2} S_1^\dagger S_{2,j_1}^\dagger}{\sqrt{3}}$ $+S_1 S_{2,j_2} S_1^\dagger S_{2,j_1}^\dagger \mathcal{I}_{12,j_2,j_1}{}^1 \lambda_{12}^{(1)}$	
	cont.							$+\frac{2h\lambda_{T3} L_{a_3,i_1} C P_R \overline{L_{a_4,i_2}}{}^T y_{1,i_1,r_1} \bar{y}_{2,i_2,r_1} \epsilon_{a_1,a_3} \epsilon_{a_2,a_4} H_{a_1}^\dagger H_{a_2}^\dagger}{\sqrt{3}M_F}$	$-\frac{2h\bar{y}_{1,i_1,r_1} H_{a_2} S_1 y_{1,i_1,r_1} H_{a_1}^\dagger S_1^\dagger \mathcal{I}_{H1,a_1,a_2}{}^1 \lambda_{H1}^{(1)}}{\epsilon}$ $-h\bar{y}_{1,i_1,r_1} H_{a_2} S_1 y_{1,i_1,r_1} H_{a_1}^\dagger S_1^\dagger \mathcal{I}_{H1,a_1,a_2}{}^1 \lambda_{H1}^{(1)}$	$-\frac{2h\bar{y}_{1,i_1,r_1} S_1 S_{2,j_2} y_{1,i_1,r_1} S_1^\dagger S_{2,j_1}^\dagger \mathcal{I}_{12,j_2,j_1}{}^1 \lambda_{12}^{(1)}}{\epsilon}$ $-\frac{h\bar{y}_{1,i_1,r_1} S_1 S_{2,j_2} y_{1,i_1,r_1} S_1^\dagger S_{2,j_1}^\dagger \mathcal{I}_{12,j_2,j_1}{}^1 \lambda_{12}^{(1)}}{\sqrt{3}\epsilon}$ $-\frac{2h\bar{y}_{2,i_1,r_1} S_1 S_{2,j_2} y_{2,i_1,r_1} S_1^\dagger S_{2,j_1}^\dagger \mathcal{I}_{12,j_2,j_1}{}^1 \lambda_{12}^{(1)}}{\sqrt{3}}$	
	cont.										
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	$+\frac{2h\bar{y}_{1,i_1,r_1} \bar{y}_{2,i_2,r_1} L_{a_4,i_2}{}^T C P_L L_{a_2,i_1} H_{a_1} H_{a_3} \lambda_{T3} \epsilon_{a_1,a_2} \epsilon_{a_3,a_4}}{\sqrt{3}\epsilon M_F}$ $+\frac{2h\bar{y}_{1,i_1,r_1} \bar{y}_{2,i_2,r_1} L_{a_4,i_2}{}^T C P_L L_{a_2,i_1} H_{a_1} H_{a_3} \lambda_{T3} \epsilon_{a_1,a_2} \epsilon_{a_3,a_4}}{\sqrt{3}M_F}$ $+\frac{2h\lambda_{T3} L_{a_3,i_1} C P_R \overline{L_{a_4,i_2}}{}^T y_{1,i_1,r_1} \bar{y}_{2,i_2,r_1} \epsilon_{a_1,a_3} \epsilon_{a_2,a_4} H_{a_1}^\dagger H_{a_2}^\dagger}{\sqrt{3}\epsilon M_F}$	$-\frac{2h\bar{y}_{1,i_1,r_1} \overline{Y_{e,i_2,i_3}} H_{a_2} S_{1,j_2} y_{1,i_2,r_1} Y_{e,i_1,i_3} \mathcal{I}_{y1,a_1,j_2} \mathcal{I}_{y1,a_2,j_1} H_{a_1}^\dagger S_{1,j_1}^\dagger}{\epsilon}$ $-2h\bar{y}_{1,i_1,r_1} \overline{Y_{e,i_2,i_3}} H_{a_2} S_{1,j_2} y_{1,i_2,r_1} Y_{e,i_1,i_3} \mathcal{I}_{y1,a_1,j_2} \mathcal{I}_{y1,a_2,j_1} H_{a_1}^\dagger S_{1,j_1}^\dagger$ $+H_{a_2} S_{1,j_2} H_{a_1}^\dagger S_{1,j_1}^\dagger \mathcal{I}_{H1,a_1,a_2,j_1,j_2}{}^1 \lambda_{H1}^{(1)}$	$+\frac{2h\bar{y}_{1,i_1,r_1} \bar{y}_{2,i_2,r_2} S_{1,j_2} S_{2,j_1} y_{1,i_2,r_1} y_{2,i_1,r_2} S_{1,j_1}^\dagger S_{2,j_2}^\dagger}{\epsilon}$ $-\frac{2h\bar{y}_{1,i_1,r_1} \bar{y}_{2,i_2,r_2} S_{1,j_1} S_{2,j_2} y_{1,i_2,r_1} y_{2,i_1,r_2} S_{1,j_1}^\dagger S_{2,j_2}^\dagger}{\epsilon}$ $-2h\bar{y}_{1,i_1,r_1} \bar{y}_{2,i_2,r_1} S_{1,j_2} S_{2,j_1} y_{1,i_2,r_2} y_{2,i_1,r_2} S_{1,j_1}^\dagger S_{2,j_2}^\dagger$ $+2h\bar{y}_{1,i_1,r_1} \bar{y}_{2,i_2,r_1} S_{1,j_1} S_{2,j_2} y_{1,i_2,r_2} y_{2,i_1,r_2} S_{1,j_1}^\dagger S_{2,j_2}^\dagger$ $+2h\bar{y}_{1,i_1,r_1} \bar{y}_{2,i_2,r_1} S_{1,j_1} S_{2,j_2} y_{1,i_1,r_2} y_{2,i_2,r_2} S_{1,j_1}^\dagger S_{2,j_2}^\dagger$ $+S_{1,j_4} S_{2,j_3} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^1 \lambda_{12}^{(1)}$	
	cont.							$+\frac{2h\lambda_{T3} L_{a_3,i_1} C P_R \overline{L_{a_4,i_2}}{}^T y_{1,i_1,r_1} \bar{y}_{2,i_2,r_1} \epsilon_{a_1,a_3} \epsilon_{a_2,a_4} H_{a_1}^\dagger H_{a_2}^\dagger}{\sqrt{3}M_F}$	$-\frac{h\bar{y}_{1,i_1,r_1} H_{a_2} S_{1,j_2} y_{1,i_1,r_1} H_{a_1}^\dagger S_{1,j_1}^\dagger \mathcal{I}_{H1,a_1,a_2,j_1,j_2}{}^1 \lambda_{H1}^{(1)}}{\epsilon}$ $-\frac{1}{2}h\bar{y}_{1,i_1,r_1} H_{a_2} S_{1,j_2} y_{1,i_1,r_1} H_{a_1}^\dagger S_{1,j_1}^\dagger \mathcal{I}_{H1,a_1,a_2,j_1,j_2}{}^1 \lambda_{H1}^{(1)}$ $+H_{a_2} S_{1,j_2} H_{a_1}^\dagger S_{1,j_1}^\dagger \mathcal{I}_{H1,a_1,a_2,j_1,j_2}{}^2 \lambda_{H1}^{(2)}$	$-\frac{2h\bar{y}_{1,i_1,r_1} S_{1,j_4} S_{2,j_3} y_{1,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^1 \lambda_{12}^{(1)}}{\epsilon}$ $-\frac{1}{2}h\bar{y}_{1,i_1,r_1} S_{1,j_4} S_{2,j_3} y_{1,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^1 \lambda_{12}^{(1)}$ $-\frac{h\bar{y}_{2,i_1,r_1} S_{1,j_4} S_{2,j_3} y_{2,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^1 \lambda_{12}^{(1)}}{\epsilon}$ $-\frac{1}{2}h\bar{y}_{2,i_1,r_1} S_{1,j_4} S_{2,j_3} y_{2,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^1 \lambda_{12}^{(1)}$ $-\frac{1}{2}h\bar{y}_{2,i_1,r_1} S_{1,j_4} S_{2,j_3} y_{2,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^1 \lambda_{12}^{(1)}$ $-\frac{1}{2}h\bar{y}_{2,i_1,r_1} S_{1,j_4} S_{2,j_3} y_{2,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^1 \lambda_{12}^{(1)}$ $-\frac{1}{2}h\bar{y}_{2,i_1,r_1} S_{1,j_4} S_{2,j_3} y_{2,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^1 \lambda_{12}^{(1)}$ $-\frac{1}{2}h\bar{y}_{2,i_1,r_1} S_{1,j_4} S_{2,j_3} y_{2,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^1 \lambda_{12}^{(1)}$ $-\frac{1}{2}h\bar{y}_{2,i_1,r_1} S_{1,j_4} S_{2,j_3} y_{2,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^1 \lambda_{12}^{(1)}$ $-\frac{1}{2}h\bar{y}_{2,i_1,r_1} S_{1,j_4} S_{2,j_3} y_{2,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^1 \lambda_{12}^{(1)}$ $-\frac{1}{2}h\bar{y}_{2,i_1,r_1} S_{1,j_4} S_{2,j_3} y_{2,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^1 \lambda_{12}^{(1)}$ $-\frac{1}{2}h\bar{y}_{2,i_1,r_1} S_{1,j_4} S_{2,j_3} y_{2,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^1 \lambda_{12}^{(1)}$ $-\frac{1}{2}h\bar{y}_{2,i_1,r_1} S_{1,j_4} S_{2,j_3} y_{2,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^1 \lambda_{12}^{(1)}$ $-\frac{1}{2}h\bar{y}_{2,i_1,r_1} S_{1,j_4} S_{2,j_3} y_{2,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^1 \lambda_{12}^{(1)}$ $-\frac{1}{2}h\bar{y}_{2,i_1,r_1} S_{1,j_4} S_{2,j_3} y_{2,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^1 \lambda_{12}^{(1)}$ $-\frac{1}{2}h\bar{y}_{2,i_1,r_1} S_{1,j_4} S_{2,j_3} y_{2,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^1 \lambda_{12}^{(1)}$ $-\frac{1}{2}h\bar{y}_{2,i_1,r_1} S_{1,j_4} S_{2,j_3} y_{2,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^1 \lambda_{12}^{(1)}$ $-\frac{1}{2}h\bar{y}_{2,i_1,r_1} S_{1,j_4} S_{2,j_3} y_{2,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^1 \lambda_{12}^{(1)}$ $-\frac{1}{2}h\bar{y}_{2,i_1,r_1} S_{1,j_4} S_{2,j_3} y_{2,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^1 \lambda_{12}^{(1)}$ $-\frac{1}{2}h\bar{y}_{2,i_1,r_1} S_{1,j_4} S_{2,j_3} y_{2,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^1 \lambda_{12}^{(1)}$ $-\frac{1}{2}h\bar{y}_{2,i_1,r_1} S_{1,j_4} S_{2,j_3} y_{2,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^1 \lambda_{12}^{(1)}$ $-\frac{1}{2}h\bar{y}_{2,i_1,r_1} S_{1,j_4} S_{2,j_3} y_{2,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^1 \lambda_{12}^{(1)}$ $-\frac{1}{2}h\bar{y}_{2,i_1,r_1} S_{1,j_4} S_{2,j_3} y_{2,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^1 \lambda_{12}^{(1)}$ $-\frac{1}{2}h\bar{y}_{2,i_1,r_1} S_{1,j_4} S_{2,j_3} y_{2,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^1 \lambda_{12}^{(1)}$ $-\frac{1}{2}h\bar{y}_{2,i_1,r_1} S_{1,j_4} S_{2,j_3} y_{2,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^1 \lambda_{12}^{(1)}$ $-\frac{1}{2}h\bar{y}_{2,i_1,r_1} S_{1,j_4} S_{2,j_3} y_{2,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^1 \lambda_{12}^{(1)}$ $-\frac{1}{2}h\bar{y}_{2,i_1,r_1} S_{1,j_4} S_{2,j_3} y_{2,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^1 \lambda_{12}^{(1)}$ $-\frac{1}{2}h\bar{y}_{2,i_1,r_1} S_{1,j_4} S_{2,j_3} y_{2,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^1 \lambda_{12}^{(1)}$ $-\frac{1}{2}h\bar{y}_{2,i_1,r_1} S_{1,j_4} S_{2,j_3} y_{2,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^1 \lambda_{12}^{(1)}$ $-\frac{1}{2}h\bar{y}_{2,i_1,r_1} S_{1,j_4} S_{2,j_3} y_{2,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^1 \lambda_{12}^{(1)}$ $-\frac{1}{2}h\bar{y}_{2,i_1,r_1} S_{1,j_4} S_{2,j_3} y_{2,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^1 \lambda_{12}^{(1)}$ $-\frac{1}{2}h\bar{y}_{2,i_1,r_1} S_{1,j_4} S_{2,j_3} y_{2,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^1 \lambda_{12}^{(1)}$ $-\frac{1}{2}h\bar{y}_{2,i_1,r_1} S_{1,j_4} S_{2,j_3} y_{2,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^1 \lambda_{12}^{(1)}$ $-\frac{1}{2}h\bar{y}_{2,i_1,r_1} S_{1,j_4} S_{2,j_3} y_{2,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^1 \lambda_{12}^{(1)}$ $-\frac{1}{2}h\bar{y}_{2,i_1,r_1} S_{1,j_4} S_{2,j_3} y_{2,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^1 \lambda_{12}^{(1)}$ $-\frac{1}{2}h\bar{y}_{2,i_1,r_1} S_{1,j_4} S_{2,j_3} y_{2,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^1 \lambda_{12}^{(1)}$ $-\frac{1}{2}h\bar{y}_{2,i_1,r_1} S_{1,j_4} S_{2,j_3} y_{2,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^1 \lambda_{12}^{(1)}$ $-\frac{1}{2}h\bar{y}_{2,i_1,r_1} S_{1,j_4} S_{2,j_3} y_{2,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^1 \lambda_{12}^{(1)}$ $-\frac{1}{2}h\bar{y}_{2,i_1,r_1} S_{1,j_4} S_{2,j_3} y_{2,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^1 \lambda_{12}^{(1)}$ $-\frac{1}{2}h\bar{y}_{2,i_1,r_1} S_{1,j_4} S_{2,j_3} y_{2,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^1 \lambda_{12}^{(1)}$ $-\frac{1}{2}h\bar{y}_{2,i_1,r_1} S_{1,j_4} S_{2,j_3} y_{2,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^1 \lambda_{12}^{(1)}$ $-\frac{1}{2}h\bar{y}_{2,i_1,r_1} S_{1,j_4} S_{2,j_3} y_{2,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^1 \lambda_{12}^{(1)}$ $-\frac{1}{2}h\bar{y}_{2,i_1,r_1} S_{1,j_4} S_{2,j_3} y_{2,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^1 \lambda_{12}^{(1)}$ $-\frac{1}{2}h\bar{y}_{2,i_1,r_1} S_{1,j_4} S_{2,j_3} y_{2,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j$	

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$H \ell$	$S_1 H$	$S_1 S_2$
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+\frac{2h\overline{y_1, i_1, r_1 y_2, i_2, r_1} L_{a_4, i_2}{}^T C P_L L_{a_2, i_1} H_{a_1} H_{a_3} \lambda_{T3} \epsilon_{a_1, a_2} \epsilon_{a_3, a_4}}{\sqrt{3} \epsilon M_F}$ $+\frac{2h\overline{y_1, i_1, r_1 y_2, i_2, r_1} L_{a_4, i_2}{}^T C P_L L_{a_2, i_1} H_{a_1} H_{a_3} \lambda_{T3} \epsilon_{a_1, a_2} \epsilon_{a_3, a_4}}{\sqrt{3} M_F}$ $+\frac{2h\overline{\lambda_{T3} L_{a_3, i_1}} C P_R \overline{L_{a_4, i_2}}{}^T y_1, i_1, r_1 y_2, i_2, r_1 \epsilon_{a_1, a_3} \epsilon_{a_2, a_4} H_{a_1}{}^\dagger H_{a_2}{}^\dagger}{\sqrt{3} \epsilon M_F}$	$+\frac{2h\overline{y_2, i_1, r_1} H_{a_1} H_{a_2} S_{1, j_1} S_{1, j_2} y_1, i_1, r_1 \lambda_{T3} \overline{I_{y_2 a_1, j_1}} \overline{I_{y_2 a_2, j_2}}}{\sqrt{3} \epsilon}$ $+\frac{h\overline{y_2, i_1, r_1} H_{a_1} H_{a_2} S_{1, j_1} S_{1, j_2} y_1, i_1, r_1 \lambda_{T3} \overline{I_{y_2 a_1, j_1}} \overline{I_{y_2 a_2, j_2}}}{\sqrt{3}}$ $-\frac{2h\overline{y_1, i_1, r_1} Y_{\epsilon, i_2, i_3} H_{a_2} S_{1, j_2} y_1, i_2, r_1 Y_{\epsilon, i_1, i_3} \overline{I_{y_1, a_1, j_2}} \overline{I_{y_1 a_2, j_1}} H_{a_1}{}^\dagger S_{1, j_1}{}^\dagger}{\epsilon}$ $-\frac{2h\overline{y_1, i_1, r_1} Y_{\epsilon, i_2, i_3} H_{a_2} S_{1, j_2} y_1, i_2, r_1 Y_{\epsilon, i_1, i_3} \overline{I_{y_1, a_1, j_2}} \overline{I_{y_1 a_2, j_1}} H_{a_1}{}^\dagger S_{1, j_1}{}^\dagger}{\epsilon}$ $+\frac{2h\overline{y_1, i_1, r_1} \overline{\lambda_{T3} y_2, i_1, r_1} \overline{I_{y_2, a_1, j_2}} \overline{I_{y_2, a_2, j_1}} H_{a_1}{}^\dagger H_{a_2}{}^\dagger S_{1, j_1}{}^\dagger S_{1, j_2}{}^\dagger}{\sqrt{3} \epsilon}$ $+\frac{h\overline{y_1, i_1, r_1} \overline{\lambda_{T3} y_2, i_1, r_1} \overline{I_{y_2, a_1, j_2}} \overline{I_{y_2, a_2, j_1}} H_{a_1}{}^\dagger H_{a_2}{}^\dagger S_{1, j_1}{}^\dagger S_{1, j_2}{}^\dagger}{\sqrt{3}}$ $+\frac{H_{a_2} S_{1, j_2} H_{a_1}{}^\dagger S_{1, j_1}{}^\dagger \mathcal{I}_{H1, a_1, a_2, j_1, j_2}{}^1 \lambda_{H1}^{(1)}}{\sqrt{3}}$ $-\frac{h\overline{y_1, i_1, r_1} H_{a_2} S_{1, j_2} y_1, i_1, r_1 H_{a_1}{}^\dagger S_{1, j_1}{}^\dagger \mathcal{I}_{H1, a_1, a_2, j_1, j_2}{}^1 \lambda_{H1}^{(1)}}{\epsilon}$ $-\frac{1}{2} h\overline{y_1, i_1, r_1} H_{a_2} S_{1, j_2} y_1, i_1, r_1 H_{a_1}{}^\dagger S_{1, j_1}{}^\dagger \mathcal{I}_{H1, a_1, a_2, j_1, j_2}{}^1 \lambda_{H1}^{(1)}}{\epsilon}$ $+\frac{H_{a_2} S_{1, j_2} H_{a_1}{}^\dagger S_{1, j_1}{}^\dagger \mathcal{I}_{H1, a_1, a_2, j_1, j_2}{}^2 \lambda_{H1}^{(2)}}{\epsilon}$ $-\frac{h\overline{y_1, i_1, r_1} H_{a_2} S_{1, j_2} y_1, i_1, r_1 H_{a_1}{}^\dagger S_{1, j_1}{}^\dagger \mathcal{I}_{H1, a_1, a_2, j_1, j_2}{}^2 \lambda_{H1}^{(2)}}{\epsilon}$ $-\frac{1}{2} h\overline{y_1, i_1, r_1} H_{a_2} S_{1, j_2} y_1, i_1, r_1 H_{a_1}{}^\dagger S_{1, j_1}{}^\dagger \mathcal{I}_{H1, a_1, a_2, j_1, j_2}{}^2 \lambda_{H1}^{(2)}}{\epsilon}$	$+C \text{NewScalar1} \text{NewScalar2} h S_{1, j_1} S_{2, j_2} \text{Bar}(\text{cg1})_{j_1, j_2}$ $+\frac{1}{2} C_{S_1^\dagger S_2} h \overline{y_2, i_1, r_1} S_{1, j_1} S_{2, j_2} y_1, i_1, r_1 \text{Bar}(\text{cg1})_{j_1, j_2}$ $+\frac{C_{S_2^\dagger S_2} h \overline{y_2, i_1, r_1} S_{1, j_1} S_{2, j_2} y_1, i_1, r_1 \text{Bar}(\text{cg1})_{j_1, j_2}}{\epsilon}$ $-\frac{h\overline{y_2, i_1, r_1 y_2, i_2, r_2} S_{1, j_1} S_{1, j_2} S_{2, j_2} S_{2, j_4} y_1, i_1, r_2 y_1, i_2, r_2 \text{Bar}(\text{cg1})_{j_1, j_2} \text{Bar}(\text{cg1})_{j_3, j_4}}{\epsilon}$ $+h\overline{y_2, i_1, r_1 y_2, i_2, r_2} S_{1, j_1} S_{1, j_2} S_{2, j_2} S_{2, j_4} y_1, i_1, r_2 y_1, i_2, r_2 \text{Bar}(\text{cg1})_{j_1, j_2} \text{Bar}(\text{cg1})_{j_3, j_4}$ $+\frac{2h\overline{y_1, i_1, r_1 y_2, i_2, r_2} S_{1, j_1} S_{1, j_2} S_{2, j_2} y_1, i_1, r_2 y_1, i_2, r_2 \text{Bar}(\text{cg1})_{j_2, j_3} S_{1, j_1}{}^\dagger}{\epsilon}$ $-\frac{2h\overline{y_1, i_1, r_1 y_2, i_2, r_2} S_{1, j_1} S_{1, j_2} S_{2, j_2} y_1, i_1, r_2 y_1, i_2, r_2 \text{Bar}(\text{cg1})_{j_2, j_3} S_{1, j_1}{}^\dagger}{\epsilon}$ $+\frac{2h\overline{y_2, i_1, r_1 y_2, i_2, r_2} S_{2, j_1} S_{2, j_2} y_1, i_1, r_2 y_2, i_2, r_2 \text{Bar}(\text{cg1})_{j_2, j_3} S_{2, j_1}{}^\dagger}{\epsilon}$ $-\frac{2h\overline{y_2, i_1, r_1 y_2, i_2, r_2} S_{1, j_2} S_{2, j_1} S_{2, j_3} y_1, i_1, r_2 y_2, i_2, r_2 \text{Bar}(\text{cg1})_{j_2, j_3} S_{2, j_1}{}^\dagger}{\epsilon}$ $+\frac{1}{2} C_{S_2^\dagger S_2} h \overline{y_1, i_1, r_1} \text{cg1}_{j_1, j_2} y_2, i_1, r_1 S_{1, j_1}{}^\dagger S_{2, j_2}{}^\dagger$ $+\frac{C_{S_1^\dagger S_2} h \overline{y_1, i_1, r_1} \text{cg1}_{j_1, j_2} y_2, i_1, r_1 S_{1, j_1}{}^\dagger S_{2, j_2}{}^\dagger}{\epsilon}$ $+\frac{2h\overline{y_1, i_1, r_1 y_2, i_2, r_2} S_{1, j_2} S_{2, j_1} y_1, i_2, r_1 y_2, i_1, r_2 S_{1, j_1}{}^\dagger S_{2, j_2}{}^\dagger}{\epsilon}$ $-\frac{2h\overline{y_1, i_1, r_1 y_2, i_2, r_2} S_{1, j_1} S_{2, j_2} y_1, i_2, r_1 y_2, i_1, r_2 S_{1, j_1}{}^\dagger S_{2, j_2}{}^\dagger}{\epsilon}$ $-2h\overline{y_1, i_1, r_1 y_2, i_2, r_1} S_{1, j_2} S_{2, j_1} y_1, i_2, r_2 y_2, i_1, r_2 S_{1, j_1}{}^\dagger S_{2, j_2}{}^\dagger$ $+2h\overline{y_1, i_1, r_1 y_2, i_2, r_1} S_{1, j_1} S_{2, j_2} y_1, i_2, r_2 y_2, i_1, r_2 S_{1, j_1}{}^\dagger S_{2, j_2}{}^\dagger$ $-\frac{2h\overline{y_1, i_1, r_1 y_2, i_2, r_2} S_{1, j_1} S_{2, j_2} y_1, i_1, r_2 y_2, i_2, r_1 S_{1, j_1}{}^\dagger S_{2, j_2}{}^\dagger}{\epsilon}$ $+2h\overline{y_1, i_1, r_1 y_2, i_2, r_1} S_{1, j_1} S_{2, j_2} y_1, i_1, r_2 y_2, i_2, r_2 S_{1, j_1}{}^\dagger S_{2, j_2}{}^\dagger$ $+\frac{2h\overline{y_1, i_1, r_1 y_2, i_2, r_2} \text{cg1}_{j_1, j_3} S_{2, j_2} y_2, i_1, r_2 y_2, i_2, r_1 S_{1, j_1}{}^\dagger S_{2, j_3}{}^\dagger}{\epsilon}$ $-\frac{2h\overline{y_1, i_1, r_1 y_2, i_2, r_1} \text{cg1}_{j_2, j_3} S_{1, j_1} y_1, i_1, r_2 y_2, i_2, r_2 S_{1, j_1}{}^\dagger S_{2, j_3}{}^\dagger}{\epsilon}$ $+\frac{2h\overline$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	H	ℓ	S_1	H	S_1	S_2
													$-\frac{\sqrt{3}h\overline{y_{2,i_1,r_1}}S_{1,j_1}S_{2,j_2}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \lambda_{12}^{(2)}}{\epsilon}$
cont.													$-\frac{1}{2}\sqrt{3}h\overline{y_{2,i_1,r_1}}S_{1,j_1}S_{2,j_2}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \lambda_{12}^{(2)}$ $+\frac{2h\overline{y_{1,i_1,r_1}}\text{cg}1_{j_2,j_3}S_{1,j_1}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{1,j_2}^\dagger S_{2,j_3}^\dagger \lambda_{12}^{(2)}}{\sqrt{3}\epsilon}$ $+\frac{h\overline{y_{1,i_1,r_1}}\text{cg}1_{j_2,j_3}S_{1,j_1}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{1,j_2}^\dagger S_{2,j_3}^\dagger \lambda_{12}^{(2)}}{\sqrt{3}}$
cont.													$+S_{1,j_3}S_{2,j_4}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^2\lambda_{12}^{(2)}$ $-\frac{h\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^2\lambda_{12}^{(2)}}{\epsilon}$
cont.													$-\frac{1}{2}h\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^2\lambda_{12}^{(2)}$ $-\frac{h\overline{y_{2,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^2\lambda_{12}^{(2)}}{\epsilon}$
													$-\frac{1}{2}h\overline{y_{2,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^2\lambda_{12}^{(2)}$ $-\frac{h\overline{y_{2,i_1,r_1}}S_{1,j_2}S_{2,j_1}S_{2,j_3}y_{1,i_1,r_1}\text{Bar}(\text{cg}1)_{j_2,j_3}S_{2,j_1}^\dagger \lambda_{S_2}^{(1)}}{\epsilon}$
cont.													$+\frac{1}{2}h\overline{y_{2,i_1,r_1}}S_{1,j_2}S_{2,j_1}S_{2,j_3}y_{1,i_1,r_1}\text{Bar}(\text{cg}1)_{j_2,j_3}S_{2,j_1}^\dagger \lambda_{S_2}^{(1)}$ $+\frac{h\overline{y_{1,i_1,r_1}}\text{cg}1_{j_1,j_3}S_{2,j_2}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger S_{2,j_3}^\dagger \lambda_{S_2}^{(1)}}{\epsilon}$
cont.													$+\frac{1}{2}h\overline{y_{1,i_1,r_1}}\text{cg}1_{j_1,j_3}S_{2,j_2}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger S_{2,j_3}^\dagger \lambda_{S_2}^{(1)}$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$+\frac{2h\overline{y_{1,i_1,r_1}}y_{2,i_2,r_1}L_{a_4,i_2}{}^T C P_L L_{a_2,i_1}H_{a_1}H_{a_3}\lambda_{T3}\epsilon_{a_1,a_2}\epsilon_{a_3,a_4}}{\sqrt{3}\epsilon M_F}$ $+\frac{2h\overline{y_{1,i_1,r_1}}y_{2,i_2,r_1}L_{a_4,i_2}{}^T C P_L L_{a_2,i_1}H_{a_1}H_{a_3}\lambda_{T3}\epsilon_{a_1,a_2}\epsilon_{a_3,a_4}}{\sqrt{3}M_F}$ $+\frac{2h\lambda_{T3}L_{a_3,i_1}C P_R \overline{L_{a_4,i_2}}{}^T y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_3}\epsilon_{a_2,a_4}H_{a_1}^\dagger H_{a_2}^\dagger}{\sqrt{3}\epsilon M_F}$	$-\frac{2h\overline{y_{1,i_1,r_1}}Y_{e,i_2,i_3}H_{a_2}S_{1,j_2}y_{1,i_2,r_1}Y_{e,i_1,i_3}\mathcal{I}_{y_{1,a_1},j_2}\mathcal{I}_{y_{1,a_2},j_1}{}^*H_{a_1}^\dagger S_{1,j_1}^\dagger}{\epsilon}$ $-2h\overline{y_{1,i_1,r_1}}Y_{e,i_2,i_3}H_{a_2}S_{1,j_2}y_{1,i_2,r_1}Y_{e,i_1,i_3}\mathcal{I}_{y_{1,a_1},j_2}\mathcal{I}_{y_{1,a_2},j_1}{}^*H_{a_1}^\dagger S_{1,j_1}^\dagger$ $+H_{a_2}S_{1,j_2}H_{a_1}^\dagger S_{1,j_1}^\dagger \mathcal{I}_{H1,a_1,a_2,j_1,j_2}{}^1\lambda_{H1}^{(1)}$	$+\frac{2h\overline{y_{1,i_1,r_1}}y_{2,i_2,r_2}S_{1,j_2}S_{2,j_1}y_{1,i_2,r_2}y_{2,i_1,r_2}S_{1,j_1}^\dagger S_{2,j_2}^\dagger}{\epsilon}$ $-\frac{2h\overline{y_{1,i_1,r_1}}y_{2,i_2,r_2}S_{1,j_1}S_{2,j_2}y_{1,i_2,r_2}y_{2,i_1,r_2}S_{1,j_1}^\dagger S_{2,j_2}^\dagger}{\epsilon}$ $-2h\overline{y_{1,i_1,r_1}}y_{2,i_2,r_1}S_{1,j_2}S_{2,j_1}y_{1,i_2,r_2}y_{2,i_1,r_2}S_{1,j_1}^\dagger S_{2,j_2}^\dagger$ $+2h\overline{y_{1,i_1,r_1}}y_{2,i_2,r_1}S_{1,j_1}S_{2,j_2}y_{1,i_2,r_2}y_{2,i_1,r_2}S_{1,j_1}^\dagger S_{2,j_2}^\dagger$ $+2h\overline{y_{1,i_1,r_1}}y_{2,i_2,r_1}S_{1,j_1}S_{2,j_2}y_{1,i_1,r_2}y_{2,i_2,r_2}S_{1,j_1}^\dagger S_{2,j_2}^\dagger$ $+S_{1,j_4}S_{2,j_3}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^1\lambda_{12}^{(1)}$			
cont.								$-\frac{h\overline{y_{1,i_1,r_1}}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}H_{a_1}^\dagger S_{1,j_1}^\dagger \mathcal{I}_{H1,a_1,a_2,j_1,j_2}{}^1\lambda_{H1}^{(1)}}{\epsilon}$ $-\frac{1}{2}h\overline{y_{1,i_1,r_1}}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}H_{a_1}^\dagger S_{1,j_1}^\dagger \mathcal{I}_{H1,a_1,a_2,j_1,j_2}{}^1\lambda_{H1}^{(1)}$ $+H_{a_2}S_{1,j_2}H_{a_1}^\dagger S_{1,j_1}^\dagger \mathcal{I}_{H1,a_1,a_2,j_1,j_2}{}^2\lambda_{H1}^{(2)}$	$+\frac{2h\overline{y_{1,i_1,r_1}}y_{2,i_2,r_2}S_{1,j_2}S_{2,j_1}y_{1,i_2,r_2}y_{2,i_1,r_2}S_{1,j_1}^\dagger S_{2,j_2}^\dagger}{\epsilon}$ $-\frac{h\overline{y_{1,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^1\lambda_{12}^{(1)}}{\epsilon}$				
cont.								$-\frac{1}{2}h\overline{y_{1,i_1,r_1}}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}H_{a_1}^\dagger S_{1,j_1}^\dagger \mathcal{I}_{H1,a_1,a_2,j_1,j_2}{}^2\lambda_{H1}^{(2)}$	$-\frac{1}{2}h\overline{y_{1,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^1\lambda_{12}^{(1)}$ $-\frac{h\overline{y_{2,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^1\lambda_{12}^{(1)}}{\epsilon}$				
cont.													$-\frac{1}{2}h\overline{y_{2,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^1\lambda_{12}^{(1)}$ $-\sqrt{3}S_{1,j_2}S_{2,j_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \lambda_{12}^{(2)}$ $+\sqrt{3}S_{1,j_1}S_{2,j_2}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \lambda_{12}^{(2)}$
													$+\frac{\sqrt{3}h\overline{y_{1,i_1,r_1}}S_{1,j_2}S_{2,j_1}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \lambda_{12}^{(2)}}{\epsilon}$
cont.													$+\frac{1}{2}\sqrt{3}h\overline{y_{1,i_1,r_1}}S_{1,j_2}S_{2,j_1}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \lambda_{12}^{(2)}$
													$-\frac{\sqrt{3}h\overline{y_{1,i_1,r_1}}S_{1,j_1}S_{2,j_2}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \lambda_{12}^{(2)}}{\epsilon}$
cont.													$-\frac{1}{2}\sqrt{3}h\overline{y_{1,i_1,r_1}}S_{1,j_1}S_{2,j_2}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \lambda_{12}^{(2)}$ $+\frac{\sqrt{3}h\overline{y_{2,i_1,r_1}}S_{1,j_2}S_{2,j_1}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \lambda_{12}^{(2)}}{\epsilon}$
													$+\frac{1}{2}\sqrt{3}h\overline{y_{2,i_1,r_1}}S_{1,j_2}S_{2,j_1}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \lambda_{12}^{(2)}$ $-\frac{\sqrt{3}h\overline{y_{2,i_1,r_1}}S_{1,j_1}S_{2,j_2}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \lambda_{12}^{(2)}}{\epsilon}$
cont.													$-\frac{1}{2}\sqrt{3}h\overline{y_{2,i_1,r_1}}S_{1,j_1}S_{2,j_2}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \lambda_{12}^{(2)}$ $+S_{1,j_3}S_{2,j_4}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^2\lambda_{12}^{(2)}$ $-\frac{h\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^2\lambda_{12}^{(2)}}{\epsilon}$
cont.													$-\frac{1}{2}h\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^2\lambda_{12}^{(2)}$ $-\frac{h\overline{y_{2,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^2\lambda_{12}^{(2)}}{\epsilon}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	H	ℓ	S_1	H	S_1	S_2
cont.												$-\frac{1}{2}h\overline{y_{2,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4}2\lambda_{12}^{(2)}$	
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	$+\frac{2h\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}L_{a_4,i_2}{}^TC P_L L_{a_2,i_1}H_{a_1}H_{a_3}\lambda_{T3}\epsilon_{a_1,a_2}\epsilon_{a_3,a_4}}{\sqrt{3}\epsilon M_F}$ $+\frac{2h\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}L_{a_4,i_2}{}^TC P_L L_{a_2,i_1}H_{a_1}H_{a_3}\lambda_{T3}\epsilon_{a_1,a_2}\epsilon_{a_3,a_4}}{\sqrt{3}M_F}$ $+\frac{2h\lambda_{T3}L_{a_3,i_1}C P_R \overline{L_{a_4,i_2}}{}^TY_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_3}\epsilon_{a_2,a_4}H_{a_1}^\dagger H_{a_2}^\dagger}{\sqrt{3}\epsilon M_F}$		$-\frac{2h\overline{y_{1,i_1,r_1}Y_{e,i_2,i_3}}H_{a_2}S_{1,j_2}y_{1,i_2,r_1}Y_{e,i_1,i_3}\mathcal{I}_{y_1,a_1,j_2}\mathcal{I}_{y_1,a_2,j_1}H_{a_1}^\dagger S_{1,j_1}^\dagger}{\epsilon}$ $-2h\overline{y_{1,i_1,r_1}Y_{e,i_2,i_3}}H_{a_2}S_{1,j_2}y_{1,i_2,r_1}Y_{e,i_1,i_3}\mathcal{I}_{y_1,a_1,j_2}\mathcal{I}_{y_1,a_2,j_1}H_{a_1}^\dagger S_{1,j_1}^\dagger$ $+H_{a_2}S_{1,j_2}H_{a_1}^\dagger S_{1,j_1}^\dagger \mathcal{I}_{H1,a_1,a_2,j_1,j_2}1\lambda_{H1}^{(1)}$ $-\frac{h\overline{y_{1,i_1,r_1}}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}H_{a_1}^\dagger S_{1,j_1}^\dagger \mathcal{I}_{H1,a_1,a_2,j_1,j_2}1\lambda_{H1}^{(1)}}{\epsilon}$	$+\frac{2h\overline{y_{1,i_1,r_1}y_{2,i_2,r_2}}S_{1,j_2}S_{2,j_1}y_{1,i_2,r_1}y_{2,i_1,r_2}S_{1,j_1}^\dagger S_{2,j_2}^\dagger}{\epsilon}$ $-\frac{2h\overline{y_{1,i_1,r_1}y_{2,i_2,r_2}}S_{1,j_1}S_{2,j_2}y_{1,i_2,r_1}y_{2,i_1,r_2}S_{1,j_1}^\dagger S_{2,j_2}^\dagger}{\epsilon}$ $-2h\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}S_{1,j_2}S_{2,j_1}y_{1,i_2,r_2}y_{2,i_1,r_2}S_{1,j_1}^\dagger S_{2,j_2}^\dagger$ $+2h\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}S_{1,j_1}S_{2,j_2}y_{1,i_1,r_2}y_{2,i_2,r_2}S_{1,j_1}^\dagger S_{2,j_2}^\dagger$ $+S_{1,j_4}S_{2,j_3}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4}1\lambda_{12}^{(1)}$ $-\frac{h\overline{y_{1,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4}1\lambda_{12}^{(1)}}{\epsilon}$		
cont.								$+\frac{2h\lambda_{T3}L_{a_3,i_1}C P_R \overline{L_{a_4,i_2}}{}^TY_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_3}\epsilon_{a_2,a_4}H_{a_1}^\dagger H_{a_2}^\dagger}{\sqrt{3}M_F}$		$-\frac{1}{2}h\overline{y_{1,i_1,r_1}}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}H_{a_1}^\dagger S_{1,j_1}^\dagger \mathcal{I}_{H1,a_1,a_2,j_1,j_2}1\lambda_{H1}^{(1)}$ $+H_{a_2}S_{1,j_2}H_{a_1}^\dagger S_{1,j_1}^\dagger \mathcal{I}_{H1,a_1,a_2,j_1,j_2}2\lambda_{H1}^{(2)}$ $-\frac{h\overline{y_{1,i_1,r_1}}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}H_{a_1}^\dagger S_{1,j_1}^\dagger \mathcal{I}_{H1,a_1,a_2,j_1,j_2}2\lambda_{H1}^{(2)}}{\epsilon}$	$+2h\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}S_{1,j_1}S_{2,j_2}y_{1,i_1,r_2}y_{2,i_2,r_2}S_{1,j_1}^\dagger S_{2,j_2}^\dagger$ $+2h\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}S_{1,j_1}S_{2,j_2}y_{1,i_1,r_2}y_{2,i_2,r_2}S_{1,j_1}^\dagger S_{2,j_2}^\dagger$ $+S_{1,j_4}S_{2,j_3}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4}1\lambda_{12}^{(1)}$ $-\frac{h\overline{y_{1,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4}1\lambda_{12}^{(1)}}{\epsilon}$		
cont.										$-\frac{1}{2}h\overline{y_{1,i_1,r_1}}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}H_{a_1}^\dagger S_{1,j_1}^\dagger \mathcal{I}_{H1,a_1,a_2,j_1,j_2}2\lambda_{H1}^{(2)}$ $-\frac{1}{2}h\overline{y_{1,i_1,r_1}}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}H_{a_1}^\dagger S_{1,j_1}^\dagger \mathcal{I}_{H1,a_1,a_2,j_1,j_2}2\lambda_{H1}^{(2)}$	$-\frac{1}{2}h\overline{y_{1,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4}1\lambda_{12}^{(1)}$ $-\frac{h\overline{y_{2,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4}1\lambda_{12}^{(1)}}{\epsilon}$ $-\frac{1}{2}h\overline{y_{2,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4}1\lambda_{12}^{(1)}$ $-\sqrt{3}S_{1,j_2}S_{2,j_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \lambda_{12}^{(2)}$ $+\sqrt{3}S_{1,j_1}S_{2,j_2}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \lambda_{12}^{(2)}$ $+\sqrt{3h\overline{y_{1,i_1,r_1}}S_{1,j_2}S_{2,j_1}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \lambda_{12}^{(2)}}$		
cont.												$+\frac{1}{2}\sqrt{3h\overline{y_{1,i_1,r_1}}S_{1,j_2}S_{2,j_1}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \lambda_{12}^{(2)}}$ $-\frac{\sqrt{3h\overline{y_{1,i_1,r_1}}S_{1,j_1}S_{2,j_2}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \lambda_{12}^{(2)}}{\epsilon}$ $-\frac{1}{2}\sqrt{3h\overline{y_{1,i_1,r_1}}S_{1,j_1}S_{2,j_2}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \lambda_{12}^{(2)}}$ $+\frac{\sqrt{3h\overline{y_{2,i_1,r_1}}S_{1,j_2}S_{2,j_1}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \lambda_{12}^{(2)}}{\epsilon}$	
cont.												$+\frac{1}{2}\sqrt{3h\overline{y_{2,i_1,r_1}}S_{1,j_2}S_{2,j_1}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \lambda_{12}^{(2)}}$ $-\frac{\sqrt{3h\overline{y_{2,i_1,r_1}}S_{1,j_1}S_{2,j_2}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \lambda_{12}^{(2)}}{\epsilon}$ $-\frac{1}{2}\sqrt{3h\overline{y_{2,i_1,r_1}}S_{1,j_1}S_{2,j_2}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \lambda_{12}^{(2)}}$ $+\frac{\sqrt{3h\overline{y_{2,i_1,r_1}}S_{1,j_2}S_{2,j_1}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \lambda_{12}^{(2)}}{\epsilon}$	
cont.												$+\frac{1}{2}\sqrt{3h\overline{y_{2,i_1,r_1}}S_{1,j_2}S_{2,j_1}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \lambda_{12}^{(2)}}$ $-\frac{\sqrt{3h\overline{y_{2,i_1,r_1}}S_{1,j_1}S_{2,j_2}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \lambda_{12}^{(2)}}{\epsilon}$ $-\frac{1}{2}\sqrt{3h\overline{y_{2,i_1,r_1}}S_{1,j_1}S_{2,j_2}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \lambda_{12}^{(2)}}$ $+\frac{\sqrt{3h\overline{y_{2,i_1,r_1}}S_{1,j_2}S_{2,j_1}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \lambda_{12}^{(2)}}{\epsilon}$	
cont.												$-\frac{1}{2}\sqrt{3h\overline{y_{2,i_1,r_1}}S_{1,j_2}S_{2,j_1}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \lambda_{12}^{(2)}}$ $+S_{1,j_3}S_{2,j_4}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4}2\lambda_{12}^{(2)}$ $-\frac{h\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4}1\lambda_{12}^{(1)}}{\epsilon}$ $-\frac{1}{2}h\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4}2\lambda_{12}^{(2)}$ $-\frac{h\overline{y_{2,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4}2\lambda_{12}^{(2)}}{\epsilon}$ $-\frac{1}{2}h\overline{y_{2,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4}2\lambda_{12}^{(2)}$	
cont.												$-\frac{1}{2}h\overline{y_{2,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4}2\lambda_{12}^{(2)}$ $-\frac{h\overline{y_{2,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4}2\lambda_{12}^{(2)}}{\epsilon}$ $-\frac{1}{2}h\overline{y_{2,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4}2\lambda_{12}^{(2)}$ $-\frac{1}{2}h\overline{y_{2,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4}2\lambda_{12}^{(2)}$	
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	$+\frac{2h\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}L_{a_4,i_2}{}^TC P_L L_{a_2,i_1}H_{a_1}H_{a_3}\lambda_{T3}\epsilon_{a_1,a_2}\epsilon_{a_3,a_4}}{\sqrt{3}\epsilon M_F}$ $+\frac{2h\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}L_{a_4,i_2}{}^TC P_L L_{a_2,i_1}H_{a_1}H_{a_3}\lambda_{T3}\epsilon_{a_1,a_2}\epsilon_{a_3,a_4}}{\sqrt{3}M_F}$ $+\frac{2h\lambda_{T3}L_{a_3,i_1}C P_R \overline{L_{a_4,i_2}}{}^TY_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_3}\epsilon_{a_2,a_4}H_{a_1}^\dagger H_{a_2}^\dagger}{\sqrt{3}\epsilon M_F}$		$-\frac{2h\overline{y_{1,i_1,r_1}Y_{e,i_2,i_3}}H_{a_2}S_{1,j_2}y_{1,i_2,r_1}Y_{e,i_1,i_3}\mathcal{I}_{y_1,a_1,j_2}\mathcal{I}_{y_1,a_2,j_1}H_{a_1}^\dagger S_{1,j_1}^\dagger}{\epsilon}$ $-2h\overline{y_{1,i_1,r_1}Y_{e,i_2,i_3}}H_{a_2}S_{1,j_2}y_{1,i_2,r_1}Y_{e,i_1,i_3}\mathcal{I}_{y_1,a_1,j_2}\mathcal{I}_{y_1,a_2,j_1}H_{a_1}^\dagger S_{1,j_1}^\dagger$ $+H_{a_2}S_{1,j_2}H_{a_1}^\dagger S_{1,j_1}^\dagger \mathcal{I}_{H1,a_1,a_2,j_1,j_2}1\lambda_{H1}^{(1)}$ $-\frac{h\overline{y_{1,i_1,r_1}}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}H_{a_1}^\dagger S_{1,j_1}^\dagger \mathcal{I}_{H1,a_1,a_2,j_1,j_2}1\lambda_{H1}^{(1)}}{\epsilon}$	$+\frac{2h\overline{y_{1,i_1,r_1}y_{2,i_2,r_2}}S_{1,j_2}S_{2,j_1}y_{1,i_2,r_1}y_{2,i_1,r_2}S_{1,j_1}^\dagger S_{2,j_2}^\dagger}{\epsilon}$ $-\frac{2h\overline{y_{1,i_1,r_1}y_{2,i_2,r_2}}S_{1,j_1}S_{2,j_2}y_{1,i_2,r_1}y_{2,i_1,r_2}S_{1,j_1}^\dagger S_{2,j_2}^\dagger}{\epsilon}$ $-2h\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}S_{1,j_2}S_{2,j_1}y_{1,i_2,r_2}y_{2,i_1,r_2}S_{1,j_1}^\dagger S_{2,j_2}^\dagger$ $+2h\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}S_{1,j_1}S_{2,j_2}y_{1,i_1,r_2}y_{2,i_2,r_2}S_{1,j_1}^\dagger S_{2,j_2}^\dagger$ $+S_{1,j_4}S_{2,j_3}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4}1\lambda_{12}^{(1)}$ $-\frac{h\overline{y_{1,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4}1\lambda_{12}^{(1)}}{\epsilon}$		
cont.								$+\frac{2h\lambda_{T3}L_{a_3,i_1}C P_R \overline{L_{a_4,i_2}}{}^TY_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_3}\epsilon_{a_2,a_4}H_{a_1}^\dagger H_{a_2}^\dagger}{\sqrt{3}M_F}$		$-\frac{1}{2}h\overline{y_{1,i_1,r_1}}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}H_{a_1}^\dagger S_{1,j_1}^\dagger \mathcal{I}_{H1,a_1,a_2,j_1,j_2}1\lambda_{H1}^{(1)}$ $+H_{a_2}S_{1,j_2}H_{a_1}^\dagger S_{1,j_1}^\dagger \mathcal{I}_{H1,a_1,a_2,j_1,j_2}2\lambda_{H1}^{(2)}$ $-\frac{h\overline{y_{1,i_1,r_1}}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}H_{a_1}^\dagger S_{1,j_1}^\dagger \mathcal{I}_{H1,a_1,a_2,j_1,j_2}2\lambda_{H1}^{(2)}}{\epsilon}$	$+2h\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}S_{1,j_1}S_{2,j_2}y_{1,i_1,r_2}y_{2,i_2,r_2}S_{1,j_1}^\dagger S_{2,j_2}^\dagger$ $+2h\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}S_{1,j_1}S_{2,j_2}y_{1,i_1,r_2}y_{2,i_2,r_2}S_{1,j_1}^\dagger S_{2,j_2}^\dagger$ $+S_{1,j_4}S_{2,j_3}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4}1\lambda_{12}^{(1)}$ $-\frac{h\overline{y_{1,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4}1\lambda_{12}^{(1)}}{\epsilon}$		
cont.										$-\frac{1}{2}h\overline{y_{1,i_1,r_1}}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}H_{a_1}^\dagger S_{1,j_1}^\dagger \mathcal{I}_{H1,a_1,a_2,j_1,j_2}2\lambda_{H1}^{(2)}$ $-\frac{1}{2}h\overline{y_{1,i_1,r_1}}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}H_{a_1}^\dagger S_{1,j_1}^\dagger \mathcal{I}_{H1,a_1,a_2,j_1,j_2}2\lambda_{H1}^{(2)}$	$-\frac{1}{2}h\overline{y_{1,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4}1\lambda_{12}^{(1)}$ $-\frac{h\overline{y_{2,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4}1\lambda_{12}^{(1)}}{\epsilon}$ $-\frac{1}{2}h\overline{y_{2,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4}1\lambda_{12}^{(1)}$ $-\sqrt{3}S_{1,j_2}S_{2,j_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \lambda_{12}^{(2)}$ $+\sqrt{3}S_{1,j_1}S_{2,j_2}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \lambda_{12}^{(2)}$ $+\sqrt{3h\overline{y_{1,i_1,r_1}}S_{1,j_2}S_{2,j_1}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \lambda_{12}^{(2)}}$		
cont.												$+\frac{1}{2}\sqrt{3h\overline{y_{1,i_1,r_1}}S_{1,j_2}S_{2,j_1}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \lambda_{12}^{(2)}}$ $-\frac{\sqrt{3h\overline{y_{1,i_1,r_1}}S_{1,j_1}S_{2,j_2}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \lambda_{12}^{(2)}}{\epsilon}$ $-\frac{1}{2}\sqrt{3h\overline{y_{1,i_1,r_1}}S_{1,j_1}S_{2,j_2}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \lambda_{12}^{(2)}}$ $+\frac{\sqrt{3h\overline{y_{2,i_1,r_1}}S_{1,j_2}S_{2,j_1}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \lambda_{12}^{(2)}}{\epsilon}$	
cont.												$+\frac{1}{2}\sqrt{3h\overline{y_{2,i_1,r_1}}S_{1,j_2}S_{2,j_1}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \lambda_{12}^{(2)}}$ $-\frac{\sqrt{3h\overline{y_{2,i_1,r_1}}S_{1,j_1}S_{2,j_2}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \lambda_{12}^{(2)}}{\epsilon}$ $-\frac{1}{2}\sqrt{3h\overline{y_{2,i_1,r_1}}S_{1,j_1}S_{2,j_2}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \lambda_{12}^{(2)}}$ $+\frac{\sqrt{3h\overline{y_{2,i_1,r_1}}S_{1,j_2}S_{2,j_1}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \lambda_{12}^{(2)}}{\epsilon}$	
cont.												$+\frac{1}{2}\sqrt{3h\overline{y_{2,i_1,r_1}}S_{1,j_2}S_{2,j_1}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \lambda_{12}^{(2)}}$ $-\frac{\sqrt{3h\overline{y_{2,i_1,r_1}}S_{1,j_1}S_{2,j_2}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \lambda_{12}^{(2)}}{\epsilon}$ $-\frac{1}{2}\sqrt{3h\overline{y_{2,i_1,r_1}}S_{1,j_1}S_{2,j_2}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \lambda_{12}^{(2)}}$ $+\frac{\sqrt{3h\overline{y_{2,i_1,r_1}}S_{1,j_2}S_{2,j_1}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \lambda_{12}^{(2)}}{\epsilon}$	
cont.												$+\frac{1}{2}\sqrt{3h\overline{y_{2,i_1,r_1}}S_{1,j_2}S_{2,j_1}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \lambda_{12}^{(2)}}$ $-\frac{\sqrt{3h\overline{y_{2,i_1,r_1}}S_{1,j_1}S_{2,j_2}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \lambda_{12}^{(2)}}{\epsilon}$ $-\frac{1}{2}\sqrt{3h\overline{y_{2,i_1,r_1}}S_{1,j_1}S_{2,j_2}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \lambda_{12}^{(2)}}$ $+\frac{\sqrt{3h\overline{y_{2,i_1,r_1}}S_{1,j_2}S_{2,j_1}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \lambda_{12}^{(2)}}{\epsilon}$	
cont.												$+\frac{1}{2}\sqrt{3h\overline{y_{2,i_1,r_1}}S_{1,j_2}S_{2,j_1}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \lambda_{12}^{(2)}}$ $-\frac{\sqrt{3h\overline{y_{2,i_1,r_1}}S_{1,j_1}S_{2,j_2}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \lambda_{12}^{(2)}}{\epsilon}$ $-\frac{1}{2}\sqrt{3h\overline{y_{2,i_1,r_1}}S_{1,j_1}S_{2,j_2}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \lambda_{12}^{(2)}}$ $+\frac{\sqrt{3h\overline{y_{2,i_1,r_1}}S_{1,j_2}S_{2,j_1}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \lambda_{12}^{(2)}}{\epsilon}$	
cont.												$+\frac{1}{2}\sqrt{3h\overline{y_{2,i_1,r_1}}S_{1,j_2}S_{2,j_1}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \lambda_{12}^{(2)}}$ $-\frac{\sqrt{3h\overline{y_{2,i_1,r_1}}S_{1,j_1}S_{2,j_2}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \lambda_{12}^{(2)}}{\epsilon}$ $-\frac{1}{2}\sqrt{3h\overline{y_{2,i_1,r_1}}S_{1,j_1}S_{2,j_2}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \lambda_{12}^{(2)}}$ $+\frac{\sqrt{3h\overline{y_{2,i_1,r_1}}S_{1,j_2}S_{2,j_1}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \lambda_{12}^{(2)}}{\epsilon}$	
cont.												$+\frac{1}{2}\sqrt{3h\overline{y_{2,i_1,r_1}}S_{1,j_2}S_{2,j_1}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \lambda_{12}^{(2)}}$ $-\frac{\sqrt{3h\overline{y_{2,i_1,r_1}}S_{1,j_1}S_{2,j_2}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \lambda_{12}^{(2)}}{\epsilon}$ $-\frac{1}{2}\sqrt{3h\overline{y_{2,i_1,r_1}}S_{1,j_1}S_{2,j_2}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \lambda_{12}^{(2)}}$ $+\frac{\sqrt{3h\overline{y_{2,i_1,r_1}}S_{1,j_2}S_{2,j_1}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \lambda_{12}^{(2)}}{\epsilon}$	
cont.												$+\frac{1}{2}\sqrt{3h\overline{y_{2,i_1,r_1}}S_{1,j_2}S_{2,j_1}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \lambda_{12}^{(2)}}$ $-\frac{\sqrt{3h\overline{y_{2,i_1,r_1}}S_{1,j_1}S_{2,j_2}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \lambda_{12}^{(2)}}{\epsilon}$ $-\frac{1}{2}\sqrt{3h\overline{y_{2,i_1,r_1}}S_{1,j_1}S_{2,j_2}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \lambda_{12}^{(2)}}$ $+\frac{\sqrt{3h\overline{y_{2,i_1,r_1}}S_{1,j_2}S_{2,j_1}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \lambda_{12}^{(2)}}{\epsilon}$	
cont.												$+\frac{1}{2}\sqrt{3h\overline{y_{2,i_1,r_1}}S_{1,j_2}S_{2,j_1}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \lambda_{12}^{(2)}}$ $-\frac{\sqrt{3h\overline{y_{2,i_1,r_1}}S_{1,j_1}S_{2,j_2}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \lambda_{12}^{(2)}}{\epsilon}$ $-\frac{1}{2}\sqrt{3h\overline{y_{2,i_1,r_1}}S_{1,j_1}S_{2,j_2}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \lambda_{12}^{(2)}}$ $+\frac{\sqrt{3h\overline{y_{2,i_1,r_1}}S_{1,j_2}S_{2,j_1}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \lambda_{12}^{(2)}}{\epsilon}$	
cont.												$+\frac{1}{2}\sqrt{3h\overline{y_{2,i_1,r_1}}S_{1,j_2}S_{2,j_1}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \lambda_{12}^{(2)}}$ $-\frac{\sqrt{3h\overline{y_{2,i_1,r_1}}S_{1,j_1}S_{2,j_2}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \lambda_{12}^{(2)}}{\epsilon}$ $-\frac{1}{2}\sqrt{3h\overline{y_{2,i_1,r_1}}S_{1,j_1}S_{2,j_2}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{$	

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	H	ℓ	S_1	H	S_1	S_2
cont.													$+ \frac{\sqrt{3}h\bar{y}_{1,i_1,r_1}S_{1,j_2}S_{2,j_1}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \lambda_{12}^{(2)}}{\epsilon}$
													$+ \frac{1}{2} \frac{\sqrt{3}h\bar{y}_{1,i_1,r_1}S_{1,j_2}S_{2,j_1}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \lambda_{12}^{(2)}}{\epsilon}$
													$- \frac{1}{2} \frac{\sqrt{3}h\bar{y}_{1,i_1,r_1}S_{1,j_1}S_{2,j_2}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \lambda_{12}^{(2)}}{\epsilon}$
cont.													$+ \frac{\sqrt{3}h\bar{y}_{2,i_1,r_1}S_{1,j_2}S_{2,j_1}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \lambda_{12}^{(2)}}{\epsilon}$
													$+ \frac{1}{2} \frac{\sqrt{3}h\bar{y}_{2,i_1,r_1}S_{1,j_2}S_{2,j_1}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \lambda_{12}^{(2)}}{\epsilon}$
													$- \frac{1}{2} \frac{\sqrt{3}h\bar{y}_{2,i_1,r_1}S_{1,j_1}S_{2,j_2}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \lambda_{12}^{(2)}}{\epsilon}$
cont.													$- \frac{1}{2} \frac{\sqrt{3}h\bar{y}_{2,i_1,r_1}S_{1,j_1}S_{2,j_2}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \lambda_{12}^{(2)}}{\epsilon}$
													$+ \frac{S_{1,j_3}S_{2,j_4}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4}^2 \lambda_{12}^{(2)}}{h\bar{y}_{1,i_1,r_1}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4}^2 \lambda_{12}^{(2)}}$
													$- \frac{1}{2} \frac{h\bar{y}_{1,i_1,r_1}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4}^2 \lambda_{12}^{(2)}}{\epsilon}$
cont.													$- \frac{1}{2} \frac{h\bar{y}_{2,i_1,r_1}S_{1,j_3}S_{2,j_4}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4}^2 \lambda_{12}^{(2)}}{\epsilon}$
													$+ \frac{2h\bar{y}_{1,i_1,r_1}y_{2,i_2,r_2}S_{1,j_2}S_{2,j_1}y_{1,i_2,r_2}y_{2,i_1,r_2}S_{1,j_1}^\dagger S_{2,j_2}^\dagger}{3\epsilon}$
													$- \frac{10h\bar{y}_{1,i_1,r_1}y_{2,i_2,r_2}S_{1,j_1}S_{2,j_2}y_{1,i_2,r_2}y_{2,i_1,r_2}S_{1,j_1}^\dagger S_{2,j_2}^\dagger}{3\epsilon}$
cont.													$- \frac{10}{3} \frac{h\bar{y}_{1,i_1,r_1}y_{2,i_2,r_2}S_{1,j_2}S_{2,j_1}y_{1,i_2,r_2}y_{2,i_1,r_2}S_{1,j_1}^\dagger S_{2,j_2}^\dagger}{\epsilon}$
													$+ \frac{8}{3} \frac{h\bar{y}_{1,i_1,r_1}y_{2,i_2,r_2}S_{1,j_2}S_{2,j_1}y_{1,i_1,r_2}y_{2,i_2,r_2}S_{1,j_1}^\dagger S_{2,j_2}^\dagger}{\epsilon}$
													$+ \frac{2}{3} \frac{h\bar{y}_{1,i_1,r_1}y_{2,i_2,r_2}S_{1,j_1}S_{2,j_2}y_{1,i_1,r_2}y_{2,i_2,r_2}S_{1,j_1}^\dagger S_{2,j_2}^\dagger}{\epsilon}$
cont.													$- \frac{1}{2} \frac{\sqrt{3}h\bar{y}_{1,i_1,r_1}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}H_{a_1}^\dagger S_{1,j_1}^\dagger \mathcal{I}_{H1,a_1,a_2,j_1,j_2}^1 \lambda_{H1}^{(1)}}{\epsilon}$
													$- \frac{1}{2} \frac{\sqrt{3}h\bar{y}_{1,i_1,r_1}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}H_{a_1}^\dagger S_{1,j_1}^\dagger \mathcal{I}_{H1,a_1,a_2,j_1,j_2}^2 \lambda_{H1}^{(2)}}{\epsilon}$
													$- \frac{1}{2} \frac{\sqrt{3}h\bar{y}_{1,i_1,r_1}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}H_{a_1}^\dagger S_{1,j_1}^\dagger \mathcal{I}_{H1,a_1,a_2,j_1,j_2}^2 \lambda_{H1}^{(2)}}{\epsilon}$
cont.													$+ \frac{3h\bar{y}_{1,i_1,r_1}S_{1,j_2}S_{2,j_1}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \lambda_{12}^{(2)}}{\epsilon}$
													$+ \frac{3}{2} \frac{h\bar{y}_{1,i_1,r_1}S_{1,j_2}S_{2,j_1}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \lambda_{12}^{(2)}}{\epsilon}$
													$- \frac{3}{2} \frac{h\bar{y}_{1,i_1,r_1}S_{1,j_1}S_{2,j_2}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \lambda_{12}^{(2)}}{\epsilon}$
cont.													$- \frac{3}{2} \frac{h\bar{y}_{1,i_1,r_1}S_{1,j_1}S_{2,j_2}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \lambda_{12}^{(2)}}{\epsilon}$
													$+ \frac{3h\bar{y}_{2,i_1,r_1}S_{1,j_2}S_{2,j_1}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \lambda_{12}^{(2)}}{\epsilon}$
													$- \frac{3}{2} \frac{h\bar{y}_{2,i_1,r_1}S_{1,j_1}S_{2,j_2}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \lambda_{12}^{(2)}}{\epsilon}$
cont.													$- \frac{3}{2} \frac{h\bar{y}_{2,i_1,r_1}S_{1,j_1}S_{2,j_2}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \lambda_{12}^{(2)}}{\epsilon}$
													$+ \frac{3}{2} \frac{h\bar{y}_{2,i_1,r_1}S_{1,j_2}S_{2,j_1}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \lambda_{12}^{(2)}}{\epsilon}$
													$- \frac{3}{2} \frac{h\bar{y}_{2,i_1,r_1}S_{1,j_1}S_{2,j_2}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \lambda_{12}^{(2)}}{\epsilon}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$H \ell$	S_1	H	S_1	S_2
												$+S_{1,j_3}S_{2,j_4}S_{1,j_1}^\dagger S_{2,j_2}^\dagger T_{12,j_2,j_3,j_1,j_4}2\lambda_{12}^{(2)}$ $-\frac{\sqrt{3}h\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger T_{12,j_2,j_3,j_1,j_4}2\lambda_{12}^{(2)}}{\epsilon}$ $-\frac{1}{2}\frac{\sqrt{3}h\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger T_{12,j_2,j_3,j_1,j_4}2\lambda_{12}^{(2)}}{\sqrt{3}h\overline{y_{2,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger T_{12,j_2,j_3,j_1,j_4}2\lambda_{12}^{(2)}}$ $-\frac{1}{2}\sqrt{3}h\overline{y_{2,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger T_{12,j_2,j_3,j_1,j_4}2\lambda_{12}^{(2)}$
cont.												
												$+\text{CNewScalar1NewScalar2}hS_{1,j_1}S_{2,j_2}\text{Bar}(\text{cg3})_{j_1,j_2}$ $+\frac{1}{2}C_S^\dagger s_2\overline{h y_{2,i_1,r_1}}S_{1,j_1}S_{2,j_2}y_{1,i_1,r_1}\text{Bar}(\text{cg3})_{j_1,j_2}$ $+\frac{C_S^\dagger S_2\overline{h y_{2,i_1,r_1}}S_{1,j_1}S_{2,j_2}y_{1,i_1,r_1}\text{Bar}(\text{cg3})_{j_1,j_2}}{\epsilon}$ $+\frac{5h\overline{y_{2,i_1,r_1}}y_{2,i_2,r_2}S_{1,j_1}S_{1,j_2}S_{2,j_2}S_{2,j_4}y_{1,i_1,r_2}y_{1,i_2,r_1}\text{Bar}(\text{cg3})_{j_1,j_2}\text{Bar}(\text{cg3})_{j_2,j_4}}{9\epsilon}$ $+\frac{1}{9}h\overline{y_{1,i_1,r_1}}y_{2,i_2,r_2}S_{1,j_1}S_{1,j_2}S_{2,j_2}S_{2,j_4}y_{1,i_1,r_2}y_{1,i_2,r_1}2\text{Bar}(\text{cg3})_{j_1,j_2}\text{Bar}(\text{cg3})_{j_2,j_4}$ $+\frac{10h\overline{y_{1,i_1,r_1}}y_{2,i_2,r_2}S_{1,j_1}S_{1,j_2}S_{2,j_2}y_{1,i_1,r_2}y_{1,i_2,r_1}\text{Bar}(\text{cg3})_{j_2,j_3}S_{1,j_1}^\dagger}{3\sqrt{3}\epsilon}$ $+\frac{2h\overline{y_{1,i_1,r_1}}y_{2,i_2,r_2}S_{1,j_1}S_{1,j_2}S_{2,j_2}y_{1,i_1,r_2}y_{1,i_2,r_2}\text{Bar}(\text{cg3})_{j_2,j_3}S_{1,j_1}^\dagger}{3\sqrt{3}}$ $+\frac{10h\overline{y_{2,i_1,r_1}}y_{2,i_2,r_2}S_{1,j_2}S_{2,j_2}y_{1,i_1,r_2}y_{2,i_2,r_1}\text{Bar}(\text{cg3})_{j_2,j_3}S_{2,j_1}^\dagger}{3\sqrt{3}\epsilon}$ $-\frac{2h\overline{y_{2,i_1,r_1}}y_{2,i_2,r_2}S_{1,j_2}S_{2,j_2}y_{1,i_1,r_2}y_{2,i_2,r_2}\text{Bar}(\text{cg3})_{j_2,j_3}S_{2,j_1}^\dagger}{3\sqrt{3}}$ $+\tilde{h}\text{CNewScalar1NewScalar2}cg3_{j_1,j_2}S_{1,j_1}^\dagger S_{2,j_2}^\dagger$ $+\frac{1}{2}C_S^\dagger S_2\overline{h y_{1,i_1,r_1}}cg3_{j_1,j_2}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger$ $+\frac{C_S^\dagger S_2\overline{h y_{1,i_1,r_1}}cg3_{j_1,j_2}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger}{\epsilon}$ $+\frac{2h\overline{y_{1,i_1,r_1}}y_{2,i_2,r_2}S_{1,j_2}S_{2,j_2}y_{1,i_2,r_1}y_{2,i_1,r_2}S_{1,j_1}^\dagger S_{2,j_2}^\dagger}{3\epsilon}$ $+\frac{10h\overline{y_{1,i_1,r_1}}y_{2,i_2,r_2}S_{1,j_1}S_{2,j_2}y_{1,i_2,r_1}y_{2,i_1,r_2}S_{1,j_1}^\dagger S_{2,j_2}^\dagger}{3\epsilon}$ $-\frac{10}{3}\frac{h\overline{y_{1,i_1,r_1}}y_{2,i_2,r_2}S_{1,j_2}S_{2,j_2}y_{1,i_2,r_2}y_{2,i_1,r_2}S_{1,j_1}^\dagger S_{2,j_2}^\dagger}{h\overline{y_{1,i_1,r_1}}y_{2,i_2,r_2}S_{1,j_1}S_{2,j_2}y_{1,i_2,r_2}y_{2,i_1,r_2}S_{1,j_1}^\dagger S_{2,j_2}^\dagger}$ $+\frac{2}{3}h\overline{y_{1,i_1,r_1}}y_{2,i_2,r_2}S_{1,j_1}S_{2,j_2}y_{1,i_1,r_2}y_{2,i_2,r_2}S_{1,j_1}^\dagger S_{2,j_2}^\dagger$ $+\frac{8h\overline{y_{1,i_1,r_1}}y_{2,i_2,r_2}S_{1,j_2}S_{2,j_2}y_{1,i_1,r_2}y_{2,i_2,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger}{3\epsilon}$ $+\frac{10h\overline{y_{1,i_1,r_1}}y_{2,i_2,r_2}S_{1,j_1}S_{2,j_2}y_{1,i_1,r_2}y_{2,i_2,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger}{3\epsilon}$ $+\frac{8}{3}h\overline{y_{1,i_1,r_1}}y_{2,i_2,r_2}S_{1,j_2}S_{2,j_2}y_{1,i_1,r_2}y_{2,i_2,r_2}S_{1,j_1}^\dagger S_{2,j_2}^\dagger$ $+\frac{2}{3}h\overline{y_{1,i_1,r_1}}y_{2,i_2,r_2}S_{1,j_1}S_{2,j_2}y_{1,i_1,r_2}y_{2,i_2,r_2}S_{1,j_1}^\dagger S_{2,j_2}^\dagger$ $+\frac{10h\overline{y_{1,i_1,r_1}}y_{1,i_2,r_2}cg3_{j_2,j_3}S_{1,j_1}y_{1,i_1,r_2}y_{2,i_2,r_2}S_{1,j_1}^\dagger S_{1,j_2}^\dagger S_{2,j_3}^\dagger}{3\sqrt{3}\epsilon}$ $-\frac{2h\overline{y_{1,i_1,r_1}}y_{1,i_2,r_2}cg3_{j_2,j_3}S_{1,j_1}y_{1,i_1,r_2}y_{2,i_2,r_2}S_{1,j_1}^\dagger S_{1,j_2}^\dagger S_{2,j_3}^\dagger}{3\sqrt{3}}$ $+\frac{10h\overline{y_{1,i_1,r_1}}y_{2,i_2,r_2}cg3_{j_1,j_2}S_{2,j_2}y_{2,i_1,r_2}y_{2,i_2,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger S_{2,j_3}^\dagger}{3\sqrt{3}\epsilon}$ $-\frac{2h\overline{y_{1,i_1,r_1}}y_{2,i_2,r_2}cg3_{j_1,j_2}S_{2,j_2}y_{2,i_1,r_2}y_{2,i_2,r_2}S_{1,j_1}^\dagger S_{2,j_2}^\dagger S_{2,j_3}^\dagger}{3\sqrt{3}}$ $-\frac{5h\overline{y_{1,i_1,r_1}}y_{1,i_2,r_2}cg3_{j_1,j_3}cg3_{j_2,j_4}y_{2,i_1,r_2}y_{2,i_2,r_1}S_{1,j_1}^\dagger S_{1,j_2}^\dagger S_{2,j_3}^\dagger S_{2,j_4}^\dagger}{9\epsilon}$ $+\frac{1}{9}h\overline{y_{1,i_1,r_1}}y_{1,i_2,r_2}cg3_{j_1,j_3}cg3_{j_2,j_4}y_{2,i_1,r_2}y_{2,i_2,r_2}S_{1,j_1}^\dagger S_{1,j_2}^\dagger S_{2,j_3}^\dagger S_{2,j_4}^\dagger$ $+\frac{S_{1,j_4}S_{2,j_3}S_{1,j_1}^\dagger S_{2,j_2}^\dagger T_{12,j_2,j_3,j_1,j_4}^\dagger\lambda_{12}^{(1)}}{\sqrt{3}h\overline{y_{1,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger T_{12,j_2,j_3,j_1,j_4}^\dagger\lambda_{12}^{(1)}}$ $-\frac{1}{2}\sqrt{3}h\overline{y_{1,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger T_{12,j_2,j_3,j_1,j_4}^\dagger\lambda_{12}^{(1)}$ $-\frac{1}{2}\frac{\sqrt{3}h\overline{y_{2,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger T_{12,j_2,j_3,j_1,j_4}^\dagger\lambda_{12}^{(1)}}{\sqrt{3}h\overline{y_{2,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger T_{12,j_2,j_3,j_1,j_4}^\dagger\lambda_{12}^{(1)}}$ $-\frac{1}{2}\sqrt{3}h\overline{y_{2,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger T_{12,j_2,j_3,j_1,j_4}^\dagger$

[illegible]

[illegible]

[illegible]

[illegible]

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	H	ℓ	S_1	H	S_2
$cont.$										$-\frac{h\bar{y}_{1,i_1,r_1}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}H_{a_1}^\dagger S_{1,j_1}^\dagger \mathcal{I}_{H1,a_1,a_2,j_2,j_1}2\lambda_{H1}^{(2)}}{\sqrt{3}}$	$-\frac{2h\bar{y}_{2,i_1,r_1}S_2S_{1,j_2}y_{2,i_1,r_1}S_2^\dagger S_{1,j_1}^\dagger \mathcal{I}_{12,j_2,j_1}1\lambda_{12}^{(1)}}{\epsilon}$	
											$-\frac{h\bar{y}_{2,i_1,r_1}S_2S_{1,j_2}y_{2,i_1,r_1}S_2^\dagger S_{1,j_1}^\dagger \mathcal{I}_{12,j_2,j_1}1\lambda_{12}^{(1)}}{\epsilon}$	
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$			$-\frac{2\sqrt{\frac{2}{3}}h\bar{y}_{1,i_1,r_1}y_{2,i_2,r_1}L_{a_4,i_2}^T C P_L L_{a_2,i_1}H_{a_1}H_{a_3}\lambda_{T3}\epsilon_{a_1,a_2}\epsilon_{a_3,a_4}}{\epsilon M_F}$	$-\frac{2h\bar{y}_{1,i_1,r_1}Y_{e,i_2,i_3}H_{a_1}S_{1,j_1}y_{1,i_2,r_1}Y_{e,i_1,i_3}H_{a_1}^\dagger S_{1,j_1}^\dagger}{\sqrt{3}\epsilon}$	$-\frac{4h\bar{y}_{1,i_1,r_1}y_{2,i_2,r_2}S_{1,j_2}S_{2,j_2}y_{1,i_2,r_1}y_{2,i_1,r_2}S_{1,j_1}^\dagger S_{2,j_1}^\dagger}{3\epsilon}$
										$-\frac{2\sqrt{\frac{2}{3}}h\bar{y}_{1,i_1,r_1}L_{a_4,i_2}^T C P_L L_{a_2,i_1}H_{a_1}H_{a_3}\lambda_{T3}\epsilon_{a_1,a_2}\epsilon_{a_3,a_4}}{M_F}$	$-\frac{2h\bar{y}_{1,i_1,r_1}^\dagger Y_{e,i_2,i_3}H_{a_1}S_{1,j_1}y_{1,i_2,r_1}Y_{e,i_1,i_3}H_{a_1}^\dagger S_{1,j_1}^\dagger}{\sqrt{3}}$	$-\frac{4}{3}h\bar{y}_{1,i_1,r_1}y_{2,i_2,r_1}S_{1,j_2}S_{2,j_2}y_{1,i_2,r_2}y_{2,i_1,r_2}S_{1,j_1}^\dagger S_{2,j_1}^\dagger$
										$-\frac{2\sqrt{\frac{2}{3}}h\lambda_{T3}L_{a_3,i_1}C P_R \bar{L}_{a_4,i_2}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_3}\epsilon_{a_2,a_4}H_{a_1}^\dagger H_{a_2}^\dagger}{\epsilon M_F}$	$-2i\sqrt{\frac{2}{3}}h\bar{y}_{1,i_1,r_1}Y_{e,i_2,i_3}H_{a_2}S_{1,j_2}y_{1,i_2,r_1}Y_{e,i_1,i_3}H_{a_1}^\dagger S_{1,j_1}^\dagger \mathcal{I}_{H1,a_1,a_2,j_2,j_1}1$	$-\frac{4}{3}h\bar{y}_{1,i_1,r_1}y_{2,i_2,r_1}S_{1,j_2}S_{2,j_2}y_{1,i_1,r_2}y_{2,i_2,r_2}S_{1,j_1}^\dagger S_{2,j_1}^\dagger$
$cont.$										$-2i\sqrt{\frac{2}{3}}h\bar{y}_{1,i_1,r_1}Y_{e,i_2,i_3}H_{a_2}S_{1,j_2}y_{1,i_2,r_1}Y_{e,i_1,i_3}H_{a_1}^\dagger S_{1,j_1}^\dagger \mathcal{I}_{H1,a_1,a_2,j_2,j_1}1$	$-\frac{4h\bar{y}_{1,i_1,r_1}y_{2,i_2,r_2}S_{1,j_2}S_{2,j_2}y_{1,i_2,r_1}y_{2,i_1,r_2}S_{1,j_1}^\dagger S_{2,j_1}^\dagger}{3\epsilon}$	$+\frac{4h\bar{y}_{1,i_1,r_1}y_{2,i_2,r_2}S_{1,j_2}S_{2,j_2}y_{1,i_2,r_1}y_{2,i_1,r_2}S_{1,j_1}^\dagger S_{2,j_1}^\dagger}{3\epsilon}$
										$-\frac{2\sqrt{\frac{2}{3}}h\lambda_{T3}L_{a_3,i_1}C P_R \bar{L}_{a_4,i_2}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_3}\epsilon_{a_2,a_4}H_{a_1}^\dagger H_{a_2}^\dagger}{M_F}$	$+\frac{H_{a_2}S_{1,j_2}H_{a_1}^\dagger S_{1,j_1}^\dagger \mathcal{I}_{H1,a_1,a_2,j_2,j_1}2\lambda_{H1}^{(2)}}{2h\bar{y}_{1,i_1,r_1}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}H_{a_1}^\dagger S_{1,j_1}^\dagger \mathcal{I}_{H1,a_1,a_2,j_2,j_1}2\lambda_{H1}^{(2)}}$	$-\frac{4h\bar{y}_{1,i_1,r_1}y_{2,i_2,r_2}S_{1,j_2}S_{2,j_2}y_{1,i_2,r_1}y_{2,i_1,r_2}S_{1,j_1}^\dagger S_{2,j_1}^\dagger}{3\epsilon}$
$cont.$										$-\frac{h\bar{y}_{1,i_1,r_1}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}H_{a_1}^\dagger S_{1,j_1}^\dagger \mathcal{I}_{H1,a_1,a_2,j_2,j_1}2\lambda_{H1}^{(2)}}{\sqrt{3}}$	$+\frac{4}{3}h\bar{y}_{1,i_1,r_1}y_{2,i_2,r_1}S_{1,j_2}S_{2,j_2}y_{1,i_1,r_2}y_{2,i_2,r_2}S_{1,j_1}^\dagger S_{2,j_1}^\dagger$	
											$+\frac{4}{3}h\bar{y}_{1,i_1,r_1}y_{2,i_2,r_1}S_{1,j_2}S_{2,j_2}y_{1,i_1,r_2}y_{2,i_2,r_2}S_{1,j_1}^\dagger S_{2,j_1}^\dagger$	$+\frac{4}{3}h\bar{y}_{1,i_1,r_1}y_{2,i_2,r_1}S_{1,j_2}S_{2,j_2}y_{1,i_1,r_2}y_{2,i_2,r_2}S_{1,j_1}^\dagger S_{2,j_1}^\dagger$
$cont.$										$-\frac{2h\bar{y}_{1,i_1,r_1}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_3,j_1,j_4,j_2}1\lambda_{12}^{(1)}}{\sqrt{3}\epsilon}$	$-\frac{h\bar{y}_{1,i_1,r_1}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_3,j_1,j_4,j_2}1\lambda_{12}^{(1)}}{\sqrt{3}}$	
											$-\frac{2h\bar{y}_{2,i_1,r_1}S_{1,j_3}S_{2,j_4}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_3,j_1,j_4,j_2}1\lambda_{12}^{(1)}}{\sqrt{3}\epsilon}$	$-\frac{2h\bar{y}_{2,i_1,r_1}S_{1,j_3}S_{2,j_4}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_3,j_1,j_4,j_2}1\lambda_{12}^{(1)}}{\sqrt{3}}$
$cont.$										$-\frac{h\bar{y}_{2,i_1,r_1}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_3,j_1,j_4,j_2}1\lambda_{12}^{(1)}}{\sqrt{3}}$	$-\frac{h\bar{y}_{2,i_1,r_1}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_3,j_1,j_4,j_2}1\lambda_{12}^{(1)}}{\sqrt{3}}$	
											$-\frac{h\bar{y}_{2,i_1,r_1}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_3,j_1,j_4,j_2}1\lambda_{12}^{(1)}}{\sqrt{3}}$	$-\frac{h\bar{y}_{2,i_1,r_1}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_3,j_1,j_4,j_2}1\lambda_{12}^{(1)}}{\sqrt{3}}$
$cont.$										$-\frac{h\bar{y}_{2,i_1,r_1}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_3,j_1,j_4,j_2}1\lambda_{12}^{(1)}}{\sqrt{3}}$	$-\frac{h\bar{y}_{2,i_1,r_1}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_3,j_1,j_4,j_2}1\lambda_{12}^{(1)}}{\sqrt{3}}$	
											$-\frac{h\bar{y}_{2,i_1,r_1}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_3,j_1,j_4,j_2}1\lambda_{12}^{(1)}}{\sqrt{3}}$	$-\frac{h\bar{y}_{2,i_1,r_1}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_3,j_1,j_4,j_2}1\lambda_{12}^{(1)}}{\sqrt{3}}$
$cont.$										$-\frac{h\bar{y}_{2,i_1,r_1}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_3,j_1,j_4,j_2}1\lambda_{12}^{(1)}}{\sqrt{3}}$	$-\frac{h\bar{y}_{2,i_1,r_1}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_3,j_1,j_4,j_2}1\lambda_{12}^{(1)}}{\sqrt{3}}$	
											$-\frac{h\bar{y}_{2,i_1,r_1}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_3,j_1,j_4,j_2}1\lambda_{12}^{(1)}}{\sqrt{3}}$	$-\frac{h\bar{y}_{2,i_1,r_1}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_3,j_1,j_4,j_2}1\lambda_{12}^{(1)}}{\sqrt{3}}$
$cont.$										$-\frac{h\bar{y}_{2,i_1,r_1}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_3,j_1,j_4,j_2}1\lambda_{12}^{(1)}}{\sqrt{3}}$	$-\frac{h\bar{y}_{2,i_1,r_1}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_3,j_1,j_4,j_2}1\lambda_{12}^{(1)}}{\sqrt{3}}$	
											$-\frac{h\bar{y}_{2,i_1,r_1}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_3,j_1,j_4,j_2}1\lambda_{12}^{(1)}}{\sqrt{3}}$	$-\frac{h\bar{y}_{2,i_1,r_1}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_3,j_1,j_4,j_2}1\lambda_{12}^{(1)}}{\sqrt{3}}$
$cont.$										$-\frac{h\bar{y}_{2,i_1,r_1}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_3,j_1,j_4,j_2}1\lambda_{12}^{(1)}}{\sqrt{3}}$	$-\frac{h\bar{y}_{2,i_1,r_1}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_3,j_1,j_4,j_2}1\lambda_{12}^{(1)}}{\sqrt{3}}$	
											$-\frac{h\bar{y}_{2,i_1,r_1}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_3,j_1,j_4,j_2}1\lambda_{12}^{(1)}}{\sqrt{3}}$	$-\frac{h\bar{y}_{2,i_1,r_1}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_3,j_1,j_4,j_2}1\lambda_{12}^{(1)}}{\sqrt{3}}$
$cont.$										$-\frac{h\bar{y}_{2,i_1,r_1}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_3,j_1,j_4,j_2}1\lambda_{12}^{(1)}}{\sqrt{3}}$	$-\frac{h\bar{y}_{2,i_1,r_1}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_3,j_1,j_4,j_2}1\lambda_{12}^{(1)}}{\sqrt{3}}$	
											$-\frac{h\bar{y}_{2,i_1,r_1}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_3,j_1,j_4,j_2}1\lambda_{12}^{(1)}}{\sqrt{3}}$	$-\frac{h\bar{y}_{2,i_1,r_1}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_3,j_1,j_4,j_2}1\lambda_{12}^{(1)}}{\sqrt{3}}$
$cont.$										$-\frac{h\bar{y}_{2,i_1,r_1}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_3,j_1,j_4,j_2}1\lambda_{12}^{(1)}}{\sqrt{3}}$	$-\frac{h\bar{y}_{2,i_1,r_1}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_3,j_1,j_4,j_2}1\lambda_{12}^{(1)}}{\sqrt{3}}$	
											$-\frac{h\bar{y}_{2,i_1,r_1}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_3,j_1,j_4,j_2}1\lambda_{12}^{(1)}}{\sqrt{3}}$	$-\frac{h\bar{y}_{2,i_1,r_1}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_3,j_1,j_4,j_2}1\lambda_{12}^{(1)}}{\sqrt{3}}$
$cont.$										$-\frac{h\bar{y}_{2,i_1,r_1}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_3,j_1,j_4,j_2}1\lambda_{12}^{(1)}}{\sqrt{3}}$	$-\frac{h\bar{y}_{2,i_1,r_1}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_3,j_1,j_4,j_2}1\lambda_{12}^{(1)}}{\sqrt{3}}$	
											$-\frac{h\bar{y}_{2,i_1,r_1}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_3,j_1,j_4,j_2}1\lambda_{12}^{(1)}}{\sqrt{3}}$	$-\frac{h\bar{y}_{2,i_1,r_1}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_3,j_1,j_4,j_2}1\lambda_{12}^{(1)}}{\sqrt{3}}$
$cont.$										$-\frac{h\bar{y}_{2,i_1,r_1}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_3,j_1,j_4,j_2}1\lambda_{12}^{(1)}}{\sqrt{3}}$	$-\frac{h\bar{y}_{2,i_1,r_1}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_3,j_1,j_4,j_2}1\lambda_{12}^{(1)}}{\sqrt{3}}$	
											$-\frac{h\bar{y}_{2,i_1,r_1}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_3,j_1,j_4,j_2}1\lambda_{12}^{(1)}}{\sqrt{3}}$	$-\frac{h\bar{y}_{2,i_1,r_1}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_3,j_1,j_4,j_2}1\lambda_{12}^{(1)}}{\sqrt{3}}$
$cont.$										$-\frac{h\bar{y}_{2,i_1,r_1}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_3,j_1,j_4,j_2}1\lambda_{12}^{(1)}}{\sqrt{3}}$	$-\frac{h\bar{y}_{2,i_1,r_1}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_3,j_1,j_4,j_2}1\lambda_{12}^{(1)}}{\sqrt{3}}$	
											$-\frac{h\bar{y}_{2,i_1,r_1}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_3,j_1,j_4,j_2}1\lambda_{12}^{(1)}}{\sqrt{3}}$	$-\frac{h\bar{y}_{2,i_1,r_1}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_3,j_1,j_4,j_2}1\lambda_{12}^{(1)}}{\sqrt{3}}$
$cont.$										$-\frac{h\bar{y}_{2,i_1,r_1}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_3,j_1,j_4,j_2}1\lambda_{12}^{(1)}}{\sqrt{3}}$	$-\frac{h\bar{y}_{2,i_1,r_1}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_3,j_1,j_4,j_2}1\lambda_{12}^{(1)}}{\sqrt{3}}$	
											$-\frac{h\bar{y}_{2,i_1,r_1}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_3,j_1,j_4,j_2}1\lambda_{12}^{(1)}}{\sqrt{3}}$	$-\frac{h\bar{y}_{2,i_1,r_1}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_3,j_1,j_4,j_2}1\lambda_{12}^{(1)}}{\sqrt{3}}$
$cont.$										$-\frac{h\bar{y}_{2,i_1,r_1}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_3,j_1,j_4,j_2}1\lambda_{12}^{(1)}}{\sqrt{3}}$	$-\frac{h\bar{y}_{2,i_1,r_1}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_3,j_1,j_4,j_2}1\lambda_{12}^{(1)}}{\sqrt{3}}$	
											$-\frac{h\bar{y}_{2,i_1,r_1}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_3,j_1,j_4,j_2}1\lambda_{12}^{(1)}}{\sqrt{3}}$	$-\frac{h\bar{y}_{2,i_1,r_1}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_3,j_1,j_4,j_2}1\lambda_{12}^{(1)}}{\sqrt{3}}$
$cont.$										$-\frac{h\bar{y}_{2,i_1,r_1}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_3,j_1,j_4,j_2}1\lambda_{12}^{(1)}}{\sqrt{3}}$	$-\frac{h\bar{y}_{2,i_1,r_1}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_3,j_1,j_4,j_2}1\lambda_{12}^{(1)}}{\sqrt{3}}$	
											$-\frac{h\bar{y}_{2,i_1,r_1}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_3,j_1,j_4,j_2}1\lambda_{12}^{(1)}}{\sqrt{3}}$	$-\frac{h\bar{y}_{2,i_1,r_1}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_3,j_1,j_4,j_2}1\lambda_{12}^{(1)}}{\sqrt{3}}$
$cont.$										$-\frac{h\bar{y}_{2,i_1,r_1}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_3,j_1,j_4,j_2}1\lambda_{12}^{(1)}}{\sqrt{3}}$	$-\frac{h\bar{y}_{2,i_1,r_1}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_3,j_1,j_4,j_2}1\lambda_{12}^{(1)}}{\sqrt{3}}$	
											$-\frac{h\bar{y}_{2,i_1,r_1}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_3,j_1,j_4,j_2}1\lambda_{12}^{(1)}}{\sqrt{3}}$	$-\frac{h\bar{y}_{2,i_1,r_1}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_3,j_1,j_4,j_2}1\lambda_{12}^{(1)}}{\sqrt{3}}$
$cont.$										$-\frac{h\bar{y}_{2,i_1,r_1}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_3,j_1,j_4,j_2}1\lambda_{12}^{(1)}}{\sqrt{3}}$	$-\frac{h\bar{y}_{2,i_1,r_1}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_3,j_1,j_4,j_2}1\lambda_{12}^{(1)}}{\sqrt{3}}$	
											$-\frac{h\bar{y}_{2,i_1,r_1}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_3,j_1,j_4,j_2}1\lambda_{12}^{(1)}}{\sqrt{3}}$	$-\frac{h\bar{y}_{2,i_1,r_1}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_3,j_1,j_4,j_2}1\lambda_{12}^{(1)}}{\sqrt{3}}$
$cont.$										$-\frac{h\bar{y}_{2,i_1,r_1}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_3,j_1,j_4,j_2}1\lambda_{12}^{(1)}}{\sqrt{3}}$	$-\frac{h\bar{y}_{2,i_1,r_1}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_3,j_1,j_4,j_2}1\lambda_{12}^{(1)}}{\sqrt{3}}$	
											$-\frac{h\bar{y}_{2,i_1,r_1}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_3,j_1,j_4,j_2}1\lambda_{12}^{(1)}}{\sqrt{3}}$	$-\frac{h\bar{y}_{2,i_1,r_1}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_3,j_1,j_4,j_2}1\lambda_{12}^{(1)}}{\sqrt{3}}$
$cont.$										$-\frac{h\bar{y}_{2,i_1,r_1}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_3,j_1,j_4,j_2}1\lambda_{12}^{(1)}}{\sqrt{3}}$	$-\frac{h\bar{y}_{2,i_1,r_1}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_3,j_1,j_4,j_2}1\lambda_{12}^{(1)}}{\sqrt{3}}$	
											$-\frac{h\bar{y}_{2,i_1,r_1}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_3,j_1,j_4,j_2}1\lambda_{12}^{(1)}}{\sqrt{3}}$	$-\frac{h\bar{y}_{2,i_1,r_1}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_3,j_1,j_4,j_2}1\lambda_{12}^{(1)}}{\sqrt{3}}$
$cont.$										$-\frac{h\bar{y}_{2,i_1,r_1}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{$		

[illegible]

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$H \ell$	$S_1 H$	$S_1 S_2$			
cont.										$\begin{aligned} & + S_{1,j_4} S_{2,j_3} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_1,j_3,j_2,j_4} {}^2\lambda_{12}^{(2)} \\ & - \frac{2h\bar{y}_{1,i_1,r_1} S_{1,j_4} S_{2,j_3} y_{1,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_1,j_3,j_2,j_4} {}^2\lambda_{12}^{(2)}}{\sqrt{3}\epsilon} \\ & - \frac{h\bar{y}_{1,i_1,r_1} S_{1,j_4} S_{2,j_3} y_{1,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_1,j_3,j_2,j_4} {}^2\lambda_{12}^{(2)}}{\sqrt{3}} \end{aligned}$			
										$\begin{aligned} & - \frac{2h\bar{y}_{2,i_1,r_1} S_{1,j_4} S_{2,j_3} y_{2,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_1,j_3,j_2,j_4} {}^2\lambda_{12}^{(2)}}{\sqrt{3}\epsilon} \\ & - \frac{h\bar{y}_{2,i_1,r_1} S_{1,j_4} S_{2,j_3} y_{2,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_1,j_3,j_2,j_4} {}^2\lambda_{12}^{(2)}}{\sqrt{3}} \end{aligned}$			
											$\begin{aligned} & - S_{1,j_4} S_{2,j_3} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_1,j_3,j_4,j_2} {}^2\lambda_{12}^{(2)} \\ & + S_{1,j_3} S_{2,j_4} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_1,j_3,j_4,j_2} {}^2\lambda_{12}^{(2)} \\ & + \frac{2h\bar{y}_{1,i_1,r_1} S_{1,j_4} S_{2,j_3} y_{1,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_1,j_3,j_4,j_2} {}^2\lambda_{12}^{(2)}}{\sqrt{3}\epsilon} \\ & + \frac{h\bar{y}_{1,i_1,r_1} S_{1,j_4} S_{2,j_3} y_{1,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_1,j_3,j_4,j_2} {}^2\lambda_{12}^{(2)}}{\sqrt{3}} \end{aligned}$		
cont.										$\begin{aligned} & - \frac{2h\bar{y}_{2,i_1,r_1} S_{1,j_3} S_{2,j_4} y_{2,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_1,j_3,j_4,j_2} {}^2\lambda_{12}^{(2)}}{\sqrt{3}\epsilon} \\ & - \frac{h\bar{y}_{2,i_1,r_1} S_{1,j_3} S_{2,j_4} y_{2,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_1,j_3,j_4,j_2} {}^2\lambda_{12}^{(2)}}{\sqrt{3}} \end{aligned}$			
										$\begin{aligned} & + \frac{2h\bar{y}_{2,i_1,r_1} S_{1,j_3} S_{2,j_4} y_{2,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_1,j_3,j_4,j_2} {}^2\lambda_{12}^{(2)}}{\sqrt{3}\epsilon} \\ & + \frac{h\bar{y}_{2,i_1,r_1} S_{1,j_3} S_{2,j_4} y_{2,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_1,j_3,j_4,j_2} {}^2\lambda_{12}^{(2)}}{\sqrt{3}} \end{aligned}$			
											$\begin{aligned} & - \frac{2h\bar{y}_{2,i_1,r_1} S_{1,j_3} S_{2,j_4} y_{2,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_1,j_3,j_4,j_2} {}^2\lambda_{12}^{(2)}}{\sqrt{3}\epsilon} \\ & - \frac{h\bar{y}_{2,i_1,r_1} S_{1,j_3} S_{2,j_4} y_{2,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_1,j_3,j_4,j_2} {}^2\lambda_{12}^{(2)}}{\sqrt{3}} \end{aligned}$		
cont.										$\begin{aligned} & + S_{1,j_3} S_{2,j_4} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_3,j_1,j_4,j_2} {}^3\lambda_{12}^{(3)} \\ & - \frac{2h\bar{y}_{1,i_1,r_1} S_{1,j_3} S_{2,j_4} y_{1,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_3,j_1,j_4,j_2} {}^3\lambda_{12}^{(3)}}{\sqrt{3}\epsilon} \\ & - \frac{h\bar{y}_{1,i_1,r_1} S_{1,j_3} S_{2,j_4} y_{1,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_3,j_1,j_4,j_2} {}^3\lambda_{12}^{(3)}}{\sqrt{3}} \end{aligned}$			
										$\begin{aligned} & - \frac{2h\bar{y}_{2,i_1,r_1} S_{1,j_3} S_{2,j_4} y_{2,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_3,j_1,j_4,j_2} {}^3\lambda_{12}^{(3)}}{\sqrt{3}\epsilon} \\ & - \frac{h\bar{y}_{2,i_1,r_1} S_{1,j_3} S_{2,j_4} y_{2,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_3,j_1,j_4,j_2} {}^3\lambda_{12}^{(3)}}{\sqrt{3}} \end{aligned}$			
											$\begin{aligned} & + S_{1,j_3} S_{2,j_4} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_3,j_1,j_4,j_2} {}^3\lambda_{12}^{(3)} \\ & - \frac{2h\bar{y}_{1,i_1,r_1} S_{1,j_3} S_{2,j_4} y_{1,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_3,j_1,j_4,j_2} {}^3\lambda_{12}^{(3)}}{\sqrt{3}\epsilon} \\ & - \frac{h\bar{y}_{1,i_1,r_1} S_{1,j_3} S_{2,j_4} y_{1,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_3,j_1,j_4,j_2} {}^3\lambda_{12}^{(3)}}{\sqrt{3}} \end{aligned}$		
T3-E	0	3	0	3	1	2	$\frac{1}{2}$		$\begin{aligned} & - \frac{2\sqrt{\frac{2}{3}}h\bar{y}_{1,i_1,r_1} y_{2,i_2,r_1} L_{a_4,i_2} {}^T C P_L L_{a_2,i_1} H_{a_1} H_{a_3} \lambda_{T3} \epsilon_{a_1,a_2} \epsilon_{a_3,a_4}}{\epsilon M_F} \\ & - \frac{2\sqrt{\frac{2}{3}}h\bar{y}_{1,i_1,r_1} y_{2,i_2,r_1} L_{a_4,i_2} {}^T C P_L L_{a_2,i_1} H_{a_1} H_{a_3} \lambda_{T3} \epsilon_{a_1,a_2} \epsilon_{a_3,a_4}}{M_F} \\ & - \frac{2\sqrt{\frac{2}{3}}h\lambda_{T3} L_{a_3,i_1} C P_R \overline{L_{a_4,i_2}}^T y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_3} \epsilon_{a_2,a_4} H_{a_1}^\dagger H_{a_2}^\dagger}{\epsilon M_F} \end{aligned}$	$\begin{aligned} & - \frac{2h\bar{y}_{1,i_1,r_1} \overline{Y_{e,i_2,i_3}} H_{a_1} S_{1,j_1} y_{1,i_2,r_1} Y_{e,i_1,i_3} H_{a_1}^\dagger S_{1,j_1}^\dagger}{\sqrt{3}\epsilon} \\ & - \frac{2h\bar{y}_{1,i_1,r_1} \overline{Y_{e,i_2,i_3}} H_{a_1} S_{1,j_1} y_{1,i_2,r_1} Y_{e,i_1,i_3} H_{a_1}^\dagger S_{1,j_1}^\dagger}{\sqrt{3}} \\ & - \frac{2i\sqrt{\frac{2}{3}}h\bar{y}_{1,i_1,r_1} \overline{Y_{e,i_2,i_3}} H_{a_2} S_{1,j_2} y_{1,i_2,r_1} Y_{e,i_1,i_3} H_{a_1}^\dagger S_{1,j_1}^\dagger \mathcal{I}_{H1,a_1,a_2,j_2,j_1}^\dagger}{\epsilon} \end{aligned}$	$\begin{aligned} & - \frac{4h\bar{y}_{1,i_1,r_1} y_{2,i_2,r_2} S_{1,j_2} S_{2,j_2} y_{1,i_2,r_2} y_{2,i_1,r_2} S_{1,j_1}^\dagger S_{2,j_1}^\dagger}{3\epsilon} \\ & - \frac{4}{3}h\bar{y}_{1,i_1,r_1} y_{2,i_2,r_2} S_{1,j_2} S_{2,j_2} y_{1,i_2,r_2} y_{2,i_1,r_2} S_{1,j_1}^\dagger S_{2,j_1}^\dagger \\ & - \frac{4}{3}h\bar{y}_{1,i_1,r_1} y_{2,i_2,r_2} S_{1,j_2} S_{2,j_2} y_{1,i_1,r_2} y_{2,i_2,r_2} S_{1,j_1}^\dagger S_{2,j_1}^\dagger \\ & - \frac{4h\bar{y}_{1,i_1,r_1} y_{2,i_2,r_2} S_{1,j_2} S_{2,j_2} y_{1,i_2,r_1} y_{2,i_1,r_2} S_{1,j_1}^\dagger S_{2,j_2}^\dagger}{3\epsilon} \end{aligned}$		
cont.										$\begin{aligned} & - \frac{2i\sqrt{\frac{2}{3}}h\bar{y}_{1,i_1,r_1} \overline{Y_{e,i_2,i_3}} H_{a_2} S_{1,j_2} y_{1,i_2,r_1} Y_{e,i_1,i_3} H_{a_1}^\dagger S_{1,j_1}^\dagger \mathcal{I}_{H1,a_1,a_2,j_2,j_1}^\dagger}{\sqrt{3}} \\ & - \frac{2\sqrt{\frac{2}{3}}h\lambda_{T3} L_{a_3,i_1} C P_R \overline{L_{a_4,i_2}}^T y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_3} \epsilon_{a_2,a_4} H_{a_1}^\dagger H_{a_2}^\dagger}{M_F} \end{aligned}$	$\begin{aligned} & + H_{a_2} S_{1,j_2} H_{a_1}^\dagger S_{1,j_1}^\dagger \mathcal{I}_{H1,a_1,a_2,j_2,j_1} {}^2\lambda_{H1}^{(2)} \\ & - \frac{2h\bar{y}_{1,i_1,r_1} H_{a_2} S_{1,j_2} y_{1,i_1,r_1} H_{a_1}^\dagger S_{1,j_1}^\dagger \mathcal{I}_{H1,a_1,a_2,j_2,j_1} {}^2\lambda_{H1}^{(2)}}{\sqrt{3}\epsilon} \end{aligned}$	$\begin{aligned} & - \frac{4h\bar{y}_{1,i_1,r_1} y_{2,i_2,r_2} S_{1,j_1} S_{2,j_2} y_{1,i_2,r_1} y_{2,i_1,r_2} S_{1,j_1}^\dagger S_{2,j_2}^\dagger}{3\epsilon} \\ & - \frac{4h\bar{y}_{1,i_1,r_1} y_{2,i_2,r_2} S_{1,j_1} S_{2,j_2} y_{1,i_2,r_1} y_{2,i_1,r_2} S_{1,j_1}^\dagger S_{2,j_2}^\dagger}{3\epsilon} \\ & - \frac{4}{3}h\bar{y}_{1,i_1,r_1} y_{2,i_2,r_2} S_{1,j_2} S_{2,j_2} y_{1,i_1,r_2} y_{2,i_2,r_2} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \\ & + \frac{4}{3}h\bar{y}_{1,i_1,r_1} y_{2,i_2,r_2} S_{1,j_1} S_{2,j_2} y_{1,i_2,r_2} y_{2,i_1,r_2} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \end{aligned}$	
											$\begin{aligned} & + \frac{4}{3}h\bar{y}_{1,i_1,r_1} y_{2,i_2,r_2} S_{1,j_2} S_{2,j_2} y_{1,i_1,r_2} y_{2,i_2,r_2} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \\ & + \frac{4}{3}h\bar{y}_{1,i_1,r_1} y_{2,i_2,r_2} S_{1,j_1} S_{2,j_2} y_{1,i_1,r_2} y_{2,i_2,r_2} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \\ & + \frac{4}{3}h\bar{y}_{1,i_1,r_1} y_{2,i_2,r_2} S_{1,j_1} S_{2,j_2} y_{1,i_1,r_2} y_{2,i_2,r_2} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \\ & + \frac{4}{3}h\bar{y}_{1,i_1,r_1} y_{2,i_2,r_2} S_{1,j_1} S_{2,j_2} y_{1,i_1,r_2} y_{2,i_2,r_2} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \end{aligned}$		
											$\begin{aligned} & + S_{1,j_3} S_{2,j_4} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_3,j_1,j_4,j_2} {}^1\lambda_{12}^{(1)} \\ & - \frac{2h\bar{y}_{1,i_1,r_1} S_{1,j_3} S_{2,j_4} y_{1,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_3,j_1,j_4,j_2} {}^1\lambda_{12}^{(1)}}{\sqrt{3}\epsilon} \\ & - \frac{h\bar{y}_{1,i_1,r_1} S_{1,j_3} S_{2,j_4} y_{1,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_3,j_1,j_4,j_2} {}^1\lambda_{12}^{(1)}}{\sqrt{3}} \end{aligned}$		
cont.										$\begin{aligned} & - \frac{2h\bar{y}_{2,i_1,r_1} S_{1,j_3} S_{2,j_4} y_{2,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_3,j_1,j_4,j_2} {}^1\lambda_{12}^{(1)}}{\sqrt{3}\epsilon} \\ & - \frac{h\bar{y}_{2,i_1,r_1} S_{1,j_3} S_{2,j_4} y_{2,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_3,j_1,j_4,j_2} {}^1\lambda_{12}^{(1)}}{\sqrt{3}} \end{aligned}$	$\begin{aligned} & - \frac{2h\bar{y}_{2,i_1,r_1} S_{1,j_3} S_{2,j_4} y_{2,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_3,j_1,j_4,j_2} {}^1\lambda_{12}^{(1)}}{\sqrt{3}\epsilon} \\ & - \frac{h\bar{y}_{2,i_1,r_1} S_{1,j_3} S_{2,j_4} y_{2,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_3,j_1,j_4,j_2} {}^1\lambda_{12}^{(1)}}{\sqrt{3}} \end{aligned}$	$\begin{aligned} & - \frac{2h\bar{y}_{2,i_1,r_1} S_{1,j_3} S_{2,j_4} y_{2,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_3,j_1,j_4,j_2} {}^1\lambda_{12}^{(1)}}{\sqrt{3}\epsilon} \\ & - \frac{h\bar{y}_{2,i_1,r_1} S_{1,j_3} S_{2,j_4} y_{2,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_3,j_1,j_4,j_2} {}^1\lambda_{12}^{(1)}}{\sqrt{3}} \end{aligned}$	
											$\begin{aligned} & - \frac{2h\bar{y}_{2,i_1,r_1} S_{1,j_3} S_{2,j_4} y_{2,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_3,j_1,j_4,j_2} {}^1\lambda_{12}^{(1)}}{\sqrt{3}\epsilon} \\ & - \frac{h\bar{y}_{2,i_1,r_1} S_{1,j_3} S_{2,j_4} y_{2,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_3,j_1,j_4,j_2} {}^1\lambda_{12}^{(1)}}{\sqrt{3}} \end{aligned}$	$\begin{aligned} & - \frac{2h\bar{y}_{2,i_1,r_1} S_{1,j_3} S_{2,j_4} y_{2,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_3,j_1,j_4,j_2} {}^1\lambda_{12}^{(1)}}{\sqrt{3}\epsilon} \\ & - \frac{h\bar{y}_{2,i_1,r_1} S_{1,j_3} S_{2,j_4} y_{2,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_3,j_1,j_4,j_2} {}^1\lambda_{12}^{(1)}}{\sqrt{3}} \end{aligned}$	$\begin{aligned} & - \frac{2h\bar{y}_{2,i_1,r_1} S_{1,j_3} S_{2,j_4} y_{2,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_3,j_1,j_4,j_2} {}^1\lambda_{12}^{(1)}}{\sqrt{3}\epsilon} \\ & - \frac{h\bar{y}_{2,i_1,r_1} S_{1,j_3} S_{2,j_4} y_{2,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_3,j_1,j_4,j_2} {}^1\lambda_{12}^{(1)}}{\sqrt{3}} \end{aligned}$
											$\begin{aligned} & - S_{1,j_4} S_{2,j_3} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_1,j_2,j_3,j_4} {}^2\lambda_{12}^{(2)} \\ & - \frac{2h\bar{y}_{1,i_1,r_1} S_{1,j_4} S_{2,j_3} y_{1,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_1,j_2,j_3,j_4} {}^2\lambda_{12}^{(2)}}{\sqrt{3}\epsilon} \\ & - \frac{h\bar{y}_{1,i_1,r_1} S_{1,j_4} S_{2,j_3} y_{1,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_1,j_2,j_3,j_4} {}^2\lambda_{12}^{(2)}}{\sqrt{3}} \end{aligned}$		

[illegible]

[illegible]

[illegible]

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	H	ℓ	S_1	H	S_1	S_2
													$-\frac{2h\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_1,j_3,j_4,j_2}{}^2\lambda_{12}^{(2)}}{\sqrt{3}e}$
cont.													$-\frac{h\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_1,j_3,j_4,j_2}{}^2\lambda_{12}^{(2)}}{\sqrt{3}}$ $+\frac{2h\overline{y_{2,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_1,j_3,j_4,j_2}{}^2\lambda_{12}^{(2)}}{\sqrt{3}e}$ $+\frac{h\overline{y_{2,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_1,j_3,j_4,j_2}{}^2\lambda_{12}^{(2)}}{\sqrt{3}}$
cont.													$-\frac{2h\overline{y_{2,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_1,j_3,j_4,j_2}{}^2\lambda_{12}^{(2)}}{\sqrt{3}e}$ $-\frac{h\overline{y_{2,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_1,j_3,j_4,j_2}{}^2\lambda_{12}^{(2)}}{\sqrt{3}}$ $+S_{1,j_3}S_{2,j_4}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_3,j_1,j_4,j_2}{}^3\lambda_{12}^{(3)}$
cont.													$-\frac{2h\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_3,j_1,j_4,j_2}{}^3\lambda_{12}^{(3)}}{\sqrt{3}e}$ $-\frac{h\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_3,j_1,j_4,j_2}{}^3\lambda_{12}^{(3)}}{\sqrt{3}}$ $-\frac{2h\overline{y_{2,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_3,j_1,j_4,j_2}{}^3\lambda_{12}^{(3)}}{\sqrt{3}e}$
cont.													$-\frac{h\overline{y_{2,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_3,j_1,j_4,j_2}{}^3\lambda_{12}^{(3)}}{\sqrt{3}}$

Field combinations 4 of 10

[illegible]

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	S_1	F	S_1	ℓ	S_2	H
cont.													$-\frac{h\overline{y_{2,i_1,r_1}}H_{a_2}S_{2,j_2}y_{2,i_1,r_1}H_{a_1}^\dagger S_{2,j_1}^\dagger \mathcal{I}_{H2,a_1,a_2,j_1,j_2}{}^2\lambda_{H2}^{(2)}}{\sqrt{3}}$
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	$-h\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\frac{-C_{S_1^\dagger S_1}h\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)}{\overline{y_{1,i_1,r_1}y_{1,i_2,r_2}}S_{1,j_1}S_{1,j_2}y_{1,i_1,r_1}S_{1,j_1}^\dagger}$	$-h\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\frac{-C_{S_1^\dagger S_1}h\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)}{\overline{y_{1,i_1,r_1}y_{1,i_2,r_2}}S_{1,j_1}S_{1,j_2}y_{1,i_1,r_1}S_{1,j_1}^\dagger}$	$-h\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\frac{-C_{S_1^\dagger S_1}h\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)}{\overline{y_{1,i_1,r_1}y_{1,i_2,r_2}}S_{1,j_1}S_{1,j_2}y_{1,i_1,r_1}S_{1,j_1}^\dagger}$	$-h\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\frac{-C_{S_1^\dagger S_1}h\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)}{\overline{y_{1,i_1,r_1}y_{1,i_2,r_2}}S_{1,j_1}S_{1,j_2}y_{1,i_1,r_1}S_{1,j_1}^\dagger}$	$-\frac{2h\overline{y_{2,i_1,r_1}}\overline{y_{e,i_2,i_3}}H_{a_2}S_{2,j_2}y_{2,i_2,r_1}Y_{e,i_1,i_3}\mathcal{I}_{y2,a_1,j_1}\mathcal{I}_{y2,a_2,j_2}H_{a_1}^\dagger S_{2,j_1}^\dagger}{\epsilon}$	
cont.													$-\frac{2h\overline{y_{2,i_1,r_1}}\overline{y_{e,i_2,i_3}}H_{a_2}S_{2,j_2}y_{2,i_2,r_1}Y_{e,i_1,i_3}\mathcal{I}_{y2,a_1,j_1}\mathcal{I}_{y2,a_2,j_2}H_{a_1}^\dagger S_{2,j_1}^\dagger}{\epsilon}$
cont.													$-\frac{h\overline{y_{2,i_1,r_1}}H_{a_2}S_{2,j_2}y_{2,i_1,r_1}H_{a_1}^\dagger S_{2,j_1}^\dagger \mathcal{I}_{H2,a_1,a_2,j_1,j_2}{}^1\lambda_{H2}^{(1)}}{\epsilon}$
cont.													$-\frac{1}{2}h\overline{y_{2,i_1,r_1}}H_{a_2}S_{2,j_2}y_{2,i_1,r_1}H_{a_1}^\dagger S_{2,j_1}^\dagger \mathcal{I}_{H2,a_1,a_2,j_1,j_2}{}^1\lambda_{H2}^{(1)}$
													$+\frac{H_{a_2}S_{2,j_2}H_{a_1}^\dagger S_{2,j_1}^\dagger \mathcal{I}_{H2,a_1,a_2,j_1,j_2}{}^2\lambda_{H2}^{(2)}}{\epsilon}$
cont.													$-\frac{h\overline{y_{2,i_1,r_1}}H_{a_2}S_{2,j_2}y_{2,i_1,r_1}H_{a_1}^\dagger S_{2,j_1}^\dagger \mathcal{I}_{H2,a_1,a_2,j_1,j_2}{}^2\lambda_{H2}^{(2)}}{\epsilon}$
													$-\frac{1}{2}h\overline{y_{2,i_1,r_1}}H_{a_2}S_{2,j_2}y_{2,i_1,r_1}H_{a_1}^\dagger S_{2,j_1}^\dagger \mathcal{I}_{H2,a_1,a_2,j_1,j_2}{}^2\lambda_{H2}^{(2)}$
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-h\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\frac{-C_{S_1^\dagger S_1}h\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)}{\overline{y_{1,i_1,r_1}y_{1,i_2,r_2}}S_{1,j_1}S_{1,j_2}y_{1,i_1,r_1}S_{1,j_1}^\dagger}$	$-h\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\frac{-C_{S_1^\dagger S_1}h\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)}{\overline{y_{1,i_1,r_1}y_{1,i_2,r_2}}S_{1,j_1}S_{1,j_2}y_{1,i_1,r_1}S_{1,j_1}^\dagger}$	$-h\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\frac{-C_{S_1^\dagger S_1}h\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)}{\overline{y_{1,i_1,r_1}y_{1,i_2,r_2}}S_{1,j_1}S_{1,j_2}y_{1,i_1,r_1}S_{1,j_1}^\dagger}$	$-h\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\frac{-C_{S_1^\dagger S_1}h\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)}{\overline{y_{1,i_1,r_1}y_{1,i_2,r_2}}S_{1,j_1}S_{1,j_2}y_{1,i_1,r_1}S_{1,j_1}^\dagger}$	$-\frac{2h\overline{y_{2,i_1,r_1}}\overline{y_{e,i_2,i_3}}H_{a_2}S_{2,j_2}y_{2,i_2,r_1}Y_{e,i_1,i_3}\mathcal{I}_{y2,a_1,j_1}\mathcal{I}_{y2,a_2,j_2}H_{a_1}^\dagger S_{2,j_1}^\dagger}{\epsilon}$	
cont.													$-\frac{2h\overline{y_{2,i_1,r_1}}\overline{y_{e,i_2,i_3}}H_{a_2}S_{2,j_2}y_{2,i_2,r_1}Y_{e,i_1,i_3}\mathcal{I}_{y2,a_1,j_1}\mathcal{I}_{y2,a_2,j_2}H_{a_1}^\dagger S_{2,j_1}^\dagger}{\epsilon}$
cont.													$-\frac{h\overline{y_{2,i_1,r_1}}H_{a_2}S_{2,j_2}y_{2,i_1,r_1}H_{a_1}^\dagger S_{2,j_1}^\dagger \mathcal{I}_{H2,a_1,a_2,j_1,j_2}{}^1\lambda_{H2}^{(1)}}{\epsilon}$
cont.													$-\frac{1}{2}h\overline{y_{2,i_1,r_1}}H_{a_2}S_{2,j_2}y_{2,i_1,r_1}H_{a_1}^\dagger S_{2,j_1}^\dagger \mathcal{I}_{H2,a_1,a_2,j_1,j_2}{}^1\lambda_{H2}^{(1)}$
													$+\frac{H_{a_2}S_{2,j_2}H_{a_1}^\dagger S_{2,j_1}^\dagger \mathcal{I}_{H2,a_1,a_2,j_1,j_2}{}^2\lambda_{H2}^{(2)}}{\epsilon}$
cont.													$-\frac{h\overline{y_{2,i_1,r_1}}H_{a_2}S_{2,j_2}y_{2,i_1,r_1}H_{a_1}^\dagger S_{2,j_1}^\dagger \mathcal{I}_{H2,a_1,a_2,j_1,j_2}{}^2\lambda_{H2}^{(2)}}{\epsilon}$
													$-\frac{1}{2}h\overline{y_{2,i_1,r_1}}H_{a_2}S_{2,j_2}y_{2,i_1,r_1}H_{a_1}^\dagger S_{2,j_1}^\dagger \mathcal{I}_{H2,a_1,a_2,j_1,j_2}{}^2\lambda_{H2}^{(2)}$

[illegible]

[illegible]

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	S_1	F	$S_1 \ell$	S_2	H
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	$-\sqrt{3}C_{S_1^\dagger S_1} \hbar \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} S_{1,j_1} y_{1,i_1,r_1} S_{1,j_1}^\dagger}$			$-\frac{2\hbar \overline{y_{2,i_1,r_1} Y_{e,i_2,i_3}} H_{a_1} S_{2,j_2} y_{2,i_2,r_1} Y_{e,i_1,i_3} H_{a_1}^\dagger S_{2,j_2}^\dagger}{\sqrt{3}\epsilon}$	
								$-\frac{5}{3} \hbar \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{1,i_2,r_2} S_{1,j_1} S_{1,j_2} y_{1,i_1,r_2} y_{1,i_2,r_1} S_{1,j_1}^\dagger S_{1,j_2}^\dagger}$			$-\frac{2\hbar \overline{y_{2,i_1,r_1} Y_{e,i_2,i_3}} H_{a_1} S_{2,j_2} y_{2,i_2,r_1} Y_{e,i_1,i_3} H_{a_1}^\dagger S_{2,j_2}^\dagger}{\sqrt{3}}$	
								$-\sqrt{3} \hbar \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} S_{1,j_2} S_{1,j_4} y_{1,i_1,r_1} S_{1,j_1}^\dagger S_{1,j_2}^\dagger \mathcal{I}_{S_1,j_1,j_2,j_4}} \lambda_{S_1}^{(1)}$			$-\frac{2\hbar \overline{y_{2,i_1,r_1} Y_{e,i_2,i_3}} H_{a_2} S_{2,j_2} y_{2,i_2,r_1} Y_{e,i_1,i_3} H_{a_1}^\dagger S_{2,j_1}^\dagger \mathcal{I}_{H,a_1,a_2,j_1,j_2}}{\sqrt{3}\epsilon}$	
cont.								$-\frac{2\hbar \overline{y_{2,i_1,r_1} Y_{e,i_2,i_3}} H_{a_2} S_{2,j_2} y_{2,i_2,r_1} Y_{e,i_1,i_3} H_{a_1}^\dagger S_{2,j_1}^\dagger \mathcal{I}_{H,a_1,a_2,j_1,j_2}}{\sqrt{3}\epsilon}$			$-\frac{2\hbar \overline{y_{2,i_1,r_1} Y_{e,i_2,i_3}} H_{a_2} S_{2,j_2} y_{2,i_2,r_1} Y_{e,i_1,i_3} H_{a_1}^\dagger S_{2,j_1}^\dagger \mathcal{I}_{H,a_1,a_2,j_1,j_2}}{\sqrt{3}\epsilon}$	
cont.								$-\frac{2\hbar \overline{y_{2,i_1,r_1} Y_{e,i_2,i_3}} H_{a_2} S_{2,j_2} y_{2,i_2,r_1} Y_{e,i_1,i_3} H_{a_1}^\dagger S_{2,j_1}^\dagger \mathcal{I}_{H,a_1,a_2,j_1,j_2}}{\sqrt{3}\epsilon}$			$-\frac{2\hbar \overline{y_{2,i_1,r_1} Y_{e,i_2,i_3}} H_{a_2} S_{2,j_2} y_{2,i_2,r_1} Y_{e,i_1,i_3} H_{a_1}^\dagger S_{2,j_1}^\dagger \mathcal{I}_{H,a_1,a_2,j_1,j_2}}{\sqrt{3}\epsilon}$	
cont.								$-\frac{2\hbar \overline{y_{2,i_1,r_1} Y_{e,i_2,i_3}} H_{a_2} S_{2,j_2} y_{2,i_2,r_1} Y_{e,i_1,i_3} H_{a_1}^\dagger S_{2,j_1}^\dagger \mathcal{I}_{H,a_1,a_2,j_1,j_2}}{\sqrt{3}\epsilon}$			$-\frac{2\hbar \overline{y_{2,i_1,r_1} Y_{e,i_2,i_3}} H_{a_2} S_{2,j_2} y_{2,i_2,r_1} Y_{e,i_1,i_3} H_{a_1}^\dagger S_{2,j_1}^\dagger \mathcal{I}_{H,a_1,a_2,j_1,j_2}}{\sqrt{3}\epsilon}$	
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$+\frac{2}{3} \hbar \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{1,i_2,r_2} S_{1,j_2}^2 y_{1,i_1,r_2} y_{1,i_2,r_1} (S_{1,j_1}^\dagger)^2}$			$-\frac{2\hbar \overline{y_{2,i_1,r_1} Y_{e,i_2,i_3}} H_{a_1} S_{2,y_2,i_2,r_1} Y_{e,i_1,i_3} H_{a_1}^\dagger S_2^\dagger}{\epsilon}$	
								$-\frac{2C_{S_1^\dagger S_1} \hbar \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} S_{1,j_1} y_{1,i_1,r_1} S_{1,j_1}^\dagger}}{\sqrt{3}}$			$-2\hbar \overline{y_{2,i_1,r_1} Y_{e,i_2,i_3}} H_{a_1} S_{2,y_2,i_2,r_1} Y_{e,i_1,i_3} H_{a_1}^\dagger S_2^\dagger$	
								$-\frac{4}{3} \hbar \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{1,i_2,r_2} S_{1,j_1} S_{1,j_2} y_{1,i_1,r_2} y_{1,i_2,r_1} S_{1,j_1}^\dagger S_{1,j_2}^\dagger}$			$+H_{a_2} S_2 H_{a_1}^\dagger S_2^\dagger \mathcal{I}_{H2,a_1,a_2}} \lambda_{H2}^{(1)}$	
cont.								$-\frac{2\hbar \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} S_{1,j_2} S_{1,j_4} y_{1,i_1,r_1} S_{1,j_1}^\dagger S_{1,j_2}^\dagger \mathcal{I}_{S_1,j_2,j_3,j_4,j_2}} \lambda_{S_1}^{(1)}}{\sqrt{3}}$			$-\frac{2\hbar \overline{y_{2,i_1,r_1} H_{a_2} S_{2,y_2,i_1,r_1} H_{a_1}^\dagger S_2^\dagger \mathcal{I}_{H2,a_1,a_2}} \lambda_{H2}^{(1)}}{\epsilon}$	
cont.								$-\frac{2\hbar \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} S_{1,j_2} S_{1,j_4} y_{1,i_1,r_1} S_{1,j_1}^\dagger S_{1,j_2}^\dagger \mathcal{I}_{S_1,j_2,j_3,j_4,j_2}} \lambda_{S_1}^{(2)}}{\sqrt{3}}$			$-\hbar \overline{y_{2,i_1,r_1} H_{a_2} S_{2,y_2,i_1,r_1} H_{a_1}^\dagger S_2^\dagger \mathcal{I}_{H2,a_1,a_2}} \lambda_{H2}^{(1)}$	
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	$+\frac{2}{3} \hbar \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{1,i_2,r_2} S_{1,j_2}^2 y_{1,i_1,r_2} y_{1,i_2,r_1} (S_{1,j_1}^\dagger)^2}$			$-\frac{2\hbar \overline{y_{2,i_1,r_1} Y_{e,i_2,i_3}} H_{a_1} S_{2,y_2,i_2,r_1} Y_{e,i_1,i_3} H_{a_1}^\dagger S_2^\dagger}{\epsilon}$	
								$-\frac{2C_{S_1^\dagger S_1} \hbar \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} S_{1,j_1} y_{1,i_1,r_1} S_{1,j_1}^\dagger}}{\sqrt{3}}$			$-2\hbar \overline{y_{2,i_1,r_1} Y_{e,i_2,i_3}} H_{a_1} S_{2,y_2,i_2,r_1} Y_{e,i_1,i_3} H_{a_1}^\dagger S_2^\dagger$	
								$-\frac{4}{3} \hbar \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{1,i_2,r_2} S_{1,j_1} S_{1,j_2} y_{1,i_1,r_2} y_{1,i_2,r_1} S_{1,j_1}^\dagger S_{1,j_2}^\dagger}$			$+H_{a_2} S_2 H_{a_1}^\dagger S_2^\dagger \mathcal{I}_{H2,a_1,a_2}} \lambda_{H2}^{(1)}$	
cont.								$-\frac{2\hbar \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} S_{1,j_2} S_{1,j_4} y_{1,i_1,r_1} S_{1,j_1}^\dagger S_{1,j_2}^\dagger \mathcal{I}_{S_1,j_2,j_3,j_4,j_2}} \lambda_{S_1}^{(1)}}{\sqrt{3}}$			$-\frac{2\hbar \overline{y_{2,i_1,r_1} H_{a_2} S_{2,y_2,i_1,r_1} H_{a_1}^\dagger S_2^\dagger \mathcal{I}_{H2,a_1,a_2}} \lambda_{H2}^{(1)}}{\epsilon}$	
cont.								$-\frac{2\hbar \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} S_{1,j_2} S_{1,j_4} y_{1,i_1,r_1} S_{1,j_1}^\dagger S_{1,j_2}^\dagger \mathcal{I}_{S_1,j_2,j_3,j_4,j_2}} \lambda_{S_1}^{(2)}}{\sqrt{3}}$			$-\hbar \overline{y_{2,i_1,r_1} H_{a_2} S_{2,y_2,i_1,r_1} H_{a_1}^\dagger S_2^\dagger \mathcal{I}_{H2,a_1,a_2}} \lambda_{H2}^{(1)}$	
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$+\frac{2}{3} \hbar \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{1,i_2,r_2} S_{1,j_2}^2 y_{1,i_1,r_2} y_{1,i_2,r_1} (S_{1,j_1}^\dagger)^2}$			$-\frac{2\hbar \overline{y_{2,i_1,r_1} Y_{e,i_2,i_3}} H_{a_1} S_{2,y_2,i_2,r_1} Y_{e,i_1,i_3} H_{a_1}^\dagger S_2^\dagger}{\epsilon}$	
								$-\frac{2C_{S_1^\dagger S_1} \hbar \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} S_{1,j_1} y_{1,i_1,r_1} S_{1,j_1}^\dagger}}{\sqrt{3}}$			$-2\hbar \overline{y_{2,i_1,r_1} Y_{e,i_2,i_3}} H_{a_1} S_{2,y_2,i_2,r_1} Y_{e,i_1,i_3} H_{a_1}^\dagger S_2^\dagger$	
								$-\frac{4}{3} \hbar \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{1,i_2,r_2} S_{1,j_1} S_{1,j_2} y_{1,i_1,r_2} y_{1,i_2,r_1} S_{1,j_1}^\dagger S_{1,j_2}^\dagger}$			$+H_{a_2} S_2 H_{a_1}^\dagger S_2^\dagger \mathcal{I}_{H2,a_1,a_2}} \lambda_{H2}^{(1)}$	
cont.								$-\frac{2\hbar \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} S_{1,j_2} S_{1,j_4} y_{1,i_1,r_1} S_{1,j_1}^\dagger S_{1,j_2}^\dagger \mathcal{I}_{S_1,j_2,j_3,j_4,j_2}} \lambda_{S_1}^{(1)}}{\sqrt{3}}$			$-\frac{2\hbar \overline{y_{2,i_1,r_1} H_{a_2} S_{2,y_2,i_1,r_1} H_{a_1}^\dagger S_2^\dagger \mathcal{I}_{H2,a_1,a_2}} \lambda_{H2}^{(1)}}{\epsilon}$	
cont.								$-\frac{2\hbar \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} S_{1,j_2} S_{1,j_4} y_{1,i_1,r_1} S_{1,j_1}^\dagger S_{1,j_2}^\dagger \mathcal{I}_{S_1,j_2,j_3,j_4,j_2}} \lambda_{S_1}^{(2)}}{\sqrt{3}}$			$-\hbar \overline{y_{2,i_1,r_1} H_{a_2} S_{2,y_2,i_1,r_1} H_{a_1}^\dagger S_2^\dagger \mathcal{I}_{H2,a_1,a_2}} \lambda_{H2}^{(1)}$	
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	$+\frac{2}{3} \hbar \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{1,i_2,r_2} S_{1,j_2}^2 y_{1,i_1,r_2} y_{1,i_2,r_1} (S_{1,j_1}^\dagger)^2}$			$-\frac{2\hbar \overline{y_{2,i_1,r_1} Y_{e,i_2,i_3}} H_{a_1} S_{2,y_2,i_2,r_1} Y_{e,i_1,i_3} H_{a_1}^\dagger S_2^\dagger}{\epsilon}$	
								$-\frac{2C_{S_1^\dagger S_1} \hbar \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} S_{1,j_1} y_{1,i_1,r_1} S_{1,j_1}^\dagger}}{\sqrt{3}}$			$-2\hbar \overline{y_{2,i_1,r_1} Y_{e,i_2,i_3}} H_{a_1} S_{2,y_2,i_2,r_1} Y_{e,i_1,i_3} H_{a_1}^\dagger S_2^\dagger$	
								$-\frac{4}{3} \hbar \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{1,i_2,r_2} S_{1,j_1} S_{1,j_2} y_{1,i_1,r_2} y_{1,i_2,r_1} S_{1,j_1}^\dagger S_{1,j_2}^\dagger}$			$+H_{a_2} S_2 H_{a_1}^\dagger S_2^\dagger \mathcal{I}_{H2,a_1,a_2}} \lambda_{H2}^{(1)}$	
cont.								$-\frac{2\hbar \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} S_{1,j_2} S_{1,j_4} y_{1,i_1,r_1} S_{1,j_1}^\dagger S_{1,j_2}^\dagger \mathcal{I}_{S_1,j_2,j_3,j_4,j_2}} \lambda_{S_1}^{(1)}}{\sqrt{3}}$			$-\frac{2\hbar \overline{y_{2,i_1,r_1} H_{a_2} S_{2,y_2,i_1,r_1} H_{a_1}^\dagger S_2^\dagger \mathcal{I}_{H2,a_1,a_2}} \lambda_{H2}^{(1)}}{\epsilon}$	
cont.								$-\frac{2\hbar \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} S_{1,j_2} S_{1,j_4} y_{1,i_1,r_1} S_{1,j_1}^\dagger S_{1,j_2}^\dagger \mathcal{I}_{S_1,j_2,j_3,j_4,j_2}} \lambda_{S_1}^{(2)}}{\sqrt{3}}$			$-\hbar \overline{y_{2,i_1,r_1} H_{a_2} S_{2,y_2,i_1,r_1} H_{a_1}^\dagger S_2^\dagger \mathcal{I}_{H2,a_1,a_2}} \lambda_{H2}^{(1)}$	
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	$+\frac{2}{3} \hbar \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{1,i_2,r_2} S_{1,j_2}^2 y_{1,i_1,r_2} y_{1,i_2,r_1} (S_{1,j_1}^\dagger)^2}$			$-\frac{2\hbar \overline{y_{2,i_1,r_1} Y_{e,i_2,i_3}} H_{a_1} S_{2,y_2,i_2,r_1} Y_{e,i_1,i_3} H_{a_1}^\dagger S_2^\dagger}{\epsilon}$	
								$-\frac{2C_{S_1^\dagger S_1} \hbar \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} S_{1,j_1} y_{1,i_1,r_1} S_{1,j_1}^\dagger}}{\sqrt{3}}$			$-2\hbar \overline{y_{2,i_1,r_1} Y_{e,i_2,i_3}} H_{a_1} S_{2,y_2,i_2,r_1} Y_{e,i_1,i_3} H_{a_1}^\dagger S_2^\dagger$	
								$-\frac{4}{3} \hbar \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{1,i_2,r_2} S_{1,j_1} S_{1,j_2} y_{1,i_1,r_2} y_{1,i_2,r_1} S_{1,j_1}^\dagger S_{1,j_2}^\dagger}$			$+H_{a_2} S_2 H_{a_1}^\dagger S_2^\dagger \mathcal{I}_{H2,a_1,a_2}} \lambda_{H2}^{(1)}$	

[illegible]

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	S_1	F	S_1	ℓ	S_2	H
$cont.$													$-\frac{\overline{hy_{2,i_1,r_1}}H_{a_2}S_{2,j_2}y_{2,i_1,r_1}H_{a_1}{}^\dagger S_{2,j_1}{}^\dagger T_{H^2,a_1,a_2,j_2,j_1}}{\sqrt{3}}{}^2\lambda_{H^2}^{(2)}$

Field combinations 5 of 10

[illegible]

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$S_2 F$	$S_2 \ell$	$F B_{\mu\nu}$
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$\begin{aligned} & -C_{S_2^\dagger S_2} \hbar \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1} S_{2,j_1} y_{2,i_1,r_1} S_{2,j_1}^\dagger} \\ & -\hbar \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1} y_{2,i_2,r_2} S_{2,j_1} S_{2,j_2} y_{2,i_1,r_2} y_{2,i_2,r_1} S_{2,j_1}^\dagger S_{2,j_2}^\dagger} \\ & -\hbar \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1} S_{2,j_3} S_{2,j_4} y_{2,i_1,r_1} S_{2,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{S_2,j_1,j_3,j_2,j_4}} \lambda_{S_2}^{(1)} \end{aligned}$	$\begin{aligned} & + \frac{3\hbar \overline{y_{2,i_1,r_1} y_{2,i_2,r_2} L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_{2,j_1} S_{2,j_2} \mathcal{I}_{y_{2,a_1,j_1}} \mathcal{I}_{y_{2,a_2,j_2}} g_2^2}{8c M_F} \\ & + \frac{\hbar \overline{y_{2,i_1,r_1} y_{2,i_2,r_2} L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_{2,j_1} S_{2,j_2} \mathcal{I}_{y_{2,a_1,j_1}} \mathcal{I}_{y_{2,a_2,j_2}} g_2^2}{8M_F} \\ & + \frac{3\hbar \overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T y_{2,i_1,r_1} y_{2,i_2,r_1} \mathcal{I}_{y_{2,a_1,j_1}} \mathcal{I}_{y_{2,a_2,j_2} S_{2,j_1}^\dagger S_{2,j_2}^\dagger} g_2^2}{8c M_F} \\ & + \frac{\hbar \overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T y_{2,i_1,r_1} y_{2,i_2,r_1} \mathcal{I}_{y_{2,a_1,j_1}} \mathcal{I}_{y_{2,a_2,j_2} S_{2,j_1}^\dagger S_{2,j_2}^\dagger} g_2^2}{8M_F} \\ & + \frac{3\hbar \overline{y_{2,i_1,r_1} y_{2,i_2,r_2} L_{a_2,i_2}}^T C P_L L_{a_1,i_1} g_Y^2 S_{2,j_1} S_{2,j_2} \mathcal{I}_{y_{2,a_1,j_1}} \mathcal{I}_{y_{2,a_2,j_2}}}{8c M_F} \\ & + \frac{\hbar \overline{y_{2,i_1,r_1} y_{2,i_2,r_2} L_{a_2,i_2}}^T C P_L L_{a_1,i_1} g_Y^2 S_{2,j_1} S_{2,j_2} \mathcal{I}_{y_{2,a_1,j_1}} \mathcal{I}_{y_{2,a_2,j_2}}}{8M_F} \\ & + \frac{\overline{y_{2,i_1,r_1} y_{2,i_2,r_2} L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_{2,j_1} S_{2,j_2} \mathcal{I}_{y_{2,a_1,j_1}} \mathcal{I}_{y_{2,a_2,j_2}}}{2M_F} \\ & - \frac{\hbar \overline{y_{1,i_1,r_1} y_{2,i_2,r_2} y_{2,i_3,r_2} L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_{2,j_1} S_{2,j_2} y_{1,i_3,r_1} \mathcal{I}_{y_{2,a_1,j_1}} \mathcal{I}_{y_{2,a_2,j_2}}}{4c M_F} \\ & - \frac{3\hbar \overline{y_{1,i_1,r_1} y_{2,i_2,r_2} y_{2,i_3,r_2} L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_{2,j_1} S_{2,j_2} y_{1,i_3,r_1} \mathcal{I}_{y_{2,a_1,j_1}} \mathcal{I}_{y_{2,a_2,j_2}}}{8M_F} \\ & - \frac{\hbar \overline{y_{2,i_1,r_1} y_{2,i_2,r_2} y_{2,i_3,r_1} L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_{2,j_1} S_{2,j_2} y_{2,i_3,r_2} \mathcal{I}_{y_{2,a_1,j_1}} \mathcal{I}_{y_{2,a_2,j_2}}}{4c M_F} \\ & - \frac{3\hbar \overline{y_{2,i_1,r_1} y_{2,i_2,r_2} y_{2,i_3,r_1} L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_{2,j_1} S_{2,j_2} y_{2,i_3,r_2} \mathcal{I}_{y_{2,a_1,j_1}} \mathcal{I}_{y_{2,a_2,j_2}}}{8M_F} \\ & - \frac{\hbar \overline{y_{2,i_1,r_1} y_{2,i_2,r_1} y_{2,i_3,r_2} L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_{2,j_1} S_{2,j_2} y_{2,i_3,r_2} \mathcal{I}_{y_{2,a_1,j_1}} \mathcal{I}_{y_{2,a_2,j_2}}}{2c M_F} \\ & - \frac{\hbar \overline{y_{2,i_1,r_1} y_{2,i_2,r_1} y_{2,i_3,r_2} L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_{2,j_1} S_{2,j_2} y_{2,i_3,r_2} \mathcal{I}_{y_{2,a_1,j_1}} \mathcal{I}_{y_{2,a_2,j_2}}}{4M_F} \\ & + \frac{3\hbar \overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T g_Y^2 y_{2,i_1,r_1} y_{2,i_3,r_1} \mathcal{I}_{y_{2,a_1,j_1}} \mathcal{I}_{y_{2,a_2,j_2} S_{2,j_1}^\dagger S_{2,j_2}^\dagger}}{8c M_F} \\ & + \frac{\hbar \overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T g_Y^2 y_{2,i_1,r_1} y_{2,i_3,r_1} \mathcal{I}_{y_{2,a_1,j_1}} \mathcal{I}_{y_{2,a_2,j_2} S_{2,j_1}^\dagger S_{2,j_2}^\dagger}}{8M_F} \\ & + \frac{\overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T y_{2,i_1,r_1} y_{2,i_3,r_1} \mathcal{I}_{y_{2,a_1,j_1}} \mathcal{I}_{y_{2,a_2,j_2} S_{2,j_1}^\dagger S_{2,j_2}^\dagger}}{2M_F} \end{aligned}$	
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T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$\begin{aligned} & -C_{S_2^\dagger S_2} \hbar \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1} S_{2,j_1} y_{2,i_1,r_1} S_{2,j_1}^\dagger} \\ & -\hbar \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1} y_{2,i_2,r_2} S_{2,j_1} S_{2,j_2} y_{2,i_1,r_2} y_{2,i_2,r_1} S_{2,j_1}^\dagger S_{2,j_2}^\dagger} \\ & -\hbar \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1} S_{2,j_3} S_{2,j_4} y_{2,i_1,r_1} S_{2,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{S_2,j_1,j_3,j_2,j_4}} \lambda_{S_2}^{(1)} \end{aligned}$	$-\frac{1}{4} \hbar \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) B_{\mu_1 \mu_2}^2$	
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	$\begin{aligned} & -C_{S_2^\dagger S_2} \hbar \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1} S_{2,j_1} y_{2,i_1,r_1} S_{2,j_1}^\dagger} \\ & -\hbar \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1} y_{2,i_2,r_2} S_{2,j_1} S_{2,j_2} y_{2,i_1,r_2} y_{2,i_2,r_1} S_{2,j_1}^\dagger S_{2,j_2}^\dagger} \\ & -\hbar \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1} S_{2,j_3} S_{2,j_4} y_{2,i_1,r_1} S_{2,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{S_2,j_1,j_3,j_2,j_4}} \lambda_{S_2}^{(1)} \end{aligned}$	$-\hbar \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) B_{\mu_1 \mu_2}^2$	
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	$\begin{aligned} & -C_{S_2^\dagger S_2} \hbar \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1} S_{2,j_1} y_{2,i_1,r_1} S_{2,j_1}^\dagger} \\ & -\hbar \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1} y_{2,i_2,r_2} S_{2,j_1} S_{2,j_2} y_{2,i_1,r_2} y_{2,i_2,r_1} S_{2,j_1}^\dagger S_{2,j_2}^\dagger} \\ & -\hbar \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1} S_{2,j_3} S_{2,j_4} y_{2,i_1,r_1} S_{2,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{S_2,j_1,j_3,j_2,j_4}} \lambda_{S_2}^{(1)} \end{aligned}$	$-\frac{9}{4} \hbar \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) B_{\mu_1 \mu_2}^2$	
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	$\begin{aligned} & -\sqrt{3} C_{S_2^\dagger S_2} \hbar \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1} S_{2,j_1} y_{2,i_1,r_1} S_{2,j_1}^\dagger} \\ & -\frac{5}{3} \hbar \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1} y_{2,i_2,r_2} S_{2,j_1} S_{2,j_2} y_{2,i_1,r_2} y_{2,i_2,r_1} S_{2,j_1}^\dagger S_{2,j_2}^\dagger} \\ & -\sqrt{3} \hbar \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1} S_{2,j_3} S_{2,j_4} y_{2,i_1,r_1} S_{2,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{S_2,j_1,j_3,j_2,j_4}} \lambda_{S_2}^{(1)} \end{aligned}$	$-\frac{3}{4} \hbar \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) B_{\mu_1 \mu_2}^2$	

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$S_2 F$	$S_2 \ell$	$F B_{\mu\nu}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$\begin{aligned} & -\sqrt{3}C_{S_1^\dagger S_2} h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1}} S_{2,j_1} y_{2,i_1,r_1} S_{2,j_1}^\dagger \\ & -\frac{5}{3} h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1} y_{2,i_2,r_2}} S_{2,j_1} S_{2,j_2} y_{2,i_1,r_2} y_{2,i_2,r_1} S_{2,j_1}^\dagger S_{2,j_2}^\dagger \\ & -\sqrt{3} h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1}} S_{2,j_3} S_{2,j_4} y_{2,i_1,r_1} S_{2,j_1}^\dagger S_{2,j_2}^\dagger I_{S_2,j_1,j_3,j_2,j_4}^{(1)} \lambda_{S_2}^{(1)} \end{aligned}$	$\begin{aligned} & + \frac{3h\overline{y_{2,i_1,r_1} y_{2,i_2,r_1}} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_1} S_{2,j_2} \text{Bar}(cg1)_{a_1,j_1,a_2,j_2} g_2^2}{8M_F} \\ & + \frac{h\overline{y_{2,i_1,r_1} y_{2,i_2,r_1}} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_1} S_{2,j_2} \text{Bar}(cg1)_{a_1,j_1,a_2,j_2} g_2^2}{8M_F} \\ & + \frac{3h\overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}{}^T cg1_{a_1,j_1,a_2,j_2} y_{2,i_1,r_1} y_{2,i_2,r_1} S_{2,j_1}^\dagger S_{2,j_2}^\dagger g_2^2}{8M_F} \\ & + \frac{h\overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}{}^T cg1_{a_1,j_1,a_2,j_2} y_{2,i_1,r_1} y_{2,i_2,r_1} S_{2,j_1}^\dagger S_{2,j_2}^\dagger g_2^2}{8M_F} \\ & + \frac{3h\overline{y_{2,i_1,r_1} y_{2,i_2,r_1}} L_{a_2,i_2}{}^T C P_L L_{a_3,i_3} g_1^2 S_{2,j_1} S_{2,j_2} \text{Bar}(cg1)_{a_1,j_1,a_2,j_2}}{8M_F} \\ & + \frac{h\overline{y_{2,i_1,r_1} y_{2,i_2,r_1}} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} g_2^2 S_{2,j_1} S_{2,j_2} \text{Bar}(cg1)_{a_1,j_1,a_2,j_2}}{8M_F} \\ & + \frac{\overline{y_{2,i_1,r_1} y_{2,i_2,r_1}} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_1} S_{2,j_2} \text{Bar}(cg1)_{a_1,j_1,a_2,j_2}}{2M_F} \\ & - \frac{\sqrt{3}h\overline{y_{1,i_1,r_1} y_{2,i_2,r_2} y_{2,i_3,r_2}} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_1} S_{2,j_2} y_{1,i_3,r_1} \text{Bar}(cg1)_{a_1,j_1,a_2,j_2}}{4M_F} \\ & - \frac{3\sqrt{3}h\overline{y_{1,i_1,r_1} y_{2,i_2,r_2} y_{2,i_3,r_2}} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_1} S_{2,j_2} y_{1,i_3,r_1} \text{Bar}(cg1)_{a_1,j_1,a_2,j_2}}{8M_F} \\ & - \frac{\sqrt{3}h\overline{y_{2,i_1,r_1} y_{2,i_2,r_2} y_{2,i_3,r_1}} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_1} S_{2,j_2} y_{2,i_3,r_2} \text{Bar}(cg1)_{a_1,j_1,a_2,j_2}}{4M_F} \\ & - \frac{3\sqrt{3}h\overline{y_{2,i_1,r_1} y_{2,i_2,r_2} y_{2,i_3,r_1}} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_1} S_{2,j_2} y_{2,i_3,r_2} \text{Bar}(cg1)_{a_1,j_1,a_2,j_2}}{8M_F} \\ & - \frac{\sqrt{3}h\overline{y_{2,i_1,r_1} y_{2,i_2,r_2} y_{2,i_3,r_2}} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_1} S_{2,j_2} y_{2,i_3,r_2} \text{Bar}(cg1)_{a_1,j_1,a_2,j_2}}{2M_F} \\ & - \frac{\sqrt{3}h\overline{y_{2,i_1,r_1} y_{2,i_2,r_2} y_{2,i_3,r_2}} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_1} S_{2,j_2} y_{2,i_3,r_2} \text{Bar}(cg1)_{a_1,j_1,a_2,j_2}}{4M_F} \\ & + \frac{3h\overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}{}^T cg1_{a_1,j_1,a_2,j_2} g_1^2 y_{2,i_1,r_1} y_{2,i_2,r_1} S_{2,j_1}^\dagger S_{2,j_2}^\dagger}{8M_F} \\ & + \frac{h\overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}{}^T cg1_{a_1,j_1,a_2,j_2} g_1^2 y_{2,i_1,r_1} y_{2,i_2,r_1} S_{2,j_1}^\dagger S_{2,j_2}^\dagger}{8M_F} \\ & + \frac{\overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}{}^T cg1_{a_1,j_1,a_2,j_2} y_{2,i_1,r_1} y_{2,i_2,r_1} S_{2,j_1}^\dagger S_{2,j_2}^\dagger}{2M_F} \end{aligned}$	
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T3-C	0	2	0	2	1	3	$\frac{1}{2}$	$\begin{aligned} & -\sqrt{3}C_{S_1^\dagger S_2} h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1}} S_{2,j_1} y_{2,i_1,r_1} S_{2,j_1}^\dagger \\ & -\frac{5}{3} h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1} y_{2,i_2,r_2}} S_{2,j_1} S_{2,j_2} y_{2,i_1,r_2} y_{2,i_2,r_1} S_{2,j_1}^\dagger S_{2,j_2}^\dagger \\ & -\sqrt{3} h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1}} S_{2,j_3} S_{2,j_4} y_{2,i_1,r_1} S_{2,j_1}^\dagger S_{2,j_2}^\dagger I_{S_2,j_1,j_3,j_2,j_4}^{(1)} \lambda_{S_2}^{(1)} \end{aligned}$	$\begin{aligned} & -\frac{3}{4} \hbar \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) B_{\mu_1 \mu_2}^2 \end{aligned}$	
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	$\begin{aligned} & -\sqrt{3}C_{S_1^\dagger S_2} h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1}} S_{2,j_1} y_{2,i_1,r_1} S_{2,j_1}^\dagger \\ & -\frac{5}{3} h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1} y_{2,i_2,r_2}} S_{2,j_1} S_{2,j_2} y_{2,i_1,r_2} y_{2,i_2,r_1} S_{2,j_1}^\dagger S_{2,j_2}^\dagger \\ & -\sqrt{3} h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1}} S_{2,j_3} S_{2,j_4} y_{2,i_1,r_1} S_{2,j_1}^\dagger S_{2,j_2}^\dagger I_{S_2,j_1,j_3,j_2,j_4}^{(1)} \lambda_{S_2}^{(1)} \end{aligned}$	$\begin{aligned} & -3h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) B_{\mu_1 \mu_2}^2 \end{aligned}$	
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	$\begin{aligned} & -\sqrt{3}C_{S_1^\dagger S_2} h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1}} S_{2,j_1} y_{2,i_1,r_1} S_{2,j_1}^\dagger \\ & -\frac{5}{3} h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1} y_{2,i_2,r_2}} S_{2,j_1} S_{2,j_2} y_{2,i_1,r_2} y_{2,i_2,r_1} S_{2,j_1}^\dagger S_{2,j_2}^\dagger \\ & -\sqrt{3} h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1}} S_{2,j_3} S_{2,j_4} y_{2,i_1,r_1} S_{2,j_1}^\dagger S_{2,j_2}^\dagger I_{S_2,j_1,j_3,j_2,j_4}^{(1)} \lambda_{S_2}^{(1)} \end{aligned}$	$\begin{aligned} & -\frac{27}{4} \hbar \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) B_{\mu_1 \mu_2}^2 \end{aligned}$	
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$\begin{aligned} & -2\hbar \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1} y_{2,i_2,r_2}} S_{2,j_2}^2 y_{2,i_1,r_2} y_{2,i_2,r_1} \left(S_{2,j_1}^\dagger \right)^2 \\ & -2\hbar \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1}} S_{2,j_2}^2 y_{2,i_1,r_1} \left(S_{2,j_1}^\dagger \right)^2 \lambda_{S_2}^{(1)} \end{aligned}$	$\begin{aligned} & -\frac{1}{2} \hbar \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) B_{\mu_1 \mu_2}^2 \end{aligned}$	

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	S_2	F	S_2	ℓ	F	$B_{\mu\nu}$
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	$-2\hbar \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \frac{\overline{y_{2,i_1,r_1} y_{2,i_2,r_2}}}{S_2^2 y_{2,i_1,r_2} y_{2,i_2,r_1}} \left(S_2^\dagger \right)^2$					
								$-2C_{S_2^\dagger S_2} \hbar \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \frac{\overline{y_{2,i_1,r_1}}}{S_2 y_{2,i_1,r_1}} S_2^\dagger$					
								$-2\hbar \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \frac{\overline{y_{2,i_1,r_1}}}{S_2^2 y_{2,i_1,r_1}} \left(S_2^\dagger \right)^2 \lambda_{S_2}^{(1)}$					
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$-2\hbar \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \frac{\overline{y_{2,i_1,r_1} y_{2,i_2,r_2}}}{S_2^2 y_{2,i_1,r_2} y_{2,i_2,r_1}} \left(S_2^\dagger \right)^2$			$-\frac{1}{2} \hbar \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) B_{\mu_1 \mu_2}^2$		
								$-2C_{S_2^\dagger S_2} \hbar \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \frac{\overline{y_{2,i_1,r_1}}}{S_2 y_{2,i_1,r_1}} S_2^\dagger$					
								$-2\hbar \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \frac{\overline{y_{2,i_1,r_1}}}{S_2^2 y_{2,i_1,r_1}} \left(S_2^\dagger \right)^2 \lambda_{S_2}^{(1)}$					
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	$-2\hbar \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \frac{\overline{y_{2,i_1,r_1} y_{2,i_2,r_2}}}{S_2^2 y_{2,i_1,r_2} y_{2,i_2,r_1}} \left(S_2^\dagger \right)^2$			$-2\hbar \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) B_{\mu_1 \mu_2}^2$		
								$-2C_{S_2^\dagger S_2} \hbar \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \frac{\overline{y_{2,i_1,r_1}}}{S_2 y_{2,i_1,r_1}} S_2^\dagger$					
								$-2\hbar \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \frac{\overline{y_{2,i_1,r_1}}}{S_2^2 y_{2,i_1,r_1}} \left(S_2^\dagger \right)^2 \lambda_{S_2}^{(1)}$					
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	$-2\hbar \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \frac{\overline{y_{2,i_1,r_1} y_{2,i_2,r_2}}}{S_2^2 y_{2,i_1,r_2} y_{2,i_2,r_1}} \left(S_2^\dagger \right)^2$			$-\frac{9}{2} \hbar \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) B_{\mu_1 \mu_2}^2$		
								$-2C_{S_2^\dagger S_2} \hbar \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \frac{\overline{y_{2,i_1,r_1}}}{S_2 y_{2,i_1,r_1}} S_2^\dagger$					
								$-2\hbar \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \frac{\overline{y_{2,i_1,r_1}}}{S_2^2 y_{2,i_1,r_1}} \left(S_2^\dagger \right)^2 \lambda_{S_2}^{(1)}$					
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	$+\frac{2}{3} \hbar \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \frac{\overline{y_{2,i_1,r_1} y_{2,i_2,r_2}}}{S_2^2 y_{2,i_1,r_2} y_{2,i_2,r_1}} \left(S_{2,j_1}^\dagger \right)^2$			$-\frac{1}{2} \hbar \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) B_{\mu_1 \mu_2}^2$		
								$-\frac{2C_{S_2^\dagger S_2} \hbar \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1}} S_{2,j_1} y_{2,i_1,r_1} S_{2,j_1}^\dagger}{\sqrt{3}}$					
								$-\frac{\frac{4}{3} \hbar \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1} y_{2,i_2,r_2}} S_{2,j_1} S_{2,j_2} y_{2,i_1,r_2} y_{2,i_2,r_1} S_{2,j_1}^\dagger S_{2,j_2}^\dagger + 2\hbar \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1}} S_{2,j_3} S_{2,j_4} y_{2,i_1,r_1} S_{2,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{S_2,j_3,j_1,j_4,j_2} \lambda_{S_2}^{(1)}}{\sqrt{3}}$					
cont.								$2\hbar \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \frac{\overline{y_{2,i_1,r_1}}}{S_2 y_{2,i_1,r_1}} S_{2,j_3} S_{2,j_4} y_{2,i_1,r_1} S_{2,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{S_2,j_3,j_1,j_4,j_2} \lambda_{S_2}^{(2)}$					
								$-\frac{\sqrt{3}}{\sqrt{3}}$					
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$+\frac{2}{3} \hbar \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \frac{\overline{y_{2,i_1,r_1} y_{2,i_2,r_2}}}{S_2^2 y_{2,i_1,r_2} y_{2,i_2,r_1}} \left(S_{2,j_1}^\dagger \right)^2$					
								$-\frac{2C_{S_2^\dagger S_2} \hbar \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1}} S_{2,j_1} y_{2,i_1,r_1} S_{2,j_1}^\dagger}{\sqrt{3}}$					
								$-\frac{\frac{4}{3} \hbar \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1} y_{2,i_2,r_2}} S_{2,j_1} S_{2,j_2} y_{2,i_1,r_2} y_{2,i_2,r_1} S_{2,j_1}^\dagger S_{2,j_2}^\dagger + 2\hbar \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1}} S_{2,j_3} S_{2,j_4} y_{2,i_1,r_1} S_{2,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{S_2,j_3,j_1,j_4,j_2} \lambda_{S_2}^{(1)}}{\sqrt{3}}$					
cont.								$2\hbar \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \frac{\overline{y_{2,i_1,r_1}}}{S_2 y_{2,i_1,r_1}} S_{2,j_3} S_{2,j_4} y_{2,i_1,r_1} S_{2,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{S_2,j_3,j_1,j_4,j_2} \lambda_{S_2}^{(2)}$					
								$-\frac{\sqrt{3}}{\sqrt{3}}$					
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+\frac{2}{3} \hbar \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \frac{\overline{y_{2,i_1,r_1} y_{2,i_2,r_2}}}{S_2^2 y_{2,i_1,r_2} y_{2,i_2,r_1}} \left(S_{2,j_1}^\dagger \right)^2$			$-\frac{1}{2} \hbar \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) B_{\mu_1 \mu_2}^2$		
								$-\frac{2C_{S_2^\dagger S_2} \hbar \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1}} S_{2,j_1} y_{2,i_1,r_1} S_{2,j_1}^\dagger}{\sqrt{3}}$					
								$-\frac{\frac{4}{3} \hbar \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1} y_{2,i_2,r_2}} S_{2,j_1} S_{2,j_2} y_{2,i_1,r_2} y_{2,i_2,r_1} S_{2,j_1}^\dagger S_{2,j_2}^\dagger + 2\hbar \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1}} S_{2,j_3} S_{2,j_4} y_{2,i_1,r_1} S_{2,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{S_2,j_3,j_1,j_4,j_2} \lambda_{S_2}^{(1)}}{\sqrt{3}}$					
cont.								$2\hbar \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \frac{\overline{y_{2,i_1,r_1}}}{S_2 y_{2,i_1,r_1}} S_{2,j_3} S_{2,j_4} y_{2,i_1,r_1} S_{2,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{S_2,j_3,j_1,j_4,j_2} \lambda_{S_2}^{(2)}$					
								$-\frac{\sqrt{3}}{\sqrt{3}}$					

[illegible]

Field combinations 6 of 10

[illegible]

[illegible]

[illegible]

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$FW_{\mu\nu}$	$H \ell e$	$S_1 S_2 H$
cont.									$+\frac{\hbar \overline{y_{2,i_1,r_1}} \overline{Y_{e,i_3,i_2}} \overline{e_{i_2}} P_L L_{a_1,i_1} y_{2,i_3,r_1} H_{a_1}^\dagger}{4\epsilon}$	$-\frac{\hbar \overline{y_{1,i_1,r_1}} H_{a_1} H_{a_2} S_{1,j_2} y_{1,i_1,r_1} \mathcal{I}_{T3,a_1,a_2,j_2,j_1} \lambda_{T3} S_{2,j_1}^\dagger}{2\epsilon}$
									$+\frac{3}{8} \frac{\hbar \overline{y_{2,i_1,r_1}} \overline{Y_{e,i_3,i_2}} \overline{e_{i_2}} P_L L_{a_1,i_1} y_{2,i_3,r_1} H_{a_1}^\dagger}{4\epsilon}$	$-\frac{\frac{1}{4} \hbar \overline{y_{1,i_1,r_1}} H_{a_1} H_{a_2} S_{1,j_2} y_{1,i_1,r_1} \mathcal{I}_{T3,a_1,a_2,j_2,j_1} \lambda_{T3} S_{2,j_1}^\dagger}{2\epsilon}$
cont.										$-\frac{\hbar \overline{y_{2,i_1,r_1}} H_{a_1} H_{a_2} S_{1,j_2} y_{2,i_1,r_1} \mathcal{I}_{T3,a_1,a_2,j_2,j_1} \lambda_{T3} S_{2,j_1}^\dagger}{2\epsilon}$
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1		$+\frac{\hbar \overline{y_{1,i_3,r_1}} \overline{L_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{1,i_2,r_1} Y_{e,i_3,i_1}}{4\epsilon}$	$+\overline{\lambda_{T3} S_{2,j_2}} \mathcal{I}_{T3,a_1,a_2,j_1,j_2} H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger$
									$+\frac{3}{8} \frac{\hbar \overline{y_{1,i_1,r_1}} \overline{L_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{1,i_2,r_1} Y_{e,i_3,i_1}}{4\epsilon}$	$-\frac{\hbar \overline{y_{1,i_1,r_1}} \overline{\lambda_{T3} S_{2,j_2}} y_{1,i_1,r_1} \mathcal{I}_{T3,a_1,a_2,j_1,j_2} H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger}{2\epsilon}$
cont.									$+\frac{\hbar \overline{y_{2,i_3,r_1}} \overline{L_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{2,i_2,r_1} Y_{e,i_3,i_1}}{4\epsilon}$	$-\frac{1}{4} \hbar \overline{y_{1,i_1,r_1}} \overline{\lambda_{T3} S_{2,j_2}} y_{1,i_1,r_1} \mathcal{I}_{T3,a_1,a_2,j_1,j_2} H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger$
									$+\frac{3}{8} \frac{\hbar \overline{y_{2,i_3,r_1}} \overline{L_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{2,i_2,r_1} Y_{e,i_3,i_1}}{4\epsilon}$	$-\frac{\hbar \overline{y_{2,i_1,r_1}} \overline{\lambda_{T3} S_{2,j_2}} y_{2,i_1,r_1} \mathcal{I}_{T3,a_1,a_2,j_1,j_2} H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger}{2\epsilon}$
cont.									$+\frac{\hbar \overline{y_{1,i_1,r_1}} \overline{Y_{e,i_3,i_2}} \overline{e_{i_2}} P_L L_{a_1,i_1} y_{1,i_3,r_1} H_{a_1}^\dagger}{4\epsilon}$	$-\frac{1}{4} \hbar \overline{y_{2,i_1,r_1}} \overline{\lambda_{T3} S_{2,j_2}} y_{2,i_1,r_1} \mathcal{I}_{T3,a_1,a_2,j_1,j_2} H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger$
									$+\frac{3}{8} \frac{\hbar \overline{y_{1,i_1,r_1}} \overline{Y_{e,i_3,i_2}} \overline{e_{i_2}} P_L L_{a_1,i_1} y_{1,i_3,r_1} H_{a_1}^\dagger}{4\epsilon}$	$+\overline{H_{a_1} H_{a_2} S_{1,j_2} \mathcal{I}_{T3,a_1,a_2,j_2,j_1} \lambda_{T3} S_{2,j_1}^\dagger}$
cont.									$+\frac{\hbar \overline{y_{2,i_1,r_1}} \overline{Y_{e,i_3,i_2}} \overline{e_{i_2}} P_L L_{a_1,i_1} y_{2,i_3,r_1} H_{a_1}^\dagger}{4\epsilon}$	$-\frac{\hbar \overline{y_{1,i_1,r_1}} H_{a_1} H_{a_2} S_{1,j_2} y_{1,i_1,r_1} \mathcal{I}_{T3,a_1,a_2,j_2,j_1} \lambda_{T3} S_{2,j_1}^\dagger}{2\epsilon}$
									$+\frac{3}{8} \frac{\hbar \overline{y_{2,i_1,r_1}} \overline{Y_{e,i_3,i_2}} \overline{e_{i_2}} P_L L_{a_1,i_1} y_{2,i_3,r_1} H_{a_1}^\dagger}{4\epsilon}$	$-\frac{1}{4} \hbar \overline{y_{1,i_1,r_1}} H_{a_1} H_{a_2} S_{1,j_2} y_{1,i_1,r_1} \mathcal{I}_{T3,a_1,a_2,j_2,j_1} \lambda_{T3} S_{2,j_1}^\dagger$
cont.									$+\frac{\hbar \overline{y_{2,i_1,r_1}} \overline{Y_{e,i_3,i_2}} \overline{e_{i_2}} P_L L_{a_1,i_1} y_{2,i_3,r_1} H_{a_1}^\dagger}{4\epsilon}$	$-\frac{\hbar \overline{y_{2,i_1,r_1}} H_{a_1} H_{a_2} S_{1,j_2} y_{2,i_1,r_1} \mathcal{I}_{T3,a_1,a_2,j_2,j_1} \lambda_{T3} S_{2,j_1}^\dagger}{2\epsilon}$
									$+\frac{3}{8} \frac{\hbar \overline{y_{2,i_1,r_1}} \overline{Y_{e,i_3,i_2}} \overline{e_{i_2}} P_L L_{a_1,i_1} y_{2,i_3,r_1} H_{a_1}^\dagger}{4\epsilon}$	$-\frac{1}{4} \hbar \overline{y_{2,i_1,r_1}} H_{a_1} H_{a_2} S_{1,j_2} y_{2,i_1,r_1} \mathcal{I}_{T3,a_1,a_2,j_2,j_1} \lambda_{T3} S_{2,j_1}^\dagger$
T3-B	2	2	1	2	2	1	$\frac{3}{2}$		$+\frac{\hbar \overline{y_{1,i_3,r_1}} \overline{L_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{1,i_2,r_1} Y_{e,i_3,i_1}}{4\epsilon}$	$+\overline{\lambda_{T3} S_{2,j_2}} \mathcal{I}_{T3,a_1,a_2,j_1,j_2} H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger$
									$+\frac{3}{8} \frac{\hbar \overline{y_{1,i_3,r_1}} \overline{L_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{1,i_2,r_1} Y_{e,i_3,i_1}}{4\epsilon}$	$-\frac{\hbar \overline{y_{1,i_1,r_1}} \overline{\lambda_{T3} S_{2,j_2}} y_{1,i_1,r_1} \mathcal{I}_{T3,a_1,a_2,j_1,j_2} H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger}{2\epsilon}$
cont.									$+\frac{\hbar \overline{y_{2,i_3,r_1}} \overline{L_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{2,i_2,r_1} Y_{e,i_3,i_1}}{4\epsilon}$	$-\frac{1}{4} \hbar \overline{y_{1,i_1,r_1}} \overline{\lambda_{T3} S_{2,j_2}} y_{1,i_1,r_1} \mathcal{I}_{T3,a_1,a_2,j_1,j_2} H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger$
									$+\frac{3}{8} \frac{\hbar \overline{y_{2,i_3,r_1}} \overline{L_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{2,i_2,r_1} Y_{e,i_3,i_1}}{4\epsilon}$	$-\frac{\hbar \overline{y_{2,i_1,r_1}} \overline{\lambda_{T3} S_{2,j_2}} y_{2,i_1,r_1} \mathcal{I}_{T3,a_1,a_2,j_1,j_2} H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger}{2\epsilon}$
cont.									$+\frac{\hbar \overline{y_{1,i_1,r_1}} \overline{Y_{e,i_3,i_2}} \overline{e_{i_2}} P_L L_{a_1,i_1} y_{1,i_3,r_1} H_{a_1}^\dagger}{4\epsilon}$	$-\frac{1}{4} \hbar \overline{y_{2,i_1,r_1}} \overline{\lambda_{T3} S_{2,j_2}} y_{2,i_1,r_1} \mathcal{I}_{T3,a_1,a_2,j_1,j_2} H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger$
									$+\frac{3}{8} \frac{\hbar \overline{y_{1,i_1,r_1}} \overline{Y_{e,i_3,i_2}} \overline{e_{i_2}} P_L L_{a_1,i_1} y_{1,i_3,r_1} H_{a_1}^\dagger}{4\epsilon}$	$+\overline{H_{a_1} H_{a_2} S_{1,j_2} \mathcal{I}_{T3,a_1,a_2,j_2,j_1} \lambda_{T3} S_{2,j_1}^\dagger}$
cont.									$+\frac{\hbar \overline{y_{2,i_1,r_1}} \overline{Y_{e,i_3,i_2}} \overline{e_{i_2}} P_L L_{a_1,i_1} y_{2,i_3,r_1} H_{a_1}^\dagger}{4\epsilon}$	$-\frac{\hbar \overline{y_{1,i_1,r_1}} H_{a_1} H_{a_2} S_{1,j_2} y_{1,i_1,r_1} \mathcal{I}_{T3,a_1,a_2,j_2,j_1} \lambda_{T3} S_{2,j_1}^\dagger}{2\epsilon}$
									$+\frac{3}{8} \frac{\hbar \overline{y_{2,i_1,r_1}} \overline{Y_{e,i_3,i_2}} \overline{e_{i_2}} P_L L_{a_1,i_1} y_{2,i_3,r_1} H_{a_1}^\dagger}{4\epsilon}$	$-\frac{1}{4} \hbar \overline{y_{1,i_1,r_1}} H_{a_1} H_{a_2} S_{1,j_2} y_{1,i_1,r_1} \mathcal{I}_{T3,a_1,a_2,j_2,j_1} \lambda_{T3} S_{2,j_1}^\dagger$
cont.									$+\frac{\hbar \overline{y_{2,i_1,r_1}} \overline{Y_{e,i_3,i_2}} \overline{e_{i_2}} P_L L_{a_1,i_1} y_{2,i_3,r_1} H_{a_1}^\dagger}{4\epsilon}$	$-\frac{\hbar \overline{y_{2,i_1,r_1}} H_{a_1} H_{a_2} S_{1,j_2} y_{2,i_1,r_1} \mathcal{I}_{T3,a_1,a_2,j_2,j_1} \lambda_{T3} S_{2,j_1}^\dagger}{2\epsilon}$
									$+\frac{3}{8} \frac{\hbar \overline{y_{2,i_1,r_1}} \overline{Y_{e,i_3,i_2}} \overline{e_{i_2}} P_L L_{a_1,i_1} y_{2,i_3,r_1} H_{a_1}^\dagger}{4\epsilon}$	$-\frac{1}{4} \hbar \overline{y_{2,i_1,r_1}} H_{a_1} H_{a_2} S_{1,j_2} y_{2,i_1,r_1} \mathcal{I}_{T3,a_1,a_2,j_2,j_1} \lambda_{T3} S_{2,j_1}^\dagger$
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	$-2\hbar \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \left(W_{\mu_1 \mu_2}^{A_1} \right)^2$	$+\frac{\sqrt{3} \hbar \overline{y_{1,i_3,r_1}} \overline{L_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{1,i_2,r_1} Y_{e,i_3,i_1}}{4\epsilon}$	$+\overline{\lambda_{T3} S_{2,j_2}} \mathcal{I}_{T3,a_1,a_2,j_1,j_2} H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger$
									$+\frac{3}{8} \frac{\sqrt{3} \hbar \overline{y_{1,i_3,r_1}} \overline{L_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{1,i_2,r_1} Y_{e,i_3,i_1}}{4\epsilon}$	$-\frac{\sqrt{3} \hbar \overline{y_{1,i_1,r_1}} \overline{\lambda_{T3} S_{2,j_2}} y_{1,i_1,r_1} \mathcal{I}_{T3,a_1,a_2,j_1,j_2} H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger}{2\epsilon}$
cont.									$+\frac{\sqrt{3} \hbar \overline{y_{2,i_3,r_1}} \overline{L_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{2,i_2,r_1} Y_{e,i_3,i_1}}{4\epsilon}$	$-\frac{1}{4} \sqrt{3} \hbar \overline{y_{1,i_1,r_1}} \overline{\lambda_{T3} S_{2,j_2}} y_{1,i_1,r_1} \mathcal{I}_{T3,a_1,a_2,j_1,j_2} H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger$
									$+\frac{3}{8} \frac{\sqrt{3} \hbar \overline{y_{2,i_3,r_1}} \overline{L_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{2,i_2,r_1} Y_{e,i_3,i_1}}{4\epsilon}$	$-\frac{\sqrt{3} \hbar \overline{y_{2,i_1,r_1}} \overline{\lambda_{T3} S_{2,j_2}} y_{2,i_1,r_1} \mathcal{I}_{T3,a_1,a_2,j_1,j_2} H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger}{2\epsilon}$
cont.									$+\frac{\sqrt{3} \hbar \overline{y_{1,i_1,r_1}} \overline{Y_{e,i_3,i_2}} \overline{e_{i_2}} P_L L_{a_1,i_1} y_{1,i_3,r_1} H_{a_1}^\dagger}{4\epsilon}$	$-\frac{1}{4} \sqrt{3} \hbar \overline{y_{2,i_1,r_1}} \overline{\lambda_{T3} S_{2,j_2}} y_{2,i_1,r_1} \mathcal{I}_{T3,a_1,a_2,j_1,j_2} H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger$
									$+\frac{3}{8} \frac{\sqrt{3} \hbar \overline{y_{1,i_1,r_1}} \overline{Y_{e,i_3,i_2}} \overline{e_{i_2}} P_L L_{a_1,i_1} y_{1,i_3,r_1} H_{a_1}^\dagger}{4\epsilon}$	$+\overline{H_{a_1} H_{a_2} S_{1,j_2} \mathcal{I}_{T3,a_1,a_2,j_2,j_1} \lambda_{T3} S_{2,j_1}^\dagger}$
cont.									$+\frac{\sqrt{3} \hbar \overline{y_{2,i_1,r_1}} \overline{Y_{e,i_3,i_2}} \overline{e_{i_2}} P_L L_{a_1,i_1} y_{2,i_3,r_1} H_{a_1}^\dagger}{4\epsilon}$	$-\frac{\sqrt{3} \hbar \overline{y_{1,i_1,r_1}} H_{a_1} H_{a_2} S_{1,j_2} y_{1,i_1,r_1} \mathcal{I}_{T3,a_1,a_2,j_2,j_1} \lambda_{T3} S_{2,j_1}^\dagger}{2\epsilon}$
									$+\frac{3}{8} \frac{\sqrt{3} \hbar \overline{y_{2,i_1,r_1}} \overline{Y_{e,i_3,i_2}} \overline{e_{i_2}} P_L L_{a_1,i_1} y_{2,i_3,r_1} H_{a_1}^\dagger}{4\epsilon}$	$-\frac{1}{4} \sqrt{3} \hbar \overline{y_{1,i_1,r_1}} H_{a_1} H_{a_2} S_{1,j_2} y_{1,i_1,r_1} \mathcal{I}_{T3,a_1,a_2,j_2,j_1} \lambda_{T3} S_{2,j_1}^\dagger$
cont.									$+\frac{\sqrt{3} \hbar \overline{y_{2,i_1,r_1}} \overline{Y_{e,i_3,i_2}} \overline{e_{i_2}} P_L L_{a_1,i_1} y_{2,i_3,r_1} H_{a_1}^\dagger}{4\epsilon}$	$-\frac{\sqrt{3} \hbar \overline{y_{2,i_1,r_1}} H_{a_1} H_{a_2} S_{1,j_2} y_{2,i_1,r_1} \mathcal{I}_{T3,a_1,a_2,j_2,j_1} \lambda_{T3} S_{2,j_1}^\dagger}{2\epsilon}$
									$+\frac{3}{8} \frac{\sqrt{3} \hbar \overline{y_{2,i_1,r_1}} \overline{Y_{e,i_3,i_2}} \overline{e_{i_2}} P_L L_{a_1,i_1} y_{2,i_3,r_1} H_{a_1}^\dagger}{4\epsilon}$	$-\frac{1}{4} \sqrt{3} \hbar \overline{y_{2,i_1,r_1}} H_{a_1} H_{a_2} S_{1,j_2} y_{2,i_1,r_1} \mathcal{I}_{T3,a_1,a_2,j_2,j_1} \lambda_{T3} S_{2,j_1}^\dagger$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$FW\mu\nu$	$H\ell e$	$S_1 S_2 H$
<i>cont.</i>										$-\frac{1}{4}\sqrt{3h\bar{y}_{2,i_1,r_1}}H_{a_1}H_{a_2}S_{1,j_2}y_{2,i_1,r_1}\mathcal{I}T_{3,a_1,a_2,j_2,j_1}\lambda T_3S_{2,j_1}^\dagger$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$-\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\left(W_{\mu_1\mu_2}^{A_1}\right)^2$	$\begin{aligned}&+\frac{\sqrt{3h\bar{y}_{1,i_3,r_1}}\overline{L_{a_1,i_2}}P_R e_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}}{4\epsilon}\\&+\frac{3}{8}\sqrt{3h\bar{y}_{1,i_3,r_1}}\overline{L_{a_1,i_2}}P_R e_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}\\&+\frac{\sqrt{3h\bar{y}_{2,i_3,r_1}}\overline{L_{a_1,i_2}}P_R e_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}}{4\epsilon}\\&+\frac{3}{8}\sqrt{3h\bar{y}_{2,i_3,r_1}}\overline{L_{a_1,i_2}}P_R e_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}\end{aligned}$	$\begin{aligned}&-\frac{2h\bar{y}_{2,i_1,r_1}\overline{Y_{e,i_2,i_3}}H_{a_2}S_{1,j_2}S_{2,j_1}y_{1,i_2,r_1}Y_{e,i_1,i_3}\text{Bar}(cg4)_{a_2,j_1,a_1,j_2}H_{a_1}^\dagger}{\epsilon}\\&-\frac{2h\bar{y}_{2,i_1,r_1}\overline{Y_{e,i_2,i_3}}H_{a_2}S_{1,j_2}S_{2,j_1}y_{1,i_2,r_1}Y_{e,i_1,i_3}\text{Bar}(cg4)_{a_2,j_1,a_1,j_2}H_{a_1}^\dagger}{\epsilon+\lambda T_3S_{2,j_2}T_3^{*a_1,a_2,j_1,j_2}H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger}\end{aligned}$
<i>cont.</i>									$\begin{aligned}&+\frac{\sqrt{3h\bar{y}_{1,i_1,r_1}}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}P_L L_{a_1,i_1}y_{1,i_3,r_1}H_{a_1}^\dagger}{4\epsilon}\\&+\frac{3}{8}\sqrt{3h\bar{y}_{1,i_1,r_1}}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}P_L L_{a_1,i_1}y_{1,i_3,r_1}H_{a_1}^\dagger\end{aligned}$	$\begin{aligned}&-\frac{1}{4}\sqrt{3h\bar{y}_{1,i_1,r_1}}\overline{\lambda T_3}S_{2,j_2}y_{1,i_1,r_1}\mathcal{I}T_{3,a_1,a_2,j_1,j_2}H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger\\&-\frac{\sqrt{3h\bar{y}_{2,i_1,r_1}}\overline{\lambda T_3}S_{2,j_2}y_{2,i_1,r_1}\mathcal{I}T_{3,a_1,a_2,j_1,j_2}H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger}{2\epsilon}\end{aligned}$
<i>cont.</i>									$\begin{aligned}&+\frac{\sqrt{3h\bar{y}_{2,i_1,r_1}}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}P_L L_{a_1,i_1}y_{2,i_3,r_1}H_{a_1}^\dagger}{4\epsilon}\\&+\frac{3}{8}\sqrt{3h\bar{y}_{2,i_1,r_1}}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}P_L L_{a_1,i_1}y_{2,i_3,r_1}H_{a_1}^\dagger\end{aligned}$	$\begin{aligned}&-\frac{1}{4}\sqrt{3h\bar{y}_{2,i_1,r_1}}\overline{\lambda T_3}S_{2,j_2}y_{2,i_1,r_1}\mathcal{I}T_{3,a_1,a_2,j_1,j_2}H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger\\&\quad +H_{a_1}H_{a_2}S_{1,j_2}\mathcal{I}T_{3,a_1,a_2,j_2,j_1}\lambda T_3S_{2,j_1}^\dagger\\&-\frac{\sqrt{3h\bar{y}_{1,i_1,r_1}}H_{a_1}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}\mathcal{I}T_{3,a_1,a_2,j_2,j_1}\lambda T_3S_{2,j_1}^\dagger}{2\epsilon}\end{aligned}$
<i>cont.</i>										$\begin{aligned}&-\frac{1}{4}\sqrt{3h\bar{y}_{1,i_1,r_1}}H_{a_1}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}\mathcal{I}T_{3,a_1,a_2,j_2,j_1}\lambda T_3S_{2,j_1}^\dagger\\&-\frac{\sqrt{3h\bar{y}_{2,i_1,r_1}}H_{a_1}H_{a_2}S_{1,j_2}y_{2,i_1,r_1}\mathcal{I}T_{3,a_1,a_2,j_2,j_1}\lambda T_3S_{2,j_1}^\dagger}{2\epsilon}\end{aligned}$
<i>cont.</i>										$\begin{aligned}&-\frac{1}{4}\sqrt{3h\bar{y}_{2,i_1,r_1}}H_{a_1}H_{a_2}S_{1,j_2}y_{2,i_1,r_1}\mathcal{I}T_{3,a_1,a_2,j_2,j_1}\lambda T_3S_{2,j_1}^\dagger\\&-\frac{2h\bar{y}_{1,i_1,r_1}\overline{Y_{e,i_2,i_3}}cg^4_{a_1,j_2,a_2,j_1}H_{a_2}y_{2,i_2,r_1}Y_{e,i_1,i_3}H_{a_1}^\dagger S_{1,j_1}^\dagger S_{2,j_2}^\dagger}{\epsilon}\end{aligned}$
<i>cont.</i>										$\begin{aligned}&-\frac{2h\bar{y}_{1,i_1,r_1}\overline{Y_{e,i_2,i_3}}cg^4_{a_1,j_2,a_2,j_1}H_{a_2}y_{2,i_2,r_1}Y_{e,i_1,i_3}H_{a_1}^\dagger S_{1,j_1}^\dagger S_{2,j_2}^\dagger}{h\bar{y}_{1,i_1,r_1}H_{a_1}S_{1,j_1}S_{2,j_2}y_{1,i_1,r_1}\text{Bar}(cg3)_{j_1,j_2}H_{a_1}^\dagger\lambda_{H_2}^{(1)}}\\&\quad -\frac{1}{2}h\bar{y}_{2,i_1,r_1}\overline{H_{a_1}S_{1,j_1}S_{2,j_2}y_{1,i_1,r_1}\text{Bar}(cg3)_{j_1,j_2}H_{a_1}^\dagger\lambda_{H_2}^{(1)}}\\&-\frac{3h\bar{y}_{2,i_1,r_1}H_{a_2}S_{1,j_2}S_{2,j_1}y_{1,i_1,r_1}\text{Bar}(cg4)_{a_2,j_1,a_1,j_2}H_{a_1}^\dagger\lambda_{H_2}^{(1)}}{\epsilon}\end{aligned}$
<i>cont.</i>										$\begin{aligned}&-\frac{3}{2}h\bar{y}_{2,i_1,r_1}H_{a_2}S_{1,j_2}S_{2,j_1}y_{1,i_1,r_1}\text{Bar}(cg4)_{a_2,j_1,a_1,j_2}H_{a_1}^\dagger\lambda_{H_2}^{(1)}\\&-\frac{h\bar{y}_{1,i_1,r_1}cg3_{j_1,j_2}H_{a_1}y_{2,i_1,r_1}H_{a_1}^\dagger S_{1,j_1}^\dagger S_{2,j_2}^\dagger\lambda_{H_2}^{(1)}}{\epsilon}\end{aligned}$
<i>cont.</i>										$\begin{aligned}&-\frac{1}{2}h\bar{y}_{1,i_1,r_1}cg3_{j_1,j_2}H_{a_1}y_{2,i_1,r_1}H_{a_1}^\dagger S_{1,j_1}^\dagger S_{2,j_2}^\dagger\lambda_{H_2}^{(1)}\\&-\frac{3h\bar{y}_{1,i_1,r_1}cg4_{a_1,j_2,a_2,j_1}H_{a_2}y_{2,i_1,r_1}H_{a_1}^\dagger S_{1,j_1}^\dagger S_{2,j_2}^\dagger\lambda_{H_2}^{(1)}}{\epsilon}\end{aligned}$
<i>cont.</i>										$\begin{aligned}&-\frac{3}{2}h\bar{y}_{1,i_1,r_1}cg4_{a_1,j_2,a_2,j_1}H_{a_2}y_{2,i_1,r_1}H_{a_1}^\dagger S_{1,j_1}^\dagger S_{2,j_2}^\dagger\lambda_{H_2}^{(1)}\\&+\frac{\sqrt{3h\bar{y}_{2,i_1,r_1}}H_{a_2}S_{1,j_1}S_{2,j_2}y_{1,i_1,r_1}\text{Bar}(cg4)_{a_2,j_1,a_1,j_2}H_{a_1}^\dagger\lambda_{H_2}^{(2)}}{\epsilon}\end{aligned}$
<i>cont.</i>										$\begin{aligned}&+\frac{1}{2}\sqrt{3h\bar{y}_{2,i_1,r_1}}H_{a_2}S_{1,j_1}S_{2,j_2}y_{1,i_1,r_1}\text{Bar}(cg4)_{a_2,j_1,a_1,j_2}H_{a_1}^\dagger\lambda_{H_2}^{(2)}\\&+\frac{\sqrt{3h\bar{y}_{1,i_1,r_1}}cg4_{a_1,j_1,a_2,j_2}H_{a_2}y_{2,i_1,r_1}H_{a_1}^\dagger S_{1,j_1}^\dagger S_{2,j_2}^\dagger\lambda_{H_2}^{(2)}}{\epsilon}\end{aligned}$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	$-2\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\left(W_{\mu_1\mu_2}^{A_1}\right)^2$	$\begin{aligned}&+\frac{\sqrt{3h\bar{y}_{1,i_3,r_1}}\overline{L_{a_1,i_2}}P_R e_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}}{4\epsilon}\\&+\frac{3}{8}\sqrt{3h\bar{y}_{1,i_3,r_1}}\overline{L_{a_1,i_2}}P_R e_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}\\&+\frac{\sqrt{3h\bar{y}_{2,i_3,r_1}}\overline{L_{a_1,i_2}}P_R e_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}}{4\epsilon}\\&+\frac{3}{8}\sqrt{3h\bar{y}_{2,i_3,r_1}}\overline{L_{a_1,i_2}}P_R e_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}\end{aligned}$	$\begin{aligned}&\quad +\overline{\lambda T_3}S_{2,j_2}T_3^{*a_1,a_2,j_1,j_2}H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger\\&-\frac{\sqrt{3h\bar{y}_{1,i_1,r_1}}\overline{\lambda T_3}S_{2,j_2}y_{1,i_1,r_1}\mathcal{I}T_{3,a_1,a_2,j_1,j_2}H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger}{2\epsilon}\\&-\frac{1}{4}\sqrt{3h\bar{y}_{1,i_1,r_1}}\overline{\lambda T_3}S_{2,j_2}y_{1,i_1,r_1}\mathcal{I}T_{3,a_1,a_2,j_1,j_2}H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger\\&-\frac{\sqrt{3h\bar{y}_{2,i_1,r_1}}\overline{\lambda T_3}S_{2,j_2}y_{2,i_1,r_1}\mathcal{I}T_{3,a_1,a_2,j_1,j_2}H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger}{2\epsilon}\\&-\frac{1}{4}\sqrt{3h\bar{y}_{2,i_1,r_1}}\overline{\lambda T_3}S_{2,j_2}y_{2,i_1,r_1}\mathcal{I}T_{3,a_1,a_2,j_1,j_2}H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger\\&\quad +H_{a_1}H_{a_2}S_{1,j_2}\mathcal{I}T_{3,a_1,a_2,j_2,j_1}\lambda T_3S_{2,j_1}^\dagger\\&-\frac{\sqrt{3h\bar{y}_{1,i_1,r_1}}H_{a_1}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}\mathcal{I}T_{3,a_1,a_2,j_2,j_1}\lambda T_3S_{2,j_1}^\dagger}{2\epsilon}\\&-\frac{1}{4}\sqrt{3h\bar{y}_{1,i_1,r_1}}H_{a_1}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}\mathcal{I}T_{3,a_1,a_2,j_2,j_1}\lambda T_3S_{2,j_1}^\dagger\\&-\frac{\sqrt{3h\bar{y}_{2,i_1,r_1}}H_{a_1}H_{a_2}S_{1,j_2}y_{2,i_1,r_1}\mathcal{I}T_{3,a_1,a_2,j_2,j_1}\lambda T_3S_{2,j_1}^\dagger}{2\epsilon}\end{aligned}$
<i>cont.</i>									$\begin{aligned}&+\frac{\sqrt{3h\bar{y}_{1,i_1,r_1}}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}P_L L_{a_1,i_1}y_{1,i_3,r_1}H_{a_1}^\dagger}{4\epsilon}\\&+\frac{3}{8}\sqrt{3h\bar{y}_{1,i_1,r_1}}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}P_L L_{a_1,i_1}y_{1,i_3,r_1}H_{a_1}^\dagger\end{aligned}$	$\begin{aligned}&-\frac{1}{4}\sqrt{3h\bar{y}_{1,i_1,r_1}}\overline{\lambda T_3}S_{2,j_2}y_{1,i_1,r_1}\mathcal{I}T_{3,a_1,a_2,j_1,j_2}H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger\\&-\frac{\sqrt{3h\bar{y}_{2,i_1,r_1}}\overline{\lambda T_3}S_{2,j_2}y_{2,i_1,r_1}\mathcal{I}T_{3,a_1,a_2,j_1,j_2}H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger}{2\epsilon}\end{aligned}$
<i>cont.</i>									$\begin{aligned}&+\frac{\sqrt{3h\bar{y}_{2,i_1,r_1}}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}P_L L_{a_1,i_1}y_{2,i_3,r_1}H_{a_1}^\dagger}{4\epsilon}\\&+\frac{3}{8}\sqrt{3h\bar{y}_{2,i_1,r_1}}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}P_L L_{a_1,i_1}y_{2,i_3,r_1}H_{a_1}^\dagger\end{aligned}$	$\begin{aligned}&-\frac{1}{4}\sqrt{3h\bar{y}_{2,i_1,r_1}}\overline{\lambda T_3}S_{2,j_2}y_{2,i_1,r_1}\mathcal{I}T_{3,a_1,a_2,j_1,j_2}H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger\\&\quad +H_{a_1}H_{a_2}S_{1,j_2}\mathcal{I}T_{3,a_1,a_2,j_2,j_1}\lambda T_3S_{2,j_1}^\dagger\\&-\frac{\sqrt{3h\bar{y}_{1,i_1,r_1}}H_{a_1}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}\mathcal{I}T_{3,a_1,a_2,j_2,j_1}\lambda T_3S_{2,j_1}^\dagger}{2\epsilon}\end{aligned}$
<i>cont.</i>										$\begin{aligned}&-\frac{1}{4}\sqrt{3h\bar{y}_{1,i_1,r_1}}H_{a_1}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}\mathcal{I}T_{3,a_1,a_2,j_2,j_1}\lambda T_3S_{2,j_1}^\dagger\\&-\frac{\sqrt{3h\bar{y}_{2,i_1,r_1}}H_{a_1}H_{a_2}S_{1,j_2}y_{2,i_1,r_1}\mathcal{I}T_{3,a_1,a_2,j_2,j_1}\lambda T_3S_{2,j_1}^\dagger}{2\epsilon}\end{aligned}$

[illegible]

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$FW_{\mu\nu}$	$H\ell e$	S_1S_2H
cont.									$+\frac{3}{8}\frac{\hbar\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_2}}}{4\epsilon}P_R e_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}$ $+\frac{\sqrt{3}\hbar\overline{y_{1,i_1,r_1}}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}}{4\epsilon}P_L L_{a_1,i_1}y_{1,i_3,r_1}H_{a_1}^\dagger$ $+\frac{3}{8}\sqrt{3}\frac{\hbar\overline{y_{1,i_1,r_1}}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}}{4\epsilon}P_L L_{a_1,i_1}y_{1,i_3,r_1}H_{a_1}^\dagger$	$-\frac{\hbar\overline{y_{2,i_1,r_1}}H_{a_1}H_{a_2}S_{1,j_1}y_{2,i_1,r_1}\mathcal{I}_{T3,a_1,a_2,j_1}\lambda_{T3}S_2^\dagger}{\epsilon}$ $-\frac{1}{2}\frac{\hbar\overline{y_{2,i_1,r_1}}H_{a_1}H_{a_2}S_{1,j_1}y_{2,i_1,r_1}\mathcal{I}_{T3,a_1,a_2,j_1}\lambda_{T3}S_2^\dagger}{\epsilon}$ $+\frac{\overline{\lambda_{T3}}S_2\mathcal{I}_{T3}^*S_{a_1,a_2,j_1}H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger}{\epsilon}$
									$+\frac{\hbar\overline{y_{2,i_1,r_1}}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}}{4\epsilon}P_L L_{a_1,i_1}y_{2,i_3,r_1}H_{a_1}^\dagger$ $+\frac{\hbar\overline{y_{2,i_1,r_1}}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}}{4\epsilon}P_L L_{a_1,i_1}y_{2,i_3,r_1}H_{a_1}^\dagger$	$-\frac{\hbar\overline{y_{1,i_1,r_1}}\overline{\lambda_{T3}}S_2y_{1,i_1,r_1}\mathcal{I}_{T3}^*S_{a_1,a_2,j_1}H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger}{\sqrt{3}\epsilon}$ $-\frac{\hbar\overline{y_{1,i_1,r_1}}\overline{\lambda_{T3}}S_2y_{1,i_1,r_1}\mathcal{I}_{T3}^*S_{a_1,a_2,j_1}H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger}{2\sqrt{3}}$ $-\frac{\hbar\overline{y_{2,i_1,r_1}}\overline{\lambda_{T3}}S_2y_{2,i_1,r_1}\mathcal{I}_{T3}^*S_{a_1,a_2,j_1}H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger}{\epsilon}$
										$-\frac{1}{2}\frac{\hbar\overline{y_{2,i_1,r_1}}\overline{\lambda_{T3}}S_2y_{2,i_1,r_1}\mathcal{I}_{T3}^*S_{a_1,a_2,j_1}H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger}{\epsilon}$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$-\frac{1}{2}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\left(W_{\mu_1\mu_2}^{A_1}\right)^2$	$+\frac{\sqrt{3}\hbar\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_2}}}{4\epsilon}P_R e_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}$ $+\frac{3}{8}\sqrt{3}\frac{\hbar\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_2}}}{4\epsilon}P_R e_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}$ $+\frac{\hbar\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_2}}}{4\epsilon}P_R e_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}$ $+\frac{3}{8}\frac{\hbar\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_2}}}{4\epsilon}P_R e_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}$	$+H_{a_1}H_{a_2}S_{1,j_1}\mathcal{I}_{T3,a_1,a_2,j_1}\lambda_{T3}S_2^\dagger$ $-\frac{\hbar\overline{y_{1,i_1,r_1}}H_{a_1}H_{a_2}S_{1,j_1}y_{1,i_1,r_1}\mathcal{I}_{T3,a_1,a_2,j_1}\lambda_{T3}S_2^\dagger}{\sqrt{3}\epsilon}$ $-\frac{\hbar\overline{y_{1,i_1,r_1}}H_{a_1}H_{a_2}S_{1,j_1}y_{1,i_1,r_1}\mathcal{I}_{T3,a_1,a_2,j_1}\lambda_{T3}S_2^\dagger}{2\sqrt{3}}$
cont.									$+\frac{\hbar\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_2}}}{4\epsilon}P_R e_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}$ $+\frac{\sqrt{3}\hbar\overline{y_{1,i_1,r_1}}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}}{4\epsilon}P_L L_{a_1,i_1}y_{1,i_3,r_1}H_{a_1}^\dagger$ $+\frac{3}{8}\sqrt{3}\frac{\hbar\overline{y_{1,i_1,r_1}}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}}{4\epsilon}P_L L_{a_1,i_1}y_{1,i_3,r_1}H_{a_1}^\dagger$	$-\frac{\hbar\overline{y_{2,i_1,r_1}}H_{a_1}H_{a_2}S_{1,j_1}y_{2,i_1,r_1}\mathcal{I}_{T3,a_1,a_2,j_1}\lambda_{T3}S_2^\dagger}{\epsilon}$ $-\frac{1}{2}\frac{\hbar\overline{y_{2,i_1,r_1}}H_{a_1}H_{a_2}S_{1,j_1}y_{2,i_1,r_1}\mathcal{I}_{T3,a_1,a_2,j_1}\lambda_{T3}S_2^\dagger}{\epsilon}$ $+\frac{\overline{\lambda_{T3}}S_2\mathcal{I}_{T3}^*S_{a_1,a_2,j_1}H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger}{\epsilon}$
									$+\frac{\hbar\overline{y_{2,i_1,r_1}}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}}{4\epsilon}P_L L_{a_1,i_1}y_{2,i_3,r_1}H_{a_1}^\dagger$ $+\frac{\hbar\overline{y_{2,i_1,r_1}}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}}{4\epsilon}P_L L_{a_1,i_1}y_{2,i_3,r_1}H_{a_1}^\dagger$	$-\frac{\hbar\overline{y_{1,i_1,r_1}}\overline{\lambda_{T3}}S_2y_{1,i_1,r_1}\mathcal{I}_{T3}^*S_{a_1,a_2,j_1}H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger}{\sqrt{3}\epsilon}$ $-\frac{\hbar\overline{y_{1,i_1,r_1}}\overline{\lambda_{T3}}S_2y_{1,i_1,r_1}\mathcal{I}_{T3}^*S_{a_1,a_2,j_1}H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger}{2\sqrt{3}}$ $-\frac{\hbar\overline{y_{2,i_1,r_1}}\overline{\lambda_{T3}}S_2y_{2,i_1,r_1}\mathcal{I}_{T3}^*S_{a_1,a_2,j_1}H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger}{\epsilon}$
										$-\frac{1}{2}\frac{\hbar\overline{y_{2,i_1,r_1}}\overline{\lambda_{T3}}S_2y_{2,i_1,r_1}\mathcal{I}_{T3}^*S_{a_1,a_2,j_1}H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger}{\epsilon}$
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	$-\frac{1}{2}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\left(W_{\mu_1\mu_2}^{A_1}\right)^2$	$+\frac{\sqrt{3}\hbar\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_2}}}{4\epsilon}P_R e_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}$ $+\frac{3}{8}\sqrt{3}\frac{\hbar\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_2}}}{4\epsilon}P_R e_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}$ $+\frac{\hbar\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_2}}}{4\epsilon}P_R e_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}$ $+\frac{3}{8}\frac{\hbar\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_2}}}{4\epsilon}P_R e_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}$	$+H_{a_1}H_{a_2}S_{1,j_1}\mathcal{I}_{T3,a_1,a_2,j_1}\lambda_{T3}S_2^\dagger$ $-\frac{\hbar\overline{y_{1,i_1,r_1}}H_{a_1}H_{a_2}S_{1,j_1}y_{1,i_1,r_1}\mathcal{I}_{T3,a_1,a_2,j_1}\lambda_{T3}S_2^\dagger}{\sqrt{3}\epsilon}$ $-\frac{\hbar\overline{y_{1,i_1,r_1}}H_{a_1}H_{a_2}S_{1,j_1}y_{1,i_1,r_1}\mathcal{I}_{T3,a_1,a_2,j_1}\lambda_{T3}S_2^\dagger}{2\sqrt{3}}$
cont.									$+\frac{\hbar\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_2}}}{4\epsilon}P_R e_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}$ $+\frac{\sqrt{3}\hbar\overline{y_{1,i_1,r_1}}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}}{4\epsilon}P_L L_{a_1,i_1}y_{1,i_3,r_1}H_{a_1}^\dagger$ $+\frac{3}{8}\sqrt{3}\frac{\hbar\overline{y_{1,i_1,r_1}}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}}{4\epsilon}P_L L_{a_1,i_1}y_{1,i_3,r_1}H_{a_1}^\dagger$	$-\frac{\hbar\overline{y_{2,i_1,r_1}}H_{a_1}H_{a_2}S_{1,j_1}y_{2,i_1,r_1}\mathcal{I}_{T3,a_1,a_2,j_1}\lambda_{T3}S_2^\dagger}{\epsilon}$ $-\frac{1}{2}\frac{\hbar\overline{y_{2,i_1,r_1}}H_{a_1}H_{a_2}S_{1,j_1}y_{2,i_1,r_1}\mathcal{I}_{T3,a_1,a_2,j_1}\lambda_{T3}S_2^\dagger}{\epsilon}$ $+\frac{\overline{\lambda_{T3}}S_2\mathcal{I}_{T3}^*S_{a_1,a_2,j_1}H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger}{\epsilon}$
									$+\frac{\hbar\overline{y_{2,i_1,r_1}}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}}{4\epsilon}P_L L_{a_1,i_1}y_{2,i_3,r_1}H_{a_1}^\dagger$ $+\frac{\hbar\overline{y_{2,i_1,r_1}}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}}{4\epsilon}P_L L_{a_1,i_1}y_{2,i_3,r_1}H_{a_1}^\dagger$	$-\frac{\hbar\overline{y_{1,i_1,r_1}}\overline{\lambda_{T3}}S_2y_{1,i_1,r_1}\mathcal{I}_{T3}^*S_{a_1,a_2,j_1}H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger}{\sqrt{3}\epsilon}$ $-\frac{\hbar\overline{y_{1,i_1,r_1}}\overline{\lambda_{T3}}S_2y_{1,i_1,r_1}\mathcal{I}_{T3}^*S_{a_1,a_2,j_1}H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger}{2\sqrt{3}}$ $-\frac{\hbar\overline{y_{2,i_1,r_1}}\overline{\lambda_{T3}}S_2y_{2,i_1,r_1}\mathcal{I}_{T3}^*S_{a_1,a_2,j_1}H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger}{\epsilon}$
										$-\frac{1}{2}\frac{\hbar\overline{y_{2,i_1,r_1}}\overline{\lambda_{T3}}S_2y_{2,i_1,r_1}\mathcal{I}_{T3}^*S_{a_1,a_2,j_1}H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger}{\epsilon}$
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	$-\frac{1}{2}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\left(W_{\mu_1\mu_2}^{A_1}\right)^2$	$+\frac{\sqrt{3}\hbar\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_2}}}{4\epsilon}P_R e_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}$ $+\frac{3}{8}\sqrt{3}\frac{\hbar\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_2}}}{4\epsilon}P_R e_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}$ $+\frac{\hbar\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_2}}}{4\epsilon}P_R e_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}$ $+\frac{3}{8}\frac{\hbar\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_2}}}{4\epsilon}P_R e_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}$	$+H_{a_1}H_{a_2}S_{1,j_1}\mathcal{I}_{T3,a_1,a_2,j_1}\lambda_{T3}S_2^\dagger$ $-\frac{\hbar\overline{y_{1,i_1,r_1}}H_{a_1}H_{a_2}S_{1,j_1}y_{1,i_1,r_1}\mathcal{I}_{T3,a_1,a_2,j_1}\lambda_{T3}S_2^\dagger}{\sqrt{3}\epsilon}$ $-\frac{\hbar\overline{y_{1,i_1,r_1}}H_{a_1}H_{a_2}S_{1,j_1}y_{1,i_1,r_1}\mathcal{I}_{T3,a_1,a_2,j_1}\lambda_{T3}S_2^\dagger}{2\sqrt{3}}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$FW_{\mu\nu}$	$H\ell e$	S_1S_2H
cont.									$+\frac{3}{8}\overline{hy_{2,i_3,r_1}}\overline{L_{a_1,i_2}}P_R e_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}$ $+\frac{\sqrt{3}\overline{hy_{1,i_1,r_1}}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}P_L L_{a_1,i_1}y_{1,i_3,r_1}H_{a_1}^\dagger}{4\epsilon}$ $+\frac{3}{8}\sqrt{3}\overline{hy_{1,i_1,r_1}}Y_{e,i_3,i_2}\overline{e_{i_2}}P_L L_{a_1,i_1}y_{1,i_3,r_1}H_{a_1}^\dagger$ $+\frac{\overline{hy_{2,i_1,r_1}}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}P_L L_{a_1,i_1}y_{2,i_3,r_1}H_{a_1}^\dagger}{4\epsilon}$	$-\frac{\overline{hy_{2,i_1,r_1}}H_{a_1}H_{a_2}S_{1,j_1}y_{2,i_1,r_1}\mathcal{I}_{T3,a_1,a_2,j_1}\lambda_{T3}S_2^\dagger}{\epsilon}$ $-\frac{1}{2}\overline{hy_{2,i_1,r_1}}H_{a_1}H_{a_2}S_{1,j_1}y_{2,i_1,r_1}\mathcal{I}_{T3,a_1,a_2,j_1}\lambda_{T3}S_2^\dagger$ $+\overline{\lambda_{T3}}S_2\mathcal{I}_{T3}^*S_{a_1,a_2,j_1}H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger$ $-\frac{\overline{hy_{1,i_1,r_1}}\overline{\lambda_{T3}}S_2y_{1,i_1,r_1}\mathcal{I}_{T3}^*S_{a_1,a_2,j_1}H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger}{\sqrt{3}\epsilon}$
	cont.								$+\frac{3}{8}\overline{hy_{2,i_1,r_1}}Y_{e,i_3,i_2}\overline{e_{i_2}}P_L L_{a_1,i_1}y_{2,i_3,r_1}H_{a_1}^\dagger$ $+\frac{\overline{hy_{2,i_1,r_1}}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}P_L L_{a_1,i_1}y_{2,i_3,r_1}H_{a_1}^\dagger}{4\epsilon}$	$-\frac{\overline{hy_{1,i_1,r_1}}\overline{\lambda_{T3}}S_2y_{1,i_1,r_1}\mathcal{I}_{T3}^*S_{a_1,a_2,j_1}H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger}{\sqrt{3}\epsilon}$ $-\frac{\overline{hy_{2,i_1,r_1}}\overline{\lambda_{T3}}S_2y_{2,i_1,r_1}\mathcal{I}_{T3}^*S_{a_1,a_2,j_1}H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger}{2\sqrt{3}}$
	cont.									$+\frac{3}{8}\overline{hy_{2,i_1,r_1}}Y_{e,i_3,i_2}\overline{e_{i_2}}P_L L_{a_1,i_1}y_{2,i_3,r_1}H_{a_1}^\dagger$ $+\frac{3}{8}\overline{hy_{2,i_1,r_1}}Y_{e,i_3,i_2}\overline{e_{i_2}}P_L L_{a_1,i_1}y_{2,i_3,r_1}H_{a_1}^\dagger$
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	$-\frac{1}{2}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\left(W_{\mu_1\mu_2}^{A_1}\right)^2$	$+\frac{\sqrt{3}\overline{hy_{1,i_3,r_1}}\overline{L_{a_1,i_2}}P_R e_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}}{4\epsilon}$ $+\frac{3}{8}\sqrt{3}\overline{hy_{1,i_3,r_1}}\overline{L_{a_1,i_2}}P_R e_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}$ $+\frac{\sqrt{3}\overline{hy_{2,i_3,r_1}}\overline{L_{a_1,i_2}}P_R e_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}}{4\epsilon}$ $+\frac{3}{8}\sqrt{3}\overline{hy_{2,i_3,r_1}}\overline{L_{a_1,i_2}}P_R e_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}$ $+\frac{\sqrt{3}\overline{hy_{1,i_1,r_1}}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}P_L L_{a_1,i_1}y_{1,i_3,r_1}H_{a_1}^\dagger}{4\epsilon}$ $+\frac{3}{8}\sqrt{3}\overline{hy_{1,i_1,r_1}}Y_{e,i_3,i_2}\overline{e_{i_2}}P_L L_{a_1,i_1}y_{1,i_3,r_1}H_{a_1}^\dagger$ $+\frac{\sqrt{3}\overline{hy_{2,i_1,r_1}}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}P_L L_{a_1,i_1}y_{2,i_3,r_1}H_{a_1}^\dagger}{4\epsilon}$ $+\frac{3}{8}\sqrt{3}\overline{hy_{2,i_1,r_1}}Y_{e,i_3,i_2}\overline{e_{i_2}}P_L L_{a_1,i_1}y_{2,i_3,r_1}H_{a_1}^\dagger$	$+\overline{\lambda_{T3}}S_{2,j_2}\mathcal{I}_{T3}^*S_{a_1,a_2,j_1,j_2}H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger$ $-\frac{\overline{hy_{1,i_1,r_1}}\overline{\lambda_{T3}}S_{2,j_2}y_{1,i_1,r_1}\mathcal{I}_{T3}^*S_{a_1,a_2,j_1,j_2}H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger}{\sqrt{3}\epsilon}$ $-\frac{\overline{hy_{1,i_1,r_1}}\overline{\lambda_{T3}}S_{2,j_2}y_{1,i_1,r_1}\mathcal{I}_{T3}^*S_{a_1,a_2,j_1,j_2}H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger}{2\sqrt{3}}$ $-\frac{\overline{hy_{2,i_1,r_1}}\overline{\lambda_{T3}}S_{2,j_2}y_{2,i_1,r_1}\mathcal{I}_{T3}^*S_{a_1,a_2,j_1,j_2}H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger}{\sqrt{3}\epsilon}$ $-\frac{\overline{hy_{2,i_1,r_1}}\overline{\lambda_{T3}}S_{2,j_2}y_{2,i_1,r_1}\mathcal{I}_{T3}^*S_{a_1,a_2,j_1,j_2}H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger}{2\sqrt{3}}$ $+\frac{H_{a_1}H_{a_2}S_{1,j_2}\mathcal{I}_{T3,a_1,a_2,j_2,j_1}\lambda_{T3}S_{2,j_1}^\dagger}{\overline{hy_{1,i_1,r_1}}H_{a_1}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}\mathcal{I}_{T3,a_1,a_2,j_2,j_1}\lambda_{T3}S_{2,j_1}^\dagger}$ $-\frac{\overline{hy_{1,i_1,r_1}}H_{a_1}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}\mathcal{I}_{T3,a_1,a_2,j_2,j_1}\lambda_{T3}S_{2,j_1}^\dagger}{\sqrt{3}\epsilon}$ $-\frac{\overline{hy_{1,i_1,r_1}}H_{a_1}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}\mathcal{I}_{T3,a_1,a_2,j_2,j_1}\lambda_{T3}S_{2,j_1}^\dagger}{2\sqrt{3}}$ $-\frac{\overline{hy_{2,i_1,r_1}}H_{a_1}H_{a_2}S_{1,j_2}y_{2,i_1,r_1}\mathcal{I}_{T3,a_1,a_2,j_2,j_1}\lambda_{T3}S_{2,j_1}^\dagger}{\sqrt{3}\epsilon}$ $-\frac{\overline{hy_{2,i_1,r_1}}H_{a_1}H_{a_2}S_{1,j_2}y_{2,i_1,r_1}\mathcal{I}_{T3,a_1,a_2,j_2,j_1}\lambda_{T3}S_{2,j_1}^\dagger}{2\sqrt{3}}$
	cont.								$+\frac{\sqrt{3}\overline{hy_{1,i_3,r_1}}\overline{L_{a_1,i_2}}P_R e_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}}{4\epsilon}$ $+\frac{3}{8}\sqrt{3}\overline{hy_{1,i_3,r_1}}\overline{L_{a_1,i_2}}P_R e_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}$ $+\frac{\sqrt{3}\overline{hy_{2,i_3,r_1}}\overline{L_{a_1,i_2}}P_R e_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}}{4\epsilon}$ $+\frac{3}{8}\sqrt{3}\overline{hy_{2,i_3,r_1}}\overline{L_{a_1,i_2}}P_R e_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}$ $+\frac{\sqrt{3}\overline{hy_{1,i_1,r_1}}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}P_L L_{a_1,i_1}y_{1,i_3,r_1}H_{a_1}^\dagger}{4\epsilon}$ $+\frac{3}{8}\sqrt{3}\overline{hy_{1,i_1,r_1}}Y_{e,i_3,i_2}\overline{e_{i_2}}P_L L_{a_1,i_1}y_{1,i_3,r_1}H_{a_1}^\dagger$ $+\frac{\sqrt{3}\overline{hy_{2,i_1,r_1}}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}P_L L_{a_1,i_1}y_{2,i_3,r_1}H_{a_1}^\dagger}{4\epsilon}$ $+\frac{3}{8}\sqrt{3}\overline{hy_{2,i_1,r_1}}Y_{e,i_3,i_2}\overline{e_{i_2}}P_L L_{a_1,i_1}y_{2,i_3,r_1}H_{a_1}^\dagger$	$+\overline{\lambda_{T3}}S_{2,j_2}\mathcal{I}_{T3}^*S_{a_1,a_2,j_1,j_2}H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger$ $-\frac{\overline{hy_{1,i_1,r_1}}\overline{\lambda_{T3}}S_{2,j_2}y_{1,i_1,r_1}\mathcal{I}_{T3}^*S_{a_1,a_2,j_1,j_2}H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger}{\sqrt{3}\epsilon}$ $-\frac{\overline{hy_{1,i_1,r_1}}\overline{\lambda_{T3}}S_{2,j_2}y_{1,i_1,r_1}\mathcal{I}_{T3}^*S_{a_1,a_2,j_1,j_2}H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger}{2\sqrt{3}}$ $-\frac{\overline{hy_{2,i_1,r_1}}\overline{\lambda_{T3}}S_{2,j_2}y_{2,i_1,r_1}\mathcal{I}_{T3}^*S_{a_1,a_2,j_1,j_2}H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger}{\sqrt{3}\epsilon}$ $-\frac{\overline{hy_{2,i_1,r_1}}\overline{\lambda_{T3}}S_{2,j_2}y_{2,i_1,r_1}\mathcal{I}_{T3}^*S_{a_1,a_2,j_1,j_2}H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger}{2\sqrt{3}}$ $+\frac{H_{a_1}H_{a_2}S_{1,j_2}\mathcal{I}_{T3,a_1,a_2,j_2,j_1}\lambda_{T3}S_{2,j_1}^\dagger}{\overline{hy_{1,i_1,r_1}}H_{a_1}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}\mathcal{I}_{T3,a_1,a_2,j_2,j_1}\lambda_{T3}S_{2,j_1}^\dagger}$ $-\frac{\overline{hy_{1,i_1,r_1}}H_{a_1}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}\mathcal{I}_{T3,a_1,a_2,j_2,j_1}\lambda_{T3}S_{2,j_1}^\dagger}{\sqrt{3}\epsilon}$ $-\frac{\overline{hy_{1,i_1,r_1}}H_{a_1}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}\mathcal{I}_{T3,a_1,a_2,j_2,j_1}\lambda_{T3}S_{2,j_1}^\dagger}{2\sqrt{3}}$ $-\frac{\overline{hy_{2,i_1,r_1}}H_{a_1}H_{a_2}S_{1,j_2}y_{2,i_1,r_1}\mathcal{I}_{T3,a_1,a_2,j_2,j_1}\lambda_{T3}S_{2,j_1}^\dagger}{\sqrt{3}\epsilon}$ $-\frac{\overline{hy_{2,i_1,r_1}}H_{a_1}H_{a_2}S_{1,j_2}y_{2,i_1,r_1}\mathcal{I}_{T3,a_1,a_2,j_2,j_1}\lambda_{T3}S_{2,j_1}^\dagger}{2\sqrt{3}}$
	cont.									$+\frac{\sqrt{3}\overline{hy_{1,i_3,r_1}}\overline{L_{a_1,i_2}}P_R e_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}}{4\epsilon}$ $+\frac{3}{8}\sqrt{3}\overline{hy_{1,i_3,r_1}}\overline{L_{a_1,i_2}}P_R e_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}$ $+\frac{\sqrt{3}\overline{hy_{2,i_3,r_1}}\overline{L_{a_1,i_2}}P_R e_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}}{4\epsilon}$ $+\frac{3}{8}\sqrt{3}\overline{hy_{2,i_3,r_1}}\overline{L_{a_1,i_2}}P_R e_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}$ $+\frac{\sqrt{3}\overline{hy_{1,i_1,r_1}}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}P_L L_{a_1,i_1}y_{1,i_3,r_1}H_{a_1}^\dagger}{4\epsilon}$ $+\frac{3}{8}\sqrt{3}\overline{hy_{1,i_1,r_1}}Y_{e,i_3,i_2}\overline{e_{i_2}}P_L L_{a_1,i_1}y_{1,i_3,r_1}H_{a_1}^\dagger$ $+\frac{\sqrt{3}\overline{hy_{2,i_1,r_1}}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}P_L L_{a_1,i_1}y_{2,i_3,r_1}H_{a_1}^\dagger}{4\epsilon}$ $+\frac{3}{8}\sqrt{3}\overline{hy_{2,i_1,r_1}}Y_{e,i_3,i_2}\overline{e_{i_2}}P_L L_{a_1,i_1}y_{2,i_3,r_1}H_{a_1}^\dagger$
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$-\frac{1}{2}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\left(W_{\mu_1\mu_2}^{A_1}\right)^2$	$+\frac{\sqrt{3}\overline{hy_{1,i_3,r_1}}\overline{L_{a_1,i_2}}P_R e_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}}{4\epsilon}$ $+\frac{3}{8}\sqrt{3}\overline{hy_{1,i_3,r_1}}\overline{L_{a_1,i_2}}P_R e_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}$ $+\frac{\sqrt{3}\overline{hy_{2,i_3,r_1}}\overline{L_{a_1,i_2}}P_R e_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}}{4\epsilon}$ $+\frac{3}{8}\sqrt{3}\overline{hy_{2,i_3,r_1}}\overline{L_{a_1,i_2}}P_R e_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}$ $+\frac{\sqrt{3}\overline{hy_{1,i_1,r_1}}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}P_L L_{a_1,i_1}y_{1,i_3,r_1}H_{a_1}^\dagger}{4\epsilon}$ $+\frac{3}{8}\sqrt{3}\overline{hy_{1,i_1,r_1}}Y_{e,i_3,i_2}\overline{e_{i_2}}P_L L_{a_1,i_1}y_{1,i_3,r_1}H_{a_1}^\dagger$ $+\frac{\sqrt{3}\overline{hy_{2,i_1,r_1}}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}P_L L_{a_1,i_1}y_{2,i_3,r_1}H_{a_1}^\dagger}{4\epsilon}$ $+\frac{3}{8}\sqrt{3}\overline{hy_{2,i_1,r_1}}Y_{e,i_3,i_2}\overline{e_{i_2}}P_L L_{a_1,i_1}y_{2,i_3,r_1}H_{a_1}^\dagger$	$+\overline{\lambda_{T3}}S_{2,j_2}\mathcal{I}_{T3}^*S_{a_1,a_2,j_1,j_2}H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger$ $-\frac{\overline{hy_{1,i_1,r_1}}\overline{\lambda_{T3}}S_{2,j_2}y_{1,i_1,r_1}\mathcal{I}_{T3}^*S_{a_1,a_2,j_1,j_2}H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger}{\sqrt{3}\epsilon}$ $-\frac{\overline{hy_{1,i_1,r_1}}\overline{\lambda_{T3}}S_{2,j_2}y_{1,i_1,r_1}\mathcal{I}_{T3}^*S_{a_1,a_2,j_1,j_2}H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger}{2\sqrt{3}}$ $-\frac{\overline{hy_{2,i_1,r_1}}\overline{\lambda_{T3}}S_{2,j_2}y_{2,i_1,r_1}\mathcal{I}_{T3}^*S_{a_1,a_2,j_1,j_2}H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger}{\sqrt{3}\epsilon}$ $-\frac{\overline{hy_{2,i_1,r_1}}\overline{\lambda_{T3}}S_{2,j_2}y_{2,i_1,r_1}\mathcal{I}_{T3}^*S_{a_1,a_2,j_1,j_2}H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger}{2\sqrt{3}}$ $+\frac{H_{a_1}H_{a_2}S_{1,j_2}\mathcal{I}_{T3,a_1,a_2,j_2,j_1}\lambda_{T3}S_{2,j_1}^\dagger}{\overline{hy_{1,i_1,r_1}}H_{a_1}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}\mathcal{I}_{T3,a_1,a_2,j_2,j_1}\lambda_{T3}S_{2,j_1}^\dagger}$ $-\frac{\overline{hy_{1,i_1,r_1}}H_{a_1}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}\mathcal{I}_{T3,a_1,a_2,j_2,j_1}\lambda_{T3}S_{2,j_1}^\dagger}{\sqrt{3}\epsilon}$ $-\frac{\overline{hy_{1,i_1,r_1}}H_{a_1}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}\mathcal{I}_{T3,a_1,a_2,j_2,j_1}\lambda_{T3}S_{2,j_1}^\dagger}{2\sqrt{3}}$ $-\frac{\overline{hy_{2,i_1,r_1}}H_{a_1}H_{a_2}S_{1,j_2}y_{2,i_1,r_1}\mathcal{I}_{T3,a_1,a_2,j_2,j_1}\lambda_{T3}S_{2,j_1}^\dagger}{\sqrt{3}\epsilon}$ $-\frac{\overline{hy_{2,i_1,r_1}}H_{a_1}H_{a_2}S_{1,j_2}y_{2,i_1,r_1}\mathcal{I}_{T3,a_1,a_2,j_2,j_1}\lambda_{T3}S_{2,j_1}^\dagger}{2\sqrt{3}}$
	cont.								$+\frac{\sqrt{3}\overline{hy_{1,i_3,r_1}}\overline{L_{a_1,i_2}}P_R e_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}}{4\epsilon}$ $+\frac{3}{8}\sqrt{3}\overline{hy_{1,i_3,r_1}}\overline{L_{a_1,i_2}}P_R e_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}$ $+\frac{\sqrt{3}\overline{hy_{2,i_3,r_1}}\overline{L_{a_1,i_2}}P_R e_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}}{4\epsilon}$ $+\frac{3}{8}\sqrt{3}\overline{hy_{2,i_3,r_1}}\overline{L_{a_1,i_2}}P_R e_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}$ $+\frac{\sqrt{3}\overline{hy_{1,i_1,r_1}}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}P_L L_{a_1,i_1}y_{1,i_3,r_1}H_{a_1}^\dagger}{4\epsilon}$ $+\frac{3}{8}\sqrt{3}\overline{hy_{1,i_1,r_1}}Y_{e,i_3,i_2}\overline{e_{i_2}}P_L L_{a_1,i_1}y_{1,i_3,r_1}H_{a_1}^\dagger$ $+\frac{\sqrt{3}\overline{hy_{2,i_1,r_1}}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}P_L L_{a_1,i_1}y_{2,i_3,r_1}H_{a_1}^\dagger}{4\epsilon}$ $+\frac{3}{8}\sqrt{3}\overline{hy_{2,i_1,r_1}}Y_{e,i_3,i_2}\overline{e_{i_2}}P_L L_{a_1,i_1}y_{2,i_3,r_1}H_{a_1}^\dagger$	$+\overline{\lambda_{T3}}S_{2,j_2}\mathcal{I}_{T3}^*S_{a_1,a_2,j_1,j_2}H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger$ $-\frac{\overline{hy_{1,i_1,r_1}}\overline{\lambda_{T3}}S_{2,j_2}y_{1,i_1,r_1}\mathcal{I}_{T3}^*S_{a_1,a_2,j_1,j_2}H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger}{\sqrt{3}\epsilon}$ $-\frac{\overline{hy_{1,i_1,r_1}}\overline{\lambda_{T3}}S_{2,j_2}y_{1,i_1,r_1}\mathcal{I}_{T3}^*S_{a_1,a_2,j_1,j_2}H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger}{2\sqrt{3}}$ $-\frac{\overline{hy_{2,i_1,r_1}}\overline{\lambda_{T3}}S_{2,j_2}y_{2,i_1,r_1}\mathcal{I}_{T3}^*S_{a_1,a_2,j_1,j_2}H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger}{\sqrt{3}\epsilon}$ $-\frac{\overline{hy_{2,i_1,r_1}}\overline{\lambda_{T3}}S_{2,j_2}y_{2,i_1,r_1}\mathcal{I}_{T3}^*S_{a_1,a_2,j_1,j_2}H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger}{2\sqrt{3}}$ $+\frac{H_{a_1}H_{a_2}S_{1,j_2}\mathcal{I}_{T3,a_1,a_2,j_2,j_1}\lambda_{T3}S_{2,j_1}^\dagger}{\overline{hy_{1,i_1,r_1}}H_{a_1}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}\mathcal{I}_{T3,a_1,a_2,j_2,j_1}\lambda_{T3}S_{2,j_1}^\dagger}$ $-\frac{\overline{hy_{1,i_1,r_1}}H_{a_1}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}\mathcal{I}_{T3,a_1,a_2,j_2,j_1}\lambda_{T3}S_{2,j_1}^\dagger}{\sqrt{3}\epsilon}$ $-\frac{\overline{hy_{1,i_1,r_1}}H_{a_1}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}\mathcal{I}_{T3,a_1,a_2,j_2,j_1}\lambda_{T3}S_{2,j_1}^\dagger}{2\sqrt{3}}$ $-\frac{\overline{hy_{2,i_1,r_1}}H_{a_1}H_{a_2}S_{1,j_2}y_{2,i_1,r_1}\mathcal{I}_{T3,a_1,a_2,j_2,j_1}\lambda_{T3}S_{2,j_1}^\dagger}{\sqrt{3}\epsilon}$ $-\frac{\overline{hy_{2,i_1,r_1}}H_{a_1}H_{a_2}S_{1,j_2}y_{2,i_1,r_1}\mathcal{I}_{T3,a_1,a_2,j_2,j_1}\lambda_{T3}S_{2,j_1}^\dagger}{2\sqrt{3}}$
	cont.									$+\frac{\sqrt{3}\overline{hy_{1,i_3,r_1}}\overline{L_{a_1,i_2}}P_R e_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}}{4\epsilon}$ $+\frac{3}{8}\sqrt{3}\overline{hy_{1,i_3,r_1}}\overline{L_{a_1,i_2}}P_R e_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}$ $+\frac{\sqrt{3}\overline{hy_{2,i_3,r_1}}\overline{L_{a_1,i_2}}P_R e_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}}{4\epsilon}$ $+\frac{3}{8}\sqrt{3}\overline{hy_{2,i_3,r_1}}\overline{L_{a_1,i_2}}P_R e_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}$ $+\frac{\sqrt{3}\overline{hy_{1,i_1,r_1}}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}P_L L_{a_1,i_1}y_{1,i_3,r_1}H_{a_1}^\dagger}{4\epsilon}$ $+\frac{3}{8}\sqrt{3}\overline{hy_{1,i_1,r_1}}Y_{e,i_3,i_2}\overline{e_{i_2}}P_L L_{a_1,i_1}y_{1,i_3,r_1}H_{a_1}^\dagger$ $+\frac{\sqrt{3}\overline{hy_{2,i_1,r_1}}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}P_L L_{a_1,i_1}y_{2,i_3,r_1}H_{a_1}^\dagger}{4\epsilon}$ $+\frac{3}{8}\sqrt{3}\overline{hy_{2,i_1,r_1}}Y_{e,i_3,i_2}\overline{e_{i_2}}P_L L_{a_1,i_1}y_{2,i_3,r_1}H_{a_1}^\dagger$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{1}{2}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\left(W_{\mu_1\mu_2}^{A_1}\right)^2$	$+\frac{\sqrt{3}\$	

[illegible]

Field combinations 7 of 10

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$S_1 S_2 F$	$S_1 S_2 \ell$	$S_1 F H$
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	$-\frac{4\hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_2}} S_1 S_{2,j_1} y_{1,i_2,r_1} y_{2,i_1,r_2} S_1^\dagger S_{2,j_1}^\dagger}{\sqrt{3}}$ $-2\hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{1,i_1,r_1} S_1 S_{2,j_2} y_{1,i_1,r_1} S_1^\dagger S_{2,j_1}^\dagger} \mathcal{I}_{12,j_2,j_1}^{-1} \lambda_{12}^{(1)}$ $-2\hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{2,i_1,r_1} S_1 S_{2,j_2} y_{2,i_1,r_1} S_1^\dagger S_{2,j_1}^\dagger} \mathcal{I}_{12,j_2,j_1}^{-1} \lambda_{12}^{(1)}$	$+\frac{\overline{y_{1,i_1,r_1} y_{2,i_2,r_1} L_{a_2,i_2}}}{M_F} \mathcal{C} P_L L_{a_1,i_1} S_{2,j_1} \mathcal{I} T_{3,a_1,a_2,j_1} S_1^\dagger$ $-\frac{\hbar y_{1,i_1,r_1} \overline{y_{1,i_3,r_2} y_{2,i_2,r_2}}}{4\epsilon M_F} \mathcal{C} P_L L_{a_1,i_1} S_{2,j_1} y_{1,i_3,r_1} \mathcal{I} T_{3,a_1,a_2,j_1} S_1^\dagger$ $-\frac{3\hbar \overline{y_{1,i_1,r_1} y_{1,i_3,r_2} y_{2,i_2,r_2}}}{8M_F} \mathcal{C} P_L L_{a_1,i_1} S_{2,j_1} y_{1,i_3,r_1} \mathcal{I} T_{3,a_1,a_2,j_1} S_1^\dagger$	$-2\hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{1,i_1,r_1} Y_{e,i_2,i_3}} H_{a_1} S_1 y_{1,i_2,r_1} Y_{e,i_1,i_3} H_{a_1}^\dagger S_1^\dagger$ $-2\hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{1,i_1,r_1} H_{a_2} S_1 y_{1,i_1,r_1} H_{a_1}^\dagger} S_1^\dagger \mathcal{I} H_{1,a_1,a_2}^{-1} \lambda_{H1}^{(1)}$
cont.								$-\frac{\hbar \overline{y_{1,i_1,r_1} y_{1,i_3,r_2} y_{2,i_2,r_1} L_{a_2,i_2}}}{\epsilon M_F} \mathcal{C} P_L L_{a_1,i_1} S_{2,j_1} y_{1,i_3,r_2} \mathcal{I} T_{3,a_1,a_2,j_1} S_1^\dagger$ $-\frac{\hbar y_{1,i_1,r_1} \overline{y_{1,i_3,r_2} y_{2,i_2,r_1} L_{a_2,i_2}}}{2M_F} \mathcal{C} P_L L_{a_1,i_1} S_{2,j_1} y_{1,i_3,r_2} \mathcal{I} T_{3,a_1,a_2,j_1} S_1^\dagger$ $-\frac{\hbar y_{1,i_1,r_1} \overline{y_{1,i_2,r_2} y_{2,i_3,r_1} L_{a_2,i_2}}}{4\epsilon M_F} \mathcal{C} P_L L_{a_1,i_1} S_{2,j_1} y_{1,i_3,r_2} \mathcal{I} T_{3,a_1,a_2,j_1} S_1^\dagger$		
cont.								$-\frac{3\hbar \overline{y_{1,i_1,r_1} y_{1,i_2,r_2} y_{2,i_3,r_1} L_{a_2,i_2}}}{8M_F} \mathcal{C} P_L L_{a_1,i_1} S_{2,j_1} y_{1,i_3,r_2} \mathcal{I} T_{3,a_1,a_2,j_1} S_1^\dagger$ $-\frac{\sqrt{3}\hbar \overline{y_{1,i_3,r_1} y_{2,i_1,r_2} y_{2,i_2,r_1} L_{a_2,i_2}}}{4\epsilon M_F} \mathcal{C} P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \mathcal{I} T_{3,a_1,a_2,j_1} S_1^\dagger$		
cont.								$-\frac{3\sqrt{3}\hbar \overline{y_{1,i_3,r_1} y_{2,i_1,r_2} y_{2,i_2,r_1} L_{a_2,i_2}}}{8M_F} \mathcal{C} P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \mathcal{I} T_{3,a_1,a_2,j_1} S_1^\dagger$ $-\frac{\sqrt{3}\hbar \overline{y_{1,i_1,r_1} y_{2,i_2,r_2} y_{2,i_3,r_1} L_{a_2,i_2}}}{4\epsilon M_F} \mathcal{C} P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \mathcal{I} T_{3,a_1,a_2,j_1} S_1^\dagger$		
cont.								$-\frac{\hbar \overline{y_{1,i_1,r_1} y_{2,i_2,r_2} y_{2,i_3,r_1} L_{a_2,i_2}}}{\sqrt{3}\epsilon M_F} \mathcal{C} P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \mathcal{I} T_{3,a_1,a_2,j_1} S_1^\dagger$ $-\frac{\hbar y_{1,i_1,r_1} \overline{y_{2,i_2,r_2} y_{2,i_3,r_1} L_{a_2,i_2}}}{2\sqrt{3}M_F} \mathcal{C} P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \mathcal{I} T_{3,a_1,a_2,j_1} S_1^\dagger$		
cont.								$-\frac{\hbar \overline{y_{1,i_3,r_1} L_{a_1,i_1}}}{4\epsilon M_F} \mathcal{C} P_R \overline{L_{a_2,i_2}}^T S_1 y_{1,i_2,r_1} y_{1,i_3,r_2} y_{2,i_1,r_2} \mathcal{I} T_{3,a_1,a_2,j_1} S_{2,j_1}^\dagger$ $-\frac{3\hbar \overline{y_{1,i_3,r_1} L_{a_1,i_1}}}{8M_F} \mathcal{C} P_R \overline{L_{a_2,i_2}}^T S_1 y_{1,i_2,r_1} y_{1,i_3,r_2} y_{2,i_1,r_2} \mathcal{I} T_{3,a_1,a_2,j_1} S_{2,j_1}^\dagger$ $+\frac{\overline{L_{a_1,i_1}}}{M_F} \mathcal{C} P_R \overline{L_{a_2,i_2}}^T S_1 y_{1,i_1,r_1} y_{2,i_2,r_2} \mathcal{I} T_{3,a_1,a_2,j_1} S_{2,j_1}^\dagger$		
cont.								$-\frac{\sqrt{3}\hbar \overline{y_{2,i_3,r_1} L_{a_1,i_1}}}{4\epsilon M_F} \mathcal{C} P_R \overline{L_{a_2,i_2}}^T S_1 y_{1,i_3,r_2} y_{2,i_1,r_2} y_{2,i_2,r_1} \mathcal{I} T_{3,a_1,a_2,j_1} S_{2,j_1}^\dagger$ $-\frac{3\sqrt{3}\hbar \overline{y_{2,i_3,r_1} L_{a_1,i_1}}}{8M_F} \mathcal{C} P_R \overline{L_{a_2,i_2}}^T S_1 y_{1,i_3,r_2} y_{2,i_1,r_2} y_{2,i_2,r_1} \mathcal{I} T_{3,a_1,a_2,j_1} S_{2,j_1}^\dagger$ $-\frac{\hbar \overline{y_{1,i_3,r_1} L_{a_1,i_1}}}{\epsilon M_F} \mathcal{C} P_R \overline{L_{a_2,i_2}}^T S_1 y_{1,i_1,r_2} y_{1,i_3,r_1} y_{2,i_2,r_2} \mathcal{I} T_{3,a_1,a_2,j_1} S_{2,j_1}^\dagger$		
cont.								$-\frac{\hbar \overline{y_{1,i_3,r_1} L_{a_1,i_1}}}{2M_F} \mathcal{C} P_R \overline{L_{a_2,i_2}}^T S_1 y_{1,i_1,r_2} y_{1,i_3,r_1} y_{2,i_2,r_2} \mathcal{I} T_{3,a_1,a_2,j_1} S_{2,j_1}^\dagger$ $-\frac{\hbar \overline{y_{2,i_3,r_1} L_{a_1,i_1}}}{\sqrt{3}\epsilon M_F} \mathcal{C} P_R \overline{L_{a_2,i_2}}^T S_1 y_{1,i_1,r_2} y_{2,i_2,r_2} y_{2,i_3,r_1} \mathcal{I} T_{3,a_1,a_2,j_1} S_{2,j_1}^\dagger$ $-\frac{\hbar \overline{y_{2,i_3,r_1} L_{a_1,i_1}}}{2\sqrt{3}M_F} \mathcal{C} P_R \overline{L_{a_2,i_2}}^T S_1 y_{1,i_1,r_2} y_{2,i_2,r_2} y_{2,i_3,r_1} \mathcal{I} T_{3,a_1,a_2,j_1} S_{2,j_1}^\dagger$		
cont.								$-\frac{\hbar \overline{y_{1,i_3,r_1} L_{a_1,i_1}}}{4\epsilon M_F} \mathcal{C} P_R \overline{L_{a_2,i_2}}^T S_1 y_{1,i_1,r_2} y_{1,i_3,r_1} y_{2,i_2,r_2} \mathcal{I} T_{3,a_1,a_2,j_1} S_{2,j_1}^\dagger$ $-\frac{3\hbar \overline{y_{1,i_3,r_1} L_{a_1,i_1}}}{8M_F} \mathcal{C} P_R \overline{L_{a_2,i_2}}^T S_1 y_{1,i_1,r_2} y_{1,i_3,r_1} y_{2,i_2,r_2} \mathcal{I} T_{3,a_1,a_2,j_1} S_{2,j_1}^\dagger$ $-\frac{\sqrt{3}\hbar \overline{y_{2,i_3,r_1} L_{a_1,i_1}}}{4\epsilon M_F} \mathcal{C} P_R \overline{L_{a_2,i_2}}^T S_1 y_{1,i_1,r_2} y_{2,i_2,r_2} y_{2,i_3,r_1} \mathcal{I} T_{3,a_1,a_2,j_1} S_{2,j_1}^\dagger$		
cont.								$-\frac{3\sqrt{3}\hbar \overline{y_{2,i_3,r_1} L_{a_1,i_1}}}{8M_F} \mathcal{C} P_R \overline{L_{a_2,i_2}}^T S_1 y_{1,i_1,r_2} y_{2,i_2,r_2} y_{2,i_3,r_1} \mathcal{I} T_{3,a_1,a_2,j_1} S_{2,j_1}^\dagger$ $-\frac{\hbar \overline{y_{1,i_1,r_1} y_{2,i_2,r_1} L_{a_2,i_2}}}{\epsilon M_F} \mathcal{C} P_L L_{a_1,i_1} S_{2,j_1} \mathcal{I} T_{3,a_1,a_2,j_1} S_1^\dagger \lambda_{12}^{(1)}$ $-\frac{\hbar \overline{y_{1,i_1,r_1} y_{2,i_2,r_1} L_{a_2,i_2}}}{M_F} \mathcal{C} P_L L_{a_1,i_1} S_{2,j_1} \mathcal{I} T_{3,a_1,a_2,j_1} S_1^\dagger \lambda_{12}^{(1)}$		
cont.								$-\frac{\hbar \overline{L_{a_1,i_1}}}{\epsilon M_F} \mathcal{C} P_R \overline{L_{a_2,i_2}}^T S_1 y_{1,i_1,r_1} y_{2,i_2,r_1} \mathcal{I} T_{3,a_1,a_2,j_1} S_{2,j_1}^\dagger \lambda_{12}^{(1)}$ $-\frac{\hbar \overline{L_{a_1,i_1}}}{\epsilon M_F} \mathcal{C} P_R \overline{L_{a_2,i_2}}^T S_1 y_{1,i_1,r_1} y_{2,i_2,r_1} \mathcal{I} T_{3,a_1,a_2,j_1} S_{2,j_1}^\dagger \lambda_{12}^{(1)}$		
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$-\frac{4\hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_2}} S_1 S_{2,j_1} y_{1,i_2,r_1} y_{2,i_1,r_2} S_1^\dagger S_{2,j_1}^\dagger}{\sqrt{3}}$ $-2\hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{1,i_1,r_1} S_1 S_{2,j_2} y_{1,i_1,r_1} S_1^\dagger S_{2,j_1}^\dagger} \mathcal{I}_{12,j_2,j_1}^{-1} \lambda_{12}^{(1)}$ $-2\hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{2,i_1,r_1} S_1 S_{2,j_2} y_{2,i_1,r_1} S_1^\dagger S_{2,j_1}^\dagger} \mathcal{I}_{12,j_2,j_1}^{-1} \lambda_{12}^{(1)}$	$3\hbar \overline{y_{1,i_1,r_1} y_{2,i_2,r_1} L_{a_2,i_2}} \mathcal{C} P_L L_{a_1,i_1} S_{2,j_1} \mathcal{I} T_{3,a_1,a_2,j_1} S_1^\dagger g_Y^2$ $+\frac{\hbar y_{1,i_1,r_1} \overline{y_{1,i_3,r_2} y_{2,i_2,r_1} L_{a_2,i_2}}}{4\epsilon M_F} \mathcal{C} P_L L_{a_1,i_1} S_{2,j_1} \mathcal{I} T_{3,a_1,a_2,j_1} S_1^\dagger g_Y^2$ $+\frac{3\hbar \overline{L_{a_1,i_1}}}{4\epsilon M_F} \mathcal{C} P_R \overline{L_{a_2,i_2}}^T S_1 y_{1,i_1,r_1} y_{2,i_2,r_1} \mathcal{I} T_{3,a_1,a_2,j_1} S_{2,j_1}^\dagger g_Y^2$	$-2\hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{1,i_1,r_1} Y_{e,i_2,i_3}} H_{a_1} S_1 y_{1,i_2,r_1} Y_{e,i_1,i_3} H_{a_1}^\dagger S_1^\dagger$ $-2\hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{1,i_1,r_1} H_{a_2} S_1 y_{1,i_1,r_1} H_{a_1}^\dagger} S_1^\dagger \mathcal{I} H_{1,a_1,a_2}^{-1} \lambda_{H1}^{(1)}$
cont.								$+\frac{\hbar \overline{L_{a_1,i_1}}}{4M_F} \mathcal{C} P_R \overline{L_{a_2,i_2}}^T S_1 y_{1,i_1,r_1} y_{2,i_2,r_1} \mathcal{I} T_{3,a_1,a_2,j_1} S_{2,j_1}^\dagger g_Y^2$ $+\frac{\overline{y_{1,i_1,r_1} y_{2,i_2,r_1} L_{a_2,i_2}}}{M_F} \mathcal{C} P_L L_{a_1,i_1} S_{2,j_1} \mathcal{I} T_{3,a_1,a_2,j_1} S_1^\dagger$ $-\frac{\hbar \overline{y_{1,i_1,r_1} y_{1,i_3,r_2} y_{2,i_2,r_1} L_{a_2,i_2}}}{4\epsilon M_F} \mathcal{C} P_L L_{a_1,i_1} S_{2,j_1} y_{1,i_3,r_1} \mathcal{I} T_{3,a_1,a_2,j_1} S_1^\dagger$		

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	S_1	S_2	F	S_1	S_2	ℓ	S_1	F	H
cont.													$\frac{3h\overline{y}_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_2}L_{a_2,i_2}}{8M_F}{}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{1,i_3,r_1}\overline{I}T_{3,a_1,a_2,j_1}S_1^\dagger$			
													$\frac{h\overline{y}_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_2}L_{a_2,i_2}}{\epsilon M_F}{}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{1,i_3,r_2}\overline{I}T_{3,a_1,a_2,j_1}S_1^\dagger$			
cont.													$\frac{h\overline{y}_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_1}L_{a_2,i_2}}{2M_F}{}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{1,i_3,r_2}\overline{I}T_{3,a_1,a_2,j_1}S_1^\dagger$			
													$\frac{h\overline{y}_{1,i_1,r_1}y_{1,i_2,r_2}y_{2,i_3,r_1}L_{a_2,i_2}}{4\epsilon M_F}{}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{1,i_3,r_2}\overline{I}T_{3,a_1,a_2,j_1}S_1^\dagger$			
cont.													$\frac{3h\overline{y}_{1,i_1,r_1}y_{1,i_2,r_2}y_{2,i_3,r_1}L_{a_2,i_2}}{8M_F}{}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{1,i_3,r_2}\overline{I}T_{3,a_1,a_2,j_1}S_1^\dagger$			
													$\frac{3h\overline{y}_{1,i_1,r_1}y_{1,i_2,r_2}y_{2,i_3,r_1}L_{a_2,i_2}}{4\epsilon M_F}{}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{1,i_3,r_2}\overline{I}T_{3,a_1,a_2,j_1}S_1^\dagger$			
cont.													$\frac{\sqrt{3}h\overline{y}_{1,i_1,r_1}y_{2,i_2,r_2}y_{2,i_3,r_1}L_{a_2,i_2}}{4\epsilon M_F}{}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{2,i_3,r_2}\overline{I}T_{3,a_1,a_2,j_1}S_1^\dagger$			
													$\frac{\sqrt{3}h\overline{y}_{1,i_1,r_1}y_{2,i_2,r_2}y_{2,i_3,r_1}L_{a_2,i_2}}{4\epsilon M_F}{}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{2,i_3,r_2}\overline{I}T_{3,a_1,a_2,j_1}S_1^\dagger$			
cont.													$\frac{3\sqrt{3}h\overline{y}_{1,i_1,r_1}y_{2,i_2,r_2}y_{2,i_3,r_1}L_{a_2,i_2}}{8M_F}{}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{2,i_3,r_2}\overline{I}T_{3,a_1,a_2,j_1}S_1^\dagger$			
													$\frac{h\overline{y}_{1,i_1,r_1}y_{2,i_2,r_2}y_{2,i_3,r_2}L_{a_2,i_2}}{\sqrt{3}\epsilon M_F}{}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{2,i_3,r_2}\overline{I}T_{3,a_1,a_2,j_1}S_1^\dagger$			
cont.													$\frac{h\overline{y}_{1,i_1,r_1}y_{2,i_2,r_1}y_{2,i_3,r_2}L_{a_2,i_2}}{2\sqrt{3}M_F}{}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{2,i_3,r_2}\overline{I}T_{3,a_1,a_2,j_1}S_1^\dagger$			
													$\frac{h\overline{y}_{1,i_3,r_1}\overline{L}_{a_1,i_1}}{4\epsilon M_F}{}^T C P_R \overline{L}_{a_2,i_2}{}^T S_1 y_{1,i_2,r_1}y_{1,i_3,r_2}y_{2,i_1,r_2}\overline{I}T_{3,a_1,a_2,j_1}S_{2,j_1}^\dagger$			
cont.													$\frac{3h\overline{y}_{1,i_3,r_1}\overline{L}_{a_1,i_1}}{8M_F}{}^T C P_R \overline{L}_{a_2,i_2}{}^T S_1 y_{1,i_2,r_1}y_{1,i_3,r_2}y_{2,i_1,r_2}\overline{I}T_{3,a_1,a_2,j_1}S_{2,j_1}^\dagger$			
													$\frac{3h\overline{y}_{1,i_3,r_1}\overline{L}_{a_1,i_1}}{M_F}{}^T C P_R \overline{L}_{a_2,i_2}{}^T S_1 y_{1,i_1,r_1}y_{1,i_2,r_2}y_{2,i_2,r_1}\overline{I}T_{3,a_1,a_2,j_1}S_{2,j_1}^\dagger$			
cont.													$\frac{\sqrt{3}h\overline{y}_{2,i_3,r_1}\overline{L}_{a_1,i_1}}{4\epsilon M_F}{}^T C P_R \overline{L}_{a_2,i_2}{}^T S_1 y_{1,i_3,r_2}y_{2,i_2,r_1}y_{2,i_2,r_2}\overline{I}T_{3,a_1,a_2,j_1}S_{2,j_1}^\dagger$			
													$\frac{3\sqrt{3}h\overline{y}_{2,i_3,r_1}\overline{L}_{a_1,i_1}}{8M_F}{}^T C P_R \overline{L}_{a_2,i_2}{}^T S_1 y_{1,i_3,r_2}y_{2,i_2,r_1}y_{2,i_2,r_2}\overline{I}T_{3,a_1,a_2,j_1}S_{2,j_1}^\dagger$			
cont.													$\frac{h\overline{y}_{1,i_3,r_1}\overline{L}_{a_1,i_1}}{\epsilon M_F}{}^T C P_R \overline{L}_{a_2,i_2}{}^T S_1 y_{1,i_1,r_2}y_{1,i_3,r_1}y_{2,i_2,r_2}\overline{I}T_{3,a_1,a_2,j_1}S_{2,j_1}^\dagger$			
													$\frac{h\overline{y}_{1,i_3,r_1}\overline{L}_{a_1,i_1}}{2M_F}{}^T C P_R \overline{L}_{a_2,i_2}{}^T S_1 y_{1,i_1,r_2}y_{2,i_2,r_2}y_{2,i_2,r_1}\overline{I}T_{3,a_1,a_2,j_1}S_{2,j_1}^\dagger$			
cont.													$\frac{h\overline{y}_{2,i_3,r_1}\overline{L}_{a_1,i_1}}{4\epsilon M_F}{}^T C P_R \overline{L}_{a_2,i_2}{}^T S_1 y_{1,i_1,r_2}y_{2,i_2,r_2}y_{2,i_2,r_1}\overline{I}T_{3,a_1,a_2,j_1}S_{2,j_1}^\dagger$			
													$\frac{h\overline{y}_{2,i_3,r_1}\overline{L}_{a_1,i_1}}{2\sqrt{3}\epsilon M_F}{}^T C P_R \overline{L}_{a_2,i_2}{}^T S_1 y_{1,i_1,r_2}y_{2,i_2,r_2}y_{2,i_2,r_1}\overline{I}T_{3,a_1,a_2,j_1}S_{2,j_1}^\dagger$			
cont.													$\frac{h\overline{y}_{1,i_3,r_1}\overline{L}_{a_1,i_1}}{4\epsilon M_F}{}^T C P_R \overline{L}_{a_2,i_2}{}^T S_1 y_{1,i_1,r_2}y_{1,i_2,r_1}y_{2,i_2,r_2}\overline{I}T_{3,a_1,a_2,j_1}S_{2,j_1}^\dagger$			
													$\frac{3h\overline{y}_{1,i_3,r_1}\overline{L}_{a_1,i_1}}{8M_F}{}^T C P_R \overline{L}_{a_2,i_2}{}^T S_1 y_{1,i_1,r_2}y_{1,i_2,r_1}y_{2,i_2,r_2}\overline{I}T_{3,a_1,a_2,j_1}S_{2,j_1}^\dagger$			
cont.													$\frac{\sqrt{3}h\overline{y}_{2,i_3,r_1}\overline{L}_{a_1,i_1}}{4\epsilon M_F}{}^T C P_R \overline{L}_{a_2,i_2}{}^T S_1 y_{1,i_1,r_2}y_{2,i_2,r_1}y_{2,i_2,r_2}\overline{I}T_{3,a_1,a_2,j_1}S_{2,j_1}^\dagger$			
													$\frac{3\sqrt{3}h\overline{y}_{2,i_3,r_1}\overline{L}_{a_1,i_1}}{8M_F}{}^T C P_R \overline{L}_{a_2,i_2}{}^T S_1 y_{1,i_1,r_2}y_{2,i_2,r_1}y_{2,i_2,r_2}\overline{I}T_{3,a_1,a_2,j_1}S_{2,j_1}^\dagger$			
cont.													$\frac{h\overline{y}_{1,i_1,r_1}y_{2,i_2,r_1}L_{a_2,i_2}}{\epsilon M_F}{}^T C P_L L_{a_1,i_1}S_{2,j_1}\overline{I}T_{3,a_1,a_2,j_1}S_1^\dagger\lambda_{12}^{(1)}$			
													$\frac{h\overline{y}_{1,i_1,r_1}y_{2,i_2,r_1}L_{a_2,i_2}}{M_F}{}^T C P_L L_{a_1,i_1}S_{2,j_1}\overline{I}T_{3,a_1,a_2,j_1}S_1^\dagger\lambda_{12}^{(1)}$			
cont.													$\frac{h\overline{L}_{a_1,i_1}}{\epsilon M_F}{}^T C P_R \overline{L}_{a_2,i_2}{}^T S_1 y_{1,i_1,r_1}y_{2,i_2,r_1}\overline{I}T_{3,a_1,a_2,j_1}S_{2,j_1}^\dagger\lambda_{12}^{(1)}$			
													$\frac{h\overline{L}_{a_1,i_1}}{M_F}{}^T C P_R \overline{L}_{a_2,i_2}{}^T S_1 y_{1,i_1,r_1}y_{2,i_2,r_1}\overline{I}T_{3,a_1,a_2,j_1}S_{2,j_1}^\dagger\lambda_{12}^{(1)}$			
T3-A	0	1	0	3	1	2	$\frac{1}{2}$						$\frac{4h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y}_{1,i_1,r_1}y_{2,i_2,r_2}S_1S_{2,j_1}y_{1,i_2,r_1}y_{2,i_1,r_2}S_1^\dagger S_{2,j_1}^\dagger}{\sqrt{3}}$			
cont.													$\frac{h\overline{y}_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_2}L_{a_2,i_2}}{4\epsilon M_F}{}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{1,i_3,r_1}\overline{I}T_{3,a_1,a_2,j_1}S_1^\dagger$			
													$\frac{3h\overline{y}_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_2}L_{a_2,i_2}}{8M_F}{}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{1,i_3,r_1}\overline{I}T_{3,a_1,a_2,j_1}S_1^\dagger$			
cont.													$\frac{h\overline{y}_{1,i_1,r_1}y_{1,i_2,r_2}y_{2,i_3,r_1}L_{a_2,i_2}}{2M_F}{}^T C P_L L_{a_1,i_1}S_{2,j_1}\overline{I}T_{3,a_1,a_2,j_1}S_1^\dagger\lambda_{12}^{(1)}$			
													$\frac{h\overline{y}_{1,i_1,r_1}y_{1,i_2,r_2}y_{2,i_3,r_1}L_{a_2,i_2}}{\epsilon M_F}{}^T C P_L L_{a_1,i_1}S_{2,j_1}\overline{I}T_{3,a_1,a_2,j_1}S_1^\dagger\lambda_{12}^{(1)}$			
cont.													$\frac{h\overline{y}_{1,i_1,r_1}y_{1,i_2,r_2}y_{2,i_3,r_1}L_{a_2,i_2}}{4\epsilon M_F}{}^T C P_L L_{a_1,i_1}S_{2,j_1}\overline{I}T_{3,a_1,a_2,j_1}S_1^\dagger$			
													$\frac{3h\overline{y}_{1,i_1,r_1}y_{1,i_2,r_2}y_{2,i_3,r_1}L_{a_2,i_2}}{8M_F}{}^T C P_L L_{a_1,i_1}S_{2,j_1}\overline{I}T_{3,a_1,a_2,j_1}S_1^\dagger$			
cont.													$\frac{h\overline{y}_{1,i_1,r_1}y_{1,i_2,r_2}y_{2,i_3,r_1}L_{a_2,i_2}}{2M_F}{}^T C P_L L_{a_1,i_1}S_{2,j_1}\overline{I}T_{3,a_1,a_2,j_1}S_1^\dagger$			
													$\frac{h\overline{y}_{1,i_1,r_1}y_{1,i_2,r_2}y_{2,i_3,r_1}L_{a_2,i_2}}{4\epsilon M_F}{}^T C P_L L_{a_1,i_1}S_{2,j_1}\overline{I}T_{3,a_1,a_2,j_1}S_1^\dagger$			
cont.													$\frac{3h\overline{y}_{1,i_1,r_1}y_{1,i_2,r_2}y_{2,i_3,r_1}L_{a_2,i_2}}{8M_F}{}^T C P_L L_{a_1,i_1}S_{2,j_1}\overline{I}T_{3,a_1,a_2,j_1}S_1^\dagger$			
													$\frac{3h\overline{y}_{1,i_1,r_1}y_{1,i_2,r_2}y_{2,i_3,r_1}L_{a_2,i_2}}{4\epsilon M_F}{}^T C P_L L_{a_1,i_1}S_{2,j_1}\overline{I}T_{3,a_1,a_2,j_1}S_1^\dagger$			
cont.													$\frac{h\overline{y}_{1,i_1,r_1}y_{1,i_2,r_2}y_{2,i_3,r_1}L_{a_2,i_2}}{2M_F}{}^T C P_L L_{a_1,i_1}S_{2,j_1}\overline{I}T_{3,a_1,a_2,j_1}S_1^\dagger$			
													$\frac{h\overline{y}_{1,i_1,r_1}y_{1,i_2,r_2}y_{2,i_3,r_1}L_{a_2,i_2}}{4\epsilon M_F}{}^T C P_L L_{a_1,i_1}S_{2,j_1}\overline{I}T_{3,a_1,a_2,j_1}S_1^\dagger$			
cont.													$\frac{3h\overline{y}_{1,i_1,r_1}y_{1,i_2,r_2}y_{2,i_3,r_1}L_{a_2,i_2}}{8M_F}{}^T C P_L L_{a_1,i_1}S_{2,j_1}\overline{I}T_{3,a_1,a_2,j_1}S_1^\dagger$			
													$\frac{3h\overline{y}_{1,i_1,r_1}y_{1,i_2,r_2}y_{2,i_3,r_1}L_{a_2,i_2}}{4\epsilon M_F}{}^T C P_L L_{a_1,i_1}S_{2,j_1}\overline{I}T_{3,a_1,a_2,j_1}S_1^\dagger$			
cont.													$\frac{h\overline{y}_{1,i_1,r_1}y_{1,i_2,r_2}y_{2,i_3,r_1}L_{a_2,i_2}}{2M_F}{}^T C P_L L_{a_1,i_1}S_{2,j_1}\overline{I}T_{3,a_1,a_2,j_1}S_1^\dagger$			
													$\frac{h\overline{y}_{1,i_1,r_1}y_{1,i_2,r_2}y_{2,i_3,r_1}L_{a_2,i_2}}{4\epsilon M_F}{}^T C P_L L_{a_1,i_1}S_{2,j_1}\overline{I}T_{3,a_1,a_2,j_1}S_1^\dagger$			
cont.													$\frac{3h\overline{y}_{1,i_1,r_1}y_{1,i_2,r_2}y_{2,i_3,r_1}L_{a_2,i_2}}{8M_F}{}^T C P_L L_{a_1,i_1}S_{2,j_1}\overline{I}T_{3,a_1,a_2,j_1}S_1^\dagger$			
													$\frac{3h\overline{y}_{1,i_1,r_1}y_{1,i_2,r_2}y_{2,i_3,r_1}L_{a_2,i_2}}{4\epsilon M_F}{}^T C P_L L_{a_1,i_1}S_{2,j_1}\overline{I}T_{3,a_1,a_2,j_1}S_1^\dagger$			
cont.													$\frac{h\overline{y}_{1,i_1,r_1}y_{1,i_2,r_2}y_{2,i_3,r_1}L_{a_2,i_2}}{2M_F}{}^T C P_L L_{a_1,i_1}S_{2,j_1}\overline{I}T_{3,a_1,a_2,j_1}S_1^\dagger$			
													$\frac{h\overline{y}_{1,i_1,r_1}y_{1,i_2,r_2}y_{2,i_3,r_1}L_{a_2,i_2}}{4\epsilon M_F}{}^T C P_L L_{a_1,i_1}S_{2,j_1}\overline{I}T_{3,a_1,a_2,j_1}S_1^\dagger$			
cont.													$\frac{3h\overline{y}_{1,i_1,r_1}y_{1,i_2,r_2}y_{2,i_3,r_1}L_{a_2,i_2}}{8M_F}{}^T C P_L L_{a_1,i_1}S_{2,j_1}\overline{I}T_{3,a_1,a_2,j_1}S_1^\dagger$			
													$\frac{3h\overline{y}_{1,i_1,r_1}y_{1,i_2,r_2}y_{2,i_3,r_1}L_{a_2,i_2}}{4\epsilon M_F}{}^T C P_L L_{a_1,i_1}S_{2,j_1}\overline{I}T_{3,a_1,a_2,j_1}S_1^\dagger$			
cont.													$\frac{h\overline{y}_{1,i_1,r_1}y_{1,i_2,r_2}y_{2,i_3,r_1}L_{a_2,i_2}}{2M_F}{}^T C P_L L_{a_1,i_1}S_{2,j_1}\overline{I}T_{3,a_1,a_2,j_1}S_1^\dagger$			
													$\frac{h\overline{y}_{1,i_1,r_1}y_{1,i_2,r_2}y_{2,i_3,r_1}L_{a_2,i_2}}{4\epsilon M_F}{}^T C P_L L_{a_1,i_1}S_{2,j_1}\overline{I}T_{3,a_1,a_2,j_1}S_1^\dagger$			
cont.													$\frac{3h\overline{y}_{1,i_1,r_1}y_{1,i_2,r_2}y_{2,i_3,r_1}L_{a_2,i_2}}{8M_F}{}^T C P_L L_{a_1,i_1}S_{2,j_1}\overline{I}T_{3,a_1,a_2,j_1}S_1^\dagger$			
													$\frac{3h\overline{y}_{1,i_1,r_1}y_{1,i_2,r_2}y_{2,i_3,r_1}L_{a_2,i_2}}{4\epsilon M_F}{}^T C P_L L_{a_1,i_1}S_{2,j_1}\overline{I}T_{3,a_1,a_2,j_1}S_1^\dagger$			
cont.													$\frac{h\overline{y}_{1,i_1,r_1}y_{1,i_2,r_2}y_{2,i_3,r_1}L_{a_2,i_2}}{2M_F}{}^T C P_L L_{a_1,i_1}S_{2,j_1}\overline{I}T_{3,a_1,a_2,j_1}S_1^\dagger$			
													$\frac{h\overline{y}_{1,i_1,r_1}y_{1,i_2,r_2}y_{2,i_3,r_1}L_{a_2,i_2}}{4\epsilon M_F}{}^T C P_L L_{a_1,i_1}S_{2,j_1}\overline{I}T_{3,a_1,a_2,j_1}S_1^\dagger$			
cont.													$\frac{3h\overline{y}_{1,i_1,r_1}y_{1,i_2,r_2}y_{2,i_3,r_1}L_{a_2,i_2}}{8M_F}{}^T C P_L L_{a_1,i_1}S_{2,j_1}\overline{I}T_{3,a_1,a_2,j_1}S_1^\dagger$			
													$\frac{3h\overline{y}_{1,i_1,r_1}y_{1,i_2,r_2}y_{2,i_3,r_1}L_{a_2,i_2}}{4\epsilon M_F}{}^T C P_L L_{a_1,i_1}S_{2,j_1}\overline{I}T_{3,a_1,a_2,j_1}S_1^\dagger$			
cont.													$\frac{h\overline{y}_{1,i_1,r_1}y_{1,i_2,r_2}y_{2,i_3,r_1}L_{a_2,i_2}}{2M_F}{}^T C P_L L_{a_1,i_1}S_{2,j_1}\overline{I}T_{3,a_1,a_2,j_1}S_1^\dagger$			
													$\frac{h\overline{y}_{1,i_1,r_1}y_{1,i_2,r_2}y_{2,i_3,r_1}L_{a_2,i_2}}{4\epsilon M_F}{}^T C P_L L_{a_1,i_1}S_{2,j_1}\overline{I}T_{3,a_1,a_2,j_1}S_1^\dagger$			
cont.													$\frac{3h\overline{y}_{1,i_1,r_1}y_{1,i_2,r_2}y_{2,i_3,r_1}L_{a_2,i_2}}{8M_F}{}^T C P_L L_{a_1,i_1}S_{2,j_1}\overline{I}T_{3,a_1,a_2,j_1}S_1^\dagger$			
													$\frac{3h\overline{y}_{1,i_1,r_1}y_{1,i_2,r_2}y_{2,i_3,r_1}L_{a_2,i_2}}{4\epsilon M_F}{}^T C P_L L_{a_1,i_1}S_{2,j_1}\overline{I}T_{3,a_1,a_2,j_1}S_1^\dagger$			
cont.													$\frac{h\overline{y}_{1,i_1,r_1}y_{1,i_2,r_2}y_{2,i_3,r_1}L_{a_2,i_2}}{2M_F}{}^T C P_L L_{a_1,i_1}S_{2,j_1}\overline{I}T_{3,a_1,a_2,j_1}S_1^\dagger$			
													$\frac{h\overline{y}_{1,i_1,r_1}y_{1,i_2,r_2}y_{2,i_3,r_1}L_{a_2,i_2}}{4\epsilon M_F}{}^T C P_L L_{a_1,i_1}S_{2,j_1}\overline{I}T_{3,a_1,a_2,j_1}S_1^\dagger$			
cont.													$\frac{3h\overline{y}_{1,i_1,r_1}y_{1,i_2,r_2$			

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	S_1	S_2	F	$S_1 S_2 \ell$	$S_1 F H$
cont.											$\frac{h\overline{y}_{1,i_1,r_1}y_{2,i_2,r_1}y_{2,i_3,r_2}L_{a_2,i_2}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{2,i_3,r_2}IT_{3,a_1,a_2,j_1}S_1^\dagger}{2\sqrt{3}M_F}$ $\frac{h\overline{y}_{1,i_3,r_1}L_{a_1,i_1}C P_R \overline{L}_{a_2,i_2}^T S_1 y_{1,i_2,r_1}y_{1,i_3,r_2}y_{2,i_4,r_2}IT_{3,a_1,a_2,j_1}S_{2,j_1}^\dagger}{4\epsilon M_F}$ $\frac{3h\overline{y}_{1,i_3,r_1}L_{a_1,i_1}C P_R \overline{L}_{a_2,i_2}^T S_1 y_{1,i_2,r_1}y_{1,i_3,r_2}y_{2,i_4,r_2}IT_{3,a_1,a_2,j_1}S_{2,j_1}^\dagger}{8M_F}$ $+\frac{\overline{L}_{a_1,i_1}C P_R \overline{L}_{a_2,i_2}^T S_1 y_{1,i_1,r_1}y_{1,i_2,r_2}IT_{3,a_1,a_2,j_1}S_{2,j_1}^\dagger}{M_F}$	
											$\frac{\sqrt{3}h\overline{y}_{2,i_3,r_1}L_{a_1,i_1}C P_R \overline{L}_{a_2,i_2}^T S_1 y_{1,i_3,r_2}y_{2,i_4,r_2}y_{2,i_5,r_2}IT_{3,a_1,a_2,j_1}S_{2,j_1}^\dagger}{4\epsilon M_F}$ $\frac{3\sqrt{3}h\overline{y}_{2,i_3,r_1}L_{a_1,i_1}C P_R \overline{L}_{a_2,i_2}^T S_1 y_{1,i_3,r_2}y_{2,i_4,r_2}y_{2,i_5,r_2}IT_{3,a_1,a_2,j_1}S_{2,j_1}^\dagger}{8M_F}$ $\frac{h\overline{y}_{1,i_3,r_1}L_{a_1,i_1}C P_R \overline{L}_{a_2,i_2}^T S_1 y_{1,i_1,r_2}y_{1,i_3,r_1}y_{2,i_4,r_2}IT_{3,a_1,a_2,j_1}S_{2,j_1}^\dagger}{\epsilon M_F}$	
												$\frac{h\overline{y}_{1,i_3,r_1}L_{a_1,i_1}C P_R \overline{L}_{a_2,i_2}^T S_1 y_{1,i_1,r_2}y_{1,i_3,r_1}y_{2,i_4,r_2}IT_{3,a_1,a_2,j_1}S_{2,j_1}^\dagger}{2M_F}$ $\frac{h\overline{y}_{2,i_3,r_1}L_{a_1,i_1}C P_R \overline{L}_{a_2,i_2}^T S_1 y_{1,i_1,r_2}y_{2,i_4,r_2}y_{2,i_5,r_2}IT_{3,a_1,a_2,j_1}S_{2,j_1}^\dagger}{\sqrt{3}M_F}$ $\frac{h\overline{y}_{2,i_3,r_1}L_{a_1,i_1}C P_R \overline{L}_{a_2,i_2}^T S_1 y_{1,i_1,r_2}y_{2,i_4,r_2}y_{2,i_5,r_1}IT_{3,a_1,a_2,j_1}S_{2,j_1}^\dagger}{2\sqrt{3}M_F}$
cont.											$\frac{h\overline{y}_{1,i_3,r_1}L_{a_1,i_1}C P_R \overline{L}_{a_2,i_2}^T S_1 y_{1,i_1,r_2}y_{1,i_2,r_1}y_{2,i_3,r_2}IT_{3,a_1,a_2,j_1}S_{2,j_1}^\dagger}{4\epsilon M_F}$ $\frac{3h\overline{y}_{1,i_3,r_1}L_{a_1,i_1}C P_R \overline{L}_{a_2,i_2}^T S_1 y_{1,i_1,r_2}y_{1,i_2,r_1}y_{2,i_3,r_2}IT_{3,a_1,a_2,j_1}S_{2,j_1}^\dagger}{8M_F}$ $\frac{\sqrt{3}h\overline{y}_{2,i_3,r_1}L_{a_1,i_1}C P_R \overline{L}_{a_2,i_2}^T S_1 y_{1,i_1,r_2}y_{2,i_4,r_1}y_{2,i_5,r_2}IT_{3,a_1,a_2,j_1}S_{2,j_1}^\dagger}{4\epsilon M_F}$	
											$\frac{3\sqrt{3}h\overline{y}_{2,i_3,r_1}L_{a_1,i_1}C P_R \overline{L}_{a_2,i_2}^T S_1 y_{1,i_1,r_2}y_{2,i_4,r_1}y_{2,i_5,r_2}IT_{3,a_1,a_2,j_1}S_{2,j_1}^\dagger}{8M_F}$ $-\frac{h\overline{y}_{1,i_1,r_1}y_{2,i_2,r_1}L_{a_2,i_2}^T C P_L L_{a_1,i_1}S_{2,j_1}IT_{3,a_1,a_2,j_1}S_1^\dagger\lambda_{12}^{(1)}}{\epsilon M_F}$ $-\frac{h\overline{y}_{1,i_1,r_1}y_{2,i_2,r_1}L_{a_2,i_2}^T C P_L L_{a_1,i_1}S_{2,j_1}IT_{3,a_1,a_2,j_1}S_1^\dagger\lambda_{12}^{(1)}}{M_F}$	
												$-\frac{h\overline{L}_{a_1,i_1}C P_R \overline{L}_{a_2,i_2}^T S_1 y_{1,i_1,r_1}y_{2,i_2,r_2}IT_{3,a_1,a_2,j_1}S_{2,j_1}^\dagger\lambda_{12}^{(1)}}{\epsilon M_F}$ $-\frac{h\overline{L}_{a_1,i_1}C P_R \overline{L}_{a_2,i_2}^T S_1 y_{1,i_1,r_1}y_{2,i_2,r_1}IT_{3,a_1,a_2,j_1}S_{2,j_1}^\dagger\lambda_{12}^{(1)}}{M_F}$
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1				$-\frac{4h\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{y}_{1,i_1,r_1}y_{2,i_2,r_2}S_1S_{2,j_1}y_{1,i_2,r_1}y_{2,i_4,r_2}S_1^\dagger S_{2,j_1}^\dagger}{\sqrt{3}}$ $-2h\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\frac{\overline{y}_{1,i_1,r_1}S_1S_{2,j_2}y_{1,i_1,r_1}S_1^\dagger S_{2,j_1}^\dagger\mathcal{I}_{12,j_2,j_1}^1\lambda_{12}^{(1)}}{\overline{y}_{1,i_1,r_1}S_1S_{2,j_2}y_{2,i_1,r_1}S_1^\dagger S_{2,j_1}^\dagger\mathcal{I}_{12,j_2,j_1}^1\lambda_{12}^{(1)}}$ $-2h\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\frac{\overline{y}_{2,i_1,r_1}S_1S_{2,j_2}y_{2,i_1,r_1}S_1^\dagger S_{2,j_1}^\dagger\mathcal{I}_{12,j_2,j_1}^1\lambda_{12}^{(1)}}{\sqrt{3}}$	$-2h\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\frac{\overline{y}_{1,i_1,r_1}Y_{e,i_2,i_3}H_{a_1}S_1y_{1,i_2,r_1}Y_{e,i_1,i_3}H_{a_1}^\dagger S_1^\dagger}{M_F^2}$ $-2h\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\frac{\overline{y}_{1,i_1,r_1}H_{a_2}S_1y_{1,i_1,r_1}H_{a_1}^\dagger S_1^\dagger\mathcal{I}_{H1,a_1,a_2}^1\lambda_{12}^{(1)}}{M_F^2}$
											$\frac{h\overline{y}_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_2}L_{a_2,i_2}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{1,i_3,r_1}IT_{3,a_1,a_2,j_1}S_1^\dagger}{4\epsilon M_F}$ $\frac{h\overline{y}_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_2}L_{a_2,i_2}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{1,i_3,r_1}IT_{3,a_1,a_2,j_1}S_1^\dagger}{8M_F}$ $\frac{h\overline{y}_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_1}L_{a_2,i_2}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{1,i_3,r_2}IT_{3,a_1,a_2,j_1}S_1^\dagger}{\epsilon M_F}$ $\frac{h\overline{y}_{1,i_1,r_1}y_{1,i_2,r_2}y_{2,i_3,r_1}L_{a_2,i_2}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{1,i_2,r_2}IT_{3,a_1,a_2,j_1}S_1^\dagger}{2M_F}$ $\frac{h\overline{y}_{1,i_1,r_1}y_{1,i_2,r_2}y_{2,i_3,r_1}L_{a_2,i_2}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{1,i_2,r_2}IT_{3,a_1,a_2,j_1}S_1^\dagger}{4\epsilon M_F}$ $\frac{3h\overline{y}_{1,i_1,r_1}y_{1,i_2,r_2}y_{2,i_3,r_1}L_{a_2,i_2}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{1,i_2,r_2}IT_{3,a_1,a_2,j_1}S_1^\dagger}{8M_F}$ $\frac{\sqrt{3}h\overline{y}_{1,i_1,r_1}y_{1,i_2,r_2}y_{2,i_3,r_1}L_{a_2,i_2}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{1,i_2,r_2}IT_{3,a_1,a_2,j_1}S_1^\dagger}{4M_F}$ $\frac{3\sqrt{3}h\overline{y}_{1,i_1,r_1}y_{1,i_2,r_2}y_{2,i_3,r_1}L_{a_2,i_2}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{1,i_2,r_2}IT_{3,a_1,a_2,j_1}S_1^\dagger}{8M_F}$ $\frac{\sqrt{3}h\overline{y}_{1,i_1,r_1}y_{1,i_2,r_2}y_{2,i_3,r_1}L_{a_2,i_2}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{1,i_2,r_2}IT_{3,a_1,a_2,j_1}S_1^\dagger}{4M_F}$ $\frac{3\sqrt{3}h\overline{y}_{1,i_1,r_1}y_{1,i_2,r_2}y_{2,i_3,r_1}L_{a_2,i_2}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{1,i_2,r_2}IT_{3,a_1,a_2,j_1}S_1^\dagger}{8M_F}$ $-\frac{h\overline{y}_{1,i_1,r_1}y_{2,i_2,r_1}y_{2,i_3,r_2}L_{a_2,i_2}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{2,i_3,r_2}IT_{3,a_1,a_2,j_1}S_1^\dagger}{\sqrt{3}\epsilon M_F}$ $-\frac{h\overline{y}_{1,i_1,r_1}y_{2,i_2,r_1}y_{2,i_3,r_2}L_{a_2,i_2}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{2,i_3,r_2}IT_{3,a_1,a_2,j_1}S_1^\dagger}{2\sqrt{3}M_F}$ $\frac{h\overline{y}_{1,i_3,r_1}L_{a_1,i_1}C P_R \overline{L}_{a_2,i_2}^T S_1 y_{1,i_2,r_1}y_{1,i_3,r_2}y_{2,i_4,r_2}IT_{3,a_1,a_2,j_1}S_{2,j_1}^\dagger}{4\epsilon M_F}$ $\frac{3h\overline{y}_{1,i_3,r_1}L_{a_1,i_1}C P_R \overline{L}_{a_2,i_2}^T S_1 y_{1,i_2,r_1}y_{1,i_3,r_2}y_{2,i_4,r_2}IT_{3,a_1,a_2,j_1}S_{2,j_1}^\dagger}{8M_F}$ $-\frac{9h\overline{L}_{a_1,i_1}C P_R \overline{L}_{a_2,i_2}^T g_V^2 S_1 y_{1,i_1,r_1}y_{2,i_2,r_1}IT_{3,a_1,a_2,j_1}S_{2,j_1}^\dagger}{4\epsilon M_F}$ $-\frac{3h\overline{L}_{a_1,i_1}C P_R \overline{L}_{a_2,i_2}^T g_V^2 S_1 y_{1,i_1,r_1}y_{2,i_2,r_1}IT_{3,a_1,a_2,j_1}S_{2,j_1}^\dagger}{4M_F}$ $+\frac{\overline{L}_{a_1,i_1}C P_R \overline{L}_{a_2,i_2}^T S_1 y_{1,i_1,r_1}y_{1,i_2,r_2}IT_{3,a_1,a_2,j_1}S_{2,j_1}^\dagger}{M_F}$ $\frac{\sqrt{3}h\overline{y}_{2,i_3,r_1}L_{a_1,i_1}C P_R \overline{L}_{a_2,i_2}^T S_1 y_{1,i_3,r_2}y_{2,i_4,r_2}y_{2,i_5,r_2}IT_{3,a_1,a_2,j_1}S_{2,j_1}^\dagger}{4\epsilon M_F}$	
											$-\frac{h\overline{y}_{1,i_1,r_1}y_{2,i_2,r_1}y_{2,i_3,r_2}L_{a_2,i_2}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{2,i_3,r_2}IT_{3,a_1,a_2,j_1}S_1^\dagger}{4\epsilon M_F}$ $\frac{h\overline{y}_{1,i_1,r_1}y_{2,i_2,r_1}y_{2,i_3,r_2}L_{a_2,i_2}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{2,i_3,r_2}IT_{3,a_1,a_2,j_1}S_1^\dagger}{8M_F}$ $\frac{h\overline{y}_{1,i_1,r_1}y_{2,i_2,r_1}y_{2,i_3,r_1}L_{a_2,i_2}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{2,i_3,r_2}IT_{3,a_1,a_2,j_1}S_1^\dagger}{\epsilon M_F}$ $\frac{h\overline{y}_{1,i_1,r_1}y_{2,i_2,r_1}y_{2,i_3,r_1}L_{a_2,i_2}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{2,i_3,r_2}IT_{3,a_1,a_2,j_1}S_1^\dagger}{2M_F}$ $\frac{h\overline{y}_{1,i_1,r_1}y_{2,i_2,r_1}y_{2,i_3,r_1}L_{a_2,i_2}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{2,i_3,r_2}IT_{3,a_1,a_2,j_1}S_1^\dagger}{4\epsilon M_F}$ $\frac{3h\overline{y}_{1,i_1,r_1}y_{2,i_2,r_1}y_{2,i_3,r_1}L_{a_2,i_2}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{2,i_3,r_2}IT_{3,a_1,a_2,j_1}S_1^\dagger}{8M_F}$ $\frac{\sqrt{3}h\overline{y}_{1,i_1,r_1}y_{2,i_2,r_1}y_{2,i_3,r_1}L_{a_2,i_2}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{2,i_3,r_2}IT_{3,a_1,a_2,j_1}S_1^\dagger}{4M_F}$ $\frac{3\sqrt{3}h\overline{y}_{1,i_1,r_1}y_{2,i_2,r_1}y_{2,i_3,r_1}L_{a_2,i_2}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{2,i_3,r_2}IT_{3,a_1,a_2,j_1}S_1^\dagger}{8M_F}$ $-\frac{h\overline{y}_{1,i_1,r_1}y_{2,i_2,r_1}y_{2,i_3,r_2}L_{a_2,i_2}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{2,i_3,r_2}IT_{3,a_1,a_2,j_1}S_1^\dagger}{\sqrt{3}\epsilon M_F}$ $-\frac{h\overline{y}_{1,i_1,r_1}y_{2,i_2,r_1}y_{2,i_3,r_2}L_{a_2,i_2}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{2,i_3,r_2}IT_{3,a_1,a_2,j_1}S_1^\dagger}{2\sqrt{3}M_F}$ $\frac{h\overline{y}_{1,i_3,r_1}L_{a_1,i_1}C P_R \overline{L}_{a_2,i_2}^T S_1 y_{1,i_2,r_1}y_{1,i_3,r_2}y_{2,i_4,r_2}IT_{3,a_1,a_2,j_1}S_{2,j_1}^\dagger}{4\epsilon M_F}$ $\frac{3h\overline{y}_{1,i_3,r_1}L_{a_1,i_1}C P_R \overline{L}_{a_2,i_2}^T S_1 y_{1,i_2,r_1}y_{1,i_3,r_2}y_{2,i_4,r_2}IT_{3,a_1,a_2,j_1}S_{2,j_1}^\dagger}{8M_F}$ $-\frac{9h\overline{L}_{a_1,i_1}C P_R \overline{L}_{a_2,i_2}^T g_V^2 S_1 y_{1,i_1,r_1}y_{2,i_2,r_1}IT_{3,a_1,a_2,j_1}S_{2,j_1}^\dagger}{4\epsilon M_F}$ $-\frac{3h\overline{L}_{a_1,i_1}C P_R \overline{L}_{a_2,i_2}^T g_V^2 S_1 y_{1,i_1,r_1}y_{2,i_2,r_1}IT_{3,a_1,a_2,j_1}S_{2,j_1}^\dagger}{4M_F}$ $+\frac{\overline{L}_{a_1,i_1}C P_R \overline{L}_{a_2,i_2}^T S_1 y_{1,i_1,r_1}y_{1,i_2,r_2}IT_{3,a_1,a_2,j_1}S_{2,j_1}^\dagger}{M_F}$ $\frac{\sqrt{3}h\overline{y}_{2,i_3,r_1}L_{a_1,i_1}C P_R \overline{L}_{a_2,i_2}^T S_1 y_{1,i_3,r_2}y_{2,i_4,r_2}y_{2,i_5,r_2}IT_{3,a_1,a_2,j_1}S_{2,j_1}^\dagger}{4\epsilon M_F}$	
											$-\frac{h\overline{y}_{1,i_1,r_1}y_{2,i_2,r_1}y_{2,i_3,r_2}L_{a_2,i_2}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{2,i_3,r_2}IT_{3,a_1,a_2,j_1}S_1^\dagger}{4\epsilon M_F}$ $\frac{h\overline{y}_{1,i_1,r_1}y_{2,i_2,r_1}y_{2,i_3,r_2}L_{a_2,i_2}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{2,i_3,r_2}IT_{3,a_1,a_2,j_1}S_1^\dagger}{8M_F}$ $\frac{h\overline{y}_{1,i_1,r_1}y_{2,i_2,r_1}y_{2,i_3,r_1}L_{a_2,i_2}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{2,i_3,r_2}IT_{3,a_1,a_2,j_1}S_1^\dagger}{\epsilon M_F}$ $\frac{h\overline{y}_{1,i_1,r_1}y_{2,i_2,r_1}y_{2,i_3,r_1}L_{a_2,i_2}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{2,i_3,r_2}IT_{3,a_1,a_2,j_1}S_1^\dagger}{2M_F}$ $\frac{h\overline{y}_{1,i_1,r_1}y_{2,i_2,r_1}y_{2,i_3,r_1}L_{a_2,i_2}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{2,i_3,r_2}IT_{3,a_1,a_2,j_1}S_1^\dagger}{4\epsilon M_F}$ $\frac{3h\overline{y}_{1,i_1,r_1}y_{2,i_2,r_1}y_{2,i_3,r_1}L_{a_2,i_2}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{2,i_3,r_2}IT_{3,a_1,a_2,j_1}S_1^\dagger}{8M_F}$ $\frac{\sqrt{3}h\overline{y}_{1,i_1,r_1}y_{2,i_2,r_1}y_{2,i_3,r_1}L_{a_2,i_2}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{2,i_3,r_2}IT_{3,a_1,a_2,j_1}S_1^\dagger}{4M_F}$ $\frac{3\sqrt{3}h\overline{y}_{1,i_1,r_1}y_{2,i_2,r_1}y_{2,i_3,r_1}L_{a_2,i_2}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{2,i_3,r_2}IT_{3,a_1,a_2,j_1}S_1^\dagger}{8M_F}$ $-\frac{h\overline{y}_{1,i_1,r_1}y_{2,i_2,r_1}y_{2,i_3,r_2}L_{a_2,i_2}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{2,i_3,r_2}IT_{3,a_1,a_2,j_1}S_1^\dagger}{\sqrt{3}\epsilon M_F}$ $-\frac{h\overline{y}_{1,i_1,r_1}y_{2,i_2,r_1}y_{2,i_3,r_2}L_{a_2,i_2}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{2,i_3,r_2}IT_{3,a_1,a_2,j_1}S_1^\dagger}{2\sqrt{3}M_F}$ $\frac{h\overline{y}_{1,i_3,r_1}L_{a_1,i_1}C P_R \overline{L}_{a_2,i_2}^T S_1 y_{1,i_2,r_1}y_{1,i_3,r_2}y_{2,i_4,r_2}IT_{3,a_1,a_2,j_1}S_{2,j_1}^\dagger}{4\epsilon M_F}$ $\frac{3h\overline{y}_{1,i_3,r_1}L_{a_1,i_1}C P_R \overline{L}_{a_2,i_2}^T S_1 y_{1,i_2,r_1}y_{1,i_3,r_2}y_{2,i_4,r_2}IT_{3,a_1,a_2,j_1}S_{2,j_1}^\dagger}{8M_F}$ $-\frac{9h\overline{L}_{a_1,i_1}C P_R \overline{L}_{a_2,i_2}^T g_V^2 S_1 y_{1,i_1,r_1}y_{2,i_2,r_1}IT_{3,a_1,a_2,j_1}S_{2,j_1}^\dagger}{4\epsilon M_F}$ $-\frac{3h\overline{L}_{a_1,i_1}C P_R \overline{L}_{a_2,i_2}^T g_V^2 S_1 y_{1,i_1,r_1}y_{2,i_2,r_1}IT_{3,a_1,a_2,j_1}S_{2,j_1}^\dagger}{4M_F}$ $+\frac{\overline{L}_{a_1,i_1}C P_R \overline{L}_{a_2,i_2}^T S_1 y_{1,i_1,r_1}y_{1,i_2,r_2}IT_{3,a_1,a_2,j_1}S_{2,j_1}^\dagger}{M_F}$ $\frac{\sqrt{3}h\overline{y}_{2,i_3,r_1}L_{a_1,i_1}C P_R \overline{L}_{a_2,i_2}^T S_1 y_{1,i_3,r_2}y_{2,i_4,r_2}y_{2,i_5,r_2}IT_{3,a_1,a_2,j_1}S_{2,j_1}^\dagger}{4\epsilon M_F}$	

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	S_1	S_2	F	S_1	S_2	ℓ	S_1	F	H
cont.													$-\frac{3\sqrt{3}\overline{h\overline{y_2}}_{i_3,r_1}L_{a_1,i_1}CPR\overline{L_{a_2,i_2}}^TS_1y_{1,i_3,r_2}y_{2,i_1,r_2}y_{2,i_2,r_1}\overline{IT_3}_{a_1,a_2,j_1}S_{2,j_1}^\dagger}{8M_F}$ $-\frac{h\overline{y_{1,i_3,r_1}}L_{a_1,i_1}CPR\overline{L_{a_2,i_2}}^TS_1y_{1,i_1,r_2}y_{1,i_3,r_1}y_{2,i_2,r_2}\overline{IT_3}_{a_1,a_2,j_1}S_{2,j_1}^\dagger}{\epsilon M_F}$			
													$-\frac{h\overline{y_{1,i_3,r_1}}L_{a_1,i_1}CPR\overline{L_{a_2,i_2}}^TS_1y_{1,i_1,r_2}y_{1,i_3,r_1}y_{2,i_2,r_2}\overline{IT_3}_{a_1,a_2,j_1}S_{2,j_1}^\dagger}{2M_F}$ $-\frac{h\overline{y_2}_{i_3,r_1}L_{a_1,i_1}CPR\overline{L_{a_2,i_2}}^TS_1y_{1,i_1,r_2}y_{2,i_2,r_2}y_{2,i_3,r_1}\overline{IT_3}_{a_1,a_2,j_1}S_{2,j_1}^\dagger}{\sqrt{3}\epsilon M_F}$			
cont.													$-\frac{h\overline{y_2}_{i_3,r_1}L_{a_1,i_1}CPR\overline{L_{a_2,i_2}}^TS_1y_{1,i_1,r_2}y_{2,i_2,r_2}y_{2,i_3,r_1}\overline{IT_3}_{a_1,a_2,j_1}S_{2,j_1}^\dagger}{2\sqrt{3}M_F}$ $-\frac{h\overline{y_{1,i_3,r_1}}L_{a_1,i_1}CPR\overline{L_{a_2,i_2}}^TS_1y_{1,i_1,r_2}y_{1,i_2,r_1}y_{2,i_3,r_2}\overline{IT_3}_{a_1,a_2,j_1}S_{2,j_1}^\dagger}{4\epsilon M_F}$			
													$-\frac{3h\overline{y_{1,i_3,r_1}}L_{a_1,i_1}CPR\overline{L_{a_2,i_2}}^TS_1y_{1,i_1,r_2}y_{1,i_2,r_1}y_{2,i_3,r_2}\overline{IT_3}_{a_1,a_2,j_1}S_{2,j_1}^\dagger}{8M_F}$ $-\frac{\sqrt{3}h\overline{y_2}_{i_3,r_1}L_{a_1,i_1}CPR\overline{L_{a_2,i_2}}^TS_1y_{1,i_1,r_2}y_{2,i_2,r_1}y_{2,i_3,r_2}\overline{IT_3}_{a_1,a_2,j_1}S_{2,j_1}^\dagger}{4\epsilon M_F}$			
cont.													$-\frac{3\sqrt{3}h\overline{y_2}_{i_3,r_1}L_{a_1,i_1}CPR\overline{L_{a_2,i_2}}^TS_1y_{1,i_1,r_2}y_{2,i_2,r_1}y_{2,i_3,r_2}\overline{IT_3}_{a_1,a_2,j_1}S_{2,j_1}^\dagger}{8M_F}$ $-\frac{h\overline{y_{1,i_1,r_1}}y_{2,i_2,r_1}L_{a_2,i_2}^TCPL_{a_1,i_1}S_{2,j_1}\overline{IT_3}_{a_1,a_2,j_1}S_1^\dagger\lambda_{12}^{(1)}}{\epsilon M_F}$			
													$-\frac{h\overline{y_{1,i_1,r_1}}y_{2,i_2,r_1}L_{a_2,i_2}^TCPL_{a_1,i_1}S_{2,j_1}\overline{IT_3}_{a_1,a_2,j_1}S_1^\dagger\lambda_{12}^{(1)}}{M_F}$ $-\frac{h\overline{L_{a_1,i_1}}CPR\overline{L_{a_2,i_2}}^TS_1y_{1,i_1,r_1}y_{2,i_2,r_1}\overline{IT_3}_{a_1,a_2,j_1}S_{2,j_1}^\dagger\lambda_{12}^{(1)}}{\epsilon M_F}$			
cont.													$-\frac{h\overline{L_{a_1,i_1}}CPR\overline{L_{a_2,i_2}}^TS_1y_{1,i_1,r_1}y_{2,i_2,r_1}\overline{IT_3}_{a_1,a_2,j_1}S_{2,j_1}^\dagger\lambda_{12}^{(1)}}{\epsilon M_F}$ $-\frac{h\overline{L_{a_1,i_1}}CPR\overline{L_{a_2,i_2}}^TS_1y_{1,i_1,r_1}y_{2,i_2,r_1}\overline{IT_3}_{a_1,a_2,j_1}S_{2,j_1}^\dagger\lambda_{12}^{(1)}}{M_F}$			
T3-A	2	1	1	3	2	2	$\frac{3}{2}$						$-\frac{4\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}}y_{2,i_2,r_2}S_1S_{2,j_1}y_{1,i_2,r_1}y_{2,i_1,r_2}S_1^\dagger S_{2,j_1}^\dagger}{\sqrt{3}}$ $-2\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}}S_1S_{2,j_2}y_{1,i_1,r_1}S_1^\dagger S_{2,j_1}^\dagger I_{12,j_2,j_1}^1\lambda_{12}^{(1)}$			
													$-2\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}}S_1S_{2,j_2}y_{2,i_1,r_1}S_1^\dagger S_{2,j_1}^\dagger I_{12,j_2,j_1}^1\lambda_{12}^{(1)}$ $-\frac{6h\overline{y_{1,i_1,r_1}}y_{2,i_2,r_1}L_{a_2,i_2}^TCPL_{a_1,i_1}g_Y^2S_{2,j_1}\overline{IT_3}_{a_1,a_2,j_1}S_1^\dagger}{\epsilon M_F}$	$-2\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}}\overline{Y_{e,i_2,i_3}}H_{a_1}S_1y_{1,i_2,r_1}Y_{e,i_1,i_3}H_{a_1}^\dagger S_1^\dagger$		
cont.													$-\frac{2h\overline{y_{1,i_1,r_1}}y_{2,i_2,r_1}L_{a_2,i_2}^TCPL_{a_1,i_1}g_Y^2S_{2,j_1}\overline{IT_3}_{a_1,a_2,j_1}S_1^\dagger}{M_F}$ $+\frac{\overline{y_{1,i_1,r_1}}y_{2,i_2,r_1}L_{a_2,i_2}^TCPL_{a_1,i_1}S_{2,j_1}\overline{IT_3}_{a_1,a_2,j_1}S_1^\dagger}{M_F}$	$-2\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}}H_{a_2}S_1y_{1,i_1,r_1}H_{a_1}^\dagger S_1^\dagger\overline{IT_{H1,a_1,a_2}}^1\lambda_{H1}^{(1)}$		
													$-\frac{h\overline{y_{1,i_1,r_1}}y_{1,i_3,r_2}y_{2,i_2,r_2}L_{a_2,i_2}^TCPL_{a_1,i_1}S_{2,j_1}y_{1,i_3,r_1}\overline{IT_3}_{a_1,a_2,j_1}S_1^\dagger}{4\epsilon M_F}$ $-\frac{3h\overline{y_{1,i_1,r_1}}y_{1,i_3,r_2}y_{2,i_2,r_2}L_{a_2,i_2}^TCPL_{a_1,i_1}S_{2,j_1}y_{1,i_3,r_1}\overline{IT_3}_{a_1,a_2,j_1}S_1^\dagger}{8M_F}$			
cont.													$-\frac{h\overline{y_{1,i_1,r_1}}y_{1,i_3,r_2}y_{2,i_2,r_2}L_{a_2,i_2}^TCPL_{a_1,i_1}S_{2,j_1}y_{1,i_3,r_2}\overline{IT_3}_{a_1,a_2,j_1}S_1^\dagger}{8M_F}$ $-\frac{h\overline{y_{1,i_1,r_1}}y_{1,i_3,r_2}y_{2,i_2,r_1}L_{a_2,i_2}^TCPL_{a_1,i_1}S_{2,j_1}y_{1,i_3,r_2}\overline{IT_3}_{a_1,a_2,j_1}S_1^\dagger}{\epsilon M_F}$			
													$-\frac{h\overline{y_{1,i_1,r_1}}y_{1,i_3,r_2}y_{2,i_2,r_1}L_{a_2,i_2}^TCPL_{a_1,i_1}S_{2,j_1}y_{1,i_3,r_2}\overline{IT_3}_{a_1,a_2,j_1}S_1^\dagger}{2M_F}$ $-\frac{h\overline{y_{1,i_1,r_1}}y_{1,i_2,r_2}y_{2,i_3,r_1}L_{a_2,i_2}^TCPL_{a_1,i_1}S_{2,j_1}y_{1,i_3,r_2}\overline{IT_3}_{a_1,a_2,j_1}S_1^\dagger}{4\epsilon M_F}$			
cont.													$-\frac{3h\overline{y_{1,i_1,r_1}}y_{1,i_2,r_2}y_{2,i_3,r_1}L_{a_2,i_2}^TCPL_{a_1,i_1}S_{2,j_1}y_{1,i_3,r_2}\overline{IT_3}_{a_1,a_2,j_1}S_1^\dagger}{8M_F}$ $-\frac{\sqrt{3}h\overline{y_{1,i_1,r_1}}y_{2,i_1,r_2}y_{2,i_2,r_1}L_{a_2,i_2}^TCPL_{a_1,i_1}S_{2,j_1}y_{2,i_3,r_2}\overline{IT_3}_{a_1,a_2,j_1}S_1^\dagger}{8M_F}$			
													$-\frac{\sqrt{3}h\overline{y_{1,i_1,r_1}}y_{2,i_2,r_2}y_{2,i_3,r_1}L_{a_2,i_2}^TCPL_{a_1,i_1}S_{2,j_1}y_{2,i_3,r_2}\overline{IT_3}_{a_1,a_2,j_1}S_1^\dagger}{8M_F}$ $-\frac{3\sqrt{3}h\overline{y_{1,i_1,r_1}}y_{2,i_2,r_2}y_{2,i_3,r_1}L_{a_2,i_2}^TCPL_{a_1,i_1}S_{2,j_1}y_{2,i_3,r_2}\overline{IT_3}_{a_1,a_2,j_1}S_1^\dagger}{8M_F}$			
cont.													$-\frac{h\overline{y_{1,i_1,r_1}}y_{2,i_2,r_1}y_{2,i_3,r_2}L_{a_2,i_2}^TCPL_{a_1,i_1}S_{2,j_1}y_{2,i_3,r_2}\overline{IT_3}_{a_1,a_2,j_1}S_1^\dagger}{\sqrt{3}\epsilon M_F}$ $-\frac{h\overline{y_{1,i_1,r_1}}y_{2,i_2,r_1}y_{2,i_3,r_2}L_{a_2,i_2}^TCPL_{a_1,i_1}S_{2,j_1}y_{2,i_3,r_2}\overline{IT_3}_{a_1,a_2,j_1}S_1^\dagger}{\epsilon M_F}$			
													$-\frac{h\overline{y_{1,i_3,r_1}}L_{a_1,i_1}CPR\overline{L_{a_2,i_2}}^TS_1y_{1,i_2,r_1}y_{1,i_3,r_2}y_{2,i_4,r_1}\overline{IT_3}_{a_1,a_2,j_1}S_{2,j_1}^\dagger}{4\epsilon M_F}$ $-\frac{3h\overline{y_{1,i_3,r_1}}L_{a_1,i_1}CPR\overline{L_{a_2,i_2}}^TS_1y_{1,i_2,r_1}y_{1,i_3,r_2}y_{2,i_4,r_1}\overline{IT_3}_{a_1,a_2,j_1}S_{2,j_1}^\dagger}{8M_F}$			
cont.													$-\frac{6h\overline{L_{a_1,i_1}}CPR\overline{L_{a_2,i_2}}^Tg_Y^2S_1y_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_4,r_1}\overline{IT_3}_{a_1,a_2,j_1}S_{2,j_1}^\dagger}{\epsilon M_F}$ $-\frac{2h\overline{L_{a_1,i_1}}CPR\overline{L_{a_2,i_2}}^Tg_Y^2S_1y_{1,i_1,r_1}y_{2,i_2,r_1}\overline{IT_3}_{a_1,a_2,j_1}S_{2,j_1}^\dagger}{M_F}$			
													$-\frac{\overline{L_{a_1,i_1}}CPR\overline{L_{a_2,i_2}}^TS_1y_{1,i_1,r_1}y_{2,i_2,r_1}\overline{IT_3}_{a_1,a_2,j_1}S_{2,j_1}^\dagger}{M_F}$ $+\frac{\overline{L_{a_1,i_1}}CPR\overline{L_{a_2,i_2}}^TS_1y_{1,i_1,r_1}y_{2,i_2,r_1}\overline{IT_3}_{a_1,a_2,j_1}S_{2,j_1}^\dagger}{M_F}$			
cont.													$-\frac{\sqrt{3}h\overline{y_2}_{i_3,r_1}L_{a_1,i_1}CPR\overline{L_{a_2,i_2}}^TS_1y_{1,i_3,r_2}y_{2,i_1,r_2}y_{2,i_2,r_1}\overline{IT_3}_{a_1,a_2,j_1}S_{2,j_1}^\dagger}{4\epsilon M_F}$ $-\frac{3\sqrt{3}h\overline{y_2}_{i_3,r_1}L_{a_1,i_1}CPR\overline{L_{a_2,i_2}}^TS_1y_{1,i_3,r_2}y_{2,i_1,r_2}y_{2,i_2,r_1}\overline{IT_3}_{a_1,a_2,j_1}S_{2,j_1}^\dagger}{8M_F}$			
													$-\frac{h\overline{y_{1,i_3,r_1}}L_{a_1,i_1}CPR\overline{L_{a_2,i_2}}^TS_1y_{1,i_1,r_2}y_{1,i_3,r_1}y_{2,i_4,r_2}\overline{IT_3}_{a_1,a_2,j_1}S_{2,j_1}^\dagger}{\epsilon M_F}$ $-\frac{h\overline{y_{1,i_3,r_1}}L_{a_1,i_1}CPR\overline{L_{a_2,i_2}}^TS_1y_{1,i_1,r_2}y_{1,i_3,r_1}y_{2,i_4,r_2}\overline{IT_3}_{a_1,a_2,j_1}S_{2,j_1}^\dagger}{2M_F}$			
cont.													$-\frac{h\overline{y_2}_{i_3,r_1}L_{a_1,i_1}CPR\overline{L_{a_2,i_2}}^TS_1y_{1,i_1,r_2}y_{2,i_2,r_2}y_{2,i_3,r_1}\overline{IT_3}_{a_1,a_2,j_1}S_{2,j_1}^\dagger}{\sqrt{3}\epsilon M_F}$ $-\frac{h\overline{y_2}_{i_3,r_1}L_{a_1,i_1}CPR\overline{L_{a_2,i_2}}^TS_1y_{1,i_1,r_2}y_{2,i_2,r_2}y_{2,i_3,r_1}\overline{IT_3}_{a_1,a_2,j_1}S_{2,j_1}^\dagger}{2\sqrt{3}M_F}$			
													$-\frac{h\overline{y_{1,i_3,r_1}}L_{a_1,i_1}CPR\overline{L_{a_2,i_2}}^TS_1y_{1,i_1,r_2}y_{1,i_2,r_1}y_{2,i_3,r_2}\overline{IT_3}_{a_1,a_2,j_1}S_{2,j_1}^\dagger}{4\epsilon M_F}$ $-\frac{h\overline{y_{1,i_3,r_1}}L_{a_1,i_1}CPR\overline{L_{a_2,i_2}}^TS_1y_{1,i_1,r_2}y_{1,i_2,r_1}y_{2,i_3,r_2}\overline{IT_3}_{a_1,a_2,j_1}S_{2,j_1}^\dagger}{4\epsilon M_F}$			

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$S_1 S_2 F$	$S_1 S_2 \ell$	$S_1 F H$
									$-\frac{3h\overline{y}_{1,i_1,r_1}\overline{L}_{a_1,i_1}CPr\overline{L}_{a_2,i_2}^T S_1 y_{1,i_1,r_2} y_{1,i_2,r_1} y_{2,i_3,r_2} \overline{I} T \overline{y}_{a_1,a_2,j_1}^* S_{2,j_1}^\dagger}{8M_F}$	
cont.									$-\frac{\sqrt{3}h\overline{y}_{2,i_3,r_1}\overline{L}_{a_1,i_1}CPr\overline{L}_{a_2,i_2}^T S_1 y_{1,i_1,r_2} y_{2,i_2,r_1} y_{2,i_3,r_2} \overline{I} T \overline{y}_{a_1,a_2,j_1}^* S_{2,j_1}^\dagger}{4\epsilon M_F}$	
									$-\frac{3\sqrt{3}h\overline{y}_{2,i_3,r_1}\overline{L}_{a_1,i_1}CPr\overline{L}_{a_2,i_2}^T S_1 y_{1,i_1,r_2} y_{2,i_2,r_1} y_{2,i_3,r_2} \overline{I} T \overline{y}_{a_1,a_2,j_1}^* S_{2,j_1}^\dagger}{8M_F}$	
									$-\frac{h\overline{y}_{1,i_1,r_1}\overline{y}_{2,i_2,r_1}\overline{L}_{a_2,i_2}^T CPr\overline{L}_{a_1,i_1}S_{2,j_1}\overline{I} T \overline{y}_{3,a_1,a_2,j_1}^* S_1^\dagger \lambda_{12}^{(1)}}{\epsilon M_F}$	
cont.									$-\frac{h\overline{y}_{1,i_1,r_1}\overline{y}_{2,i_2,r_1}\overline{L}_{a_2,i_2}^T CPr\overline{L}_{a_1,i_1}S_{2,j_1}\overline{I} T \overline{y}_{3,a_1,a_2,j_1}^* S_1^\dagger \lambda_{12}^{(1)}}{M_F}$	
									$-\frac{h\overline{L}_{a_1,i_1}CPr\overline{L}_{a_2,i_2}^T S_1 y_{1,i_1,r_1} y_{2,i_2,r_1} \overline{I} T \overline{y}_{a_1,a_2,j_1}^* S_{2,j_1}^\dagger \lambda_{12}^{(1)}}{\epsilon M_F}$	
cont.									$-\frac{h\overline{L}_{a_1,i_1}CPr\overline{L}_{a_2,i_2}^T S_1 y_{1,i_1,r_1} y_{2,i_2,r_1} \overline{I} T \overline{y}_{a_1,a_2,j_1}^* S_{2,j_1}^\dagger \lambda_{12}^{(1)}}{M_F}$	
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	$+2h\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{y}_{1,i_1,r_1}\overline{y}_{2,i_2,r_2}S_{1,j_2}S_{2,j_1}y_{1,i_2,r_1}y_{2,i_1,r_2}S_{1,j_1}^\dagger S_{2,j_2}^\dagger$	$+3h\overline{y}_{1,i_2,r_1}\overline{y}_{2,i_1,r_1}\overline{L}_{a_2,i_2}^T CPr\overline{L}_{a_1,i_1}g_2^2S_{2,j_2}\overline{I} y_{1,a_1,j_1}\overline{I} y_{2,a_2,j_2}^* S_{1,j_1}^\dagger$	$-2h\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{y}_{1,i_1,r_1}\overline{Y}_{e,i_2,i_3}H_{a_2}S_{1,j_2}y_{1,i_2,r_1}Y_{e,i_1,i_3}\overline{I} y_{1,a_1,j_2}\overline{I} y_{1,a_2,j_1}^* H_{a_1}^\dagger S_{1,j_1}^\dagger$
								$-2h\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{y}_{1,i_1,r_1}\overline{y}_{2,i_2,r_2}S_{1,j_1}S_{2,j_2}y_{1,i_2,r_1}y_{2,i_1,r_2}S_{1,j_1}^\dagger S_{2,j_2}^\dagger$	$+\frac{h\overline{y}_{1,i_2,r_1}\overline{y}_{2,i_1,r_1}\overline{L}_{a_2,i_2}^T CPr\overline{L}_{a_1,i_1}g_2^2S_{2,j_2}\overline{I} y_{1,a_1,j_1}\overline{I} y_{2,a_2,j_2}^* S_{1,j_1}^\dagger}{4M_F}$	
cont.								$-h\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{y}_{1,i_1,r_1}S_{1,j_4}S_{2,j_3}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \overline{I} T_{12,j_2,j_3,j_1,j_4}{}^{1,\lambda_{12}^{(1)}}$	$+\frac{\overline{y}_{1,i_1,r_1}\overline{y}_{2,i_2,r_1}\overline{L}_{a_2,i_2}^T CPr\overline{L}_{a_1,i_1}S_{2,j_2}\overline{I} y_{1,a_1,j_1}\overline{I} y_{2,a_2,j_2}^* S_{1,j_1}^\dagger}{M_F}$	$-h\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{y}_{1,i_1,r_1}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}H_{a_1}^\dagger S_{1,j_1}^\dagger \overline{I} T_{H1,a_1,a_2,j_1,j_2}{}^{1,\lambda_{12}^{(1)}}$
								$-h\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{y}_{2,i_1,r_1}S_{1,j_4}S_{2,j_3}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \overline{I} T_{12,j_2,j_3,j_1,j_4}{}^{1,\lambda_{12}^{(1)}}$	$+\frac{h\overline{y}_{1,i_1,r_1}\overline{y}_{1,i_3,r_2}\overline{y}_{2,i_2,r_2}\overline{L}_{a_2,i_2}^T CPr\overline{L}_{a_1,i_1}S_{2,j_2}y_{1,i_3,r_1}\overline{I} y_{1,a_1,j_1}^* \overline{I} T_{a_2,j_2}^* S_{1,j_1}^\dagger}{4M_F}$	
								$+\sqrt{3}h\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{y}_{1,i_1,r_1}S_{1,j_2}S_{2,j_1}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \lambda_{12}^{(2)}$	$-\frac{3h\overline{y}_{1,i_1,r_1}\overline{y}_{1,i_3,r_2}\overline{y}_{2,i_2,r_2}\overline{L}_{a_2,i_2}^T CPr\overline{L}_{a_1,i_1}S_{2,j_2}y_{1,i_3,r_1}\overline{I} y_{1,a_1,j_1}^* \overline{I} T_{a_2,j_2}^* S_{1,j_1}^\dagger}{8M_F}$	
								$-\sqrt{3}h\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{y}_{1,i_1,r_1}S_{1,j_1}S_{2,j_2}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \lambda_{12}^{(2)}$	$-\frac{h\overline{y}_{1,i_1,r_1}\overline{y}_{1,i_3,r_2}\overline{y}_{2,i_2,r_2}\overline{L}_{a_2,i_2}^T CPr\overline{L}_{a_1,i_1}S_{2,j_2}y_{1,i_3,r_1}\overline{I} y_{1,a_1,j_1}^* \overline{I} T_{a_2,j_2}^* S_{1,j_1}^\dagger}{2\epsilon M_F}$	
cont.								$+\sqrt{3}h\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{y}_{2,i_1,r_1}S_{1,j_2}S_{2,j_1}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \lambda_{12}^{(2)}$	$-\frac{h\overline{y}_{1,i_1,r_1}\overline{y}_{1,i_3,r_2}\overline{y}_{2,i_2,r_2}\overline{L}_{a_2,i_2}^T CPr\overline{L}_{a_1,i_1}S_{2,j_2}y_{1,i_3,r_1}\overline{I} y_{1,a_1,j_1}^* \overline{I} T_{a_2,j_2}^* S_{1,j_1}^\dagger}{4M_F}$	

[illegible]

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$S_1 S_2 F$	$S_1 S_2 \ell$	$S_1 F H$
cont.									$-\frac{h\overline{Y_{1,i1,r1}}\overline{Y_{2,i2,r1}}\overline{Y_{2,i3,r2}}\overline{L_{a2,i2}}^T C P_L L_{a1,i1} S_{2,j2} Y_{2,i3,r1} T_{y1,a1,j1} I_{y2,a2,j2} S_{1,j1}^\dagger}{4M_F}$ $+\frac{3h\overline{L_{a1,i1}} C P_R \overline{L_{a2,i2}}^T g_2^2 S_{1,j2} Y_{1,i1,r1} Y_{2,i2,r1} T_{y1,a1,j2} \overline{I_{y2,a2,j1}} S_{2,j1}^\dagger}{4\epsilon M_F}$ $+\frac{h\overline{L_{a1,i1}} C P_R \overline{L_{a2,i2}}^T g_2^2 S_{1,j2} Y_{1,i1,r1} Y_{2,i2,r1} T_{y1,a1,j2} \overline{I_{y2,a2,j1}} S_{2,j1}^\dagger}{4M_F}$ $+\frac{\overline{L_{a1,i1}} C P_R \overline{L_{a2,i2}}^T S_{1,j2} Y_{1,i1,r1} Y_{2,i2,r1} \overline{I_{y1,a1,j2}} \overline{I_{y2,a2,j1}} S_{2,j1}^\dagger}{M_F}$	
cont.									$\frac{h\overline{Y_{1,i3,r1}}\overline{L_{a1,i1}} C P_R \overline{L_{a2,i2}}^T S_{1,j2} Y_{1,i1,r2} Y_{1,i3,r1} Y_{2,i2,r2} \overline{I_{y1,a1,j2}} \overline{I_{y2,a2,j1}} S_{2,j1}^\dagger}{2\epsilon M_F}$ $\frac{h\overline{Y_{1,i3,r1}}\overline{L_{a1,i1}} C P_R \overline{L_{a2,i2}}^T S_{1,j2} Y_{1,i1,r2} Y_{1,i3,r1} Y_{2,i2,r2} \overline{I_{y1,a1,j2}} \overline{I_{y2,a2,j1}} S_{2,j1}^\dagger}{4M_F}$ $-\frac{h\overline{Y_{1,i3,r1}}\overline{L_{a1,i1}} C P_R \overline{L_{a2,i2}}^T S_{1,j2} Y_{1,i1,r2} Y_{1,i3,r2} Y_{2,i2,r2} \overline{I_{y1,a1,j2}} \overline{I_{y2,a2,j1}} S_{2,j1}^\dagger}{4\epsilon M_F}$	
cont.									$-\frac{3h\overline{Y_{1,i3,r1}}\overline{L_{a1,i1}} C P_R \overline{L_{a2,i2}}^T S_{1,j2} Y_{1,i1,r1} Y_{1,i3,r2} Y_{2,i2,r2} \overline{I_{y1,a1,j2}} \overline{I_{y2,a2,j1}} S_{2,j1}^\dagger}{8M_F}$ $-\frac{h\overline{Y_{2,i3,r1}}\overline{L_{a1,i1}} C P_R \overline{L_{a2,i2}}^T S_{1,j2} Y_{1,i3,r2} Y_{2,i2,r1,r1} Y_{2,i2,r2} \overline{I_{y1,a1,j2}} \overline{I_{y2,a2,j1}} S_{2,j1}^\dagger}{4\epsilon M_F}$ $-\frac{3h\overline{Y_{2,i3,r1}}\overline{L_{a1,i1}} C P_R \overline{L_{a2,i2}}^T S_{1,j2} Y_{1,i3,r2} Y_{2,i2,r1,r1} Y_{2,i2,r2} \overline{I_{y1,a1,j2}} \overline{I_{y2,a2,j1}} S_{2,j1}^\dagger}{8M_F}$ $-\frac{h\overline{Y_{2,i3,r1}}\overline{L_{a1,i1}} C P_R \overline{L_{a2,i2}}^T S_{1,j2} Y_{1,i3,r1} Y_{2,i2,i2,r2} \overline{I_{y1,a1,j2}} \overline{I_{y2,a2,j1}} S_{2,j1}^\dagger}{\epsilon M_F}$	
cont.									$-\frac{h\overline{Y_{2,i3,r1}}\overline{L_{a1,i1}} C P_R \overline{L_{a2,i2}}^T S_{1,j2} Y_{1,i3,r2} Y_{2,i2,i2,r2} \overline{I_{y1,a1,j2}} \overline{I_{y2,a2,j1}} S_{2,j1}^\dagger}{2M_F}$ $-\frac{h\overline{Y_{2,i3,r1}}\overline{L_{a1,i1}} C P_R \overline{L_{a2,i2}}^T S_{1,j2} Y_{1,i3,r2} Y_{2,i2,i2,r2} Y_{2,i3,r1} \overline{I_{y1,a1,j2}} \overline{I_{y2,a2,j1}} S_{2,j1}^\dagger}{2\epsilon M_F}$ $-\frac{h\overline{Y_{2,i3,r1}}\overline{L_{a1,i1}} C P_R \overline{L_{a2,i2}}^T S_{1,j2} Y_{1,i3,r2} Y_{2,i2,i2,r2} Y_{2,i3,r1} \overline{I_{y1,a1,j2}} \overline{I_{y2,a2,j1}} S_{2,j1}^\dagger}{4M_F}$	
cont.									$-\frac{h\overline{Y_{1,i1,r1}}\overline{L_{a1,i1}} C P_R \overline{L_{a2,i2}}^T S_{1,j2} Y_{1,i1,r2} Y_{2,i2,r2} Y_{2,i3,r1} \overline{I_{y1,a1,j2}} \overline{I_{y2,a2,j1}} S_{2,j1}^\dagger}{4M_F}$ $-\frac{h\overline{Y_{1,i1,r1}}\overline{L_{a1,i1}} C P_R \overline{L_{a2,i2}}^T S_{1,j2} Y_{1,i1,r2} Y_{2,i2,r2} Y_{2,i3,r1} \overline{I_{y1,a1,j2}} \overline{I_{y2,a2,j1}} S_{2,j1}^\dagger}{4\epsilon M_F}$ $-\frac{3h\overline{Y_{2,i3,r1}}\overline{L_{a1,i1}} C P_R \overline{L_{a2,i2}}^T S_{1,j2} Y_{1,i1,r2} Y_{2,i2,i2,r2} Y_{2,i3,r1} \overline{I_{y1,a1,j2}} \overline{I_{y2,a2,j1}} S_{2,j1}^\dagger}{8M_F}$	
cont.									$+\frac{h\overline{Y_{1,i1,r1}}\overline{Y_{2,i2,r1}}\overline{L_{a2,i2}}^T C P_L L_{a1,i1} S_{2,j1} \epsilon_{a1,a2} S_{1,j1}^\dagger \lambda_{12}^{(1)}}{\epsilon M_F}$ $+\frac{h\overline{Y_{1,i1,r1}}\overline{Y_{2,i2,r1}}\overline{L_{a2,i2}}^T C P_L L_{a1,i1} S_{2,j1} \epsilon_{a1,a2} S_{1,j1}^\dagger \lambda_{12}^{(1)}}{M_F}$ $+\frac{h\overline{L_{a1,i1}} C P_R \overline{L_{a2,i2}}^T S_{1,j1} Y_{1,i1,r1} Y_{2,i2,r1} \epsilon_{a1,a2} S_{2,j1}^\dagger \lambda_{12}^{(1)}}{\epsilon M_F}$ $+\frac{h\overline{L_{a1,i1}} C P_R \overline{L_{a2,i2}}^T S_{1,j1} Y_{1,i1,r1} Y_{2,i2,r1} \epsilon_{a1,a2} S_{2,j1}^\dagger \lambda_{12}^{(1)}}{M_F}$	
cont.									$-\frac{h\overline{Y_{1,i2,r1}}\overline{Y_{2,i1,r1}}\overline{L_{a2,i2}}^T C P_L L_{a1,i1} S_{2,j2} \overline{I_{y1,a1,j1}} \overline{I_{y2,a2,j2}} S_{1,j1}^\dagger \lambda_{12}^{(2)}}{\sqrt{3}\epsilon M_F}$ $-\frac{h\overline{Y_{1,i2,r1}}\overline{Y_{2,i1,r1}}\overline{L_{a2,i2}}^T C P_L L_{a1,i1} S_{2,j2} \overline{I_{y1,a1,j1}} \overline{I_{y2,a2,j2}} S_{1,j1}^\dagger \lambda_{12}^{(2)}}{\sqrt{3}M_F}$ $-\frac{h\overline{Y_{1,i1,r1}}\overline{Y_{2,i2,r1}}\overline{L_{a2,i2}}^T C P_L L_{a1,i1} S_{2,j2} \overline{I_{y1,a1,j1}} \overline{I_{y2,a2,j2}} S_{1,j1}^\dagger \lambda_{12}^{(2)}}{\sqrt{3}\epsilon M_F}$ $-\frac{h\overline{Y_{1,i1,r1}}\overline{Y_{2,i2,r1}}\overline{L_{a2,i2}}^T C P_L L_{a1,i1} S_{2,j2} \overline{I_{y1,a1,j1}} \overline{I_{y2,a2,j2}} S_{1,j1}^\dagger \lambda_{12}^{(2)}}{\sqrt{3}M_F}$	
cont.									$-\frac{h\overline{L_{a1,i1}} C P_R \overline{L_{a2,i2}}^T S_{1,j2} Y_{1,i2,r1} Y_{2,i1,r1} \overline{I_{y1,a1,j2}} \overline{I_{y2,a2,j1}} S_{2,j1}^\dagger \lambda_{12}^{(2)}}{\sqrt{3}\epsilon M_F}$ $-\frac{h\overline{L_{a1,i1}} C P_R \overline{L_{a2,i2}}^T S_{1,j2} Y_{1,i2,r1} Y_{2,i1,r1} \overline{I_{y1,a1,j2}} \overline{I_{y2,a2,j1}} S_{2,j1}^\dagger \lambda_{12}^{(2)}}{\sqrt{3}M_F}$ $-\frac{h\overline{L_{a1,i1}} C P_R \overline{L_{a2,i2}}^T S_{1,j2} Y_{1,i1,r1} Y_{2,i2,r2} \overline{I_{y1,a1,j2}} \overline{I_{y2,a2,j1}} S_{2,j1}^\dagger \lambda_{12}^{(2)}}{\sqrt{3}\epsilon M_F}$ $-\frac{h\overline{L_{a1,i1}} C P_R \overline{L_{a2,i2}}^T S_{1,j2} Y_{1,i1,r1} Y_{2,i2,r2} \overline{I_{y1,a1,j2}} \overline{I_{y2,a2,j1}} S_{2,j1}^\dagger \lambda_{12}^{(2)}}{\sqrt{3}M_F}$	
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$+2h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{Y_{1,i1,r1}} \overline{Y_{2,i2,r2}} S_{1,j2} S_{2,j1} Y_{1,i2,r1} Y_{2,i1,r2} S_{1,j1}^\dagger S_{2,j2}^\dagger$ $-2h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{Y_{1,i1,r1}} \overline{Y_{2,i2,r2}} S_{1,j1} S_{2,j2} Y_{1,i2,r1} Y_{2,i1,r2} S_{1,j1}^\dagger S_{2,j2}^\dagger$ $-h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{Y_{1,i1,r1}} S_{1,j4} S_{2,j3} Y_{1,i1,r1} S_{1,j1}^\dagger S_{2,j2}^\dagger \overline{I_{12,j2,j3,j1,j4}} \lambda_{12}^{(1)}$ $-h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{Y_{2,i1,r1}} S_{1,j4} S_{2,j3} Y_{2,i1,r1} S_{1,j1}^\dagger S_{2,j2}^\dagger \overline{I_{12,j2,j3,j1,j4}} \lambda_{12}^{(1)}$	$+\frac{3h\overline{Y_{1,i2,r1}}\overline{Y_{2,i1,r1}}\overline{L_{a2,i2}}^T C P_L L_{a1,i1} S_{2,j2} \overline{I_{y1,a1,j1}} \overline{I_{y2,a2,j2}} S_{1,j1}^\dagger \lambda_{12}^{(2)}}{4\epsilon M_F}$ $+\frac{h\overline{Y_{1,i2,r1}}\overline{Y_{2,i1,r1}}\overline{L_{a2,i2}}^T C P_L L_{a1,i1} S_{2,j2} \overline{I_{y1,a1,j1}} \overline{I_{y2,a2,j2}} S_{1,j1}^\dagger \lambda_{12}^{(2)}}{4M_F}$ $+\frac{\overline{Y_{1,i1,r1}}\overline{Y_{2,i2,r1}}\overline{L_{a2,i2}}^T C P_L L_{a1,i1} S_{2,j2} \overline{I_{y1,a1,j1}} \overline{I_{y2,a2,j2}} S_{1,j1}^\dagger \lambda_{12}^{(2)}}{M_F}$ $-\frac{h\overline{Y_{1,i1,r1}}\overline{Y_{1,i3,r2}}\overline{Y_{2,i2,r2}}\overline{L_{a2,i2}}^T C P_L L_{a1,i1} S_{2,j2} Y_{1,i3,r1} \overline{I_{y1,a1,j1}} \overline{I_{y2,a2,j2}} S_{1,j1}^\dagger \lambda_{12}^{(2)}}{4\epsilon M_F}$ $-\frac{3h\overline{Y_{1,i1,r1}}\overline{Y_{1,i3,r2}}\overline{Y_{2,i2,r2}}\overline{L_{a2,i2}}^T C P_L L_{a1,i1} S_{2,j2} Y_{1,i3,r1} \overline{I_{y1,a1,j1}} \overline{I_{y2,a2,j2}} S_{1,j1}^\dagger \lambda_{12}^{(2)}}{8M_F}$ $-\frac{h\overline{Y_{1,i1,r1}}\overline{Y_{1,i3,r2}}\overline{Y_{2,i2,r2}}\overline{L_{a2,i2}}^T C P_L L_{a1,i1} S_{2,j2} Y_{1,i3,r2} \overline{I_{y1,a1,j1}} \overline{I_{y2,a2,j2}} S_{1,j1}^\dagger \lambda_{12}^{(2)}}{2\epsilon M_F}$ $-\frac{h\overline{Y_{1,i1,r1}}\overline{Y_{1,i3,r2}}\overline{Y_{2,i2,r2}}\overline{L_{a2,i2}}^T C P_L L_{a1,i1} S_{2,j2} Y_{1,i3,r2} \overline{I_{y1,a1,j1}} \overline{I_{y2,a2,j2}} S_{1,j1}^\dagger \lambda_{12}^{(2)}}{4M_F}$ $-\frac{h\overline{Y_{1,i1,r1}}\overline{Y_{1,i3,r2}}\overline{Y_{2,i2,r2}}\overline{L_{a2,i2}}^T C P_L L_{a1,i1} S_{2,j2} Y_{1,i3,r2} \overline{I_{y1,a1,j1}} \overline{I_{y2,a2,j2}} S_{1,j1}^\dagger \lambda_{12}^{(2)}}{4\epsilon M_F}$ $-\frac{h\overline{Y_{1,i1,r1}}\overline{Y_{1,i3,r2}}\overline{Y_{2,i2,r2}}\overline{L_{a2,i2}}^T C P_L L_{a1,i1} S_{2,j2} Y_{1,i3,r2} \overline{I_{y1,a1,j1}} \overline{I_{y2,a2,j2}} S_{1,j1}^\dagger \lambda_{12}^{(2)}}{8M_F}$ $-\frac{h\overline{Y_{1,i1,r1}}\overline{Y_{1,i3,r2}}\overline{Y_{2,i2,r2}}\overline{L_{a2,i2}}^T C P_L L_{a1,i1} S_{2,j2} Y_{1,i3,r2} \overline{I_{y1,a1,j1}} \overline{I_{y2,a2,j2}} S_{1,j1}^\dagger \lambda_{12}^{(2)}}{4\epsilon M_F}$ $-\frac{h\overline{Y_{1,i1,r1}}\overline{Y_{1,i3,r2}}\overline{Y_{2,i2,r2}}\overline{L_{a2,i2}}^T C P_L L_{a1,i1} S_{2,j2} Y_{1,i3,r2} \overline{I_{y1,a1,j1}} \overline{I_{y2,a2,j2}} S_{1,j1}^\dagger \lambda_{12}^{(2)}}{4\epsilon M_F}$ $-\frac{h\overline{Y_{1,i1,r1}}\overline{Y_{1,i3,r2}}\overline{Y_{2,i2,r2}}\overline{L_{a2,i2}}^T C P_L L_{a1,i1} S_{2,j2} Y_{1,i3,r2} \overline{I_{y1,a1,j1}} \overline{I_{y2,a2,j2}} S_{1,j1}^\dagger \lambda_{12}^{(2)}}{8M_F}$ $-\frac{h\overline{Y_{1,i1,r1}}\overline{Y_{1,i3,r2}}\overline{Y_{2,i2,r2}}\overline{L_{a2,i2}}^T C P_L L_{a1,i1} S_{2,j2} Y_{1,i3,r2} \overline{I_{y1,a1,j1}} \overline{I_{y2,a2,j2}} S_{1,j1}^\dagger \lambda_{12}^{(2)}}{4\epsilon M_F}$ $-\frac{h\overline{Y_{1,i1,r1}}\overline{Y_{1,i3,r2}}\overline{Y_{2,i2,r2}}\overline{L_{a2,i2}}^T C P_L L_{a1,i1} S_{2,j2} Y_{1,i3,r2} \overline{I_{y1,a1,j1}} \overline{I_{y2,a2,j2}} S_{1,j1}^\dagger \lambda_{12}^{(2)}}{4\epsilon M_F}$ $-\frac{h\overline{Y_{1,i1,r1}}\overline{Y_{1,i3,r2}}\overline{Y_{2,i2,r2}}\overline{L_{a2,i2}}^T C P_L L_{a1,i1} S_{2,j2} Y_{1,i3,r2} \overline{I_{y1,a1,j1}} \overline{I_{y2,a2,j2}} S_{1,j1}^\dagger \lambda_{12}^{(2)}}{8M_F}$ $-\frac{h\overline{Y_{1,i1,r1}}\overline{Y_{1,i3,r2}}\overline{Y_{2,i2,r2}}\overline{L_{a2,i2}}^T C P_L L_{a1,i1} S_{2,j2} Y_{1,i3,r2} \overline{I_{y1,a1,j1}} \overline{I_{y2,a2,j2}} S_{1,j1}^\dagger \lambda_{12}^{(2)}}{4\epsilon M_F}$ $-\frac{h\overline{Y_{1,i1,r1}}\overline{Y_{1,i3,r2}}\overline{Y_{2,i2,r2}}\overline{L_{a2,i2}}^T C P_L L_{a1,i1} S_{2,j2} Y_{1,i3,r2} \overline{I_{y1,a1,j1}} \overline{I_{y2,a2,j2}} S_{1,j1}^\dagger \lambda_{12}^{(2)}}{4\epsilon M_F}$ $-\frac{h\overline{Y_{1,i1,r1}}\overline{Y_{1,i3,r2}}\overline{Y_{2,i2,r2}}\overline{L_{a2,i2}}^T C P_L L_{a1,i1} S_{2,j2} Y_{1,i3,r2} \overline{I_{y1,a1,j1}} \overline{I_{y2,a2,j2}} S_{1,j1}^\dagger \lambda_{12}^{(2)}}{8M_F}$ $-\frac{h\overline{Y_{1,i1,r1}}\overline{Y_{1,i3,r2}}\overline{Y_{2,i2,r2}}\overline{L_{a2,i2}}^T C P_L L_{a1,i1} S_{2,j2} Y_{1,i3,r2} \overline{I_{y1,a1,j1}} \overline{I_{y2,a2,j2}} S_{1,j1}^\dagger \lambda_{12}^{(2)}}{4\epsilon M_F}$ $-\frac{h\overline{Y_{1,i1,r1}}\overline{Y_{1,i3,r2}}\overline{Y_{2,i2,r2}}\overline{L_{a2,i2}}^T C P_L L_{a1,i1} S_{2,j2} Y_{1,i3,r2} \overline{I_{y1,a1,j1}} \overline{I_{y2,a2,j2}} S_{1,j1}^\dagger \lambda_{12}^{(2)}}{4\epsilon M_F}$ $-\frac{h\overline{Y_{1,i1,r1}}\overline{Y_{1,i3,r2}}\overline{Y_{2,i2,r2}}\overline{L_{a2,i2}}^T C P_L L_{a1,i1} S_{2,j2} Y_{1,i3,r2} \overline{I_{y1,a1,j1}} \overline{I_{y2,a2,j2}} S_{1,j1}^\dagger \lambda_{12}^{(2)}}{8M_F}$ $-\frac{h\overline{Y_{1,i1,r1}}\overline{Y_{1,i3,r2}}\overline{Y_{2,i2,r2}}\overline{L_{a2,i2}}^T C P_L L_{a1,i1} S_{2,j2} Y_{1,i3,r2} \overline{I_{y1,a1,j1}} \overline{I_{y2,a2,j2}} S_{1,j1}^\dagger \lambda_{12}^{(2)}}{4\epsilon M_F}$ $-\frac{h\overline{Y_{1,i1,r1}}\overline{Y_{1,i3,r2}}\overline{Y_{2,i2,r2}}\overline{L_{a2,i2}}^T C P_L L_{a1,i1} S_{2,j2} Y_{1,i3,r2} \overline{I_{y1,a1,j1}} \overline{I_{y2,a2,j2}} S_{1,j1}^\dagger \lambda_{12}^{(2)}}{4\epsilon M_F}$ $-\frac{h\overline{Y_{1,i1,r1}}\overline{Y_{1,i3,r2}}\overline{Y_{2,i2,r2}}\overline{L_{a2,i2}}^T C P_L L_{a1,i1} S_{2,j2} Y_{1,i3,r2} \overline{I_{y1,a1,j1}} \overline{I_{y2,a2,j2}} S_{1,j1}^\dagger \lambda_{12}^{(2)}}{8M_F}$ $-\frac{h\overline{Y_{1,i1,r1}}\overline{Y_{1,i3,r2}}\overline{Y_{2,i2,r2}}\overline{L_{a2,i2}}^T C P_L L_{a1,i1} S_{2,j2} Y_{1,i3,r2} \overline{I_{y1,a1,j1}} \overline{I_{y2,a2,j2}} S_{1,j1}^\dagger \lambda_{12}^{(2)}}{4\epsilon M_F}$ $-\frac{h\overline{Y_{1,i1,r1}}\overline{Y_{1,i3,r2}}\overline{Y_{2,i2,r2}}\overline{L_{a2,i2}}^T C P_L L_{a1,i1} S_{2,j2} Y_{1,i3,r2} \overline{I_{y1,a1,j1}} \overline{I_{y2,a2,j2}} S_{1,j1}^\dagger \lambda_{12}^{(2)}}{4\epsilon M_F}$ $-\frac{h\overline{Y_{1,i1,r1}}\overline{Y_{1,i3,r2}}\overline{Y_{2,i2,r2}}\overline{L_{a2,i2}}^T C P_L L_{a1,i1} S_{2,j2} Y_{1,i3,r2} \overline{I_{y1,a1,j1}} \overline{I_{y2,a2,j2}} S_{1,j1}^\dagger \lambda_{12}^{(2)}}{8M_F}$ $-\frac{h\overline{Y_{1,i1,r1}}\overline{Y_{1,i3,r2}}\overline{Y_{2,i2,r2}}\overline{L_{a2,i2}}^T C P_L L_{a1,i1} S_{2,j2} Y_{1,i3,r2} \overline{I_{y1,a1,j1}} \overline{I_{y2,a2,j2}} S_{1,j1}^\dagger \lambda_{12}^{(2)}}{4\epsilon M_F}$ $-\frac{h\overline{Y_{1,i1,r1}}\overline{Y_{1,i3,r2}}\overline{Y_{2,i2,r2}}\overline{L_{a2,i2}}^T C P_L L_{a1,i1} S_{2,j2} Y_{1,i3,r2} \overline{I_{y1,a1,j1}} \overline{I_{y2,a2,j2}} S_{1,j1}^\dagger \lambda_{12}^{(2)}}{4\epsilon M_F}$ $-\frac{h\overline{Y_{1,i1,r1}}\overline{Y_{1,i3,r2}}\overline{Y_{2,i2,r2}}\overline{L_{a2,i2}}^T C P_L L_{a1,i1} S_{2,j2} Y_{1,i3,r2} \overline{I_{y1,a1,j1}} \overline{I_{y2,a2,j2}} S_{1,j1}^\dagger \lambda_{12}^{(2)}}{8M_F}$ $-\frac{h\overline{Y_{1,i1,r1}}\overline{Y_{1,i3,r2}}\overline{Y_{2,i2,r2}}\overline{L_{a2,i2}}^T C P_L L_{a1,i1} S_{2,j2} Y_{1,i3,r2} \overline{I_{y1,a1,j1}} \overline{I_{y2,a2,j2}} S_{1,j1}^\dagger \lambda_{12}^{(2)}}{4\epsilon M_F}$ $-\frac{h\overline{Y_{1,i1,r1}}\overline{Y_{1,i3,r2}}\overline{Y_{2,i2,r2}}\overline{L_{a2,i2}}^T C P_L L_{a1,i1} S_{2,j2} Y_{1,i3,r2} \overline{I_{y1,a1,j1}} \overline{I_{y2,a2,j2}} S_{1,j1}^\dagger \lambda_{12}^{(2)}}{4\epsilon M_F}$ $-\frac{h\overline{Y_{1,i1,r1}}\overline{Y_{1,i3,r2}}\overline{Y_{2,i2,r2}}\overline{L_{a2,i2}}^T C P_L L_{a1,i1} S_{2,j2} Y_{1,i3,r2} \overline{I_{y1,a1,j1}} \overline{I_{y2,a2,j2}} S_{1,j1}^\dagger \lambda_{12}^{(2)}}{8M_F}$ $-\frac{h\overline{Y_{1,i1,r1}}\overline{Y_{1,i3,r2}}\overline{Y_{2,i2,r2}}\overline{L_{a2,i2}}^T C P_L L_{a1,i1} S_{2,j2} Y_{1,i3,r2} \overline{I_{y1,a1,j1}} \overline{I_{y2,a2,j2}} S_{1,j1}^\dagger \lambda_{12}^{(2)}}{4\epsilon M_F}$ $-\frac{h\overline{Y_{1,i1,r1}}\overline{Y_{1,i3,r2}}\overline{Y_{2,i2,r2}}\overline{L_{a2,i2}}^T C P_L L_{a1,i1} S_{2,j2} Y_{1,i3,r2} \overline{I_{y1,a1,j1}} \overline{I_{y2,a2,j2}} S_{1,j1}^\dagger \lambda_{12}^{(2)}}{4\epsilon M_F}$ $-\frac{h\overline{Y_{1,i1,r1}}\overline{Y_{1,i3,r2}}\overline{Y_{2,i2,r2}}\overline{L_{a2,i2}}^T C P_L L_{a1,i1} S_{2,j2} Y_{1,i3,r2} \overline{I_{y1,a1,j1}} \overline{I_{y2,a2,j2}} S_{1,j1}^\dagger \lambda_{12}^{(2)}}{8M_F}$ $-\frac{h\overline{Y_{1,i1,r1}}\overline{Y_{1,i3,r2}}\overline{Y_{2,i2,r2}}\overline{L_{a2,i2}}^T C P_L L_{a1,i1} S_{2,j2} Y_{1,i3,r2} \overline{I_{y1,a1,j1}} \overline{I_{y2,a2,j2}} S_{1,j1}^\dagger \lambda_{12}^{(2)}}{4\epsilon M_F}$ $-\frac{h\overline{Y_{1,i1,r1}}\overline{Y_{1,i3,r2}}\overline{Y_{2,i2,r2}}\overline{L_{a2,i2}}^T C P_L L_{a1,i1} S_{2,j2} Y_{1,i3,r2} \overline{I_{y1,a1,j1}} \overline{I_{y2,a2,j2}} S_{1,j1}^\dagger \lambda_{12}^{(2)}}{4\epsilon M_F}$ $-\frac{h\overline{Y_{1,i1,r1}}\overline{Y_{1,i3,r2}}\overline{Y_{2,i2,r2}}\overline{L_{a2,i2}}^T C P_L L_{a1,i1} S_{2,j2} Y_{1,i3,r2} \overline{I_{y1,a1,j1}} \overline{I_{y2,a2,j2}} S_{1,j1}^\dagger \lambda_{12}^{(2)}}{8M_F}$ $-\frac{h\overline{Y_{1,i1,r1}}\overline{Y_{1,i3,r2}}\overline{Y_{2,i2,r2}}\overline{L_{a2,i2}}^T C P_L L_{a1,i1} S_{2,j2} Y_{1,i3,r2} \overline{I_{y1,a1,j1}} \overline{I_{y2,a2,j2}} S_{1,j1}^\dagger \lambda_{12}^{(2)}}{4\epsilon M_F}$ $-\frac{h\overline{Y_{1,i1,r1}}\overline{Y_{1,i3,r2}}\overline{Y_{2,i2,r2}}\overline{L_{a2,i2}}^T C P_L L_{a1,i1} S_{2,j2} Y_{1,i3,r2} \overline{I_{y1,a1,j1}} \overline{I_{y2,a2,j2}} S_{1,j1}^\dagger \lambda_{12}^{(2)}}{4\epsilon M_F}$ $-\frac{h\overline{Y_{1,i1,r1}}\overline{Y_{1,i3,r2}}\overline{Y_{2,i2,r2}}\overline{L_{a2,i2}}^T C P_L L_{a1,i1} S_{2,j2} Y_{1,i3,r2} \overline{I_{y1,a1,j1}} \overline{I_{y2,a2,j2}} S_{1,j1}^\dagger \lambda_{12}^{(2)}}{8M_F}$ $-\frac{h\overline{Y_{1,i1,r1}}\overline{Y_{1,i3,r2}}\overline{Y_{2,i2,r2}}\overline{L_{a2,i2}}^T C P_L L_{a1,i1} S_{2,j2} Y_{1,i3,r2} \overline{I_{y1,a1,j1}} \overline{I_{y2,a2,j2}} S_{1,j1}^\dagger \lambda_{12}^{(2)}}{4\epsilon M_F}$ $-\frac{h\overline{Y_{1,i1,r1}}\overline{Y_{1,i3,r2}}\overline{Y_{2,i2,r2}}\overline{L_{a2,i2}}^T C P_L L_{a1,i1} S_{2,j2} Y_{1,i3,r2} \overline{I_{y1,a1,j1}} \overline{I_{y2,a2,j2}} S_{1,j1}^\dagger \lambda_{12}^{(2)}}{4\epsilon M_F}$ $-\frac{h\overline{Y_{1,i1,r1}}\overline{Y_{1,i3,r2}}\overline{Y_{2,i2,r2}}\overline{L_{a2,i2}}^T C P_L L_{a1,i1} S_{2,j2} Y_{1,i3,r2} \overline{I_{y1,a1,j1}} \overline{I_{y2,a2,j2}} S_{1,j1}^\dagger \lambda_{12}^{(2)}}{8M_F}$ $-\frac{h\overline{Y_{1,i1,r1}}\overline{Y_{1,i3,r2}}\overline{Y_{2,i2,r2}}\overline{L_{a2,i2}}^T C P_L L_{a1,i1} S_{2,j2} Y_{1,i3,r2} \overline{I_{y1,a1,j1}} \overline{I_{y2,a2,j2}} S_{1,j1}^\dagger \lambda_{12}^{(2)}}{4\epsilon M_F}$ $-\frac{h\overline{Y_{1,i1,r1}}\overline{Y_{1,i3,r2}}\overline{Y_{2,i2,r2}}\overline{L_{a2,i2}}^T C P_L L_{a1,i1} S_{2,j2} Y_{1,i3,r2} \overline{I_{y1,a1,j1}} \overline{I_{y2,a2,j2}} S_{1,j1}^\dagger \lambda_{12}^{(2)}}{4\epsilon M_F}$ $-\frac{h\overline{Y_{1,i1,r1}}\overline{Y_{1,i3,r2}}\overline{Y_{2,i2,r2}}\overline{L_{a2,i2}}^T C P_L L_{a1,i1} S_{2,j2} Y_{1,i3,r2} \overline{I_{y1,a1,j1}} \overline{I_{y2,a2,j2}} S_{1,j1}^\dagger \lambda_{12}^{(2)}}{8M_F}$ $-\frac{h\overline{Y_{1,i1,r1}}\overline{Y_{1,i3,r2}}\overline{Y_{2,i2,r2}}\overline{L_{a2,i2}}^T C P_L L_{a1,i1} S_{2,j2} Y_{1,i3,r2} \overline{I_{y1,a1,j1}} \overline{I_{y2,a2,j2}} S_{1,j1}^\dagger \lambda_{12}^{(2)}}{4\epsilon M_F}$ $-\frac{h\overline{Y_{1,i1,r1}}\overline{Y_{1,i3,r2}}\overline{Y_{2,i2,r2}}\overline{L_{a2,i2}}^T C P_L L_{a1,i1} S_{2,j2} Y_{1,i3,r2} \overline{I_{y1,a1,j1}} \overline{I_{y2,a2,j2}} S_{1,j1}^\dagger \lambda_{12}^{(2)}}{4\epsilon M_F}$ $-\frac{h\overline{Y_{1,i1,r1}}\overline{Y_{1,i3,r2}}\overline{Y_{2,i2,r2}}\overline{L_{a2,i2}}^T C P_L L_{a1,i1} S_{2,j2} Y_{1,i3,r2} \overline{I_{y1,a1,j1}} \overline{I_{y2,a2,j2}} S_{1,j1}^\dagger \lambda_{12}^{(2)}}{8M_F}$ $-\frac{h\overline{Y_{1,i1,r1}}\overline{Y_{1,i3,r2}}\overline{Y_{2,i2,r2}}\overline{L_{a2,i2}}^T C P_L L_{a1,i1} S_{2,j2} Y_{1,i3,r2} \overline{I_{y1,a1,j1}} \overline{I_{y2,a2,j2}} S_{1,j1}^\dagger \lambda_{12}^{(2)}}{4\epsilon M_F}$ $-\frac{h\overline{Y_{1,i1,r1}}\overline{Y_{1,i3,r2}}\overline{Y_{2,i2,r2}}\overline{L_{a2,i2}}^T C P_L L_{a1,i1} S_{2,j2} Y_{1,i3,r2} \overline{I_{y1,a1,j1}} \overline{I_{y2,a2,j2}} S_{1,j1}^\dagger \lambda_{12}^{(2)}}{4\epsilon M_F}$ $-\frac{h\overline{Y_{1,i1,r1}}\overline{Y_{1,i3,r2}}\overline{Y_{2,i2,r2}}\overline{L_{a2,i2}}^T C P_L L_{a1,i1} S_{2,j2} Y_{1,i3,r2} \overline{I_{y1,a1,j1}} \overline{I_{y2,a2,j2}} S_{1,j1}^\dagger \lambda_{12}^{(2)}}{8M_F}$ $-\frac{h\overline{Y_{1,i1,r1}}\overline{Y_{1,i3,r2}}\overline{Y_{2,i2,r2}}\overline{L_{a2,i2}}^T C P_L L_{a1,i1} S_{2,j2} Y_{1,i3,r2} \overline{I_{y1,a1,j1}} \overline{I_{y2,a2,j2}} S_{1,j1}^\dagger \lambda_{12}^{(2)}}{4\epsilon M_F}$ $-\frac{h\overline{Y_{1,i1,r1}}\overline{Y_{1,i3,r2}}\overline{Y_{2,i2,r2}}\overline{L_{a2,i2}}^T C P_L L_{a1,i1} S_{2,j2} Y_{1,i3,r2} \overline{I_{y1,a1,j1}} \overline{I_{y2,a2,j2}} S_{1,j1}^\dagger \lambda_{12}^{(2)}}{4\epsilon M_F}$ $-\frac{h\overline{Y_{1,i1,r1}}\overline{Y_{1,i3,r2}}\overline{Y_{2,i2,r2}}\overline{L_{a2,i2}}^T C P_L L_{a1,i1} S_{2,j2} Y_{1,i3,r2} \overline{I_{y1,a1,j1}} \overline{I_{y2,a2,j2}} S_{1,j1}^\dagger \lambda_{12}^{(2)}}{8M_F}$ $-\frac{h\overline{Y_{1,i1,r1}}\overline{Y_{1,i3,r2}}\overline{Y_{2,i2,r2}}\overline{L_{a2,i2}}^T C P_L L_{a1,i1} S_{2,j2} Y_{1,i3,r2} \overline{I_{y1,a1,j1}} \overline{I_{y2,a2,j2}} S_{1,j1}^\dagger \lambda_{12}^{(2)}}{4\epsilon M_F}$ $-\frac{h\overline{Y_{1,i1,r1}}\overline{Y_{1,i3,r2}}\overline{Y_{2,i2,r2}}\overline{L_{a2,i2}}^T C P_L L_{a1,i1} S_{2,j2} Y_{1,i3,r2} \overline{I_{y1,a1,j1}} \overline{I_{y2,a2,j2}} S_{1,j1}^\dagger \lambda_{12}^{(2)}}{4\epsilon M_F}$ $-\frac{h\overline{Y_{1,i1,r1}}\overline{Y_{1,i3,r2}}\overline{Y_{2,i2,r2}}\overline{L_{a2,i2}}^T C P_L L_{a1,i1} S_{2,j2} Y_{1,i3,r2} \overline{I_{y1,a1,j1}} \overline{I_{y2,a2,j2}} S_{1,j1}^\dagger \lambda_{12}^{(2)}}{8M_F}$ $-\frac{h\over$	

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$S_1 S_2 F$	$S_1 S_2 \ell$	$S_1 F H$
cont.									$\frac{3h\overline{y}_{1,i_1,r_1}\overline{y}_{2,i_2,r_2}\overline{y}_{2,i_3,r_3}L_{a_2,i_2}^T C P_L L_{a_1,i_1}S_{2,j_2}y_{2,i_3,r_2}I_{y_1,a_1,j_1}^*I_{y_2,a_2,j_2}^*S_{1,j_1}^\dagger}{8M_F}$	
									$\frac{h\overline{y}_{1,i_1,r_1}\overline{y}_{2,i_2,r_1}\overline{y}_{2,i_3,r_2}L_{a_2,i_2}^T C P_L L_{a_1,i_1}S_{2,j_2}y_{2,i_3,r_2}I_{y_1,a_1,j_1}^*I_{y_2,a_2,j_2}^*S_{1,j_1}^\dagger}{2\epsilon M_F}$	
									$\frac{h\overline{y}_{1,i_1,r_1}\overline{y}_{2,i_2,r_1}\overline{y}_{2,i_3,r_2}L_{a_2,i_2}^T C P_L L_{a_1,i_1}S_{2,j_2}y_{2,i_3,r_2}I_{y_1,a_1,j_1}^*I_{y_2,a_2,j_2}^*S_{1,j_1}^\dagger}{4M_F}$	
									$+\frac{3h\overline{y}_{1,i_1,r_1}\overline{y}_{2,i_2,r_1}L_{a_2,i_2}^T C P_L L_{a_1,i_1}g_2^2S_{2,j_1}\epsilon_{a_1,a_2}S_{1,j_1}^\dagger}{4\epsilon M_F}$	
cont.									$+\frac{h\overline{y}_{1,i_1,r_1}\overline{y}_{2,i_2,r_1}L_{a_2,i_2}^T C P_L L_{a_1,i_1}g_2^2S_{2,j_1}\epsilon_{a_1,a_2}S_{1,j_1}^\dagger}{4M_F}$	
									$+\frac{h\overline{L}_{a_1,i_1}C P_R \overline{L}_{a_2,i_2}^T g_2^2S_{1,j_2}y_{1,i_2,r_1}y_{2,i_1,r_1}I_{y_1,a_1,j_2}I_{y_2,a_2,j_1}S_{2,j_1}^\dagger}{4M_F}$	
cont.									$+\frac{h\overline{L}_{a_1,i_1}C P_R \overline{L}_{a_2,i_2}^T g_2^2S_{1,j_2}y_{1,i_2,r_1}y_{2,i_1,r_1}I_{y_1,a_1,j_2}I_{y_2,a_2,j_1}S_{2,j_1}^\dagger}{4M_F}$	
									$+\frac{\overline{L}_{a_1,i_1}C P_R \overline{L}_{a_2,i_2}^T S_{1,j_2}y_{1,i_1,r_1}y_{2,i_3,r_1}I_{y_1,a_1,j_2}I_{y_2,a_2,j_1}S_{2,j_1}^\dagger}{M_F}$	
cont.									$\frac{h\overline{y}_{1,i_3,r_1}\overline{L}_{a_1,i_1}C P_R \overline{L}_{a_2,i_2}^T S_{1,j_2}y_{1,i_1,r_2}y_{1,i_3,r_1}y_{2,i_2,r_2}I_{y_1,a_1,j_2}I_{y_2,a_2,j_1}S_{2,j_1}^\dagger}{2\epsilon M_F}$	
									$\frac{h\overline{y}_{1,i_3,r_1}\overline{L}_{a_1,i_1}C P_R \overline{L}_{a_2,i_2}^T S_{1,j_2}y_{1,i_1,r_2}y_{1,i_3,r_1}y_{2,i_2,r_2}I_{y_1,a_1,j_2}I_{y_2,a_2,j_1}S_{2,j_1}^\dagger}{4M_F}$	
									$\frac{h\overline{y}_{1,i_3,r_1}\overline{L}_{a_1,i_1}C P_R \overline{L}_{a_2,i_2}^T S_{1,j_2}y_{1,i_1,r_2}y_{1,i_3,r_1}y_{2,i_2,r_2}I_{y_1,a_1,j_2}I_{y_2,a_2,j_1}S_{2,j_1}^\dagger}{4\epsilon M_F}$	
									$\frac{3h\overline{y}_{1,i_3,r_1}\overline{L}_{a_1,i_1}C P_R \overline{L}_{a_2,i_2}^T S_{1,j_2}y_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_2}I_{y_1,a_1,j_2}I_{y_2,a_2,j_1}S_{2,j_1}^\dagger}{8M_F}$	
cont.									$\frac{h\overline{y}_{2,i_3,r_1}\overline{L}_{a_1,i_1}C P_R \overline{L}_{a_2,i_2}^T S_{1,j_2}y_{1,i_3,r_2}y_{2,i_2,r_2}I_{y_1,a_1,j_2}I_{y_2,a_2,j_1}S_{2,j_1}^\dagger}{8M_F}$	
									$\frac{h\overline{y}_{2,i_3,r_1}\overline{L}_{a_1,i_1}C P_R \overline{L}_{a_2,i_2}^T S_{1,j_2}y_{1,i_3,r_2}y_{2,i_2,r_2}I_{y_1,a_1,j_2}I_{y_2,a_2,j_1}S_{2,j_1}^\dagger}{4M_F}$	
									$\frac{h\overline{y}_{2,i_3,r_1}\overline{L}_{a_1,i_1}C P_R \overline{L}_{a_2,i_2}^T S_{1,j_2}y_{1,i_3,r_2}y_{2,i_2,r_2}I_{y_1,a_1,j_2}I_{y_2,a_2,j_1}S_{2,j_1}^\dagger}{4M_F}$	
									$\frac{3h\overline{y}_{1,i_3,r_1}\overline{L}_{a_1,i_1}C P_R \overline{L}_{a_2,i_2}^T S_{1,j_2}y_{1,i_1,r_2}y_{1,i_3,r_2}y_{2,i_2,r_2}I_{y_1,a_1,j_2}I_{y_2,a_2,j_1}S_{2,j_1}^\dagger}{8M_F}$	
cont.									$\frac{3h\overline{y}_{1,i_3,r_1}\overline{L}_{a_1,i_1}C P_R \overline{L}_{a_2,i_2}^T S_{1,j_2}y_{1,i_1,r_2}y_{1,i_3,r_2}y_{2,i_2,r_2}I_{y_1,a_1,j_2}I_{y_2,a_2,j_1}S_{2,j_1}^\dagger}{8M_F}$	
									$+\frac{h\overline{L}_{a_1,i_1}C P_R \overline{L}_{a_2,i_2}^T g_2^2S_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger}{4M_F}$	
									$+\frac{h\overline{L}_{a_1,i_1}C P_R \overline{L}_{a_2,i_2}^T g_2^2S_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger}{4M_F}$	
									$+\frac{h\overline{y}_{1,i_1,r_1}\overline{y}_{2,i_2,r_1}L_{a_2,i_2}^T C P_L L_{a_1,i_1}S_{2,j_1}\epsilon_{a_1,a_2}S_{1,j_1}^\dagger\lambda_{12}^{(1)}}{\epsilon M_F}$	
cont.									$+\frac{h\overline{y}_{1,i_1,r_1}\overline{y}_{2,i_2,r_1}L_{a_2,i_2}^T C P_L L_{a_1,i_1}S_{2,j_1}\epsilon_{a_1,a_2}S_{1,j_1}^\dagger\lambda_{12}^{(1)}}{\epsilon M_F}$	
									$+\frac{h\overline{L}_{a_1,i_1}C P_R \overline{L}_{a_2,i_2}^T S_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger\lambda_{12}^{(1)}}{\epsilon M_F}$	
									$+\frac{h\overline{L}_{a_1,i_1}C P_R \overline{L}_{a_2,i_2}^T S_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger\lambda_{12}^{(1)}}{M_F}$	
cont.									$\frac{h\overline{y}_{1,i_2,r_1}\overline{y}_{2,i_1,r_1}L_{a_2,i_2}^T C P_L L_{a_1,i_1}S_{2,j_2}I_{y_1,a_1,j_1}^*I_{y_2,a_2,j_2}^*S_{1,j_1}^\dagger\lambda_{12}^{(2)}}{\sqrt{3}\epsilon M_F}$	
									$\frac{h\overline{y}_{1,i_2,r_1}\overline{y}_{2,i_1,r_1}L_{a_2,i_2}^T C P_L L_{a_1,i_1}S_{2,j_2}I_{y_1,a_1,j_1}^*I_{y_2,a_2,j_2}^*S_{1,j_1}^\dagger\lambda_{12}^{(2)}}{\sqrt{3}M_F}$	
									$\frac{h\overline{y}_{1,i_1,r_1}\overline{y}_{2,i_2,r_1}L_{a_2,i_2}^T C P_L L_{a_1,i_1}S_{2,j_2}I_{y_1,a_1,j_1}^*I_{y_2,a_2,j_2}^*S_{1,j_1}^\dagger\lambda_{12}^{(2)}}{\sqrt{3}\epsilon M_F}$	
									$\frac{h\overline{y}_{1,i_1,r_1}\overline{y}_{2,i_2,r_1}L_{a_2,i_2}^T C P_L L_{a_1,i_1}S_{2,j_2}I_{y_1,a_1,j_1}^*I_{y_2,a_2,j_2}^*S_{1,j_1}^\dagger\lambda_{12}^{(2)}}{\sqrt{3}M_F}$	
cont.									$\frac{h\overline{L}_{a_1,i_1}C P_R \overline{L}_{a_2,i_2}^T S_{1,j_2}y_{1,i_2,r_1}y_{2,i_1,r_1}I_{y_1,a_1,j_2}I_{y_2,a_2,j_1}S_{2,j_1}^\dagger\lambda_{12}^{(2)}}{\sqrt{3}\epsilon M_F}$	
									$\frac{h\overline{L}_{a_1,i_1}C P_R \overline{L}_{a_2,i_2}^T S_{1,j_2}y_{1,i_2,r_1}y_{2,i_1,r_1}I_{y_1,a_1,j_2}I_{y_2,a_2,j_1}S_{2,j_1}^\dagger\lambda_{12}^{(2)}}{\sqrt{3}M_F}$	
									$\frac{h\overline{L}_{a_1,i_1}C P_R \overline{L}_{a_2,i_2}^T S_{1,j_2}y_{1,i_1,r_1}y_{2,i_2,r_1}I_{y_1,a_1,j_2}I_{y_2,a_2,j_1}S_{2,j_1}^\dagger\lambda_{12}^{(2)}}{\sqrt{3}\epsilon M_F}$	
									$\frac{h\overline{L}_{a_1,i_1}C P_R \overline{L}_{a_2,i_2}^T S_{1,j_2}y_{1,i_1,r_1}y_{2,i_2,r_1}I_{y_1,a_1,j_2}I_{y_2,a_2,j_1}S_{2,j_1}^\dagger\lambda_{12}^{(2)}}{\sqrt{3}M_F}$	
cont.									$\frac{h\overline{L}_{a_1,i_1}C P_R \overline{L}_{a_2,i_2}^T S_{1,j_2}y_{1,i_1,r_1}y_{2,i_2,r_1}I_{y_1,a_1,j_2}I_{y_2,a_2,j_1}S_{2,j_1}^\dagger\lambda_{12}^{(2)}}{\sqrt{3}\epsilon M_F}$	
									$\frac{h\overline{L}_{a_1,i_1}C P_R \overline{L}_{a_2,i_2}^T S_{1,j_2}y_{1,i_1,r_1}y_{2,i_2,r_1}I_{y_1,a_1,j_2}I_{y_2,a_2,j_1}S_{2,j_1}^\dagger\lambda_{12}^{(2)}}{\sqrt{3}M_F}$	
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	$+2h\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{y}_{1,i_1,r_1}\overline{y}_{2,i_2,r_2}S_{1,j_2}S_{2,j_1}y_{1,i_2,r_1}y_{2,i_1,r_2}S_{1,j_1}^\dagger S_{2,j_2}^\dagger$	$+\frac{3h\overline{y}_{1,i_2,r_1}\overline{y}_{2,i_1,r_1}L_{a_2,i_2}^T C P_L L_{a_1,i_1}g_2^2S_{2,j_2}I_{y_1,a_1,j_1}^*I_{y_2,a_2,j_2}^*S_{1,j_1}^\dagger}{4\epsilon M_F}$	$-2h\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{y}_{1,i_1,r_1}\overline{y}_{2,i_2,r_2}H_{a_2}S_{1,j_2}y_{1,i_2,r_1}y_{2,i_1,r_2}I_{y_1,a_1,j_2}I_{y_2,a_2,j_1}^*H_{a_1}^\dagger S_{1,j_1}^\dagger$
								$-2h\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{y}_{1,i_1,r_1}\overline{y}_{2,i_2,r_2}S_{1,j_1}S_{2,j_2}y_{1,i_2,r_1}y_{2,i_1,r_2}S_{1,j_1}^\dagger S_{2,j_2}^\dagger$	$+\frac{h\overline{y}_{1,i_2,r_1}\overline{y}_{2,i_1,r_1}L_{a_2,i_2}^T C P_L L_{a_1,i_1}g_2^2S_{2,j_2}I_{y_1,a_1,j_1}^*I_{y_2,a_2,j_2}^*S_{1,j_1}^\dagger}{4M_F}$	$-h\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{y}_{1,i_1,r_1}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}H_{a_1}^\dagger S_{1,j_1}^\dagger I_{H1,a_1,a_2,j_1,j_2}^1\lambda_{12}^{(1)}$
								$-h\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{y}_{1,i_1,r_1}S_{1,j_4}S_{2,j_3}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger I_{12,j_2,j_3,j_1,j_4}^1\lambda_{12}^{(1)}$	$-\frac{9h\overline{y}_{1,i_1,r_1}\overline{y}_{2,i_2,r_1}L_{a_2,i_2}^T C P_L L_{a_1,i_1}g_2^2S_{2,j_2}I_{y_1,a_1,j_1}^*I_{y_2,a_2,j_2}^*S_{1,j_1}^\dagger}{4\epsilon M_F}$	$-h\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{y}_{1,i_1,r_1}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}H_{a_1}^\dagger S_{1,j_1}^\dagger I_{H1,a_1,a_2,j_1,j_2}^2\lambda_{12}^{(2)}$
								$-h\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{y}_{2,i_1,r_1}S_{1,j_4}S_{2,j_3}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger I_{12,j_2,j_3,j_1,j_4}^1\lambda_{12}^{(1)}$	$-\frac{3h\overline{y}_{1,i_1,r_1}\overline{y}_{2,i_2,r_1}L_{a_2,i_2}^T C P_L L_{a_1,i_1}g_2^2S_{2,j_2}I_{y_1,a_1,j_1}^*I_{y_2,a_2,j_2}^*S_{1,j_1}^\dagger}{4M_F}$	
cont.								$+\sqrt{3}h\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{y}_{1,i_1,r_1}S_{1,j_1}S_{2,j_2}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\lambda_{12}^{(2)}$	$+\frac{\overline{y}_{1,i_1,r_1}\overline{y}_{2,i_2,r_1}L_{a_2,i_2}^T C P_L L_{a_1,i_1}S_{2,j_2}I_{y_1,a_1,j_1}^*I_{y_2,a_2,j_2}^*S_{1,j_1}^\dagger}{M_F}$	
								$-\sqrt{3}h\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{y}_{1,i_1,r_1}S_{1,j_1}S_{2,j_2}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\lambda_{12}^{(2)}$	$\frac{h\overline{y}_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_2}L_{a_2,i_2}^T C P_L L_{a_1,i_1}S_{2,j_2}y_{1,i_3,r_1}I_{y_1,a_1,j_2}I_{y_2,a_2,j_1}S_{2,j_1}^\dagger}{4\epsilon M_F}$	

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	S_1	S_2	F	S_1	S_2	ℓ	S_1	F	H
cont.										$+\sqrt{3}h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}}S_{1,j_2}S_{2,j_1}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\lambda_{12}^{(2)}$			$\frac{3h\overline{y_{1,i_1,r_1}}y_{1,i_3,r_2}y_{2,i_2,r_1}L_{a_2,i_2}{}^TCPL_{a_1,i_1}S_{2,j_2}y_{1,i_3,r_1}x_{y_{1,a_1,j_1}}{}^*I_{y_{2,a_2,j_2}}{}^*S_{1,j_1}^\dagger}{8M_F}$			
										$-\sqrt{3}h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}}S_{1,j_1}S_{2,j_2}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\lambda_{12}^{(2)}$			$\frac{h\overline{y_{1,i_1,r_1}}y_{1,i_3,r_2}y_{2,i_2,r_1}L_{a_2,i_2}{}^TCPL_{a_1,i_1}S_{2,j_2}y_{1,i_3,r_2}x_{y_{1,a_1,j_1}}{}^*I_{y_{2,a_2,j_2}}{}^*S_{1,j_1}^\dagger}{24M_F}$			
cont.										$-\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^2\lambda_{12}^{(2)}$			$\frac{h\overline{y_{1,i_1,r_1}}y_{1,i_3,r_2}y_{2,i_2,r_1}L_{a_2,i_2}{}^TCPL_{a_1,i_1}S_{2,j_2}y_{1,i_3,r_2}x_{y_{1,a_1,j_1}}{}^*I_{y_{2,a_2,j_2}}{}^*S_{1,j_1}^\dagger}{4M_F}$			
													$\frac{h\overline{y_{1,i_1,r_1}}y_{1,i_2,r_2}y_{2,i_3,r_1}L_{a_2,i_2}{}^TCPL_{a_1,i_1}S_{2,j_2}y_{1,i_3,r_2}x_{y_{1,a_1,j_1}}{}^*I_{y_{2,a_2,j_2}}{}^*S_{1,j_1}^\dagger}{4M_F}$			
cont.										$-\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^2\lambda_{12}^{(2)}$			$\frac{3h\overline{y_{1,i_1,r_1}}y_{1,i_2,r_2}y_{2,i_3,r_1}L_{a_2,i_2}{}^TCPL_{a_1,i_1}S_{2,j_2}y_{1,i_3,r_2}x_{y_{1,a_1,j_1}}{}^*I_{y_{2,a_2,j_2}}{}^*S_{1,j_1}^\dagger}{8M_F}$			
													$\frac{h\overline{y_{1,i_1,r_1}}y_{2,i_1,r_2}y_{2,i_2,r_1}L_{a_2,i_2}{}^TCPL_{a_1,i_1}S_{2,j_2}y_{2,i_3,r_2}x_{y_{1,a_1,j_1}}{}^*I_{y_{2,a_2,j_2}}{}^*S_{1,j_1}^\dagger}{4M_F}$			
cont.													$\frac{3h\overline{y_{1,i_3,r_1}}y_{2,i_1,r_2}y_{2,i_2,r_1}L_{a_2,i_2}{}^TCPL_{a_1,i_1}S_{2,j_2}y_{2,i_3,r_2}x_{y_{1,a_1,j_1}}{}^*I_{y_{2,a_2,j_2}}{}^*S_{1,j_1}^\dagger}{8M_F}$			
													$\frac{h\overline{y_{1,i_1,r_1}}y_{2,i_1,r_2}y_{2,i_2,r_1}L_{a_2,i_2}{}^TCPL_{a_1,i_1}S_{2,j_2}y_{2,i_3,r_2}x_{y_{1,a_1,j_1}}{}^*I_{y_{2,a_2,j_2}}{}^*S_{1,j_1}^\dagger}{4M_F}$			
cont.													$\frac{3h\overline{y_{1,i_1,r_1}}y_{2,i_1,r_2}y_{2,i_3,r_1}L_{a_2,i_2}{}^TCPL_{a_1,i_1}S_{2,j_2}y_{2,i_3,r_2}x_{y_{1,a_1,j_1}}{}^*I_{y_{2,a_2,j_2}}{}^*S_{1,j_1}^\dagger}{8M_F}$			
													$\frac{h\overline{y_{1,i_1,r_1}}y_{2,i_1,r_2}y_{2,i_3,r_2}L_{a_2,i_2}{}^TCPL_{a_1,i_1}S_{2,j_2}y_{2,i_3,r_2}x_{y_{1,a_1,j_1}}{}^*I_{y_{2,a_2,j_2}}{}^*S_{1,j_1}^\dagger}{8M_F}$			
cont.													$\frac{h\overline{y_{1,i_1,r_1}}y_{2,i_1,r_2}y_{2,i_3,r_2}L_{a_2,i_2}{}^TCPL_{a_1,i_1}S_{2,j_2}y_{2,i_3,r_2}x_{y_{1,a_1,j_1}}{}^*I_{y_{2,a_2,j_2}}{}^*S_{1,j_1}^\dagger}{4M_F}$			
													$\frac{h\overline{y_{1,i_1,r_1}}y_{2,i_1,r_2}y_{2,i_3,r_2}L_{a_2,i_2}{}^TCPL_{a_1,i_1}S_{2,j_2}y_{2,i_3,r_2}x_{y_{1,a_1,j_1}}{}^*I_{y_{2,a_2,j_2}}{}^*S_{1,j_1}^\dagger}{4M_F}$			
cont.													$\frac{h\overline{y_{1,i_1,r_1}}y_{2,i_1,r_2}y_{2,i_3,r_2}L_{a_2,i_2}{}^TCPL_{a_1,i_1}S_{2,j_2}y_{2,i_3,r_2}x_{y_{1,a_1,j_1}}{}^*I_{y_{2,a_2,j_2}}{}^*S_{1,j_1}^\dagger}{4M_F}$			
													$\frac{h\overline{y_{1,i_1,r_1}}y_{2,i_1,r_2}y_{2,i_3,r_2}L_{a_2,i_2}{}^TCPL_{a_1,i_1}S_{2,j_2}y_{2,i_3,r_2}x_{y_{1,a_1,j_1}}{}^*I_{y_{2,a_2,j_2}}{}^*S_{1,j_1}^\dagger}{4M_F}$			
cont.													$\frac{h\overline{y_{1,i_1,r_1}}y_{2,i_1,r_2}y_{2,i_3,r_2}L_{a_2,i_2}{}^TCPL_{a_1,i_1}S_{2,j_2}y_{2,i_3,r_2}x_{y_{1,a_1,j_1}}{}^*I_{y_{2,a_2,j_2}}{}^*S_{1,j_1}^\dagger}{4M_F}$			
													$\frac{h\overline{y_{1,i_1,r_1}}y_{2,i_1,r_2}y_{2,i_3,r_2}L_{a_2,i_2}{}^TCPL_{a_1,i_1}S_{2,j_2}y_{2,i_3,r_2}x_{y_{1,a_1,j_1}}{}^*I_{y_{2,a_2,j_2}}{}^*S_{1,j_1}^\dagger}{4M_F}$			
cont.													$\frac{h\overline{y_{1,i_1,r_1}}y_{2,i_1,r_2}y_{2,i_3,r_2}L_{a_2,i_2}{}^TCPL_{a_1,i_1}S_{2,j_2}y_{2,i_3,r_2}x_{y_{1,a_1,j_1}}{}^*I_{y_{2,a_2,j_2}}{}^*S_{1,j_1}^\dagger}{4M_F}$			
													$\frac{h\overline{y_{1,i_1,r_1}}y_{2,i_1,r_2}y_{2,i_3,r_2}L_{a_2,i_2}{}^TCPL_{a_1,i_1}S_{2,j_2}y_{2,i_3,r_2}x_{y_{1,a_1,j_1}}{}^*I_{y_{2,a_2,j_2}}{}^*S_{1,j_1}^\dagger}{4M_F}$			
cont.													$\frac{h\overline{y_{1,i_1,r_1}}y_{2,i_1,r_2}y_{2,i_3,r_2}L_{a_2,i_2}{}^TCPL_{a_1,i_1}S_{2,j_2}y_{2,i_3,r_2}x_{y_{1,a_1,j_1}}{}^*I_{y_{2,a_2,j_2}}{}^*S_{1,j_1}^\dagger}{4M_F}$			
													$\frac{h\overline{y_{1,i_1,r_1}}y_{2,i_1,r_2}y_{2,i_3,r_2}L_{a_2,i_2}{}^TCPL_{a_1,i_1}S_{2,j_2}y_{2,i_3,r_2}x_{y_{1,a_1,j_1}}{}^*I_{y_{2,a_2,j_2}}{}^*S_{1,j_1}^\dagger}{4M_F}$			
cont.													$\frac{h\overline{y_{1,i_1,r_1}}y_{2,i_1,r_2}y_{2,i_3,r_2}L_{a_2,i_2}{}^TCPL_{a_1,i_1}S_{2,j_2}y_{2,i_3,r_2}x_{y_{1,a_1,j_1}}{}^*I_{y_{2,a_2,j_2}}{}^*S_{1,j_1}^\dagger}{4M_F}$			
													$\frac{h\overline{y_{1,i_1,r_1}}y_{2,i_1,r_2}y_{2,i_3,r_2}L_{a_2,i_2}{}^TCPL_{a_1,i_1}S_{2,j_2}y_{2,i_3,r_2}x_{y_{1,a_1,j_1}}{}^*I_{y_{2,a_2,j_2}}{}^*S_{1,j_1}^\dagger}{4M_F}$			
cont.													$\frac{h\overline{y_{1,i_1,r_1}}y_{2,i_1,r_2}y_{2,i_3,r_2}L_{a_2,i_2}{}^TCPL_{a_1,i_1}S_{2,j_2}y_{2,i_3,r_2}x_{y_{1,a_1,j_1}}{}^*I_{y_{2,a_2,j_2}}{}^*S_{1,j_1}^\dagger}{4M_F}$			
													$\frac{h\overline{y_{1,i_1,r_1}}y_{2,i_1,r_2}y_{2,i_3,r_2}L_{a_2,i_2}{}^TCPL_{a_1,i_1}S_{2,j_2}y_{2,i_3,r_2}x_{y_{1,a_1,j_1}}{}^*I_{y_{2,a_2,j_2}}{}^*S_{1,j_1}^\dagger}{4M_F}$			
cont.													$\frac{h\overline{y_{1,i_1,r_1}}y_{2,i_1,r_2}y_{2,i_3,r_2}L_{a_2,i_2}{}^TCPL_{a_1,i_1}S_{2,j_2}y_{2,i_3,r_2}x_{y_{1,a_1,j_1}}{}^*I_{y_{2,a_2,j_2}}{}^*S_{1,j_1}^\dagger}{4M_F}$			
													$\frac{h\overline{y_{1,i_1,r_1}}y_{2,i_1,r_2}y_{2,i_3,r_2}L_{a_2,i_2}{}^TCPL_{a_1,i_1}S_{2,j_2}y_{2,i_3,r_2}x_{y_{1,a_1,j_1}}{}^*I_{y_{2,a_2,j_2}}{}^*S_{1,j_1}^\dagger}{4M_F}$			
cont.													$\frac{h\overline{y_{1,i_1,r_1}}y_{2,i_1,r_2}y_{2,i_3,r_2}L_{a_2,i_2}{}^TCPL_{a_1,i_1}S_{2,j_2}y_{2,i_3,r_2}x_{y_{1,a_1,j_1}}{}^*I_{y_{2,a_2,j_2}}{}^*S_{1,j_1}^\dagger}{4M_F}$			
													$\frac{h\overline{y_{1,i_1,r_1}}y_{2,i_1,r_2}y_{2,i_3,r_2}L_{a_2,i_2}{}^TCPL_{a_1,i_1}S_{2,j_2}y_{2,i_3,r_2}x_{y_{1,a_1,j_1}}{}^*I_{y_{2,a_2,j_2}}{}^*S_{1,j_1}^\dagger}{4M_F}$			
cont.													$\frac{h\overline{y_{1,i_1,r_1}}y_{2,i_1,r_2}y_{2,i_3,r_2}L_{a_2,i_2}{}^TCPL_{a_1,i_1}S_{2,j_2}y_{2,i_3,r_2}x_{y_{1,a_1,j_1}}{}^*I_{y_{2,a_2,j_2}}{}^*S_{1,j_1}^\dagger}{4M_F}$			
													$\frac{h\overline{y_{1,i_1,r_1}}y_{2,i_1,r_2}y_{2,i_3,r_2}L_{a_2,i_2}{}^TCPL_{a_1,i_1}S_{2,j_2}y_{2,i_3,r_2}x_{y_{1,a_1,j_1}}{}^*I_{y_{2,a_2,j_2}}{}^*S_{1,j_1}^\dagger}{4M_F}$			
cont.													$\frac{h\overline{y_{1,i_1,r_1}}y_{2,i_1,r_2}y_{2,i_3,r_2}L_{a_2,i_2}{}^TCPL_{a_1,i_1}S_{2,j_2}y_{2,i_3,r_2}x_{y_{1,a_1,j_1}}{}^*I_{y_{2,a_2,j_2}}{}^*S_{1,j_1}^\dagger}{4M_F}$			
													$\frac{h\overline{y_{1,i_1,r_1}}y_{2,i_1,r_2}y_{2,i_3,r_2}L_{a_2,i_2}{}^TCPL_{a_1,i_1}S_{2,j_2}y_{2,i_3,r_2}x_{y_{1,a_1,j_1}}{}^*I_{y_{2,a_2,j_2}}{}^*S_{1,j_1}^\dagger}{4M_F}$			
cont.																

[illegible]

[illegible]

[illegible]

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$S_1 S_2 F$	$S_1 S_2 \ell$	$S_1 F H$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+C_{\frac{1}{2}g_2}^2 h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1} S_{1,j_1} S_{2,j_2} y_{1,i_1,r_1} \text{Bar}(\text{cg3})_{j_1,j_2}}$ $+\frac{5}{9} h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1} y_{2,i_2,r_2} S_{1,j_1} S_{1,j_3} S_{2,j_2} S_{2,j_4} y_{1,i_1,r_2} y_{1,i_2,r_1} \text{Bar}(\text{cg3})_{j_1,j_2} \text{Bar}(\text{cg3})_{j_3,j_4}}$ $+\frac{10h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_2} S_{1,j_1} S_{1,j_2} S_{2,j_3} y_{1,i_1,r_2} y_{1,i_2,r_1} \text{Bar}(\text{cg3})_{j_2,j_3} S_{2,j_4}^\dagger}}{3\sqrt{3}}$ $+\frac{10h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1} y_{2,i_2,r_2} S_{1,j_2} S_{2,j_1} S_{2,j_3} y_{1,i_1,r_2} y_{2,i_2,r_1} \text{Bar}(\text{cg3})_{j_2,j_3} S_{2,j_4}^\dagger}}{3\sqrt{3}}$	$-\frac{3h \overline{y_{1,i_1,r_1} y_{2,i_2,r_1} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_2} \overline{I_{T3,a_1,a_2,j_2,j_1} S_{1,j_1}^\dagger} g_2^2}}{8M_F}$ $-\frac{h \overline{y_{1,i_1,r_1} y_{2,i_2,r_1} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_2} \overline{I_{T3,a_1,a_2,j_2,j_1} S_{1,j_1}^\dagger} g_2^2}}{8M_F}$ $+\frac{9\sqrt{3}h \overline{y_{1,i_1,r_1} y_{2,i_2,r_1} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_1} \epsilon_{a_1,a_2} S_{1,j_1}^\dagger} g_2^2}{8M_F}$ $+\frac{3\sqrt{3}h \overline{y_{1,i_1,r_1} y_{2,i_2,r_1} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_1} \epsilon_{a_1,a_2} S_{1,j_1}^\dagger} g_2^2}{8M_F}$	$-2\sqrt{3}h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1} H_{a_1} H_{a_2} S_{1,j_1} S_{1,j_2} y_{1,i_1,r_1} \lambda_{T3} \text{Bar}(\text{cg1})_{a_1,j_1,a_2,j_2}}$ $-\frac{4h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} Y_{e,i_2,i_3} H_{a_1} S_{1,j_1} y_{1,i_2,r_1} Y_{e,i_1,i_3} H_{a_1}^\dagger S_{1,j_1}^\dagger}}{\sqrt{3}}$
cont.								$+C_{\frac{1}{2}g_2}^2 h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} \text{cg3}_{j_1,j_2} y_{2,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger}$ $+\frac{2}{3} h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_2} S_{1,j_2} S_{2,j_1} y_{1,i_1,r_2} y_{2,i_1,r_2} S_{1,j_1}^\dagger S_{2,j_2}^\dagger}$ $-\frac{10}{3} h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_2} S_{1,j_1} S_{2,j_2} y_{1,i_2,r_1} y_{2,i_1,r_2} S_{1,j_1}^\dagger S_{2,j_2}^\dagger}$ $+\frac{8}{3} h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_2} S_{1,j_2} S_{2,j_1} y_{1,i_1,r_2} y_{2,i_2,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger}$ $-\frac{10}{3} h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_2} S_{1,j_1} S_{2,j_2} y_{1,i_1,r_2} y_{2,i_2,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger}$ $+\frac{10h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{1,i_2,r_2} \text{cg3}_{j_2,j_3} S_{1,j_1} y_{1,i_1,r_2} y_{2,i_2,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger}}{3\sqrt{3}}$ $+\frac{10h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_2} \text{cg3}_{j_1,j_2} S_{2,j_2} y_{1,i_1,r_2} y_{2,i_2,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger}}{3\sqrt{3}}$	$-\frac{3h L_{a_1,i_1} C P_R \overline{L_{a_2,i_2}^T S_{1,j_2} y_{1,i_1,r_1} y_{2,i_2,r_1} \overline{I_{T3,a_1,a_2,j_2,j_1} S_{2,j_1}^\dagger} g_2^2}}{8M_F}$ $-\frac{h L_{a_1,i_1} C P_R \overline{L_{a_2,i_2}^T S_{1,j_2} y_{1,i_1,r_1} y_{2,i_2,r_1} \overline{I_{T3,a_1,a_2,j_2,j_1} S_{2,j_1}^\dagger} g_2^2}}{8M_F}$ $+\frac{9\sqrt{3}h L_{a_1,i_1} C P_R \overline{L_{a_2,i_2}^T S_{1,j_1} y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_2} S_{2,j_1}^\dagger} g_2^2}{8M_F}$ $+\frac{3\sqrt{3}h L_{a_1,i_1} C P_R \overline{L_{a_2,i_2}^T S_{1,j_1} y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_2} S_{2,j_1}^\dagger} g_2^2}{8M_F}$	$-\sqrt{3}h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} H_{a_2} S_{1,j_2} y_{1,i_1,r_1} H_{a_1}^\dagger S_{1,j_1}^\dagger \overline{I_{H1,a_1,a_2,j_1,j_2}^{-1} \lambda_{11}^{(1)}}}$ $-\sqrt{3}h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} H_{a_2} S_{1,j_2} y_{1,i_1,r_1} H_{a_1}^\dagger S_{1,j_1}^\dagger \overline{I_{H1,a_1,a_2,j_1,j_2}^{-1} \lambda_{12}^{(1)}}}$
cont.								$+\frac{8}{3} h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_2} S_{1,j_2} S_{2,j_1} y_{1,i_1,r_2} y_{2,i_2,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger}$ $-\frac{10}{3} h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_2} S_{1,j_1} S_{2,j_2} y_{1,i_1,r_2} y_{2,i_2,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger}$ $+\frac{10h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{1,i_2,r_2} \text{cg3}_{j_2,j_3} S_{1,j_1} y_{1,i_1,r_2} y_{2,i_2,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger}}{3\sqrt{3}}$ $+\frac{10h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_2} \text{cg3}_{j_1,j_2} S_{2,j_2} y_{1,i_1,r_2} y_{2,i_2,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger}}{3\sqrt{3}}$	$-\frac{3h \overline{y_{1,i_1,r_1} y_{2,i_2,r_1} L_{a_2,i_2}^T C P_L L_{a_1,i_1} g_2^2 S_{2,j_2} \overline{I_{T3,a_1,a_2,j_2,j_1} S_{1,j_1}^\dagger}}}{8M_F}$ $-\frac{h \overline{y_{1,i_1,r_1} y_{2,i_2,r_1} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_2} \overline{I_{T3,a_1,a_2,j_2,j_1} S_{1,j_1}^\dagger}}}{2M_F}$ $+\frac{\sqrt{3}h \overline{y_{1,i_1,r_1} y_{1,i_3,r_2} y_{2,i_2,r_2} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{1,i_3,r_1} \overline{I_{T3,a_1,a_2,j_2,j_1} S_{1,j_1}^\dagger}}}{8M_F}$ $+\frac{3\sqrt{3}h \overline{y_{1,i_1,r_1} y_{1,i_3,r_2} y_{2,i_2,r_2} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{1,i_3,r_1} \overline{I_{T3,a_1,a_2,j_2,j_1} S_{1,j_1}^\dagger}}}{16M_F}$ $+\frac{\sqrt{3}h \overline{y_{1,i_1,r_1} y_{1,i_3,r_2} y_{2,i_2,r_2} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{1,i_3,r_2} \overline{I_{T3,a_1,a_2,j_2,j_1} S_{1,j_1}^\dagger}}}{4M_F}$	
cont.								$+\frac{2h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1} S_{1,j_1} S_{2,j_2} S_{2,j_3} y_{1,i_1,r_1} \text{Bar}(\text{cg3})_{j_2,j_3} S_{1,j_1}^\dagger \lambda_{12}^{(2)}}}{\sqrt{3}}$ $+3h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} S_{1,j_2} S_{2,j_1} y_{1,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \lambda_{12}^{(2)}}$ $-3h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} S_{1,j_1} S_{2,j_2} y_{1,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \lambda_{12}^{(2)}}$ $+3h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1} S_{1,j_2} S_{2,j_1} y_{2,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \lambda_{12}^{(2)}}$ $-3h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1} S_{1,j_1} S_{2,j_2} y_{2,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \lambda_{12}^{(2)}}$ $+2h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} \text{cg3}_{j_2,j_3} S_{1,j_1} y_{2,i_1,r_1} S_{1,j_1}^\dagger S_{1,j_2}^\dagger S_{2,j_3}^\dagger \lambda_{12}^{(2)}}$ $+\frac{2h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} \text{cg3}_{j_2,j_3} S_{1,j_1} y_{2,i_1,r_1} S_{1,j_1}^\dagger S_{1,j_2}^\dagger S_{2,j_3}^\dagger \lambda_{12}^{(2)}}}{\sqrt{3}}$	$+\frac{\sqrt{3}h \overline{y_{1,i_1,r_1} y_{2,i_2,r_1} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{1,i_3,r_1} \overline{I_{T3,a_1,a_2,j_2,j_1} S_{1,j_1}^\dagger}}}{4M_F}$ $+\frac{3\sqrt{3}h \overline{y_{1,i_1,r_1} y_{2,i_2,r_1} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{1,i_3,r_2} \overline{I_{T3,a_1,a_2,j_2,j_1} S_{1,j_1}^\dagger}}}{16M_F}$ $+\frac{\sqrt{3}h \overline{y_{1,i_1,r_1} y_{2,i_2,r_1} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{1,i_3,r_2} \overline{I_{T3,a_1,a_2,j_2,j_1} S_{1,j_1}^\dagger}}}{2M_F}$ $+\frac{\sqrt{3}h \overline{y_{1,i_1,r_1} y_{2,i_2,r_1} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{1,i_3,r_2} \overline{I_{T3,a_1,a_2,j_2,j_1} S_{1,j_1}^\dagger}}}{16M_F}$ $+\frac{\sqrt{3}h \overline{y_{1,i_1,r_1} y_{2,i_2,r_1} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{1,i_3,r_2} \overline{I_{T3,a_1,a_2,j_2,j_1} S_{1,j_1}^\dagger}}}{16M_F}$ $+\frac{\sqrt{3}h \overline{y_{1,i_1,r_1} y_{2,i_2,r_1} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{1,i_3,r_2} \overline{I_{T3,a_1,a_2,j_2,j_1} S_{1,j_1}^\dagger}}}{16M_F}$ $+\frac{\sqrt{3}h \overline{y_{1,i_1,r_1} y_{2,i_2,r_1} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{1,i_3,r_2} \overline{I_{T3,a_1,a_2,j_2,j_1} S_{1,j_1}^\dagger}}}{16M_F}$ $+\frac{\sqrt{3}h \overline{y_{1,i_1,r_1} y_{2,i_2,r_1} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{1,i_3,r_2} \overline{I_{T3,a_1,a_2,j_2,j_1} S_{1,j_1}^\dagger}}}{16M_F}$ $+\frac{\sqrt{3}h \overline{y_{1,i_1,r_1} y_{2,i_2,r_1} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{1,i_3,r_2} \overline{I_{T3,a_1,a_2,j_2,j_1} S_{1,j_1}^\dagger}}}{16M_F}$ $+\frac{\sqrt{3}h \overline{y_{1,i_1,r_1} y_{2,i_2,r_1} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{1,i_3,r_2} \overline{I_{T3,a_1,a_2,j_2,j_1} S_{1,j_1}^\dagger}}}{16M_F}$ $+\frac{\sqrt{3}h \overline{y_{1,i_1,r_1} y_{2,i_2,r_1} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{1,i_3,r_2} \overline{I_{T3,a_1,a_2,j_2,j_1} S_{1,j_1}^\dagger}}}{16M_F}$ $+\frac{\sqrt{3}h \overline{y_{1,i_1,r_1} y_{2,i_2,r_1} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{1,i_3,r_2} \overline{I_{T3,a_1,a_2,j_2,j_1} S_{1,j_1}^\dagger}}}{16M_F}$ $+\frac{\sqrt{3}h \overline{y_{1,i_1,r_1} y_{2,i_2,r_1} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{1,i_3,r_2} \overline{I_{T3,a_1,a_2,j_2,j_1} S_{1,j_1}^\dagger}}}{16M_F}$ $+\frac{\sqrt{3}h \overline{y_{1,i_1,r_1} y_{2,i_2,r_1} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{1,i_3,r_2} \overline{I_{T3,a_1,a_2,j_2,j_1} S_{1,j_1}^\dagger}}}{16M_F}$ $+\frac{\sqrt{3}h \overline{y_{1,i_1,r_1} y_{2,i_2,r_1} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{1,i_3,r_2} \overline{I_{T3,a_1,a_2,j_2,j_1} S_{1,j_1}^\dagger}}}{16M_F}$ $+\frac{\sqrt{3}h \overline{y_{1,i_1,r_1} y_{2,i_2,r_1} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{1,i_3,r_2} \overline{I_{T3,a_1,a_2,j_2,j_1} S_{1,j_1}^\dagger}}}{16M_F}$ $+\frac{\sqrt{3}h \overline{y_{1,i_1,r_1} y_{2,i_2,r_1} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{1,i_3,r_2} \overline{I_{T3,a_1,a_2,j_2,j_1} S_{1,j_1}^\dagger}}}{16M_F}$ $+\frac{\sqrt{3}h \overline{y_{1,i_1,r_1} y_{2,i_2,r_1} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{1,i_3,r_2} \overline{I_{T3,a_1,a_2,j_2,j_1} S_{1,j_1}^\dagger}}}{16M_F}$ $+\frac{\sqrt{3}h \overline{y_{1,i_1,r_1} y_{2,i_2,r_1} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{1,i_3,r_2} \overline{I_{T3,a_1,a_2,j_2,j_1} S_{1,j_1}^\dagger}}}{16M_F}$ $+\frac{\sqrt{3}h \overline{y_{1,i_1,r_1} y_{2,i_2,r_1} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{1,i_3,r_2} \overline{I_{T3,a_1,a_2,j_2,j_1} S_{1,j_1}^\dagger}}}{16M_F}$ $+\frac{\sqrt{3}h \overline{y_{1,i_1,r_1} y_{2,i_2,r_1} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{1,i_3,r_2} \overline{I_{T3,a_1,a_2,j_2,j_1} S_{1,j_1}^\dagger}}}{16M_F}$ $+\frac{\sqrt{3}h \overline{y_{1,i_1,r_1} y_{2,i_2,r_1} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{1,i_3,r_2} \overline{I_{T3,a_1,a_2,j_2,j_1} S_{1,j_1}^\dagger}}}{16M_F}$ $+\frac{\sqrt{3}h \overline{y_{1,i_1,r_1} y_{2,i_2,r_1} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{1,i_3,r_2} \overline{I_{T3,a_1,a_2,j_2,j_1} S_{1,j_1}^\dagger}}}{16M_F}$ $+\frac{\sqrt{3}h \overline{y_{1,i_1,r_1} y_{2,i_2,r_1} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{1,i_3,r_2} \overline{I_{T3,a_1,a_2,j_2,j_1} S_{1,j_1}^\dagger}}}{16M_F}$ $+\frac{\sqrt{3}h \overline{y_{1,i_1,r_1} y_{2,i_2,r_1} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{1,i_3,r_2} \overline{I_{T3,a_1,a_2,j_2,j_1} S_{1,j_1}^\dagger}}}{16M_F}$ $+\frac{\sqrt{3}h \overline{y_{1,i_1,r_1} y_{2,i_2,r_1} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{1,i_3,r_2} \overline{I_{T3,a_1,a_2,j_2,j_1} S_{1,j_1}^\dagger}}}{16M_F}$ $+\frac{\sqrt{3}h \overline{y_{1,i_1,r_1} y_{2,i_2,r_1} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{1,i_3,r_2} \overline{I_{T3,a_1,a_2,j_2,j_1} S_{1,j_1}^\dagger}}}{16M_F}$ $+\frac{\sqrt{3}h \overline{y_{1,i_1,r_1} y_{2,i_2,r_1} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{1,i_3,r_2} \overline{I_{T3,a_1,a_2,j_2,j_1} S_{1,j_1}^\dagger}}}{16M_F}$ $+\frac{\sqrt{3}h \overline{y_{1,i_1,r_1} y_{2,i_2,r_1} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{1,i_3,r_2} \overline{I_{T3,a_1,a_2,j_2,j_1} S_{1,j_1}^\dagger}}}{16M_F}$ $+\frac{\sqrt{3}h \overline{y_{1,i_1,r_1} y_{2,i_2,r_1} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{1,i_3,r_2} \overline{I_{T3,a_1,a_2,j_2,j_1} S_{1,j_1}^\dagger}}}{16M_F}$ $+\frac{\sqrt{3}h \overline{y_{1,i_1,r_1} y_{2,i_2,r_1} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{1,i_3,r_2} \overline{I_{T3,a_1,a_2,j_2,j_1} S_{1,j_1}^\dagger}}}{16M_F}$ $+\frac{\sqrt{3}h \overline{y_{1,i_1,r_1} y_{2,i_2,r_1} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{1,i_3,r_2} \overline{I_{T3,a_1,a_2,j_2,j_1} S_{1,j_1}^\dagger}}}{16M_F}$ $+\frac{\sqrt{3}h \overline{y_{1,i_1,r_1} y_{2,i_2,r_1} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{1,i_3,r_2} \overline{I_{T3,a_1,a_2,j_2,j_1} S_{1,j_1}^\dagger}}}{16M_F}$ $+\frac{\sqrt{3}h \overline{y_{1,i_1,r_1} y_{2,i_2,r_1} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{1,i_3,r_2} \overline{I_{T3,a_1,a_2,j_2,j_1} S_{1,j_1}^\dagger}}}{16M_F}$ $+\frac{\sqrt{3}h \overline{y_{1,i_1,r_1} y_{2,i_2,r_1} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{1,i_3,r_2} \overline{I_{T3,a_1,a_2,j_2,j_1} S_{1,j_1}^\dagger}}}{16M_F}$ $+\frac{\sqrt{3}h \overline{y_{1,i_1,r_1} y_{2,i_2,r_1} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{1,i_3,r_2} \overline{I_{T3,a_1,a_2,j_2,j_1} S_{1,j_1}^\dagger}}}{16M_F}$ $+\frac{\sqrt{3}h \overline{y_{1,i_1,r_1} y_{2,i_2,r_1} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{1,i_3,r_2} \overline{I_{T3,a_1,a_2,j_2,j_1} S_{1,j_1}^\dagger}}}{16M_F}$ $+\$	

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$S_1 S_2 F$	$S_1 S_2 \ell$	$S_1 F H$	
cont.									$+\frac{3h\overline{y}_{1,i_1,r_1}\overline{y}_{2,i_2,r_1}\overline{y}_{2,i_3,r_2}L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}^\dagger}{8M_F}$ $+\frac{\sqrt{3}h\overline{y}_{1,i_3,r_1} \overline{L}_{a_1,i_1} C P_R \overline{L}_{a_2,i_2}^T S_{1,j_2} y_{1,i_2,r_1} y_{1,i_3,r_2} y_{2,i_1,r_2} \overline{y}_{2,i_1,r_2} \overline{I} T \tau_{a_1,a_2,j_1,j_2}^* S_{2,j_1}^\dagger}{8cM_F}$ $+\frac{3\sqrt{3}h\overline{y}_{1,i_3,r_1} \overline{L}_{a_1,i_1} C P_R \overline{L}_{a_2,i_2}^T S_{1,j_2} y_{1,i_2,r_1} y_{1,i_3,r_2} y_{2,i_1,r_2} \overline{I} T \tau_{a_1,a_2,j_1,j_2}^* S_{2,j_1}^\dagger}{16M_F}$		
								$+\frac{3hL_{a_1,i_1} C P_R \overline{L}_{a_2,i_2}^T g_Y^2 S_{1,j_2} y_{1,i_1,r_1} y_{2,i_2,r_1} \overline{I} T \tau_{a_1,a_2,j_1,j_2}^* S_{2,j_1}^\dagger}{8cM_F}$ $-\frac{h\overline{L}_{a_1,i_1} C P_R \overline{L}_{a_2,i_2}^T g_Y^2 S_{1,j_2} y_{1,i_1,r_1} y_{2,i_2,r_1} \overline{I} T \tau_{a_1,a_2,j_1,j_2}^* S_{2,j_1}^\dagger}{8M_F}$ $-\frac{\overline{L}_{a_1,i_1} C P_R \overline{L}_{a_2,i_2}^T S_{1,j_2} y_{1,i_1,r_1} y_{2,i_2,r_1} \overline{I} T \tau_{a_1,a_2,j_1,j_2}^* S_{2,j_1}^\dagger}{2M_F}$			
									$+\frac{\sqrt{3}h\overline{y}_{2,i_3,r_1} \overline{L}_{a_1,i_1} C P_R \overline{L}_{a_2,i_2}^T S_{1,j_2} y_{1,i_3,r_2} y_{2,i_1,r_2} y_{2,i_2,r_1} \overline{I} T \tau_{a_1,a_2,j_1,j_2}^* S_{2,j_1}^\dagger}{8cM_F}$ $+\frac{3\sqrt{3}h\overline{y}_{2,i_3,r_1} \overline{L}_{a_1,i_1} C P_R \overline{L}_{a_2,i_2}^T S_{1,j_2} y_{1,i_3,r_2} y_{2,i_1,r_2} y_{2,i_2,r_1} \overline{I} T \tau_{a_1,a_2,j_1,j_2}^* S_{2,j_1}^\dagger}{16M_F}$ $+\frac{\sqrt{3}h\overline{y}_{1,i_3,r_1} \overline{L}_{a_1,i_1} C P_R \overline{L}_{a_2,i_2}^T S_{1,j_2} y_{1,i_1,r_2} y_{1,i_3,r_1} y_{2,i_2,r_2} \overline{I} T \tau_{a_1,a_2,j_1,j_2}^* S_{2,j_1}^\dagger}{4cM_F}$		
cont.									$+\frac{\sqrt{3}h\overline{y}_{1,i_3,r_1} \overline{L}_{a_1,i_1} C P_R \overline{L}_{a_2,i_2}^T S_{1,j_2} y_{1,i_1,r_2} y_{1,i_3,r_1} y_{2,i_2,r_2} \overline{I} T \tau_{a_1,a_2,j_1,j_2}^* S_{2,j_1}^\dagger}{8M_F}$ $+\frac{\sqrt{3}h\overline{y}_{2,i_3,r_1} \overline{L}_{a_1,i_1} C P_R \overline{L}_{a_2,i_2}^T S_{1,j_2} y_{1,i_3,r_2} y_{2,i_1,r_2} y_{2,i_2,r_1} \overline{I} T \tau_{a_1,a_2,j_1,j_2}^* S_{2,j_1}^\dagger}{2cM_F}$ $+\frac{\sqrt{3}h\overline{y}_{2,i_3,r_1} \overline{L}_{a_1,i_1} C P_R \overline{L}_{a_2,i_2}^T S_{1,j_2} y_{1,i_3,r_2} y_{2,i_1,r_2} y_{2,i_2,r_1} \overline{I} T \tau_{a_1,a_2,j_1,j_2}^* S_{2,j_1}^\dagger}{4M_F}$		
								$+\frac{\sqrt{3}h\overline{y}_{2,i_3,r_1} \overline{L}_{a_1,i_1} C P_R \overline{L}_{a_2,i_2}^T S_{1,j_2} y_{1,i_1,r_2} y_{2,i_2,r_2} y_{2,i_3,r_1} \overline{I} T \tau_{a_1,a_2,j_1,j_2}^* S_{2,j_1}^\dagger}{4cM_F}$ $+\frac{\sqrt{3}h\overline{y}_{1,i_3,r_1} \overline{L}_{a_1,i_1} C P_R \overline{L}_{a_2,i_2}^T S_{1,j_2} y_{1,i_1,r_2} y_{2,i_2,r_2} y_{2,i_3,r_1} \overline{I} T \tau_{a_1,a_2,j_1,j_2}^* S_{2,j_1}^\dagger}{8M_F}$ $+\frac{\sqrt{3}h\overline{y}_{1,i_3,r_1} \overline{L}_{a_1,i_1} C P_R \overline{L}_{a_2,i_2}^T S_{1,j_2} y_{1,i_1,r_2} y_{2,i_2,r_2} y_{2,i_3,r_1} \overline{I} T \tau_{a_1,a_2,j_1,j_2}^* S_{2,j_1}^\dagger}{8M_F}$			
									$+\frac{\sqrt{3}h\overline{y}_{2,i_3,r_1} \overline{L}_{a_1,i_1} C P_R \overline{L}_{a_2,i_2}^T S_{1,j_2} y_{1,i_1,r_2} y_{2,i_2,r_2} y_{2,i_3,r_1} \overline{I} T \tau_{a_1,a_2,j_1,j_2}^* S_{2,j_1}^\dagger}{8M_F}$ $+\frac{\sqrt{3}h\overline{y}_{1,i_3,r_1} \overline{L}_{a_1,i_1} C P_R \overline{L}_{a_2,i_2}^T S_{1,j_2} y_{1,i_1,r_2} y_{2,i_2,r_2} y_{2,i_3,r_1} \overline{I} T \tau_{a_1,a_2,j_1,j_2}^* S_{2,j_1}^\dagger}{8M_F}$ $+\frac{3\sqrt{3}h\overline{y}_{1,i_3,r_1} \overline{L}_{a_1,i_1} C P_R \overline{L}_{a_2,i_2}^T S_{1,j_2} y_{1,i_1,r_2} y_{2,i_2,r_2} y_{2,i_3,r_2} \overline{I} T \tau_{a_1,a_2,j_1,j_2}^* S_{2,j_1}^\dagger}{16M_F}$		
cont.									$-\frac{3h\overline{y}_{1,i_3,r_1} \overline{L}_{a_1,i_1} C P_R \overline{L}_{a_2,i_2}^T S_{1,j_1} y_{1,i_1,r_2} y_{1,i_3,r_2} y_{2,i_1,r_2} \epsilon_{a_1,a_2} S_{2,j_1}^\dagger}{8cM_F}$ $-\frac{3h\overline{y}_{1,i_3,r_1} \overline{L}_{a_1,i_1} C P_R \overline{L}_{a_2,i_2}^T S_{1,j_1} y_{1,i_1,r_2} y_{1,i_3,r_2} y_{2,i_1,r_2} \epsilon_{a_1,a_2} S_{2,j_1}^\dagger}{8cM_F}$ $-\frac{9h\overline{y}_{1,i_3,r_1} \overline{L}_{a_1,i_1} C P_R \overline{L}_{a_2,i_2}^T S_{1,j_1} y_{1,i_1,r_2} y_{2,i_2,r_2} y_{2,i_3,r_1} \epsilon_{a_1,a_2} S_{2,j_1}^\dagger}{16M_F}$		
									$+\frac{3h\overline{y}_{1,i_3,r_1} \overline{L}_{a_1,i_1} C P_R \overline{L}_{a_2,i_2}^T S_{1,j_1} y_{1,i_1,r_2} y_{1,i_3,r_1} y_{2,i_2,r_2} \epsilon_{a_1,a_2} S_{2,j_1}^\dagger}{4cM_F}$ $+\frac{3h\overline{y}_{1,i_3,r_1} \overline{L}_{a_1,i_1} C P_R \overline{L}_{a_2,i_2}^T S_{1,j_1} y_{1,i_1,r_2} y_{1,i_3,r_1} y_{2,i_2,r_2} \epsilon_{a_1,a_2} S_{2,j_1}^\dagger}{8M_F}$ $+\frac{3h\overline{y}_{2,i_3,r_1} \overline{L}_{a_1,i_1} C P_R \overline{L}_{a_2,i_2}^T S_{1,j_2} y_{1,i_1,r_2} y_{2,i_2,r_2} y_{2,i_3,r_1} \epsilon_{a_1,a_2} S_{2,j_1}^\dagger}{4cM_F}$		
cont.									$+\frac{3h\overline{y}_{2,i_3,r_1} \overline{L}_{a_1,i_1} C P_R \overline{L}_{a_2,i_2}^T S_{1,j_2} y_{1,i_1,r_2} y_{2,i_2,r_2} y_{2,i_3,r_1} \epsilon_{a_1,a_2} S_{2,j_1}^\dagger}{8M_F}$ $+\frac{3h\overline{y}_{1,i_3,r_1} \overline{L}_{a_1,i_1} C P_R \overline{L}_{a_2,i_2}^T S_{1,j_1} y_{1,i_1,r_2} y_{1,i_3,r_1} y_{2,i_2,r_2} \epsilon_{a_1,a_2} S_{2,j_1}^\dagger}{8cM_F}$ $+\frac{9h\overline{y}_{1,i_3,r_1} \overline{L}_{a_1,i_1} C P_R \overline{L}_{a_2,i_2}^T S_{1,j_1} y_{1,i_1,r_2} y_{1,i_3,r_1} y_{2,i_2,r_2} \epsilon_{a_1,a_2} S_{2,j_1}^\dagger}{16M_F}$		
								$+\frac{3h\overline{y}_{2,i_3,r_1} \overline{L}_{a_1,i_1} C P_R \overline{L}_{a_2,i_2}^T S_{1,j_2} y_{1,i_1,r_2} y_{2,i_2,r_2} y_{2,i_3,r_1} \epsilon_{a_1,a_2} S_{2,j_1}^\dagger}{8cM_F}$ $+\frac{3h\overline{y}_{1,i_3,r_1} \overline{L}_{a_1,i_1} C P_R \overline{L}_{a_2,i_2}^T S_{1,j_1} y_{1,i_1,r_2} y_{1,i_3,r_1} y_{2,i_2,r_2} \epsilon_{a_1,a_2} S_{2,j_1}^\dagger}{8cM_F}$ $+\frac{9h\overline{y}_{1,i_3,r_1} \overline{L}_{a_1,i_1} C P_R \overline{L}_{a_2,i_2}^T S_{1,j_1} y_{1,i_1,r_2} y_{1,i_3,r_1} y_{2,i_2,r_2} \epsilon_{a_1,a_2} S_{2,j_1}^\dagger}{16M_F}$			
cont.									$+\frac{3h\overline{y}_{2,i_3,r_1} \overline{L}_{a_1,i_1} C P_R \overline{L}_{a_2,i_2}^T S_{1,j_2} y_{1,i_1,r_2} y_{2,i_2,r_2} y_{2,i_3,r_1} \epsilon_{a_1,a_2} S_{2,j_1}^\dagger}{8cM_F}$ $+\frac{9h\overline{y}_{2,i_3,r_1} \overline{L}_{a_1,i_1} C P_R \overline{L}_{a_2,i_2}^T S_{1,j_2} y_{1,i_1,r_2} y_{2,i_2,r_2} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{2,j_1}^\dagger}{16M_F}$ $+\frac{\sqrt{3}h\overline{y}_{1,i_1,r_1} \overline{y}_{2,i_2,r_1} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_1} \epsilon_{a_1,a_2} S_{1,j_1}^\dagger \lambda_{12}^{(1)}}{\epsilon M_F}$		
								$+\frac{\sqrt{3}h\overline{y}_{1,i_1,r_1} \overline{y}_{2,i_2,r_1} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_1} \epsilon_{a_1,a_2} S_{1,j_1}^\dagger \lambda_{12}^{(1)}}{M_F}$ $+\frac{\sqrt{3}h\overline{L}_{a_1,i_1} C P_R \overline{L}_{a_2,i_2}^T S_{1,j_1} y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_2} S_{2,j_1}^\dagger \lambda_{12}^{(1)}}{\epsilon M_F}$ $+\frac{\sqrt{3}h\overline{L}_{a_1,i_1} C P_R \overline{L}_{a_2,i_2}^T S_{1,j_1} y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_2} S_{2,j_1}^\dagger \lambda_{12}^{(1)}}{M_F}$			
cont.									$+\frac{h\overline{y}_{1,i_1,r_1} \overline{y}_{2,i_2,r_1} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_2} \overline{I} T \tau_{a_1,a_2,j_2,j_1} S_{1,j_1}^\dagger \lambda_{12}^{(2)}}{\sqrt{3}\epsilon M_F}$ $+\frac{h\overline{y}_{1,i_1,r_1} \overline{y}_{2,i_2,r_1} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_2} \overline{I} T \tau_{a_1,a_2,j_2,j_1} S_{1,j_1}^\dagger \lambda_{12}^{(2)}}{\sqrt{3}M_F}$ $+\frac{h\overline{L}_{a_1,i_1} C P_R \overline{L}_{a_2,i_2}^T S_{1,j_2} y_{1,i_1,r_1} y_{2,i_2,r_1} \overline{I} T \tau_{a_1,a_2,j_1,j_2}^* S_{2,j_1}^\dagger \lambda_{12}^{(2)}}{\sqrt{3}\epsilon M_F}$		
								$+\frac{h\overline{L}_{a_1,i_1} C P_R \overline{L}_{a_2,i_2}^T S_{1,j_2} y_{1,i_1,r_1} y_{2,i_2,r_1} \overline{I} T \tau_{a_1,a_2,j_1,j_2}^* S_{2,j_1}^\dagger \lambda_{12}^{(2)}}{\sqrt{3}M_F}$ $+\frac{h\overline{L}_{a_1,i_1} C P_R \overline{L}_{a_2,i_2}^T S_{1,j_2} y_{1,i_1,r_1} y_{2,i_2,r_1} \overline{I} T \tau_{a_1,a_2,j_1,j_2}^* S_{2,j_1}^\dagger \lambda_{12}^{(2)}}{\sqrt{3}M_F}$			

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$S_1 S_2 F$	$S_1 S_2 \ell$	$S_1 F H$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	$+\frac{2}{3}h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\frac{1}{\overline{y_{1,i_1,r_1}y_{2,i_2,r_2}S_{1,j_2}S_{2,j_1}y_{1,i_2,r_1}y_{2,i_1,r_2}S_{1,j_1}^\dagger S_{2,j_2}^\dagger}}$ $-\frac{10}{3}h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\frac{1}{\overline{y_{1,i_1,r_1}y_{2,i_2,r_2}S_{1,j_1}S_{2,j_2}y_{1,i_2,r_1}y_{2,i_1,r_2}S_{1,j_1}^\dagger S_{2,j_2}^\dagger}}$ $-\sqrt{3}h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\frac{1}{\overline{y_{1,i_1,r_1}S_{1,j_4}S_{2,j_3}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\mathcal{I}_{12,j_2,j_3,j_1,j_4}^{(1)}}}$ $-\sqrt{3}h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\frac{1}{\overline{y_{2,i_1,r_1}S_{1,j_4}S_{2,j_3}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\mathcal{I}_{12,j_2,j_3,j_1,j_4}^{(1)}}}$	$-\frac{3h\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}L_{a_2,i_2}}}{8M_F}{}^T C P_L L_{a_1,i_1}g_2^2S_{2,j_2}\mathcal{I}T_{3,a_1,a_2,j_2,j_1}S_{1,j_1}^\dagger$ $-\frac{h\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}L_{a_2,i_2}}}{8M_F}{}^T C P_L L_{a_1,i_1}g_2^2S_{2,j_2}\mathcal{I}T_{3,a_1,a_2,j_2,j_1}S_{1,j_1}^\dagger$ $-\frac{\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}L_{a_2,i_2}}}{2M_F}{}^T C P_L L_{a_1,i_1}S_{2,j_2}\mathcal{I}T_{3,a_1,a_2,j_2,j_1}S_{1,j_1}^\dagger$	$-\frac{4h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)}{\sqrt{3}}\frac{1}{\overline{y_{1,i_1,r_1}Y_{e,i_2,i_3}H_{a_1}S_{1,j_1}y_{1,i_2,r_1}Y_{e,i_1,i_3}H_{a_1}^\dagger S_{1,j_1}^\dagger}}$ $+\frac{2h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)}{\sqrt{3}}\frac{1}{\overline{y_{1,i_1,r_1}Y_{e,i_2,i_3}H_{a_2}S_{1,j_2}y_{1,i_2,r_1}Y_{e,i_1,i_3}H_{a_1}^\dagger S_{1,j_1}^\dagger\mathcal{I}H_{1,a_1,a_2,j_1,j_2}^{(1)}}}$ $-\sqrt{3}h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\frac{1}{\overline{y_{1,i_1,r_1}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}H_{a_1}^\dagger S_{1,j_1}^\dagger\mathcal{I}H_{1,a_1,a_2,j_1,j_2}^{(1)}}}$
cont.								$+\frac{3h\overline{y_{1,i_1,r_1}S_{1,j_4}S_{2,j_3}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger}\lambda_{12}^{(2)}}{M_F^2}$ $+\frac{3h\overline{y_{1,i_1,r_1}S_{1,j_4}S_{2,j_3}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger}\lambda_{12}^{(2)}}{M_F^2}$ $+\frac{3h\overline{y_{1,i_1,r_1}S_{1,j_4}S_{2,j_3}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger}\lambda_{12}^{(2)}}{M_F^2}$	$+\frac{\sqrt{3}h\overline{y_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_1}L_{a_2,i_2}}}{8M_F}{}^T C P_L L_{a_1,i_1}S_{2,j_2}y_{1,i_3,r_1}\mathcal{I}T_{3,a_1,a_2,j_2,j_1}S_{1,j_1}^\dagger$ $+\frac{3\sqrt{3}h\overline{y_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_1}L_{a_2,i_2}}}{16M_F}{}^T C P_L L_{a_1,i_1}S_{2,j_2}y_{1,i_3,r_1}\mathcal{I}T_{3,a_1,a_2,j_2,j_1}S_{1,j_1}^\dagger$ $+\frac{\sqrt{3}h\overline{y_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_1}L_{a_2,i_2}}}{4M_F}{}^T C P_L L_{a_1,i_1}S_{2,j_2}y_{1,i_3,r_1}\mathcal{I}T_{3,a_1,a_2,j_2,j_1}S_{1,j_1}^\dagger$	$-\sqrt{3}h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\frac{1}{\overline{y_{1,i_1,r_1}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}H_{a_1}^\dagger S_{1,j_1}^\dagger\mathcal{I}H_{1,a_1,a_2,j_1,j_2}^{(2)}}}$
cont.								$+\frac{3h\overline{y_{1,i_1,r_1}S_{1,j_4}S_{2,j_3}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger}\lambda_{12}^{(2)}}{M_F^2}$ $+\frac{3h\overline{y_{1,i_1,r_1}S_{1,j_4}S_{2,j_3}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger}\lambda_{12}^{(2)}}{M_F^2}$ $-\sqrt{3}h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\frac{1}{\overline{y_{2,i_1,r_1}S_{1,j_3}S_{2,j_4}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\mathcal{I}_{12,j_2,j_3,j_1,j_4}^{(2)}}}$	$+\frac{\sqrt{3}h\overline{y_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_1}L_{a_2,i_2}}}{8M_F}{}^T C P_L L_{a_1,i_1}S_{2,j_2}y_{1,i_3,r_1}\mathcal{I}T_{3,a_1,a_2,j_2,j_1}S_{1,j_1}^\dagger$ $+\frac{\sqrt{3}h\overline{y_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_1}L_{a_2,i_2}}}{8M_F}{}^T C P_L L_{a_1,i_1}S_{2,j_2}y_{1,i_3,r_1}\mathcal{I}T_{3,a_1,a_2,j_2,j_1}S_{1,j_1}^\dagger$ $+\frac{3\sqrt{3}h\overline{y_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_1}L_{a_2,i_2}}}{16M_F}{}^T C P_L L_{a_1,i_1}S_{2,j_2}y_{1,i_3,r_1}\mathcal{I}T_{3,a_1,a_2,j_2,j_1}S_{1,j_1}^\dagger$	
cont.								$-\sqrt{3}h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\frac{1}{\overline{y_{2,i_1,r_1}S_{1,j_3}S_{2,j_4}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\mathcal{I}_{12,j_2,j_3,j_1,j_4}^{(2)}}}$	$+\frac{\sqrt{3}h\overline{y_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_1}L_{a_2,i_2}}}{8M_F}{}^T C P_L L_{a_1,i_1}S_{2,j_2}y_{1,i_3,r_1}\mathcal{I}T_{3,a_1,a_2,j_2,j_1}S_{1,j_1}^\dagger$ $+\frac{3\sqrt{3}h\overline{y_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_1}L_{a_2,i_2}}}{16M_F}{}^T C P_L L_{a_1,i_1}S_{2,j_2}y_{1,i_3,r_1}\mathcal{I}T_{3,a_1,a_2,j_2,j_1}S_{1,j_1}^\dagger$ $+\frac{3\sqrt{3}h\overline{y_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_1}L_{a_2,i_2}}}{16M_F}{}^T C P_L L_{a_1,i_1}S_{2,j_2}y_{1,i_3,r_1}\mathcal{I}T_{3,a_1,a_2,j_2,j_1}S_{1,j_1}^\dagger$	
cont.								$+\frac{\sqrt{3}h\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}y_{2,i_3,r_2}L_{a_2,i_2}}}{4M_F}{}^T C P_L L_{a_1,i_1}S_{2,j_2}y_{2,i_3,r_2}\mathcal{I}T_{3,a_1,a_2,j_2,j_1}S_{1,j_1}^\dagger$ $+\frac{\sqrt{3}h\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}y_{2,i_3,r_2}L_{a_2,i_2}}}{4M_F}{}^T C P_L L_{a_1,i_1}S_{2,j_2}y_{2,i_3,r_2}\mathcal{I}T_{3,a_1,a_2,j_2,j_1}S_{1,j_1}^\dagger$ $+\frac{9\sqrt{3}h\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}L_{a_2,i_2}}}{8M_F}{}^T C P_L L_{a_1,i_1}g_2^2S_{2,j_1}\epsilon_{a_1,a_2}S_{1,j_1}^\dagger$	$+\frac{3\sqrt{3}h\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}L_{a_2,i_2}}}{8M_F}{}^T C P_L L_{a_1,i_1}g_2^2S_{2,j_1}\epsilon_{a_1,a_2}S_{1,j_1}^\dagger$ $-\frac{\sqrt{3}\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}L_{a_2,i_2}}}{8M_F}{}^T C P_L L_{a_1,i_1}S_{2,j_1}\epsilon_{a_1,a_2}S_{1,j_1}^\dagger$	
cont.								$+\frac{3h\overline{y_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_2}L_{a_2,i_2}}}{8M_F}{}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{1,i_3,r_1}\epsilon_{a_1,a_2}S_{1,j_1}^\dagger$ $+\frac{9h\overline{y_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_2}L_{a_2,i_2}}}{16M_F}{}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{1,i_3,r_1}\epsilon_{a_1,a_2}S_{1,j_1}^\dagger$ $+\frac{3h\overline{y_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_1}L_{a_2,i_2}}}{4M_F}{}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{1,i_3,r_2}\epsilon_{a_1,a_2}S_{1,j_1}^\dagger$	$+\frac{3h\overline{y_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_1}L_{a_2,i_2}}}{4M_F}{}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{1,i_3,r_2}\epsilon_{a_1,a_2}S_{1,j_1}^\dagger$ $+\frac{3h\overline{y_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_1}L_{a_2,i_2}}}{8M_F}{}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{1,i_3,r_2}\epsilon_{a_1,a_2}S_{1,j_1}^\dagger$ $+\frac{9h\overline{y_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_1}L_{a_2,i_2}}}{16M_F}{}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{1,i_3,r_2}\epsilon_{a_1,a_2}S_{1,j_1}^\dagger$	
cont.								$+\frac{3h\overline{y_{1,i_1,r_1}y_{1,i_2,r_2}y_{2,i_3,r_1}L_{a_2,i_2}}}{8M_F}{}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{1,i_3,r_2}\epsilon_{a_1,a_2}S_{1,j_1}^\dagger$ $+\frac{9h\overline{y_{1,i_1,r_1}y_{1,i_2,r_2}y_{2,i_3,r_1}L_{a_2,i_2}}}{16M_F}{}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{1,i_3,r_2}\epsilon_{a_1,a_2}S_{1,j_1}^\dagger$ $+\frac{3h\overline{y_{1,i_1,r_1}y_{2,i_2,r_2}y_{2,i_3,r_1}L_{a_2,i_2}}}{8M_F}{}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{2,i_3,r_2}\epsilon_{a_1,a_2}S_{1,j_1}^\dagger$	$+\frac{3h\overline{y_{1,i_1,r_1}y_{2,i_2,r_2}y_{2,i_3,r_1}L_{a_2,i_2}}}{8M_F}{}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{2,i_3,r_2}\epsilon_{a_1,a_2}S_{1,j_1}^\dagger$ $+\frac{9h\overline{y_{1,i_1,r_1}y_{2,i_2,r_2}y_{2,i_3,r_1}L_{a_2,i_2}}}{16M_F}{}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{2,i_3,r_2}\epsilon_{a_1,a_2}S_{1,j_1}^\dagger$ $+\frac{9h\overline{y_{1,i_1,r_1}y_{2,i_2,r_2}y_{2,i_3,r_1}L_{a_2,i_2}}}{16M_F}{}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{2,i_3,r_2}\epsilon_{a_1,a_2}S_{1,j_1}^\dagger$	
cont.								$+\frac{3h\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}y_{2,i_3,r_2}L_{a_2,i_2}}}{4M_F}{}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{2,i_3,r_2}\epsilon_{a_1,a_2}S_{1,j_1}^\dagger$ $+\frac{3h\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}y_{2,i_3,r_2}L_{a_2,i_2}}}{4M_F}{}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{2,i_3,r_2}\epsilon_{a_1,a_2}S_{1,j_1}^\dagger$ $+\frac{3h\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}y_{2,i_3,r_2}L_{a_2,i_2}}}{4M_F}{}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{2,i_3,r_2}\epsilon_{a_1,a_2}S_{1,j_1}^\dagger$	$+\frac{3h\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}y_{2,i_3,r_2}L_{a_2,i_2}}}{4M_F}{}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{2,i_3,r_2}\epsilon_{a_1,a_2}S_{1,j_1}^\dagger$ $+\frac{3h\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}y_{2,i_3,r_2}L_{a_2,i_2}}}{4M_F}{}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{2,i_3,r_2}\epsilon_{a_1,a_2}S_{1,j_1}^\dagger$ $+\frac{3h\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}y_{2,i_3,r_2}L_{a_2,i_2}}}{4M_F}{}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{2,i_3,r_2}\epsilon_{a_1,a_2}S_{1,j_1}^\dagger$	
cont.								$+\frac{3h\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}y_{2,i_3,r_2}L_{a_2,i_2}}}{4M_F}{}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{2,i_3,r_2}\epsilon_{a_1,a_2}S_{1,j_1}^\dagger$ $+\frac{3h\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}y_{2,i_3,r_2}L_{a_2,i_2}}}{4M_F}{}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{2,i_3,r_2}\epsilon_{a_1,a_2}S_{1,j_1}^\dagger$ $+\frac{3h\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}y_{2,i_3,r_2}L_{a_2,i_2}}}{4M_F}{}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{2,i_3,r_2}\epsilon_{a_1,a_2}S_{1,j_1}^\dagger$	$+\frac{3h\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}y_{2,i_3,r_2}L_{a_2,i_2}}}{4M_F}{}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{2,i_3,r_2}\epsilon_{a_1,a_2}S_{1,j_1}^\dagger$ $+\frac{3h\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}y_{2,i_3,r_2}L_{a_2,i_2}}}{4M_F}{}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{2,i_3,r_2}\epsilon_{a_1,a_2}S_{1,j_1}^\dagger$ $+\frac{3h\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}y_{2,i_3,r_2}L_{a_2,i_2}}}{4M_F}{}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{2,i_3,r_2}\epsilon_{a_1,a_2}S_{1,j_1}^\dagger$	
cont.								$+\frac{3h\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}y_{2,i_3,r_2}L_{a_2,i_2}}}{4M_F}{}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{2,i_3,r_2}\epsilon_{a_1,a_2}S_{1,j_1}^\dagger$ $+\frac{3h\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}y_{2,i_3,r_2}L_{a_2,i_2}}}{4M_F}{}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{2,i_3,r_2}\epsilon_{a_1,a_2}S_{1,j_1}^\dagger$ $+\frac{3h\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}y_{2,i_3,r_2}L_{a_2,i_2}}}{4M_F}{}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{2,i_3,r_2}\epsilon_{a_1,a_2}S_{1,j_1}^\dagger$	$+\frac{3h\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}y_{2,i_3,r_2}L_{a_2,i_2}}}{4M_F}{}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{2,i_3,r_2}\epsilon_{a_1,a_2}S_{1,j_1}^\dagger$ $+\frac{3h\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}y_{2,i_3,r_2}L_{a_2,i_2}}}{4M_F}{}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{2,i_3,r_2}\epsilon_{a_1,a_2}S_{1,j_1}^\dagger$ $+\frac{3h\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}y_{2,i_3,r_2}L_{a_2,i_2}}}{4M_F}{}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{2,i_3,r_2}\epsilon_{a_1,a_2}S_{1,j_1}^\dagger$	
cont.								$+\frac{3h\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}y_{2,i_3,r_2}L_{a_2,i_2}}}{4M_F}{}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{2,i_3,r_2}\epsilon_{a_1,a_2}S_{1,j_1}^\dagger$ $+\frac{3h\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}y_{2,i_3,r_2}L_{a_2,i_2}}}{4M_F}{}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{2,i_3,r_2}\epsilon_{a_1,a_2}S_{1,j_1}^\dagger$ $+\frac{3h\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}y_{2,i_3,r_2}L_{a_2,i_2}}}{4M_F}{}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{2,i_3,r_2}\epsilon_{a_1,a_2}S_{1,j_1}^\dagger$	$+\frac{3h\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}y_{2,i_3,r_2}L_{a_2,i_2}}}{4M_F}{}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{2,i_3,r_2}\epsilon_{a_1,a_2}S_{1,j_1}^\dagger$ $+\frac{3h\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}y_{2,i_3,r_2}L_{a_2,i_2}}}{4M_F}{}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{2,i_3,r_2}\epsilon_{a_1,a_2}S_{1,j_1}^\dagger$ $+\frac{3h\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}y_{2,i_3,r_2}L_{a_2,i_2}}}{4M_F}{}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{2,i_3,r_2}\epsilon_{a_1,a_2}S_{1,j_1}^\dagger$	
cont.								$+\frac{3h\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}y_{2,i_3,r_2}L_{a_2,i_2}}}{4M_F}{}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{2,i_3,r_2}\epsilon_{a_1,a_2}S_{1,j_1}^\dagger$ $+\frac{3h\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}y_{2,i_3,r_2}L_{a_2,i_2}}}{4M_F}{}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{2,i_3,r_2}\epsilon_{a_1,a_2}S_{1,j_1}^\dagger$ $+\frac{3h\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}y_{2,i_3,r_2}L_{a_2,i_2}}}{4M_F}{}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{2,i_3,r_2}\epsilon_{a_1,a_2}S_{1,j_1}^\dagger$	$+\frac{3h\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}y_{2,i_3,r_2}L_{a_2,i_2}}}{4M_F}{}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{2,i_3,r_2}\epsilon_{a_1,a_2}S_{1,j_1}^\dagger$ $+\frac{3h\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}y_{2,i_3,r_2}L_{a_2,i_2}}}{4M_F}{}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{2,i_3,r_2}\epsilon_{a_1,a_2}S_{1,j_1}^\dagger$ $+\frac{3h\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}y_{2,i_3,r_2}L_{a_2,i_2}}}{4M_F}{}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{2,i_3,r_2}\epsilon_{a_1,a_2}S_{1,j_1}^\dagger$	
cont.								$+\frac{3h\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}y_{2,i_3,r_2}L_{a_2,i_2}}}{4M_F}{}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{2,i_3,r_2}\epsilon_{a_1,a_2}S_{1,j_1}^\dagger$ $+\frac{3h\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}y_{2,i_3,r_2}L_{a_2,i_2}}}{4M_F}{}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{2,i_3,r_2}\epsilon_{a_1,a_2}S_{1,j_1}^\dagger$ $+\frac{3h\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}y_{2,i_3,r_2}L_{a_2,i_2}}}{4M_F}{}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{2,i_3,r_2}\epsilon_{a_1,a_2}S_{1,j_1}^\dagger$	$+\frac{3h\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}y_{2,i_3,r_2}L_{a_2,i_2}}}{4M_F}{}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{2,i_3,r_2}\epsilon_{a_1,a_2}S_{1,j_1}^\dagger$ $+\frac{3h\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}y_{2,i_3,r_2}L_{a_2,i_2}}}{4M_F}{}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{2,i_3,r_2}\epsilon_{a_1,a_2}S_{1,j_1}^\dagger$ $+\frac{3h\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}y_{2,i_3,r_2}L_{a_2,i_2}}}{4M_F}{}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{2,i_3,r_2}\epsilon_{a_1,a_2}S_{1,j_1}^\dagger$	
cont.								$+\frac{3h\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}y_{2,i_3,r_2}L_{a_2,i_2}}}{4M_F}{}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{2,i_3,r_2}\epsilon_{a_1,a_2}S_{1,j_1}^\dagger$ $+\frac{3h\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}y_{2,i_3,r_2}L_{a_2,i_2}}}{4M_F}{}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{2,i_3,r_2}\epsilon_{a_1,a_2}S_{1,j_1}^\dagger$ $+\frac{3h\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}y_{2,i_3,r_2}L_{a_2,i_2}}}{4M_F}{}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{2,i_3,r_2}\epsilon_{a_1,a_2}S_{1,j_1}^\dagger$	$+\frac{3h\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}y_{2,i_3,r_2}L_{a_2,i_2}}}{4M_F}{}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{2,i_3,r_2}\epsilon_{a_1,a_2}S_{1,j_1}^\dagger$ $+\frac{3h\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}y_{2,i_3,r_2}L_{a_2,i_2}}}{4M_F}{}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{2,i_3,r_2}\epsilon_{a_1,a_2}S_{1,j_1}^\dagger$ $+\frac{3h\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}y_{2,i_3,r_2}L_{a_2,i_2}}}{4M_F}{}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{2,i_3,r_2}\epsilon_{a_1,a_2}S_{1,j_1}^\dagger$	
cont.								$+\frac{3h\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}y_{2,i_3,r_2}L_{a_2,i_2}}}{4M_F}{}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{2,i_3,r_2}\epsilon_{a_1,a_2}S_{1,j_1}^\dagger$ $+\frac{3h\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}y_{2,i_3,r_2}L_{a_2,i_2}}}{4M_F}{}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{2,i_3,r_2}\epsilon_{a_1,a_2}S_{1,j_1}^\dagger$ $+\frac{3h\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}y_{2,i_3,r_2}L_{a_2,i_2}}}{4M_F}{}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{2,i_3,r_2}\epsilon_{a_1,a_2}S_{1,j_1}^\dagger$	$+\frac{3h\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}y_{2,i_3,r_2}L_{a_2,i_2}}}{4M_F}{}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{2,i_3,r_2}\epsilon_{a_1,a_2}S_{1,j_1}^\dagger$ $+\frac{3h\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}y_{2,i_3,r_2}L_{a_2,i_2}}}{4M_F}{}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{2,i_3,r_2}\epsilon_{a_1,a_2}S_{1,j_1}^\dagger$ $+\frac{3h\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}y_{2,i_3,r_2}L_{a_2,i_2}}}{4M_F}{}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{2,i_3,r_2}\epsilon_{a_1,a_2}S_{1,j_1}^\dagger$	
cont.								$+\frac{3h\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}y_{2,i_3,r_2}L_{a_2,i_2}}}{4M_F}{}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{2,i_3,r_2}\epsilon_{a_1,a_2}S_{1,j_1}^\dagger$ $+\frac{3h\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}y_{2,i_3,r_2}L_{a_2,i_2}}}{4M_F}{}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{2,i_3,r_2}\epsilon_{a_1,a_2}S_{1,j_1}^\dagger$ $+\frac{3h\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}y_{2,i_3,r_2}L_{a_2,i_2}}}{4M_F}{}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{2,i_3,r_2}\epsilon_{a_1,a_2}S_{1,j_1}^\dagger$	$+\frac{3h\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}y_{2,i_3,r_2}L_{a_2,i_2}}}{4M_F}{}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{2,i_3,r_2}\epsilon_{a_1,a_2}S_{1,j_1}^\dagger$ $+\frac{3h\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}y_{2,i_3,r_2}L_{a_2,i_2}}}{4M_F}{}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{2,i_3,r_2}\epsilon_{a_1,a_2}S_{1,j_1}^\dagger$ $+\frac{3h\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}y_{2,i_3,r_2}L_{a_2,i_2}}}{4M_F}{}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{2,i_3,r_2}\epsilon_{a_1,a_2}S_{1,j_1}^\dagger$	
cont.								$+\frac{3h\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}y_{2,i_3,r_2}L_{a_2,i_2}}}{4M_F}{}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{2,i_3,r_2}\epsilon_{a_1,a_2}S_{1,j_1}^\dagger$ $+\frac{3h\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}y_{2,i_3,r_2}L_{a_2,i_2}}}{4M_F}{}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{2,i_3,r_2}\epsilon_{a_1,a_2}S_{1,j_1}^\dagger$ $+\frac{3h\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}y_{2,i_3,r_2}L_{a_2,i_2}}}{4M_F}{}^T C P_L L_{a_1,i_1$		

[illegible]

[illegible]

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	S_1	S_2	F	S_1	S_2	ℓ	S_1	F	H
													$-\frac{\sqrt{3L_{a_1, i_1}} C P_R L_{a_2, i_2}^T S_{1, j_1} y_{1, i_1, r_1} y_{2, i_2, r_1} \epsilon_{a_1, a_2} S_{2, j_1}^\dagger}{2M_F}$			
cont.													$-\frac{3h\overline{y_{2, i_3, r_1}} L_{a_1, i_1} C P_R L_{a_2, i_2}^T S_{1, j_1} y_{1, i_3, r_2} y_{2, i_1, r_2} y_{2, i_2, r_2} \epsilon_{a_1, a_2} S_{2, j_1}^\dagger}{8\epsilon M_F}$ $-\frac{9h\overline{y_{2, i_3, r_1}} L_{a_1, i_1} C P_R L_{a_2, i_2}^T S_{1, j_1} y_{1, i_3, r_2} y_{2, i_1, r_2} y_{2, i_2, r_2} \epsilon_{a_1, a_2} S_{2, j_1}^\dagger}{16M_F}$ $+\frac{3h\overline{y_{1, i_3, r_1}} L_{a_1, i_1} C P_R L_{a_2, i_2}^T S_{1, j_1} y_{1, i_4, r_2} y_{1, i_3, r_1} y_{2, i_2, r_2} \epsilon_{a_1, a_2} S_{2, j_1}^\dagger}{4\epsilon M_F}$			
cont.													$+\frac{3h\overline{y_{1, i_3, r_1}} L_{a_1, i_1} C P_R L_{a_2, i_2}^T S_{1, j_1} y_{1, i_1, r_2} y_{1, i_3, r_1} y_{2, i_2, r_2} \epsilon_{a_1, a_2} S_{2, j_1}^\dagger}{8M_F}$ $+\frac{3h\overline{y_{2, i_3, r_1}} L_{a_1, i_1} C P_R L_{a_2, i_2}^T S_{1, j_1} y_{1, i_1, r_2} y_{2, i_2, r_2} y_{2, i_3, r_1} \epsilon_{a_1, a_2} S_{2, j_1}^\dagger}{4\epsilon M_F}$ $+\frac{3h\overline{y_{2, i_3, r_1}} L_{a_1, i_1} C P_R L_{a_2, i_2}^T S_{1, j_1} y_{2, i_2, r_2} y_{2, i_3, r_1} \epsilon_{a_1, a_2} S_{2, j_1}^\dagger}{8M_F}$			
cont.													$+\frac{3h\overline{y_{1, i_3, r_1}} L_{a_1, i_1} C P_R L_{a_2, i_2}^T S_{1, j_1} y_{1, i_1, r_2} y_{1, i_2, r_1} y_{2, i_3, r_2} \epsilon_{a_1, a_2} S_{2, j_1}^\dagger}{8\epsilon M_F}$ $+\frac{9h\overline{y_{1, i_3, r_1}} L_{a_1, i_1} C P_R L_{a_2, i_2}^T S_{1, j_1} y_{1, i_1, r_2} y_{1, i_2, r_1} y_{2, i_3, r_2} \epsilon_{a_1, a_2} S_{2, j_1}^\dagger}{16M_F}$ $+\frac{3h\overline{y_{2, i_3, r_1}} L_{a_1, i_1} C P_R L_{a_2, i_2}^T S_{1, j_1} y_{1, i_1, r_2} y_{1, i_2, r_1} y_{2, i_3, r_2} \epsilon_{a_1, a_2} S_{2, j_1}^\dagger}{8\epsilon M_F}$			
cont.													$+\frac{9h\overline{y_{2, i_3, r_1}} L_{a_1, i_1} C P_R L_{a_2, i_2}^T S_{1, j_1} y_{1, i_1, r_2} y_{2, i_2, r_1} y_{2, i_3, r_2} \epsilon_{a_1, a_2} S_{2, j_1}^\dagger}{16M_F}$ $+\frac{\sqrt{3}h\overline{y_{1, i_1, r_1}} y_{2, i_2, r_1} L_{a_2, i_2}^T C P_L L_{a_1, i_1} S_{2, j_1} \epsilon_{a_1, a_2} S_{1, j_1}^\dagger \lambda_{12}^{(1)}}{\epsilon M_F}$ $+\frac{\sqrt{3}h\overline{y_{1, i_1, r_1}} y_{2, i_2, r_1} L_{a_2, i_2}^T C P_L L_{a_1, i_1} S_{2, j_1} \epsilon_{a_1, a_2} S_{1, j_1}^\dagger \lambda_{12}^{(1)}}{M_F}$			
cont.													$+\frac{\sqrt{3}h\overline{L_{a_1, i_1}} C P_R L_{a_2, i_2}^T S_{1, j_1} y_{1, i_1, r_1} y_{2, i_2, r_1} \epsilon_{a_1, a_2} S_{2, j_1}^\dagger \lambda_{12}^{(1)}}{\epsilon M_F}$ $+\frac{\sqrt{3}h\overline{L_{a_1, i_1}} C P_R L_{a_2, i_2}^T S_{1, j_1} y_{1, i_1, r_1} y_{2, i_2, r_1} \epsilon_{a_1, a_2} S_{2, j_1}^\dagger \lambda_{12}^{(1)}}{M_F}$ $+\frac{h\overline{y_{1, i_1, r_1}} y_{2, i_2, r_1} L_{a_2, i_2}^T C P_L L_{a_1, i_1} S_{2, j_2} \mathcal{I} T_{3, a_1, a_2, j_2, j_1} S_{1, j_1}^\dagger \lambda_{12}^{(2)}}{\sqrt{3}\epsilon M_F}$			
cont.													$+\frac{h\overline{y_{1, i_1, r_1}} y_{2, i_2, r_1} L_{a_2, i_2}^T C P_L L_{a_1, i_1} S_{2, j_2} \mathcal{I} T_{3, a_1, a_2, j_2, j_1} S_{1, j_1}^\dagger \lambda_{12}^{(2)}}{\sqrt{3}M_F}$ $+\frac{h\overline{L_{a_1, i_1}} C P_R L_{a_2, i_2}^T S_{1, j_2} y_{1, i_1, r_1} y_{2, i_2, r_1} \mathcal{I} T_{3, a_1, a_2, j_1, j_2} S_{2, j_1}^\dagger \lambda_{12}^{(2)}}{\sqrt{3}\epsilon M_F}$ $+\frac{h\overline{L_{a_1, i_1}} C P_R L_{a_2, i_2}^T S_{1, j_2} y_{1, i_1, r_1} y_{2, i_2, r_1} \mathcal{I} T_{3, a_1, a_2, j_1, j_2} S_{2, j_1}^\dagger \lambda_{12}^{(2)}}{\sqrt{3}M_F}$			
T3-C	2	2	1	2	2	3	$\frac{3}{2}$						$+\frac{2}{3}h\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\frac{\overline{y_{1, i_1, r_1}} y_{2, i_2, r_2} S_{1, j_2} S_{2, j_1} y_{1, i_2, r_1} y_{2, i_1, r_2} S_{1, j_1}^\dagger S_{2, j_2}^\dagger}{M_F^2}$ $-\frac{10}{3}h\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\frac{\overline{y_{1, i_1, r_1}} y_{2, i_2, r_2} S_{1, j_1} S_{2, j_2} y_{1, i_2, r_1} y_{2, i_1, r_2} S_{1, j_1}^\dagger S_{2, j_2}^\dagger}{M_F^2}$		$-\frac{4h\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{y_{1, i_1, r_1}} Y_{e, i_2, i_3} H_{a_1} S_{1, j_1} y_{1, i_2, r_1} Y_{e, i_1, i_3} H_{a_1}^\dagger S_{1, j_1}^\dagger}{\sqrt{3}}$	
													$-\frac{h\overline{y_{1, i_1, r_1}} y_{2, i_2, r_1} L_{a_2, i_2}^T C P_L L_{a_1, i_1} g_Y^2 S_{2, j_2} \mathcal{I} T_{3, a_1, a_2, j_2, j_1} S_{1, j_1}^\dagger}{8M_F}$ $+\frac{3h\overline{y_{1, i_1, r_1}} y_{2, i_2, r_1} L_{a_2, i_2}^T C P_L L_{a_1, i_1} g_Y^2 S_{2, j_2} \mathcal{I} T_{3, a_1, a_2, j_2, j_1} S_{1, j_1}^\dagger}{\epsilon M_F}$		$+\frac{2h\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{y_{1, i_1, r_1}} Y_{e, i_2, i_3} H_{a_2} S_{1, j_2} y_{1, i_2, r_1} Y_{e, i_1, i_3} H_{a_1}^\dagger S_{1, j_1}^\dagger \mathcal{I} H_{1, a_1, a_2, j_1, j_2}^{-1}}{\sqrt{3}}$	
cont.													$+\frac{h\overline{y_{1, i_1, r_1}} y_{2, i_2, r_1} L_{a_2, i_2}^T C P_L L_{a_1, i_1} g_Y^2 S_{2, j_2} \mathcal{I} T_{3, a_1, a_2, j_2, j_1} S_{1, j_1}^\dagger}{M_F}$ $+\frac{h\overline{y_{1, i_1, r_1}} y_{2, i_2, r_1} L_{a_2, i_2}^T C P_L L_{a_1, i_1} g_Y^2 S_{2, j_2} \mathcal{I} T_{3, a_1, a_2, j_2, j_1} S_{1, j_1}^\dagger}{\epsilon M_F}$		$-\sqrt{3}h\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\frac{\overline{y_{1, i_1, r_1}} H_{a_2} S_{1, j_2} y_{1, i_1, r_1} H_{a_1}^\dagger S_{1, j_1}^\dagger \mathcal{I} H_{1, a_1, a_2, j_1, j_2}^{-1} \lambda_{H1}^{(1)}}{\sqrt{3}}$	
cont.													$+\frac{h\overline{y_{1, i_1, r_1}} y_{2, i_2, r_1} L_{a_2, i_2}^T C P_L L_{a_1, i_1} g_Y^2 S_{2, j_2} \mathcal{I} T_{3, a_1, a_2, j_2, j_1} S_{1, j_1}^\dagger}{M_F}$ $+\frac{h\overline{y_{1, i_1, r_1}} y_{2, i_2, r_1} L_{a_2, i_2}^T C P_L L_{a_1, i_1} g_Y^2 S_{2, j_2} \mathcal{I} T_{3, a_1, a_2, j_2, j_1} S_{1, j_1}^\dagger}{\epsilon M_F}$		$-\sqrt{3}h\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\frac{\overline{y_{1, i_1, r_1}} H_{a_2} S_{1, j_2} y_{1, i_1, r_1} H_{a_1}^\dagger S_{1, j_1}^\dagger \mathcal{I} H_{1, a_1, a_2, j_1, j_2}^{-2} \lambda_{H1}^{(2)}}{\sqrt{3}}$	
cont.													$+\frac{h\overline{y_{1, i_1, r_1}} y_{2, i_2, r_1} L_{a_2, i_2}^T C P_L L_{a_1, i_1} g_Y^2 S_{2, j_2} \mathcal{I} T_{3, a_1, a_2, j_2, j_1} S_{1, j_1}^\dagger}{M_F}$ $+\frac{h\overline{y_{1, i_1, r_1}} y_{2, i_2, r_1} L_{a_2, i_2}^T C P_L L_{a_1, i_1} g_Y^2 S_{2, j_2} \mathcal{I} T_{3, a_1, a_2, j_2, j_1} S_{1, j_1}^\dagger}{\epsilon M_F}$		$-\sqrt{3}h\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\frac{\overline{y_{1, i_1, r_1}} H_{a_2} S_{1, j_2} y_{1, i_1, r_1} H_{a_1}^\dagger S_{1, j_1}^\dagger \mathcal{I} H_{1, a_1, a_2, j_1, j_2}^{-2} \lambda_{H1}^{(2)}}{\sqrt{3}}$	
cont.													$+\frac{h\overline{y_{1, i_1, r_1}} y_{2, i_2, r_1} L_{a_2, i_2}^T C P_L L_{a_1, i_1} g_Y^2 S_{2, j_2} \mathcal{I} T_{3, a_1, a_2, j_2, j_1} S_{1, j_1}^\dagger}{M_F}$ $+\frac{h\overline{y_{1, i_1, r_1}} y_{2, i_2, r_1} L_{a_2, i_2}^T C P_L L_{a_1, i_1} g_Y^2 S_{2, j_2} \mathcal{I} T_{3, a_1, a_2, j_2, j_1} S_{1, j_1}^\dagger}{\epsilon M_F}$		$-\sqrt{3}h\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\frac{\overline{y_{1, i_1, r_1}} H_{a_2} S_{1, j_2} y_{1, i_1, r_1} H_{a_1}^\dagger S_{1, j_1}^\dagger \mathcal{I} H_{1, a_1, a_2, j_1, j_2}^{-2} \lambda_{H1}^{(2)}}{\sqrt{3}}$	
cont.													$+\frac{h\overline{y_{1, i_1, r_1}} y_{2, i_2, r_1} L_{a_2, i_2}^T C P_L L_{a_1, i_1} g_Y^2 S_{2, j_2} \mathcal{I} T_{3, a_1, a_2, j_2, j_1} S_{1, j_1}^\dagger}{M_F}$ $+\frac{h\overline{y_{1, i_1, r_1}} y_{2, i_2, r_1} L_{a_2, i_2}^T C P_L L_{a_1, i_1} g_Y^2 S_{2, j_2} \mathcal{I} T_{3, a_1, a_2, j_2, j_1} S_{1, j_1}^\dagger}{\epsilon M_F}$		$-\sqrt{3}h\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\frac{\overline{y_{1, i_1, r_1}} H_{a_2} S_{1, j_2} y_{1, i_1, r_1} H_{a_1}^\dagger S_{1, j_1}^\dagger \mathcal{I} H_{1, a_1, a_2, j_1, j_2}^{-2} \lambda_{H1}^{(2)}}{\sqrt{3}}$	
cont.													$+\frac{h\overline{y_{1, i_1, r_1}} y_{2, i_2, r_1} L_{a_2, i_2}^T C P_L L_{a_1, i_1} g_Y^2 S_{2, j_2} \mathcal{I} T_{3, a_1, a_2, j_2, j_1} S_{1, j_1}^\dagger}{M_F}$ $+\frac{h\overline{y_{1, i_1, r_1}} y_{2, i_2, r_1} L_{a_2, i_2}^T C P_L L_{a_1, i_1} g_Y^2 S_{2, j_2} \mathcal{I} T_{3, a_1, a_2, j_2, j_1} S_{1, j_1}^\dagger}{\epsilon M_F}$		$-\sqrt{3}h\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\frac{\overline{y_{1, i_1, r_1}} H_{a_2} S_{1, j_2} y_{1, i_1, r_1} H_{a_1}^\dagger S_{1, j_1}^\dagger \mathcal{I} H_{1, a_1, a_2, j_1, j_2}^{-2} \lambda_{H1}^{(2)}}{\sqrt{3}}$	
cont.													$+\frac{h\overline{y_{1, i_1, r_1}} y_{2, i_2, r_1} L_{a_2, i_2}^T C P_L L_{a_1, i_1} g_Y^2 S_{2, j_2} \mathcal{I} T_{3, a_1, a_2, j_2, j_1} S_{1, j_1}^\dagger}{M_F}$ $+\frac{h\overline{y_{1, i_1, r_1}} y_{2, i_2, r_1} L_{a_2, i_2}^T C P_L L_{a_1, i_1} g_Y^2 S_{2, j_2} \mathcal{I} T_{3, a_1, a_2, j_2, j_1} S_{1, j_1}^\dagger}{\epsilon M_F}$		$-\sqrt{3}h\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\frac{\overline{y_{1, i_1, r_1}} H_{a_2} S_{1, j_2} y_{1, i_1, r_1} H_{a_1}^\dagger S_{1, j_1}^\dagger \mathcal{I} H_{1, a_1, a_2, j_1, j_2}^{-2} \lambda_{H1}^{(2)}}{\sqrt{3}}$	
cont.													$+\frac{h\overline{y_{1, i_1, r_1}} y_{2, i_2, r_1} L_{a_2, i_2}^T C P_L L_{a_1, i_1} g_Y^2 S_{2, j_2} \mathcal{I} T_{3, a_1, a_2, j_2, j_1} S_{1, j_1}^\dagger}{M_F}$ $+\frac{h\overline{y_{1, i_1, r_1}} y_{2, i_2, r_1} L_{a_2, i_2}^T C P_L L_{a_1, i_1} g_Y^2 S_{2, j_2} \mathcal{I} T_{3, a_1, a_2, j_2, j_1} S_{1, j_1}^\dagger}{\epsilon M_F}$		$-\sqrt{3}h\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\frac{\overline{y_{1, i_1, r_1}} H_{a_2} S_{1, j_2} y_{1, i_1, r_1} H_{a_1}^\dagger S_{1, j_1}^\dagger \mathcal{I} H_{1, a_1, a_2, j_1, j_2}^{-2} \lambda_{H1}^{(2)}}{\sqrt{3}}$	
cont.													$+\frac{h\overline{y_{1, i_1, r_1}} y_{2, i_2, r_1} L_{a_2, i_2}^T C P_L L_{a_1, i_1} g_Y^2 S_{2, j_2} \mathcal{I} T_{3, a_1, a_2, j_2, j_1} S_{1, j_1}^\dagger}{M_F}$ $+\frac{h\overline{y_{1, i_1, r_1}} y_{2, i_2, r_1} L_{a_2, i_2}^T C P_L L_{a_1, i_1} g_Y^2 S_{2, j_2} \mathcal{I} T_{3, a_1, a_2, j_2, j_1} S_{1, j_1}^\dagger}{\epsilon M_F}$		$-\sqrt{3}h\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\frac{\overline{y_{1, i_1, r_1}} H_{a_2} S_{1, j_2} y_{1, i_1, r_1} H_{a_1}^\dagger S_{1, j_1}^\dagger \mathcal{I} H_{1, a_1, a_2, j_1, j_2}^{-2} \lambda_{H1}^{(2)}}{\sqrt{3}}$	
cont.													$+\frac{h\overline{y_{1, i_1, r_1}} y_{2, i_2, r_1} L_{a_2, i_2}^T C P_L L_{a_1, i_1} g_Y^2 S_{2, j_2} \mathcal{I} T_{3, a_1, a_2, j_2, j_1} S_{1, j_1}^\dagger}{M_F}$ $+\frac{h\overline{y_{1, i_1, r_1}} y_{2, i_2, r_1} L_{a_2, i_2}^T C P_L L_{a_1, i_1} g_Y^2 S_{2, j_2} \mathcal{I} T_{3, a_1, a_2, j_2, j_1} S_{1, j_1}^\dagger}{\epsilon M_F}$		$-\sqrt{3}h\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\frac{\overline{y_{1, i_1, r_1}} H_{a_2} S_{1, j_2} y_{1, i_1, r_1} H_{a_1}^\dagger S_{1, j_1}^\dagger \mathcal{I} H_{1, a_1, a_2, j_1, j_2}^{-2} \lambda_{H1}^{(2)}}{\sqrt{3}}$	
cont.													$+\frac{h\overline{y_{1, i_1, r_1}} y_{2, i_2, r_1} L_{a_2, i_2}^T C P_L L_{a_1, i_1} g_Y^2 S_{2, j_2} \mathcal{I} T_{3, a_1, a_2, j_2, j_1} S_{1, j_1}^\dagger}{M_F}$ $+\frac{h\overline{y_{1, i_1, r_1}} y_{2, i_2, r_1} L_{a_2, i_2}^T C P_L L_{a_1, i_1} g_Y^2 S_{2, j_2} \mathcal{I} T_{3, a_1, a_2, j_2, j_1} S_{1, j_1}^\dagger}{\epsilon M_F}$		$-\sqrt{3}h\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\frac{\overline{y_{1, i_1, r_1}} H_{a_2} S_{1, j_2} y_{1, i_1, r_1} H_{a_1}^\dagger S_{1, j_1}^\dagger \mathcal{I} H_{1, a_1, a_2, j_1, j_2}^{-2} \lambda_{H1}^{(2)}}{\sqrt{3}}$	
cont.													$+\frac{h\overline{y_{1, i_1, r_1}} y_{2, i_2, r_1} L_{a_2, i_2}^T C P_L L_{a_1, i_1} g_Y^2 S_{2, j_2} \mathcal{I} T_{3, a_1, a_2, j_2, j_1} S_{1, j_1}^\dagger}{M_F}$ $+\frac{h\overline{y_{1, i_1, r_1}} y_{2, i_2, r_1} L_{a_2, i_2}^T C P_L L_{a_1, i_1} g_Y^2 S_{2, j_2} \mathcal{I} T_{3, a_1, a_2, j_2, j_1} S_{1, j_1}^\dagger}{\epsilon M_F}$		$-\sqrt{3}h\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\frac{\overline{y_{1, i_1, r_1}} H_{a_2} S_{1, j_2} y_{1, i_1, r_1} H_{a_1}^\dagger S_{1, j_1}^\dagger \mathcal{I} H_{1, a_1, a_2, j_1, j_2}^{-2} \lambda_{H1}^{(2)}}{\sqrt{3}}$	
cont.													$+\frac{h\overline{y_{1, i_1, r_1}} y_{2, i_2, r_1} L_{a_2, i_2}^T C P_L L_{a_1, i_1} g_Y^2 S_{2, j_2} \mathcal{I} T_{3, a_1, a_2, j_2, j_1} S_{1, j_1}^\dagger}{M_F}$ $+\frac{h\overline{y_{1, i_1, r_1}} y_{2, i_2, r_1} L_{a_2, i_2}^T C P_L L_{a_1, i_1} g_Y^2 S_{2, j_2} \mathcal{I} T_{3, a_1, a_2, j_2, j_1} S_{1, j_1}^\dagger}{\epsilon M_F}$		$-\sqrt{3}h\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\frac{\overline{y_{1, i_1, r_1}} H_{a_2} S_{1, j_2} y_{1, i_1, r_1} H_{a_1}^\dagger S_{1, j_1}^\dagger \mathcal{I} H_{1, a_1, a_2, j_1, j_2}^{-2} \lambda_{H1}^{(2)}}{\sqrt{3}}$	
cont.													$+\frac{h\overline{y_{1, i_1, r_1}} y_{2, i_2, r_1} L_{a_2, i_2}^T C P_L L_{a_1, i_1} g_Y^2 S_{2, j_2} \mathcal{I} T_{3, a_1, a_2, j_2, j_1} S_{1, j_1}^\dagger}{M_F}$ $+\frac{h\overline{y_{1, i_1, r_1}} y_{2, i_2, r_1} L_{a_2, i_2}^T C P_L L_{a_1, i_1} g_Y^2 S_{2, j_2} \mathcal{I} T_{3, a_1, a_2, j_2, j_1} S_{1, j_1}^\dagger}{\epsilon M_F}$		$-\sqrt{3}h\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\frac{\overline{y_{1, i_1, r_1}} H_{a_2} S_{1, j_2} y_{1, i_1, r_1} H_{a_1}^\dagger S_{1, j_1}^\dagger \mathcal{I} H_{1, a_1, a_2, j_1, j_2}^{-2} \lambda_{H1}^{(2)}}{\sqrt{3}}$	
cont.													$+\frac{h\overline{y_{1, i_1, r_1}} y_{2, i_2, r_1} L_{a_2, i_2}^T C P_L L_{a_1, i_1} g_Y^2 S_{2, j_2} \mathcal{I} T_{3, a_1, a_2, j_2, j_1} S_{1, j_1}^\dagger}{M_F}$ $+\frac{h\overline{y_{1, i_1, r_1}} y_{2, i_2, r_1} L_{a_2, i_2}^T C P_L L_{a_1, i_1} g_Y^2 S_{2, j_2} \mathcal{I} T_{3, a_1, a_2, j_2, j_1} S_{1, j_1}^\dagger}{\epsilon M_F}$		$-\sqrt{3}h\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\frac{\overline{y_{1, i_1, r_1}} H_{a_2} S_{1, j_2} y_{1, i_1, r_1} H_{a_1}^\dagger S_{1, j_1}^\dagger \mathcal{I} H_{1, a_1, a_2, j_1, j_2}^{-2} \lambda_{H1}^{(2)}}{\sqrt{3}}$	
cont.													$+\frac{h\overline{y_{1, i_1, r_1}} y_{2, i_2, r_1} L_{a_2, i_2}^T C P_L L_{a_1, i_1} g_Y^2 S_{2, j_2} \mathcal{I} T_{3, a_1, a_2, j_2, j_1} S_{1, j_1}^\dagger}{M_F}$ $+\frac{h\overline{y_{1, i_1, r_1}} y_{2, i_2, r_1} L_{a_2, i_2}^T C P_L L_{a_1, i_1} g_Y^2 S_{2, j_2} \mathcal{I} T_{3, a_1, a_2, j_2, j_1} S_{1, j_1}^\dagger}{\epsilon M_F}$		$-\sqrt{3}h\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\frac{\overline{y_{1, i_1, r_1}} H_{a_2} S_{1, j_2} y_{1, i_1, r_1} H_{a_1}^\dagger S_{1, j_1}^\dagger \mathcal{I} H_{1, a_1, a_2, j_1, j_2}^{-2} \lambda_{H1}^{(2)}}{\sqrt{3}}$	
cont.													$+\frac{h\overline{y_{1, i_1, r_1}} y_{2, i_2, r_1} L_{a_2, i_2}^T C P_L L_{a_1, i_1} g_Y^2 S_{2, j_2} \mathcal{I} T_{3, a_1, a_2, j_2, j_1} S_{1, j_1}^\dagger}{M_F}$ $+\frac{h\overline{y_{1, i_1, r_1}} y_{2, i_2, r_1} L_{a_2, i_2}^T C P_L L_{a_1, i_1} g_Y^2 S_{2, j_2} \mathcal{I} T_{3, a_1, a_2, j_2, j_1} S_{1, j_1}^\dagger}{\epsilon M_F}$		$-\sqrt{3}h\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\frac{\overline{y_{1, i_1, r_1}} H_{a_2} S_{1, j_2} y_{1, i_1, r_1} H_{a_1}^\dagger S_{1, j_1}^\dagger \mathcal{I} H_{1, a_1, a_2, j_1, j_2}^{-2} \lambda_{H1}^{(2)}}{\sqrt{3}}$	
cont.													$+\frac{h\overline{y_{1, i_1, r_1}} y_{2, i_2, r_1} L_{a_2, i_2}^T C P_L L_{a_1, i_1} g_Y^2 S_{2, j_2} \mathcal{I} T_{3, a_1, a_2, j_2, j_1} S_{1, j_1}^\dagger}{M_F}$ $+\frac{h\overline{y_{1, i_1, r_1}} y_{2, i_2, r_1} L_{a_2, i_2}^T C P_L L_{a_1, i_1} g_Y^2 S_{2, j_2} \mathcal{I} T_{3, a_1, a_2, j_2, j_1} S_{1, j_1}^\dagger}{\epsilon M_F}$		$-\sqrt{3}h\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\frac{\overline{y_{1, i_1, r_1}} H_{a_2} S_{1, j_2} y_{1, i_1, r_1} H_{a_1}^\dagger S_{1, j_1}^\dagger \mathcal{I} H_{1, a_1, a_2, j_1, j_2}^{-2} \lambda_{H1}^{(2)}}{\sqrt{3}}$	
cont.													$+\frac{h\overline{y_{1, i_1, r_1}} y_{2, i_2, r_1} L_{a_2, i_2}^T C P_L L_{a_1, i_1} g_Y^2 S_{2, j_2} \mathcal{I} T_{3, a_1, a_2, j_2, j_1} S_{1, j_1}^\dagger}{M_F}$ $+\frac{h\overline{y_{1, i_1, r_1}} y_{2, i_2, r_1} L_{a_2, i_2}^T C P_L L_{a_1, i_1} g_Y^2 S_{2, j_2} \mathcal{I} T_{3, a_1, a_2, j_2, j_1} S_{1, j_1}^\dagger}{\epsilon M_F}$		$-\sqrt{3}h\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\frac{\overline{y_{1, i_1, r_1}} H_{a_2} S_{1, j_2} y_{1, i$	

[illegible]

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$S_1 S_2 F$	$S_1 S_2 \ell$	$S_1 F H$
									$+ \frac{3h\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_1}}}{8M_F} C P_R \overline{L_{a_2,i_2}^T} S_{1,j_1} y_{1,i_1,r_2} y_{2,i_2,r_2} y_{2,i_3,r_1} \epsilon_{a_1,a_2} S_{2,j_1}^\dagger$	
cont.									$+ \frac{3h\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_1}}}{8\epsilon M_F} C P_R \overline{L_{a_2,i_2}^T} S_{1,j_1} y_{1,i_1,r_2} y_{1,i_2,r_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{2,j_1}^\dagger$	
									$+ \frac{9h\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_1}}}{16M_F} C P_R \overline{L_{a_2,i_2}^T} S_{1,j_1} y_{1,i_1,r_2} y_{1,i_2,r_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{2,j_1}^\dagger$	
									$+ \frac{3h\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_1}}}{8\epsilon M_F} C P_R \overline{L_{a_2,i_2}^T} S_{1,j_1} y_{1,i_1,r_2} y_{2,i_2,r_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{2,j_1}^\dagger$	
cont.									$+ \frac{9h\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_1}}}{16M_F} C P_R \overline{L_{a_2,i_2}^T} S_{1,j_1} y_{1,i_1,r_2} y_{2,i_2,r_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{2,j_1}^\dagger$	
									$+ \frac{\sqrt{3}h\overline{y_{1,i_1,r_1}}\overline{y_{2,i_2,r_1}}\overline{L_{a_2,i_2}^T} C P_L L_{a_1,i_1} S_{2,j_1} \epsilon_{a_1,a_2} S_{1,j_1}^\dagger \lambda_{12}^{(1)}}{\epsilon M_F}$	
									$+ \frac{\sqrt{3}h\overline{y_{1,i_1,r_1}}\overline{y_{2,i_2,r_1}}\overline{L_{a_2,i_2}^T} C P_L L_{a_1,i_1} S_{2,j_1} \epsilon_{a_1,a_2} S_{1,j_1}^\dagger \lambda_{12}^{(1)}}{M_F}$	
cont.									$+ \frac{\sqrt{3}h\overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}^T} S_{1,j_1} y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_2} S_{2,j_1}^\dagger \lambda_{12}^{(1)}}{\epsilon M_F}$	
									$+ \frac{\sqrt{3}h\overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}^T} S_{1,j_1} y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_2} S_{2,j_1}^\dagger \lambda_{12}^{(1)}}{M_F}$	
									$+ \frac{h\overline{y_{1,i_1,r_1}}\overline{y_{2,i_2,r_1}}\overline{L_{a_2,i_2}^T} C P_L L_{a_1,i_1} S_{2,j_2} \overline{\mathcal{I}T3.a_1.a_2,j_2,j_1} S_{1,j_1}^\dagger \lambda_{12}^{(2)}}{\sqrt{3}\epsilon M_F}$	
cont.									$+ \frac{h\overline{y_{1,i_1,r_1}}\overline{y_{2,i_2,r_1}}\overline{L_{a_2,i_2}^T} C P_L L_{a_1,i_1} S_{2,j_2} \overline{\mathcal{I}T3.a_1.a_2,j_2,j_1} S_{1,j_1}^\dagger \lambda_{12}^{(2)}}{\sqrt{3}M_F}$	
									$+ \frac{h\overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}^T} S_{1,j_2} y_{1,i_1,r_1} y_{2,i_2,r_1} \overline{\mathcal{I}T3.a_1.a_2,j_1,j_2} S_{2,j_1}^\dagger \lambda_{12}^{(2)}}{\sqrt{3}\epsilon M_F}$	
cont.									$+ \frac{h\overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}^T} S_{1,j_2} y_{1,i_1,r_1} y_{2,i_2,r_1} \overline{\mathcal{I}T3.a_1.a_2,j_1,j_2} S_{2,j_1}^\dagger \lambda_{12}^{(2)}}{\sqrt{3}M_F}$	
									$- \frac{4h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1}} \overline{y_{2,i_2,r_2}} S_2 S_{1,j_1} y_{1,i_2,r_1} y_{2,i_1,r_2} S_2^\dagger S_{1,j_1}^\dagger}{\sqrt{3}}$	
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$- \frac{2h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1}} S_2 S_{1,j_2} y_{1,i_1,r_1} S_2^\dagger S_{1,j_1}^\dagger \overline{\mathcal{I}1_2.j_2,j_1}^\dagger \lambda_{12}^{(1)}}{\sqrt{3}}$	$- \frac{3\sqrt{3}h\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_1}}}{4\epsilon M_F} C P_R \overline{L_{a_2,i_2}^T} S_{1,j_1} y_{1,i_2,r_1} y_{1,i_3,r_2} y_{2,i_2,i_1,r_2} \overline{\mathcal{I}T3.a_1.a_2,j_1} S_2^\dagger$	$- \frac{2h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1}} \overline{Y_{e,i_2,i_3}} H_{a_1} S_{1,j_1} y_{1,i_2,r_1} Y_{e,i_1,i_3} H_{a_1}^\dagger S_{1,j_1}^\dagger}{\sqrt{3}}$
								$- \frac{2h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1}} S_2 S_{1,j_2} y_{1,i_1,r_1} S_2^\dagger S_{1,j_1}^\dagger \overline{\mathcal{I}1_2.j_2,j_1}^\dagger \lambda_{12}^{(1)}}{\sqrt{3}}$	$+ \frac{\overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}^T} S_{1,j_1} y_{1,i_1,r_1} y_{2,i_2,r_1} \overline{\mathcal{I}T3.a_1.a_2,j_1} S_2^\dagger}{M_F}$	$- 2i\sqrt{\frac{2}{3}} h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \frac{\overline{y_{1,i_1,r_1}} \overline{Y_{e,i_2,i_3}} H_{a_2} S_{1,j_2} y_{1,i_2,r_1} Y_{e,i_1,i_3} H_{a_1}^\dagger S_{1,j_1}^\dagger \overline{\mathcal{I}H_{1,a_1,a_2,j_2,j_1}^\dagger}}{\sqrt{3}}$
								$- 2h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1}} S_2 S_{1,j_2} y_{2,i_1,r_1} S_2^\dagger S_{1,j_1}^\dagger \overline{\mathcal{I}1_2.j_2,j_1}^\dagger \lambda_{12}^{(1)}}{\sqrt{3}}$	$+ \frac{\overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}^T} S_{1,j_2} y_{1,i_1,r_1} y_{2,i_2,r_1} \overline{\mathcal{I}T3.a_1.a_2,j_1,j_2} S_{2,j_1}^\dagger \lambda_{12}^{(2)}}{\sqrt{3}\epsilon M_F}$	$- \frac{2h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1}} \overline{\mathcal{I}H_{a_2} S_{1,j_2} y_{1,i_1,r_1} H_{a_1}^\dagger S_{1,j_1}^\dagger \overline{\mathcal{I}H_{1,a_1,a_2,j_2,j_1}^\dagger}}}{\sqrt{3}}$
cont.									$- \frac{h\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_1}}}{4\epsilon M_F} C P_R \overline{L_{a_2,i_2}^T} S_{1,j_1} y_{1,i_3,r_2} y_{2,i_2,i_1,r_2} y_{2,i_3,r_1} \overline{\mathcal{I}T3.a_1.a_2,j_1} S_2^\dagger$	
									$- \frac{3h\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_1}}}{8M_F} C P_R \overline{L_{a_2,i_2}^T} S_{1,j_1} y_{1,i_3,r_2} y_{2,i_2,i_1,r_2} y_{2,i_3,r_1} \overline{\mathcal{I}T3.a_1.a_2,j_1} S_2^\dagger$	
									$- \frac{h\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_1}}}{\sqrt{3}\epsilon M_F} C P_R \overline{L_{a_2,i_2}^T} S_{1,j_1} y_{1,i_1,r_2} y_{1,i_3,r_1} y_{2,i_2,r_2} \overline{\mathcal{I}T3.a_1.a_2,j_1} S_2^\dagger$	
									$- \frac{h\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_1}}}{\sqrt{3}M_F} C P_R \overline{L_{a_2,i_2}^T} S_{1,j_1} y_{1,i_1,r_2} y_{1,i_3,r_1} y_{2,i_2,r_2} \overline{\mathcal{I}T3.a_1.a_2,j_1} S_2^\dagger$	
cont.									$- \frac{h\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_1}}}{\epsilon M_F} C P_R \overline{L_{a_2,i_2}^T} S_{1,j_1} y_{1,i_1,r_2} y_{2,i_2,r_2} y_{2,i_3,r_1} \overline{\mathcal{I}T3.a_1.a_2,j_1} S_2^\dagger$	
									$- \frac{h\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_1}}}{\epsilon M_F} C P_R \overline{L_{a_2,i_2}^T} S_{1,j_1} y_{1,i_1,r_2} y_{2,i_2,r_2} y_{2,i_3,r_1} \overline{\mathcal{I}T3.a_1.a_2,j_1} S_2^\dagger$	
									$- \frac{3h\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_1}}}{4\epsilon M_F} C P_R \overline{L_{a_2,i_2}^T} S_{1,j_1} y_{1,i_1,r_2} y_{1,i_3,r_1} y_{2,i_3,r_2} \overline{\mathcal{I}T3.a_1.a_2,j_1} S_2^\dagger$	
cont.									$- \frac{3h\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_1}}}{8M_F} C P_R \overline{L_{a_2,i_2}^T} S_{1,j_1} y_{1,i_1,r_2} y_{1,i_3,r_1} y_{2,i_3,r_2} \overline{\mathcal{I}T3.a_1.a_2,j_1} S_2^\dagger$	
									$- \frac{h\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_1}}}{4\epsilon M_F} C P_R \overline{L_{a_2,i_2}^T} S_{1,j_1} y_{1,i_1,r_2} y_{2,i_2,r_1} y_{2,i_3,r_2} \overline{\mathcal{I}T3.a_1.a_2,j_1} S_2^\dagger$	
									$- \frac{h\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_1}}}{4M_F} C P_R \overline{L_{a_2,i_2}^T} S_{1,j_1} y_{1,i_1,r_2} y_{2,i_2,r_1} y_{2,i_3,r_2} \overline{\mathcal{I}T3.a_1.a_2,j_1} S_2^\dagger$	
cont.									$+ \frac{\overline{y_{1,i_1,r_1}}\overline{y_{2,i_2,r_1}}\overline{L_{a_2,i_2}^T} C P_L L_{a_1,i_1} S_2 \overline{\mathcal{I}T3.a_1.a_2,j_1} S_{1,j_1}^\dagger}{M_F}$	
									$+ \frac{\sqrt{3}h\overline{y_{1,i_1,r_1}}\overline{y_{1,i_3,r_2}}\overline{y_{2,i_2,r_2}}\overline{L_{a_2,i_2}^T} C P_L L_{a_1,i_1} S_2 y_{1,i_3,r_1} \overline{\mathcal{I}T3.a_1.a_2,j_1} S_{1,j_1}^\dagger}{4\epsilon M_F}$	
									$+ \frac{3\sqrt{3}h\overline{y_{1,i_1,r_1}}\overline{y_{1,i_3,r_2}}\overline{y_{2,i_2,r_2}}\overline{L_{a_2,i_2}^T} C P_L L_{a_1,i_1} S_2 y_{1,i_3,r_1} \overline{\mathcal{I}T3.a_1.a_2,j_1} S_{1,j_1}^\dagger}{8M_F}$	
									$- \frac{h\overline{y_{1,i_1,r_1}}\overline{y_{1,i_3,r_2}}\overline{y_{2,i_2,r_1}}\overline{L_{a_2,i_2}^T} C P_L L_{a_1,i_1} S_2 y_{1,i_3,r_2} \overline{\mathcal{I}T3.a_1.a_2,j_1} S_{1,j_1}^\dagger}{\sqrt{3}\epsilon M_F}$	
									$- \frac{h\overline{y_{1,i_1,r_1}}\overline{y_{1,i_3,r_2}}\overline{y_{2,i_2,r_1}}\overline{L_{a_2,i_2}^T} C P_L L_{a_1,i_1} S_2 y_{1,i_3,r_2} \overline{\mathcal{I}T3.a_1.a_2,j_1} S_{1,j_1}^\dagger}{\sqrt{3}M_F}$	
									$- \frac{\sqrt{3}h\overline{y_{1,i_1,r_1}}\overline{y_{1,i_2,r_2}}\overline{y_{2,i_3,r_1}}\overline{L_{a_2,i_2}^T} C P_L L_{a_1,i_1} S_2 y_{1,i_3,r_2} \overline{\mathcal{I}T3.a_1.a_2,j_1} S_{1,j_1}^\dagger}{4\epsilon M_F}$	
cont.									$+ \frac{3\sqrt{3}h\overline{y_{1,i_1,r_1}}\overline{y_{1,i_2,r_2}}\overline{y_{2,i_3,r_1}}\overline{L_{a_2,i_2}^T} C P_L L_{a_1,i_1} S_2 y_{1,i_3,r_2} \overline{\mathcal{I}T3.a_1.a_2,j_1} S_{1,j_1}^\dagger}{8M_F}$	
									$- \frac{h\overline{y_{1,i_3,r_1}}\overline{y_{2,i_1,r_2}}\overline{y_{2,i_2,r_1}}\overline{L_{a_2,i_2}^T} C P_L L_{a_1,i_1} S_2 y_{2,i_3,r_2} \overline{\mathcal{I}T3.a_1.a_2,j_1} S_{1,j_1}^\dagger}{4\epsilon M_F}$	
									$- \frac{3h\overline{y_{1,i_3,r_1}}\overline{y_{2,i_1,r_2}}\overline{y_{2,i_2,r_1}}\overline{L_{a_2,i_2}^T} C P_L L_{a_1,i_1} S_2 y_{2,i_3,r_2} \overline{\mathcal{I}T3.a_1.a_2,j_1} S_{1,j_1}^\dagger}{8M_F}$	
cont.									$- \frac{h\overline{y_{1,i_1,r_1}}\overline{y_{2,i_2,r_2}}\overline{y_{2,i_3,r_1}}\overline{L_{a_2,i_2}^T} C P_L L_{a_1,i_1} S_2 y_{2,i_3,r_2} \overline{\mathcal{I}T3.a_1.a_2,j_1} S_{1,j_1}^\dagger}{4\epsilon M_F}$	
									$- \frac{3h\overline{y_{1,i_1,r_1}}\overline{y_{2,i_2,r_2}}\overline{y_{2,i_3,r_1}}\overline{L_{a_2,i_2}^T} C P_L L_{a_1,i_1} S_2 y_{2,i_3,r_2} \overline{\mathcal{I}T3.a_1.a_2,j_1} S_{1,j_1}^\dagger}{8M_F}$	
									$+ \frac{3h\overline{y_{1,i_1,r_1}}\overline{y_{2,i_2,r_2}}\overline{y_{2,i_3,r_1}}\overline{L_{a_2,i_2}^T} C P_L L_{a_1,i_1} S_2 y_{2,i_3,r_2} \overline{\mathcal{I}T3.a_1.a_2,j_1} S_{1,j_1}^\dagger}{8M_F}$	
cont.									$- \frac{h\overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}^T} S_{1,j_1} y_{1,i_1,r_1} y_{2,i_2,r_1} \overline{\mathcal{I}T3.a_1.a_2,j_1} S_2^\dagger \lambda_{12}^{(1)}}{\epsilon M_F}$	

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	S_1	S_2	F	S_1	S_2	ℓ	S_1	F	H
cont.													$-\frac{\hbar \overline{L_{a_1, i_1}} C P_R \overline{L_{a_2, i_2}}^T S_{1, j_1} y_{1, i_1, r_1} y_{2, i_2, r_1} \mathcal{I} T 3_{a_1, a_2, j_1}^* S_2^\dagger \lambda_{12}^{(1)}}{M_F}$			
													$-\frac{\hbar \overline{y_{1, i_1, r_1}} y_{2, i_2, r_1} \overline{L_{a_2, i_2}}^T C P_L \overline{L_{a_1, i_1}} S_2 \mathcal{I} T 3_{a_1, a_2, j_1} S_{1, j_1}^\dagger \lambda_{12}^{(1)}}{\epsilon M_F}$			
													$-\frac{\hbar \overline{y_{1, i_1, r_1}} y_{2, i_2, r_1} \overline{L_{a_2, i_2}}^T C P_L \overline{L_{a_1, i_1}} S_2 \mathcal{I} T 3_{a_1, a_2, j_1} S_{1, j_1}^\dagger \lambda_{12}^{(1)}}{M_F}$			
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0			$-\frac{4\hbar \log\left(\frac{\hat{\mu}^2}{M_F^2}\right) \overline{y_{1, i_1, r_1}} y_{2, i_2, r_2} S_2 S_{1, j_1} y_{1, i_2, r_1} y_{2, i_1, r_2} S_2^\dagger S_{1, j_1}^\dagger}{\sqrt{3}}$			$-\frac{3\hbar \overline{L_{a_1, i_1}} C P_R \overline{L_{a_2, i_2}}^T S_{1, j_1} y_{1, i_1, r_1} y_{2, i_2, r_1} \mathcal{I} T 3_{a_1, a_2, j_1}^* S_2^\dagger g_Y^2}{4\epsilon M_F}$		$-\frac{2\hbar \log\left(\frac{\hat{\mu}^2}{M_F^2}\right) \overline{y_{1, i_1, r_1}} y_{2, i_2, r_2} H_{a_1} S_{1, j_1} y_{1, i_2, r_1} y_{2, i_1, r_2} Y_{e, i_1, i_3} H_{a_1}^\dagger S_{1, j_1}^\dagger}{\sqrt{3}}$	
										$-\frac{2\hbar \log\left(\frac{\hat{\mu}^2}{M_F^2}\right) \overline{y_{1, i_1, r_1}} S_2 S_{1, j_2} y_{2, i_1, r_1} S_2^\dagger S_{1, j_1}^\dagger \mathcal{I}_{12, j_2, j_1} \lambda_{12}^{(1)}}{\sqrt{3}}$			$+\frac{\hbar \overline{L_{a_1, i_1}} C P_R \overline{L_{a_2, i_2}}^T S_{1, j_1} y_{1, i_1, r_1} y_{2, i_2, r_1} \mathcal{I} T 3_{a_1, a_2, j_1}^* S_2^\dagger g_Y^2}{4M_F}$		$-2i\sqrt{\frac{2}{3}}\hbar \log\left(\frac{\hat{\mu}^2}{M_F^2}\right) \frac{\overline{y_{1, i_1, r_1}} y_{2, i_2, r_2} H_{a_2} S_{1, j_2} y_{1, i_2, r_1} y_{2, i_1, r_2} Y_{e, i_1, i_3} H_{a_1}^\dagger S_{1, j_1}^\dagger \mathcal{I}_{H1, a_1, a_2, j_2, j_1}^\dagger}{\sqrt{3}}$	
											$-2\hbar \log\left(\frac{\hat{\mu}^2}{M_F^2}\right) \overline{y_{2, i_1, r_1}} S_2 S_{1, j_2} y_{2, i_1, r_1} S_2^\dagger S_{1, j_1}^\dagger \mathcal{I}_{12, j_2, j_1} \lambda_{12}^{(1)}$			$+\frac{3\hbar \overline{y_{1, i_1, r_1}} y_{2, i_2, r_1} \overline{L_{a_2, i_2}}^T C P_L \overline{L_{a_1, i_1}} S_2 \mathcal{I} T 3_{a_1, a_2, j_1} S_{1, j_1}^\dagger g_Y^2}{4\epsilon M_F}$		$-\frac{2\hbar \log\left(\frac{\hat{\mu}^2}{M_F^2}\right) \overline{y_{1, i_1, r_1}} H_{a_2} S_{1, j_2} y_{1, i_1, r_1} H_{a_1}^\dagger S_{1, j_1}^\dagger \mathcal{I}_{H1, a_1, a_2, j_2, j_1} 2\lambda_{12}^{(2)}}{\sqrt{3}}$
cont.													$+\frac{\hbar \overline{y_{1, i_1, r_1}} y_{2, i_2, r_1} \overline{L_{a_2, i_2}}^T C P_L \overline{L_{a_1, i_1}} S_2 \mathcal{I} T 3_{a_1, a_2, j_1} S_{1, j_1}^\dagger g_Y^2}{4M_F}$			
													$-\frac{\sqrt{3}\hbar \overline{y_{1, i_3, r_1}} \overline{L_{a_1, i_1}} C P_R \overline{L_{a_2, i_2}}^T S_{1, j_1} y_{1, i_2, r_1} y_{1, i_3, r_2} y_{2, i_1, r_2} \mathcal{I} T 3_{a_1, a_2, j_1}^* S_2^\dagger}{4\epsilon M_F}$			
													$-\frac{3\sqrt{3}\hbar \overline{y_{1, i_3, r_1}} \overline{L_{a_1, i_1}} C P_R \overline{L_{a_2, i_2}}^T S_{1, j_1} y_{1, i_2, r_1} y_{1, i_3, r_2} y_{2, i_1, r_2} \mathcal{I} T 3_{a_1, a_2, j_1}^* S_2^\dagger}{8M_F}$			
cont.													$+\frac{\overline{L_{a_1, i_1}} C P_R \overline{L_{a_2, i_2}}^T S_{1, j_1} y_{1, i_1, r_1} y_{2, i_2, r_1} \mathcal{I} T 3_{a_1, a_2, j_1}^* S_2^\dagger}{M_F}$			
													$-\frac{\hbar \overline{y_{2, i_3, r_1}} \overline{L_{a_1, i_1}} C P_R \overline{L_{a_2, i_2}}^T S_{1, j_1} y_{1, i_3, r_2} y_{2, i_1, r_2} y_{2, i_2, r_1} \mathcal{I} T 3_{a_1, a_2, j_1}^* S_2^\dagger}{4\epsilon M_F}$			
													$-\frac{3\hbar \overline{y_{2, i_3, r_1}} \overline{L_{a_1, i_1}} C P_R \overline{L_{a_2, i_2}}^T S_{1, j_1} y_{1, i_3, r_2} y_{2, i_1, r_2} y_{2, i_2, r_1} \mathcal{I} T 3_{a_1, a_2, j_1}^* S_2^\dagger}{8M_F}$			
cont.													$-\frac{\hbar \overline{y_{1, i_3, r_1}} \overline{L_{a_1, i_1}} C P_R \overline{L_{a_2, i_2}}^T S_{1, j_1} y_{1, i_1, r_2} y_{1, i_3, r_1} y_{2, i_2, r_2} \mathcal{I} T 3_{a_1, a_2, j_1}^* S_2^\dagger}{\sqrt{3}\epsilon M_F}$			
													$-\frac{\hbar \overline{y_{1, i_3, r_1}} \overline{L_{a_1, i_1}} C P_R \overline{L_{a_2, i_2}}^T S_{1, j_1} y_{1, i_1, r_2} y_{1, i_3, r_1} y_{2, i_2, r_2} \mathcal{I} T 3_{a_1, a_2, j_1}^* S_2^\dagger}{2\sqrt{3}M_F}$			
													$-\frac{\hbar \overline{y_{2, i_3, r_1}} \overline{L_{a_1, i_1}} C P_R \overline{L_{a_2, i_2}}^T S_{1, j_1} y_{1, i_1, r_2} y_{2, i_2, r_2} y_{2, i_3, r_1} \mathcal{I} T 3_{a_1, a_2, j_1}^* S_2^\dagger}{2M_F}$			
cont.													$-\frac{\hbar \overline{y_{2, i_3, r_1}} \overline{L_{a_1, i_1}} C P_R \overline{L_{a_2, i_2}}^T S_{1, j_1} y_{1, i_1, r_2} y_{2, i_2, r_2} y_{2, i_3, r_1} \mathcal{I} T 3_{a_1, a_2, j_1}^* S_2^\dagger}{2M_F}$			
													$-\frac{\sqrt{3}\hbar \overline{y_{1, i_3, r_1}} \overline{L_{a_1, i_1}} C P_R \overline{L_{a_2, i_2}}^T S_{1, j_1} y_{1, i_1, r_2} y_{1, i_2, r_1} y_{2, i_3, r_2} \mathcal{I} T 3_{a_1, a_2, j_1}^* S_2^\dagger}{4\epsilon M_F}$			
													$-\frac{3\sqrt{3}\hbar \overline{y_{1, i_3, r_1}} \overline{L_{a_1, i_1}} C P_R \overline{L_{a_2, i_2}}^T S_{1, j_1} y_{1, i_1, r_2} y_{1, i_2, r_1} y_{2, i_3, r_2} \mathcal{I} T 3_{a_1, a_2, j_1}^* S_2^\dagger}{8M_F}$			
cont.													$-\frac{\hbar \overline{y_{2, i_3, r_1}} \overline{L_{a_1, i_1}} C P_R \overline{L_{a_2, i_2}}^T S_{1, j_1} y_{1, i_1, r_2} y_{2, i_2, r_1} y_{2, i_3, r_2} \mathcal{I} T 3_{a_1, a_2, j_1}^* S_2^\dagger}{4\epsilon M_F}$			
													$-\frac{3\hbar \overline{y_{2, i_3, r_1}} \overline{L_{a_1, i_1}} C P_R \overline{L_{a_2, i_2}}^T S_{1, j_1} y_{1, i_1, r_2} y_{2, i_2, r_1} y_{2, i_3, r_2} \mathcal{I} T 3_{a_1, a_2, j_1}^* S_2^\dagger}{8M_F}$			
													$+\frac{\overline{y_{1, i_1, r_1}} y_{2, i_2, r_1} \overline{L_{a_2, i_2}}^T C P_L \overline{L_{a_1, i_1}} S_2 \mathcal{I} T 3_{a_1, a_2, j_1} S_{1, j_1}^\dagger}{M_F}$			
cont.													$-\frac{\sqrt{3}\hbar \overline{y_{1, i_1, r_1}} y_{2, i_2, r_1} \overline{L_{a_2, i_2}}^T C P_L \overline{L_{a_1, i_1}} S_2 y_{1, i_3, r_1} \mathcal{I} T 3_{a_1, a_2, j_1} S_{1, j_1}^\dagger}{4\epsilon M_F}$			
													$-\frac{3\sqrt{3}\hbar \overline{y_{1, i_1, r_1}} y_{2, i_2, r_1} \overline{L_{a_2, i_2}}^T C P_L \overline{L_{a_1, i_1}} S_2 y_{1, i_3, r_1} \mathcal{I} T 3_{a_1, a_2, j_1} S_{1, j_1}^\dagger}{8M_F}$			
													$-\frac{\hbar \overline{y_{1, i_3, r_1}} y_{2, i_1, r_2} y_{2, i_2, r_1} \overline{L_{a_2, i_2}}^T C P_L \overline{L_{a_1, i_1}} S_2 y_{2, i_3, r_2} \mathcal{I} T 3_{a_1, a_2, j_1} S_{1, j_1}^\dagger}{2\sqrt{3}M_F}$			
cont.													$-\frac{\sqrt{3}\hbar \overline{y_{1, i_1, r_1}} y_{1, i_2, r_2} y_{2, i_3, r_1} \overline{L_{a_2, i_2}}^T C P_L \overline{L_{a_1, i_1}} S_2 y_{1, i_3, r_2} \mathcal{I} T 3_{a_1, a_2, j_1} S_{1, j_1}^\dagger}{4\epsilon M_F}$			
													$-\frac{3\sqrt{3}\hbar \overline{y_{1, i_1, r_1}} y_{1, i_2, r_2} y_{2, i_3, r_1} \overline{L_{a_2, i_2}}^T C P_L \overline{L_{a_1, i_1}} S_2 y_{1, i_3, r_2} \mathcal{I} T 3_{a_1, a_2, j_1} S_{1, j_1}^\dagger}{8M_F}$			
													$-\frac{\hbar \overline{y_{1, i_3, r_1}} y_{2, i_1, r_2} y_{2, i_2, r_1} \overline{L_{a_2, i_2}}^T C P_L \overline{L_{a_1, i_1}} S_2 y_{2, i_3, r_2} \mathcal{I} T 3_{a_1, a_2, j_1} S_{1, j_1}^\dagger}{4\epsilon M_F}$			
cont.													$-\frac{3\hbar \overline{y_{1, i_3, r_1}} y_{2, i_1, r_2} y_{2, i_2, r_1} \overline{L_{a_2, i_2}}^T C P_L \overline{L_{a_1, i_1}} S_2 y_{2, i_3, r_2} \mathcal{I} T 3_{a_1, a_2, j_1} S_{1, j_1}^\dagger}{8M_F}$			
													$-\frac{\hbar \overline{y_{1, i_1, r_1}} y_{2, i_2, r_1} y_{2, i_3, r_2} \overline{L_{a_2, i_2}}^T C P_L \overline{L_{a_1, i_1}} S_2 y_{2, i_3, r_2} \mathcal{I} T 3_{a_1, a_2, j_1} S_{1, j_1}^\dagger}{4\epsilon M_F}$			
													$-\frac{3\hbar \overline{y_{1, i_1, r_1}} y_{2, i_2, r_1} y_{2, i_3, r_2} \overline{L_{a_2, i_2}}^T C P_L \overline{L_{a_1, i_1}} S_2 y_{2, i_3, r_2} \mathcal{I} T 3_{a_1, a_2, j_1} S_{1, j_1}^\dagger}{8M_F}$			
cont.													$-\frac{\hbar \overline{y_{1, i_1, r_1}} y_{2, i_2, r_1} y_{2, i_3, r_2} \overline{L_{a_2, i_2}}^T C P_L \overline{L_{a_1, i_1}} S_2 y_{2, i_3, r_2} \mathcal{I} T 3_{a_1, a_2, j_1} S_{1, j_1}^\dagger}{\epsilon M_F}$			
													$-\frac{\hbar \overline{y_{1, i_1, r_1}} y_{2, i_2, r_1} y_{2, i_3, r_2} \overline{L_{a_2, i_2}}^T C P_L \overline{L_{a_1, i_1}} S_2 y_{2, i_3, r_2} \mathcal{I} T 3_{a_1, a_2, j_1} S_{1, j_1}^\dagger}{2M_F}$			
													$-\frac{\hbar \overline{L_{a_1, i_1}} C P_R \overline{L_{a_2, i_2}}^T S_{1, j_1} y_{1, i_1, r_1} y_{2, i_2, r_1} \mathcal{I} T 3_{a_1, a_2, j_1}^* S_2^\dagger \lambda_{12}^{(1)}}{\epsilon M_F}$			
cont.													$-\frac{\hbar \overline{L_{a_1, i_1}} C P_R \overline{L_{a_2, i_2}}^T S_{1, j_1} y_{1, i_1, r_1} y_{2, i_2, r_1} \mathcal{I} T 3_{a_1, a_2, j_1}^* S_2^\dagger \lambda_{12}^{(1)}}{M_F}$			
													$-\frac{\hbar \overline{y_{1, i_1, r_1}} y_{2, i_2, r_1} \overline{L_{a_2, i_2}}^T C P_L \overline{L_{a_1, i_1}} S_2 \mathcal{I} T 3_{a_1, a_2, j_1} S_{1, j_1}^\dagger \lambda_{12}^{(1)}}{\epsilon M_F}$			
													$-\frac{\hbar \overline{y_{1, i_1, r_1}} y_{2, i_2, r_1} \overline{L_{a_2, i_2}}^T C P_L \overline{L_{a_1, i_1}} S_2 \mathcal{I} T 3_{a_1, a_2, j_1} S_{1, j_1}^\dagger \lambda_{12}^{(1)}}{M_F}$			

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$S_1 S_2 F$	$S_1 S_2 \ell$	$S_1 F H$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$-\frac{4h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_2} S_2 S_{1,j_1} y_{1,i_2,r_1} y_{2,i_1,r_2} S_2^\dagger S_{1,j_1}^\dagger}}{\sqrt{3}}$	$-\frac{\sqrt{3} h \overline{y_{1,i_3,r_1} L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_2,r_1} y_{1,i_3,r_2} y_{2,i_1,r_2} \overline{y_{2,i_2,r_1} S_{1,j_1}^\dagger}}{4\epsilon M_F}$	$-\frac{2h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} Y_{e,i_2,i_3} H_{a_1} S_{1,j_2} y_{1,i_2,r_1} Y_{e,i_1,i_3} H_{a_1}^\dagger S_{1,j_1}^\dagger}}{\sqrt{3}}$
								$-\frac{2h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} S_2 S_{1,j_2} y_{1,i_1,r_1} S_2^\dagger S_{1,j_1}^\dagger} \overline{I_{12,j_2,j_1}^1} \lambda_{12}^{(1)}}{\sqrt{3}}$	$-\frac{3\sqrt{3} h \overline{y_{1,i_3,r_1} L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_2,r_1} y_{1,i_3,r_2} y_{2,i_1,r_2} \overline{y_{2,i_2,r_1} S_{1,j_1}^\dagger}}{8M_F}$	$-2i\sqrt{\frac{2}{3}} h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} Y_{e,i_2,i_3} H_{a_2} S_{1,j_2} y_{1,i_2,r_1} Y_{e,i_1,i_3} H_{a_1}^\dagger S_{1,j_1}^\dagger} \overline{I_{H1,a_1,\omega_2,j_2,j_1}^{(2)}}$
								$-2h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1} S_2 S_{1,j_2} y_{2,i_1,r_1} S_2^\dagger S_{1,j_1}^\dagger} \overline{I_{12,j_2,j_1}^1} \lambda_{12}^{(1)}$	$+\frac{\overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_1,r_1} y_{2,i_2,r_1} \overline{I_{T3,a_1,\omega_2,j_1} S_2^\dagger}}{M_F}$	$-\frac{2h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} H_{a_2} S_{1,j_2} y_{1,i_1,r_1} H_{a_1}^\dagger S_{1,j_1}^\dagger} \overline{I_{H1,a_1,\omega_2,j_2,j_1}^{(2)}}}{\sqrt{3}}$
cont.								$-\frac{h \overline{y_{2,i_3,r_1} L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_3,r_2} y_{2,i_1,r_2} y_{2,i_2,r_1} \overline{I_{T3,a_1,\omega_2,j_1} S_2^\dagger}}{4\epsilon M_F}$	$-\frac{3h \overline{y_{2,i_3,r_1} L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_3,r_2} y_{2,i_1,r_2} y_{2,i_2,r_1} \overline{I_{T3,a_1,\omega_2,j_1} S_2^\dagger}}{8M_F}$	
								$-\frac{h \overline{y_{1,i_3,r_1} L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_1,r_2} y_{1,i_3,r_1} y_{2,i_2,r_1} \overline{I_{T3,a_1,\omega_2,j_1} S_2^\dagger}}{4\epsilon M_F}$	$-\frac{\sqrt{3}\epsilon M_F}{2\sqrt{3}M_F}$	
cont.								$-\frac{h \overline{y_{1,i_3,r_1} L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_1,r_2} y_{1,i_3,r_1} y_{2,i_2,r_1} \overline{I_{T3,a_1,\omega_2,j_1} S_2^\dagger}}{2\sqrt{3}M_F}$	$-\frac{h \overline{y_{2,i_3,r_1} L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_1,r_2} y_{2,i_2,r_1} \overline{I_{T3,a_1,\omega_2,j_1} S_2^\dagger}}{\epsilon M_F}$	
								$-\frac{h \overline{y_{2,i_3,r_1} L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_1,r_2} y_{2,i_2,r_1} \overline{I_{T3,a_1,\omega_2,j_1} S_2^\dagger}}{2M_F}$	$-\frac{h \overline{y_{2,i_3,r_1} L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_1,r_2} y_{2,i_2,r_1} \overline{I_{T3,a_1,\omega_2,j_1} S_2^\dagger}}{4\epsilon M_F}$	
cont.								$-\frac{\sqrt{3} h \overline{y_{1,i_3,r_1} L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_1,r_2} y_{1,i_3,r_1} y_{2,i_2,r_1} \overline{I_{T3,a_1,\omega_2,j_1} S_2^\dagger}}{4\epsilon M_F}$	$-\frac{3\sqrt{3} h \overline{y_{1,i_3,r_1} L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_1,r_2} y_{1,i_3,r_1} y_{2,i_2,r_1} \overline{I_{T3,a_1,\omega_2,j_1} S_2^\dagger}}{8M_F}$	
								$-\frac{h \overline{y_{2,i_3,r_1} L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_1,r_2} y_{2,i_2,r_1} y_{2,i_3,r_1} \overline{I_{T3,a_1,\omega_2,j_1} S_2^\dagger}}{8M_F}$	$-\frac{3h \overline{y_{2,i_3,r_1} L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_1,r_2} y_{2,i_2,r_1} y_{2,i_3,r_1} \overline{I_{T3,a_1,\omega_2,j_1} S_2^\dagger}}{8M_F}$	
								$+\frac{\overline{y_{1,i_1,r_1} y_{2,i_2,r_1} L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_2 \overline{I_{T3,a_1,\omega_2,j_1} S_{1,j_1}^\dagger}}{M_F}$	$-\frac{\sqrt{3} h \overline{y_{1,i_1,r_1} y_{1,i_3,r_2} y_{2,i_2,r_2} L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_2 y_{1,i_3,r_1} \overline{I_{T3,a_1,\omega_2,j_1} S_{1,j_1}^\dagger}}{4\epsilon M_F}$	
								$-\frac{\sqrt{3} h \overline{y_{1,i_1,r_1} y_{1,i_3,r_2} y_{2,i_2,r_2} L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_2 y_{1,i_3,r_1} \overline{I_{T3,a_1,\omega_2,j_1} S_{1,j_1}^\dagger}}{4\epsilon M_F}$	$-\frac{3\sqrt{3} h \overline{y_{1,i_1,r_1} y_{1,i_3,r_2} y_{2,i_2,r_2} L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_2 y_{1,i_3,r_1} \overline{I_{T3,a_1,\omega_2,j_1} S_{1,j_1}^\dagger}}{8M_F}$	
cont.								$-\frac{h \overline{y_{1,i_1,r_1} y_{1,i_3,r_2} y_{2,i_2,r_1} L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_2 y_{1,i_3,r_2} \overline{I_{T3,a_1,\omega_2,j_1} S_{1,j_1}^\dagger}}{\sqrt{3}\epsilon M_F}$	$-\frac{h \overline{y_{1,i_1,r_1} y_{1,i_3,r_2} y_{2,i_2,r_1} L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_2 y_{1,i_3,r_2} \overline{I_{T3,a_1,\omega_2,j_1} S_{1,j_1}^\dagger}}{4\epsilon M_F}$	
								$-\frac{h \overline{y_{1,i_1,r_1} y_{1,i_3,r_2} y_{2,i_2,r_1} L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_2 y_{1,i_3,r_2} \overline{I_{T3,a_1,\omega_2,j_1} S_{1,j_1}^\dagger}}{4\epsilon M_F}$	$-\frac{2\sqrt{3}M_F}{4\epsilon M_F}$	
cont.								$-\frac{\sqrt{3} h \overline{y_{1,i_1,r_1} y_{1,i_2,r_2} y_{2,i_3,r_1} L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_2 y_{1,i_3,r_2} \overline{I_{T3,a_1,\omega_2,j_1} S_{1,j_1}^\dagger}}{4\epsilon M_F}$	$-\frac{3\sqrt{3} h \overline{y_{1,i_1,r_1} y_{1,i_2,r_2} y_{2,i_3,r_1} L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_2 y_{1,i_3,r_2} \overline{I_{T3,a_1,\omega_2,j_1} S_{1,j_1}^\dagger}}{8M_F}$	
								$-\frac{h \overline{y_{1,i_1,r_1} y_{2,i_2,r_1} y_{2,i_3,r_1} L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_2 y_{2,i_3,r_2} \overline{I_{T3,a_1,\omega_2,j_1} S_{1,j_1}^\dagger}}{4\epsilon M_F}$	$-\frac{h \overline{y_{1,i_1,r_1} y_{2,i_2,r_1} y_{2,i_3,r_1} L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_2 y_{2,i_3,r_2} \overline{I_{T3,a_1,\omega_2,j_1} S_{1,j_1}^\dagger}}{8M_F}$	
cont.								$-\frac{h \overline{y_{1,i_1,r_1} y_{2,i_2,r_1} y_{2,i_3,r_1} L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_2 y_{2,i_3,r_2} \overline{I_{T3,a_1,\omega_2,j_1} S_{1,j_1}^\dagger}}{4\epsilon M_F}$	$-\frac{3h \overline{y_{1,i_1,r_1} y_{2,i_2,r_1} y_{2,i_3,r_1} L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_2 y_{2,i_3,r_2} \overline{I_{T3,a_1,\omega_2,j_1} S_{1,j_1}^\dagger}}{8M_F}$	
								$-\frac{h \overline{y_{1,i_1,r_1} y_{2,i_2,r_1} y_{2,i_3,r_1} L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_2 y_{2,i_3,r_2} \overline{I_{T3,a_1,\omega_2,j_1} S_{1,j_1}^\dagger}}{4\epsilon M_F}$	$-\frac{3h \overline{y_{1,i_1,r_1} y_{2,i_2,r_1} y_{2,i_3,r_1} L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_2 y_{2,i_3,r_2} \overline{I_{T3,a_1,\omega_2,j_1} S_{1,j_1}^\dagger}}{8M_F}$	
cont.								$-\frac{h \overline{y_{1,i_1,r_1} y_{2,i_2,r_1} y_{2,i_3,r_2} L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_2 y_{2,i_3,r_2} \overline{I_{T3,a_1,\omega_2,j_1} S_{1,j_1}^\dagger}}{2M_F}$	$-\frac{h \overline{y_{1,i_1,r_1} y_{2,i_2,r_1} y_{2,i_3,r_2} L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_2 y_{2,i_3,r_2} \overline{I_{T3,a_1,\omega_2,j_1} S_{1,j_1}^\dagger}}{4\epsilon M_F}$	
								$-\frac{h \overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_1,r_1} y_{2,i_2,r_1} \overline{I_{T3,a_1,\omega_2,j_1} S_2^\dagger} \lambda_{12}^{(1)}}{\epsilon M_F}$	$-\frac{h \overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_1,r_1} y_{2,i_2,r_1} \overline{I_{T3,a_1,\omega_2,j_1} S_2^\dagger} \lambda_{12}^{(1)}}{\epsilon M_F}$	
cont.								$-\frac{h \overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_1,r_1} y_{2,i_2,r_1} \overline{I_{T3,a_1,\omega_2,j_1} S_2^\dagger} \lambda_{12}^{(1)}}{M_F}$	$-\frac{h \overline{y_{1,i_1,r_1} y_{2,i_2,r_1} L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_2 \overline{I_{T3,a_1,\omega_2,j_1} S_{1,j_1}^\dagger} \lambda_{12}^{(1)}}{\epsilon M_F}$	
								$-\frac{h \overline{y_{1,i_1,r_1} y_{2,i_2,r_1} L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_2 \overline{I_{T3,a_1,\omega_2,j_1} S_{1,j_1}^\dagger} \lambda_{12}^{(1)}}{\epsilon M_F}$	$-\frac{h \overline{y_{1,i_1,r_1} y_{2,i_2,r_1} L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_2 \overline{I_{T3,a_1,\omega_2,j_1} S_{1,j_1}^\dagger} \lambda_{12}^{(1)}}{M_F}$	
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	$-\frac{4h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_2} S_2 S_{1,j_1} y_{1,i_2,r_1} y_{2,i_1,r_2} S_2^\dagger S_{1,j_1}^\dagger}}{\sqrt{3}}$	$-\frac{\sqrt{3} h \overline{y_{1,i_3,r_1} L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_2,r_1} y_{1,i_3,r_2} y_{2,i_1,r_2} \overline{y_{2,i_2,r_1} S_{1,j_1}^\dagger}}{4\epsilon M_F}$	$-\frac{2h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} Y_{e,i_2,i_3} H_{a_1} S_{1,j_2} y_{1,i_2,r_1} Y_{e,i_1,i_3} H_{a_1}^\dagger S_{1,j_1}^\dagger}}{\sqrt{3}}$
								$-\frac{2h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} S_2 S_{1,j_2} y_{1,i_1,r_1} S_2^\dagger S_{1,j_1}^\dagger} \overline{I_{12,j_2,j_1}^1} \lambda_{12}^{(1)}}{\sqrt{3}}$	$-\frac{3\sqrt{3} h \overline{y_{1,i_3,r_1} L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_2,r_1} y_{1,i_3,r_2} y_{2,i_1,r_2} \overline{y_{2,i_2,r_1} S_{1,j_1}^\dagger}}{8M_F}$	$-2i\sqrt{\frac{2}{3}} h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} Y_{e,i_2,i_3} H_{a_2} S_{1,j_2} y_{1,i_2,r_1} Y_{e,i_1,i_3} H_{a_1}^\dagger S_{1,j_1}^\dagger} \overline{I_{H1,a_1,\omega_2,j_2,j_1}^{(2)}}$
								$-2h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1} S_2 S_{1,j_2} y_{2,i_1,r_1} S_2^\dagger S_{1,j_1}^\dagger} \overline{I_{12,j_2,j_1}^1} \lambda_{12}^{(1)}$	$-\frac{9h \overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T g_{\tilde{g}}^2 S_{1,j_1} y_{1,i_1,r_1} y_{2,i_2,r_1} \overline{I_{T3,a_1,\omega_2,j_1} S_2^\dagger}}{4\epsilon M_F}$	$-\frac{2h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} H_{a_2} S_{1,j_2} y_{1,i_1,r_1} H_{a_1}^\dagger S_{1,j_1}^\dagger} \overline{I_{H1,a_1,\omega_2,j_2,j_1}^{(2)}}}{\sqrt{3}}$
cont.								$-\frac{3h \overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T g_{\tilde{g}}^2 S_{1,j_1} y_{1,i_1,r_1} y_{2,i_2,r_1} \overline{I_{T3,a_1,\omega_2,j_1} S_2^\dagger}}{4M_F}$	$-\frac{h \overline{y_{2,i_3,r_1} L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_3,r_2} y_{2,i_1,r_2} y_{2,i_2,r_1} \overline{I_{T3,a_1,\omega_2,j_1} S_2^\dagger}}{4\epsilon M_F}$	
								$+\frac{\overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_1,r_1} y_{2,i_2,r_1} \overline{I_{T3,a_1,\omega_2,j_1} S_2^\dagger}}{M_F}$	$-\frac{3h \overline{y_{2,i_3,r_1} L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_3,r_2} y_{2,i_1,r_2} y_{2,i_2,r_1} \overline{I_{T3,a_1,\omega_2,j_1} S_2^\dagger}}{8M_F}$	
cont.								$-\frac{h \overline{y_{2,i_3,r_1} L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_3,r_2} y_{2,i_1,r_2} y_{2,i_2,r_1} \overline{I_{T3,a_1,\omega_2,j_1} S_2^\dagger}}{4\epsilon M_F}$	$-\frac{3h \overline{y_{2,i_3,r_1} L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_3,r_2} y_{2,i_1,r_2} y_{2,i_2,r_1} \overline{I_{T3,a_1,\omega_2,j_1} S_2^\dagger}}{8M_F}$	
								$-\frac{h \overline{y_{1,i_3,r_1} L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_1,r_2} y_{1,i_3,r_1} y_{2,i_2,r_1} \overline{I_{T3,a_1,\omega_2,j_1} S_2^\dagger}}{4\epsilon M_F}$	$-\frac{\sqrt{3}\epsilon M_F}{2\sqrt{3}M_F}$	
cont.								$-\frac{h \overline{y_{1,i_3,r_1} L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_1,r_2} y_{1,i_3,r_1} y_{2,i_2,r_1} \overline{I_{T3,a_1,\omega_2,j_1} S_2^\dagger}}{2\sqrt{3}M_F}$	$-\frac{h \overline{y_{2,i_3,r_1} L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_1,r_2} y_{2,i_2,r_1} \overline{I_{T3,a_1,\omega_2,j_1} S_2^\dagger}}{2\sqrt{3}M_F}$	

[illegible]

[illegible]

[illegible]

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$S_1 S_2 F$	$S_1 S_2 \ell$	$S_1 F H$
cont.									$-\frac{3h\overline{y}_{1,i_3,r_1}L_{a_1,i_1}CPR\overline{L}_{a_2,i_2}^TS_{1,j_1}y_{1,i_1,r_2}y_{1,i_2,r_1}y_{2,i_3,r_2}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger}{8M_F}$	
									$-\frac{h\overline{y}_{2,i_3,r_1}L_{a_1,i_1}CPR\overline{L}_{a_2,i_2}^TS_{1,j_1}y_{1,i_1,r_2}y_{2,i_2,r_2}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger}{4\epsilon M_F}$	
									$-\frac{3h\overline{y}_{2,i_3,r_1}L_{a_1,i_1}CPR\overline{L}_{a_2,i_2}^TS_{1,j_1}y_{1,i_1,r_2}y_{2,i_2,r_1}y_{2,i_3,r_2}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger}{8M_F}$	
cont.									$+\frac{\sqrt{\frac{2}{3}}h\overline{y}_{1,i_1,r_1}y_{2,i_2,r_1}L_{a_2,i_2}^TCPL_{a_1,i_1}S_{2,j_2}\mathcal{I}T_{3,a_1,a_2,j_1,j_2}S_{1,j_1}^\dagger\lambda_{12}^{(1)}}{\epsilon M_F}$	
									$+\frac{\sqrt{\frac{2}{3}}h\overline{y}_{1,i_1,r_1}y_{2,i_2,r_1}L_{a_2,i_2}^TCPL_{a_1,i_1}S_{2,j_2}\mathcal{I}T_{3,a_1,a_2,j_1,j_2}S_{1,j_1}^\dagger\lambda_{12}^{(1)}}{M_F}$	
									$-\frac{h\overline{y}_{1,i_1,r_1}y_{2,i_2,r_1}L_{a_2,i_2}^TCPL_{a_1,i_1}S_{2,j_1}\epsilon_{a_1,a_2}S_{1,j_1}^\dagger\lambda_{12}^{(1)}}{\sqrt{3}\epsilon M_F}$	
cont.									$-\frac{h\overline{y}_{1,i_1,r_1}y_{2,i_2,r_1}L_{a_2,i_2}^TCPL_{a_1,i_1}S_{2,j_1}\epsilon_{a_1,a_2}S_{1,j_1}^\dagger\lambda_{12}^{(1)}}{\sqrt{3}M_F}$	
									$+\frac{\sqrt{\frac{2}{3}}hL_{a_1,i_1}CPR\overline{L}_{a_2,i_2}^TS_{1,j_2}y_{1,i_1,r_1}y_{2,i_2,r_1}\mathcal{I}T_{3,a_1,a_2,j_2,j_1}S_{2,j_1}^\dagger\lambda_{12}^{(1)}}{\epsilon M_F}$	
									$+\frac{\sqrt{\frac{2}{3}}hL_{a_1,i_1}CPR\overline{L}_{a_2,i_2}^TS_{1,j_2}y_{1,i_1,r_1}y_{2,i_2,r_1}\mathcal{I}T_{3,a_1,a_2,j_2,j_1}S_{2,j_1}^\dagger\lambda_{12}^{(1)}}{M_F}$	
cont.									$-\frac{h\overline{L}_{a_1,i_1}CPR\overline{L}_{a_2,i_2}^TS_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger\lambda_{12}^{(1)}}{\sqrt{3}\epsilon M_F}$	
									$-\frac{h\overline{L}_{a_1,i_1}CPR\overline{L}_{a_2,i_2}^TS_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger\lambda_{12}^{(1)}}{\sqrt{3}M_F}$	
									$-\frac{h\overline{L}_{a_1,i_1}CPR\overline{L}_{a_2,i_2}^TS_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger\lambda_{12}^{(1)}}{\sqrt{3}M_F}$	
cont.									$-\frac{3\sqrt{\frac{2}{3}}h\overline{y}_{1,i_1,r_1}y_{2,i_2,r_1}L_{a_2,i_2}^TCPL_{a_1,i_1}S_{2,j_2}\mathcal{I}T_{3,a_1,a_2,j_1,j_2}S_{1,j_1}^\dagger\lambda_{12}^{(2)}}{\epsilon M_F}$	
									$-\frac{3\sqrt{\frac{2}{3}}h\overline{y}_{1,i_1,r_1}y_{2,i_2,r_1}L_{a_2,i_2}^TCPL_{a_1,i_1}S_{2,j_2}\mathcal{I}T_{3,a_1,a_2,j_1,j_2}S_{1,j_1}^\dagger\lambda_{12}^{(2)}}{M_F}$	
									$-\frac{h\overline{y}_{1,i_1,r_1}y_{2,i_2,r_1}L_{a_2,i_2}^TCPL_{a_1,i_1}S_{2,j_1}\epsilon_{a_1,a_2}S_{1,j_1}^\dagger\lambda_{12}^{(2)}}{\sqrt{11}\epsilon M_F}$	
cont.									$-\frac{h\overline{y}_{1,i_1,r_1}y_{2,i_2,r_1}L_{a_2,i_2}^TCPL_{a_1,i_1}S_{2,j_1}\epsilon_{a_1,a_2}S_{1,j_1}^\dagger\lambda_{12}^{(2)}}{\sqrt{11}M_F}$	
									$-\frac{h\overline{y}_{1,i_1,r_1}y_{2,i_2,r_1}L_{a_2,i_2}^TCPL_{a_1,i_1}S_{2,j_1}\epsilon_{a_1,a_2}S_{1,j_1}^\dagger\lambda_{12}^{(2)}}{\sqrt{11}M_F}$	
									$-\frac{h\overline{y}_{1,i_1,r_1}y_{2,i_2,r_1}L_{a_2,i_2}^TCPL_{a_1,i_1}S_{2,j_1}\epsilon_{a_1,a_2}S_{1,j_1}^\dagger\lambda_{12}^{(2)}}{\sqrt{11}M_F}$	
cont.									$-\frac{3\sqrt{\frac{2}{3}}hL_{a_1,i_1}CPR\overline{L}_{a_2,i_2}^TS_{1,j_2}y_{1,i_1,r_1}y_{2,i_2,r_1}\mathcal{I}T_{3,a_1,a_2,j_2,j_1}S_{2,j_1}^\dagger\lambda_{12}^{(2)}}{\epsilon M_F}$	
									$-\frac{3\sqrt{\frac{2}{3}}hL_{a_1,i_1}CPR\overline{L}_{a_2,i_2}^TS_{1,j_2}y_{1,i_1,r_1}y_{2,i_2,r_1}\mathcal{I}T_{3,a_1,a_2,j_2,j_1}S_{2,j_1}^\dagger\lambda_{12}^{(2)}}{M_F}$	
									$-\frac{h\overline{L}_{a_1,i_1}CPR\overline{L}_{a_2,i_2}^TS_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger\lambda_{12}^{(2)}}{\sqrt{11}\epsilon M_F}$	
cont.									$-\frac{h\overline{L}_{a_1,i_1}CPR\overline{L}_{a_2,i_2}^TS_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger\lambda_{12}^{(2)}}{\sqrt{11}M_F}$	
									$-\frac{h\overline{L}_{a_1,i_1}CPR\overline{L}_{a_2,i_2}^TS_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger\lambda_{12}^{(2)}}{\sqrt{11}M_F}$	
									$-\frac{h\overline{L}_{a_1,i_1}CPR\overline{L}_{a_2,i_2}^TS_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger\lambda_{12}^{(2)}}{\sqrt{11}M_F}$	
cont.									$-\frac{7h\overline{y}_{1,i_1,r_1}y_{2,i_2,r_1}L_{a_2,i_2}^TCPL_{a_1,i_1}S_{2,j_1}\epsilon_{a_1,a_2}S_{1,j_1}^\dagger\lambda_{12}^{(3)}}{\sqrt{23}\epsilon M_F}$	
									$-\frac{7h\overline{y}_{1,i_1,r_1}y_{2,i_2,r_1}L_{a_2,i_2}^TCPL_{a_1,i_1}S_{2,j_1}\epsilon_{a_1,a_2}S_{1,j_1}^\dagger\lambda_{12}^{(3)}}{\sqrt{23}\epsilon M_F}$	
									$-\frac{7h\overline{y}_{1,i_1,r_1}y_{2,i_2,r_1}L_{a_2,i_2}^TCPL_{a_1,i_1}S_{2,j_1}\epsilon_{a_1,a_2}S_{1,j_1}^\dagger\lambda_{12}^{(3)}}{\sqrt{23}\epsilon M_F}$	
cont.									$-\frac{7h\overline{L}_{a_1,i_1}CPR\overline{L}_{a_2,i_2}^TS_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger\lambda_{12}^{(3)}}{\sqrt{23}\epsilon M_F}$	
									$-\frac{7h\overline{L}_{a_1,i_1}CPR\overline{L}_{a_2,i_2}^TS_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger\lambda_{12}^{(3)}}{\sqrt{23}\epsilon M_F}$	
									$-\frac{7h\overline{L}_{a_1,i_1}CPR\overline{L}_{a_2,i_2}^TS_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger\lambda_{12}^{(3)}}{\sqrt{23}\epsilon M_F}$	
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$-\frac{4}{3}h\log\left(\frac{\hat{\mu}^2}{M_F^2}\right)\overline{y}_{1,i_1,r_1}y_{2,i_2,r_2}S_{1,j_2}S_{2,j_2}y_{1,i_2,r_2}y_{2,i_1,r_2}S_{1,j_1}^\dagger S_{2,j_1}^\dagger$	$-\frac{\sqrt{6}h\overline{y}_{1,i_1,r_1}y_{2,i_2,r_1}L_{a_2,i_2}^TCPL_{a_1,i_1}S_{2,j_2}\mathcal{I}T_{3,a_1,a_2,j_1,j_2}S_{1,j_1}^\dagger g_2^2}{\epsilon M_F}$	$-\frac{2h\log\left(\frac{\hat{a}^2}{M_F^2}\right)\overline{y}_{1,i_1,r_1}Y_{e,i_2,i_3}H_{a_1}S_{1,j_1}y_{1,i_2,r_1}Y_{e,i_1,i_3}H_{a_1}^\dagger S_{1,j_1}^\dagger}{\sqrt{3}}$
								$+\frac{4}{3}h\log\left(\frac{\hat{\mu}^2}{M_F^2}\right)\overline{y}_{1,i_1,r_1}y_{2,i_2,r_2}S_{1,j_2}S_{2,j_1}y_{1,i_2,r_1}y_{2,i_1,r_2}S_{1,j_1}^\dagger S_{2,j_2}^\dagger$	$-\frac{\sqrt{3}h\overline{y}_{1,i_1,r_1}y_{2,i_2,r_1}L_{a_2,i_2}^TCPL_{a_1,i_1}S_{2,j_2}\mathcal{I}T_{3,a_1,a_2,j_1,j_2}S_{1,j_1}^\dagger g_2^2}{M_F}$	$-2i\left(\sqrt{\frac{3}{2}}h\log\left(\frac{\hat{\mu}^2}{M_F^2}\right)\overline{y}_{1,i_1,r_1}Y_{e,i_2,i_3}H_{a_2}S_{1,j_2}y_{1,i_2,r_1}Y_{e,i_1,i_3}H_{a_1}^\dagger S_{1,j_1}^\dagger\mathcal{I}H_{1,a_1,a_2,j_2,j_1}^\dagger\right.$
								$-\frac{4}{3}h\log\left(\frac{\hat{\mu}^2}{M_F^2}\right)\overline{y}_{1,i_1,r_1}y_{2,i_2,r_2}S_{1,j_1}S_{2,j_2}y_{1,i_2,r_1}y_{2,i_1,r_2}S_{1,j_1}^\dagger S_{2,j_2}^\dagger$	$-\frac{2\sqrt{3}h\overline{y}_{1,i_1,r_1}y_{2,i_2,r_1}L_{a_2,i_2}^TCPL_{a_1,i_1}S_{2,j_1}\epsilon_{a_1,a_2}S_{1,j_1}^\dagger g_2^2}{\epsilon M_F}$	$-\frac{2h\log\left(\frac{\hat{a}^2}{M_F^2}\right)\overline{y}_{1,i_1,r_1}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}H_{a_1}^\dagger S_{1,j_1}^\dagger\mathcal{I}H_{1,a_1,a_2,j_2,j_1}^\dagger 2\lambda_{12}^{(2)}}{\sqrt{3}}$
cont.								$-\frac{2h\log\left(\frac{\hat{\mu}^2}{M_F^2}\right)\overline{y}_{1,i_1,r_1}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\mathcal{I}T_{12,j_3,j_1,j_4,j_2}^\dagger\lambda_{12}^{(1)}}{\sqrt{3}}$	$-\frac{2h\overline{y}_{1,i_1,r_1}y_{2,i_2,r_1}L_{a_2,i_2}^TCPL_{a_1,i_1}S_{2,j_1}\epsilon_{a_1,a_2}S_{1,j_1}^\dagger g_2^2}{\sqrt{3}M_F}$	
								$-\frac{2h\log\left(\frac{\hat{\mu}^2}{M_F^2}\right)\overline{y}_{2,i_1,r_1}S_{1,j_3}S_{2,j_4}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\mathcal{I}T_{12,j_3,j_1,j_4,j_2}^\dagger\lambda_{12}^{(1)}}{\sqrt{3}}$	$-\frac{\sqrt{6}h\overline{L}_{a_1,i_1}CPR\overline{L}_{a_2,i_2}^TS_{1,j_2}y_{1,i_1,r_1}y_{2,i_2,r_1}\mathcal{I}T_{3,a_1,a_2,j_2,j_1}S_{2,j_1}^\dagger g_2^2}{\epsilon M_F}$	
								$+\frac{2h\log\left(\frac{\hat{\mu}^2}{M_F^2}\right)\overline{y}_{1,i_1,r_1}S_{1,j_4}S_{2,j_3}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\mathcal{I}T_{12,j_1,j_2,j_3,j_4}^\dagger 2\lambda_{12}^{(2)}}{\sqrt{3}}$	$-\frac{\sqrt{\frac{2}{3}}h\overline{L}_{a_1,i_1}CPR\overline{L}_{a_2,i_2}^TS_{1,j_2}y_{1,i_1,r_1}y_{2,i_2,r_1}\mathcal{I}T_{3,a_1,a_2,j_2,j_1}S_{2,j_1}^\dagger g_2^2}{M_F}$	
cont.								$-\frac{2h\log\left(\frac{\hat{\mu}^2}{M_F^2}\right)\overline{y}_{1,i_1,r_1}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\mathcal{I}T_{12,j_1,j_2,j_3,j_4}^\dagger 2\lambda_{12}^{(2)}}{\sqrt{3}}$	$-\frac{2\sqrt{3}h\overline{L}_{a_1,i_1}CPR\overline{L}_{a_2,i_2}^TS_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger g_2^2}{\epsilon M_F}$	
								$-\frac{2h\log\left(\frac{\hat{\mu}^2}{M_F^2}\right)\overline{y}_{1,i_1,r_1}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\mathcal{I}T_{12,j_1,j_2,j_3,j_4}^\dagger 2\lambda_{12}^{(2)}}{\sqrt{3}}$	$-\frac{2h\overline{L}_{a_1,i_1}CPR\overline{L}_{a_2,i_2}^TS_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger g_2^2}{\sqrt{3}M_F}$	
								$+\frac{2h\log\left(\frac{\hat{\mu}^2}{M_F^2}\right)\overline{y}_{2,i_1,r_1}S_{1,j_4}S_{2,j_3}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\mathcal{I}T_{12,j_1,j_2,j_3,j_4}^\dagger 2\lambda_{12}^{(2)}}{\sqrt{3}}$	$-\frac{\sqrt{\frac{2}{3}}h\overline{y}_{1,i_1,r_1}y_{2,i_2,r_1}L_{a_2,i_2}^TCPL_{a_1,i_1}g_1^2S_{2,j_2}\mathcal{I}T_{3,a_1,a_2,j_1,j_2}S_{1,j_1}^\dagger}{2\epsilon M_F}$	
cont.								$-\frac{2h\log\left(\frac{\hat{\mu}^2}{M_F^2}\right)\overline{y}_{1,i_1,r_1}S_{1,j_4}S_{2,j_3}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\mathcal{I}T_{12,j_1,j_2,j_3,j_4}^\dagger 2\lambda_{12}^{(2)}}{\sqrt{3}}$	$+\frac{h\overline{y}_{1,i_1,r_1}y_{2,i_2,r_1}L_{a_2,i_2}^TCPL_{a_1,i_1}g_1^2S_{2,j_2}\mathcal{I}T_{3,a_1,a_2,j_1,j_2}S_{1,j_1}^\dagger}{2\sqrt{6}M_F}$	
								$-\frac{2h\log\left(\frac{\hat{\mu}^2}{M_F^2}\right)\overline{y}_{2,i_1,r_1}S_{1,j_4}S_{2,j_3}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\mathcal{I}T_{12,j_1,j_2,j_3,j_4}^\dagger 2\lambda_{12}^{(2)}}{\sqrt{3}}$	$+\frac{\sqrt{\frac{2}{3}}h\overline{y}_{1,i_1,r_1}y_{2,i_2,r_1}L_{a_2,i_2}^TCPL_{a_1,i_1}S_{2,j_2}\mathcal{I}T_{3,a_1,a_2,j_1,j_2}S_{1,j_1}^\dagger}{M_F}$	
								$+\frac{2h\log\left(\frac{\hat{\mu}^2}{M_F^2}\right)\overline{y}_{1,i_1,r_1}S_{1,j_4}S_{2,j_3}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\mathcal{I}T_{12,j_1,j_2,j_3,j_4}^\dagger 2\lambda_{12}^{(2)}}{\sqrt{3}}$	$-\frac{h\overline{y}_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_2}L_{a_2,i_2}^TCPL_{a_1,i_1}S_{2,j_2}y_{1,i_3,r_1}\mathcal{I}T_{3,a_1,a_2,j_1,j_2}S_{1,j_1}^\dagger}{2\sqrt{2}\epsilon M_F}$	

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$S_1 S_2 F$	$S_1 S_2 \ell$	$S_1 F H$	
cont.								$-\frac{2h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_1,r_1}} S_{1,j_3} S_{2,j_4} y_{1,i_1,r_1} S_{1,j_1} {}^\dagger S_{2,j_2} {}^\dagger I_{12,j_1,j_3,j_4,j_2} {}^\lambda \Lambda_{12}^{(2)}}{\sqrt{3}}$	$-\frac{3h \overline{y_{1,i_1,r_1}} y_{1,i_3,r_2} \overline{y_{2,i_2,r_1}} L_{a_2,i_2} {}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{1,i_3,r_1} \overline{I_{T3,a_1,a_2,j_1,j_2} S_{1,j_1} }^\dagger}{4\sqrt{2} M_F}$	$-\frac{\sqrt{2} h \overline{y_{1,i_1,r_1}} y_{1,i_3,r_2} \overline{y_{2,i_2,r_1}} L_{a_2,i_2} {}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{1,i_3,r_2} \overline{I_{T3,a_1,a_2,j_1,j_2} S_{1,j_1} }^\dagger}{36 M_F}$	
								$+\frac{2h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{2,i_1,r_1}} S_{1,j_4} S_{2,j_3} y_{2,i_1,r_1} S_{1,j_1} {}^\dagger S_{2,j_2} {}^\dagger I_{12,j_1,j_3,j_4,j_2} {}^\lambda \Lambda_{12}^{(2)}}{\sqrt{3}}$	$-\frac{h \overline{y_{1,i_1,r_1}} y_{1,i_3,r_2} \overline{y_{2,i_2,r_1}} L_{a_2,i_2} {}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{1,i_3,r_2} \overline{I_{T3,a_1,a_2,j_1,j_2} S_{1,j_1} }^\dagger}{3\sqrt{2} M_F}$		
								$-\frac{2h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{2,i_1,r_1}} S_{1,j_3} S_{2,j_4} y_{2,i_1,r_1} S_{1,j_1} {}^\dagger S_{2,j_2} {}^\dagger I_{12,j_1,j_3,j_4,j_2} {}^\lambda \Lambda_{12}^{(2)}}{\sqrt{3}}$	$-\frac{h \overline{y_{1,i_1,r_1}} y_{1,i_2,r_2} \overline{y_{2,i_3,r_1}} L_{a_2,i_2} {}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{1,i_2,r_2} \overline{I_{T3,a_1,a_2,j_1,j_2} S_{1,j_1} }^\dagger}{2\sqrt{2} M_F}$		
cont.								$-\frac{2h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_1,r_1}} S_{1,j_3} S_{2,j_4} y_{1,i_1,r_1} S_{1,j_1} {}^\dagger S_{2,j_2} {}^\dagger I_{12,j_3,j_1,j_4,j_2} {}^\lambda \Lambda_{12}^{(3)}}{\sqrt{3}}$	$-\frac{3h \overline{y_{1,i_1,r_1}} y_{1,i_2,r_2} \overline{y_{2,i_3,r_1}} L_{a_2,i_2} {}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{1,i_2,r_2} \overline{I_{T3,a_1,a_2,j_1,j_2} S_{1,j_1} }^\dagger}{4\sqrt{2} M_F}$	$-\frac{h \overline{y_{1,i_3,r_1}} y_{2,i_1,r_2} \overline{y_{2,i_2,r_1}} L_{a_2,i_2} {}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{2,i_3,r_2} \overline{I_{T3,a_1,a_2,j_1,j_2} S_{1,j_1} }^\dagger}{2\sqrt{2} M_F}$	
								$-\frac{2h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{2,i_1,r_1}} S_{1,j_3} S_{2,j_4} y_{2,i_1,r_1} S_{1,j_1} {}^\dagger S_{2,j_2} {}^\dagger I_{12,j_3,j_1,j_4,j_2} {}^\lambda \Lambda_{12}^{(3)}}{\sqrt{3}}$	$-\frac{3h \overline{y_{1,i_3,r_1}} y_{2,i_1,r_2} \overline{y_{2,i_2,r_1}} L_{a_2,i_2} {}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{2,i_3,r_2} \overline{I_{T3,a_1,a_2,j_1,j_2} S_{1,j_1} }^\dagger}{4\sqrt{2} M_F}$	$-\frac{h \overline{y_{1,i_1,r_1}} y_{2,i_2,r_2} \overline{y_{2,i_3,r_1}} L_{a_2,i_2} {}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{2,i_3,r_2} \overline{I_{T3,a_1,a_2,j_1,j_2} S_{1,j_1} }^\dagger}{2\sqrt{2} M_F}$	
								$-\frac{2h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{2,i_1,r_1}} S_{1,j_3} S_{2,j_4} y_{2,i_1,r_1} S_{1,j_1} {}^\dagger S_{2,j_2} {}^\dagger I_{12,j_3,j_1,j_4,j_2} {}^\lambda \Lambda_{12}^{(3)}}{\sqrt{3}}$	$-\frac{3h \overline{y_{1,i_3,r_1}} y_{2,i_1,r_2} \overline{y_{2,i_2,r_1}} L_{a_2,i_2} {}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{2,i_3,r_2} \overline{I_{T3,a_1,a_2,j_1,j_2} S_{1,j_1} }^\dagger}{4\sqrt{2} M_F}$	$-\frac{h \overline{y_{1,i_1,r_1}} y_{2,i_2,r_2} \overline{y_{2,i_3,r_1}} L_{a_2,i_2} {}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{2,i_3,r_2} \overline{I_{T3,a_1,a_2,j_1,j_2} S_{1,j_1} }^\dagger}{2\sqrt{2} M_F}$	
cont.									$-\frac{h \overline{y_{1,i_1,r_1}} y_{2,i_2,r_2} \overline{y_{2,i_3,r_1}} L_{a_2,i_2} {}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{2,i_3,r_2} \overline{I_{T3,a_1,a_2,j_1,j_2} S_{1,j_1} }^\dagger}{2\sqrt{2} M_F}$	$-\frac{3h \overline{y_{1,i_1,r_1}} y_{2,i_2,r_2} \overline{y_{2,i_3,r_1}} L_{a_2,i_2} {}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{2,i_3,r_2} \overline{I_{T3,a_1,a_2,j_1,j_2} S_{1,j_1} }^\dagger}{4\sqrt{2} M_F}$	$-\frac{\sqrt{2} h \overline{y_{1,i_1,r_1}} y_{2,i_2,r_2} \overline{y_{2,i_3,r_1}} L_{a_2,i_2} {}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{2,i_3,r_2} \overline{I_{T3,a_1,a_2,j_1,j_2} S_{1,j_1} }^\dagger}{36 M_F}$
									$-\frac{h \overline{y_{1,i_1,r_1}} y_{2,i_2,r_2} \overline{y_{2,i_3,r_1}} L_{a_2,i_2} {}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{2,i_3,r_2} \overline{I_{T3,a_1,a_2,j_1,j_2} S_{1,j_1} }^\dagger}{3\sqrt{2} M_F}$	$+\frac{\sqrt{3} h \overline{y_{1,i_1,r_1}} y_{2,i_2,r_2} {}^T C P_L L_{a_1,i_1} g_V^2 S_{2,j_1} \epsilon_{a_1,a_2} S_{1,j_1} }^\dagger}{4\sqrt{3} M_F}$	
									$+\frac{h \overline{y_{1,i_1,r_1}} y_{2,i_2,r_2} {}^T C P_L L_{a_1,i_1} g_V^2 S_{2,j_1} \epsilon_{a_1,a_2} S_{1,j_1} }^\dagger}{4\sqrt{3} M_F}$	$+\frac{\overline{y_{1,i_1,r_1}} y_{2,i_2,r_1} L_{a_2,i_2} {}^T C P_L L_{a_1,i_1} S_{2,j_1} \epsilon_{a_1,a_2} S_{1,j_1} }^\dagger}{\sqrt{3} M_F}$	
cont.									$-\frac{h \overline{y_{1,i_1,r_1}} y_{1,i_3,r_2} \overline{y_{2,i_2,r_1}} L_{a_2,i_2} {}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{1,i_3,r_1} \epsilon_{a_1,a_2} S_{1,j_1} }^\dagger}{4\epsilon M_F}$	$-\frac{3h \overline{y_{1,i_1,r_1}} y_{1,i_3,r_2} \overline{y_{2,i_2,r_1}} L_{a_2,i_2} {}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{1,i_3,r_1} \epsilon_{a_1,a_2} S_{1,j_1} }^\dagger}{8 M_F}$	$-\frac{h \overline{y_{1,i_1,r_1}} y_{1,i_3,r_2} \overline{y_{2,i_2,r_1}} L_{a_2,i_2} {}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{1,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1} }^\dagger}{36 M_F}$
									$-\frac{h \overline{y_{1,i_1,r_1}} y_{1,i_3,r_2} \overline{y_{2,i_2,r_1}} L_{a_2,i_2} {}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{1,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1} }^\dagger}{6 M_F}$	$-\frac{h \overline{y_{1,i_1,r_1}} y_{1,i_3,r_2} \overline{y_{2,i_2,r_1}} L_{a_2,i_2} {}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{1,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1} }^\dagger}{4\epsilon M_F}$	$-\frac{3h \overline{y_{1,i_1,r_1}} y_{1,i_3,r_2} \overline{y_{2,i_2,r_1}} L_{a_2,i_2} {}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{1,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1} }^\dagger}{8 M_F}$
									$-\frac{h \overline{y_{1,i_1,r_1}} y_{1,i_3,r_2} \overline{y_{2,i_2,r_1}} L_{a_2,i_2} {}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{1,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1} }^\dagger}{4\epsilon M_F}$	$-\frac{h \overline{y_{1,i_1,r_1}} y_{1,i_3,r_2} \overline{y_{2,i_2,r_1}} L_{a_2,i_2} {}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{1,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1} }^\dagger}{6 M_F}$	$-\frac{3h \overline{y_{1,i_1,r_1}} y_{1,i_3,r_2} \overline{y_{2,i_2,r_1}} L_{a_2,i_2} {}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{1,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1} }^\dagger}{8 M_F}$
cont.									$-\frac{h \overline{y_{1,i_1,r_1}} y_{1,i_3,r_2} \overline{y_{2,i_2,r_1}} L_{a_2,i_2} {}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{1,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1} }^\dagger}{6 M_F}$	$-\frac{h \overline{y_{1,i_1,r_1}} y_{1,i_3,r_2} \overline{y_{2,i_2,r_1}} L_{a_2,i_2} {}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{1,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1} }^\dagger}{4\epsilon M_F}$	$-\frac{3h \overline{y_{1,i_1,r_1}} y_{1,i_3,r_2} \overline{y_{2,i_2,r_1}} L_{a_2,i_2} {}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{1,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1} }^\dagger}{8 M_F}$
									$-\frac{h \overline{y_{1,i_1,r_1}} y_{1,i_3,r_2} \overline{y_{2,i_2,r_1}} L_{a_2,i_2} {}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{1,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1} }^\dagger}{4\epsilon M_F}$	$-\frac{h \overline{y_{1,i_1,r_1}} y_{1,i_3,r_2} \overline{y_{2,i_2,r_1}} L_{a_2,i_2} {}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{1,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1} }^\dagger}{6 M_F}$	$-\frac{3h \overline{y_{1,i_1,r_1}} y_{1,i_3,r_2} \overline{y_{2,i_2,r_1}} L_{a_2,i_2} {}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{1,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1} }^\dagger}{8 M_F}$
									$-\frac{h \overline{y_{1,i_1,r_1}} y_{1,i_3,r_2} \overline{y_{2,i_2,r_1}} L_{a_2,i_2} {}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{1,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1} }^\dagger}{4\epsilon M_F}$	$-\frac{h \overline{y_{1,i_1,r_1}} y_{1,i_3,r_2} \overline{y_{2,i_2,r_1}} L_{a_2,i_2} {}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{1,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1} }^\dagger}{6 M_F}$	$-\frac{3h \overline{y_{1,i_1,r_1}} y_{1,i_3,r_2} \overline{y_{2,i_2,r_1}} L_{a_2,i_2} {}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{1,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1} }^\dagger}{8 M_F}$
cont.									$-\frac{h \overline{y_{1,i_1,r_1}} y_{1,i_3,r_2} \overline{y_{2,i_2,r_1}} L_{a_2,i_2} {}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{1,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1} }^\dagger}{4\epsilon M_F}$	$-\frac{h \overline{y_{1,i_1,r_1}} y_{1,i_3,r_2} \overline{y_{2,i_2,r_1}} L_{a_2,i_2} {}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{1,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1} }^\dagger}{6 M_F}$	$-\frac{3h \overline{y_{1,i_1,r_1}} y_{1,i_3,r_2} \overline{y_{2,i_2,r_1}} L_{a_2,i_2} {}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{1,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1} }^\dagger}{8 M_F}$
									$-\frac{h \overline{y_{1,i_1,r_1}} y_{1,i_3,r_2} \overline{y_{2,i_2,r_1}} L_{a_2,i_2} {}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{1,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1} }^\dagger}{4\epsilon M_F}$	$-\frac{h \overline{y_{1,i_1,r_1}} y_{1,i_3,r_2} \overline{y_{2,i_2,r_1}} L_{a_2,i_2} {}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{1,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1} }^\dagger}{6 M_F}$	$-\frac{3h \overline{y_{1,i_1,r_1}} y_{1,i_3,r_2} \overline{y_{2,i_2,r_1}} L_{a_2,i_2} {}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{1,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1} }^\dagger}{8 M_F}$
									$-\frac{h \overline{y_{1,i_1,r_1}} y_{1,i_3,r_2} \overline{y_{2,i_2,r_1}} L_{a_2,i_2} {}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{1,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1} }^\dagger}{4\epsilon M_F}$	$-\frac{h \overline{y_{1,i_1,r_1}} y_{1,i_3,r_2} \overline{y_{2,i_2,r_1}} L_{a_2,i_2} {}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{1,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1} }^\dagger}{6 M_F}$	$-\frac{3h \overline{y_{1,i_1,r_1}} y_{1,i_3,r_2} \overline{y_{2,i_2,r_1}} L_{a_2,i_2} {}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{1,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1} }^\dagger}{8 M_F}$
cont.									$-\frac{h \overline{y_{1,i_1,r_1}} y_{1,i_3,r_2} \overline{y_{2,i_2,r_1}} L_{a_2,i_2} {}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{1,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1} }^\dagger}{4\epsilon M_F}$	$-\frac{h \overline{y_{1,i_1,r_1}} y_{1,i_3,r_2} \overline{y_{2,i_2,r_1}} L_{a_2,i_2} {}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{1,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1} }^\dagger}{6 M_F}$	$-\frac{3h \overline{y_{1,i_1,r_1}} y_{1,i_3,r_2} \overline{y_{2,i_2,r_1}} L_{a_2,i_2} {}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{1,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1} }^\dagger}{8 M_F}$
									$-\frac{h \overline{y_{1,i_1,r_1}} y_{1,i_3,r_2} \overline{y_{2,i_2,r_1}} L_{a_2,i_2} {}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{1,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1} }^\dagger}{4\epsilon M_F}$	$-\frac{h \overline{y_{1,i_1,r_1}} y_{1,i_3,r_2} \overline{y_{2,i_2,r_1}} L_{a_2,i_2} {}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{1,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1} }^\dagger}{6 M_F}$	$-\frac{3h \overline{y_{1,i_1,r_1}} y_{1,i_3,r_2} \overline{y_{2,i_2,r_1}} L_{a_2,i_2} {}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{1,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1} }^\dagger}{8 M_F}$
									$-\frac{h \overline{y_{1,i_1,r_1}} y_{1,i_3,r_2} \overline{y_{2,i_2,r_1}} L_{a_2,i_2} {}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{1,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1} }^\dagger}{4\epsilon M_F}$	$-\frac{h \overline{y_{1,i_1,r_1}} y_{1,i_3,r_2} \overline{y_{2,i_2,r_1}} L_{a_2,i_2} {}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{1,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1} }^\dagger}{6 M_F}$	$-\frac{3h \overline{y_{1,i_1,r_1}} y_{1,i_3,r_2} \overline{y_{2,i_2,r_1}} L_{a_2,i_2} {}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{1,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1} }^\dagger}{8 M_F}$
cont.									$-\frac{h \overline{y_{1,i_1,r_1}} y_{1,i_3,r_2} \overline{y_{2,i_2,r_1}} L_{a_2,i_2} {}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{1,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1} }^\dagger}{4\epsilon M_F}$	$-\frac{h \overline{y_{1,i_1,r_1}} y_{1,i_3,r_2} \overline{y_{2,i_2,r_1}} L_{a_2,i_2} {}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{1,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1} }^\dagger}{6 M_F}$	$-\frac{3h \overline{y_{1,i_1,r_1}} y_{1,i_3,r_2} \overline{y_{2,i_2,r_1}} L_{a_2,i_2} {}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{1,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1} }^\dagger}{8 M_F}$
									$-\frac{h \overline{y_{1,i_1,r_1}} y_{1,i_3,r_2} \overline{y_{2,i_2,r_1}} L_{a_2,i_2} {}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{1,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1} }^\dagger}{4\epsilon M_F}$	$-\frac{h \overline{y_{1,i_1,r_1}} y_{1,i_3,r_2} \overline{y_{2,i_2,r_1}} L_{a_2,i_2} {}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{1,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1} }^\dagger}{6 M_F}$	$-\frac{3h \overline{y_{1,i_1,r_1}} y_{1,i_3,r_2} \overline{y_{2,i_2,r_1}} L_{a_2,i_2} {}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{1,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1} }^\dagger}{8 M_F}$
									$-\frac{h \overline{y_{1,i_1,r_1}} y_{1,i_3,r_2} \overline{y_{2,i_2,r_1}} L_{a_2,i_2} {}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{1,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1} }^\dagger}{4\epsilon M_F}$	$-\frac{h \overline{y_{1,i_1,r_1}} y_{1,i_3,r_2} \overline{y_{2,i_2,r_1}} L_{a_2,i_2} {}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{1,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1} }^\dagger}{6 M_F}$	$-\frac{3h \overline{y_{1,i_1,r_1}} y_{1,i_3,r_2} \overline{y_{2,i_2,r_1}} L_{a_2,i_2} {}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{1,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1} }^\dagger}{8 M_F}$
cont.									$-\frac{h \overline{y_{1,i_1,r_1}} y_{1,i_3,r_2} \overline{y_{2,i_2,r_1}} L_{a_2,i_2} {}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{1,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1} }^\dagger}{4\epsilon M_F}$	$-\frac{h \overline{y_{1,i_1,r_1}} y_{1,i_3,r_2} \overline{y_{2,i_2,r_1}} L_{a_2,i_2} {}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{1,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1} }^\dagger}{6 M_F}$	$-\frac{3h \overline{y_{1,i_1,r_1}} y_{1,i_3,r_2} \overline{y_{2,i_2,r_1}} L_{a_2,i_2} {}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{1,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1} }^\dagger}{8 M_F}$
									$-\frac{h \overline{y_{1,i_1,r_1}} y_{1,i_3,r_2} \overline{y_{2,i_2,r_1}} L_{a_2,i_2} {}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{1,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1} }^\dagger}{4\epsilon M_F}$	$-\frac{h \overline{y_{1,i_1,r_1}} y_{1,i_3,r_2} \overline{y_{2,i_2,r_1}} L_{a_2,i_2} {}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{1,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1} }^\dagger}{6 M_F}$	$-\frac{3h \overline{y_{1,i_1,r_1}} y_{1,i_3,r_2} \overline{y_{2,i_2,r_1}} L_{a_2,i_2} {}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{1,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1} }^\dagger}{8 M_F}$
									$-\frac{h \overline{y_{1,i_1,r_1}} y_{1,i_3,r_2} \overline{y_{2,i_2,r_1}} L_{a_2,i_2} {}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{1,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1} }^\dagger}{4\epsilon M_F}$	$-\frac{h \overline{y_{1,i_1,r_1}} y_{1,i_3,r_2} \overline{y_{2,i_2,r_1}} L_{a_2,i_2} {}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{1,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1} }^\dagger}{6 M_F}$	$-\frac{3h \overline{y_{1,i_1,r_1}} y_{1,i_3,r_2} \overline{y_{2,i_2,r_1}} L_{a_2,i_2} {}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{1,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1} }^\dagger}{8 M_F}$
cont.									$-\frac{h \overline{y_{1,i_1,r_1}} y_{1,i_3,r_2} \overline{y_{2,i_2,r_1}} L_{a_2,i_2} {}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{1,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1} }^\dagger}{4\epsilon M_F}$	$-\frac{h \overline{y_{1,i_1,r_1}} y_{1,i_3,r_2} \overline{y_{2,i_2,r_1}} L_{a_2,i_2} {}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{1,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1} }^\dagger}{6 M_F}$	$-\frac{3h \overline{y_{1,i_1,r_1}} y_{1,i_3,r_2} \overline{y_{2,i_2,r_1}} L_{a_2,i_2} {}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{1,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1} }^\dagger}{8 M_F}$
									$-\frac{h \overline{y_{1,i_1,r_1}} y_{1,i_3,r_2} \overline{y_{2,i_2,r_1}} L_{a_2,i_2} {}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{1,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1} }^\dagger}{4\epsilon M_F}$	$-\frac{h \overline{y_{1,i_1,r_1}} y_{1,i_3,r_2} \overline{y_{2,i_2,r_1}} L_{a_2,i_2} {}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{1,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1} }^\dagger}{6 M_F}$	$-\frac{3h \overline{y_{1,i_1,r_1}} y_{1,i_3,r_2} \overline{y_{2,i_2,r_1}} L_{a_2,i_2} {}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{1,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1} }^\dagger}{8 M_F}$
									$-\frac{h \overline{y_{1,i_1,r_1}} y_{1,i_3,r_2} \overline{y_{2,i_2,r_1}} L_{a_2,i_2} {}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{1,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1} }^\dagger}{4\epsilon M_F}$	$-\frac{h \overline{y_{1,i_1,r_1}} y_{1,i_3,r_2} \overline{y_{2,i_2,r_1}} L_{a_2,i_2} {}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{1,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1} }^\dagger}{6 M_F}$	$-\frac{3h \overline{y_{1,i_1,r_1}} y_{1,i_3,r_2} \overline{y_{2,i_2,r_1}} L_{a_2,i_2} {}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{1,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1} }^\dagger}{8 M_F}$
cont.									$-\frac{h \overline{y_{1,i_1,r_1}} y_{1,i_3,r_2} \overline{y_{2,i_2,r_1}} L_{a_2,i_2} {}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{1,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1} }^\dagger}{4\epsilon M_F}$	$-\frac{h \overline{y_{1,i_1,r_1}} y_{1,i_3,r_2} \overline{y_{2,i_2,r_1}} L_{a_2,i_2} {}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{1,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1} }^\dagger}{6 M_F}$	$-\frac{3h \overline{y_{1,i_1,r_1}} y_{1,i_3,r_2} \overline{y_{2,i_2,r_1}} L_{a_2,i_2} {}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{1,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1} }^\dagger}{8 M_F}$
									$-\frac{h \overline{y_{1,i_1,r_1}} y$		

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	S_1	S_2	F	S_1	S_2	ℓ	S_1	F	H
													$+ \frac{3h\overline{y}_{1,i_3,r_1}\overline{L}_{a_1,i_1}}{8M_F}CPR\overline{L}_{a_2,i_2}^TS_{1,j_1}y_{1,i_2,r_1}y_{1,i_3,r_2}y_{2,i_2,i_1,r_2}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger$			
cont.													$+ \frac{\sqrt{3}h\overline{L}_{a_1,i_1}}{4\epsilon M_F}CPR\overline{L}_{a_2,i_2}^Tg_2^2S_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger$			
													$+ \frac{h\overline{L}_{a_1,i_1}}{4\sqrt{3}M_F}CPR\overline{L}_{a_2,i_2}^Tg_2^2S_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger$			
													$+ \frac{\overline{L}_{a_1,i_1}}{\sqrt{3}M_F}CPR\overline{L}_{a_2,i_2}^TS_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger$			
cont.													$+ \frac{h\overline{y}_{2,i_3,r_1}\overline{L}_{a_1,i_1}}{4\epsilon M_F}CPR\overline{L}_{a_2,i_2}^TS_{1,j_1}y_{1,i_3,r_2}y_{2,i_1,r_2}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger$			
													$+ \frac{3h\overline{y}_{2,i_3,r_1}\overline{L}_{a_1,i_1}}{8M_F}CPR\overline{L}_{a_2,i_2}^TS_{1,j_1}y_{1,i_3,r_2}y_{2,i_1,r_2}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger$			
cont.													$- \frac{h\overline{y}_{1,i_3,r_1}\overline{L}_{a_1,i_1}}{3\epsilon M_F}CPR\overline{L}_{a_2,i_2}^TS_{1,j_1}y_{1,i_1,r_2}y_{1,i_3,r_1}y_{2,i_2,r_2}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger$			
													$- \frac{h\overline{y}_{1,i_3,r_1}\overline{L}_{a_1,i_1}}{6M_F}CPR\overline{L}_{a_2,i_2}^TS_{1,j_1}y_{1,i_1,r_2}y_{1,i_3,r_1}y_{2,i_2,r_2}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger$			
													$- \frac{h\overline{y}_{2,i_3,r_1}\overline{L}_{a_1,i_1}}{3\epsilon M_F}CPR\overline{L}_{a_2,i_2}^TS_{1,j_1}y_{1,i_1,r_2}y_{2,i_2,r_2}y_{2,i_3,r_1}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger$			
cont.													$- \frac{h\overline{y}_{2,i_3,r_1}\overline{L}_{a_1,i_1}}{6M_F}CPR\overline{L}_{a_2,i_2}^TS_{1,j_1}y_{1,i_1,r_2}y_{2,i_2,r_2}y_{2,i_3,r_1}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger$			
													$- \frac{h\overline{y}_{1,i_3,r_1}\overline{L}_{a_1,i_1}}{4\epsilon M_F}CPR\overline{L}_{a_2,i_2}^TS_{1,j_1}y_{1,i_1,r_2}y_{1,i_2,r_1}y_{2,i_3,r_2}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger$			
													$- \frac{3h\overline{y}_{1,i_3,r_1}\overline{L}_{a_1,i_1}}{8M_F}CPR\overline{L}_{a_2,i_2}^TS_{1,j_1}y_{1,i_1,r_2}y_{1,i_2,r_1}y_{2,i_3,r_2}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger$			
cont.													$- \frac{h\overline{y}_{2,i_3,r_1}\overline{L}_{a_1,i_1}}{4\epsilon M_F}CPR\overline{L}_{a_2,i_2}^TS_{1,j_1}y_{1,i_1,r_2}y_{2,i_2,r_1}y_{2,i_3,r_2}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger$			
													$- \frac{3h\overline{y}_{2,i_3,r_1}\overline{L}_{a_1,i_1}}{8M_F}CPR\overline{L}_{a_2,i_2}^TS_{1,j_1}y_{1,i_1,r_2}y_{2,i_2,r_1}y_{2,i_3,r_2}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger$			
													$+ \sqrt{\frac{2}{3}}\frac{h\overline{y}_{1,i_1,r_1}\overline{y}_{2,i_2,r_1}}{\epsilon M_F}L_{a_1,i_1}S_{2,j_2}\mathcal{I}T_{3,a_1,a_2,j_1,j_2}S_{1,j_1}^\dagger\lambda_{12}^{(1)}$			
cont.													$+ \sqrt{\frac{2}{3}}\frac{h\overline{y}_{1,i_1,r_1}\overline{y}_{2,i_2,r_1}}{M_F}L_{a_1,i_1}S_{2,j_2}\mathcal{I}T_{3,a_1,a_2,j_1,j_2}S_{1,j_1}^\dagger\lambda_{12}^{(1)}$			
													$- \frac{h\overline{y}_{1,i_1,r_1}\overline{y}_{2,i_2,r_1}}{\sqrt{3}\epsilon M_F}L_{a_2,i_2}^TCPL_{a_1,i_1}S_{2,j_1}\epsilon_{a_1,a_2}S_{1,j_1}^\dagger\lambda_{12}^{(1)}$			
													$- \frac{h\overline{y}_{1,i_1,r_1}\overline{y}_{2,i_2,r_1}}{\sqrt{3}M_F}L_{a_2,i_2}^TCPL_{a_1,i_1}S_{2,j_1}\epsilon_{a_1,a_2}S_{1,j_1}^\dagger\lambda_{12}^{(1)}$			
cont.													$+ \sqrt{\frac{2}{3}}\frac{h\overline{L}_{a_1,i_1}}{\epsilon M_F}CPR\overline{L}_{a_2,i_2}^TS_{1,j_2}y_{1,i_1,r_1}y_{2,i_2,r_1}\mathcal{I}T_{3,a_1,a_2,j_2,j_1}^\dagger\lambda_{12}^{(1)}$			
													$+ \sqrt{\frac{2}{3}}\frac{h\overline{L}_{a_1,i_1}}{M_F}CPR\overline{L}_{a_2,i_2}^TS_{1,j_2}y_{1,i_1,r_1}y_{2,i_2,r_1}\mathcal{I}T_{3,a_1,a_2,j_2,j_1}^\dagger\lambda_{12}^{(1)}$			
													$- \frac{h\overline{L}_{a_1,i_1}}{\sqrt{3}\epsilon M_F}CPR\overline{L}_{a_2,i_2}^TS_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger\lambda_{12}^{(1)}$			
cont.													$- \frac{h\overline{L}_{a_1,i_1}}{\sqrt{3}M_F}CPR\overline{L}_{a_2,i_2}^TS_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger\lambda_{12}^{(1)}$			
													$- \frac{h\overline{L}_{a_1,i_1}}{\sqrt{3}M_F}CPR\overline{L}_{a_2,i_2}^TS_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger\lambda_{12}^{(1)}$			
cont.													$- \frac{h\overline{L}_{a_1,i_1}}{\sqrt{3}M_F}CPR\overline{L}_{a_2,i_2}^TS_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger\lambda_{12}^{(1)}$			
													$- \frac{h\overline{L}_{a_1,i_1}}{\sqrt{3}M_F}CPR\overline{L}_{a_2,i_2}^TS_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger\lambda_{12}^{(1)}$			
cont.													$- \frac{h\overline{L}_{a_1,i_1}}{\sqrt{3}M_F}CPR\overline{L}_{a_2,i_2}^TS_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger\lambda_{12}^{(1)}$			
													$- \frac{h\overline{L}_{a_1,i_1}}{\sqrt{3}M_F}CPR\overline{L}_{a_2,i_2}^TS_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger\lambda_{12}^{(1)}$			
cont.													$- \frac{h\overline{L}_{a_1,i_1}}{\sqrt{3}M_F}CPR\overline{L}_{a_2,i_2}^TS_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger\lambda_{12}^{(1)}$			
													$- \frac{h\overline{L}_{a_1,i_1}}{\sqrt{3}M_F}CPR\overline{L}_{a_2,i_2}^TS_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger\lambda_{12}^{(1)}$			
cont.													$- \frac{h\overline{L}_{a_1,i_1}}{\sqrt{3}M_F}CPR\overline{L}_{a_2,i_2}^TS_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger\lambda_{12}^{(1)}$			
													$- \frac{h\overline{L}_{a_1,i_1}}{\sqrt{3}M_F}CPR\overline{L}_{a_2,i_2}^TS_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger\lambda_{12}^{(1)}$			
cont.													$- \frac{h\overline{L}_{a_1,i_1}}{\sqrt{3}M_F}CPR\overline{L}_{a_2,i_2}^TS_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger\lambda_{12}^{(1)}$			
													$- \frac{h\overline{L}_{a_1,i_1}}{\sqrt{3}M_F}CPR\overline{L}_{a_2,i_2}^TS_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger\lambda_{12}^{(1)}$			
cont.													$- \frac{h\overline{L}_{a_1,i_1}}{\sqrt{3}M_F}CPR\overline{L}_{a_2,i_2}^TS_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger\lambda_{12}^{(1)}$			
													$- \frac{h\overline{L}_{a_1,i_1}}{\sqrt{3}M_F}CPR\overline{L}_{a_2,i_2}^TS_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger\lambda_{12}^{(1)}$			
cont.													$- \frac{h\overline{L}_{a_1,i_1}}{\sqrt{3}M_F}CPR\overline{L}_{a_2,i_2}^TS_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger\lambda_{12}^{(1)}$			
													$- \frac{h\overline{L}_{a_1,i_1}}{\sqrt{3}M_F}CPR\overline{L}_{a_2,i_2}^TS_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger\lambda_{12}^{(1)}$			
cont.													$- \frac{h\overline{L}_{a_1,i_1}}{\sqrt{3}M_F}CPR\overline{L}_{a_2,i_2}^TS_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger\lambda_{12}^{(1)}$			
													$- \frac{h\overline{L}_{a_1,i_1}}{\sqrt{3}M_F}CPR\overline{L}_{a_2,i_2}^TS_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger\lambda_{12}^{(1)}$			
cont.													$- \frac{h\overline{L}_{a_1,i_1}}{\sqrt{3}M_F}CPR\overline{L}_{a_2,i_2}^TS_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger\lambda_{12}^{(1)}$			
													$- \frac{h\overline{L}_{a_1,i_1}}{\sqrt{3}M_F}CPR\overline{L}_{a_2,i_2}^TS_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger\lambda_{12}^{(1)}$			
cont.													$- \frac{h\overline{L}_{a_1,i_1}}{\sqrt{3}M_F}CPR\overline{L}_{a_2,i_2}^TS_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger\lambda_{12}^{(1)}$			
													$- \frac{h\overline{L}_{a_1,i_1}}{\sqrt{3}M_F}CPR\overline{L}_{a_2,i_2}^TS_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger\lambda_{12}^{(1)}$			
cont.													$- \frac{h\overline{L}_{a_1,i_1}}{\sqrt{3}M_F}CPR\overline{L}_{a_2,i_2}^TS_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger\lambda_{12}^{(1)}$			
													$- \frac{h\overline{L}_{a_1,i_1}}{\sqrt{3}M_F}CPR\overline{L}_{a_2,i_2}^TS_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger\lambda_{12}^{(1)}$			
cont.													$- \frac{h\overline{L}_{a_1,i_1}}{\sqrt{3}M_F}CPR\overline{L}_{a_2,i_2}^TS_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger\lambda_{12}^{(1)}$			
													$- \frac{h\overline{L}_{a_1,i_1}}{\sqrt{3}M_F}CPR\overline{L}_{a_2,i_2}^TS_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger\lambda_{12}^{(1)}$			
cont.													$- \frac{h\overline{L}_{a_1,i_1}}{\sqrt{3}M_F}CPR\overline{L}_{a_2,i_2}^TS_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger\lambda_{12}^{(1)}$			
													$- \frac{h\overline{L}_{a_1,i_1}}{\sqrt{3}M_F}CPR\overline{L}_{a_2,i_2}^TS_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger\lambda_{12}^{(1)}$			
cont.													$- \frac{h\overline{L}_{a_1,i_1}}{\sqrt{3}M_F}CPR\overline{L}_{a_2,i_2}^TS_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger\lambda_{12}^{(1)}$			
													$- \frac{h\overline{L}_{a_1,i_1}}{\sqrt{3}M_F}CPR\overline{L}_{a_2,i_2}^TS_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger\lambda_{12}^{(1)}$			
cont.													$- \frac{h\overline{L}_{a_1,i_1}}{\sqrt{3}M_F}CPR\overline{L}_{a_2,i_2}^TS_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger\lambda_{12}^{(1)}$			
													$- \frac{h\overline{L}_{a_1,i_1}}{\sqrt{3}M_F}CPR\overline{L}_{a_2,i_2}^TS_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger\lambda_{12}^{(1)}$			
cont.													$- \frac{h\overline{L}_{a_1,i_1}}{\sqrt{3}M_F}CPR\overline{L}_{a_2,i_2}^TS_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger\lambda_{12}^{(1)}$			
													$- \frac{h\overline{L}_{a_1,i_1}}{\sqrt{3}M_F}CPR\overline{L}_{a_2,i_2}^TS_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger\lambda_{12}^{(1)}$			
cont.													$- \frac{h\overline{L}_{a_1,i_1}}{\sqrt{3}M_F}CPR\overline{L}_{a_2,i_2}^TS_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger\lambda_{12}^{(1)}$			
													$- \frac{h\overline{L}_{a_1,i_1}}{\sqrt{3}M_F}CPR\overline{L}_{a_2,i_2}^TS_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger\lambda_{12}^{(1)}$			
cont.													$- \frac{h\overline{L}_{a_1,i_1}}{\sqrt{3}M_F}CPR\overline{L}_{a_2,i_2}^TS_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger\lambda_{12}^{(1)}$			
													$- \frac{h\overline{L}_{a_1,i_1}}{\sqrt{3}M_F}CPR\overline{L}_{a_2,i_2}^TS_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger\lambda_{12}^{(1)}$			
cont.													$- \frac{h\overline{L}_{a_1,i_1}}{\sqrt{3}M_F}CPR\overline{L}_{a_2,i_2}^TS_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger\lambda_{12}^{(1)}$			
													$- \frac{h\overline{L}_{a_1,i_1}}{\sqrt{3}M_F}CPR\overline{L}_{a_2,i_2}^TS_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger\lambda_{12}^{(1)}$			
cont.													$- \frac{h\overline{L}_{a_1,i_1}}{\sqrt{3}M_F}CPR\overline{L}_{a_2,i_2}^TS_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger\lambda_{12}^{(1)}$			
													$- \frac{h\overline{L}_{a_1,i_1}}{\sqrt{3}M_F}CPR\overline{L}_{a_2,i_2}^TS_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger\lambda_{12}^{(1)}$			
cont.													$- \frac{h\overline{L}_{a_1,i_1}}{\sqrt{3}M_F}CPR\overline{L}_{a_2,i_2}^TS_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger\lambda_{12}^{(1)}$			
													$- \frac{h\overline{L}_{a_1,i_1}}{\sqrt{3}M_F}CPR\overline{L}_{a_2,i_2}^TS_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger\lambda_{12}^{(1)}$			
cont.													$- \frac{h\overline{L}_{a_1,i_1}}{\sqrt{3}M_F}CPR\overline{L}_{a_2,i_2}^TS_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger\lambda_{12}^{(1)}$			
													$- \frac{h\overline{L}_{a_1,i_1}}{\sqrt{3}M_F}CPR\overline{L}_{a_2,i_2}^TS_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger\lambda_{12}^{(1)}$			
cont.													$- \frac{h\overline{L}_{a_1,i_1}}{\sqrt{3}M_F}CPR\overline{L}_{a_2,i_2}^TS_{1,j_1}y_{1,i_1,r_1}$			

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$S_1 S_2 F$	$S_1 S_2 \ell$	$S_1 F H$
cont.								$-\frac{2h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_1,r_1}} S_{1,j_3} S_{2,j_4} y_{1,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger I_{12,j_3,j_1,j_4,j_2} \lambda_{12}^{(1)}}{\sqrt{3}}$	$-\frac{h \overline{y_{1,i_1,r_1}} y_{1,i_3,r_2} \overline{y_{2,i_2,r_2}} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{1,i_3,r_1} \overline{I_{T3,a_1,a_2,j_1,j_2}} S_{1,j_1}^\dagger}{2\sqrt{2} M_F}$	
								$-\frac{2h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{2,i_1,r_1}} S_{1,j_3} S_{2,j_4} y_{2,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger I_{12,j_3,j_1,j_4,j_2} \lambda_{12}^{(1)}}{\sqrt{3}}$	$-\frac{3h \overline{y_{1,i_1,r_1}} y_{1,i_3,r_2} \overline{y_{2,i_2,r_2}} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{1,i_3,r_1} \overline{I_{T3,a_1,a_2,j_1,j_2}} S_{1,j_1}^\dagger}{4\sqrt{2} M_F}$	
								$+\frac{2h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_1,r_1}} S_{1,j_4} S_{2,j_3} y_{1,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger I_{12,j_1,j_2,j_3,j_4} \lambda_{12}^{(2)}}{\sqrt{3}}$	$-\frac{\sqrt{2} h \overline{y_{1,i_1,r_1}} y_{1,i_3,r_2} \overline{y_{2,i_2,r_1}} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{1,i_3,r_2} \overline{I_{T3,a_1,a_2,j_1,j_2}} S_{1,j_1}^\dagger}{3\epsilon M_F}$	
cont.								$-\frac{2h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_1,r_1}} S_{1,j_3} S_{2,j_4} y_{1,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger I_{12,j_1,j_2,j_3,j_4} \lambda_{12}^{(2)}}{\sqrt{3}}$	$-\frac{h \overline{y_{1,i_1,r_1}} y_{1,i_3,r_2} \overline{y_{2,i_2,r_1}} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{1,i_3,r_2} \overline{I_{T3,a_1,a_2,j_1,j_2}} S_{1,j_1}^\dagger}{3\sqrt{2} M_F}$	
								$+\frac{2h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{2,i_1,r_1}} S_{1,j_3} S_{2,j_3} y_{2,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger I_{12,j_1,j_2,j_3,j_4} \lambda_{12}^{(2)}}{\sqrt{3}}$	$-\frac{h \overline{y_{1,i_1,r_1}} y_{1,i_3,r_2} \overline{y_{2,i_3,r_1}} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{1,i_3,r_2} \overline{I_{T3,a_1,a_2,j_1,j_2}} S_{1,j_1}^\dagger}{2\sqrt{2} \epsilon M_F}$	
								$-\frac{2h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{2,i_1,r_1}} S_{1,j_3} S_{2,j_4} y_{2,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger I_{12,j_1,j_2,j_3,j_4} \lambda_{12}^{(2)}}{\sqrt{3}}$	$-\frac{3h \overline{y_{1,i_1,r_1}} y_{1,i_2,r_2} \overline{y_{2,i_3,r_1}} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{1,i_3,r_2} \overline{I_{T3,a_1,a_2,j_1,j_2}} S_{1,j_1}^\dagger}{4\sqrt{2} M_F}$	
cont.								$-\frac{2h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_1,r_1}} S_{1,j_4} S_{2,j_3} y_{1,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger I_{12,j_1,j_2,j_3,j_4} \lambda_{12}^{(2)}}{\sqrt{3}}$	$-\frac{h \overline{y_{1,i_3,r_1}} y_{2,i_1,r_2} \overline{y_{2,i_2,r_1}} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{2,i_3,r_2} \overline{I_{T3,a_1,a_2,j_1,j_2}} S_{1,j_1}^\dagger}{2\sqrt{2} \epsilon M_F}$	
								$+\frac{2h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{2,i_1,r_1}} S_{1,j_4} S_{2,j_3} y_{2,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger I_{12,j_1,j_2,j_3,j_4} \lambda_{12}^{(2)}}{\sqrt{3}}$	$-\frac{3h \overline{y_{1,i_3,r_1}} y_{2,i_1,r_2} \overline{y_{2,i_2,r_1}} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{2,i_3,r_2} \overline{I_{T3,a_1,a_2,j_1,j_2}} S_{1,j_1}^\dagger}{4\sqrt{2} M_F}$	
								$-\frac{2h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_1,r_1}} S_{1,j_3} S_{2,j_4} y_{1,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger I_{12,j_1,j_2,j_3,j_4} \lambda_{12}^{(2)}}{\sqrt{3}}$	$-\frac{h \overline{y_{1,i_1,r_1}} y_{2,i_2,r_2} \overline{y_{2,i_3,r_1}} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{2,i_3,r_2} \overline{I_{T3,a_1,a_2,j_1,j_2}} S_{1,j_1}^\dagger}{2\sqrt{2} \epsilon M_F}$	
cont.								$+\frac{2h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_1,r_1}} S_{1,j_4} S_{2,j_3} y_{1,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger I_{12,j_1,j_2,j_3,j_4} \lambda_{12}^{(2)}}{\sqrt{3}}$	$-\frac{3h \overline{y_{1,i_1,r_1}} y_{2,i_2,r_2} \overline{y_{2,i_3,r_1}} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{2,i_3,r_2} \overline{I_{T3,a_1,a_2,j_1,j_2}} S_{1,j_1}^\dagger}{4\sqrt{2} M_F}$	
								$-\frac{2h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{2,i_1,r_1}} S_{1,j_3} S_{2,j_4} y_{2,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger I_{12,j_1,j_2,j_3,j_4} \lambda_{12}^{(2)}}{\sqrt{3}}$	$-\frac{\sqrt{2} h \overline{y_{1,i_1,r_1}} y_{2,i_2,r_2} \overline{y_{2,i_3,r_1}} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{2,i_3,r_2} \overline{I_{T3,a_1,a_2,j_1,j_2}} S_{1,j_1}^\dagger}{3\epsilon M_F}$	
								$+\frac{2h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{2,i_1,r_1}} S_{1,j_3} S_{2,j_4} y_{2,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger I_{12,j_1,j_2,j_3,j_4} \lambda_{12}^{(2)}}{\sqrt{3}}$	$-\frac{h \overline{y_{1,i_1,r_1}} y_{2,i_2,r_2} \overline{y_{2,i_3,r_1}} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{2,i_3,r_2} \overline{I_{T3,a_1,a_2,j_1,j_2}} S_{1,j_1}^\dagger}{3\sqrt{2} M_F}$	
cont.								$-\frac{2h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_1,r_1}} S_{1,j_3} S_{2,j_4} y_{1,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger I_{12,j_3,j_1,j_4,j_2} \lambda_{12}^{(3)}}{\sqrt{3}}$	$-\frac{2\sqrt{3} h \overline{y_{1,i_1,r_1}} y_{2,i_2,r_2} \overline{y_{2,i_3,r_1}} L_{a_2,i_2}^T C P_L L_{a_1,i_1} y_{1,i_2}^2 S_{2,j_1} \epsilon_{a_1,a_2} S_{1,j_1}^\dagger}{\epsilon M_F}$	
cont.								$-\frac{2h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{2,i_1,r_1}} S_{1,j_3} S_{2,j_4} y_{2,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger I_{12,j_3,j_1,j_4,j_2} \lambda_{12}^{(3)}}{\sqrt{3}}$	$-\frac{2h \overline{y_{1,i_1,r_1}} y_{2,i_2,r_2} \overline{y_{2,i_3,r_1}} L_{a_2,i_2}^T C P_L L_{a_1,i_1} y_{1,i_2}^2 S_{2,j_1} \epsilon_{a_1,a_2} S_{1,j_1}^\dagger}{\sqrt{3} M_F}$	
								$+\frac{\overline{y_{1,i_1,r_1}} y_{2,i_2,r_2} \overline{y_{2,i_3,r_1}} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_1} \epsilon_{a_1,a_2} S_{1,j_1}^\dagger}{\sqrt{3} M_F}$	$-\frac{h \overline{y_{1,i_1,r_1}} y_{1,i_3,r_2} \overline{y_{2,i_2,r_2}} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{1,i_3,r_1} \epsilon_{a_1,a_2} S_{1,j_1}^\dagger}{4\epsilon M_F}$	
								$-\frac{3h \overline{y_{1,i_1,r_1}} y_{1,i_3,r_2} \overline{y_{2,i_2,r_2}} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{1,i_3,r_1} \epsilon_{a_1,a_2} S_{1,j_1}^\dagger}{8 M_F}$	$-\frac{h \overline{y_{1,i_1,r_1}} y_{1,i_3,r_2} \overline{y_{2,i_2,r_1}} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{1,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}^\dagger}{4\epsilon M_F}$	
cont.								$-\frac{h \overline{y_{1,i_1,r_1}} y_{1,i_3,r_2} \overline{y_{2,i_2,r_1}} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{1,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}^\dagger}{6 M_F}$	$-\frac{h \overline{y_{1,i_1,r_1}} y_{1,i_2,r_2} \overline{y_{2,i_3,r_1}} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{1,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}^\dagger}{4\epsilon M_F}$	
								$+\frac{3h \overline{y_{1,i_1,r_1}} y_{1,i_2,r_2} \overline{y_{2,i_3,r_1}} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{1,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}^\dagger}{8 M_F}$	$-\frac{h \overline{y_{1,i_3,r_1}} y_{2,i_1,r_2} \overline{y_{2,i_2,r_1}} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}^\dagger}{4\epsilon M_F}$	
								$-\frac{h \overline{y_{1,i_1,r_1}} y_{2,i_1,r_2} \overline{y_{2,i_3,r_1}} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}^\dagger}{4\epsilon M_F}$	$-\frac{3h \overline{y_{1,i_3,r_1}} y_{2,i_1,r_2} \overline{y_{2,i_2,r_1}} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}^\dagger}{8 M_F}$	
cont.								$-\frac{h \overline{y_{1,i_1,r_1}} y_{2,i_2,r_2} \overline{y_{2,i_3,r_1}} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}^\dagger}{3\epsilon M_F}$	$-\frac{h \overline{y_{1,i_1,r_1}} y_{2,i_2,r_2} \overline{y_{2,i_3,r_1}} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}^\dagger}{4\epsilon M_F}$	
								$+\frac{3h \overline{y_{1,i_1,r_1}} y_{2,i_2,r_2} \overline{y_{2,i_3,r_1}} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}^\dagger}{8 M_F}$	$-\frac{h \overline{y_{1,i_1,r_1}} y_{2,i_2,r_2} \overline{y_{2,i_3,r_1}} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}^\dagger}{4\epsilon M_F}$	
								$-\frac{h \overline{y_{1,i_1,r_1}} y_{2,i_2,r_2} \overline{y_{2,i_3,r_1}} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}^\dagger}{3\epsilon M_F}$	$-\frac{3h \overline{y_{1,i_1,r_1}} y_{2,i_2,r_2} \overline{y_{2,i_3,r_1}} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}^\dagger}{8 M_F}$	
cont.								$-\frac{h \overline{y_{1,i_1,r_1}} y_{2,i_2,r_2} \overline{y_{2,i_3,r_1}} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}^\dagger}{3\epsilon M_F}$	$-\frac{h \overline{y_{1,i_1,r_1}} y_{2,i_2,r_2} \overline{y_{2,i_3,r_1}} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}^\dagger}{4\epsilon M_F}$	
								$+\frac{3h \overline{y_{1,i_1,r_1}} y_{2,i_2,r_2} \overline{y_{2,i_3,r_1}} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}^\dagger}{8 M_F}$	$-\frac{h \overline{y_{1,i_1,r_1}} y_{2,i_2,r_2} \overline{y_{2,i_3,r_1}} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}^\dagger}{4\epsilon M_F}$	
								$-\frac{h \overline{y_{1,i_1,r_1}} y_{2,i_2,r_2} \overline{y_{2,i_3,r_1}} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}^\dagger}{3\epsilon M_F}$	$-\frac{3h \overline{y_{1,i_1,r_1}} y_{2,i_2,r_2} \overline{y_{2,i_3,r_1}} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}^\dagger}{8 M_F}$	
cont.								$-\frac{h \overline{y_{1,i_1,r_1}} y_{2,i_2,r_2} \overline{y_{2,i_3,r_1}} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}^\dagger}{3\epsilon M_F}$	$-\frac{h \overline{y_{1,i_1,r_1}} y_{2,i_2,r_2} \overline{y_{2,i_3,r_1}} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}^\dagger}{4\epsilon M_F}$	
								$+\frac{3h \overline{y_{1,i_1,r_1}} y_{2,i_2,r_2} \overline{y_{2,i_3,r_1}} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}^\dagger}{8 M_F}$	$-\frac{h \overline{y_{1,i_1,r_1}} y_{2,i_2,r_2} \overline{y_{2,i_3,r_1}} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}^\dagger}{4\epsilon M_F}$	
								$-\frac{h \overline{y_{1,i_1,r_1}} y_{2,i_2,r_2} \overline{y_{2,i_3,r_1}} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}^\dagger}{3\epsilon M_F}$	$-\frac{3h \overline{y_{1,i_1,r_1}} y_{2,i_2,r_2} \overline{y_{2,i_3,r_1}} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}^\dagger}{8 M_F}$	
cont.								$-\frac{h \overline{y_{1,i_1,r_1}} y_{2,i_2,r_2} \overline{y_{2,i_3,r_1}} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}^\dagger}{3\epsilon M_F}$	$-\frac{h \overline{y_{1,i_1,r_1}} y_{2,i_2,r_2} \overline{y_{2,i_3,r_1}} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}^\dagger}{4\epsilon M_F}$	
								$+\frac{3h \overline{y_{1,i_1,r_1}} y_{2,i_2,r_2} \overline{y_{2,i_3,r_1}} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}^\dagger}{8 M_F}$	$-\frac{h \overline{y_{1,i_1,r_1}} y_{2,i_2,r_2} \overline{y_{2,i_3,r_1}} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}^\dagger}{4\epsilon M_F}$	
								$-\frac{h \overline{y_{1,i_1,r_1}} y_{2,i_2,r_2} \overline{y_{2,i_3,r_1}} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}^\dagger}{3\epsilon M_F}$	$-\frac{3h \overline{y_{1,i_1,r_1}} y_{2,i_2,r_2} \overline{y_{2,i_3,r_1}} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}^\dagger}{8 M_F}$	
cont.								$-\frac{h \overline{y_{1,i_1,r_1}} y_{2,i_2,r_2} \overline{y_{2,i_3,r_1}} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}^\dagger}{3\epsilon M_F}$	$-\frac{h \overline{y_{1,i_1,r_1}} y_{2,i_2,r_2} \overline{y_{2,i_3,r_1}} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}^\dagger}{4\epsilon M_F}$	
								$+\frac{3h \overline{y_{1,i_1,r_1}} y_{2,i_2,r_2} \overline{y_{2,i_3,r_1}} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}^\dagger}{8 M_F}$	$-\frac{h \overline{y_{1,i_1,r_1}} y_{2,i_2,r_2} \overline{y_{2,i_3,r_1}} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}^\dagger}{4\epsilon M_F}$	
								$-\frac{h \overline{y_{1,i_1,r_1}} y_{2,i_2,r_2} \overline{y_{2,i_3,r_1}} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}^\dagger}{3\epsilon M_F}$	$-\frac{3h \overline{y_{1,i_1,r_1}} y_{2,i_2,r_2} \overline{y_{2,i_3,r_1}} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}^\dagger}{8 M_F}$	
cont.								$-\frac{h \overline{y_{1,i_1,r_1}} y_{2,i_2,r_2} \overline{y_{2,i_3,r_1}} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}^\dagger}{3\epsilon M_F}$	$-\frac{h \overline{y_{1,i_1,r_1}} y_{2,i_2,r_2} \overline{y_{2,i_3,r_1}} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}^\dagger}{4\epsilon M_F}$	
								$+\frac{3h \overline{y_{1,i_1,r_1}} y_{2,i_2,r_2} \overline{y_{2,i_3,r_1}} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}^\dagger}{8 M_F}$	$-\frac{h \overline{y_{1,i_1,r_1}} y_{2,i_2,r_2} \overline{y_{2,i_3,r_1}} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}^\dagger}{4\epsilon M_F}$	
								$-\frac{h \overline{y_{1,i_1,r_1}} y_{2,i_2,r_2} \overline{y_{2,i_3,r_1}} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}^\dagger}{3\epsilon M_F}$	$-\frac{3h \overline{y_{1,i_1,r_1}} y_{2,i_2,r_2} \overline{y_{2,i_3,r_1}} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}^\dagger}{8 M_F}$	
cont.								$-\frac{h \overline{y_{1,i_1,r_1}} y_{2,i_2,r_2} \overline{y_{2,i_3,r_1}} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}^\dagger}{3\epsilon M_F}$	$-\frac{h \overline{y_{1,i_1,r_1}} y_{2,i_2,r_2} \overline{y_{2,i_3,r_1}} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}^\dagger}{4\epsilon M_F}$	
								$+\frac{3h \overline{y_{1,i_1,r_1}} y_{2,i_2,r_2} \overline{y_{2,i_3,r_1}} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}^\dagger}{8 M_F}$	$-\frac{h \overline{y_{1,i_1,r_1}} y_{2,i_2,r_2} \overline{y_{2,i_3,r_1}} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}^\dagger}{4\epsilon M_F}$	
								$-\frac{h \overline{y_{1,i_1,r_1}} y_{2,i_2,r_2} \overline{y_{2,i_3,r_1}} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}^\dagger}{3\epsilon M_F}$	$-\frac{3h \overline{y_{1,i_1,r_1}} y_{2,i_2,r_2} \overline{y_{2,i_3,r_1}} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}^\dagger}{8 M_F}$	
cont.								$-\frac{h \overline{y_{1,i_1,r_1}} y_{2,i_2,r_2} \overline{y_{2,i_3,r_1}} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}^\dagger}{3\epsilon M_F}$	$-\frac{h \overline{y_{1,i_1,r_1}} y_{2,i_2,r_2} \overline{y_{2,i_3,r_1}} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}^\dagger}{4\epsilon M_F}$	
								$+\frac{3h \overline{y_{1,i_1,r_1}} y_{2,i_2,r_2} \overline{y_{2,i_3,r_1}} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}^\dagger}{8 M_F}$	$-\frac{h \overline{y_{1,i_1,r_1}} y_{2,i_2,r_2} \overline{y_{2,i_3,r_1}} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}^\dagger}{4\epsilon M_F}$	
								$-\frac{h \overline{y_{1,i_1,r_1}} y_{2,i_2,r_2} \overline{y_{2,i_3,r_1}} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}^\dagger}{3\epsilon M_F}$	$-\frac{3h \overline{y_{1,i_1,r_1}} y_{2,i_2,r_2} \overline{y_{2,i_3,r_1}} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}^\dagger}{8 M_F}$	

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	S_1	S_2	F	S_1	S_2	ℓ	S_1	F	H
cont.													$-\frac{\sqrt{2}h\overline{y_{1,13,r_1}}\overline{L_{a_1,i_1}}CPr\overline{L_{a_2,i_2}}^TS_{1,j_2}y_{1,i_1,r_2}y_{2,i_2,r_2}y_{2,i_3,r_1}\overline{IT\tau_{a_1}^*}\omega_{2,j_2,j_1}S_{2,j_1}^\dagger}{3\epsilon M_F}$			
													$-\frac{h\overline{y_{2,13,r_1}}\overline{L_{a_1,i_1}}CPr\overline{L_{a_2,i_2}}^TS_{1,j_2}y_{1,i_1,r_2}y_{2,i_2,r_2}y_{2,i_3,r_1}\overline{IT\tau_{a_1}^*}\omega_{2,j_2,j_1}S_{2,j_1}^\dagger}{3\sqrt{2}M_F}$			
													$-\frac{h\overline{y_{1,13,r_1}}\overline{L_{a_1,i_1}}CPr\overline{L_{a_2,i_2}}^TS_{1,j_2}y_{1,i_1,r_2}y_{1,i_2,r_1}y_{2,i_3,r_2}\overline{IT\tau_{a_1}^*}\omega_{2,j_2,j_1}S_{2,j_1}^\dagger}{2\sqrt{2}\epsilon M_F}$			
cont.													$-\frac{3h\overline{y_{1,13,r_1}}\overline{L_{a_1,i_1}}CPr\overline{L_{a_2,i_2}}^TS_{1,j_2}y_{1,i_1,r_2}y_{1,i_2,r_1}y_{2,i_3,r_2}\overline{IT\tau_{a_1}^*}\omega_{2,j_2,j_1}S_{2,j_1}^\dagger}{4\sqrt{2}M_F}$			
													$+\frac{h\overline{y_{1,13,r_1}}\overline{L_{a_1,i_1}}CPr\overline{L_{a_2,i_2}}^TS_{1,j_1}y_{1,i_2,r_1}y_{1,i_3,r_2}y_{2,i_4,r_2}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger}{4\epsilon M_F}$			
													$+\frac{3h\overline{y_{1,13,r_1}}\overline{L_{a_1,i_1}}CPr\overline{L_{a_2,i_2}}^TS_{1,j_1}y_{1,i_2,r_1}y_{1,i_3,r_2}y_{2,i_4,r_2}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger}{8M_F}$			
cont.													$-\frac{2\sqrt{3}h\overline{L_{a_1,i_1}}CPr\overline{L_{a_2,i_2}}^Tg_2^2S_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger}{\epsilon M_F}$			
													$-\frac{2h\overline{L_{a_1,i_1}}CPr\overline{L_{a_2,i_2}}^Tg_2^2S_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger}{\sqrt{3}M_F}$			
													$+\frac{\overline{L_{a_1,i_1}}CPr\overline{L_{a_2,i_2}}^TS_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger}{\sqrt{3}M_F}$			
cont.													$+\frac{h\overline{y_{2,13,r_1}}\overline{L_{a_1,i_1}}CPr\overline{L_{a_2,i_2}}^TS_{1,j_1}y_{1,i_3,r_2}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger}{4\epsilon M_F}$			
													$+\frac{3h\overline{y_{2,13,r_1}}\overline{L_{a_1,i_1}}CPr\overline{L_{a_2,i_2}}^TS_{1,j_1}y_{1,i_3,r_2}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger}{8M_F}$			
													$-\frac{h\overline{y_{1,13,r_1}}\overline{L_{a_1,i_1}}CPr\overline{L_{a_2,i_2}}^TS_{1,j_1}y_{1,i_1,r_2}y_{1,i_3,r_1}y_{2,i_2,r_2}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger}{3\epsilon M_F}$			
cont.													$-\frac{h\overline{y_{1,13,r_1}}\overline{L_{a_1,i_1}}CPr\overline{L_{a_2,i_2}}^TS_{1,j_1}y_{1,i_1,r_2}y_{1,i_3,r_1}y_{2,i_2,r_2}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger}{6M_F}$			
													$-\frac{h\overline{y_{2,13,r_1}}\overline{L_{a_1,i_1}}CPr\overline{L_{a_2,i_2}}^TS_{1,j_1}y_{1,i_1,r_2}y_{2,i_2,r_2}y_{2,i_3,r_1}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger}{3\epsilon M_F}$			
													$-\frac{h\overline{y_{2,13,r_1}}\overline{L_{a_1,i_1}}CPr\overline{L_{a_2,i_2}}^TS_{1,j_1}y_{1,i_1,r_2}y_{2,i_2,r_2}y_{2,i_3,r_1}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger}{6M_F}$			
cont.													$-\frac{h\overline{y_{1,13,r_1}}\overline{L_{a_1,i_1}}CPr\overline{L_{a_2,i_2}}^TS_{1,j_1}y_{1,i_2,r_1}y_{2,i_3,r_2}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger}{4\epsilon M_F}$			
													$-\frac{3h\overline{y_{1,13,r_1}}\overline{L_{a_1,i_1}}CPr\overline{L_{a_2,i_2}}^TS_{1,j_1}y_{1,i_2,r_1}y_{2,i_3,r_2}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger}{8M_F}$			
													$-\frac{h\overline{y_{2,13,r_1}}\overline{L_{a_1,i_1}}CPr\overline{L_{a_2,i_2}}^TS_{1,j_1}y_{1,i_1,r_2}y_{2,i_2,r_1}y_{2,i_3,r_2}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger}{4\epsilon M_F}$			
cont.													$+\frac{3h\overline{y_{2,13,r_1}}\overline{L_{a_1,i_1}}CPr\overline{L_{a_2,i_2}}^TS_{1,j_1}y_{1,i_1,r_2}y_{2,i_2,r_1}y_{2,i_3,r_2}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger}{8M_F}$			
													$+\frac{\sqrt{\frac{2}{3}}h\overline{y_{1,i_1,r_1}}\overline{y_{2,i_2,r_1}}\overline{L_{a_2,i_2}}^TCPl_L\overline{L_{a_1,i_1}}S_{2,j_2}\overline{IT\tau_{a_1}^*}\omega_{2,j_1,j_2}S_{1,j_1}^\dagger\lambda_{12}^{(1)}}{\epsilon M_F}$			
													$+\frac{\sqrt{\frac{2}{3}}h\overline{y_{1,i_1,r_1}}\overline{y_{2,i_2,r_1}}\overline{L_{a_2,i_2}}^TCPl_L\overline{L_{a_1,i_1}}S_{2,j_2}\overline{IT\tau_{a_1}^*}\omega_{2,j_1,j_2}S_{1,j_1}^\dagger\lambda_{12}^{(1)}}{M_F}$			
cont.													$-\frac{h\overline{y_{1,i_1,r_1}}\overline{y_{2,i_2,r_1}}\overline{L_{a_2,i_2}}^TCPl_L\overline{L_{a_1,i_1}}S_{2,j_1}\epsilon_{a_1,a_2}S_{1,j_1}^\dagger\lambda_{12}^{(1)}}{\sqrt{3}\epsilon M_F}$			
													$-\frac{h\overline{y_{1,i_1,r_1}}\overline{y_{2,i_2,r_1}}\overline{L_{a_2,i_2}}^TCPl_L\overline{L_{a_1,i_1}}S_{2,j_1}\epsilon_{a_1,a_2}S_{1,j_1}^\dagger\lambda_{12}^{(1)}}{\sqrt{3}M_F}$			
													$+\frac{\sqrt{\frac{2}{3}}h\overline{L_{a_1,i_1}}CPr\overline{L_{a_2,i_2}}^TS_{1,j_2}y_{1,i_1,r_1}y_{2,i_2,r_1}\overline{IT\tau_{a_1}^*}\omega_{2,j_2,j_1}S_{2,j_1}^\dagger\lambda_{12}^{(1)}}{\epsilon M_F}$			
cont.													$+\frac{\sqrt{\frac{2}{3}}h\overline{L_{a_1,i_1}}CPr\overline{L_{a_2,i_2}}^TS_{1,j_2}y_{1,i_1,r_1}y_{2,i_2,r_1}\overline{IT\tau_{a_1}^*}\omega_{2,j_2,j_1}S_{2,j_1}^\dagger\lambda_{12}^{(1)}}{M_F}$			
													$-\frac{h\overline{L_{a_1,i_1}}CPr\overline{L_{a_2,i_2}}^TS_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger\lambda_{12}^{(1)}}{\sqrt{3}\epsilon M_F}$			
													$-\frac{h\overline{L_{a_1,i_1}}CPr\overline{L_{a_2,i_2}}^TS_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger\lambda_{12}^{(1)}}{\sqrt{3}M_F}$			
cont.													$-\frac{3\sqrt{\frac{2}{11}}h\overline{y_{1,i_1,r_1}}\overline{y_{2,i_2,r_1}}\overline{L_{a_2,i_2}}^TCPl_L\overline{L_{a_1,i_1}}S_{2,j_2}\overline{IT\tau_{a_1}^*}\omega_{2,j_1,j_2}S_{1,j_1}^\dagger\lambda_{12}^{(2)}}{\epsilon M_F}$			
													$-\frac{3\sqrt{\frac{2}{11}}h\overline{y_{1,i_1,r_1}}\overline{y_{2,i_2,r_1}}\overline{L_{a_2,i_2}}^TCPl_L\overline{L_{a_1,i_1}}S_{2,j_2}\overline{IT\tau_{a_1}^*}\omega_{2,j_1,j_2}S_{1,j_1}^\dagger\lambda_{12}^{(2)}}{M_F}$			
													$-\frac{h\overline{y_{1,i_1,r_1}}\overline{y_{2,i_2,r_1}}\overline{L_{a_2,i_2}}^TCPl_L\overline{L_{a_1,i_1}}S_{2,j_1}\epsilon_{a_1,a_2}S_{1,j_1}^\dagger\lambda_{12}^{(2)}}{\sqrt{11}\epsilon M_F}$			
cont.													$-\frac{h\overline{y_{1,i_1,r_1}}\overline{y_{2,i_2,r_1}}\overline{L_{a_2,i_2}}^TCPl_L\overline{L_{a_1,i_1}}S_{2,j_1}\epsilon_{a_1,a_2}S_{1,j_1}^\dagger\lambda_{12}^{(2)}}{\sqrt{11}M_F}$			
													$-\frac{3\sqrt{\frac{2}{11}}h\overline{L_{a_1,i_1}}CPr\overline{L_{a_2,i_2}}^TS_{1,j_2}y_{1,i_1,r_1}y_{2,i_2,r_1}\overline{IT\tau_{a_1}^*}\omega_{2,j_2,j_1}S_{2,j_1}^\dagger\lambda_{12}^{(2)}}{\epsilon M_F}$			
													$-\frac{3\sqrt{\frac{2}{11}}h\overline{L_{a_1,i_1}}CPr\overline{L_{a_2,i_2}}^TS_{1,j_2}y_{1,i_1,r_1}y_{2,i_2,r_1}\overline{IT\tau_{a_1}^*}\omega_{2,j_2,j_1}S_{2,j_1}^\dagger\lambda_{12}^{(2)}}{M_F}$			
cont.													$-\frac{h\overline{L_{a_1,i_1}}CPr\overline{L_{a_2,i_2}}^TS_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger\lambda_{12}^{(2)}}{\sqrt{11}\epsilon M_F}$			
													$-\frac{h\overline{L_{a_1,i_1}}CPr\overline{L_{a_2,i_2}}^TS_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger\lambda_{12}^{(2)}}{\sqrt{11}M_F}$			
													$-\frac{7h\overline{y_{1,i_1,r_1}}\overline{y_{2,i_2,r_1}}\overline{L_{a_2,i_2}}^TCPl_L\overline{L_{a_1,i_1}}S_{2,j_1}\epsilon_{a_1,a_2}S_{1,j_1}^\dagger\lambda_{12}^{(3)}}{\sqrt{23}\epsilon M_F}$			
cont.													$-\frac{7h\overline{y_{1,i_1,r_1}}\overline{y_{2,i_2,r_1}}\overline{L_{a_2,i_2}}^TCPl_L\overline{L_{a_1,i_1}}S_{2,j_1}\epsilon_{a_1,a_2}S_{1,j_1}^\dagger\lambda_{12}^{(3)}}{\sqrt{23}M_F}$			
													$-\frac{7h\overline{y_{1,i_1,r_1}}\overline{y_{2,i_2,r_1}}\overline{L_{a_2,i_2}}^TCPl_L\overline{L_{a_1,i_1}}S_{2,j_1}\epsilon_{a_1,a_2}S_{1,j_1}^\dagger\lambda_{12}^{(3)}}{\sqrt{23}M_F}$			

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$S_1 S_2 F$	$S_1 S_2 \ell$	$S_1 F H$
$cont.$									$\frac{7h\overline{L_{a_1, i_1}} C P_R \overline{L_{a_2, i_2}}^T S_{1, j_1} Y_{1, i_1, r_1} Y_{2, i_2, r_1} \epsilon_{a_1, a_2} S_{2, j_1}^\dagger \lambda_{12}^{(3)}}{\sqrt{23}\epsilon M_F}$	$\frac{7h\overline{L_{a_1, i_1}} C P_R \overline{L_{a_2, i_2}}^T S_{1, j_1} Y_{1, i_1, r_1} Y_{2, i_2, r_1} \epsilon_{a_1, a_2} S_{2, j_1}^\dagger \lambda_{12}^{(3)}}{\sqrt{23}M_F}$
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$-\frac{4}{3}h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1, i_1, r_1} y_{2, i_2, r_2}} S_{1, j_2} S_{2, j_2} y_{1, i_2, r_1} y_{2, i_1, r_2} S_{1, j_1}^\dagger S_{2, j_1}^\dagger$ $+\frac{4}{3}h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1, i_1, r_1} y_{2, i_2, r_2}} S_{1, j_2} S_{2, j_1} y_{1, i_2, r_1} y_{2, i_1, r_2} S_{1, j_1}^\dagger S_{2, j_2}^\dagger$ $-\frac{4}{3}h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1, i_1, r_1} y_{2, i_2, r_2}} S_{1, j_1} S_{2, j_2} y_{1, i_2, r_1} y_{2, i_1, r_2} S_{1, j_1}^\dagger S_{2, j_2}^\dagger$ $2h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1, i_1, r_1} S_{1, j_3} S_{2, j_4}} y_{1, i_1, r_1} S_{1, j_1}^\dagger S_{2, j_2}^\dagger \mathbb{I}_{12, j_3, j_1, j_4, j_2}^{-1} \chi_{12}^{(1)}$	$-\frac{\sqrt{6}h\overline{y_{1, i_1, r_1} y_{2, i_2, r_1} L_{a_2, i_2}}^T C P_L L_{a_1, i_1} g_Y^2 S_{2, j_2} \mathbb{I}_{T3, a_1, a_2, j_1, j_2} S_{1, j_1}^\dagger}{\epsilon M_F}$ $-\frac{\sqrt{\frac{2}{3}}h\overline{y_{1, i_1, r_1} y_{2, i_2, r_1} L_{a_2, i_2}}^T C P_L L_{a_1, i_1} g_Y^2 S_{2, j_2} \mathbb{I}_{T3, a_1, a_2, j_1, j_2} S_{1, j_1}^\dagger}{M_F}$ $3\sqrt{\frac{2}{3}}h\overline{y_{1, i_1, r_1} y_{2, i_2, r_1} L_{a_2, i_2}}^T C P_L L_{a_1, i_1} g_Y^2 S_{2, j_2} \mathbb{I}_{T3, a_1, a_2, j_1, j_2} S_{1, j_1}^\dagger$ $-\frac{\sqrt{\frac{2}{3}}h\overline{y_{1, i_1, r_1} y_{2, i_2, r_1} L_{a_2, i_2}}^T C P_L L_{a_1, i_1} g_Y^2 S_{2, j_2} \mathbb{I}_{T3, a_1, a_2, j_1, j_2} S_{1, j_1}^\dagger}{2M_F}$	$-\frac{2h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1, i_1, r_1} Y_{e, i_2, i_3}} H_{a_2} S_{1, j_1} y_{1, i_2, r_1} Y_{e, i_1, i_3} H_{a_1}^\dagger S_{1, j_1}^\dagger}{\sqrt{3}}$ $-2i\sqrt{\frac{2}{3}}h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1, i_1, r_1} Y_{e, i_2, i_3}} H_{a_2} S_{1, j_2} y_{1, i_2, r_1} Y_{e, i_1, i_3} H_{a_1}^\dagger S_{1, j_1}^\dagger \mathbb{I}_{H1, a_1, a_2, j_2, j_1}^{-1}$ $-\frac{2h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1, i_1, r_1} H_{a_2} S_{1, j_2}} y_{1, i_1, r_1} H_{a_1}^\dagger S_{1, j_1}^\dagger \mathbb{I}_{H1, a_1, a_2, j_2, j_1}^{-1} \chi_{H1}^{(2)}}{\sqrt{3}}$
$cont.$								$-\frac{2h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2, i_1, r_1} S_{1, j_3} S_{2, j_4}} y_{2, i_1, r_1} S_{1, j_1}^\dagger S_{2, j_2}^\dagger \mathbb{I}_{12, j_3, j_1, j_4, j_2}^{-1} \chi_{12}^{(1)}}{\sqrt{3}}$ $2h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1, i_1, r_1} S_{1, j_4} S_{2, j_3}} y_{1, i_1, r_1} S_{1, j_1}^\dagger S_{2, j_2}^\dagger \mathbb{I}_{12, j_1, j_2, j_3, j_4}^{-2} \chi_{12}^{(2)}$ $+\frac{2h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1, i_1, r_1} S_{1, j_3} S_{2, j_4}} y_{1, i_1, r_1} S_{1, j_1}^\dagger S_{2, j_2}^\dagger \mathbb{I}_{12, j_1, j_2, j_3, j_4}^{-2} \chi_{12}^{(2)}}{\sqrt{3}}$	$-\frac{\sqrt{\frac{2}{3}}h\overline{y_{1, i_1, r_1} y_{2, i_2, r_1} L_{a_2, i_2}}^T C P_L L_{a_1, i_1} g_Y^2 S_{2, j_2} \mathbb{I}_{T3, a_1, a_2, j_1, j_2} S_{1, j_1}^\dagger}{2M_F}$ $+\frac{\sqrt{\frac{2}{3}}h\overline{y_{1, i_1, r_1} y_{2, i_2, r_1} L_{a_2, i_2}}^T C P_L L_{a_1, i_1} S_{2, j_2} \mathbb{I}_{T3, a_1, a_2, j_1, j_2} S_{1, j_1}^\dagger}{M_F}$ $\frac{h\overline{y_{1, i_1, r_1} y_{1, i_3, r_2} y_{2, i_2, r_2}}^T C P_L L_{a_1, i_1} S_{2, j_2} y_{1, i_3, r_1} \mathbb{I}_{T3, a_1, a_2, j_1, j_2} S_{1, j_1}^\dagger}{2\sqrt{2}\epsilon M_F}$	
$cont.$								$-\frac{2h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1, i_1, r_1} S_{1, j_3} S_{2, j_4}} y_{1, i_1, r_1} S_{1, j_1}^\dagger S_{2, j_2}^\dagger \mathbb{I}_{12, j_1, j_2, j_3, j_4}^{-2} \chi_{12}^{(2)}}{\sqrt{3}}$ $+\frac{2h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2, i_1, r_1} S_{1, j_4} S_{2, j_3}} y_{2, i_1, r_1} S_{1, j_1}^\dagger S_{2, j_2}^\dagger \mathbb{I}_{12, j_1, j_2, j_3, j_4}^{-2} \chi_{12}^{(2)}}{\sqrt{3}}$ $2h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1, i_1, r_1} S_{1, j_3} S_{2, j_4}} y_{1, i_1, r_1} S_{1, j_1}^\dagger S_{2, j_2}^\dagger \mathbb{I}_{12, j_1, j_2, j_3, j_4}^{-2} \chi_{12}^{(2)}$	$3h\overline{y_{1, i_1, r_1} y_{1, i_3, r_2} y_{2, i_2, r_2}}^T C P_L L_{a_1, i_1} S_{2, j_2} y_{1, i_3, r_1} \mathbb{I}_{T3, a_1, a_2, j_1, j_2} S_{1, j_1}^\dagger$ $4\sqrt{2}M_F$ $\frac{\sqrt{2}h\overline{y_{1, i_1, r_1} y_{1, i_3, r_2} y_{2, i_2, r_1} L_{a_2, i_2}}^T C P_L L_{a_1, i_1} S_{2, j_2} y_{1, i_3, r_2} \mathbb{I}_{T3, a_1, a_2, j_1, j_2} S_{1, j_1}^\dagger}{3\epsilon M_F}$ $\frac{h\overline{y_{1, i_1, r_1} y_{1, i_3, r_2} y_{2, i_2, r_1} L_{a_2, i_2}}^T C P_L L_{a_1, i_1} S_{2, j_2} y_{1, i_3, r_2} \mathbb{I}_{T3, a_1, a_2, j_1, j_2} S_{1, j_1}^\dagger}{3\sqrt{2}M_F}$	
$cont.$								$-\frac{2h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1, i_1, r_1} S_{1, j_4} S_{2, j_3}} y_{1, i_1, r_1} S_{1, j_1}^\dagger S_{2, j_2}^\dagger \mathbb{I}_{12, j_1, j_2, j_3, j_4}^{-2} \chi_{12}^{(2)}}{\sqrt{3}}$ $2h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1, i_1, r_1} S_{1, j_4} S_{2, j_3}} y_{1, i_1, r_1} S_{1, j_1}^\dagger S_{2, j_2}^\dagger \mathbb{I}_{12, j_1, j_2, j_3, j_4}^{-2} \chi_{12}^{(2)}$	$\frac{h\overline{y_{1, i_1, r_1} y_{1, i_2, r_2} y_{2, i_2, r_1} L_{a_2, i_2}}^T C P_L L_{a_1, i_1} S_{2, j_2} y_{1, i_2, r_1} \mathbb{I}_{T3, a_1, a_2, j_1, j_2} S_{1, j_1}^\dagger}{2\sqrt{3}\epsilon M_F}$ $3h\overline{y_{1, i_1, r_1} y_{1, i_2, r_2} y_{2, i_2, r_1} L_{a_2, i_2}}^T C P_L L_{a_1, i_1} S_{2, j_2} y_{1, i_2, r_1} \mathbb{I}_{T3, a_1, a_2, j_1, j_2} S_{1, j_1}^\dagger$ $4\sqrt{3}M_F$ $\frac{h\overline{y_{1, i_1, r_1} y_{2, i_2, r_1} y_{2, i_2, r_2} y_{1, i_3, r_2}}^T C P_L L_{a_1, i_1} S_{2, j_2} y_{1, i_3, r_2} \mathbb{I}_{T3, a_1, a_2, j_1, j_2} S_{1, j_1}^\dagger}{2\sqrt{2}\epsilon M_F}$	
$cont.$								$-\frac{2h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1, i_1, r_1} S_{1, j_3} S_{2, j_4}} y_{1, i_1, r_1} S_{1, j_1}^\dagger S_{2, j_2}^\dagger \mathbb{I}_{12, j_1, j_3, j_4, j_2}^{-2} \chi_{12}^{(2)}}{\sqrt{3}}$ $+\frac{2h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1, i_1, r_1} S_{1, j_4} S_{2, j_3}} y_{1, i_1, r_1} S_{1, j_1}^\dagger S_{2, j_2}^\dagger \mathbb{I}_{12, j_1, j_3, j_4, j_2}^{-2} \chi_{12}^{(2)}}{\sqrt{3}}$ $2h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1, i_1, r_1} S_{1, j_3} S_{2, j_4}} y_{1, i_1, r_1} S_{1, j_1}^\dagger S_{2, j_2}^\dagger \mathbb{I}_{12, j_1, j_3, j_4, j_2}^{-2} \chi_{12}^{(2)}$	$3h\overline{y_{1, i_3, r_1} y_{2, i_1, r_2} y_{2, i_2, r_1} L_{a_2, i_2}}^T C P_L L_{a_1, i_1} S_{2, j_2} y_{2, i_3, r_2} \mathbb{I}_{T3, a_1, a_2, j_1, j_2} S_{1, j_1}^\dagger$ $4\sqrt{2}M_F$ $\frac{h\overline{y_{1, i_1, r_1} y_{2, i_2, r_2} y_{2, i_2, r_1} L_{a_2, i_2}}^T C P_L L_{a_1, i_1} S_{2, j_2} y_{2, i_2, r_2} \mathbb{I}_{T3, a_1, a_2, j_1, j_2} S_{1, j_1}^\dagger}{2\sqrt{2}\epsilon M_F}$ $3h\overline{y_{1, i_1, r_1} y_{2, i_2, r_2} y_{2, i_3, r_1} L_{a_2, i_2}}^T C P_L L_{a_1, i_1} S_{2, j_2} y_{2, i_3, r_2} \mathbb{I}_{T3, a_1, a_2, j_1, j_2} S_{1, j_1}^\dagger$ $4\sqrt{2}M_F$	
$cont.$								$-\frac{2h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2, i_1, r_1} S_{1, j_3} S_{2, j_4}} y_{2, i_1, r_1} S_{1, j_1}^\dagger S_{2, j_2}^\dagger \mathbb{I}_{12, j_1, j_3, j_4, j_2}^{-2} \chi_{12}^{(2)}}{\sqrt{3}}$ $+\frac{2h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2, i_1, r_1} S_{1, j_4} S_{2, j_3}} y_{2, i_1, r_1} S_{1, j_1}^\dagger S_{2, j_2}^\dagger \mathbb{I}_{12, j_1, j_3, j_4, j_2}^{-2} \chi_{12}^{(2)}}{\sqrt{3}}$ $2h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2, i_1, r_1} S_{1, j_3} S_{2, j_4}} y_{2, i_1, r_1} S_{1, j_1}^\dagger S_{2, j_2}^\dagger \mathbb{I}_{12, j_1, j_3, j_4, j_2}^{-2} \chi_{12}^{(2)}$	$\frac{\sqrt{2}h\overline{y_{1, i_1, r_1} y_{2, i_2, r_1} y_{2, i_3, r_2} L_{a_2, i_2}}^T C P_L L_{a_1, i_1} S_{2, j_2} y_{2, i_3, r_2} \mathbb{I}_{T3, a_1, a_2, j_1, j_2} S_{1, j_1}^\dagger}{3\epsilon M_F}$ $\frac{h\overline{y_{1, i_1, r_1} y_{2, i_2, r_1} y_{2, i_3, r_2} L_{a_2, i_2}}^T C P_L L_{a_1, i_1} S_{2, j_2} y_{2, i_3, r_2} \mathbb{I}_{T3, a_1, a_2, j_1, j_2} S_{1, j_1}^\dagger}{3\sqrt{2}M_F}$ $\frac{2\sqrt{3}h\overline{y_{1, i_1, r_1} y_{2, i_2, r_1} L_{a_2, i_2}}^T C P_L L_{a_1, i_1} g_Y^2 S_{2, j_1} \epsilon_{a_1, a_2} S_{1, j_1}^\dagger}{\epsilon M_F}$ $\frac{2h\overline{y_{1, i_1, r_1} y_{2, i_2, r_1} L_{a_2, i_2}}^T C P_L L_{a_1, i_1} g_Y^2 S_{2, j_1} \epsilon_{a_1, a_2} S_{1, j_1}^\dagger}{\sqrt{3}M_F}$	
$cont.$								$-\frac{2h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1, i_1, r_1} S_{1, j_3} S_{2, j_4}} y_{1, i_1, r_1} S_{1, j_1}^\dagger S_{2, j_2}^\dagger \mathbb{I}_{12, j_3, j_1, j_4, j_2}^{-3} \chi_{12}^{(3)}}{\sqrt{3}}$	$3\sqrt{3}h\overline{y_{1, i_1, r_1} y_{2, i_2, r_1} L_{a_2, i_2}}^T C P_L L_{a_1, i_1} g_Y^2 S_{2, j_1} \epsilon_{a_1, a_2} S_{1, j_1}^\dagger$ $4\epsilon M_F$ $-\frac{3h\overline{y_{1, i_1, r_1} y_{2, i_2, r_1} L_{a_2, i_2}}^T C P_L L_{a_1, i_1} g_Y^2 S_{2, j_1} \epsilon_{a_1, a_2} S_{1, j_1}^\dagger}{4\epsilon M_F}$ $+\frac{\overline{y_{1, i_1, r_1} y_{2, i_2, r_1} L_{a_2, i_2}}^T C P_L L_{a_1, i_1} S_{2, j_1} \epsilon_{a_1, a_2} S_{1, j_1}^\dagger}{\sqrt{3}M_F}$	
$cont.$								$-\frac{2h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1, i_1, r_1} S_{1, j_3} S_{2, j_4}} y_{1, i_1, r_1} S_{1, j_1}^\dagger S_{2, j_2}^\dagger \mathbb{I}_{12, j_3, j_1, j_4, j_2}^{-3} \chi_{12}^{(3)}}{\sqrt{3}}$	$\frac{h\overline{y_{1, i_1, r_1} y_{1, i_3, r_2} y_{2, i_2, r_2}}^T C P_L L_{a_1, i_1} S_{2, j_1} y_{1, i_3, r_2} \mathbb{I}_{T3, a_1, a_2, j_1, j_2} S_{1, j_1}^\dagger}{4\epsilon M_F}$ $-\frac{3h\overline{y_{1, i_1, r_1} y_{1, i_3, r_2} y_{2, i_2, r_2}}^T C P_L L_{a_1, i_1} S_{2, j_1} y_{1, i_3, r_2} \mathbb{I}_{T3, a_1, a_2} S_{1, j_1}^\dagger}{4\epsilon M_F}$ $-\frac{3h\overline{y_{1, i_1, r_1} y_{1, i_3, r_2} y_{2, i_2, r_2}}^T C P_L L_{a_1, i_1} S_{2, j_1} y_{1, i_3, r_2} \mathbb{I}_{T3, a_1, a_2} S_{1, j_1}^\dagger}{4\epsilon M_F}$ $-\frac{3h\overline{y_{1, i_1, r_1} y_{1, i_3, r_2} y_{2, i_2, r_2}}^T C P_L L_{a_1, i_1} S_{2, j_1} y_{1, i_3, r_2} \mathbb{I}_{T3, a_1, a_2} S_{1, j_1}^\dagger}{4\epsilon M_F}$ $-\frac{3h\overline{y_{1, i_1, r_1} y_{1, i_3, r_2} y_{2, i_2, r_2}}^T C P_L L_{a_1, i_1} S_{2, j_1} y_{1, i_3, r_2} \mathbb{I}_{T3, a_1, a_2} S_{1, j_1}^\dagger}{4\epsilon M_F}$ $-\frac{3h\overline{y_{1, i_1, r_1} y_{1, i_3, r_2} y_{2, i_2, r_2}}^T C P_L L_{a_1, i_1} S_{2, j_1} y_{1, i_3, r_2} \mathbb{I}_{T3, a_1, a_2} S_{1, j_1}^\dagger}{4\epsilon M_F}$ $-\frac{3h\overline{y_{1, i_1, r_1} y_{1, i_3, r_2} y_{2, i_2, r_2}}^T C P_L L_{a_1, i_1} S_{2, j_1} y_{1, i_3, r_2} \mathbb{I}_{T3, a_1, a_2} S_{1, j_1}^\dagger}{4\epsilon M_F}$ $-\frac{3h\overline{y_{1, i_1, r_1} y_{1, i_3, r_2} y_{2, i_2, r_2}}^T C P_L L_{a_1, i_1} S_{2, j_1} y_{1, i_3, r_2} \mathbb{I}_{T3, a_1, a_2} S_{1, j_1}^\dagger}{4\epsilon M_F}$ $-\frac{3h\overline{y_{1, i_1, r_1} y_{1, i_3, r_2} y_{2, i_2, r_2}}^T C P_L L_{a_1, i_1} S_{2, j_1} y_{1, i_3, r_2} \mathbb{I}_{T3, a_1, a_2} S_{1, j_1}^\dagger}{4\epsilon M_F}$ $-\frac{3h\overline{y_{1, i_1, r_1} y_{1, i_3, r_2} y_{2, i_2, r_2}}^T C P_L L_{a_1, i_1} S_{2, j_1} y_{1, i_3, r_2} \mathbb{I}_{T3, a_1, a_2} S_{1, j_1}^\dagger}{4\epsilon M_F}$ $-\frac{3h\overline{y_{1, i_1, r_1} y_{1, i_3, r_2} y_{2, i_2, r_2}}^T C P_L L_{a_1, i_1} S_{2, j_1} y_{1, i_3, r_2} \mathbb{I}_{T3, a_1, a_2} S_{1, j_1}^\dagger}{4\epsilon M_F}$ $-\frac{3h\overline{y_{1, i_1, r_1} y_{1, i_3, r_2} y_{2, i_2, r_2}}^T C P_L L_{a_1, i_1} S_{2, j_1} y_{1, i_3, r_2} \mathbb{I}_{T3, a_1, a_2} S_{1, j_1}^\dagger}{4\epsilon M_F}$ $-\frac{3h\overline{y_{1, i_1, r_1} y_{1, i_3, r_2} y_{2, i_2, r_2}}^T C P_L L_{a_1, i_1} S_{2, j_1} y_{1, i_3, r_2} \mathbb{I}_{T3, a_1, a_2} S_{1, j_1}^\dagger}{4\epsilon M_F}$ $-\frac{3h\overline{y_{1, i_1, r_1} y_{1, i_3, r_2} y_{2, i_2, r_2}}^T C P_L L_{a_1, i_1} S_{2, j_1} y_{1, i_3, r_2} \mathbb{I}_{T3, a_1, a_2} S_{1, j_1}^\dagger}{4\epsilon M_F}$ $-\frac{3h\overline{y_{1, i_1, r_1} y_{1, i_3, r_2} y_{2, i_2, r_2}}^T C P_L L_{a_1, i_1} S_{2, j_1} y_{1, i_3, r_2} \mathbb{I}_{T3, a_1, a_2} S_{1, j_1}^\dagger}{4\epsilon M_F}$ $-\frac{3h\overline{y_{1, i_1, r_1} y_{1, i_3, r_2} y_{2, i_2, r_2}}^T C P_L L_{a_1, i_1} S_{2, j_1} y_{1, i_3, r_2} \mathbb{I}_{T3, a_1, a_2} S_{1, j_1}^\dagger}{4\epsilon M_F}$ $-\frac{3h\overline{y_{1, i_1, r_1} y_{1, i_3, r_2} y_{2, i_2, r_2}}^T C P_L L_{a_1, i_1} S_{2, j_1} y_{1, i_3, r_2} \mathbb{I}_{T3, a_1, a_2} S_{1, j_1}^\dagger}{4\epsilon M_F}$ $-\frac{3h\overline{y_{1, i_1, r_1} y_{1, i_3, r_2} y_{2, i_2, r_2}}^T C P_L L_{a_1, i_1} S_{2, j_1} y_{1, i_3, r_2} \mathbb{I}_{T3, a_1, a_2} S_{1, j_1}^\dagger}{4\epsilon M_F}$ $-\frac{3h\overline{y_{1, i_1, r_1} y_{1, i_3, r_2} y_{2, i_2, r_2}}^T C P_L L_{a_1, i_1} S_{2, j_1} y_{1, i_3, r_2} \mathbb{I}_{T3, a_1, a_2} S_{1, j_1}^\dagger}{4\epsilon M_F}$ $-\frac{3h\overline{y_{1, i_1, r_1} y_{1, i_3, r_2} y_{2, i_2, r_2}}^T C P_L L_{a_1, i_1} S_{2, j_1} y_{1, i_3, r_2} \mathbb{I}_{T3, a_1, a_2} S_{1, j_1}^\dagger}{4\epsilon M_F}$ $-\frac{3h\overline{y_{1, i_1, r_1} y_{1, i_3, r_2} y_{2, i_2, r_2}}^T C P_L L_{a_1, i_1} S_{2, j_1} y_{1, i_3, r_2} \mathbb{I}_{T3, a_1, a_2} S_{1, j_1}^\dagger}{4\epsilon M_F}$ $-\frac{3h\overline{y_{1, i_1, r_1} y_{1, i_3, r_2} y_{2, i_2, r_2}}^T C P_L L_{a_1, i_1} S_{2, j_1} y_{1, i_3, r_2} \mathbb{I}_{T3, a_1, a_2} S_{1, j_1}^\dagger}{4\epsilon M_F}$ $-\frac{3h\overline{y_{1, i_1, r_1} y_{1, i_3, r_2} y_{2, i_2, r_2}}^T C P_L L_{a_1, i_1} S_{2, j_1} y_{1, i_3, r_2} \mathbb{I}_{T3, a_1, a_2} S_{1, j_1}^\dagger}{4\epsilon M_F}$ $-\frac{3h\overline{y_{1, i_1, r_1} y_{1, i_3, r_2} y_{2, i_2, r_2}}^T C P_L L_{a_1, i_1} S_{2, j_1} y_{1, i_3, r_2} \mathbb{I}_{T3, a_1, a_2} S_{1, j_1}^\dagger}{4\epsilon M_F}$ $-\frac{3h\overline{y_{1, i_1, r_1} y_{1, i_3, r_2} y_{2, i_2, r_2}}^T C P_L L_{a_1, i_1} S_{2, j_1} y_{1, i_3, r_2} \mathbb{I}_{T3, a_1, a_2} S_{1, j_1}^\dagger}{4\epsilon M_F}$ $-\frac{3h\overline{y_{1, i_1, r_1} y_{1, i_3, r_2} y_{2, i_2, r_2}}^T C P_L L_{a_1, i_1} S_{2, j_1} y_{1, i_3, r_2} \mathbb{I}_{T3, a_1, a_2} S_{1, j_1}^\dagger}{4\epsilon M_F}$ $-\frac{3h\overline{y_{1, i_1, r_1} y_{1, i_3, r_2} y_{2, i_2, r_2}}^T C P_L L_{a_1, i_1} S_{2, j_1} y_{1, i_3, r_2} \mathbb{I}_{T3, a_1, a_2} S_{1, j_1}^\dagger}{4\epsilon M_F}$ $-\frac{3h\overline{y_{1, i_1, r_1} y_{1, i_3, r_2} y_{2, i_2, r_2}}^T C P_L L_{a_1, i_1} S_{2, j_1} y_{1, i_3, r_2} \mathbb{I}_{T3, a_1, a_2} S_{1, j_1}^\dagger}{4\epsilon M_F}$ $-\frac{3h\overline{y_{1, i_1, r_1} y_{1, i_3, r_2} y_{2, i_2, r_2}}^T C P_L L_{a_1, i_1} S_{2, j_1} y_{1, i_3, r_2} \mathbb{I}_{T3, a_1, a_2} S_{1, j_1}^\dagger}{4\epsilon M_F}$ $-\frac{3h\overline{y_{1, i_1, r_1} y_{1, i_3, r_2} y_{2, i_2, r_2}}^T C P_L L_{a_1, i_1} S_{2, j_1} y_{1, i_3, r_2} \mathbb{I}_{T3, a_1, a_2} S_{1, j_1}^\dagger}{4\epsilon M_F}$ $-\frac{3h\overline{y_{1, i_1, r_1} y_{1, i_3, r_2} y_{2, i_2, r_2}}^T C P_L L_{a_1, i_1} S_{2, j_1} y_{1, i_3, r_2} \mathbb{I}_{T3, a_1, a_2} S_{1, j_1}^\dagger}{4\epsilon M_F}$ $-\frac{3h\overline{y_{1, i_1, r_1} y_{1, i_3, r_2} y_{2, i_2, r_2}}^T C P_L L_{a_1, i_1} S_{2, j_1} y_{1, i_3, r_2} \mathbb{I}_{T3, a_1, a_2} S_{1, j_1}^\dagger}{4\epsilon M_F}$ $-\frac{3h\overline{y_{1, i_1, r_1} y_{1, i_3, r_2} y_{2, i_2, r_2}}^T C P_L L_{a_1, i_1} S_{2, j_1} y_{1, i_3, r_2} \mathbb{I}_{T3, a_1, a_2} S_{1, j_1}^\dagger}{4\epsilon M_F}$ $-\frac{3h\overline{y_{1, i_1, r_1} y_{1, i_3, r_2} y_{2, i_2, r_2}}^T C P_L L_{a_1, i_1} S_{2, j_1} y_{1, i_3, r_2} \mathbb{I}_{T3, a_1, a_2} S_{1, j_1}^\dagger}{4\epsilon M_F}$ $-\frac{3h\overline{y_{1, i_1, r_1} y_{1, i_3, r_2} y_{2, i_2, r_2}}^T C P_L L_{a_1, i_1} S_{2, j_1} y_{1, i_3, r_2} \mathbb{I}_{T3, a_1, a_2} S_{1, j_1}^\dagger}{4\epsilon M_F}$ $-\frac{3h\overline{y_{1, i_1, r_1} y_{1, i_3, r_2} y_{2, i_2, r_2}}^T C P_L L_{a_1, i_1} S_{2, j_1} y_{1, i_3, r_2} \mathbb{I}_{T3, a_1, a_2} S_{1, j_1}^\dagger}{4\epsilon M_F}$ $-\frac{3h\overline{y_{1, i_1, r_1} y_{1, i_3, r_2} y_{2, i_2, r_2}}^T C P_L L_{a_1, i_1} S_{2, j_1} y_{1, i_3, r_2} \mathbb{I}_{T3, a_1, a_2} S_{1, j_1}^\dagger}{4\epsilon M_F}$ $-\frac{3h\overline{y_{1, i_1, r_1} y_{1, i_3, r_2} y_{2, i_2, r_2}}^T C P_L L_{a_1, i_1} S_{2, j_1} y_{1, i_3, r_2} \mathbb{I}_{T3, a_1, a_2} S_{1, j_1}^\dagger}{4\epsilon M_F}$ $-\frac{3h\overline{y_{1, i_1, r_1} y_{1, i_3, r_2} y_{2, i_2, r_2}}^T C P_L L_{a_1, i_1} S_{2, j_1} y_{1, i_3, r_2} \mathbb{I}_{T3, a_1, a_2} S_{1, j_1}^\dagger}{4\epsilon M_F}$ $-\frac{3h\overline{y_{1, i$	

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	S_1	S_2	F	S_1	S_2	ℓ	S_1	F	H
cont.													$-\frac{h\overline{y}_{1,i_1,r_1}\overline{y}_{2,i_2,r_1}\overline{y}_{2,i_3,r_2}L_{a_2,i_2}^TCPL_{a_1,i_1}S_{2,j_1}\overline{y}_{2,i_3,r_2}\epsilon_{a_1,a_2}S_{1,j_1}^\dagger}{6M_F}$			
													$-\frac{h\overline{y}_{1,i_3,r_1}L_{a_1,i_1}CP_R\overline{L}_{a_2,i_2}^TS_{1,j_2}y_{1,i_2,r_1}y_{1,i_3,r_2}\overline{y}_{2,i_1,r_2}Tr_{a_1}^*a_{2,j_2,j_1}S_{2,j_1}^\dagger}{2\sqrt{2}M_F}$			
													$-\frac{3h\overline{y}_{1,i_3,r_1}L_{a_1,i_1}CP_R\overline{L}_{a_2,i_2}^TS_{1,j_2}y_{1,i_3,r_1}y_{1,i_3,r_2}\overline{y}_{2,i_1,r_2}Tr_{a_1}^*a_{2,j_2,j_1}S_{2,j_1}^\dagger}{4\sqrt{2}M_F}$			
cont.													$-\frac{\sqrt{6}hL_{a_1,i_1}CP_R\overline{L}_{a_2,i_2}^Tg_2^2S_{1,j_2}y_{1,i_1,r_1}y_{2,i_2,r_1}Tr_{a_1}^*a_{2,j_2,j_1}S_{2,j_1}^\dagger}{\epsilon M_F}$			
													$-\frac{\sqrt{\frac{3}{2}}hL_{a_1,i_1}CP_R\overline{L}_{a_2,i_2}^Tg_2^2S_{1,j_2}y_{1,i_1,r_1}y_{2,i_2,r_1}Tr_{a_1}^*a_{2,j_2,j_1}S_{2,j_1}^\dagger}{M_F}$			
													$-\frac{3\sqrt{\frac{3}{2}}hL_{a_1,i_1}CP_R\overline{L}_{a_2,i_2}^Tg_Y^2S_{1,j_2}y_{1,i_1,r_1}y_{2,i_2,r_1}Tr_{a_1}^*a_{2,j_2,j_1}S_{2,j_1}^\dagger}{2\epsilon M_F}$			
cont.													$-\frac{\sqrt{\frac{3}{2}}hL_{a_1,i_1}CP_R\overline{L}_{a_2,i_2}^Tg_Y^2S_{1,j_2}y_{1,i_1,r_1}y_{2,i_2,r_1}Tr_{a_1}^*a_{2,j_2,j_1}S_{2,j_1}^\dagger}{2M_F}$			
													$+\frac{\sqrt{\frac{3}{2}}L_{a_1,i_1}CP_R\overline{L}_{a_2,i_2}^TS_{1,j_2}y_{1,i_1,r_1}y_{2,i_2,r_1}Tr_{a_1}^*a_{2,j_2,j_1}S_{2,j_1}^\dagger}{M_F}$			
													$-\frac{h\overline{y}_{2,i_3,r_1}L_{a_1,i_1}CP_R\overline{L}_{a_2,i_2}^TS_{1,j_2}y_{1,i_3,r_2}\overline{y}_{2,i_1,r_2}\overline{y}_{2,i_2,r_1}Tr_{a_1}^*a_{2,j_2,j_1}S_{2,j_1}^\dagger}{2\sqrt{2}M_F}$			
cont.													$-\frac{3h\overline{y}_{2,i_3,r_1}L_{a_1,i_1}CP_R\overline{L}_{a_2,i_2}^TS_{1,j_2}y_{1,i_3,r_2}\overline{y}_{2,i_1,r_2}\overline{y}_{2,i_2,r_1}Tr_{a_1}^*a_{2,j_2,j_1}S_{2,j_1}^\dagger}{4\sqrt{2}M_F}$			
													$-\frac{\sqrt{2}h\overline{y}_{1,i_3,r_1}L_{a_1,i_1}CP_R\overline{L}_{a_2,i_2}^TS_{1,j_2}y_{1,i_1,r_2}y_{1,i_3,r_1}\overline{y}_{2,i_2,r_2}Tr_{a_1}^*a_{2,j_2,j_1}S_{2,j_1}^\dagger}{3\epsilon M_F}$			
													$-\frac{h\overline{y}_{1,i_3,r_1}L_{a_1,i_1}CP_R\overline{L}_{a_2,i_2}^TS_{1,j_2}y_{1,i_1,r_2}y_{1,i_3,r_1}\overline{y}_{2,i_2,r_2}Tr_{a_1}^*a_{2,j_2,j_1}S_{2,j_1}^\dagger}{3\sqrt{2}M_F}$			
cont.													$-\frac{\sqrt{2}h\overline{y}_{2,i_3,r_1}L_{a_1,i_1}CP_R\overline{L}_{a_2,i_2}^TS_{1,j_2}y_{1,i_1,r_2}\overline{y}_{2,i_2,r_2}\overline{y}_{2,i_3,r_1}Tr_{a_1}^*a_{2,j_2,j_1}S_{2,j_1}^\dagger}{3\epsilon M_F}$			
													$-\frac{h\overline{y}_{2,i_3,r_1}L_{a_1,i_1}CP_R\overline{L}_{a_2,i_2}^TS_{1,j_2}y_{1,i_1,r_2}\overline{y}_{2,i_2,r_2}\overline{y}_{2,i_3,r_1}Tr_{a_1}^*a_{2,j_2,j_1}S_{2,j_1}^\dagger}{3\sqrt{2}M_F}$			
													$-\frac{3h\overline{y}_{1,i_3,r_1}L_{a_1,i_1}CP_R\overline{L}_{a_2,i_2}^TS_{1,j_2}y_{1,i_1,r_2}y_{1,i_3,r_2}\overline{y}_{2,i_1,r_2}Tr_{a_1}^*a_{2,j_2,j_1}S_{2,j_1}^\dagger}{4\sqrt{2}M_F}$			
cont.													$-\frac{h\overline{y}_{2,i_3,r_1}L_{a_1,i_1}CP_R\overline{L}_{a_2,i_2}^TS_{1,j_2}y_{1,i_1,r_2}\overline{y}_{2,i_2,r_2}Tr_{a_1}^*a_{2,j_2,j_1}S_{2,j_1}^\dagger}{2\sqrt{2}M_F}$			
													$-\frac{3h\overline{y}_{1,i_3,r_1}L_{a_1,i_1}CP_R\overline{L}_{a_2,i_2}^TS_{1,j_2}y_{1,i_1,r_2}y_{1,i_3,r_2}\overline{y}_{2,i_1,r_2}Tr_{a_1}^*a_{2,j_2,j_1}S_{2,j_1}^\dagger}{4\sqrt{2}M_F}$			
													$+\frac{3h\overline{y}_{1,i_3,r_1}L_{a_1,i_1}CP_R\overline{L}_{a_2,i_2}^TS_{1,j_1}y_{1,i_2,r_1}y_{1,i_3,r_2}\overline{y}_{2,i_1,r_2}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger}{4\epsilon M_F}$			
cont.													$+\frac{3h\overline{y}_{1,i_3,r_1}L_{a_1,i_1}CP_R\overline{L}_{a_2,i_2}^TS_{1,j_1}y_{1,i_2,r_1}y_{1,i_3,r_2}\overline{y}_{2,i_1,r_2}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger}{8M_F}$			
													$-\frac{2\sqrt{3}hL_{a_1,i_1}CP_R\overline{L}_{a_2,i_2}^Tg_2^2S_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger}{\epsilon M_F}$			
													$-\frac{2hL_{a_1,i_1}CP_R\overline{L}_{a_2,i_2}^Tg_2^2S_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger}{\sqrt{3}M_F}$			
cont.													$-\frac{3\sqrt{3}hL_{a_1,i_1}CP_R\overline{L}_{a_2,i_2}^Tg_Y^2S_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger}{4\epsilon M_F}$			
													$-\frac{\sqrt{3}hL_{a_1,i_1}CP_R\overline{L}_{a_2,i_2}^Tg_Y^2S_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger}{4M_F}$			
													$+\frac{\overline{L}_{a_1,i_1}CP_R\overline{L}_{a_2,i_2}^TS_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger}{\sqrt{3}M_F}$			
cont.													$+\frac{h\overline{y}_{2,i_3,r_1}L_{a_1,i_1}CP_R\overline{L}_{a_2,i_2}^TS_{1,j_1}y_{1,i_3,r_2}\overline{y}_{2,i_1,r_2}\overline{y}_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger}{4\epsilon M_F}$			
													$+\frac{3h\overline{y}_{2,i_3,r_1}L_{a_1,i_1}CP_R\overline{L}_{a_2,i_2}^TS_{1,j_1}y_{1,i_3,r_2}\overline{y}_{2,i_1,r_2}\overline{y}_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger}{8M_F}$			
													$-\frac{h\overline{y}_{1,i_3,r_1}L_{a_1,i_1}CP_R\overline{L}_{a_2,i_2}^TS_{1,j_1}y_{1,i_1,r_2}y_{1,i_3,r_1}\overline{y}_{2,i_2,r_2}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger}{3\epsilon M_F}$			
cont.													$-\frac{h\overline{y}_{1,i_3,r_1}L_{a_1,i_1}CP_R\overline{L}_{a_2,i_2}^TS_{1,j_1}y_{1,i_1,r_2}y_{1,i_3,r_1}\overline{y}_{2,i_2,r_2}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger}{6M_F}$			
													$-\frac{h\overline{y}_{2,i_3,r_1}L_{a_1,i_1}CP_R\overline{L}_{a_2,i_2}^TS_{1,j_1}y_{1,i_1,r_2}y_{2,i_2,r_2}\overline{y}_{2,i_3,r_1}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger}{6M_F}$			
													$-\frac{h\overline{y}_{2,i_3,r_1}L_{a_1,i_1}CP_R\overline{L}_{a_2,i_2}^TS_{1,j_1}y_{1,i_1,r_2}y_{2,i_2,r_2}\overline{y}_{2,i_3,r_1}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger}{6M_F}$			
cont.													$-\frac{h\overline{y}_{1,i_3,r_1}L_{a_1,i_1}CP_R\overline{L}_{a_2,i_2}^TS_{1,j_1}y_{1,i_1,r_2}y_{1,i_3,r_1}\overline{y}_{2,i_2,r_2}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger}{4\epsilon M_F}$			
													$-\frac{3h\overline{y}_{1,i_3,r_1}L_{a_1,i_1}CP_R\overline{L}_{a_2,i_2}^TS_{1,j_1}y_{1,i_1,r_2}y_{1,i_3,r_1}\overline{y}_{2,i_2,r_2}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger}{8M_F}$			
													$-\frac{h\overline{y}_{2,i_3,r_1}L_{a_1,i_1}CP_R\overline{L}_{a_2,i_2}^TS_{1,j_1}y_{1,i_1,r_2}\overline{y}_{2,i_3,r_1}\overline{y}_{2,i_3,r_2}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger}{4\epsilon M_F}$			
cont.													$-\frac{3h\overline{y}_{2,i_3,r_1}L_{a_1,i_1}CP_R\overline{L}_{a_2,i_2}^TS_{1,j_1}y_{1,i_1,r_2}\overline{y}_{2,i_3,r_1}\overline{y}_{2,i_3,r_2}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger}{8M_F}$			
													$+\frac{\sqrt{\frac{3}{2}}h\overline{y}_{1,i_1,r_1}\overline{y}_{2,i_2,r_1}L_{a_2,i_2}^TCPL_{a_1,i_1}S_{2,j_2}Tr_{a_1}^*a_{2,j_1,j_2}S_{1,j_1}^\dagger\lambda_{12}^{(1)}}{\epsilon M_F}$			
													$+\frac{\sqrt{\frac{3}{2}}h\overline{y}_{1,i_1,r_1}\overline{y}_{2,i_2,r_1}L_{a_2,i_2}^TCPL_{a_1,i_1}S_{2,j_2}Tr_{a_1}^*a_{2,j_1,j_2}S_{1,j_1}^\dagger\lambda_{12}^{(1)}}{M_F}$			
cont.													$-\frac{h\overline{y}_{1,i_1,r_1}\overline{y}_{2,i_2,r_1}L_{a_2,i_2}^TCPL_{a_1,i_1}S_{2,j_1}\epsilon_{a_1,a_2}S_{1,j_1}^\dagger\lambda_{12}^{(1)}}{\sqrt{3}\epsilon M_F}$			
													$-\frac{h\overline{y}_{1,i_1,r_1}\overline{y}_{2,i_2,r_1}L_{a_2,i_2}^TCPL_{a_1,i_1}S_{2,j_1}\epsilon_{a_1,a_2}S_{1,j_1}^\dagger\lambda_{12}^{(1)}}{\sqrt{3}M_F}$			

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	S_1	S_2	F	S_1	S_2	ℓ	S_1	F	H
cont.													$+\frac{\sqrt{\frac{2}{3}}h\overline{L_{a_1,i_1}}CPr\overline{L_{a_2,i_2}}^TS_{1,j_2}y_{1,i_1,r_1}y_{2,i_2,r_1}\mathcal{I}T\tilde{a}_{a_1,a_2,j_2,j_1}^*S_{2,j_1}^\dagger\lambda_{12}^{(1)}}{\epsilon M_F}$			
													$+\frac{\sqrt{\frac{2}{3}}h\overline{L_{a_1,i_1}}CPr\overline{L_{a_2,i_2}}^TS_{1,j_2}y_{1,i_1,r_1}y_{2,i_2,r_1}\mathcal{I}T\tilde{a}_{a_1,a_2,j_2,j_1}^*S_{2,j_1}^\dagger\lambda_{12}^{(1)}}{M_F}$ $-\frac{h\overline{L_{a_1,i_1}}CPr\overline{L_{a_2,i_2}}^TS_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger\lambda_{12}^{(1)}}{\sqrt{3}\epsilon M_F}$ $-\frac{h\overline{L_{a_1,i_1}}CPr\overline{L_{a_2,i_2}}^TS_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger\lambda_{12}^{(1)}}{\sqrt{3}M_F}$			
													$-3\sqrt{\frac{2}{11}}h\overline{y}_{1,i_1,r_1}\overline{y}_{2,i_2,r_1}L_{a_2,i_2}^TCPL_{a_1,i_1}S_{2,j_2}\mathcal{I}T\tilde{a}_{a_1,a_2,j_1,j_2}S_{1,j_1}^\dagger\lambda_{12}^{(2)}$ $-\frac{3\sqrt{\frac{2}{11}}h\overline{y}_{1,i_1,r_1}\overline{y}_{2,i_2,r_1}L_{a_2,i_2}^TCPL_{a_1,i_1}S_{2,j_2}\mathcal{I}T\tilde{a}_{a_1,a_2,j_1,j_2}S_{1,j_1}^\dagger\lambda_{12}^{(2)}}{M_F}$ $-\frac{h\overline{y}_{1,i_1,r_1}\overline{y}_{2,i_2,r_1}L_{a_2,i_2}^TCPL_{a_1,i_1}S_{2,j_1}\epsilon_{a_1,a_2}S_{1,j_1}^\dagger\lambda_{12}^{(2)}}{\sqrt{11}\epsilon M_F}$ $-\frac{h\overline{y}_{1,i_1,r_1}\overline{y}_{2,i_2,r_1}L_{a_2,i_2}^TCPL_{a_1,i_1}S_{2,j_1}\epsilon_{a_1,a_2}S_{1,j_1}^\dagger\lambda_{12}^{(2)}}{\sqrt{11}M_F}$			
cont.													$-3\sqrt{\frac{2}{11}}h\overline{L_{a_1,i_1}}CPr\overline{L_{a_2,i_2}}^TS_{1,j_2}y_{1,i_1,r_1}y_{2,i_2,r_1}\mathcal{I}T\tilde{a}_{a_1,a_2,j_2,j_1}^*S_{2,j_1}^\dagger\lambda_{12}^{(2)}$ $-\frac{3\sqrt{\frac{2}{11}}h\overline{L_{a_1,i_1}}CPr\overline{L_{a_2,i_2}}^TS_{1,j_2}y_{1,i_1,r_1}y_{2,i_2,r_1}\mathcal{I}T\tilde{a}_{a_1,a_2,j_2,j_1}^*S_{2,j_1}^\dagger\lambda_{12}^{(2)}}{M_F}$ $-\frac{h\overline{L_{a_1,i_1}}CPr\overline{L_{a_2,i_2}}^TS_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger\lambda_{12}^{(2)}}{\sqrt{11}\epsilon M_F}$ $-\frac{h\overline{L_{a_1,i_1}}CPr\overline{L_{a_2,i_2}}^TS_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger\lambda_{12}^{(2)}}{\sqrt{11}M_F}$			
													$-7h\overline{y}_{1,i_1,r_1}\overline{y}_{2,i_2,r_1}L_{a_2,i_2}^TCPL_{a_1,i_1}S_{2,j_1}\epsilon_{a_1,a_2}S_{1,j_1}^\dagger\lambda_{12}^{(3)}$ $-\frac{7h\overline{y}_{1,i_1,r_1}\overline{y}_{2,i_2,r_1}L_{a_2,i_2}^TCPL_{a_1,i_1}S_{2,j_1}\epsilon_{a_1,a_2}S_{1,j_1}^\dagger\lambda_{12}^{(3)}}{\sqrt{23}\epsilon M_F}$ $-\frac{7h\overline{L_{a_1,i_1}}CPr\overline{L_{a_2,i_2}}^TS_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger\lambda_{12}^{(3)}}{\sqrt{23}\epsilon M_F}$ $-\frac{7h\overline{L_{a_1,i_1}}CPr\overline{L_{a_2,i_2}}^TS_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger\lambda_{12}^{(3)}}{\sqrt{23}M_F}$			
													$-7h\overline{L_{a_1,i_1}}CPr\overline{L_{a_2,i_2}}^TS_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger\lambda_{12}^{(3)}$ $-\frac{7h\overline{L_{a_1,i_1}}CPr\overline{L_{a_2,i_2}}^TS_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger\lambda_{12}^{(3)}}{\sqrt{23}\epsilon M_F}$ $-\frac{7h\overline{L_{a_1,i_1}}CPr\overline{L_{a_2,i_2}}^TS_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger\lambda_{12}^{(3)}}{\sqrt{23}M_F}$			
cont.													$-\frac{2\sqrt{3}h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y}_{1,i_1,r_1}Y_{e,i_2,i_3}H_{a_1}S_{1,j_1}y_{1,i_2,r_1}Y_{e,i_1,i_3}H_{a_1}^\dagger S_{1,j_1}^\dagger}{\sqrt{3}}$			
													$-2i\sqrt{\frac{2}{3}}h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y}_{1,i_1,r_1}Y_{e,i_2,i_3}H_{a_2}S_{1,j_2}y_{1,i_2,r_1}Y_{e,i_1,i_3}H_{a_1}^\dagger S_{1,j_1}^\dagger\mathcal{I}H_{1,a_1,a_2,j_2,j_1}^\dagger$			
													$-\frac{2h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y}_{1,i_1,r_1}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}H_{a_1}^\dagger S_{1,j_1}^\dagger\mathcal{I}H_{1,a_1,a_2,j_2,j_1}^\dagger2\lambda_{12}^{(2)}}{\sqrt{3}}$			
cont.													$-\frac{2\sqrt{6}h\overline{y}_{1,i_1,r_1}\overline{y}_{2,i_2,r_1}L_{a_2,i_2}^TCPL_{a_1,i_1}g_2^2S_{2,j_2}\mathcal{I}T\tilde{a}_{a_1,a_2,j_1,j_2}S_{1,j_1}^\dagger}{\epsilon M_F}$			
													$-\frac{2\sqrt{\frac{2}{3}}h\overline{y}_{1,i_1,r_1}\overline{y}_{2,i_2,r_1}L_{a_2,i_2}^TCPL_{a_1,i_1}g_2^2S_{2,j_2}\mathcal{I}T\tilde{a}_{a_1,a_2,j_1,j_2}S_{1,j_1}^\dagger}{M_F}$ $+\frac{\sqrt{\frac{2}{3}}h\overline{y}_{1,i_1,r_1}\overline{y}_{2,i_2,r_1}L_{a_2,i_2}^TCPL_{a_1,i_1}S_{2,j_2}\mathcal{I}T\tilde{a}_{a_1,a_2,j_1,j_2}S_{1,j_1}^\dagger}{M_F}$			
													$-\frac{h\overline{y}_{1,i_1,r_1}\overline{y}_{1,i_3,r_2}\overline{y}_{2,i_2,r_2}L_{a_2,i_2}^TCPL_{a_1,i_1}S_{2,j_2}y_{1,i_3,r_1}\mathcal{I}T\tilde{a}_{a_1,a_2,j_1,j_2}S_{1,j_1}^\dagger}{2\sqrt{2}\epsilon M_F}$			
cont.													$-\frac{3h\overline{y}_{1,i_1,r_1}\overline{y}_{1,i_3,r_2}\overline{y}_{2,i_2,r_2}L_{a_2,i_2}^TCPL_{a_1,i_1}S_{2,j_2}y_{1,i_3,r_1}\mathcal{I}T\tilde{a}_{a_1,a_2,j_1,j_2}S_{1,j_1}^\dagger}{4\sqrt{2}M_F}$			
													$-\frac{\sqrt{2}h\overline{y}_{1,i_1,r_1}\overline{y}_{1,i_3,r_2}\overline{y}_{2,i_2,r_2}L_{a_2,i_2}^TCPL_{a_1,i_1}S_{2,j_2}y_{1,i_3,r_2}\mathcal{I}T\tilde{a}_{a_1,a_2,j_1,j_2}S_{1,j_1}^\dagger}{3\epsilon M_F}$			
													$-\frac{h\overline{y}_{1,i_1,r_1}\overline{y}_{1,i_3,r_2}\overline{y}_{2,i_2,r_2}L_{a_2,i_2}^TCPL_{a_1,i_1}S_{2,j_2}y_{1,i_3,r_2}\mathcal{I}T\tilde{a}_{a_1,a_2,j_1,j_2}S_{1,j_1}^\dagger}{3\sqrt{2}M_F}$			
cont.													$-\frac{h\overline{y}_{1,i_1,r_1}\overline{y}_{1,i_3,r_2}\overline{y}_{2,i_2,r_2}L_{a_2,i_2}^TCPL_{a_1,i_1}S_{2,j_2}y_{1,i_3,r_2}\mathcal{I}T\tilde{a}_{a_1,a_2,j_1,j_2}S_{1,j_1}^\dagger}{2\sqrt{2}\epsilon M_F}$			
													$-\frac{3h\overline{y}_{1,i_1,r_1}\overline{y}_{1,i_3,r_2}\overline{y}_{2,i_2,r_2}L_{a_2,i_2}^TCPL_{a_1,i_1}S_{2,j_2}y_{1,i_3,r_2}\mathcal{I}T\tilde{a}_{a_1,a_2,j_1,j_2}S_{1,j_1}^\dagger}{4\sqrt{2}M_F}$			
													$-\frac{h\overline{y}_{1,i_3,r_1}\overline{y}_{2,i_1,r_2}\overline{y}_{1,i_1,r_1}L_{a_2,i_2}^TCPL_{a_1,i_1}S_{2,j_2}y_{2,i_3,r_1}\mathcal{I}T\tilde{a}_{a_1,a_2,j_1,j_2}S_{1,j_1}^\dagger}{2\sqrt{2}\epsilon M_F}$			
cont.													$-\frac{3h\overline{y}_{1,i_1,r_1}\overline{y}_{2,i_1,r_2}\overline{y}_{2,i_3,r_1}L_{a_2,i_2}^TCPL_{a_1,i_1}S_{2,j_2}y_{2,i_3,r_2}\mathcal{I}T\tilde{a}_{a_1,a_2,j_1,j_2}S_{1,j_1}^\dagger}{4\sqrt{2}M_F}$			
													$-\frac{h\overline{y}_{1,i_1,r_1}\overline{y}_{2,i_1,r_2}\overline{y}_{2,i_3,r_1}L_{a_2,i_2}^TCPL_{a_1,i_1}S_{2,j_2}y_{2,i_3,r_2}\mathcal{I}T\tilde{a}_{a_1,a_2,j_1,j_2}S_{1,j_1}^\dagger}{2\sqrt{2}\epsilon M_F}$			
													$-\frac{3h\overline{y}_{1,i_1,r_1}\overline{y}_{2,i_1,r_2}\overline{y}_{2,i_3,r_1}L_{a_2,i_2}^TCPL_{a_1,i_1}S_{2,j_2}y_{2,i_3,r_2}\mathcal{I}T\tilde{a}_{a_1,a_2,j_1,j_2}S_{1,j_1}^\dagger}{4\sqrt{2}M_F}$			
cont.													$-\frac{2\sqrt{3}h\overline{y}_{1,i_1,r_1}\overline{y}_{2,i_2,r_1}\overline{y}_{2,i_3,r_2}L_{a_2,i_2}^TCPL_{a_1,i_1}S_{2,j_2}y_{2,i_3,r_2}\mathcal{I}T\tilde{a}_{a_1,a_2,j_1,j_2}S_{1,j_1}^\dagger}{3\epsilon M_F}$			
													$-\frac{h\overline{y}_{1,i_1,r_1}\overline{y}_{2,i_2,r_1}\overline{y}_{2,i_3,r_2}L_{a_2,i_2}^TCPL_{a_1,i_1}S_{2,j_2}y_{2,i_3,r_2}\mathcal{I}T\tilde{a}_{a_1,a_2,j_1,j_2}S_{1,j_1}^\dagger}{3\sqrt{2}M_F}$			
													$-\frac{2\sqrt{3}h\overline{y}_{1,i_1,r_1}\overline{y}_{2,i_2,r_1}\overline{y}_{2,i_3,r_2}L_{a_2,i_2}^TCPL_{a_1,i_1}g_2^2S_{2,j_2}\epsilon_{a_1,a_2}S_{1,j_1}^\dagger}{\epsilon M_F}$			

[illegible]

[illegible]

Field combinations 8 of 10

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$S_1 F$	ℓ	$S_2 F$	ℓ
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$				
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0				
T3-A	0	1	0	3	1	2	$\frac{1}{2}$				
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1				
T3-A	2	1	1	3	2	2	$\frac{3}{2}$				
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$				
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0				
cont.											
cont.											
cont.											
T3-B	0	2	0	2	1	1	$\frac{1}{2}$				

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$S_1 F \ell$	$S_2 F H$	$S_2 F \ell$	
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1		$-2h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1}} Y_{e,i_2,i_3} H_{a_2} S_{2,j_2} y_{2,i_2,r_1} Y_{e,i_1,i_3} T_{y_{2,a_1,j_1}} I_{y_{2,a_2,j_2}} H_{a_1}^\dagger S_{2,j_1}^\dagger$ $-h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1}} H_{a_2} S_{2,j_2} y_{2,i_1,r_1} H_{a_1}^\dagger S_{2,j_1}^\dagger \mathcal{I}_{H2,a_1,a_2,j_1,j_2}^{1\lambda_{H2}^{(1)}}$ $-h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1}} H_{a_2} S_{2,j_2} y_{2,i_1,r_1} H_{a_1}^\dagger S_{2,j_1}^\dagger \mathcal{I}_{H2,a_1,a_2,j_1,j_2}^{2\lambda_{H2}^{(2)}}$		
T3-B	2	2	1	2	2	1	$\frac{3}{2}$		$-2h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1}} Y_{e,i_2,i_3} H_{a_2} S_{2,j_2} y_{2,i_2,r_1} Y_{e,i_1,i_3} T_{y_{2,a_1,j_1}} I_{y_{2,a_2,j_2}} H_{a_1}^\dagger S_{2,j_1}^\dagger$ $-h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1}} H_{a_2} S_{2,j_2} y_{2,i_1,r_1} H_{a_1}^\dagger S_{2,j_1}^\dagger \mathcal{I}_{H2,a_1,a_2,j_1,j_2}^{1\lambda_{H2}^{(1)}}$ $-h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1}} H_{a_2} S_{2,j_2} y_{2,i_1,r_1} H_{a_1}^\dagger S_{2,j_1}^\dagger \mathcal{I}_{H2,a_1,a_2,j_1,j_2}^{2\lambda_{H2}^{(2)}}$		
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$		$-\frac{2h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1}} Y_{e,i_2,i_3} H_{a_1} S_{2,j_1} y_{2,i_2,r_1} Y_{e,i_1,i_3} H_{a_1}^\dagger S_{2,j_1}^\dagger}{\sqrt{3}}$ $2h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1}} Y_{e,i_2,i_3} H_{a_2} S_{2,j_2} y_{2,i_2,r_1} Y_{e,i_1,i_3} H_{a_1}^\dagger S_{2,j_1}^\dagger \mathcal{I}_{H1,a_1,a_2,j_1,j_2}^{-1}$ $-\sqrt{3} h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1}} H_{a_2} S_{2,j_2} y_{2,i_1,r_1} H_{a_1}^\dagger S_{2,j_1}^\dagger \mathcal{I}_{H2,a_1,a_2,j_1,j_2}^{1\lambda_{H2}^{(1)}}$ $-\sqrt{3} h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1}} H_{a_2} S_{2,j_2} y_{2,i_1,r_1} H_{a_1}^\dagger S_{2,j_1}^\dagger \mathcal{I}_{H2,a_1,a_2,j_1,j_2}^{2\lambda_{H2}^{(2)}}$		
cont.											
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0		$+\frac{3h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}} c \bar{\epsilon} q_{a_1,j_1,a_2,j_2}^T S_{i,j_1} S_{i,j_2} y_{l,i_1,r_1} y_{l,i_2,r_1} y_{l,i_2,r_1}^2}{8 M_F}$ $+\frac{3h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{l,i_1,r_1}} y_{l,i_2,r_1} L_{a_1,i_1}{}^T C P_L L_{a_1,i_1} \text{Bar}(c g 2)_{a_1,j_1,a_2,j_2} S_{l,j_1}^\dagger S_{l,j_2}^\dagger g_2^2}{8 M_F}$ $+\frac{3h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}} c \bar{\epsilon} q_{a_1,j_1,a_2,j_2}^T S_{i,j_1} S_{i,j_2} y_{l,i_1,r_1} y_{l,i_2,r_1}}{8 M_F}$ $+\frac{\sqrt{3} h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{l,i_3,r_1}} L_{a_1,i_1} C P_R \overline{L_{a_2,i_2}} c \bar{\epsilon} q_{a_1,j_1,a_2,j_2}^T S_{i,j_1} S_{i,j_2} y_{l,i_1,r_2} y_{l,i_2,r_2} y_{l,i_3,r_1}}{2 M_F}$	$-\frac{2h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1}} Y_{e,i_2,i_3} H_{a_1} S_{2,j_1} y_{2,i_2,r_1} Y_{e,i_1,i_3} H_{a_1}^\dagger S_{2,j_1}^\dagger}{\sqrt{3}}$ $2h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1}} Y_{e,i_2,i_3} H_{a_2} S_{2,j_2} y_{2,i_2,r_1} Y_{e,i_1,i_3} H_{a_1}^\dagger S_{2,j_1}^\dagger \mathcal{I}_{H1,a_1,a_2,j_1,j_2}^{-1}$ $-\sqrt{3} h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1}} H_{a_2} S_{2,j_2} y_{2,i_1,r_1} H_{a_1}^\dagger S_{2,j_1}^\dagger \mathcal{I}_{H2,a_1,a_2,j_1,j_2}^{1\lambda_{H2}^{(1)}}$ $-\sqrt{3} h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1}} H_{a_2} S_{2,j_2} y_{2,i_1,r_1} H_{a_1}^\dagger S_{2,j_1}^\dagger \mathcal{I}_{H2,a_1,a_2,j_1,j_2}^{2\lambda_{H2}^{(2)}}$	$+\frac{3h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1}} y_{2,i_2,r_1} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_1} S_{2,j_2} \text{Bar}(c g 1)_{a_1,j_1,a_2,j_2} y_{2,i_2}^2}{8 M_F}$ $+\frac{3h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}} c \bar{\epsilon} q_{a_1,j_1,a_2,j_2}^T y_{2,i_1,r_1} y_{2,i_2,r_1} S_{2,j_1}^\dagger S_{2,j_2}^\dagger y_{2,i_2}^2}{8 M_F}$ $+\frac{3h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1}} y_{2,i_2,r_1} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} y_{2,i_1,r_1}^2 S_{2,j_1} S_{2,j_2} \text{Bar}(c g 1)_{a_1,j_1,a_2,j_2}}{8 M_F}$ $-\frac{\sqrt{3} h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{l,i_1,r_1}} y_{l,i_2,r_2} y_{l,i_3,r_2} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_1} S_{2,j_2} y_{l,i_3,r_1} \text{Bar}(c g 1)_{a_1,j_1,a_2,j_2}}{4 M_F}$ $-\frac{\sqrt{3} h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{l,i_1,r_1}} y_{l,i_2,r_2} y_{l,i_3,r_2} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_1} S_{2,j_2} y_{l,i_3,r_2} \text{Bar}(c g 1)_{a_1,j_1,a_2,j_2}}{4 M_F}$ $+\frac{\sqrt{3} h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{l,i_3,r_1}} L_{a_1,i_1} C P_R \overline{L_{a_2,i_2}} c \bar{\epsilon} q_{a_1,j_1,a_2,j_2}^T S_{i,j_1} S_{i,j_2} y_{l,i_1,r_2} y_{l,i_3,r_2} y_{2,i_2,r_2}}{4 M_F}$ $-\frac{\sqrt{3} h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{2,i_3,r_1}} L_{a_1,i_1} C P_R \overline{L_{a_2,i_2}} c \bar{\epsilon} q_{a_1,j_1,a_2,j_2}^T S_{i,j_1} S_{i,j_2} y_{l,i_1,r_2} y_{l,i_3,r_1} y_{2,i_2,r_2}}{M_F}$ $+\frac{\sqrt{3} h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{l,i_1,r_1}} y_{l,i_2,r_2} y_{l,i_3,r_2} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} y_{l,i_3,r_2} \text{Bar}(c g 2)_{a_1,j_1,a_2,j_2} S_{l,j_1}^\dagger S_{l,j_2}^\dagger}{4 M_F}$ $+\frac{\sqrt{3} h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{l,i_1,r_1}} y_{l,i_2,r_2} y_{l,i_3,r_2} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} y_{l,i_3,r_2} \text{Bar}(c g 2)_{a_1,j_1,a_2,j_2} S_{l,j_1}^\dagger S_{l,j_2}^\dagger}{2 M_F}$ $+\frac{\sqrt{3} h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{l,i_1,r_1}} y_{l,i_2,r_2} y_{l,i_3,r_2} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} y_{l,i_3,r_2} \text{Bar}(c g 2)_{a_1,j_1,a_2,j_2} S_{l,j_1}^\dagger S_{l,j_2}^\dagger}{M_F}$ $+\frac{\sqrt{3} h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{l,i_1,r_1}} y_{l,i_2,r_2} y_{l,i_3,r_2} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} y_{l,i_3,r_2} \text{Bar}(c g 2)_{a_1,j_1,a_2,j_2} S_{l,j_1}^\dagger S_{l,j_2}^\dagger}{4 M_F}$ $+\frac{3h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{l,i_1,r_1}} y_{l,i_2,r_1} y_{l,i_2,r_1}^2 C P_L L_{a_1,i_1} g_2^2 \text{Bar}(c g 2)_{a_1,j_1,a_2,j_2} S_{l,i_1}^\dagger S_{l,i_2}^\dagger}{8 M_F}$ $-h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}} c \bar{\epsilon} q_{a_1,j_1,a_2,j_2}^T S_{i,j_1} S_{i,j_2} y_{l,i_1,r_1} y_{l,i_2,r_1} y_{l,i_2,r_1}^2 \lambda_{H2}^{(1)}$ $-h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{l,i_1,r_1}} y_{l,i_2,r_1} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} \text{Bar}(c g 2)_{a_1,j_1,a_2,j_2} S_{l,j_1}^\dagger S_{l,j_2}^\dagger \lambda_{H2}^{(1)}$
cont.											
T3-C	0	2	0	2	1	3	$\frac{1}{2}$		$-\frac{2h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1}} Y_{e,i_2,i_3} H_{a_1} S_{2,j_1} y_{2,i_2,r_1} Y_{e,i_1,i_3} H_{a_1}^\dagger S_{2,j_1}^\dagger}{\sqrt{3}}$ $2h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1}} Y_{e,i_2,i_3} H_{a_2} S_{2,j_2} y_{2,i_2,r_1} Y_{e,i_1,i_3} H_{a_1}^\dagger S_{2,j_1}^\dagger \mathcal{I}_{H1,a_1,a_2,j_1,j_2}^{-1}$ $-\sqrt{3} h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1}} H_{a_2} S_{2,j_2} y_{2,i_1,r_1} H_{a_1}^\dagger S_{2,j_1}^\dagger \mathcal{I}_{H2,a_1,a_2,j_1,j_2}^{1\lambda_{H2}^{(1)}}$ $-\sqrt{3} h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1}} H_{a_2} S_{2,j_2} y_{2,i_1,r_1} H_{a_1}^\dagger S_{2,j_1}^\dagger \mathcal{I}_{H2,a_1,a_2,j_1,j_2}^{2\lambda_{H2}^{(2)}}$		
cont.											
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1		$-\frac{2h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1}} Y_{e,i_2,i_3} H_{a_1} S_{2,j_1} y_{2,i_2,r_1} Y_{e,i_1,i_3} H_{a_1}^\dagger S_{2,j_1}^\dagger}{\sqrt{3}}$ $2h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1}} Y_{e,i_2,i_3} H_{a_2} S_{2,j_2} y_{2,i_2,r_1} Y_{e,i_1,i_3} H_{a_1}^\dagger S_{2,j_1}^\dagger \mathcal{I}_{H1,a_1,a_2,j_1,j_2}^{-1}$ $-\sqrt{3} h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1}} H_{a_2} S_{2,j_2} y_{2,i_1,r_1} H_{a_1}^\dagger S_{2,j_1}^\dagger \mathcal{I}_{H2,a_1,a_2,j_1,j_2}^{1\lambda_{H2}^{(1)}}$ $-\sqrt{3} h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1}} H_{a_2} S_{2,j_2} y_{2,i_1,r_1} H_{a_1}^\dagger S_{2,j_1}^\dagger \mathcal{I}_{H2,a_1,a_2,j_1,j_2}^{2\lambda_{H2}^{(2)}}$		
cont.											
T3-C	2	2	1	2	2	3	$\frac{3}{2}$		$-\frac{2h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1}} Y_{e,i_2,i_3} H_{a_1} S_{2,j_1} y_{2,i_2,r_1} Y_{e,i_1,i_3} H_{a_1}^\dagger S_{2,j_1}^\dagger}{\sqrt{3}}$ $2h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1}} Y_{e,i_2,i_3} H_{a_2} S_{2,j_2} y_{2,i_2,r_1} Y_{e,i_1,i_3} H_{a_1}^\dagger S_{2,j_1}^\dagger \mathcal{I}_{H1,a_1,a_2,j_1,j_2}^{-1}$ $-\sqrt{3} h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1}} H_{a_2} S_{2,j_2} y_{2,i_1,r_1} H_{a_1}^\dagger S_{2,j_1}^\dagger \mathcal{I}_{H2,a_1,a_2,j_1,j_2}^{1\lambda_{H2}^{(1)}}$ $-\sqrt{3} h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1}} H_{a_2} S_{2,j_2} y_{2,i_1,r_1} H_{a_1}^\dagger S_{2,j_1}^\dagger \mathcal{I}_{H2,a_1,a_2,j_1,j_2}^{2\lambda_{H2}^{(2)}}$		
cont.											

[illegible]

Field combinations 9 of 10

[illegible]

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	F	H	ℓ	S_1	S_2	F	H	S_1	S_2	F	ℓ		
cont.											$\frac{h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1}} H_{a_1} H_{a_2} S_1 y_{2,i_4,r_1} \mathcal{I} T_{3,a_1,a_2,j_1} \lambda_{T_3} S_{2,j_1}^\dagger}{\sqrt{3}}$				$\frac{\sqrt{3} h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_3,r_1} y_{2,i_1,r_2} y_{2,i_2,r_1} L_{a_2,i_2}^T} C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \mathcal{I} T_{3,a_1,a_2,j_1} S_1^\dagger}{4 M_F}$					
															$\frac{h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1} y_{2,i_3,r_2} L_{a_2,i_2}^T} C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \mathcal{I} T_{3,a_1,a_2,j_1} S_1^\dagger}{4 M_F}$					
												$\frac{h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_3,r_1} L_{a_1,i_1}} C P_R L_{a_2,i_2}^T S_1 y_{1,i_1,r_2} y_{2,i_2,r_2} y_{2,i_3,r_1} \mathcal{I} T_{3,a_1,a_2,j_1} S_2^\dagger}{\sqrt{3} M_F}$								
cont.											$\frac{h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{2,i_3,r_1} L_{a_1,i_1}} C P_R L_{a_2,i_2}^T S_1 y_{1,i_1,r_2} y_{2,i_2,r_2} y_{2,i_3,r_1} \mathcal{I} T_{3,a_1,a_2,j_1} S_2^\dagger}{4 M_F}$					$\frac{\sqrt{3} h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{2,i_3,r_1} L_{a_1,i_1}} C P_R L_{a_2,i_2}^T S_1 y_{1,i_1,r_2} y_{2,i_2,r_2} y_{2,i_3,r_1} \mathcal{I} T_{3,a_1,a_2,j_1} S_2^\dagger}{4 M_F}$				
															$\frac{h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_3,r_1} L_{a_1,i_1}} C P_R L_{a_2,i_2}^T S_1 y_{1,i_1,r_2} y_{2,i_2,r_2} y_{2,i_3,r_1} \mathcal{I} T_{3,a_1,a_2,j_1} S_2^\dagger}{4 M_F}$					
												$\frac{h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_3,r_1} L_{a_1,i_1}} C P_R L_{a_2,i_2}^T S_1 y_{1,i_1,r_2} y_{2,i_2,r_2} y_{2,i_3,r_1} \mathcal{I} T_{3,a_1,a_2,j_1} S_2^\dagger}{4 M_F}$								
cont.											$\frac{h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{2,i_3,r_1} L_{a_1,i_1}} C P_R L_{a_2,i_2}^T S_1 y_{1,i_1,r_2} y_{2,i_2,r_2} y_{2,i_3,r_1} \mathcal{I} T_{3,a_1,a_2,j_1} S_2^\dagger}{4 M_F}$					$\frac{\sqrt{3} h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{2,i_3,r_1} L_{a_1,i_1}} C P_R L_{a_2,i_2}^T S_1 y_{1,i_1,r_2} y_{2,i_2,r_2} y_{2,i_3,r_1} \mathcal{I} T_{3,a_1,a_2,j_1} S_2^\dagger}{4 M_F}$				
															$\frac{h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_3,r_1} L_{a_1,i_1}} C P_R L_{a_2,i_2}^T S_1 y_{1,i_1,r_2} y_{2,i_2,r_2} y_{2,i_3,r_1} \mathcal{I} T_{3,a_1,a_2,j_1} S_2^\dagger}{4 M_F}$					
												$\frac{h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_3,r_1} L_{a_1,i_1}} C P_R L_{a_2,i_2}^T S_1 y_{1,i_1,r_2} y_{2,i_2,r_2} y_{2,i_3,r_1} \mathcal{I} T_{3,a_1,a_2,j_1} S_2^\dagger}{4 M_F}$								
cont.											$\frac{h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{2,i_3,r_1} L_{a_1,i_1}} C P_R L_{a_2,i_2}^T S_1 y_{1,i_1,r_2} y_{2,i_2,r_2} y_{2,i_3,r_1} \mathcal{I} T_{3,a_1,a_2,j_1} S_2^\dagger}{4 M_F}$					$\frac{\sqrt{3} h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{2,i_3,r_1} L_{a_1,i_1}} C P_R L_{a_2,i_2}^T S_1 y_{1,i_1,r_2} y_{2,i_2,r_2} y_{2,i_3,r_1} \mathcal{I} T_{3,a_1,a_2,j_1} S_2^\dagger}{4 M_F}$				
															$\frac{h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_3,r_1} L_{a_1,i_1}} C P_R L_{a_2,i_2}^T S_1 y_{1,i_1,r_2} y_{2,i_2,r_2} y_{2,i_3,r_1} \mathcal{I} T_{3,a_1,a_2,j_1} S_2^\dagger}{4 M_F}$					
												$\frac{h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_3,r_1} L_{a_1,i_1}} C P_R L_{a_2,i_2}^T S_1 y_{1,i_1,r_2} y_{2,i_2,r_2} y_{2,i_3,r_1} \mathcal{I} T_{3,a_1,a_2,j_1} S_2^\dagger}{4 M_F}$								
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1				$\frac{+ \frac{2 h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1}} L_{a_4,i_2}^T C P_L L_{a_2,i_1} H_{a_1} H_{a_3} \lambda_{T_3} \epsilon_{a_1,a_2} \epsilon_{a_3,a_4}}{\sqrt{3} M_F}}{+ \frac{2 h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{\lambda_{T_3} L_{a_3,i_1}} C P_R L_{a_4,i_2}^T y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_3} \epsilon_{a_2,a_4} H_{a_1}^\dagger H_{a_2}^\dagger}{\sqrt{3} M_F}}$				$\frac{- h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} \lambda_{T_3} S_{2,j_1} y_{1,i_1,r_1} \mathcal{I} T_{3,a_1,a_2,j_1} H_{a_1}^\dagger H_{a_2}^\dagger S_1^\dagger}}{4 M_F}$					
											$\frac{h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1} \lambda_{T_3} S_{2,j_1} y_{2,i_1,r_1} \mathcal{I} T_{3,a_1,a_2,j_1} H_{a_1}^\dagger H_{a_2}^\dagger S_1^\dagger}}{\sqrt{3}}$					$\frac{h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{1,i_3,r_2} y_{2,i_2,r_1} L_{a_2,i_2}^T} C P_L L_{a_1,i_1} S_{2,j_1} y_{1,i_3,r_2} \mathcal{I} T_{3,a_1,a_2,j_1} S_1^\dagger}{4 M_F}$				
												$\frac{- h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1}} H_{a_1} H_{a_2} S_1 y_{1,i_1,r_1} \mathcal{I} T_{3,a_1,a_2,j_1} \lambda_{T_3} S_{2,j_1}^\dagger}{4 M_F}$								
cont.											$\frac{h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1}} H_{a_1} H_{a_2} S_1 y_{2,i_4,r_1} \mathcal{I} T_{3,a_1,a_2,j_1} \lambda_{T_3} S_{2,j_1}^\dagger}{\sqrt{3}}$				$\frac{\sqrt{3} h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{2,i_3,r_1} L_{a_1,i_1}} C P_R L_{a_2,i_2}^T S_1 y_{1,i_1,r_2} y_{2,i_2,r_2} y_{2,i_3,r_1} \mathcal{I} T_{3,a_1,a_2,j_1} S_1^\dagger}{4 M_F}$					
															$\frac{\sqrt{3} h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{2,i_3,r_1} L_{a_1,i_1}} C P_R L_{a_2,i_2}^T S_1 y_{1,i_1,r_2} y_{2,i_2,r_2} y_{2,i_3,r_1} \mathcal{I} T_{3,a_1,a_2,j_1} S_1^\dagger}{4 M_F}$					
												$\frac{h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_3,r_1} L_{a_1,i_1}} C P_R L_{a_2,i_2}^T S_1 y_{1,i_1,r_2} y_{2,i_2,r_2} y_{2,i_3,r_1} \mathcal{I} T_{3,a_1,a_2,j_1} S_1^\dagger}{4 M_F}$								
cont.											$\frac{h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_3,r_1} L_{a_1,i_1}} C P_R L_{a_2,i_2}^T S_1 y_{1,i_1,r_2} y_{2,i_2,r_2} y_{2,i_3,r_1} \mathcal{I} T_{3,a_1,a_2,j_1} S_1^\dagger}{4 M_F}$					$\frac{\sqrt{3} h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_3,r_1} L_{a_1,i_1}} C P_R L_{a_2,i_2}^T S_1 y_{1,i_1,r_2} y_{2,i_2,r_2} y_{2,i_3,r_1} \mathcal{I} T_{3,a_1,a_2,j_1} S_1^\dagger}{4 M_F}$				
															$\frac{h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_3,r_1} L_{a_1,i_1}} C P_R L_{a_2,i_2}^T S_1 y_{1,i_1,r_2} y_{2,i_2,r_2} y_{2,i_3,r_1} \mathcal{I} T_{3,a_1,a_2,j_1} S_1^\dagger}{4 M_F}$					
												$\frac{h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_3,r_1} L_{a_1,i_1}} C P_R L_{a_2,i_2}^T S_1 y_{1,i_1,r_2} y_{2,i_2,r_2} y_{2,i_3,r_1} \mathcal{I} T_{3,a_1,a_2,j_1} S_1^\dagger}{4 M_F}$								
cont.											$\frac{h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_3,r_1} L_{a_1,i_1}} C P_R L_{a_2,i_2}^T S_1 y_{1,i_1,r_2} y_{2,i_2,r_2} y_{2,i_3,r_1} \mathcal{I} T_{3,a_1,a_2,j_1} S_1^\dagger}{4 M_F}$					$\frac{\sqrt{3} h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_3,r_1} L_{a_1,i_1}} C P_R L_{a_2,i_2}^T S_1 y_{1,i_1,r_2} y_{2,i_2,r_2} y_{2,i_3,r_1} \mathcal{I} T_{3,a_1,a_2,j_1} S_1^\dagger}{4 M_F}$				
															$\frac{h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_3,r_1} L_{a_1,i_1}} C P_R L_{a_2,i_2}^T S_1 y_{1,i_1,r_2} y_{2,i_2,r_2} y_{2,i_3,r_1} \mathcal{I} T_{3,a_1,a_2,j_1} S_1^\dagger}{4 M_F}$					
												$\frac{h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_3,r_1} L_{a_1,i_1}} C P_R L_{a_2,i_2}^T S_1 y_{1,i_1,r_2} y_{2,i_2,r_2} y_{2,i_3,r_1} \mathcal{I} T_{3,a_1,a_2,j_1} S_1^\dagger}{4 M_F}$								
cont.											$\frac{h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_3,r_1} L_{a_1,i_1}} C P_R L_{a_2,i_2}^T S_1 y_{1,i_1,r_2} y_{2,i_2,r_2} y_{2,i_3,r_1} \mathcal{I} T_{3,a_1,a_2,j_1} S_1^\dagger}{4 M_F}$					$\frac{\sqrt{3} h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_3,r_1} L_{a_1,i_1}} C P_R L_{a_2,i_2}^T S_1 y_{1,i_1,r_2} y_{2,i_2,r_2} y_{2,i_3,r_1} \mathcal{I} T_{3,a_1,a_2,j_1} S_1^\dagger}{4 M_F}$				
															$\frac{h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_3,r_1} L_{a_1,i_1}} C P_R L_{a_2,i_2}^T S_1 y_{1,i_1,r_2} y_{2,i_2,r_2} y_{2,i_3,r_1} \mathcal{I} T_{3,a_1,a_2,j_1} S_1^\dagger}{4 M_F}$					
												$\frac{h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_3,r_1} L_{a_1,i_1}} C P_R L_{a_2,i_2}^T S_1 y_{1,i_1,r_2} y_{2,i_2,r_2} y_{2,i_3,r_1} \mathcal{I} T_{3,a_1,a_2,j_1} S_1^\dagger}{4 M_F}$								
T3-A	2	1	1	3	2	2	$\frac{3}{2}$				$\frac{+ \frac{2 h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1}} L_{a_4,i_2}^T C P_L L_{a_2,i_1} H_{a_1} H_{a_3} \lambda_{T_3} \epsilon_{a_1,a_2} \epsilon_{a_3,a_4}}{\sqrt{3} M_F}}{+ \frac{2 h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{\lambda_{T_3} L_{a_3,i_1}} C P_R L_{a_4,i_2}^T y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_3} \epsilon_{a_2,a_4} H_{a_1}^\dagger H_{a_2}^\dagger}{\sqrt{3} M_F}}$				$\frac{- h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} \lambda_{T_3} S_{2,j_1} y_{1,i_1,r_1} \mathcal{I} T_{3,a_1,a_2,j_1} H_{a_1}^\dagger H_{a_2}^\dagger S_1^\dagger}}{4 M_F}$					
											$\frac{h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1} \lambda_{T_3} S_{2,j_1} y_{2,i_1,r_1} \mathcal{I} T_{3,a_1,a_2,j_1} H_{a_1}^\dagger H_{a_2}^\dagger S_1^\dagger}}{\sqrt{3}}$					$\frac{h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{1,i_3,r_2} y_{2,i_2,r_1} L_{a_2,i_2}^T} C P_L L_{a_1,i_1} S_{2,j_1} y_{1,i_3,r_2} \mathcal{I} T_{3,a_1,a_2,j_1} S_1^\dagger}{4 M_F}$				
												$\frac{- h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1}} H_{a_1} H_{a_2} S_1 y_{1,i_1,r_1} \mathcal{I} T_{3,a_1,a_2,j_1} \lambda_{T_3} S_{2,j_1}^\dagger}{4 M_F}$								
cont.											$\frac{h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1}} H_{a_1} H_{a_2} S_1 y_{2,i_4,r_1} \mathcal{I} T_{3,a_1,a_2,j_1} \lambda_{T_3} S_{2,j_1}^\dagger}{\sqrt{3}}$				$\frac{\sqrt{3} h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{2,i_3,r_1} L_{a_1,i_1}} C P_R L_{a_2,i_2}^T S_1 y_{1,i_1,r_2} y_{2,i_2,r_2} y_{2,i_3,r_1} \mathcal{I} T_{3,a_1,a_2,j_1} S_1^\dagger}{4 M_F}$					
															$\frac{\sqrt{3} h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{2,i_3,r_1} L_{a_1,i_1}} C P_R L_{a_2,i_2}^T S_1 y_{1,i_1,r_2} y_{2,i_2,r_2} y_{2,i_3,r_1} \mathcal{I} T_{3,a_1,a_2,j_1} S_1^\dagger}{4 M_F}$					
												$\frac{h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_3,r_1} L_{a_1,i_1}} C P_R L_{a_2,i_2}^T S_1 y_{1,i_1,r_2} y_{2,i_2,r_2} y_{2,i_3,r_1} \mathcal{I} T_{3,a_1,a_2,j_1} S_1^\dagger}{4 M_F}$								
cont.											$\frac{h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_3,r_1} L_{a_1,i_1}} C P_R L_{a_2,i_2}^T S_1 y_{1,i_1,r_2} y_{2,i_2,r_2} y_{2,i_3,r_1} \mathcal{I} T_{3,a_1,a_2,j_1} S_1^\dagger}{4 M_F}$					$\frac{\sqrt{3} h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_3,r_1} L_{a_1,i_1}} C P_R L_{a_2,i_2}^T S_1 y_{1,i_1,r_2} y_{2,i_2,r_2} y_{2,i_3,r_1} \mathcal{I} T_{3,a_1,a_2,j_1} S_1^\dagger}{4 M_F}$				
															$\frac{h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_3,r_1} L_{a_1,i_1}} C P_R L_{a_2,i_2}^T S_1 y_{1,i_1,r_2} y_{2,i_2,r_2} y_{2,i_3,r_1} \mathcal{I} T_{3,a_1,a_2,j_1} S_1^\dagger}{4 M_F}$					
												$\frac{h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_3,r_1} L_{a_1,i_1}} C P_R L_{a_2,i_2}^T S_1 y_{1,i_1,r_2} y_{2,i_2,r_2} y_{2,i_3,r_1} \mathcal{I} T_{3,a_1,a_2,j_1} S_1^\dagger}{4 M_F}$								
cont.											$\frac{h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_3,r_1} L_{a_1,i_1}} C P_R L_{a_2,i_2}^T S_1 y_{1,i_1,r_2} y_{2,i_2,r_2} y_{2,i_3,r_1} \mathcal{I} T_{3,a_1,a_2,j_1} S_1^\dagger}{4 M_F}$					$\frac{\sqrt{3} h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_3,r_1} L_{a_1,i_1}} C P_R L_{a_2,i_2}^T S_1 y_{1,i_1,r_2} y_{2,i_2,r_2} y_{2,i_3,r_1} \mathcal{I} T_{3,a_1,a_2,j_1} S_1^\dagger}{4 M_F}$				
															$\frac{h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_3,r_1} L_{a_1,i_1}} C P_R L_{a_2,i_2}^T S_1 y_{1,i_1,r_2} y_{2,i_2,r_2} y_{2,i_3,r_1} \mathcal{I} T_{3,a_1,a_2,j_1} S_1^\dagger}{4 M_F}$					
												$\frac{h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_3,r_1} L_{a_1,i_1}} C P_R L_{a_2,i_2}^T S_1 y_{1,i_1,r_2} y_{2,i_2,r_2} y_{2,i_3,r_1} \mathcal{I} T_{3,a_1,a_2,j_1} S_1^\dagger}{4 M_F}$								
T3-A											$\frac{- h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} \lambda_{T_3} S_{2,j_1} y_{1,i_1,r_1} \mathcal{I} T_{3,a_1,a_2,j_1} H_{a_1}^\dagger H_{a_2}^\dagger S_1^\dagger}}{4 M_F}$				$\frac{- 6 h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1} L_{a_2,i_2}^T} C P_L L_{a_1,i_1} g_{\tilde{S}_2,j_1}^2 \mathcal{I} T_{3,a_1,a_2,j_1} S_1^\dagger}{M_F}$					
											$\frac{h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1} \lambda_{T_3} S_{2,j_1} y_{2,i_1,r_1} \mathcal{I} T_{3,a_1,a_2,j_1} H_{a_1}^\dagger H_{a_2}^\dagger S_1^\dagger}}{\sqrt{3}}$					$\frac{h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{1,i_3,r_2} y_{2,i_2,r_1} L_{a_2,i_2}^T} C P_L L_{a_1,i_1} S_{2,j_1} y_{1,i_3,r_2} \mathcal{I} T_{3,a_1,a_2,j_1} S_1^\dagger}{4 M_F}$				
												$\frac{- h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1}} H_{a_1} H_{a_2} S_1 y_{1,i_1,r_1} \mathcal{I} T_{3,a_1,a_2,j_1} \lambda_{T_3} S_{2,j_1}^\dagger}{4 M_F}$								
cont.											$\frac{h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1}} H_{a_1} H_{a_2} S_1 y_{2,i_4,r_1} \mathcal{I} T_{3,a_1,a_2,j_1} \lambda_{T_3} S_{2,j_1}^\dagger}{\sqrt{3}}$				$\frac{\sqrt{3} h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{2,i_3,r_1} L_{a_1,i_1}} C P_R L_{a_2,i_2}^T S_1 y_{1,i_1,r_2} y_{2,i_2,r_2} y_{2,i_3,r_1} \mathcal{I} T_{3,a_1,a_2,j_1} S_1^\dagger}{4 M_F}$					
															$\frac{\sqrt{3} h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{2,i_3,r_1} L_{a_1,i_1}} C P_R L_{a_2,i_2}^T S_1 y_{1,i_1,r_2} y_{2,i_2,r_2} y_{2,i_3,r_1} \mathcal{I} T_{3,a_1,a_2,j_1} S_1^\dagger}{4 M_F}$					
												$\frac{h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_3,r_1} L_{a_1,i_1}} C P_R L_{a_2,i_2}^T S_1 y_{1,i_1,r_2} y_{2,i_2,r_2} y_{2,i_3,r_1} \math$								

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	F	H	ℓ	S_1	S_2	F	H
											S_1	S_2	F	H
											$\sqrt{3}h \log \left(\frac{\hat{\mu}^2}{M_F^2} \right) \overline{y_{2,1,3},r_1} \overline{L_{a_1,1,1}} C P_R \overline{L_{a_2,1,2}} \overline{S_{1,y_1,1,3},r_2 y_{2,1,1},r_2 y_{2,1,2},r_1} \overline{I_{T3,a_1,a_2,j_1}} S_{2,j_1}^\dagger$			
											$4M_F$			
cont.											$h \log \left(\frac{\hat{\mu}^2}{M_F^2} \right) \overline{y_{1,1,3},r_1} \overline{L_{a_1,1,1}} C P_R \overline{L_{a_2,1,2}} \overline{S_{1,y_1,1,1},r_2 y_{1,1,3},r_1 y_{2,1,2},r_2} \overline{I_{T3,a_1,a_2,j_1}} S_{2,j_1}^\dagger$			
											M_F			
											$h \log \left(\frac{\hat{\mu}^2}{M_F^2} \right) \overline{y_{2,1,3},r_1} \overline{L_{a_1,1,1}} C P_R \overline{L_{a_2,1,2}} \overline{S_{1,y_1,1,1},r_2 y_{2,1,2},r_2 y_{2,1,3},r_1} \overline{I_{T3,a_1,a_2,j_1}} S_{2,j_1}^\dagger$			
											$\sqrt{3}M_F$			
											$h \log \left(\frac{\hat{\mu}^2}{M_F^2} \right) \overline{y_{1,1,3},r_1} \overline{L_{a_1,1,1}} C P_R \overline{L_{a_2,1,2}} \overline{S_{1,y_1,1,1},r_2 y_{1,1,2},r_1 y_{2,1,3},r_2} \overline{I_{T3,a_1,a_2,j_1}} S_{2,j_1}^\dagger$			
											$4M_F$			
cont.											$\sqrt{3}h \log \left(\frac{\hat{\mu}^2}{M_F^2} \right) \overline{y_{2,1,3},r_1} \overline{L_{a_1,1,1}} C P_R \overline{L_{a_2,1,2}} \overline{S_{1,y_1,1,1},r_2 y_{2,1,2},r_1 y_{2,1,3},r_2} \overline{I_{T3,a_1,a_2,j_1}} S_{2,j_1}^\dagger$			
											$4M_F$			
											$h \log \left(\frac{\hat{\mu}^2}{M_F^2} \right) \overline{y_{1,1,1},r_1} \overline{y_{2,1,2},r_1} \overline{L_{a_2,1,2}} \overline{C P_L L_{a_1,1} S_{2,j_1} \overline{I_{T3,a_1,a_2,j_1}} S_{1,1}^\dagger} \lambda_{12}^{(1)}$			
cont.											$h \log \left(\frac{\hat{\mu}^2}{M_F^2} \right) \overline{L_{a_1,1,1}} C P_R \overline{L_{a_2,1,2}} \overline{S_{1,y_1,1,1},r_1 y_{2,1,2},r_1} \overline{I_{T3,a_1,a_2,j_1}} S_{2,j_1}^\dagger \lambda_{12}^{(1)}$			
											M_F			
T3-B	-2	2	-1	2	0	1	-1/2				$+\frac{2h \log \left(\frac{\hat{\mu}^2}{M_F^2} \right) \overline{y_{1,1,1},r_1} \overline{y_{2,1,2},r_1} \overline{L_{a_4,1,2}} \overline{C P_L L_{a_2,1,1} H_{a_1} H_{a_3} \lambda_{T3,a_1,a_2} \epsilon_{a_3,a_4}}}{\sqrt{3}M_F}$			
											$2h \log \left(\frac{\hat{\mu}^2}{M_F^2} \right) \overline{\lambda_{T3} L_{a_3,1,1}} C P_R \overline{L_{a_4,1,2}} \overline{y_{1,1,1},r_1} \overline{y_{2,1,2},r_1} \epsilon_{a_1,a_3} \epsilon_{a_2,a_4} H_{a_1}^\dagger H_{a_2}^\dagger$			
											$+\frac{\sqrt{3}M_F}{\sqrt{3}M_F}$			
											$-\frac{1}{2} h \log \left(\frac{\hat{\mu}^2}{M_F^2} \right) \overline{y_{1,1,1},r_1} \overline{\lambda_{T3} S_{2,j_2} y_{1,1,1},r_1} \overline{I_{T3,a_1,a_2,j_1,j_2}} H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger$			
											$-\frac{1}{2} h \log \left(\frac{\hat{\mu}^2}{M_F^2} \right) \overline{y_{2,1,1},r_1} \overline{\lambda_{T3} S_{2,j_2} y_{2,1,1},r_1} \overline{I_{T3,a_1,a_2,j_1,j_2}} H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger$			
											$-\frac{1}{2} h \log \left(\frac{\hat{\mu}^2}{M_F^2} \right) \overline{y_{1,1,1},r_1} H_{a_1} H_{a_2} S_{1,j_2} y_{1,1,1},r_1 \overline{I_{T3,a_1,a_2,j_2,j_1}} \lambda_{T3} S_{2,j_1}^\dagger$			
cont.											$-\frac{1}{2} h \log \left(\frac{\hat{\mu}^2}{M_F^2} \right) \overline{y_{2,1,1},r_1} H_{a_1} H_{a_2} S_{1,j_2} y_{2,1,1},r_1 \overline{I_{T3,a_1,a_2,j_2,j_1}} \lambda_{T3} S_{2,j_1}^\dagger$			
											$h \log \left(\frac{\hat{\mu}^2}{M_F^2} \right) \overline{y_{1,1,1},r_1} \overline{y_{2,1,2},r_1} \overline{L_{a_2,1,2}} \overline{C P_L L_{a_1,1} S_{2,j_1} \overline{I_{T3,a_1,a_2,j_1}} S_{1,1}^\dagger} \lambda_{12}^{(1)}$			
											$h \log \left(\frac{\hat{\mu}^2}{M_F^2} \right) \overline{y_{2,1,1},r_1} \overline{y_{2,1,2},r_1} \overline{L_{a_2,1,2}} \overline{C P_L L_{a_1,1} S_{2,j_1} \overline{I_{T3,a_1,a_2,j_1}} S_{1,1}^\dagger} \lambda_{12}^{(1)}$			
											$h \log \left(\frac{\hat{\mu}^2}{M_F^2} \right) \overline{y_{1,1,1},r_1} \overline{y_{2,1,2},r_1} \overline{L_{a_2,1,2}} \overline{C P_L L_{a_1,1} S_{2,j_1} \overline{I_{T3,a_1,a_2,j_1}} S_{1,1}^\dagger} \lambda_{12}^{(1)}$			
											$h \log \left(\frac{\hat{\mu}^2}{M_F^2} \right) \overline{y_{2,1,1},r_1} \overline{y_{2,1,2},r_1} \overline{L_{a_2,1,2}} \overline{C P_L L_{a_1,1} S_{2,j_1} \overline{I_{T3,a_1,a_2,j_1}} S_{1,1}^\dagger} \lambda_{12}^{(1)}$			
											$h \log \left(\frac{\hat{\mu}^2}{M_F^2} \right) \overline{y_{1,1,1},r_1} \overline{y_{2,1,2},r_1} \overline{L_{a_2,1,2}} \overline{C P_L L_{a_1,1} S_{2,j_1} \overline{I_{T3,a_1,a_2,j_1}} S_{1,1}^\dagger} \lambda_{12}^{(1)}$			
											$h \log \left(\frac{\hat{\mu}^2}{M_F^2} \right) \overline{y_{2,1,1},r_1} \overline{y_{2,1,2},r_1} \overline{L_{a_2,1,2}} \overline{C P_L L_{a_1,1} S_{2,j_1} \overline{I_{T3,a_1,a_2,j_1}} S_{1,1}^\dagger} \lambda_{12}^{(1)}$			
											$h \log \left(\frac{\hat{\mu}^2}{M_F^2} \right) \overline{y_{1,1,1},r_1} \overline{y_{2,1,2},r_1} \overline{L_{a_2,1,2}} \overline{C P_L L_{a_1,1} S_{2,j_1} \overline{I_{T3,a_1,a_2,j_1}} S_{1,1}^\dagger} \lambda_{12}^{(1)}$			
											$h \log \left(\frac{\hat{\mu}^2}{M_F^2} \right) \overline{y_{2,1,1},r_1} \overline{y_{2,1,2},r_1} \overline{L_{a_2,1,2}} \overline{C P_L L_{a_1,1} S_{2,j_1} \overline{I_{T3,a_1,a_2,j_1}} S_{1,1}^\dagger} \lambda_{12}^{(1)}$			
											$h \log \left(\frac{\hat{\mu}^2}{M_F^2} \right) \overline{y_{1,1,1},r_1} \overline{y_{2,1,2},r_1} \overline{L_{a_2,1,2}} \overline{C P_L L_{a_1,1} S_{2,j_1} \overline{I_{T3,a_1,a_2,j_1}} S_{1,1}^\dagger} \lambda_{12}^{(1)}$			
											$h \log \left(\frac{\hat{\mu}^2}{M_F^2} \right) \overline{y_{2,1,1},r_1} \overline{y_{2,1,2},r_1} \overline{L_{a_2,1,2}} \overline{C P_L L_{a_1,1} S_{2,j_1} \overline{I_{T3,a_1,a_2,j_1}} S_{1,1}^\dagger} \lambda_{12}^{(1)}$			
											$h \log \left(\frac{\hat{\mu}^2}{M_F^2} \right) \overline{y_{1,1,1},r_1} \overline{y_{2,1,2},r_1} \overline{L_{a_2,1,2}} \overline{C P_L L_{a_1,1} S_{2,j_1} \overline{I_{T3,a_1,a_2,j_1}} S_{1,1}^\dagger} \lambda_{12}^{(1)}$			
											$h \log \left(\frac{\hat{\mu}^2}{M_F^2} \right) \overline{y_{2,1,1},r_1} \overline{y_{2,1,2},r_1} \overline{L_{a_2,1,2}} \overline{C P_L L_{a_1,1} S_{2,j_1} \overline{I_{T3,a_1,a_2,j_1}} S_{1,1}^\dagger} \lambda_{12}^{(1)}$			
											$h \log \left(\frac{\hat{\mu}^2}{M_F^2} \right) \overline{y_{1,1,1},r_1} \overline{y_{2,1,2},r_1} \overline{L_{a_2,1,2}} \overline{C P_L L_{a_1,1} S_{2,j_1} \overline{I_{T3,a_1,a_2,j_1}} S_{1,1}^\dagger} \lambda_{12}^{(1)}$			
											$h \log \left(\frac{\hat{\mu}^2}{M_F^2} \right) \overline{y_{2,1,1},r_1} \overline{y_{2,1,2},r_1} \overline{L_{a_2,1,2}} \overline{C P_L L_{a_1,1} S_{2,j_1} \overline{I_{T3,a_1,a_2,j_1}} S_{1,1}^\dagger} \lambda_{12}^{(1)}$			
											$h \log \left(\frac{\hat{\mu}^2}{M_F^2} \right) \overline{y_{1,1,1},r_1} \overline{y_{2,1,2},r_1} \overline{L_{a_2,1,2}} \overline{C P_L L_{a_1,1} S_{2,j_1} \overline{I_{T3,a_1,a_2,j_1}} S_{1,1}^\dagger} \lambda_{12}^{(1)}$			
											$h \log \left(\frac{\hat{\mu}^2}{M_F^2} \right) \overline{y_{2,1,1},r_1} \overline{y_{2,1,2},r_1} \overline{L_{a_2,1,2}} \overline{C P_L L_{a_1,1} S_{2,j_1} \overline{I_{T3,a_1,a_2,j_1}} S_{1,1}^\dagger} \lambda_{12}^{(1)}$			
											$h \log \left(\frac{\hat{\mu}^2}{M_F^2} \right) \overline{y_{1,1,1},r_1} \overline{y_{2,1,2},r_1} \overline{L_{a_2,1,2}} \overline{C P_L L_{a_1,1} S_{2,j_1} \overline{I_{T3,a_1,a_2,j_1}} S_{1,1}^\dagger} \lambda_{12}^{(1)}$			
											$h \log \left(\frac{\hat{\mu}^2}{M_F^2} \right) \overline{y_{2,1,1},r_1} \overline{y_{2,1,2},r_1} \overline{L_{a_2,1,2}} \overline{C P_L L_{a_1,1} S_{2,j_1} \overline{I_{T3,a_1,a_2,j_1}} S_{1,1}^\dagger} \lambda_{12}^{(1)}$			
											$h \log \left(\frac{\hat{\mu}^2}{M_F^2} \right) \overline{y_{1,1,1},r_1} \overline{y_{2,1,2},r_1} \overline{L_{a_2,1,2}} \overline{C P_L L_{a_1,1} S_{2,j_1} \overline{I_{T3,a_1,a_2,j_1}} S_{1,1}^\dagger} \lambda_{12}^{(1)}$			
											$h \log \left(\frac{\hat{\mu}^2}{M_F^2} \right) \overline{y_{2,1,1},r_1} \overline{y_{2,1,2},r_1} \overline{L_{a_2,1,2}} \overline{C P_L L_{a_1,1} S_{2,j_1} \overline{I_{T3,a_1,a_2,j_1}} S_{1,1}^\dagger} \lambda_{12}^{(1)}$			
											$h \log \left(\frac{\hat{\mu}^2}{M_F^2} \right) \overline{y_{1,1,1},r_1} \overline{y_{2,1,2},r_1} \overline{L_{a_2,1,2}} \overline{C P_L L_{a_1,1} S_{2,j_1} \overline{I_{T3,a_1,a_2,j_1}} S_{1,1}^\dagger} \lambda_{12}^{(1)}$			
											$h \log \left(\frac{\hat{\mu}^2}{M_F^2} \right) \overline{y_{2,1,1},r_1} \overline{y_{2,1,2},r_1} \overline{L_{a_2,1,2}} \overline{C P_L L_{a_1,1} S_{2,j_1} \overline{I_{T3,a_1,a_2,j_1}} S_{1,1}^\dagger} \lambda_{12}^{(1)}$			
											$h \log \left(\frac{\hat{\mu}^2}{M_F^2} \right) \overline{y_{1,1,1},r_1} \overline{y_{2,1,2},r_1} \overline{L_{a_2,1,2}} \overline{C P_L L_{a_1,1} S_{2,j_1} \overline{I_{T3,a_1,a_2,j_1}} S_{1,1}^\dagger} \lambda_{12}^{(1)}$			
											$h \log \left(\frac{\hat{\mu}^2}{M_F^2} \right) \overline{y_{2,1,1},r_1} \overline{y_{2,1,2},r_1} \overline{L_{a_2,1,2}} \overline{C P_L L_{a_1,1} S_{2,j_1} \overline{I_{T3,a_1,a_2,j_1}} S_{1,1}^\dagger} \lambda_{12}^{(1)}$			
											$h \log \left(\frac{\hat{\mu}^2}{M_F^2} \right) \overline{y_{1,1,1},r_1} \overline{y_{2,1,2},r_1} \overline{L_{a_2,1,2}} \overline{C P_L L_{a_1,1} S_{2,j_1} \overline{I_{T3,a_1,a_2,j_1}} S_{1,1}^\dagger} \lambda_{12}^{(1)}$			
											$h \log \left(\frac{\hat{\mu}^2}{M_F^2} \right) \overline{y_{2,1,1},r_1} \overline{y_{2,1,2},r_1} \overline{L_{a_2,1,2}} \overline{C P_L L_{a_1,1} S_{2,j_1} \overline{I_{T3,a_1,a_2,j_1}} S_{1,1}^\dagger} \lambda_{12}^{(1)}$			
											$h \log \left(\frac{\hat{\mu}^2}{M_F^2} \right) \overline{y_{1,1,1},r_1} \overline{y_{2,1,2},r_1} \overline{L_{a_2,1,2}} \overline{C P_L L_{a_1,1} S_{2,j_1} \overline{I_{T3,a_1,a_2,j_1}} S_{1,1}^\dagger} \lambda_{12}^{(1)}$			
											$h \log \left(\frac{\hat{\mu}^2}{M_F^2} \right) \overline{y_{2,1,1},r_1} \overline{y_{2,1,2},r_1} \overline{L_{a_2,1,2}} \overline{C P_L L_{a_1,1} S_{2,j_1} \overline{I_{T3,a_1,a_2,j_1}} S_{1,1}^\dagger} \lambda_{12}^{(1)}$			
											$h \log \left(\frac{\hat{\mu}^2}{M_F^2} \right) \overline{y_{1,1,1},r_1} \overline{y_{2,1,2},r_1} \overline{L_{a_2,1,2}} \overline{C P_L L_{a_1,1} S_{2,j_1} \overline{I_{T3,a_1,a_2,j_1}} S_{1,1}^\dagger} \lambda_{12}^{(1)}$			
											$h \log \left(\frac{\hat{\mu}^2}{M_F^2} \right) \overline{y_{2,1,1},r_1} \overline{y_{2,1,2},r_1} \overline{L_{a_2,1,2}} \overline{C P_L L_{a_1,1} S_{2,j_1} \overline{I_{T3,a_1,a_2,j_1}} S_{1,1}^\dagger} \lambda_{12}^{(1)}$			
											$h \log \left(\frac{\hat{\mu}^2}{M_F^2} \right) \overline{y_{1,1,1},r_1} \overline{y_{2,1,2},r_1} \overline{L_{a_2,1,2}} \overline{C P_L L_{a_1,1} S_{2,j_1} \overline{I_{T3,a_1,a_2,j_1}} S_{1,1}^\dagger} \lambda_{12}^{(1)}$			
											$h \log \left(\frac{\hat{\mu}^2}{M_F^2} \right) \overline{y_{2,1,1},r_1} \overline{y_{2,1,2},r_1} \overline{L_{a_2,1,2}} \overline{C P_L L_{a_1,1} S_{2,j_1} \overline{I_{T3,a_1,a_2,j_1}} S_{1,1}^\dagger} \lambda_{12}^{(1)}$			
											$h \log \left(\frac{\hat{\mu}^2}{M_F^2} \right) \overline{y_{1,1,1},r_1} \overline{y_{2,1,2},r_1} \overline{L_{a_2,1,2}} \overline{C P_L L_{a_1,1} S_{2,j_1} \overline{I_{T3,a_1,a_2,j_1}} S_{1,1}^\dagger} \lambda_{12}^{(1)}$			
											$h \log \left(\frac{\hat{\mu}^2}{M_F^2} \right) \overline{y_{2,1,1},r_1} \overline{y_{2,1,2},r_1} \overline{L_{a_2,1,2}} \overline{C P_L L_{a_1,1} S_{2,j_1} \overline{I_{T3,a_1,a_2,j_1}} S_{1,1}^\dagger} \lambda_{12}^{(1)}$			
											$h \log \left(\frac{\hat{\mu}^2}{M_F^2} \right) \overline{y_{1,1,1},r_1} \overline{y_{2,1,2},r_1} \overline{L_{a_2,1,2}} \overline{C P_L L_{a_1,1} S_{2,j_1} \overline{I_{T3,a_1,a_2,j_1}} S_{1,1}^\dagger} \lambda_{12}^{(1)}$			
											$h \log \left(\frac{\hat{\mu}^2}{M_F^2} \right) \overline{y_{2,1,1},r_1} \overline{y_{2,1,2},r_1} \overline{L_{a_2,1,2}} \overline{C P_L L_{a_1,1} S_{2,j_1} \overline{I_{T3,a_1,a_2,j_1}} S_{1,1}^\dagger} \lambda_{12}^{(1)}$			
											$h \log \left(\frac{\hat{\mu}^2}{M_F^2} \right) \overline{y_{1,1,1},r_1} \overline{y_{2,1,2},r_1} \overline{L_{a_2,1,2}} \overline{C P_L L_{a_1,1} S_{2,j_1} \overline{I_{T3,a_1,a_2,j_1}} S_{1,1}^\dagger} \lambda_{12}^{(1)}$			
											$h \log \left(\frac{\hat{\mu}^2}{M_F^2} \right) \overline{y_{2,1,1},r_1} \overline{y_{2,1,2},r_1} \overline{L_{a_2,1,2}} \overline{C P_L L_{a_1,1} S_{2,j_1} \overline{I_{T3,a_1,a_2,j_1}} S_{1,1}^\dagger} \lambda_{12}^{(1)}$			
											$h \log \left(\frac{\hat{\mu}^2}{M_F^2} \right) \overline{y_{1,1,1},r_1} \overline{y_{2,1,2},r_1} \overline{L_{a_2,1,2}} \overline{C P_L L_{a_1,1} S_{2,j_1} \overline{I_{T3,a_1,a_2,j_1}} S_{1,1}^\dagger} \lambda_{12}^{(1)}$			
											$h \log \left(\frac{\hat{\mu}^2}{M_F^2} \right) \overline{y_{2,1,1},r_1} \overline{y_{2,1,2},r_1} \overline{L_{a_2,1,2}} \overline{C P_L L_{a_1,1} S_{2,j_1} \overline{I_{T3,a_1,a_2,j_1}} S_{1,1}^\dagger} \lambda_{12}^{(1)}$			
											$h \log \left(\frac{\hat{\mu}^2}{M_F^2} \right) \overline{y_{1,1,1},r_1} \overline{y_{2,1,2},r_1} \overline{L_{a_2,1,2}} \overline{C P_L L_{a_1,1} S_{2,j_1} \overline{I_{T3,a_1,a_2,j_1}} S_{1,1}^\dagger} \lambda_{12}^{(1)}$			
											$h \log \left(\frac{\hat{\mu}^2}{M_F^2} \right) \overline{y_{2,1,1},r_1} \overline{y_{2,1,2},r_1} \overline{L_{a_2,1,2}} \overline{C P_L L_{a_1,1} S_{2,j_1} \overline{I_{T3,a_1,a_2,j_1}} S_{1,1}^\dagger} \lambda_{12}^{(1)}$			
											$h \log \left(\frac{\hat{\mu}^2}{M_F^2} \right) \overline{y_{1,1,1},r_1} \overline{y_{2,1,2},r_1} \overline{L_{a_2,1,2}} \overline{C P_L L_{a_1,1} S_{2,j_1} \overline{I_{T3,a_1,a_2,j_1}} S_{1,1}^\dagger} \lambda_{12}^{(1)}$			
											$h \log \left(\frac{\hat{\mu}^2}{M_F^2} \right) \overline{y_{2,1,1},r_1} \overline{y_{2,1,2},r_1} \overline{L_{a_2,1,2}} \overline{C P_L L_{a_1,1} S_{2,j_1} \overline{I_{T3,a_1,a_2,j_1}} S_{1,1}^\dagger} \lambda_{12}^{(1)}$			
											$h \log \left(\frac{\hat{\mu}^2}{M_F^2} \right) \overline{y_{1,1,1},r_1} \overline{y_{2,1,2},r_1} \overline{L_{a_2,1,2}} \overline{C P_L L_{a_1,1} S_{2,j_1} \overline{I_{T3,a_1,a_2,j_1}} S_{1,1}^\dagger} \lambda_{12}^{(1)}$			
											$h \log \left(\frac{\hat{\mu}^2}{M_F^2} \right) \overline{y_{2$			

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$FH\ell$	$S_1 S_2 F H$	$S_1 S_2 F \ell$	
cont.									$+h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1}} H_{a_2} S_{1,j_2} S_{2,j_1} y_{1,i_1,r_1} \mathcal{I}_{y_{2,i_1,r_1}} \mathcal{I}_{y_{2,a_2,j_2}} H_{a_1}^\dagger \lambda_{H2}^{(1)}$	$-\frac{h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1}} y_{1,i_3,r_1} y_{1,i_3,r_2} y_{2,i_2,r_1} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{1,i_3,r_2} \mathcal{I}_{y_{a_1,i_1}} \mathcal{I}_{y_{2,a_2,j_2}} S_{2,i_1}^\dagger}{2M_F}$	
									$+h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1}} H_{a_2} y_{2,i_1,r_1} \mathcal{I}_{y_{2,a_1}} \mathcal{I}_{y_{1,a_2,j_2}} H_{a_1}^\dagger S_{1,j_1}^\dagger S_{2,j_2}^\dagger \lambda_{H2}^{(1)}$	$-\frac{h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1}} y_{1,i_2,r_1} y_{2,i_2,r_2} y_{2,i_3,r_1} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{1,i_3,r_2} \mathcal{I}_{y_{a_1,i_1}} \mathcal{I}_{y_{2,a_2,j_2}} S_{2,i_1}^\dagger}{4M_F}$	
									$+\frac{h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1}} H_{a_1} S_{1,j_1} S_{2,j_2} y_{1,i_1,r_1} \text{Bar}(\text{cg}1)_{j_1,j_2} H_{a_1}^\dagger \lambda_{H2}^{(2)}}{\sqrt{3}}$	$-\frac{h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_3,r_1}} y_{2,i_1,r_1} y_{2,i_2,r_2} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{2,i_3,r_2} \mathcal{I}_{y_{a_1,i_1}} \mathcal{I}_{y_{2,a_2,j_2}} S_{2,i_1}^\dagger}{M_F}$	
									$-\frac{h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1}} H_{a_2} S_{1,j_1} S_{2,j_2} y_{1,i_1,r_1} \mathcal{I}_{y_{1,a_1}} \mathcal{I}_{y_{2,a_2,j_2}} H_{a_1}^\dagger \lambda_{H2}^{(2)}}{\sqrt{3}}$	$-\frac{h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_3,r_1}} y_{2,i_1,r_1} y_{2,i_2,r_2} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{2,i_3,r_2} \mathcal{I}_{y_{a_1,i_1}} \mathcal{I}_{y_{2,a_2,j_2}} S_{2,i_1}^\dagger}{4M_F}$	
cont.									$+\frac{h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} \text{cg}1_{j_1,j_2}} H_{a_1} y_{2,i_1,r_1} H_{a_1}^\dagger S_{1,j_1}^\dagger S_{2,j_2}^\dagger \lambda_{H2}^{(2)}}{\sqrt{3}}$	$-\frac{h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1}} y_{2,i_2,r_2} y_{2,i_3,r_1} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{2,i_3,r_2} \mathcal{I}_{y_{a_1,i_1}} \mathcal{I}_{y_{2,a_2,j_2}} S_{2,i_1}^\dagger}{4M_F}$	
									$-h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1}} H_{a_2} y_{2,i_1,r_1} \mathcal{I}_{y_{2,a_1}} \mathcal{I}_{y_{1,a_2,j_2}} H_{a_1}^\dagger S_{1,j_1}^\dagger S_{2,j_2}^\dagger \lambda_{H2}^{(2)}$	$-\frac{h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1}} y_{2,i_2,r_2} y_{2,i_3,r_1} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{2,i_3,r_2} \mathcal{I}_{y_{a_1,i_1}} \mathcal{I}_{y_{2,a_2,j_2}} S_{2,i_1}^\dagger}{4M_F}$	
cont.										$+\frac{3h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{L_{a_1,i_1}} C P_R L_{a_2,i_2}^T g_1^T S_{1,j_2} y_{1,i_1,r_1} y_{2,i_2,r_1} \mathcal{I}_{y_{a_1,i_1}} \mathcal{I}_{y_{2,a_2,j_2}} S_{2,i_1}^\dagger}{4M_F}$	$-\frac{h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_3,r_1}} L_{a_1,i_1} C P_R L_{a_2,i_2}^T S_{1,j_2} y_{1,i_1,r_1} y_{2,i_3,r_1} y_{2,i_2,r_2} \mathcal{I}_{y_{a_1,i_1}} \mathcal{I}_{y_{2,a_2,j_2}} S_{2,i_1}^\dagger}{2M_F}$
										$-\frac{h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_3,r_1}} L_{a_1,i_1} C P_R L_{a_2,i_2}^T S_{1,j_2} y_{1,i_1,r_1} y_{1,i_3,r_2} y_{2,i_2,r_2} \mathcal{I}_{y_{a_1,i_1}} \mathcal{I}_{y_{2,a_2,j_2}} S_{2,i_1}^\dagger}{4M_F}$	$-\frac{h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{2,i_3,r_1}} L_{a_1,i_1} C P_R L_{a_2,i_2}^T S_{1,j_2} y_{1,i_3,r_2} y_{2,i_1,r_1} y_{2,i_2,r_2} \mathcal{I}_{y_{a_1,i_1}} \mathcal{I}_{y_{2,a_2,j_2}} S_{2,i_1}^\dagger}{4M_F}$
										$-\frac{h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{2,i_3,r_1}} L_{a_1,i_1} C P_R L_{a_2,i_2}^T S_{1,j_2} y_{1,i_3,r_2} y_{2,i_1,r_1} y_{2,i_2,r_2} \mathcal{I}_{y_{a_1,i_1}} \mathcal{I}_{y_{2,a_2,j_2}} S_{2,i_1}^\dagger}{4M_F}$	$-\frac{h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{2,i_3,r_1}} L_{a_1,i_1} C P_R L_{a_2,i_2}^T S_{1,j_2} y_{1,i_3,r_2} y_{2,i_1,r_1} y_{2,i_2,r_2} \mathcal{I}_{y_{a_1,i_1}} \mathcal{I}_{y_{2,a_2,j_2}} S_{2,i_1}^\dagger}{M_F}$
										$-\frac{h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{2,i_3,r_1}} L_{a_1,i_1} C P_R L_{a_2,i_2}^T S_{1,j_2} y_{1,i_1,r_1} y_{2,i_2,r_2} y_{2,i_3,r_1} \mathcal{I}_{y_{a_1,i_1}} \mathcal{I}_{y_{2,a_2,j_2}} S_{2,i_1}^\dagger}{2M_F}$	$-\frac{h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_3,r_1}} L_{a_1,i_1} C P_R L_{a_2,i_2}^T S_{1,j_2} y_{1,i_1,r_1} y_{2,i_3,r_1} y_{2,i_2,r_2} \mathcal{I}_{y_{a_1,i_1}} \mathcal{I}_{y_{2,a_2,j_2}} S_{2,i_1}^\dagger}{4M_F}$
cont.										$-\frac{h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{2,i_3,r_1}} L_{a_1,i_1} C P_R L_{a_2,i_2}^T S_{1,j_2} y_{1,i_1,r_1} y_{2,i_2,r_2} y_{2,i_3,r_1} \mathcal{I}_{y_{a_1,i_1}} \mathcal{I}_{y_{2,a_2,j_2}} S_{2,i_1}^\dagger}{4M_F}$	$+\frac{h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1}} y_{2,i_2,r_2} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_1} \epsilon_{a_1,a_2} S_{1,j_1}^\dagger \lambda_{H2}^{(1)}}{M_F}$
										$+\frac{h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{L_{a_1,i_1}} C P_R L_{a_2,i_2}^T S_{1,j_1} y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_2} S_{2,j_1}^\dagger \lambda_{H2}^{(1)}}{M_F}$	$-\frac{h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_2,r_1}} y_{2,i_1,r_1} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_2} \mathcal{I}_{y_{1,i_1}} \mathcal{I}_{y_{2,a_2,j_2}} S_{1,j_1}^\dagger \lambda_{H2}^{(1)}}{\sqrt{3} M_F}$
										$-\frac{h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1}} y_{2,i_2,r_1} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_2} \mathcal{I}_{y_{1,i_1}} \mathcal{I}_{y_{2,a_2,j_2}} S_{1,j_1}^\dagger \lambda_{H2}^{(2)}}{\sqrt{3} M_F}$	$-\frac{h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1}} y_{2,i_2,r_1} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_2} \mathcal{I}_{y_{1,i_1}} \mathcal{I}_{y_{2,a_2,j_2}} S_{1,j_1}^\dagger \lambda_{H2}^{(2)}}{\sqrt{3} M_F}$
										$-\frac{h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{L_{a_1,i_1}} C P_R L_{a_2,i_2}^T S_{1,j_2} y_{1,i_2,r_1} y_{2,i_1,r_1} \mathcal{I}_{y_{1,i_1}} \mathcal{I}_{y_{2,a_2,j_2}} S_{2,j_1}^\dagger \lambda_{H2}^{(2)}}{\sqrt{3} M_F}$	$-\frac{h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{L_{a_1,i_1}} C P_R L_{a_2,i_2}^T S_{1,j_2} y_{1,i_1,r_1} y_{2,i_2,r_1} \mathcal{I}_{y_{1,i_1}} \mathcal{I}_{y_{2,a_2,j_2}} S_{2,j_1}^\dagger \lambda_{H2}^{(2)}}{\sqrt{3} M_F}$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$+\frac{2h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1}} y_{2,i_2,r_1} L_{a_4,i_2}^T C P_L L_{a_2,i_1} H_{a_1} H_{a_3} \lambda_{T3} \epsilon_{a_1,a_2} \epsilon_{a_3,a_4}}{\sqrt{3} M_F}$	$-\frac{1}{2} h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1}} \lambda_{T3} S_{2,j_2} y_{1,i_1,r_1} \mathcal{I} T_{3,a_1,a_2,j_1,j_2} H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger$	$+\frac{3h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_2,r_1}} y_{2,i_1,r_1} L_{a_2,i_2}^T C P_L L_{a_1,i_1} g_2^T S_{2,j_2} \mathcal{I}_{y_{1,i_1}} \mathcal{I}_{y_{2,a_2,j_2}} S_{1,j_1}^\dagger}{4M_F}$	
								$+\frac{2h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \lambda_{T3} L_{a_3,i_1} C P_R L_{a_4,i_2}^T y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_3} \epsilon_{a_2,a_4} H_{a_1}^\dagger H_{a_2}^\dagger}{\sqrt{3} M_F}$	$+\frac{1}{2} h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1}} \lambda_{T3} S_{2,j_2} y_{2,i_1,r_1} \mathcal{I} T_{3,a_1,a_2,j_1,j_2} H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger$	$-\frac{h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1}} y_{1,i_3,r_1} y_{2,i_2,r_1} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{1,i_3,r_2} \mathcal{I}_{y_{a_1,i_1}} \mathcal{I}_{y_{2,a_2,j_2}} S_{1,j_1}^\dagger}{4M_F}$	
									$-\frac{1}{2} h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1}} H_{a_1} H_{a_2} S_{1,j_2} y_{1,i_1,r_1} \mathcal{I} T_{3,a_1,a_2,j_2} \lambda_{T3} S_{2,j_1}^\dagger$	$-\frac{h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1}} y_{1,i_3,r_1} y_{2,i_2,r_1} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{1,i_3,r_2} \mathcal{I}_{y_{a_1,i_1}} \mathcal{I}_{y_{2,a_2,j_2}} S_{1,j_1}^\dagger}{2M_F}$	$-\frac{h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1}} y_{1,i_3,r_1} y_{2,i_2,r_1} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{1,i_3,r_2} \mathcal{I}_{y_{a_1,i_1}} \mathcal{I}_{y_{2,a_2,j_2}} S_{1,j_1}^\dagger}{4M_F}$
									$-\frac{1}{2} h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1}} H_{a_1} H_{a_2} S_{1,j_2} y_{2,i_1,r_1} \mathcal{I} T_{3,a_1,a_2,j_2} \lambda_{T3} S_{2,j_1}^\dagger$	$-\frac{h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_3,r_1}} y_{2,i_1,r_1} y_{2,i_2,r_1} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{2,i_3,r_2} \mathcal{I}_{y_{a_1,i_1}} \mathcal{I}_{y_{2,a_2,j_2}} S_{1,j_1}^\dagger}{4M_F}$	$-\frac{h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_3,r_1}} y_{2,i_1,r_1} y_{2,i_2,r_1} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{2,i_3,r_2} \mathcal{I}_{y_{a_1,i_1}} \mathcal{I}_{y_{2,a_2,j_2}} S_{1,j_1}^\dagger}{4M_F}$
cont.										$+\frac{3h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1}} y_{2,i_2,r_1} \epsilon_{a_1,a_2} S_{2,j_1}^\dagger \lambda_{H2}^{(1)}}{4M_F}$	$-\frac{h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_3,r_1}} y_{2,i_1,r_1} y_{2,i_2,r_1} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{2,i_3,r_2} \mathcal{I}_{y_{a_1,i_1}} \mathcal{I}_{y_{2,a_2,j_2}} S_{1,j_1}^\dagger}{4M_F}$
										$+\frac{3h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{L_{a_1,i_1}} C P_R L_{a_2,i_2}^T g_2^T S_{1,j_2} y_{1,i_2,r_1} y_{2,i_1,r_1} \mathcal{I}_{y_{a_1,i_1}} \mathcal{I}_{y_{2,a_2,j_2}} S_{2,j_1}^\dagger}{4M_F}$	$-\frac{h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_3,r_1}} L_{a_1,i_1} C P_R L_{a_2,i_2}^T S_{1,j_2} y_{1,i_1,r_1} y_{2,i_3,r_1} y_{2,i_2,r_2} \mathcal{I}_{y_{a_1,i_1}} \mathcal{I}_{y_{2,a_2,j_2}} S_{2,i_1}^\dagger}{2M_F}$
										$-\frac{h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_3,r_1}} L_{a_1,i_1} C P_R L_{a_2,i_2}^T S_{1,j_2} y_{1,i_1,r_1} y_{1,i_3,r_2} y_{2,i_2,r_2} \mathcal{I}_{y_{a_1,i_1}} \mathcal{I}_{y_{2,a_2,j_2}} S_{2,i_1}^\dagger}{4M_F}$	$-\frac{h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{2,i_3,r_1}} L_{a_1,i_1} C P_R L_{a_2,i_2}^T S_{1,j_2} y_{1,i_3,r_2} y_{2,i_1,r_1} y_{2,i_2,r_2} \mathcal{I}_{y_{a_1,i_1}} \mathcal{I}_{y_{2,a_2,j_2}} S_{2,i_1}^\dagger}{4M_F}$
										$-\frac{h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{2,i_3,r_1}} L_{a_1,i_1} C P_R L_{a_2,i_2}^T S_{1,j_2} y_{1,i_1,r_1} y_{2,i_2,r_2} y_{2,i_3,r_1} \mathcal{I}_{y_{a_1,i_1}} \mathcal{I}_{y_{2,a_2,j_2}} S_{2,i_1}^\dagger}{2M_F}$	$-\frac{h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_3,r_1}} L_{a_1,i_1} C P_R L_{a_2,i_2}^T S_{1,j_2} y_{1,i_1,r_1} y_{2,i_3,r_1} y_{2,i_2,r_2} \mathcal{I}_{y_{a_1,i_1}} \mathcal{I}_{y_{2,a_2,j_2}} S_{2,i_1}^\dagger}{4M_F}$
cont.										$-\frac{h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{2,i_3,r_1}} L_{a_1,i_1} C P_R L_{a_2,i_2}^T S_{1,j_2} y_{1,i_1,r_1} y_{2,i_2,r_2} y_{2,i_3,r_1} \mathcal{I}_{y_{a_1,i_1}} \mathcal{I}_{y_{2,a_2,j_2}} S_{2,i_1}^\dagger}{4M_F}$	$+\frac{3h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1}} y_{2,i_2,r_2} \epsilon_{a_1,a_2} S_{2,j_1}^\dagger \lambda_{H2}^{(1)}}{4M_F}$
										$+\frac{3h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{L_{a_1,i_1}} C P_R L_{a_2,i_2}^T g_2^T S_{1,j_2} y_{1,i_2,r_1} y_{2,i_1,r_1} \mathcal{I}_{y_{a_1,i_1}} \mathcal{I}_{y_{2,a_2,j_2}} S_{2,j_1}^\dagger}{4M_F}$	$-\frac{h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_3,r_1}} L_{a_1,i_1} C P_R L_{a_2,i_2}^T S_{1,j_2} y_{1,i_1,r_1} y_{2,i_3,r_1} y_{2,i_2,r_2} \mathcal{I}_{y_{a_1,i_1}} \mathcal{I}_{y_{2,a_2,j_2}} S_{2,i_1}^\dagger}{2M_F}$
										$-\frac{h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_3,r_1}} L_{a_1,i_1} C P_R L_{a_2,i_2}^T S_{1,j_2} y_{1,i_1,r_1} y_{1,i_3,r_2} y_{2,i_2,r_2} \mathcal{I}_{y_{a_1,i_1}} \mathcal{I}_{y_{2,a_2,j_2}} S_{2,i_1}^\dagger}{4M_F}$	$-\frac{h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{2,i_3,r_1}} L_{a_1,i_1} C P_R L_{a_2,i_2}^T S_{1,j_2} y_{1,i_3,r_2} y_{2,i_1,r_1} y_{2,i_2,r_2} \mathcal{I}_{y_{a_1,i_1}} \mathcal{I}_{y_{2,a_2,j_2}} S_{2,i_1}^\dagger}{4M_F}$
										$-\frac{h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{2,i_3,r_1}} L_{a_1,i_1} C P_R L_{a_2,i_2}^T S_{1,j_2} y_{1,i_1,r_1} y_{2,i_2,r_2} y_{2,i_3,r_1} \mathcal{I}_{y_{a_1,i_1}} \mathcal{I}_{y_{2,a_2,j_2}} S_{2,i_1}^\dagger}{2M_F}$	$-\frac{h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_3,r_1}} L_{a_1,i_1} C P_R L_{a_2,i_2}^T S_{1,j_2} y_{1,i_1,r_1} y_{2,i_3,r_1} y_{2,i_2,r_2} \mathcal{I}_{y_{a_1,i_1}} \mathcal{I}_{y_{2,a_2,j_2}} S_{2,i_1}^\dagger}{4M_F}$
cont.										$-\frac{h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{2,i_3,r_1}} L_{a_1,i_1} C P_R L_{a_2,i_2}^T S_{1,j_2} y_{1,i_1,r_1} y_{2,i_2,r_2} y_{2,i_3,r_1} \mathcal{I}_{y_{a_1,i_1}} \mathcal{I}_{y_{2,a_2,j_2}} S_{2,i_1}^\dagger}{4M_F}$	$+\frac{3h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1}} y_{2,i_2,r_2} \epsilon_{a_1,a_2} S_{2,j_1}^\dagger \lambda_{H2}^{(1)}}{4M_F}$
										$+\frac{3h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{L_{a_1,i_1}} C P_R L_{a_2,i_2}^T g_2^T S_{1,j_2} y_{1,i_2,r_1} y_{2,i_1,r_1} \mathcal{I}_{y_{a_1,i_1}} \mathcal{I}_{y_{2,a_2,j_2}} S_{2,j_1}^\dagger}{4M_F}$	$-\frac{h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_3,r_1}} L_{a_1,i_1} C P_R L_{a_2,i_2}^T S_{1,j_2} y_{1,i_1,r_1} y_{2,i_3,r_1} y_{2,i_2,r_2} \mathcal{I}_{y_{a_1,i_1}} \mathcal{I}_{y_{2,a_2,j_2}} S_{2,i_1}^\dagger}{2M_F}$
										$-\frac{h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_3,r_1}} L_{a_1,i_1} C P_R L_{a_2,i_2}^T S_{1,j_2} y_{1,i_1,r_1} y_{1,i_3,r_2} y_{2,i_2,r_2} \mathcal{I}_{y_{a_1,i_1}} \mathcal{I}_{y_{2,a_2,j_2}} S_{2,i_1}^\dagger}{4M_F}$	$-\frac{h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{2,i_3,r_1}} L_{a_1,i_1} C P_R L_{a_2,i_2}^T S_{1,j_2} y_{1,i_3,r_2} y_{2,i_1,r_1} y_{2,i_2,r_2} \mathcal{I}_{y_{a_1,i_1}} \mathcal{I}_{y_{2,a_2,j_2}} S_{2,i_1}^\dagger}{4M_F}$
										$-\frac{h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{2,i_3,r_1}} L_{a_1,i_1} C P_R L_{a_2,i_2}^T S_{1,j_2} y_{1,i_1,r_1} y_{2,i_2,r_2} y_{2,i_3,r_1} \mathcal{I}_{y_{a_1,i_1}} \mathcal{I}_{y_{2,a_2,j_2}} S_{2,i_1}^\dagger}{2M_F}$	$-\frac{h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_3,r_1}} L_{a_1,i_1} C P_R L_{a_2,i_2}^T S_{1,j_2} y_{1,i_1,r_1} y_{2,i_3,r_1} y_{2,i_2,r_2} \mathcal{I}_{y_{a_1,i_1}} \mathcal{I}_{y_{2,a_2,j_2}} S_{2,i_1}^\dagger}{4M_F}$

[illegible]

[illegible]

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$FH\ell$	S_1S_2FH	$S_1S_2F\ell$
cont.										$-\frac{3h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1} L_{a_4,i_2}{}^T C P_L L_{a_2,i_1} H_{a_1} \lambda_{T3} \epsilon_{a_1,a_2} \epsilon_{a_3,a_4}}}{8M_F}$
										$+\frac{9\sqrt{3}h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{L_{a_1,i_1} C P_R L_{a_2,i_2}{}^T S_{1,j_1} y_{1,i_2,r_1} y_{1,i_3,r_2} y_{2,i_1,r_2} \epsilon_{a_1,a_2} S_{2,j_1}{}^\dagger}}{8M_F}$
										$-\frac{3h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{2,i_3,r_1} L_{a_1,i_1} C P_R L_{a_2,i_2}{}^T S_{1,j_1} y_{1,i_3,r_2} y_{2,i_1,r_2} y_{2,i_2,r_1} \epsilon_{a_1,a_2} S_{2,j_1}{}^\dagger}}{8M_F}$
cont.										$+\frac{3h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_3,r_1} L_{a_1,i_1} C P_R L_{a_2,i_2}{}^T S_{1,j_1} y_{1,i_1,r_2} y_{1,i_3,r_1} y_{2,i_2,r_2} \epsilon_{a_1,a_2} S_{2,j_1}{}^\dagger}}{4M_F}$
										$+\frac{3h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{2,i_3,r_1} L_{a_1,i_1} C P_R L_{a_2,i_2}{}^T S_{1,j_1} y_{1,i_1,r_2} y_{2,i_2,r_2} y_{2,i_3,r_1} \epsilon_{a_1,a_2} S_{2,j_1}{}^\dagger}}{4M_F}$
										$+\frac{3h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_3,r_1} L_{a_1,i_1} C P_R L_{a_2,i_2}{}^T S_{1,j_1} y_{1,i_1,r_2} y_{1,i_2,r_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{2,j_1}{}^\dagger}}{8M_F}$
cont.										$+\frac{3h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{2,i_3,r_1} L_{a_1,i_1} C P_R L_{a_2,i_2}{}^T S_{1,j_1} y_{1,i_1,r_2} y_{2,i_2,r_2} y_{1,i_3,r_2} \epsilon_{a_1,a_2} S_{2,j_1}{}^\dagger}}{8M_F}$
										$+\frac{\sqrt{3}h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_1} \epsilon_{a_1,a_2} S_{1,j_1}{}^\dagger \lambda_{12}^{(1)}}}{M_F}$
										$+\frac{\sqrt{3}h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{L_{a_1,i_1} C P_R L_{a_2,i_2}{}^T S_{1,j_1} y_{1,i_1,r_1} y_{2,i_2,r_2} \epsilon_{a_1,a_2} S_{2,j_1}{}^\dagger \lambda_{12}^{(1)}}}{M_F}$
cont.										$+\frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_2} I T 3_{a_1,a_2,j_2,j_1} S_{1,j_1}{}^\dagger \lambda_{12}^{(2)}}}{\sqrt{3}M_F}$
										$+\frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{L_{a_1,i_1} C P_R L_{a_2,i_2}{}^T S_{1,j_2} y_{1,i_1,r_1} y_{2,i_2,r_2} I T 3_{a_1,a_2,j_1,j_2} S_{2,j_1}{}^\dagger \lambda_{12}^{(2)}}}{\sqrt{3}M_F}$
										$-\frac{3h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_2} I T 3_{a_1,a_2,j_2,j_1} S_{1,j_1}{}^\dagger g_2^2}}{8M_F}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0		$-\frac{2h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1} L_{a_4,i_2}{}^T C P_L L_{a_2,i_1} H_{a_1} \lambda_{T3} \epsilon_{a_1,a_2} \epsilon_{a_3,a_4}}}{3M_F}$	$-2h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{2,i_1,r_1} Y_{e,i_2,i_3} H_{a_2} S_{1,j_2} S_{2,j_1} y_{1,i_2,r_1} Y_{e,i_1,i_3} \text{Bar}(\text{eg}4)_{a_2,j_1,a_1,j_2} H_{a_1}{}^\dagger}$
									$-\frac{2h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{\lambda_{T3} L_{a_3,i_1} C P_R L_{a_4,i_2}{}^T y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_3} \epsilon_{a_2,a_4} H_{a_1}{}^\dagger H_{a_2}{}^\dagger}}{3M_F}$	$-\frac{1}{2}\sqrt{3}h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} \lambda_{T3} S_{2,j_2} y_{1,i_1,r_1} I T 3_{a_1,a_2,j_1,j_2} H_{a_1}{}^\dagger H_{a_2}{}^\dagger S_{1,j_1}{}^\dagger}$
									$-\frac{1}{2}\sqrt{3}h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{2,i_1,r_1} \lambda_{T3} S_{2,j_2} y_{2,i_1,r_1} I T 3_{a_1,a_2,j_1,j_2} H_{a_1}{}^\dagger H_{a_2}{}^\dagger S_{1,j_1}{}^\dagger}$	$-\frac{1}{2}\sqrt{3}h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} H_{a_1} H_{a_2} S_{1,j_2} y_{1,i_1,r_1} I T 3_{a_1,a_2,j_2,j_1} \lambda_{T3} S_{2,j_1}{}^\dagger}$
cont.										$+\frac{9\sqrt{3}h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_1} \epsilon_{a_1,a_2} S_{1,j_1}{}^\dagger g_2^2}}{8M_F}$
										$-\frac{3h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{L_{a_1,i_1} C P_R L_{a_2,i_2}{}^T S_{1,j_2} y_{1,i_1,r_1} y_{2,i_2,r_2} I T 3_{a_1,a_2,j_1,j_2} S_{2,j_1}{}^\dagger g_2^2}}{8M_F}$
										$+\frac{9\sqrt{3}h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{L_{a_1,i_1} C P_R L_{a_2,i_2}{}^T S_{1,j_1} y_{1,i_1,r_1} y_{2,i_2,r_2} \epsilon_{a_1,a_2} S_{2,j_1}{}^\dagger g_2^2}}{8M_F}$
cont.										$-\frac{3h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} g_1^2 S_{2,j_2} I T 3_{a_1,a_2,j_2,j_1} S_{1,j_1}{}^\dagger}}{8M_F}$
										$+\frac{\sqrt{3}h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{1,i_3,r_2} y_{2,i_2,r_1} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{1,i_3,r_1} I T 3_{a_1,a_2,j_2,j_1} S_{1,j_1}{}^\dagger}}{8M_F}$
										$+\frac{\sqrt{3}h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{1,i_3,r_2} y_{2,i_2,r_1} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{1,i_3,r_2} I T 3_{a_1,a_2,j_2,j_1} S_{1,j_1}{}^\dagger}}{4M_F}$
cont.										$+\frac{\sqrt{3}h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{1,i_2,r_2} y_{2,i_3,r_1} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{1,i_3,r_2} I T 3_{a_1,a_2,j_2,j_1} S_{1,j_1}{}^\dagger}}{8M_F}$
										$+\frac{\sqrt{3}h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_3,r_1} y_{2,i_1,r_2} y_{2,i_3,r_2} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{2,i_3,r_1} I T 3_{a_1,a_2,j_2,j_1} S_{1,j_1}{}^\dagger}}{2M_F}$
										$+\frac{\sqrt{3}h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_3,r_1} y_{2,i_1,r_2} y_{2,i_3,r_1} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{2,i_3,r_2} I T 3_{a_1,a_2,j_2,j_1} S_{1,j_1}{}^\dagger}}{8M_F}$
cont.										$+\frac{\sqrt{3}h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} \text{eg}4_{a_1,j_2,a_2,j_1} H_{a_2} y_{2,i_1,r_1} H_{a_1}{}^\dagger S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger \lambda_{H2}^{(1)}}}{8M_F}$
										$+\frac{\sqrt{3}h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{2,i_1,r_1} H_{a_2} S_{1,j_1} S_{2,j_2} y_{1,i_1,r_1} \text{Bar}(\text{eg}4)_{a_2,j_1,a_1,j_2} H_{a_1}{}^\dagger \lambda_{H2}^{(2)}}}{8M_F}$
										$+\frac{\sqrt{3}h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} \text{eg}4_{a_1,j_1,a_2,j_2} H_{a_2} y_{2,i_1,r_1} H_{a_1}{}^\dagger S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger \lambda_{H2}^{(2)}}}{8M_F}$
cont.										$-\frac{3\sqrt{3}h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{1,i_3,r_1} \epsilon_{a_1,a_2} S_{1,j_1}{}^\dagger}}{8M_F}$
										$+\frac{3h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{1,i_3,r_2} y_{2,i_2,r_1} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{1,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}{}^\dagger}}{8M_F}$
										$+\frac{3h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{1,i_2,r_2} y_{2,i_3,r_1} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{1,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}{}^\dagger}}{4M_F}$
cont.										$+\frac{3h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_3,r_1} y_{2,i_1,r_2} y_{2,i_3,r_2} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}{}^\dagger}}{8M_F}$
										$+\frac{3h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_2} y_{2,i_3,r_1} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}{}^\dagger}}{8M_F}$
										$+\frac{3h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_2} y_{2,i_3,r_1} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}{}^\dagger}}{4M_F}$
cont.										$+\frac{\sqrt{3}h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{2,i_3,r_1} L_{a_1,i_1} C P_R L_{a_2,i_2}{}^T S_{1,j_2} y_{1,i_3,r_2} y_{2,i_1,r_2} y_{2,i_2,r_1} I T 3_{a_1,a_2,j_1,j_2} S_{2,j_1}{}^\dagger}}{8M_F}$
										$+\frac{\sqrt{3}h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_3,r_1} L_{a_1,i_1} C P_R L_{a_2,i_2}{}^T S_{1,j_2} y_{1,i_3,r_2} y_{2,i_1,r_2} y_{2,i_2,r_1} I T 3_{a_1,a_2,j_1,j_2} S_{2,j_1}{}^\dagger}}{8M_F}$
										$+\frac{\sqrt{3}h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_3,r_1} L_{a_1,i_1} C P_R L_{a_2,i_2}{}^T S_{1,j_2} y_{1,i_3,r_2} y_{2,i_1,r_2} y_{2,i_2,r_1} I T 3_{a_1,a_2,j_1,j_2} S_{2,j_1}{}^\dagger}}{4M_F}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$F H \ell$	$S_1 S_2 F H$	$S_1 S_2 F \ell$
										$\begin{aligned} & \sqrt{3} h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} L_{\alpha_1,i_1}} C P_R \overline{L_{\alpha_2,i_2}^\dagger} S_{1,j_2} y_{1,i_1,r_2} y_{2,i_2,r_2} y_{2,i_3,r_2} T r s_{\alpha_1,\alpha_2,j_1,j_2}^T S_{2,j_1}^\dagger \\ & + \frac{4 M_F}{8 M_F} \\ \text{cont.} & \sqrt{3} h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_3,r_1} L_{\alpha_1,i_1}} C P_R \overline{L_{\alpha_2,i_2}^\dagger} S_{1,j_2} y_{1,i_1,r_2} y_{1,i_2,r_2} y_{2,i_3,r_2} T r s_{\alpha_1,\alpha_2,j_1,j_2}^T S_{2,j_1}^\dagger \\ & + \frac{8 M_F}{8 M_F} \\ & \sqrt{3} h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{2,i_3,r_1} L_{\alpha_1,i_1}} C P_R \overline{L_{\alpha_2,i_2}^\dagger} S_{1,j_2} y_{1,i_1,r_2} y_{2,i_2,r_2} T r s_{\alpha_1,\alpha_2,j_1,j_2}^T S_{2,j_1}^\dagger \\ & + \frac{8 M_F}{8 M_F} \\ \text{cont.} & 3 \sqrt{3} h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{L_{\alpha_1,i_1}} C P_R \overline{L_{\alpha_2,i_2}^\dagger} g_2^T S_{1,j_1} y_{1,i_1,r_1} y_{2,i_2,r_1} e_{\alpha_1,\alpha_2} S_{2,j_1}^\dagger \\ & - \frac{8 M_F}{8 M_F} \\ & 3 h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{2,i_3,r_1} L_{\alpha_1,i_1}} C P_R \overline{L_{\alpha_2,i_2}^\dagger} S_{1,j_1} y_{1,i_3,r_2} y_{2,i_1,r_2} y_{2,i_2,r_1} e_{\alpha_1,\alpha_2} S_{2,j_1}^\dagger \\ & - \frac{8 M_F}{8 M_F} \\ & 3 h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_3,r_1} L_{\alpha_1,i_1}} C P_R \overline{L_{\alpha_2,i_2}^\dagger} S_{1,j_1} y_{1,i_1,r_2} y_{1,i_3,r_1} y_{2,i_2,r_2} e_{\alpha_1,\alpha_2} S_{2,j_1}^\dagger \\ & + \frac{4 M_F}{4 M_F} \\ & 3 h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{2,i_3,r_1} L_{\alpha_1,i_1}} C P_R \overline{L_{\alpha_2,i_2}^\dagger} S_{1,j_1} y_{1,i_1,r_2} y_{2,i_2,r_2} y_{2,i_3,r_1} e_{\alpha_1,\alpha_2} S_{2,j_1}^\dagger \\ & + \frac{4 M_F}{4 M_F} \\ \text{cont.} & 3 h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_3,r_1} L_{\alpha_1,i_1}} C P_R \overline{L_{\alpha_2,i_2}^\dagger} S_{1,j_1} y_{1,i_1,r_2} y_{1,i_2,r_1} y_{2,i_3,r_2} e_{\alpha_1,\alpha_2} S_{2,j_1}^\dagger \\ & + \frac{8 M_F}{8 M_F} \\ & 3 h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{2,i_3,r_1} L_{\alpha_1,i_1}} C P_R \overline{L_{\alpha_2,i_2}^\dagger} S_{1,j_1} y_{1,i_1,r_2} y_{2,i_2,r_1} y_{2,i_3,r_2} e_{\alpha_1,\alpha_2} S_{2,j_1}^\dagger \\ & + \frac{8 M_F}{8 M_F} \\ \text{cont.} & \sqrt{3} h \log \left(\frac{\mu^2}{M_F^2} \right) y_{1,i_1,r_1} y_{2,i_2,r_1} L_{\alpha_2,i_2}{}^T C P_L L_{\alpha_1,i_1} S_{2,j_1} e_{\alpha_1,\alpha_2} S_{1,j_1}^\dagger \lambda_{12}^{(1)} \\ & + \frac{M_F}{M_F} \\ & \sqrt{3} h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{L_{\alpha_1,i_1}} C P_R \overline{L_{\alpha_2,i_2}^\dagger} S_{1,j_1} y_{1,i_1,r_1} y_{2,i_2,r_1} e_{\alpha_1,\alpha_2} S_{2,j_1}^\dagger \lambda_{12}^{(1)} \\ & + \frac{M_F}{M_F} \\ & h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1} L_{\alpha_2,i_2}{}^T C P_L L_{\alpha_1,i_1} S_{2,j_2} T r s_{\alpha_1,\alpha_2,j_2,j_1} S_{1,j_1}^\dagger \lambda_{12}^{(2)}} \\ & + \frac{\sqrt{3} M_F}{\sqrt{3} M_F} \\ & h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{L_{\alpha_1,i_1}} C P_R \overline{L_{\alpha_2,i_2}^\dagger} S_{1,j_2} y_{1,i_1,r_1} y_{2,i_2,r_1} T r s_{\alpha_1,\alpha_2,j_1,j_2}^T S_{2,j_1}^\dagger \lambda_{12}^{(2)} \\ & + \frac{\sqrt{3} M_F}{\sqrt{3} M_F} \end{aligned}$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	$\begin{aligned} & - \frac{2 h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1} L_{\alpha_4,i_2}{}^T C P_L L_{\alpha_2,i_1} H_{\alpha_1} H_{\alpha_3} \lambda_{T3} e_{\alpha_1,\alpha_2} e_{\alpha_3,\alpha_4}}} {3 M_F} \\ & - \frac{2 h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{\lambda_{T3} L_{\alpha_3,i_1}} C P_R \overline{L_{\alpha_4,i_2}^\dagger} y_{1,i_1,r_1} y_{2,i_2,r_1} e_{\alpha_1,\alpha_3} e_{\alpha_2,\alpha_4} H_{\alpha_1}^\dagger H_{\alpha_2}^\dagger} {3 M_F} \end{aligned}$	$\begin{aligned} & - \frac{1}{2} \sqrt{3} h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} \overline{\lambda_{T3}} S_{2,j_2} y_{1,i_1,r_1} I T s_{\alpha_1,\alpha_2,j_1,j_2}^* H_{\alpha_1}^\dagger H_{\alpha_2}^\dagger S_{1,j_1}^\dagger} \\ & - \frac{1}{2} \sqrt{3} h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{2,i_1,r_1} \overline{\lambda_{T3}} S_{2,j_2} y_{2,i_1,r_1} I T s_{\alpha_1,\alpha_2,j_1,j_2}^* H_{\alpha_1}^\dagger H_{\alpha_2}^\dagger S_{1,j_1}^\dagger} \\ & - \frac{1}{2} \sqrt{3} h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} H_{\alpha_1} H_{\alpha_2} S_{1,j_2} y_{1,i_1,r_1} I T s_{\alpha_1,\alpha_2,j_2,j_1} \lambda_{T3} S_{2,j_1}^\dagger} \end{aligned}$	$\begin{aligned} & - \frac{3 h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1} L_{\alpha_2,i_2}{}^T C P_L L_{\alpha_1,i_1} g_2^T S_{2,T3,\alpha_1,\alpha_2,j_2,j_1} S_{1,j_1}^\dagger}} {8 M_F} \\ & + \frac{\sqrt{3} h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{1,i_3,r_2} y_{2,i_2,r_2} L_{\alpha_2,i_2}{}^T C P_L L_{\alpha_1,i_1} S_{2,j_2} y_{1,i_3,r_1} I T s_{\alpha_1,\alpha_2,j_2,j_1} S_{1,j_1}^\dagger}} {8 M_F} \\ & + \frac{\sqrt{3} h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{1,i_3,r_2} y_{2,i_2,r_2} L_{\alpha_2,i_2}{}^T C P_L L_{\alpha_1,i_1} S_{2,j_2} y_{1,i_3,r_2} I T s_{\alpha_1,\alpha_2,j_2,j_1} S_{1,j_1}^\dagger}} {4 M_F} \\ & + \frac{\sqrt{3} h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{1,i_2,r_2} y_{2,i_3,r_2} L_{\alpha_2,i_2}{}^T C P_L L_{\alpha_1,i_1} S_{2,j_2} y_{1,i_2,r_2} I T s_{\alpha_1,\alpha_2,j_2,j_1} S_{1,j_1}^\dagger}} {8 M_F} \\ & + \frac{\sqrt{3} h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{1,i_2,r_2} y_{2,i_3,r_2} L_{\alpha_2,i_2}{}^T C P_L L_{\alpha_1,i_1} S_{2,j_2} y_{1,i_2,r_2} I T s_{\alpha_1,\alpha_2,j_2,j_1} S_{1,j_1}^\dagger}} {8 M_F} \\ & + \frac{\sqrt{3} h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{1,i_2,r_2} y_{2,i_3,r_2} L_{\alpha_2,i_2}{}^T C P_L L_{\alpha_1,i_1} S_{2,j_2} y_{1,i_2,r_2} I T s_{\alpha_1,\alpha_2,j_2,j_1} S_{1,j_1}^\dagger}} {4 M_F} \\ \text{cont.} & 9 \sqrt{3} h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1} L_{\alpha_2,i_2}{}^T C P_L L_{\alpha_1,i_1} g_2^T S_{2,j_1} e_{\alpha_1,\alpha_2} S_{1,j_1}^\dagger} \\ & + \frac{8 M_F}{8 M_F} \\ & 3 h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{1,i_3,r_2} y_{2,i_2,r_2} L_{\alpha_2,i_2}{}^T C P_L L_{\alpha_1,i_1} S_{2,j_1} y_{1,i_3,r_1} e_{\alpha_1,\alpha_2} S_{1,j_1}^\dagger} \\ & + \frac{8 M_F}{8 M_F} \\ & 3 h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{1,i_3,r_2} y_{2,i_2,r_2} L_{\alpha_2,i_2}{}^T C P_L L_{\alpha_1,i_1} S_{2,j_1} y_{1$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$FH\ell$	$S_1 S_2 F H$	$S_1 S_2 F \ell$
cont.									$\frac{3h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y}_{1,i_1,r_1} L_{a_1,i_1}}{4M_F} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_1,r_2} y_{1,i_3,r_1} W_{2,i_2,r_2} \epsilon_{a_1,a_2} S_{2,j_1}^\dagger$	$\frac{3h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y}_{1,i_3,r_1} L_{a_1,i_1}}{8M_F} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_1,r_2} y_{1,i_2,r_1} W_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{2,j_1}^\dagger$
									$\frac{3h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y}_{2,i_3,r_1} L_{a_1,i_1}}{4M_F} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_1,r_2} y_{2,i_2,r_2} W_{2,i_3,r_1} \epsilon_{a_1,a_2} S_{2,j_1}^\dagger$	$\frac{3h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y}_{2,i_3,r_1} L_{a_1,i_1}}{8M_F} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_1,r_2} y_{2,i_2,r_1} W_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{2,j_1}^\dagger$
cont.									$+\frac{\sqrt{3} h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y}_{1,i_1,r_1} y_{2,i_2,r_1} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_1} \epsilon_{a_1,a_2} S_{1,j_1}^\dagger \lambda_{12}^{(1)}}{M_F}$	$+\frac{\sqrt{3} h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_2} S_{2,j_1}^\dagger \lambda_{12}^{(1)}}{M_F}$
									$+\frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y}_{1,i_1,r_1} y_{2,i_2,r_1} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_2} \mathcal{I} T_{3,a_1,a_2,j_2,j_1} S_{1,j_1}^\dagger \lambda_{12}^{(2)}}{\sqrt{3} M_F}$	$+\frac{h \log \left(\frac{\mu^2}{M_F^2} \right) L_{a_1,i_1} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_2} y_{1,i_1,r_1} y_{2,i_2,r_1} \mathcal{I} T_{3,a_1,a_2,j_1,j_2} S_{2,j_1}^\dagger \lambda_{12}^{(2)}}{\sqrt{3} M_F}$
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	$-\frac{2h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y}_{1,i_1,r_1} y_{2,i_2,r_1} L_{a_2,i_2}{}^T C P_L L_{a_2,i_1} H_{a_1} \lambda T_{3,\epsilon_{a_1},a_2,\epsilon_{a_3},a_4}}{3M_F}$	$-\frac{1}{2} \sqrt{3} h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y}_{1,i_1,r_1} \overline{\lambda T_{3,S_{2,j_2} j_2}} y_{1,i_1,r_1} \mathcal{I} T_{3,a_1,a_2,j_1,j_2} H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger$	$-\frac{3h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y}_{1,i_1,r_1} y_{2,i_2,r_1} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} g_{S_2,j_2}^2 \mathcal{I} T_{3,a_1,a_2,j_2,j_1} S_{1,j_1}^\dagger}{8M_F}$
								$-\frac{2h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{\lambda T_{3,L_{a_3,i_1}}} C P_R \overline{L_{a_4,i_2}}^T y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_3} \epsilon_{a_2,a_4} H_{a_1}^\dagger H_{a_2}^\dagger}{3M_F}$	$-\frac{1}{2} \sqrt{3} h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y}_{2,i_1,r_1} \overline{\lambda T_{3,S_{2,j_2} j_2}} y_{2,i_1,r_1} \mathcal{I} T_{3,a_1,a_2,j_1,j_2} H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger$	$+\frac{9h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y}_{1,i_1,r_1} y_{2,i_2,r_1} y_{2,i_2,r_1} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} g_{S_2,j_2}^2 \mathcal{I} T_{3,a_1,a_2,j_2,j_1} S_{1,j_1}^\dagger}{8M_F}$
cont.								$-\frac{1}{2} \sqrt{3} h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y}_{1,i_1,r_1} H_{a_1} H_{a_2} S_{1,j_2} y_{1,i_1,r_1} \mathcal{I} T_{3,a_1,a_2,j_2,j_1} \lambda T_{3,S_{2,j_1}^\dagger}$	$+\frac{\sqrt{3} h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y}_{1,i_1,r_1} y_{2,i_2,r_2} y_{2,i_2,r_1} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{1,i_3,r_1} \mathcal{I} T_{3,a_1,a_2,j_2,j_1} S_{1,j_1}^\dagger}{8M_F}$	$+\frac{\sqrt{3} h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y}_{1,i_1,r_1} y_{2,i_2,r_2} y_{2,i_2,r_1} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{1,i_3,r_2} \mathcal{I} T_{3,a_1,a_2,j_2,j_1} S_{1,j_1}^\dagger}{4M_F}$
								$-\frac{1}{2} \sqrt{3} h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y}_{2,i_1,r_1} H_{a_1} H_{a_2} S_{1,j_2} y_{2,i_1,r_1} \mathcal{I} T_{3,a_1,a_2,j_2,j_1} \lambda T_{3,S_{2,j_1}^\dagger}$	$+\frac{\sqrt{3} h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y}_{1,i_1,r_1} y_{2,i_2,r_1} y_{2,i_2,r_2} y_{2,i_2,r_1} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{1,i_3,r_2} \mathcal{I} T_{3,a_1,a_2,j_2,j_1} S_{1,j_1}^\dagger}{8M_F}$	$+\frac{9\sqrt{3} h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y}_{1,i_1,r_1} y_{2,i_2,r_1} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} g_{S_2,j_2}^2 S_{2,j_1}^\dagger \epsilon_{a_1,a_2} S_{1,j_1}^\dagger}{8M_F}$
cont.									$+\frac{9\sqrt{3} h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y}_{1,i_1,r_1} y_{2,i_2,r_2} y_{2,i_2,r_1} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} g_{S_2,j_2}^2 S_{2,j_1}^\dagger \epsilon_{a_1,a_2} S_{1,j_1}^\dagger}{8M_F}$	$+\frac{3h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y}_{1,i_1,r_1} y_{1,i_3,r_2} y_{2,i_2,r_2} y_{2,i_2,r_1} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{1,i_3,r_1} \epsilon_{a_1,a_2} S_{1,j_1}^\dagger}{8M_F}$
									$+\frac{3h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y}_{1,i_1,r_1} y_{1,i_3,r_2} y_{2,i_2,r_2} y_{2,i_2,r_1} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{1,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}^\dagger}{4M_F}$	$+\frac{3h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y}_{1,i_1,r_1} y_{1,i_2,r_2} y_{2,i_2,r_1} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{1,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}^\dagger}{8M_F}$
cont.									$+\frac{3h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y}_{1,i_3,r_1} y_{2,i_1,r_2} y_{2,i_2,r_1} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}^\dagger}{8M_F}$	$+\frac{3h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y}_{1,i_1,r_1} y_{2,i_2,r_1} y_{2,i_2,r_2} y_{2,i_2,r_1} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{1,i_3,r_2} \mathcal{I} T_{3,a_1,a_2,j_2,j_1} S_{1,j_1}^\dagger}{8M_F}$
									$+\frac{3h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y}_{1,i_1,r_1} y_{2,i_2,r_1} y_{2,i_2,r_2} y_{2,i_2,r_1} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{1,i_3,r_2} \mathcal{I} T_{3,a_1,a_2,j_2,j_1} S_{1,j_1}^\dagger}{4M_F}$	$+\frac{\sqrt{3} h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y}_{1,i_3,r_1} L_{a_1,i_1} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_2} y_{1,i_3,r_2} y_{2,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_2} S_{2,j_1}^\dagger}{8M_F}$
cont.									$-\frac{3h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T g_{S_1,j_2}^2 S_{1,j_2} y_{1,i_1,r_1} y_{2,i_2,r_1} \mathcal{I} T_{3,a_1,a_2,j_1,j_2} S_{2,j_1}^\dagger}{8M_F}$	$-\frac{9h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T g_{S_1,j_2}^2 S_{1,j_2} y_{1,i_1,r_1} y_{2,i_2,r_1} \mathcal{I} T_{3,a_1,a_2,j_1,j_2} S_{2,j_1}^\dagger}{8M_F}$
									$+\frac{\sqrt{3} h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y}_{2,i_3,r_1} L_{a_1,i_1} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_2} y_{1,i_3,r_2} y_{2,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_2} S_{2,j_1}^\dagger}{8M_F}$	$+\frac{\sqrt{3} h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y}_{2,i_3,r_1} L_{a_1,i_1} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_2} y_{1,i_3,r_2} y_{2,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_2} S_{2,j_1}^\dagger}{4M_F}$
cont.									$+\frac{\sqrt{3} h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y}_{2,i_3,r_1} L_{a_1,i_1} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_2} y_{1,i_3,r_2} y_{2,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_2} S_{2,j_1}^\dagger}{8M_F}$	$+\frac{\sqrt{3} h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y}_{2,i_3,r_1} L_{a_1,i_1} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_2} y_{1,i_3,r_2} y_{2,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_2} S_{2,j_1}^\dagger}{4M_F}$
									$+\frac{\sqrt{3} h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y}_{2,i_3,r_1} L_{a_1,i_1} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_2} y_{1,i_3,r_2} y_{2,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_2} S_{2,j_1}^\dagger}{8M_F}$	$+\frac{\sqrt{3} h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y}_{2,i_3,r_1} L_{a_1,i_1} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_2} y_{1,i_3,r_2} y_{2,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_2} S_{2,j_1}^\dagger}{4M_F}$
cont.									$+\frac{\sqrt{3} h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y}_{2,i_3,r_1} L_{a_1,i_1} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_2} y_{1,i_3,r_2} y_{2,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_2} S_{2,j_1}^\dagger}{8M_F}$	$+\frac{\sqrt{3} h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y}_{2,i_3,r_1} L_{a_1,i_1} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_2} y_{1,i_3,r_2} y_{2,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_2} S_{2,j_1}^\dagger}{4M_F}$
									$+\frac{\sqrt{3} h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y}_{2,i_3,r_1} L_{a_1,i_1} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_2} y_{1,i_3,r_2} y_{2,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_2} S_{2,j_1}^\dagger}{8M_F}$	$+\frac{\sqrt{3} h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y}_{2,i_3,r_1} L_{a_1,i_1} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_2} y_{1,i_3,r_2} y_{2,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_2} S_{2,j_1}^\dagger}{4M_F}$
cont.									$+\frac{\sqrt{3} h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y}_{2,i_3,r_1} L_{a_1,i_1} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_2} y_{1,i_3,r_2} y_{2,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_2} S_{2,j_1}^\dagger}{8M_F}$	$+\frac{\sqrt{3} h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y}_{2,i_3,r_1} L_{a_1,i_1} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_2} y_{1,i_3,r_2} y_{2,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_2} S_{2,j_1}^\dagger}{4M_F}$
									$+\frac{\sqrt{3} h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y}_{2,i_3,r_1} L_{a_1,i_1} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_2} y_{1,i_3,r_2} y_{2,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_2} S_{2,j_1}^\dagger}{8M_F}$	$+\frac{\sqrt{3} h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y}_{2,i_3,r_1} L_{a_1,i_1} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_2} y_{1,i_3,r_2} y_{2,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_2} S_{2,j_1}^\dagger}{4M_F}$
cont.									$+\frac{\sqrt{3} h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y}_{2,i_3,r_1} L_{a_1,i_1} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_2} y_{1,i_3,r_2} y_{2,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_2} S_{2,j_1}^\dagger}{8M_F}$	$+\frac{\sqrt{3} h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y}_{2,i_3,r_1} L_{a_1,i_1} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_2} y_{1,i_3,r_2} y_{2,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_2} S_{2,j_1}^\dagger}{4M_F}$
									$+\frac{\sqrt{3} h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y}_{2,i_3,r_1} L_{a_1,i_1} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_2} y_{1,i_3,r_2} y_{2,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_2} S_{2,j_1}^\dagger}{8M_F}$	$+\frac{\sqrt{3} h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y}_{2,i_3,r_1} L_{a_1,i_1} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_2} y_{1,i_3,r_2} y_{2,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_2} S_{2,j_1}^\dagger}{4M_F}$
cont.									$+\frac{\sqrt{3} h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y}_{2,i_3,r_1} L_{a_1,i_1} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_2} y_{1,i_3,r_2} y_{2,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_2} S_{2,j_1}^\dagger}{8M_F}$	$+\frac{\sqrt{3} h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y}_{2,i_3,r_1} L_{a_1,i_1} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_2} y_{1,i_3,r_2} y_{2,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_2} S_{2,j_1}^\dagger}{4M_F}$
									$+\frac{\sqrt{3} h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y}_{2,i_3,r_1} L_{a_1,i_1} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_2} y_{1,i_3,r_2} y_{2,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_2} S_{2,j_1}^\dagger}{8M_F}$	$+\frac{\sqrt{3} h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y}_{2,i_3,r_1} L_{a_1,i_1} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_2} y_{1,i_3,r_2} y_{2,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_2} S_{2,j_1}^\dagger}{4M_F}$
cont.									$+\frac{\sqrt{3} h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y}_{2,i_3,r_1} L_{a_1,i_1} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_2} y_{1,i_3,r_2} y_{2,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_2} S_{2,j_1}^\dagger}{8M_F}$	$+\frac{\sqrt{3} h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y}_{2,i_3,r_1} L_{a_1,i_1} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_2} y_{1,i_3,r_2} y_{2,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_2} S_{2,j_1}^\dagger}{4M_F}$
									$+\frac{\sqrt{3} h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y}_{2,i_3,r_1} L_{a_1,i_1} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_2} y_{1,i_3,r_2} y_{2,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_2} S_{2,j_1}^\dagger}{8M_F}$	$+\frac{\sqrt{3} h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y}_{2,i_3,r_1} L_{a_1,i_1} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_2} y_{1,i_3,r_2} y_{2,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_2} S_{2,j_1}^\dagger}{4M_F}$
cont.									$+\frac{\sqrt{3} h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y}_{2,i_3,r_1} L_{a_1,i_1} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_2} y_{1,i_3,r_2} y_{2,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_2} S_{2,j_1}^\dagger}{8M_F}$	$+\frac{\sqrt{3} h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y}_{2,i_3,r_1} L_{a_1,i_1} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_2} y_{1,i_3,r_2} y_{2,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_2} S_{2,j_1}^\dagger}{4M_F}$
									$+\frac{\sqrt{3} h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y}_{2,i_3,r_1} L_{a_1,i_1} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_2} y_{1,i_3,r_2} y_{2,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_2} S_{2,j_1}^\dagger}{8M_F}$	$+\frac{\sqrt{3} h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y}_{2,i_3,r_1} L_{a_1,i_1} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_2} y_{1,i_3,r_2} y_{2,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_2} S_{2,j_1}^\dagger}{4M_F}$
cont.									$+\frac{\sqrt{3} h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y}_{2,i_3,r_1} L_{a_1,i_1} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_2} y_{1,i_3,r_2} y_{2,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_2} S_{2,j_1}^\dagger}{8M_F}$	$+\frac{\sqrt{3} h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y}_{2,i_3,r_1} L_{a_1,i_1} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_2} y_{1,i_3,r_2} y_{2,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_2} S_{2,j_1}^\dagger}{4M_F}$
									$+\frac{\sqrt{3} h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y}_{2,i_3,r_1} L_{a_1,i_1} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_2} y_{1,i_3,r_2} y_{2,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_2} S_{2,j_1}^\dagger}{8M_F}$	$+\frac{\sqrt{3} h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y}_{2,i_3,r_1} L_{a_1,i_1} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_2} y_{1,i_3,r_2} y_{2,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_2} S_{2,j_1}^\dagger}{4M_F}$
cont.									$+\frac{\sqrt{3} h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y}_{2,i_3,r_1} L_{a_1,i_1} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_2} y_{1,i_3,r_2} y_{2,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_2} S_{2,j_1}^\dagger}{8M_F}$	$+\frac{\sqrt{3} h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y}_{2,i_3,r_1} L_{a_1,i_1} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_2} y_{1,i_3,r_2} y_{2,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_2} S_{2,j_1}^\dagger}{4M_F}$
									$+\frac{\sqrt{3} h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y}_{2,i_3,r_1} L_{a_1,i_1} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_2} y_{1,i_3,r_2} y_{2,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_2} S_{2,j_1}^\dagger}{8M_F}$	$+\frac{\sqrt{3} h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y}_{2,i_3,r_1} L_{a_1,i_1} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_2} y_{1,i_3,r_2} y_{2,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_2} S_{2,j_1}^\dagger}{4M_F}$
cont.									$+\frac{\sqrt{3} h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y}_{2,i_3,r_1} L_{a_1,i_1} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_2} y_{1,i_3,r_2} y_{2,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_2} S_{2,j_1}^\dagger}{8M_F}$	$+\frac{\sqrt{3} h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y}_{2,i_3,r_1} L_{a_1,i_1} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_2} y_{1,i_3,r_2} y_{2,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_2} S_{2,j_1}^\dagger}{4M_F}$
									$+\frac{\sqrt{3} h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y}_{2,i_3,r_1} L_{a_1,i_1} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_2} y_{1,i_3,r_2} y_{2,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_2} S_{2,j_1}^\dagger}{8M_F}$	$+\frac{\sqrt{3} h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y}_{2,i_3,r_1} L_{a_1,i_1} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_2} y_{1,i_3,r_2} y_{2,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_2} S_{2,j_1}^\dagger}{4M_F}$
cont.									$+\frac{\sqrt{3} h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y}_{2,i_3,r_1} L_{a_1,i_1} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_2} y_{1,i_3,r_2} y_{2,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_2} S_{2,j_1}^\dagger}{8M_F}$	$+\frac{\sqrt{3} h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y}_{2,i_3,r_1} L_{a_1,i_1} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_2} y_{1,i_3,r_2} y_{2,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_2} S_{2,j_1}^\dagger}{4M_F}$
									$+\frac{\sqrt{3} h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y}_{2,i_3,r_1} L_{a_1,i_1} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_2} y_{1,i_3,r_2} y_{2,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_2} S_{2,j_1}^\dagger}{8M_F}$	$+\frac{\sqrt{3} h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y}_{2,i_3,r_1} L_{a_1,i_1} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_2} y_{1,i_3,r_2} y_{2,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_2} S_{2,j_1}^\dagger}{4M_F}$
cont.										

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	F	H	ℓ	S_1	S_2	F	H	S_1	S_2	F	H	ℓ																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																	
cont.															$-\frac{3h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2,1},r_1} \overline{L_{a_1,1}} C P_R \overline{L_{a_2,2}}^{-T} S_{1,j_1} y_{1,1},r_2 y_{2,2,1},r_2 y_{2,2,2},r_1 \epsilon_{a_1,a_2} S_{2,j_1}^\dagger}{8M_F}$							$+\frac{3h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,1},r_1} \overline{L_{a_1,1}} C P_R \overline{L_{a_2,2}}^{-T} S_{1,j_1} y_{1,1},r_2 y_{1,1},r_2 y_{2,2,2},r_2 \epsilon_{a_1,a_2} S_{2,j_1}^\dagger}{4M_F}$							$+\frac{3h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2,1},r_1} \overline{L_{a_1,1}} C P_R \overline{L_{a_2,2}}^{-T} S_{1,j_1} y_{1,1},r_2 y_{2,2,2},r_2 y_{2,1},r_1 \epsilon_{a_1,a_2} S_{2,j_1}^\dagger}{4M_F}$							$+\frac{3h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,1},r_1} \overline{L_{a_1,1}} C P_R \overline{L_{a_2,2}}^{-T} S_{1,j_1} y_{1,1},r_2 y_{1,1},r_2 y_{2,1},r_2 \epsilon_{a_1,a_2} S_{2,j_1}^\dagger}{8M_F}$							$+\frac{3h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2,1},r_1} \overline{L_{a_1,1}} C P_R \overline{L_{a_2,2}}^{-T} S_{1,j_1} y_{1,1},r_2 y_{2,1},r_2 y_{2,1},r_2 \epsilon_{a_1,a_2} S_{2,j_1}^\dagger}{8M_F}$							$+\frac{\sqrt{3}h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,1},r_1} \overline{y_{2,2,2},r_1} \overline{L_{a_2,2}}^{-T} C P_L \overline{L_{a_1,1}} S_{2,j_1} \epsilon_{a_1,a_2} S_{1,j_1}^\dagger \lambda_{12}^{(1)}}{M_F}$							$+\frac{\sqrt{3}h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{L_{a_1,1}} C P_R \overline{L_{a_2,2}}^{-T} S_{1,j_1} y_{1,1},r_1 y_{2,2,2},r_1 \epsilon_{a_1,a_2} S_{2,j_1}^\dagger \lambda_{12}^{(1)}}{M_F}$							$+\frac{h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,1},r_1} \overline{y_{2,2,2},r_1} \overline{L_{a_2,2}}^{-T} C P_L \overline{L_{a_1,1}} S_{2,j_2} \mathcal{I} \mathcal{T} S_{a_1,a_2,j_2,j_1} S_{1,j_1}^\dagger \lambda_{12}^{(2)}}{\sqrt{3}M_F}$							$+\frac{h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{L_{a_1,1}} C P_R \overline{L_{a_2,2}}^{-T} S_{1,j_2} y_{1,1},r_1 y_{2,2,2},r_1 \mathcal{I} \mathcal{T} S_{a_1,a_2,j_2,j_2} S_{2,j_2}^\dagger \lambda_{12}^{(2)}}{\sqrt{3}M_F}$																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																													

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$F H \ell$	$S_1 S_2 F H$	$S_1 S_2 F \ell$
$cont.$										$- \frac{3h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2,i_3,r_1} L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}^T} S_{1,j_1} y_{1,i_3,r_2} y_{2,i_1,r_2} y_{2,i_2,r_1} \epsilon_{a_1,a_2} S_{2,j_1}^\dagger}{8M_F}$ $+ \frac{3h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_3,r_1} L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}^T} S_{1,j_1} y_{1,i_1,r_2} y_{1,i_3,r_1} y_{2,i_2,r_2} \epsilon_{a_1,a_2} S_{2,j_1}^\dagger}{4M_F}$ $+ \frac{3h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2,i_3,r_1} L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}^T} S_{1,j_1} y_{1,i_1,r_2} y_{2,i_2,r_2} y_{2,i_3,r_1} \epsilon_{a_1,a_2} S_{2,j_1}^\dagger}{4M_F}$ $+ \frac{3h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_3,r_1} L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}^T} S_{1,j_1} y_{1,i_1,r_2} y_{1,i_2,r_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{2,j_1}^\dagger}{4M_F}$
$cont.$										$+ \frac{3h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2,i_3,r_1} L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}^T} S_{1,j_1} y_{1,i_1,r_2} y_{2,i_2,r_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{2,j_1}^\dagger}{8M_F}$ $+ \frac{\sqrt{3} h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1} L_{a_2,i_2}^T} C P_L L_{a_1,i_1} S_{2,j_1} \epsilon_{a_1,a_2} S_{1,j_1}^\dagger \lambda_{12}^{(1)}}{M_F}$ $+ \frac{\sqrt{3} h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}^T} S_{1,j_1} y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_2} S_{2,j_1}^\dagger \lambda_{12}^{(1)}}{M_F}$
$cont.$										$+ \frac{h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1} L_{a_2,i_2}^T} C P_L L_{a_1,i_1} S_{2,j_2} \overline{I} T_{3,a_1,a_2,j_2,j_1} S_{1,j_1}^\dagger \lambda_{12}^{(2)}}{\sqrt{3} M_F}$ $+ \frac{h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}^T} S_{1,j_2} y_{1,i_1,r_1} y_{2,i_2,r_1} \overline{I} T_{3,a_1,a_2,j_1,j_2} S_{2,j_1}^\dagger \lambda_{12}^{(2)}}{\sqrt{3} M_F}$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$+ \frac{2h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1} L_{a_4,i_2}^T} C P_L L_{a_2,i_1} H_{a_1} H_{a_3} \lambda_{T3} \epsilon_{a_1,a_2} \epsilon_{a_3,a_4}}{\sqrt{3} M_F}$ $+ \frac{2h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{\lambda_{T3} L_{a_3,i_1}} C P_R \overline{L_{a_4,i_2}^T} y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_3} \epsilon_{a_2,a_4} H_{a_1}^\dagger H_{a_2}^\dagger}{\sqrt{3} M_F}$	$- \frac{h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1}} H_{a_1} H_{a_2} S_{1,j_1} y_{1,i_1,r_1} \overline{I} T_{3,a_1,a_2,j_1} \lambda_{T3} S_{2,j_1}^\dagger}{\sqrt{3}}$ $- h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1}} H_{a_1} H_{a_2} S_{1,j_1} y_{2,i_1,r_1} \overline{I} T_{3,a_1,a_2,j_1} \lambda_{T3} S_{2,j_1}^\dagger$ $- \frac{h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1}} \lambda_{T3} S_{2,j_1,i_1,r_1} \overline{I} T_{3,a_1,a_2,j_1} H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger}{\sqrt{3}}$	$- \frac{\sqrt{3} h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_3,r_1} L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}^T} S_{1,j_1} y_{1,i_2,r_1} y_{1,i_3,r_2} y_{2,i_1,r_2} \overline{I} T_{3,a_1,a_2,j_1} S_{2,j_1}^\dagger}{4M_F}$ $- \frac{h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2,i_3,r_1} L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}^T} S_{1,j_1} y_{1,i_3,r_2} y_{2,i_1,r_2} y_{2,i_2,r_1} \overline{I} T_{3,a_1,a_2,j_1} S_{2,j_1}^\dagger}{4M_F}$ $- \frac{h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_3,r_1} L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}^T} S_{1,j_1} y_{1,i_1,r_2} y_{2,i_1,r_3,r_1} y_{2,i_2,r_2} \overline{I} T_{3,a_1,a_2,j_1} S_{2,j_1}^\dagger}{\sqrt{3} M_F}$ $- \frac{h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2,i_3,r_1} L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}^T} S_{1,j_1} y_{1,i_1,r_2} y_{2,i_2,r_2} y_{2,i_3,r_1} \overline{I} T_{3,a_1,a_2,j_1} S_{2,j_1}^\dagger}{M_F}$ $- \frac{\sqrt{3} h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_3,r_1} L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}^T} S_{1,j_1} y_{1,i_1,r_2} y_{2,i_1,r_3,r_2} \overline{I} T_{3,a_1,a_2,j_1} S_{2,j_1}^\dagger}{4M_F}$ $- \frac{h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2,i_3,r_1} L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}^T} S_{1,j_1} y_{1,i_1,r_2} y_{2,i_2,r_1} y_{2,i_3,r_2} \overline{I} T_{3,a_1,a_2,j_1} S_{2,j_1}^\dagger}{4M_F}$ $- \frac{\sqrt{3} h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1}} y_{1,i_3,r_2} y_{2,i_2,r_2} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_1,i_3,r_1} \overline{I} T_{3,a_1,a_2,j_1} S_{1,j_1}^\dagger}{4M_F}$ $- \frac{h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1}} y_{1,i_3,r_2} y_{2,i_2,r_2} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_1,i_3,r_2} \overline{I} T_{3,a_1,a_2,j_1} S_{1,j_1}^\dagger}{\sqrt{3} M_F}$ $- \frac{\sqrt{3} h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1}} y_{1,i_2,r_2} y_{2,i_2,r_1} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_1,i_3,r_3,r_2} \overline{I} T_{3,a_1,a_2,j_1} S_{1,j_1}^\dagger}{4M_F}$ $- \frac{h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_3,r_1} y_{2,i_1,r_2} y_{2,i_2,r_1} L_{a_2,i_2}^T} C P_L L_{a_1,i_1} S_{2,j_2,i_3,r_2} \overline{I} T_{3,a_1,a_2,j_1} S_{1,j_1}^\dagger}{4M_F}$ $- \frac{h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1}} y_{2,i_2,r_2} y_{2,i_3,r_1} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_2,i_3,r_2} \overline{I} T_{3,a_1,a_2,j_1} S_{1,j_1}^\dagger}{4M_F}$ $- \frac{h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1}} y_{2,i_2,r_1} y_{2,i_3,r_2} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_2,i_3,r_2} \overline{I} T_{3,a_1,a_2,j_1} S_{1,j_1}^\dagger}{M_F}$ $- \frac{h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{L_{$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$F H \ell$	$S_1 S_2 F H$	$S_1 S_2 F \ell$
cont.										$-\frac{h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1, i_1, r_1} y_{1, i_3, r_2} y_{2, i_2, r_1} L_{a_2, i_2}}^T C P_L L_{a_1, i_1} S_2 y_{1, i_3, r_2} \mathcal{I} T_{3, a_1, a_2, j_1} S_{1, j_1}^\dagger}{\sqrt{3} M_F}$ $-\frac{\sqrt{3} h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1, i_1, r_1} y_{1, i_2, r_2} y_{2, i_3, r_1} L_{a_2, i_2}}^T C P_L L_{a_1, i_1} S_2 y_{1, i_3, r_2} \mathcal{I} T_{3, a_1, a_2, j_1} S_{1, j_1}^\dagger}{4 M_F}$ $-\frac{h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1, i_3, r_1} y_{2, i_1, r_2} y_{2, i_2, r_1} L_{a_2, i_2}}^T C P_L L_{a_1, i_1} S_2 y_{2, i_3, r_2} \mathcal{I} T_{3, a_1, a_2, j_1} S_{1, j_1}^\dagger}{4 M_F}$ $-\frac{h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1, i_1, r_1} y_{2, i_2, r_2} y_{2, i_3, r_1} L_{a_2, i_2}}^T C P_L L_{a_1, i_1} S_2 y_{2, i_3, r_2} \mathcal{I} T_{3, a_1, a_2, j_1} S_{1, j_1}^\dagger}{4 M_F}$
										$-\frac{h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1, i_1, r_1} y_{2, i_2, r_2} y_{2, i_3, r_1} L_{a_2, i_2}}^T C P_L L_{a_1, i_1} S_2 y_{2, i_3, r_2} \mathcal{I} T_{3, a_1, a_2, j_1} S_{1, j_1}^\dagger}{M_F}$ $-\frac{h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{L_{a_1, i_1}} C P_R \overline{L_{a_2, i_2}}^\dagger S_{1, j_1} y_{1, i_1, r_1} y_{2, i_2, r_1} \mathcal{I} T_{3, a_1, a_2, j_1} S_2^\dagger \lambda_{12}^2}{M_F}$ $-\frac{h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1, i_1, r_1} y_{2, i_2, r_2} y_{2, i_3, r_1} L_{a_2, i_2}}^T C P_L L_{a_1, i_1} S_2 \mathcal{I} T_{3, a_1, a_2, j_1} S_{1, j_1}^\dagger \lambda_{12}^{(1)}}{M_F}$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$+\frac{2 h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1, i_1, r_1} y_{2, i_2, r_1} L_{a_4, i_2}}^T C P_L L_{a_2, i_1} H_{a_1} H_{a_3} \lambda T_{3, a_1, a_2} \epsilon_{a_3, a_4}}{\sqrt{3} M_F}$ $+\frac{2 h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{\lambda T_{3, a_3, i_1}} C P_R \overline{L_{a_4, i_2}}^T y_{1, i_1, r_1} y_{2, i_2, r_1} \epsilon_{a_1, a_3} \epsilon_{a_2, a_4} H_{a_1}^\dagger H_{a_2}^\dagger}{\sqrt{3} M_F}$	$-\frac{h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1, i_1, r_1}} H_{a_1} H_{a_2} S_{1, j_1} y_{1, i_1, r_1} \mathcal{I} T_{3, a_1, a_2, j_1} \lambda T_{3, a_1, a_2, j_1}^\dagger}{\sqrt{3}}$ $-h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2, i_1, r_1}} H_{a_1} H_{a_2} S_{1, j_1} y_{2, i_1, r_1} \mathcal{I} T_{3, a_1, a_2, j_1} \lambda T_{3, a_1, a_2, j_1}^\dagger$ $-\frac{h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1, i_1, r_1}} \lambda T_{3, a_1, a_2, j_1} S_2 y_{1, i_1, r_1} \mathcal{I} T_{3, a_1, a_2, j_1}^\dagger H_{a_1}^\dagger H_{a_2}^\dagger S_{1, j_1}^\dagger}{\sqrt{3}}$	$-\frac{\sqrt{3} h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1, i_3, r_1}} \overline{L_{a_1, i_1}} C P_R \overline{L_{a_2, i_2}}^\dagger S_{1, j_1} y_{1, i_2, r_1} y_{1, i_3, r_2} y_{2, i_1, r_1} \mathcal{I} T_{3, a_1, a_2, j_1}^\dagger S_2^\dagger}{4 M_F}$ $-\frac{h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2, i_3, r_1}} \overline{L_{a_1, i_1}} C P_R \overline{L_{a_2, i_2}}^\dagger S_{1, j_1} y_{1, i_3, r_2} y_{2, i_1, r_1} \mathcal{I} T_{3, a_1, a_2, j_1}^\dagger S_2^\dagger}{4 M_F}$ $-\frac{h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1, i_3, r_1}} \overline{L_{a_1, i_1}} C P_R \overline{L_{a_2, i_2}}^\dagger S_{1, j_1} y_{1, i_1, r_2} y_{1, i_3, r_2} y_{2, i_1, r_1} \mathcal{I} T_{3, a_1, a_2, j_1}^\dagger S_2^\dagger}{\sqrt{3} M_F}$ $-\frac{h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2, i_3, r_1}} \overline{L_{a_1, i_1}} C P_R \overline{L_{a_2, i_2}}^\dagger S_{1, j_1} y_{1, i_1, r_2} y_{2, i_1, r_1} \mathcal{I} T_{3, a_1, a_2, j_1}^\dagger S_2^\dagger}{M_F}$ $-\frac{\sqrt{3} h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1, i_3, r_1}} \overline{L_{a_1, i_1}} C P_R \overline{L_{a_2, i_2}}^\dagger S_{1, j_1} y_{1, i_1, r_2} y_{1, i_3, r_2} y_{2, i_1, r_1} \mathcal{I} T_{3, a_1, a_2, j_1}^\dagger S_2^\dagger}{4 M_F}$ $-\frac{h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2, i_3, r_1}} \overline{L_{a_1, i_1}} C P_R \overline{L_{a_2, i_2}}^\dagger S_{1, j_1} y_{1, i_1, r_2} y_{2, i_1, r_1} \mathcal{I} T_{3, a_1, a_2, j_1}^\dagger S_2^\dagger}{4 M_F}$ $-\frac{\sqrt{3} h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1, i_1, r_1} y_{1, i_3, r_2} y_{2, i_2, r_1} L_{a_2, i_2}}^T C P_L L_{a_1, i_1} S_2 y_{1, i_3, r_2} \mathcal{I} T_{3, a_1, a_2, j_1} S_{1, j_1}^\dagger}{4 M_F}$ $-\frac{h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1, i_1, r_1} y_{1, i_3, r_2} y_{2, i_2, r_1} L_{a_2, i_2}}^T C P_L L_{a_1, i_1} S_2 y_{1, i_3, r_2} \mathcal{I} T_{3, a_1, a_2, j_1} S_{1, j_1}^\dagger}{\sqrt{3} M_F}$ $-\frac{h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1, i_1, r_1} y_{2, i_1, r_2} y_{2, i_2, r_1} L_{a_2, i_2}}^T C P_L L_{a_1, i_1} S_2 y_{2, i_3, r_2} \mathcal{I} T_{3, a_1, a_2, j_1} S_{1, j_1}^\dagger}{4 M_F}$ $-\frac{h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1, i_1, r_1} y_{2, i_2, r_2} y_{2, i_3, r_1} L_{a_2, i_2}}^T C P_L L_{a_1, i_1} S_2 y_{2, i_3, r_2} \mathcal{I} T_{3, a_1, a_2, j_1} S_{1, j_1}^\dagger}{4 M_F}$ $-\frac{h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1, i_1, r_1} y_{2, i_2, r_2} y_{2, i_3, r_1} L_{a_2, i_2}}^T C P_L L_{a_1, i_1} S_2 y_{2, i_3, r_2} \mathcal{I} T_{3, a_1, a_2, j_1} S_{1, j_1}^\dagger}{M_F}$ $-\frac{h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{L_{a_1, i_1}} C P_R \overline{L_{a_2, i_2}}^T S_{1, j_1} y_{1, i_1, r_1} y_{2, i_2, r_1} \mathcal{I} T_{3, a_1, a_2, j_1} S_2^\dagger \lambda_{12}^{(1)}}{M_F}$ $-\frac{h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1, i_1,$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$F H \ell$	$S_1 S_2 F H$	$S_1 S_2 F \ell$
cont.									$-\frac{h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_2} y_{1,i_4,i_2}^T} C P_L L_{a_1,i_1} S_2 y_{2,i_3,r_2} \mathcal{I} T_{3,a_1,a_2,j_1} S_{1,j_1}^\dagger}{4 M_F}$ $-\frac{h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1} y_{2,i_3,r_2} L_{a_2,i_2}^T} C P_L L_{a_1,i_1} S_2 y_{2,i_3,r_2} \mathcal{I} T_{3,a_1,a_2,j_1} S_{1,j_1}^\dagger}{M_F}$ $-\frac{h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{L_{a_1,i_1} C P_R L_{a_2,i_2}^T} S_{1,j_1} y_{1,i_1,r_1} y_{2,i_2,r_1} \mathcal{I} T_{3,a_1,a_2,j_1} S_2^{\dagger} \lambda_{12}^{(1)}}{M_F}$ $-\frac{h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1} L_{a_2,i_2}^T} C P_L L_{a_1,i_1} S_2 \mathcal{I} T_{3,a_1,a_2,j_1} S_{1,j_1}^\dagger \lambda_{12}^{(1)}}{M_F}$	
cont.										
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	$+\frac{2 h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1} L_{a_4,i_2}^T} C P_L L_{a_2,i_1} H_{a_1} H_{a_3} \lambda_{T3} \epsilon_{a_1,a_2} \epsilon_{a_3,a_4}}{\sqrt{3} M_F}$ $+\frac{2 h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{\lambda_{T3} L_{a_3,i_1} C P_R L_{a_4,i_2}^T} y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_3} \epsilon_{a_2,a_4} H_{a_1}^\dagger H_{a_2}^\dagger}{\sqrt{3} M_F}$	$-\frac{h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} H_{a_1} H_{a_2} S_{1,j_1} y_{1,i_1,r_1} \mathcal{I} T_{3,a_1,a_2,j_1} \lambda_{T3} S_2^\dagger}}{\sqrt{3}}$ $-\frac{h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1} H_{a_1} H_{a_2} S_{1,j_1} y_{2,i_1,r_1} \mathcal{I} T_{3,a_1,a_2,j_1} \lambda_{T3} S_2^\dagger}}{\sqrt{3}}$ $-\frac{h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} \lambda_{T3} S_2 y_{1,i_1,r_1} \mathcal{I} T_{3,a_1,a_2,j_1} H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger}}{\sqrt{3}}$	$-\sqrt{3} h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_3,r_1} L_{a_1,i_1} C P_R L_{a_2,i_2}^T} S_{1,j_1} y_{1,i_1,r_1} y_{1,i_2,r_2} y_{2,i_1,r_1} \mathcal{I} T_{3,a_1,a_2,j_1} S_2^\dagger$ $-\frac{6 h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{L_{a_1,i_1} C P_R L_{a_2,i_2}^T} g_2^* S_{1,j_1} y_{1,i_1,r_1} y_{1,i_2,r_2} y_{2,i_1,r_1} \mathcal{I} T_{3,a_1,a_2,j_1} S_2^{\dagger}}{4 M_F}$ $-\frac{h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2,i_3,r_1} L_{a_1,i_1} C P_R L_{a_2,i_2}^T} S_{1,j_1} y_{1,i_1,r_1} y_{2,i_2,r_1} y_{2,i_3,r_1} \mathcal{I} T_{3,a_1,a_2,j_1} S_2^\dagger}{4 M_F}$ $-\frac{h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_3,r_1} L_{a_1,i_1} C P_R L_{a_2,i_2}^T} S_{1,j_1} y_{1,i_1,r_1} y_{2,i_2,r_1} y_{2,i_3,r_1} \mathcal{I} T_{3,a_1,a_2,j_1} S_2^\dagger}{\sqrt{3} M_F}$ $-\frac{h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2,i_3,r_1} L_{a_1,i_1} C P_R L_{a_2,i_2}^T} S_{1,j_1} y_{1,i_1,r_1} y_{2,i_2,r_1} y_{2,i_3,r_1} \mathcal{I} T_{3,a_1,a_2,j_1} S_2^\dagger}{M_F}$ $-\sqrt{3} h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_3,r_1} L_{a_1,i_1} C P_R L_{a_2,i_2}^T} S_{1,j_1} y_{1,i_1,r_1} y_{2,i_2,r_1} y_{2,i_3,r_1} \mathcal{I} T_{3,a_1,a_2,j_1} S_2^\dagger$ $-\frac{6 h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2,i_3,r_1} L_{a_1,i_1} C P_R L_{a_2,i_2}^T} S_{1,j_1} y_{1,i_1,r_1} y_{2,i_2,r_1} y_{2,i_3,r_1} \mathcal{I} T_{3,a_1,a_2,j_1} S_2^\dagger}{4 M_F}$ $-\frac{h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2,i_3,r_1} L_{a_1,i_1} C P_R L_{a_2,i_2}^T} S_{1,j_1} y_{1,i_1,r_1} y_{2,i_2,r_1} y_{2,i_3,r_1} \mathcal{I} T_{3,a_1,a_2,j_1} S_2^\dagger}{M_F}$ $-\sqrt{3} h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_3,r_1} L_{a_1,i_1} C P_R L_{a_2,i_2}^T} S_{1,j_1} y_{1,i_1,r_1} y_{2,i_2,r_1} y_{2,i_3,r_1} \mathcal{I} T_{3,a_1,a_2,j_1} S_2^\dagger$ $-\frac{6 h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1} L_{a_2,i_2}^T} C P_L L_{a_1,i_1} g_2^* S_2 \mathcal{I} T_{3,a_1,a_2,j_1} S_{1,j_1}^\dagger}{M_F}$ $-\sqrt{3} h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{1,i_3,r_2} y_{2,i_2,r_2} L_{a_2,i_2}^T} C P_L L_{a_1,i_1} S_2 y_{1,i_3,r_1} \mathcal{I} T_{3,a_1,a_2,j_1} S_{1,j_1}^\dagger$ $-\frac{h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{1,i_3,r_2} y_{2,i_2,r_2} L_{a_2,i_2}^T} C P_L L_{a_1,i_1} S_2 y_{1,i_3,r_1} \mathcal{I} T_{3,a_1,a_2,j_1} S_{1,j_1}^\dagger}{4 M_F}$ $-\sqrt{3} h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{1,i_2,r_2} y_{2,i_2,r_2} L_{a_2,i_2}^T} C P_L L_{a_1,i_1} S_2 y_{1,i_3,r_1} \mathcal{I} T_{3,a_1,a_2,j_1} S_{1,j_1}^\dagger$ $-\frac{h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{1,i_2,r_2} y_{2,i_2,r_2} L_{a_2,i_2}^T} C P_L L_{a_1,i_1} S_2 y_{1,i_3,r_1} \mathcal{I} T_{3,a_1,a_2,j_1} S_{1,j_1}^\dagger}{4 M_F}$ $-\frac{h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1} y_{2,i_2,r_1} L_{a_2,i_2}^T} C P_L L_{a_1,i_1} S_2 y_{1,i_3,r_1} \mathcal{I} T_{3,a_1,a_2,j_1} S_{1,j_1}^\dagger}{4 M_F}$ $-\frac{h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1} y_{2,i_2,r_1} L_{a_2,i_2}^T} C P_L L_{a_1,i_1} S_2 y_{1,i_3,r_1} \mathcal{I} T_{3,a_1,a_2,j_1} S_{1,j_1}^\dagger}{M_F}$ $-\frac{h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{L_{a_1,i_1} C P_R L_{a_2,i_2}^T} S_{1,j_1} y_{1,i_1,r_1} y_{2,i_2,r_1} \mathcal{I} T_{3,a_1,a_2,j_1} S_2^{\dagger} \lambda_{12}^{(1)}}{M_F}$ $-\frac{h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1} y_{2,i_2,r_1} L_{a_2,i_2}^T} C P_L L_{a_1,i_1} S_2 \mathcal{I} T_{3,a_1,a_2,j_1} S_{1,j_1}^\dagger \lambda_{12}^{(1)}}{M_F}$
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T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	$-\frac{2 \sqrt{\frac{2}{3}} h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_2} y_{1,i_4,i_2}^T} C P_L L_{a_2,i_1} H_{a_1} H_{a_3} \lambda_{T3} \epsilon_{a_1,a_2} \epsilon_{a_3,a_4}}{M_F}$ $-\frac{2 \sqrt{\frac{2}{3}} h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{\lambda_{T3} L_{a_3,i_1} C P_R L_{a_4,i_2}^T} y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_3} \epsilon_{a_2,a_4} H_{a_1}^\dagger H_{a_2}^\dagger}{M_F}$	$-\frac{h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} \lambda_{T3} S_{2,j_2} y_{1,i_1,r_1} \mathcal{I} T_{3,a_1,a_2,j_1,j_2} H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger}}{\sqrt{3}}$ $-\frac{h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1} \lambda_{T3} S_{2,j_2} y_{2,i_1,r_1} \mathcal{I} T_{3,a_1,a_2,j_1,j_2} H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger}}{\sqrt{3}}$ $-\frac{h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} H_{a_1} H_{a_2} S_{1,j_2} y_{1,i_1,r_1} \mathcal{I} T_{3,a_1,a_2,j_2,j_1} \lambda_{T3} S_{2,j_1}^\dagger}}{\sqrt{3}}$	$-\frac{\sqrt{6} h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1} L_{a_2,i_2}^T} C P_L L_{a_1,i_1} g_2^* S_{2,j_2} \mathcal{I} T_{3,a_1,a_2,j_1,j_2} S_{1,j_1}^\dagger}{M_F}$ $-\frac{h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{1,i_3,r_2} y_{2,i_2,r_2} L_{a_2,i_2}^T} C P_L L_{a_1,i_1} S_{2,j_2} y_{1,i_3,r_1} \mathcal{I} T_{3,a_1,a_2,j_1,j_2} S_{1,j_1}^\dagger}{2 \sqrt{2} M_F}$ $-\frac{\sqrt{2} h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{1,i_3,r_2} y_{2,i_2,r_2} L_{a_2,i_2}^T} C P_L L_{a_1,i_1} S_{2,j_2} y_{1,i_3,r_1} \mathcal{I} T_{3,a_1,a_2,j_1,j_2} S_{1,j_1}^\dagger}{3 M_F}$ $-\frac{h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{1,i_2,r_2} y_{2,i_2,r_2} L_{a_2,i_2}^T} C P_L L_{a_1,i_1} S_{2,j_2} y_{1,i_3,r_1} \mathcal{I} T_{3,a_1,a_2,j_1,j_2} S_{1,j_1}^\dagger}{2 \sqrt{2} M_F}$ $-\frac{h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1} y_{2,i_2,r_1} L_{a_2,i_2}^T} C P_L L_{a_1,i_1} S_{2,j_2} y_{1,i_3,r_1} \mathcal{I} T_{3,a_1,a_2,j_1,j_2} S_{1,j_1}^\dagger}{2 \sqrt{2} M_F}$ $-\frac{2 \sqrt{3} h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1} L_{a_2,i_2}^T} C P_L L_{a_1,i_1} g_2^* S_{2,j_1} \epsilon_{a_1,a_2} S_{1,j_1}^\dagger}{M_F}$ $-\frac{h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{1,i_3,r_2} y_{2,i_2,r_2} L_{a_2,i_2}^T} C P_L L_{a_1,i_1} S_{2,j_1} y_{1,i_3,r_1} \epsilon_{a_1,a_2} S_{1,j_1}^\dagger}{4 M_F}$ $-\frac{h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{1,i_3,r_2} y_{2,i_2,r_2} L_{a_2,i_2}^T} C P_L L_{a_1,i_1} S_{2,j_1} y_{1,i_3,r_1} \epsilon_{a_1,a_2} S_{1,j_1}^\dagger}{3 M_F}$ $-\frac{h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_3,r_1} y_{2,i_1,r_1} y_{2,i_2,r_1} L_{a_2,i_2}^T} C P_L L_{a_1,i_1} S_{2,j_2} y_{1,i_3,r_1} \mathcal{I} T_{3,a_1,a_2,j_1,j_2} S_{1,j_1}^\dagger}{4 M_F}$ $-\frac{h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_3,r_1} y_{1,i_2,r_2} y_{2,i_2,r_2} L_{a_2,i_2}^T} C P_L L_{a_1,i_1} S_{2,j_2} y_{1,i_3,r_1} \mathcal{I} T_{3,a_1,a_2,j_1,j_2} S_{1,j_1}^\dagger}{4 M_F}$ $-\frac{h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_3,r_1} y_{2,i_1,r_1} y_{2,i_2,r_1} L_{a_2,i_2}^T} C P_L L_{a_1,i_1} S_{2,j_2} y_{1,i_3,r_1} \mathcal{I} T_{3,a_1,a_2,j_1,j_2} S_{1,j_1}^\dagger}{4 M_F}$ $-\frac{h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_3,r_1} y_{2,i_1,r_1} y_{2,i_2,r_1} L_{a_2,i_2}^T} C P_L L_{a_1,i_1} S_{2,j_2} y_{1,i_3,r_1} \mathcal{I} T_{3,a_1,a_2,j_1,j_2} S_{1,j_1}^\dagger}{M_F}$ $-\frac{h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_3,r_1} y_{2,i_1,r_1} y_{2,i_2,r_1} L_{a_2,i_2}^T} C P_L L_{a_1,i_1} S_{2,j_2} y_{1,i_3,r_1} \mathcal{I} T_{3,a_1,a_2,j_1,j_2} S_{1,j_1}^\dagger}{3 M_F}$ $-\frac{h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_3,r_1} y_{2,i_1,r_1} y_{2,i_2,r_1} L_{a_2,i_2}^T} C P_L L_{a_1,i_1} S_{2,j_2} y_{1,i_3,r_1} \mathcal{I} T_{3,a_1,a_2,j_1,j_2} S_{1,j_1}^\dagger}{2 \sqrt{2} M_F}$
cont.										
cont.										
cont.										
cont.										

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$FH\ell$	S_1S_2FH	$S_1S_2F\ell$	
cont.									$-\frac{\sqrt{6}h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{L_{a_1, i_1}}CPr\overline{L_{a_2, i_2}}g_2^2S_{1, j_2}Y_{i_1, r_1}Y_{2, i_2, r_1}ITr_{a_1, a_2, j_2, j_1}S_{2, j_1}^\dagger}{M_F}$	$-\frac{h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{Y_{2, i_3, r_1}L_{a_1, i_1}}CPr\overline{L_{a_2, i_2}}S_{1, j_2}Y_{i_1, r_1}Y_{2, i_2, r_1}ITr_{a_1, a_2, j_2, j_1}S_{2, j_1}^\dagger}{2\sqrt{2}M_F}$	
									$-\frac{\sqrt{2}h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{Y_{1, i_3, r_1}L_{a_1, i_1}}CPr\overline{L_{a_2, i_2}}S_{1, j_2}Y_{i_1, r_1}Y_{2, i_3, r_1}Y_{2, i_2, r_2}ITr_{a_1, a_2, j_2, j_1}S_{2, j_1}^\dagger}{3M_F}$	$-\frac{\sqrt{2}h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{Y_{2, i_3, r_1}L_{a_1, i_1}}CPr\overline{L_{a_2, i_2}}S_{1, j_2}Y_{i_1, r_1}Y_{2, i_2, r_2}Y_{2, i_3, r_2}ITr_{a_1, a_2, j_2, j_1}S_{2, j_1}^\dagger}{3M_F}$	
									$-\frac{h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{Y_{1, i_3, r_1}L_{a_1, i_1}}CPr\overline{L_{a_2, i_2}}S_{1, j_2}Y_{i_1, r_1}Y_{2, i_3, r_1}Y_{2, i_3, r_2}ITr_{a_1, a_2, j_2, j_1}S_{2, j_1}^\dagger}{2\sqrt{2}M_F}$	$-\frac{h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{Y_{2, i_3, r_1}L_{a_1, i_1}}CPr\overline{L_{a_2, i_2}}S_{1, j_2}Y_{i_1, r_1}Y_{2, i_2, r_2}Y_{2, i_3, r_2}ITr_{a_1, a_2, j_2, j_1}S_{2, j_1}^\dagger}{2\sqrt{2}M_F}$	
cont.									$+\frac{h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{Y_{1, i_3, r_1}L_{a_1, i_1}}CPr\overline{L_{a_2, i_2}}S_{1, j_1}Y_{i_2, r_2}Y_{i_3, r_2}Y_{2, i_1, r_2}\epsilon_{a_1, a_2}S_{2, j_1}^\dagger}{4M_F}$	$+\frac{2\sqrt{3}h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{L_{a_1, i_1}}CPr\overline{L_{a_2, i_2}}g_2^2S_{1, j_1}Y_{i_1, r_1}Y_{2, i_2, r_1}\epsilon_{a_1, a_2}S_{2, j_1}^\dagger}{M_F}$	
									$+\frac{h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{Y_{2, i_3, r_1}L_{a_1, i_1}}CPr\overline{L_{a_2, i_2}}S_{1, j_1}Y_{i_1, r_1}Y_{2, i_2, r_1}Y_{2, i_2, r_2}Y_{2, i_1, r_2}\epsilon_{a_1, a_2}S_{2, j_1}^\dagger}{4M_F}$	$+\frac{h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{Y_{1, i_3, r_1}L_{a_1, i_1}}CPr\overline{L_{a_2, i_2}}S_{1, j_1}Y_{i_1, r_1}Y_{2, i_2, r_2}Y_{2, i_3, r_1}\epsilon_{a_1, a_2}S_{2, j_1}^\dagger}{3M_F}$	
									$-\frac{h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{Y_{2, i_3, r_1}L_{a_1, i_1}}CPr\overline{L_{a_2, i_2}}S_{1, j_1}Y_{i_1, r_1}Y_{2, i_2, r_2}Y_{2, i_3, r_1}\epsilon_{a_1, a_2}S_{2, j_1}^\dagger}{3M_F}$	$-\frac{h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{Y_{1, i_3, r_1}L_{a_1, i_1}}CPr\overline{L_{a_2, i_2}}S_{1, j_1}Y_{i_1, r_1}Y_{2, i_2, r_2}Y_{2, i_3, r_2}\epsilon_{a_1, a_2}S_{2, j_1}^\dagger}{4M_F}$	
cont.									$-\frac{h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{Y_{2, i_3, r_1}L_{a_1, i_1}}CPr\overline{L_{a_2, i_2}}S_{1, j_1}Y_{i_1, r_1}Y_{2, i_2, r_2}Y_{2, i_3, r_2}\epsilon_{a_1, a_2}S_{2, j_1}^\dagger}{4M_F}$	$-\frac{h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{Y_{2, i_3, r_1}L_{a_1, i_1}}CPr\overline{L_{a_2, i_2}}S_{1, j_1}Y_{i_1, r_1}Y_{2, i_2, r_2}Y_{2, i_3, r_2}\epsilon_{a_1, a_2}S_{2, j_1}^\dagger}{4M_F}$	
									$+\frac{\sqrt{\frac{3}{2}}h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{Y_{1, i_1, r_1}Y_{2, i_2, r_2}L_{a_2, i_2}}^TCPl_{a_1, i_1}S_{2, j_2}ITr_{3, a_1, a_2, j_1, j_2}S_{1, j_1}^\dagger\lambda_{12}^{(1)}}{M_F}$	$+\frac{h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{Y_{1, i_1, r_1}Y_{2, i_2, r_2}L_{a_2, i_2}}^TCPl_{a_1, i_1}S_{2, j_1}\epsilon_{a_1, a_2}S_{1, j_1}^\dagger\lambda_{12}^{(1)}}{\sqrt{3}M_F}$	
									$+\frac{\sqrt{\frac{3}{2}}h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{L_{a_1, i_1}}CPr\overline{L_{a_2, i_2}}S_{1, j_2}Y_{i_1, r_1}Y_{2, i_2, r_2}ITr_{a_1, a_2, j_2, j_1}S_{2, j_1}^\dagger\lambda_{12}^{(1)}}{M_F}$	$+\frac{h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{L_{a_1, i_1}}CPr\overline{L_{a_2, i_2}}S_{1, j_1}Y_{i_1, r_1}Y_{2, i_2, r_2}Y_{2, i_3, r_2}\epsilon_{a_1, a_2}S_{2, j_1}^\dagger\lambda_{12}^{(1)}}{\sqrt{3}M_F}$	
cont.									$-\frac{3\sqrt{\frac{2}{11}}h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{Y_{1, i_1, r_1}Y_{2, i_2, r_2}L_{a_2, i_2}}^TCPl_{a_1, i_1}S_{2, j_2}ITr_{3, a_1, a_2, j_1, j_2}S_{1, j_1}^\dagger\lambda_{12}^{(2)}}{M_F}$	$-\frac{h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{Y_{1, i_1, r_1}Y_{2, i_2, r_2}L_{a_2, i_2}}^TCPl_{a_1, i_1}S_{2, j_1}\epsilon_{a_1, a_2}S_{1, j_1}^\dagger\lambda_{12}^{(2)}}{\sqrt{11}M_F}$	
									$-\frac{3\sqrt{\frac{2}{11}}h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{L_{a_1, i_1}}CPr\overline{L_{a_2, i_2}}S_{1, j_2}Y_{i_1, r_1}Y_{2, i_2, r_2}ITr_{3, a_1, a_2, j_2, j_1}S_{2, j_1}^\dagger\lambda_{12}^{(2)}}{M_F}$	$-\frac{h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{L_{a_1, i_1}}CPr\overline{L_{a_2, i_2}}S_{1, j_1}Y_{i_1, r_1}Y_{2, i_2, r_2}Y_{2, i_3, r_1}\epsilon_{a_1, a_2}S_{2, j_1}^\dagger\lambda_{12}^{(2)}}{\sqrt{11}M_F}$	
									$-\frac{7h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{Y_{1, i_1, r_1}Y_{2, i_2, r_2}L_{a_2, i_2}}^TCPl_{a_1, i_1}S_{2, j_1}\epsilon_{a_1, a_2}S_{1, j_1}^\dagger\lambda_{12}^{(3)}}{\sqrt{23}M_F}$	$-\frac{7h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{L_{a_1, i_1}}CPr\overline{L_{a_2, i_2}}S_{1, j_2}Y_{i_1, r_1}Y_{2, i_2, r_2}Y_{2, i_3, r_1}\epsilon_{a_1, a_2}S_{2, j_1}^\dagger\lambda_{12}^{(3)}}{\sqrt{23}M_F}$	
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$-\frac{2\sqrt{\frac{3}{2}}h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{Y_{1, i_1, r_1}Y_{2, i_2, r_2}L_{a_4, i_2}}^TCPl_{a_2, i_1}H_{a_1}H_{a_3}\lambda_{T3}\epsilon_{a_1, a_2}\epsilon_{a_3, a_4}}{M_F}$	$-\frac{h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{Y_{1, i_1, r_1}\lambda_{T3}S_{2, j_2}Y_{1, i_1, r_1}ITr_{3, a_1, a_2, j_1, j_2}H_{a_1}^\dagger H_{a_2}^\dagger S_{1, j_1}^\dagger}}{\sqrt{3}}$	$-\frac{\sqrt{6}h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{Y_{1, i_1, r_1}Y_{2, i_2, r_1}L_{a_2, i_2}}^TCPl_{a_1, i_1}S_{2, j_2}ITr_{3, a_1, a_2, j_1, j_2}S_{1, j_1}^\dagger g_2^2}{M_F}$	
								$-\frac{2\sqrt{\frac{3}{2}}h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{\lambda_{T3}L_{a_4, i_1}}CPr\overline{L_{a_4, i_2}}^TY_{1, i_1, r_1}Y_{2, i_2, r_1}\epsilon_{a_1, a_3}\epsilon_{a_2, a_4}H_{a_1}^\dagger H_{a_2}^\dagger}{M_F}$	$-\frac{h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{Y_{2, i_1, r_1}\lambda_{T3}S_{2, j_2}Y_{1, i_1, r_1}ITr_{3, a_1, a_2, j_1, j_2}H_{a_1}^\dagger H_{a_2}^\dagger S_{1, j_1}^\dagger}}{\sqrt{3}}$	$-\frac{2\sqrt{3}h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{Y_{1, i_1, r_1}Y_{2, i_2, r_1}L_{a_2, i_2}}^TCPl_{a_1, i_1}S_{2, j_1}\epsilon_{a_1, a_2}S_{1, j_1}^\dagger g_2^2}{M_F}$	
									$-\frac{h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{Y_{1, i_1, r_1}}H_{a_1}H_{a_2}S_{1, j_2}Y_{1, i_1, r_1}ITr_{3, a_1, a_2, j_2, j_1}\lambda_{T3}S_{2, j_1}^\dagger}{\sqrt{3}}$	$-\frac{\sqrt{6}h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{L_{a_1, i_1}}CPr\overline{L_{a_2, i_2}}^TS_{1, j_2}Y_{1, i_1, r_1}Y_{2, i_2, r_1}ITr_{3, a_1, a_2, j_2, j_1}S_{2, j_1}^\dagger 1g_2^2}{M_F}$	
cont.								$-\frac{h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{Y_{2, i_1, r_1}}H_{a_1}H_{a_2}S_{1, j_2}Y_{2, i_1, r_1}ITr_{3, a_1, a_2, j_2, j_1}\lambda_{T3}S_{2, j_1}^\dagger}{\sqrt{3}}$	$+\frac{\sqrt{\frac{3}{2}}h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{Y_{1, i_1, r_1}Y_{2, i_2, r_1}L_{a_2, i_2}}^TCPl_{a_1, i_1}g_2^2S_{2, j_2}ITr_{3, a_1, a_2, j_1, j_2}S_{1, j_1}^\dagger}{2M_F}$	$-\frac{h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{Y_{1, i_1, r_1}Y_{1, i_3}Y_{2, i_2}Y_{2, i_3}L_{a_2, i_2}}^TCPl_{a_1, i_1}S_{2, j_2}Y_{1, i_3, r_1}ITr_{3, a_1, a_2, j_1, j_2}S_{1, j_1}^\dagger}{2\sqrt{2}M_F}$	
									$-\frac{h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{Y_{1, i_1, r_1}Y_{1, i_3}Y_{2, i_2}Y_{2, i_3}L_{a_2, i_2}}^TCPl_{a_1, i_1}S_{2, j_2}Y_{1, i_3, r_1}ITr_{3, a_1, a_2, j_1, j_2}S_{1, j_1}^\dagger}{3M_F}$	$-\frac{h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{Y_{1, i_1, r_1}Y_{1, i_2}Y_{2, i_2}Y_{2, i_3}L_{a_2, i_2}}^TCPl_{a_1, i_1}S_{2, j_2}Y_{1, i_3, r_2}ITr_{3, a_1, a_2, j_1, j_2}S_{1, j_1}^\dagger}{2\sqrt{2}M_F}$	
									$-\frac{h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{Y_{1, i_3, r_1}Y_{2, i_1, r_2}Y_{2, i_2}Y_{2, i_3}L_{a_2, i_2}}^TCPl_{a_1, i_1}S_{2, j_2}Y_{2, i_3, r_2}ITr_{3, a_1, a_2, j_1, j_2}S_{1, j_1}^\dagger}{2\sqrt{2}M_F}$	$-\frac{h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{Y_{1, i_2, r_1}Y_{2, i_2}Y_{2, i_3}Y_{2, i_3}L_{a_2, i_2}}^TCPl_{a_1, i_1}S_{2, j_2}Y_{2, i_3, r_2}ITr_{3, a_1, a_2, j_1, j_2}S_{1, j_1}^\dagger}{2\sqrt{2}M_F}$	
cont.								$-\frac{\sqrt{2}h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{Y_{1, i_1, r_1}Y_{2, i_2, r_1}Y_{2, i_3, r_2}L_{a_2, i_2}}^TCPl_{a_1, i_1}L_{a_1, i_1}S_{2, j_2}Y_{2, i_3, r_2}ITr_{3, a_1, a_2, j_1, j_2}S_{1, j_1}^\dagger}{2\sqrt{2}M_F}$	$-\frac{\sqrt{2}h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{Y_{1, i_1, r_1}Y_{2, i_2, r_1}Y_{2, i_3, r_2}L_{a_2, i_2}}^TCPl_{a_1, i_1}L_{a_1, i_1}S_{2, j_2}Y_{2, i_3, r_2}ITr_{3, a_1, a_2, j_1, j_2}S_{1, j_1}^\dagger}{2\sqrt{2}M_F}$	$-\frac{\sqrt{2}h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{Y_{1, i_1, r_1}Y_{2, i_2, r_1}Y_{2, i_3, r_2}L_{a_2, i_2}}^TCPl_{a_1, i_1}L_{a_1, i_1}S_{2, j_2}Y_{2, i_3, r_2}ITr_{3, a_1, a_2, j_1, j_2}S_{1, j_1}^\dagger}{2\sqrt{2}M_F}$	
									$-\frac{\sqrt{2}h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{Y_{1, i_1, r_1}Y_{2, i_2, r_1}Y_{2, i_3, r_2}L_{a_2, i_2}}^TCPl_{a_1, i_1}L_{a_1, i_1}S_{2, j_2}Y_{2, i_3, r_2}ITr_{3, a_1, a_2, j_1, j_2}S_{1, j_1}^\dagger}{2\sqrt{2}M_F}$	$-\frac{\sqrt{2}h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{Y_{1, i_1, r_1}Y_{2, i_2, r_1}Y_{2, i_3, r_2}L_{a_2, i_2}}^TCPl_{a_1, i_1}L_{a_1, i_1}S_{2, j_2}Y_{2, i_3, r_2}ITr_{3, a_1, a_2, j_1, j_2}S_{1, j_1}^\dagger}{2\sqrt{2}M_F}$	$-\frac{\sqrt{2}h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{Y_{1, i_1, r_1}Y_{2, i_2, r_1}Y_{2, i_3, r_2}L_{a_2, i_2}}^TCPl_{a_1, i_1}L_{a_1, i_1}S_{2, j_2}Y_{2, i_3, r_2}ITr_{3, a_1, a_2, j_1, j_2}S_{1, j_1}^\dagger}{2\sqrt{2}M_F}$
									$-\frac{\sqrt{2}h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{Y_{1, i_1, r_1}Y_{2, i_2, r_1}Y_{2, i_3, r_2}L_{a_2, i_2}}^TCPl_{a_1, i_1}L_{a_1, i_1}S_{2, j_2}Y_{2, i_3, r_2}ITr_{3, a_1, a_2, j_1, j_2}S_{1, j_1}^\dagger}{2\sqrt{2}M_F}$	$-\frac{\sqrt{2}h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{Y_{1, i_1, r_1}Y_{2, i_2, r_1}Y_{2, i_3, r_2}L_{a_2, i_2}}^TCPl_{a_1, i_1}L_{a_1, i_1}S_{2, j_2}Y_{2, i_3, r_2}ITr_{3, a_1, a_2, j_1, j_2}S_{1, j_1}^\dagger}{2\sqrt{2}M_F}$	$-\frac{\sqrt{2}h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{Y_{1, i_1, r_1}Y_{2, i_2, r_1}Y_{2, i_3, r_2}L_{a_2, i_2}}^TCPl_{a_1, i_1}L_{a_1, i_1}S_{2, j_2}Y_{2, i_3, r_2}ITr_{3, a_1, a_2, j_1, j_2}S_{1, j_1}^\dagger}{2\sqrt{2}M_F}$
cont.								$-\frac{\sqrt{2}h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{Y_{1, i_1, r_1}Y_{2, i_2, r_1}Y_{2, i_3, r_2}L_{a_2, i_2}}^TCPl_{a_1, i_1}L_{a_1, i_1}S_{2, j_2}Y_{2, i_3, r_2}ITr_{3, a_1, a_2, j_1, j_2}S_{1, j_1}^\dagger}{2\sqrt{2}M_F}$	$-\frac{\sqrt{2}h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{Y_{1, i_1, r_1}Y_{2, i_2, r_1}Y_{2, i_3, r_2}L_{a_2, i_2}}^TCPl_{a_1, i_1}L_{a_1, i_1}S_{2, j_2}Y_{2, i_3, r_2}ITr_{3, a_1, a_2, j_1, j_2}S_{1, j_1}^\dagger}{2\sqrt{2}M_F}$	$-\frac{\sqrt{2}h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{Y_{1, i_1, r_1}Y_{2, i_2, r_1}Y_{2, i_3, r_2}L_{a_2, i_2}}^TCPl_{a_1, i_1}L_{a_1, i_1}S_{2, j_2}Y_{2, i_3, r_2}ITr_{3, a_1, a_2, j_1, j_2}S_{1, j_1}^\dagger}{2\sqrt{2}M_F}$	
									$-\frac{\sqrt{2}h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{Y_{1, i_1, r_1}Y_{2, i_2, r_1}Y_{2, i_3, r_2}L_{a_2, i_2}}^TCPl_{a_1, i_1}L_{a_1, i_1}S_{2, j_2}Y_{2, i_3, r_2}ITr_{3, a_1, a_2, j_1, j_2}S_{1, j_1}^\dagger}{2\sqrt{2}M_F}$	$-\frac{\sqrt{2}h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{Y_{1, i_1, r_1}Y_{2, i_2, r_1}Y_{2, i_3, r_2}L_{a_2, i_2}}^TCPl_{a_1, i_1}L_{a_1, i_1}S_{2, j_2}Y_{2, i_3, r_2}ITr_{3, a_1, a_2, j_1, j_2}S_{1, j_1}^\dagger}{2\sqrt{2}M_F}$	$-\frac{\sqrt{2}h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{Y_{1, i_1, r_1}Y_{2, i_2, r_1}Y_{2, i_3, r_2}L_{a_2, i_2}}^TCPl_{a_1, i_1}L_{a_1, i_1}S_{2, j_2}Y_{2, i_3, r_2}ITr_{3, a_1, a_2, j_1, j_2}S_{1, j_1}^\dagger}{2\sqrt{2}M_F}$
									$-\frac{\sqrt{2}h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{Y_{1, i_1, r_1}Y_{2, i_2, r_1}Y_{2, i_3, r_2}L_{a_2, i_2}}^TCPl_{a_1, i_1}L_{a_1, i_1}S_{2, j_2}Y_{2, i_3, r_2}ITr_{3, a_1, a_2, j_1, j_2}S_{1, j_1}^\dagger}{2\sqrt{2}M_F}$	$-\frac{\sqrt{2}h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{Y_{1, i_1, r_1}Y_{2, i_2, r_1}Y_{2, i_3, r_2}L_{a_2, i_2}}^TCPl_{a_1, i_1}L_{a_1, i_1}S_{2, j_2}Y_{2, i_3, r_2}ITr_{3, a_1, a_2, j_1, j_2}S_{1, j_1}^\dagger}{2\sqrt{2}M_F}$	$-\frac{\sqrt{2}h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{Y_{1, i_1, r_1}Y_{2, i_2, r_1}Y_{2, i_3, r_2}L_{a_2, i_2}}^TCPl_{a_1, i_1}L_{a_1, i_1}S_{2, j_2}Y_{2, i_3, r_2}ITr_{3, a_1, a_2, j_1, j_2}S_{1, j_1}^\dagger}{2\sqrt{2}M_F}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$F H \ell$	$S_1 S_2 F H$	$S_1 S_2 F \ell$
cont.										$\frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{1,i_3,r_2} y_{2,i_2,r_2}} L_{a_2,i_2}^{-r} C P_L L_{a_1,i_1} S_{2,j_1} y_{1,i_3,r_1} \epsilon_{a_1,a_2} S_{1,j_1}^\dagger}{4M_F}$
										$\frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{1,i_3,r_2} y_{2,i_2,r_1}} L_{a_2,i_2}^{-r} C P_L L_{a_1,i_1} S_{2,j_1} y_{1,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}^\dagger}{3M_F}$
										$\frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{1,i_2,r_2} y_{2,i_3,r_1}} L_{a_2,i_2}^{-r} C P_L L_{a_1,i_1} S_{2,j_1} y_{1,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}^\dagger}{4M_F}$
cont.										$\frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_3,r_1} y_{2,i_1,r_2} y_{2,i_2,r_1}} L_{a_2,i_2}^{-r} C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}^\dagger}{4M_F}$
										$\frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_2} y_{2,i_3,r_1}} L_{a_2,i_2}^{-r} C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}^\dagger}{4M_F}$
										$\frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1}} C P_R \overline{L_{a_3,i_3}^r} S_{1,j_2} y_{1,i_3,r_2} y_{1,i_1,r_2} y_{2,i_1,r_2} I T \gamma_{a_1,a_2,j_2,j_1}^* S_{2,j_1}^\dagger}{2\sqrt{2}M_F}$
cont.										$+ \frac{\sqrt{2} h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}^{-r}} g_Y^2 S_{1,j_2} y_{1,i_1,r_1} y_{2,i_2,r_1} I T \gamma_{a_1,a_2,j_2,j_1}^* S_{2,j_1}^\dagger}{2M_F}$
										$\frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{2,i_3,r_1}} L_{a_1,i_1} C P_R \overline{L_{a_2,i_2}^{-r}} S_{1,j_2} y_{1,i_3,r_2} y_{2,i_1,r_2} y_{2,i_2,r_1} I T \gamma_{a_1,a_2,j_2,j_1}^* S_{2,j_1}^\dagger}{2\sqrt{2}M_F}$
										$\frac{\sqrt{2} h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_3,r_1}} L_{a_1,i_1} C P_R \overline{L_{a_2,i_2}^{-r}} S_{1,j_2} y_{1,i_1,r_2} y_{1,i_3,r_1} y_{2,i_2,r_2} I T \gamma_{a_1,a_2,j_2,j_1}^* S_{2,j_1}^\dagger}{3M_F}$
cont.										$\frac{\sqrt{2} h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{2,i_3,r_1}} \overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}^{-r}} S_{1,j_2} y_{1,i_1,r_2} y_{2,i_2,r_2} y_{2,i_3,r_1} I T \gamma_{a_1,a_2,j_2,j_1}^* S_{2,j_1}^\dagger}{3M_F}$
										$\frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_3,r_1}} \overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}^{-r}} S_{1,j_2} y_{1,i_1,r_2} y_{1,i_2,r_1} y_{2,i_3,r_2} I T \gamma_{a_1,a_2,j_2,j_1}^* S_{2,j_1}^\dagger}{2\sqrt{2}M_F}$
										$\frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{2,i_3,r_1}} \overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}^{-r}} S_{1,j_2} y_{1,i_1,r_2} y_{2,i_2,r_1} y_{2,i_3,r_2} I T \gamma_{a_1,a_2,j_2,j_1}^* S_{2,j_1}^\dagger}{2\sqrt{2}M_F}$
cont.										$+ \frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_3,r_1}} \overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}^{-r}} S_{1,j_1} y_{1,i_2,r_1} y_{1,i_3,r_2} y_{2,i_1,r_2} \epsilon_{a_1,a_2} S_{2,j_1}^\dagger}{4M_F}$
										$+ \frac{\sqrt{3} h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}^{-r}} g_Y^2 S_{1,j_1} y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_2} S_{2,j_1}^\dagger}{4M_F}$
										$\frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{2,i_3,r_1}} \overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}^{-r}} S_{1,j_1} y_{1,i_3,r_2} y_{2,i_1,r_2} y_{2,i_2,r_1} \epsilon_{a_1,a_2} S_{2,j_1}^\dagger}{4M_F}$
cont.										$+ \frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_3,r_1}} \overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}^{-r}} S_{1,j_1} y_{1,i_1,r_2} y_{1,i_3,r_1} y_{2,i_2,r_2} \epsilon_{a_1,a_2} S_{2,j_1}^\dagger}{4M_F}$
										$\frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{2,i_3,r_1}} \overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}^{-r}} S_{1,j_1} y_{1,i_1,r_2} y_{1,i_2,r_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{2,j_1}^\dagger}{3M_F}$
										$\frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_3,r_1}} \overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}^{-r}} S_{1,j_1} y_{1,i_1,r_2} y_{1,i_2,r_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{2,j_1}^\dagger}{4M_F}$
cont.										$+ \frac{\sqrt{2} h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1}} L_{a_2,i_2}^{-r} C P_L L_{a_1,i_1} S_{2,j_2} I T \gamma_{a_1,a_2,j_2,j_1}^* \lambda_{12}^{(1)}}{M_F}$
										$- \frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1}} L_{a_2,i_2}^{-r} C P_L L_{a_1,i_1} \epsilon_{a_1,a_2} S_{1,j_1}^\dagger \lambda_{12}^{(1)}}{\sqrt{3}M_F}$
										$+ \frac{\sqrt{2} h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}^{-r}} S_{1,j_2} y_{1,i_1,r_1} y_{2,i_2,r_1} I T \gamma_{a_1,a_2,j_2,j_1}^* \lambda_{12}^{(1)}}{M_F}$
cont.										$+ \frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}^{-r}} S_{1,j_1} y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_2} S_{2,j_1}^\dagger \lambda_{12}^{(1)}}{\sqrt{3}M_F}$
										$- \frac{3\sqrt{\frac{2}{11}} h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1}} L_{a_2,i_2}^{-r} C P_L L_{a_1,i_1} S_{2,j_2} I T \gamma_{a_1,a_2,j_2,j_1}^* \lambda_{12}^{(2)}}{M_F}$
										$- \frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}^{-r}} S_{1,j_1} y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_2} S_{2,j_1}^\dagger \lambda_{12}^{(2)}}{\sqrt{11}M_F}$
cont.										$7 h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1}} L_{a_2,i_2}^{-r} C P_L L_{a_1,i_1} S_{2,j_1} \epsilon_{a_1,a_2} S_{1,j_1}^\dagger \lambda_{12}^{(3)}$
										$- \frac{7 h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}^{-r}} S_{1,j_1} y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_2} S_{2,j_1}^\dagger \lambda_{12}^{(3)}}{\sqrt{23}M_F}$
										$- \frac{7 h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}^{-r}} S_{1,j_1} y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_2} S_{2,j_1}^\dagger \lambda_{12}^{(3)}}{\sqrt{23}M_F}$

T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{2\sqrt{\frac{2}{3}}h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}L_{a_4,i_2}^{-r}CPL_{a_2,i_1}H_{a_1}H_{a_3}\lambda T\gamma_{a_1,a_2}^*\epsilon_{a_3,a_4}}{M_F}$	$-\frac{h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}}\lambda T\gamma_{a_1,a_2}^*S_{2,j_2}y_{1,i_1,r_1}IT\gamma_{a_1,a_2,j_2}^*H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger}{\sqrt{3}}$	$-\frac{\sqrt{6}h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}L_{a_2,i_2}^{-r}CPL_{a_1,i_1}g_Y^2S_{2,j_2}IT\gamma_{a_1,a_2,j_2}^*S_{1,j_1}^\dagger}{M_F}$
								$-\frac{2\sqrt{\frac{2}{3}}h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{\lambda T\gamma_{a_3,a_1}}CPL_{a_4,i_2}^{-r}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_3}\epsilon_{a_2,a_4}H_{a_1}^\dagger H_{a_2}^\dagger}{M_F}$	$-\frac{h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}}\lambda T\gamma_{a_1,a_2}^*S_{2,j_2}y_{2,i_1,r_1}IT\gamma_{a_1,a_2,j_2}^*H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger}{\sqrt{3}}$	$-\frac{h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_1}}L_{a_2,i_2}^{-r}CPL_{a_1,i_1}S_{2,j_2}y_{1,i_3,r_1}IT\gamma_{a_1,a_2,j_2}^*S_{1,j_1}^\dagger}{2\sqrt{2}M_F}$
								$-\frac{h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}}H_{a_1}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}IT\gamma_{a_1,a_2,j_2}^*\lambda T\gamma_{a_1,a_2,j_2}^*S_{2,j_1}^\dagger}{\sqrt{3}}$	$-\frac{\sqrt{2}h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_1}}L_{a_2,i_2}^{-r}CPL_{a_1,i_1}S_{2,j_2}y_{1,i_3,r_1}IT\gamma_{a_1,a_2,j_2}^*S_{1,j_1}^\dagger}{3M_F}$	

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	F	H	ℓ
								S_1	S_2	F
										$S_1 S_2 F$
										$h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{1,i_2,r_2} y_{2,i_3,r_3} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{1,i_3,r_2} \mathcal{I} T \mathcal{I}^* a_{1,a_2,j_1,j_2} S_{1,j_1}^\dagger}$
										$2\sqrt{2} M_F$
										$h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_3,r_1} y_{2,i_1,r_2} y_{2,i_3,r_1} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{2,i_3,r_2} \mathcal{I} T \mathcal{I}^* a_{1,a_2,j_1,j_2} S_{1,j_1}^\dagger}$
										$2\sqrt{2} M_F$
										$h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_2} y_{2,i_3,r_1} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{2,i_3,r_2} \mathcal{I} T \mathcal{I}^* a_{1,a_2,j_1,j_2} S_{1,j_1}^\dagger}$
										$2\sqrt{2} M_F$
										$\sqrt{2} h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1} y_{2,i_3,r_2} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{2,i_3,r_2} \mathcal{I} T \mathcal{I}^* a_{1,a_2,j_1,j_2} S_{1,j_1}^\dagger}$
										$3 M_F$
										$2\sqrt{3} h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1} L_{a_2,i_2}^T C P_L L_{a_1,i_1} g_{22}^* S_{2,j_1} \epsilon_{a_1,a_2} S_{1,j_1}^\dagger}$
										M_F
										$h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{1,i_3,r_2} y_{2,i_2,r_1} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{1,i_3,r_1} \epsilon_{a_1,a_2} S_{1,j_1}^\dagger}$
										$4 M_F$
										$h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{1,i_3,r_2} y_{2,i_2,r_1} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{1,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}^\dagger}$
										$3 M_F$
										$h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{1,i_3,r_2} y_{2,i_2,r_1} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{1,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}^\dagger}$
										$4 M_F$
										$h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_3,r_1} y_{2,i_1,r_2} y_{2,i_2,r_1} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}^\dagger}$
										$4 M_F$
										$h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_2} y_{2,i_3,r_1} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}^\dagger}$
										$4 M_F$
										$h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1} y_{2,i_3,r_2} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}^\dagger}$
										$3 M_F$
										$h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1} y_{2,i_3,r_2} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}^\dagger}$
										$3 M_F$
										$h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1} y_{2,i_3,r_2} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}^\dagger}$
										$3 M_F$
										$h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1} y_{2,i_3,r_2} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}^\dagger}$
										$3 M_F$
										$h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1} y_{2,i_3,r_2} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}^\dagger}$
										$2\sqrt{2} M_F$
										$\sqrt{6} h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{L_{a_1,i_1} C P_R L_{a_2,i_2}^T g_{22}^* S_{1,j_2} y_{1,i_1,r_1} y_{2,i_2,r_1} \mathcal{I} T \mathcal{I}^* a_{1,a_2,j_2,j_1} S_{2,j_1}^\dagger}$
										M_F
										$h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{2,i_3,r_1} L$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	F	H	ℓ	S_1	S_2	F	ℓ
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1				$\frac{2\sqrt{\frac{3}{2}}h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}}\overline{\lambda_{T3}}S_{2,j_2}y_{1,i_1,r_1}I_{T3}a_{a_1,a_2,j_1,j_2}^*H_{a_1}{}^\dagger H_{a_2}{}^\dagger S_{1,j_1}{}^\dagger}{M_F}$ $\frac{2\sqrt{\frac{3}{2}}h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{\lambda_{T3}}L_{a_3,i_1}C\overline{P_R}L_{a_4,i_2}{}^t y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_3}\epsilon_{a_2,a_4}H_{a_1}{}^\dagger H_{a_2}{}^\dagger}{M_F}$	$-\frac{h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}}\overline{\lambda_{T3}}S_{2,j_2}y_{1,i_1,r_1}I_{T3}a_{a_1,a_2,j_1,j_2}^*H_{a_1}{}^\dagger H_{a_2}{}^\dagger S_{1,j_1}{}^\dagger}{\sqrt{3}}$ $-\frac{h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}}\overline{\lambda_{T3}}S_{2,j_2}y_{2,i_1,r_1}I_{T3}a_{a_1,a_2,j_1,j_2}^*H_{a_1}{}^\dagger H_{a_2}{}^\dagger S_{1,j_1}{}^\dagger}{\sqrt{3}}$ $-\frac{h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}}H_{a_1}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}I_{T3,a_1,a_2,j_2,j_1}\lambda_{T3}S_{2,j_1}{}^\dagger}{\sqrt{3}}$	$-\frac{\sqrt{6}h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}}\overline{y_{2,i_2,r_1}}L_{a_2,i_2}{}^T C\overline{P_L}L_{a_1,i_1}g_{S_2,j_2}^2I_{T3,a_1,a_2,j_1,j_2}S_{1,j_1}{}^\dagger}{M_F}$ $-\frac{3\sqrt{\frac{3}{2}}h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}}\overline{y_{2,i_2,r_1}}L_{a_2,i_2}{}^T C\overline{P_L}L_{a_1,i_1}g_{S_2,j_2}^2I_{T3,a_1,a_2,j_1,j_2}S_{1,j_1}{}^\dagger}{2M_F}$ $-\frac{h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}}y_{1,i_3,r_2}y_{2,i_2,r_1}L_{a_2,i_2}{}^T C\overline{P_L}L_{a_1,i_1}S_{2,j_2}y_{1,i_3,r_1}I_{T3,a_1,a_2,j_1,j_2}S_{1,j_1}{}^\dagger}{2\sqrt{2}M_F}$	
cont.											$-\frac{\sqrt{2}h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}}y_{1,i_3,r_2}y_{2,i_2,r_1}L_{a_2,i_2}{}^T C\overline{P_L}L_{a_1,i_1}S_{2,j_2}y_{1,i_3,r_2}I_{T3,a_1,a_2,j_1,j_2}S_{1,j_1}{}^\dagger}{3M_F}$ $-\frac{h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}}y_{1,i_3,r_2}y_{2,i_2,r_1}L_{a_2,i_2}{}^T C\overline{P_L}L_{a_1,i_1}S_{2,j_2}y_{1,i_3,r_2}I_{T3,a_1,a_2,j_1,j_2}S_{1,j_1}{}^\dagger}{2\sqrt{2}M_F}$ $-\frac{h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{1,i_3,r_1}}\overline{y_{2,i_2,r_2}}y_{1,i_2,r_1}L_{a_2,i_2}{}^T C\overline{P_L}L_{a_1,i_1}S_{2,j_2}y_{2,i_3,r_2}I_{T3,a_1,a_2,j_1,j_2}S_{1,j_1}{}^\dagger}{2\sqrt{2}M_F}$	$-\frac{\sqrt{2}h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}}y_{1,i_3,r_2}y_{2,i_2,r_1}L_{a_2,i_2}{}^T C\overline{P_L}L_{a_1,i_1}S_{2,j_2}y_{2,i_3,r_2}I_{T3,a_1,a_2,j_1,j_2}S_{1,j_1}{}^\dagger}{3M_F}$ $-\frac{2\sqrt{3}h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}}\overline{y_{2,i_2,r_2}}y_{1,i_2,r_1}L_{a_2,i_2}{}^T C\overline{P_L}L_{a_1,i_1}g_{S_2,j_2}^2\epsilon_{a_1,a_2}S_{1,j_1}{}^\dagger}{M_F}$ $-\frac{3\sqrt{3}h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}}y_{1,i_2,r_2}y_{1,i_2,r_2}L_{a_2,i_2}{}^T C\overline{P_L}L_{a_1,i_1}g_{S_2,j_2}^2\epsilon_{a_1,a_2}S_{1,j_1}{}^\dagger}{4M_F}$		
cont.											$-\frac{h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}}y_{1,i_3,r_2}y_{2,i_2,r_1}L_{a_2,i_2}{}^T C\overline{P_L}L_{a_1,i_1}S_{2,j_1}y_{1,i_3,r_1}\epsilon_{a_1,a_2}S_{1,j_1}{}^\dagger}{4M_F}$ $-\frac{h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}}y_{1,i_3,r_2}y_{2,i_2,r_1}L_{a_2,i_2}{}^T C\overline{P_L}L_{a_1,i_1}S_{2,j_1}y_{1,i_3,r_2}\epsilon_{a_1,a_2}S_{1,j_1}{}^\dagger}{4M_F}$ $-\frac{h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}}y_{1,i_2,r_2}y_{2,i_2,r_2}L_{a_2,i_2}{}^T C\overline{P_L}L_{a_1,i_1}S_{2,j_1}y_{1,i_2,r_2}\epsilon_{a_1,a_2}S_{1,j_1}{}^\dagger}{4M_F}$	$-\frac{h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}}y_{1,i_3,r_2}y_{2,i_2,r_1}L_{a_2,i_2}{}^T C\overline{P_L}L_{a_1,i_1}S_{2,j_1}y_{2,i_2,r_2}\epsilon_{a_1,a_2}S_{1,j_1}{}^\dagger}{4M_F}$ $-\frac{h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{1,i_3,r_1}}L_{a_1,i_1}C\overline{P_R}L_{a_2,i_2}{}^t S_{1,j_2}y_{1,i_1}y_{1,i_3,r_2}y_{2,i_2,r_1}I_{T3}a_{a_1,a_2,j_2,j_1}^*S_{2,j_1}{}^\dagger}{2\sqrt{2}M_F}$		
cont.											$-\frac{\sqrt{6}h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{L_{a_1,i_1}}C\overline{P_R}L_{a_2,i_2}{}^t g_{S_2,j_2}^2S_{1,j_2}y_{1,i_1}y_{1,i_2,r_2,r_1}I_{T3}a_{a_1,a_2,j_2,j_1}^*S_{2,j_1}{}^\dagger}{M_F}$ $-\frac{3\sqrt{\frac{3}{2}}h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{L_{a_1,i_1}}C\overline{P_R}L_{a_2,i_2}{}^t g_{S_2,j_2}^2S_{1,j_2}y_{1,i_1}y_{1,i_2,r_2,r_1}I_{T3}a_{a_1,a_2,j_2,j_1}^*S_{2,j_1}{}^\dagger}{2M_F}$ $-\frac{h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{2,i_3,r_1}}L_{a_1,i_1}C\overline{P_R}L_{a_2,i_2}{}^t S_{1,j_2}y_{1,i_3,r_2}y_{2,i_2,r_1}I_{T3}a_{a_1,a_2,j_2,j_1}^*S_{2,j_1}{}^\dagger}{2\sqrt{2}M_F}$	$-\frac{\sqrt{2}h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{1,i_3,r_1}}L_{a_1,i_1}C\overline{P_R}L_{a_2,i_2}{}^t S_{1,j_2}y_{1,i_1}y_{1,i_3,r_2}y_{2,i_2,r_2}I_{T3}a_{a_1,a_2,j_2,j_1}^*S_{2,j_1}{}^\dagger}{3M_F}$ $-\frac{\sqrt{2}h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{1,i_3,r_1}}L_{a_1,i_1}C\overline{P_R}L_{a_2,i_2}{}^t S_{1,j_2}y_{1,i_1}y_{1,i_3,r_2}y_{2,i_2,r_1}I_{T3}a_{a_1,a_2,j_2,j_1}^*S_{2,j_1}{}^\dagger}{3M_F}$ $-\frac{h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{2,i_3,r_1}}L_{a_1,i_1}C\overline{P_R}L_{a_2,i_2}{}^t S_{1,j_2}y_{1,i_1}y_{1,i_3,r_2}y_{2,i_2,r_1}I_{T3}a_{a_1,a_2,j_2,j_1}^*S_{2,j_1}{}^\dagger}{2\sqrt{2}M_F}$		
cont.											$+\frac{h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}}L_{a_1,i_1}C\overline{P_R}L_{a_2,i_2}{}^t S_{1,j_1}y_{1,i_2,r_1}y_{1,i_3,r_2}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}{}^\dagger}{4M_F}$ $-\frac{2\sqrt{3}h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{L_{a_1,i_1}}C\overline{P_R}L_{a_2,i_2}{}^t g_{S_2,j_2}^2S_{1,j_1}y_{1,i_1}y_{1,i_2,r_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}{}^\dagger}{M_F}$ $-\frac{3\sqrt{3}h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{L_{a_1,i_1}}C\overline{P_R}L_{a_2,i_2}{}^t g_{S_2,j_2}^2S_{1,j_1}y_{1,i_1}y_{1,i_2,r_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}{}^\dagger}{4M_F}$	$+\frac{h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{2,i_3,r_1}}L_{a_1,i_1}C\overline{P_R}L_{a_2,i_2}{}^t S_{1,j_1}y_{1,i_3,r_2}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}{}^\dagger}{4M_F}$ $-\frac{h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{2,i_3,r_1}}L_{a_1,i_1}C\overline{P_R}L_{a_2,i_2}{}^t S_{1,j_1}y_{1,i_1}y_{1,i_3,r_2}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}{}^\dagger}{3M_F}$ $-\frac{h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{1,i_3,r_1}}L_{a_1,i_1}C\overline{P_R}L_{a_2,i_2}{}^t S_{1,j_1}y_{1,i_1}y_{1,i_2,r_2}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}{}^\dagger}{3M_F}$		
cont.											$-\frac{h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{2,i_3,r_1}}L_{a_1,i_1}C\overline{P_R}L_{a_2,i_2}{}^t S_{1,j_1}y_{1,i_1}y_{1,i_2,r_2}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}{}^\dagger}{4M_F}$ $-\frac{\sqrt{\frac{3}{2}}h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}}\overline{y_{2,i_2,r_1}}L_{a_2,i_2}{}^T C\overline{P_L}L_{a_1,i_1}S_{2,j_2}I_{T3,a_1,a_2,j_2,j_2}S_{1,j_1}{}^\dagger\lambda_{12}^{(1)}}{M_F}$ $-\frac{h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}}\overline{y_{2,i_2,r_1}}L_{a_2,i_2}{}^T C\overline{P_L}L_{a_1,i_1}S_{2,j_1}\epsilon_{a_1,a_2}S_{1,j_1}{}^\dagger\lambda_{12}^{(1)}}{\sqrt{3}M_F}$	$+\frac{\sqrt{\frac{3}{2}}h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{L_{a_1,i_1}}C\overline{P_R}L_{a_2,i_2}{}^t S_{1,j_2}y_{1,i_1}y_{1,i_2,r_2,r_1}I_{T3}a_{a_1,a_2,j_2,j_1}^*S_{2,j_1}{}^\dagger\lambda_{12}^{(1)}}{M_F}$ $-\frac{h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{L_{a_1,i_1}}C\overline{P_R}L_{a_2,i_2}{}^t S_{1,j_1}y_{1,i_1}y_{1,i_2,r_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}{}^\dagger\lambda_{12}^{(1)}}{\sqrt{3}M_F}$		
cont.														

[illegible]

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	F	H	ℓ
										$S_1 S_2 F$
										$S_1 S_2 F \ell$
										$\frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_3,r_1} L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_1,r_2} y_{1,i_2,r_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{2,j_1}^\dagger}{4M_F}$
cont.										$\frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{2,i_3,r_1} L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_1,r_2} y_{2,i_2,r_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{2,j_1}^\dagger}{4M_F}$
										$+ \frac{\sqrt{\frac{3}{2}} h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1} L_{a_2,i_2}}^T C P_L L_{a_3,i_1} S_{2,j_2} \mathcal{I} T 3_{a_1,a_2,j_1,j_2} S_{1,j_1}^\dagger \lambda_{12}^{(1)}}{M_F}$
										$- \frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1} L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_{2,j_1} \epsilon_{a_1,a_2} S_{1,j_1}^\dagger \lambda_{12}^{(1)}}{\sqrt{3} M_F}$
cont.										$+ \frac{\sqrt{\frac{3}{2}} h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_2} y_{1,i_1,r_1} y_{2,i_2,r_1} \mathcal{I} T 3_{a_1,a_2,j_2,j_1}^\dagger S_{2,j_1}^\dagger \lambda_{12}^{(1)}}{M_F}$
										$- \frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_2} S_{2,j_1}^\dagger \lambda_{12}^{(1)}}{\sqrt{3} M_F}$
										$3 \sqrt{\frac{2}{11}} h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1} L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_{2,j_2} \mathcal{I} T 3_{a_1,a_2,j_1,j_2} S_{1,j_1}^\dagger \lambda_{12}^{(2)}$
cont.										$- \frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1} L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_{2,j_1} \epsilon_{a_1,a_2} S_{1,j_1}^\dagger \lambda_{12}^{(2)}}{\sqrt{11} M_F}$
										$3 \sqrt{\frac{2}{11}} h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_2} y_{1,i_1,r_1} y_{2,i_2,r_1} \mathcal{I} T 3_{a_1,a_2,j_2,j_1}^\dagger S_{2,j_1}^\dagger \lambda_{12}^{(2)}$
										$- \frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_2} S_{2,j_1}^\dagger \lambda_{12}^{(2)}}{\sqrt{11} M_F}$
cont.										$7 h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1} L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_{2,j_1} \epsilon_{a_1,a_2} S_{1,j_1}^\dagger \lambda_{12}^{(3)}$
										$- \frac{7 h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_2} S_{2,j_1}^\dagger \lambda_{12}^{(3)}}{\sqrt{23} M_F}$
										$7 h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_2} S_{2,j_1}^\dagger \lambda_{12}^{(3)}$
										$- \frac{7 h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_2} S_{2,j_1}^\dagger \lambda_{12}^{(3)}}{\sqrt{23} M_F}$

Field combinations 10 of 10

[illegible]

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$F H \ell e$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$+\frac{1}{4}h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_3,r_1}L_{a_1,i_2}}P_Re_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}$
								$+\frac{1}{4}h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_3,r_1}L_{a_1,i_2}}P_Re_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}$
cont.								$+\frac{1}{4}h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}Y_{e,i_3,i_2}\bar{e}_{i_2}}P_L L_{a_1,i_1}y_{1,i_3,r_1}H_{a_1}^\dagger$
								$+\frac{1}{4}h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}Y_{e,i_3,i_2}\bar{e}_{i_2}}P_L L_{a_1,i_1}y_{2,i_3,r_1}H_{a_1}^\dagger$
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	$+\frac{1}{4}h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_3,r_1}L_{a_1,i_2}}P_Re_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}$
								$+\frac{1}{4}h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_3,r_1}L_{a_1,i_2}}P_Re_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}$
cont.								$+\frac{1}{4}h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}Y_{e,i_3,i_2}\bar{e}_{i_2}}P_L L_{a_1,i_1}y_{1,i_3,r_1}H_{a_1}^\dagger$
								$+\frac{1}{4}h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}Y_{e,i_3,i_2}\bar{e}_{i_2}}P_L L_{a_1,i_1}y_{2,i_3,r_1}H_{a_1}^\dagger$
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	$+\frac{1}{4}h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_3,r_1}L_{a_1,i_2}}P_Re_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}$
								$+\frac{1}{4}h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_3,r_1}L_{a_1,i_2}}P_Re_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}$
cont.								$+\frac{1}{4}h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}Y_{e,i_3,i_2}\bar{e}_{i_2}}P_L L_{a_1,i_1}y_{1,i_3,r_1}H_{a_1}^\dagger$
								$+\frac{1}{4}h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}Y_{e,i_3,i_2}\bar{e}_{i_2}}P_L L_{a_1,i_1}y_{2,i_3,r_1}H_{a_1}^\dagger$
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	$+\frac{1}{4}\sqrt{3}h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_3,r_1}L_{a_1,i_2}}P_Re_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}$
								$+\frac{1}{4}\sqrt{3}h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_3,r_1}L_{a_1,i_2}}P_Re_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}$
cont.								$+\frac{1}{4}\sqrt{3}h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}Y_{e,i_3,i_2}\bar{e}_{i_2}}P_L L_{a_1,i_1}y_{1,i_3,r_1}H_{a_1}^\dagger$
								$+\frac{1}{4}\sqrt{3}h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}Y_{e,i_3,i_2}\bar{e}_{i_2}}P_L L_{a_1,i_1}y_{2,i_3,r_1}H_{a_1}^\dagger$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+\frac{1}{4}\sqrt{3}h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_3,r_1}L_{a_1,i_2}}P_Re_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}$
								$+\frac{1}{4}\sqrt{3}h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_3,r_1}L_{a_1,i_2}}P_Re_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}$
cont.								$+\frac{1}{4}\sqrt{3}h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}Y_{e,i_3,i_2}\bar{e}_{i_2}}P_L L_{a_1,i_1}y_{1,i_3,r_1}H_{a_1}^\dagger$
								$+\frac{1}{4}\sqrt{3}h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}Y_{e,i_3,i_2}\bar{e}_{i_2}}P_L L_{a_1,i_1}y_{2,i$

[illegible]

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	F	H	ℓ	e
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$+\frac{1}{4}\sqrt{3}h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\frac{\overline{y_{1,i_3,r_1}L_{a_1,i_2}}}{\overline{y_{1,i_3,r_1}L_{a_1,i_2}}}P_R e_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}$			
								$+\frac{1}{4}\sqrt{3}h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\frac{\overline{y_{2,i_3,r_1}L_{a_1,i_2}}}{\overline{y_{2,i_3,r_1}L_{a_1,i_2}}}P_R e_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}$			
cont.								$+\frac{1}{4}\sqrt{3}h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\frac{\overline{y_{1,i_1,r_1}Y_{e,i_3,i_2}}}{\overline{y_{1,i_1,r_1}Y_{e,i_3,i_2}}}\overline{e_{i_2}}P_L L_{a_1,i_1}y_{1,i_3,r_1}H_{a_1}^\dagger$			
								$+\frac{1}{4}\sqrt{3}h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\frac{\overline{y_{2,i_1,r_1}Y_{e,i_3,i_2}}}{\overline{y_{2,i_1,r_1}Y_{e,i_3,i_2}}}\overline{e_{i_2}}P_L L_{a_1,i_1}y_{2,i_3,r_1}H_{a_1}^\dagger$			
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+\frac{1}{4}\sqrt{3}h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\frac{\overline{y_{1,i_3,r_1}L_{a_1,i_2}}}{\overline{y_{1,i_3,r_1}L_{a_1,i_2}}}P_R e_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}$			
								$+\frac{1}{4}\sqrt{3}h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\frac{\overline{y_{2,i_3,r_1}L_{a_1,i_2}}}{\overline{y_{2,i_3,r_1}L_{a_1,i_2}}}P_R e_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}$			
cont.								$+\frac{1}{4}\sqrt{3}h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\frac{\overline{y_{1,i_1,r_1}Y_{e,i_3,i_2}}}{\overline{y_{1,i_1,r_1}Y_{e,i_3,i_2}}}\overline{e_{i_2}}P_L L_{a_1,i_1}y_{1,i_3,r_1}H_{a_1}^\dagger$			
								$+\frac{1}{4}\sqrt{3}h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\frac{\overline{y_{2,i_1,r_1}Y_{e,i_3,i_2}}}{\overline{y_{2,i_1,r_1}Y_{e,i_3,i_2}}}\overline{e_{i_2}}P_L L_{a_1,i_1}y_{2,i_3,r_1}H_{a_1}^\dagger$			
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$+\frac{1}{4}\sqrt{3}h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\frac{\overline{y_{1,i_3,r_1}L_{a_1,i_2}}}{\overline{y_{1,i_3,r_1}L_{a_1,i_2}}}P_R e_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}$			
								$+\frac{1}{4}\sqrt{3}h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\frac{\overline{y_{2,i_3,r_1}L_{a_1,i_2}}}{\overline{y_{2,i_3,r_1}L_{a_1,i_2}}}P_R e_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}$			
cont.								$+\frac{1}{4}\sqrt{3}h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\frac{\overline{y_{1,i_1,r_1}Y_{e,i_3,i_2}}}{\overline{y_{1,i_1,r_1}Y_{e,i_3,i_2}}}\overline{e_{i_2}}P_L L_{a_1,i_1}y_{1,i_3,r_1}H_{a_1}^\dagger$			
								$+\frac{1}{4}\sqrt{3}h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\frac{\overline{y_{2,i_1,r_1}Y_{e,i_3,i_2}}}{\overline{y_{2,i_1,r_1}Y_{e,i_3,i_2}}}\overline{e_{i_2}}P_L L_{a_1,i_1}y_{2,i_3,r_1}H_{a_1}^\dagger$			
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	$+\frac{1}{4}\sqrt{3}h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\frac{\overline{y_{1,i_3,r_1}L_{a_1,i_2}}}{\overline{y_{1,i_3,r_1}L_{a_1,i_2}}}P_R e_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}$			
								$+\frac{1}{4}\sqrt{3}h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\frac{\overline{y_{2,i_3,r_1}L_{a_1,i_2}}}{\overline{y_{2,i_3,r_1}L_{a_1,i_2}}}P_R e_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}$			
cont.								$+\frac{1}{4}\sqrt{3}h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\frac{\overline{y_{1,i_1,r_1}Y_{e,i_3,i_2}}}{\overline{y_{1,i_1,r_1}Y_{e,i_3,i_2}}}\overline{e_{i_2}}P_L L_{a_1,i_1}y_{1,i_3,r_1}H_{a_1}^\dagger$			
								$+\frac{1}{4}\sqrt{3}h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\frac{\overline{y_{2,i_1,r_1}Y_{e,i_3,i_2}}}{\overline{y_{2,i_1,r_1}Y_{e,i_3,i_2}}}\overline{e_{i_2}}P_L L_{a_1,i_1}y_{2,i_3,r_1}H_{a_1}^\dagger$			